

**FANSHAWE**

Process Engineering

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# Introduction

Why are we studying this?

- Defining and identifying manufacturing processes is critical to understanding the role that automation plays and how to program an industrial process line.
- What are some examples of a manufacturing process?
- What do we have or use that is manufactured?
- What is the biggest manufacturing change that you have experienced so far?
- Processes have changed throughout the centuries and have been driven by the four Industrial Revolutions.
- First Industrial Revolution – Coal in 1765
  - This transformed society from an agricultural base to an industrial base. Processes became mechanized and rather than hand made, products were manufactured. The discovery of coal and the development of the steam engine and metal forging transformed the way goods were created.
- Second Industrial Revolution – Gas in 1870 \*The discovery of oil and electricity fueled the second industrial revolution. The discovery of oil led to the invention of the combustion engine and new steel and chemical based products entered the manufacturing streams. Technological advances in communication were driven by the invention of the telegraph and later the telephone. Modes of transportation developed with the invention of the airplane and automobile. These discoveries helped to drive the advent of mass production which forever changed how items were produced.
- Third Industrial Revolution – Electronics and Nuclear Energy in 1969 *The third industrial revolution is dominated by the development of electronics and the beginning of the age of computers.* Technological developments were centered around four pillars of science: telecommunications, biotechnology, information technology and energy engineering.

- Fourth Industrial Revolution – Internet and AI
  - The fourth industrial revolution describes the current era of digitalization, automation, and advanced manufacturing technology. It's characterized by the convergence of emerging technologies, such as robotics, the Internet of Things (IoT), and 3D printing.

Check out these videos:

**The Four Industrial Revolutions:** <https://www.youtube.com/watch?v=kpW9JcWxKq0>

and

**Industry 4.0 Explained:** <https://www.youtube.com/watch?v=okXk4Bnz2Lc>

What is a Fifth Industrial Revolution going to look like?

**The Fifth Industrial Revolution:** [https://www.youtube.com/watch?v=sGI2FFm\\_8tw](https://www.youtube.com/watch?v=sGI2FFm_8tw)

Further Reading Whitepaper 'What is Manufacturing' by Intel

| Category                  | Grade Item          | Type               | Max. Points | Weight (%) |
|---------------------------|---------------------|--------------------|-------------|------------|
| <b>Lab-Tutorial Labs</b>  | <b>Lab-Tutorial</b> | <b>Association</b> | <b>48</b>   | <b>40</b>  |
|                           | Lab-1               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-2               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-3               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-4               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-5               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-6               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-7               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-8               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-9               | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-10              | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-11              | NumericSubmissions | 48.33       | 8.33       |
|                           | Lab-12              | NumericSubmissions | 48.33       | 8.33       |
| <b>Assignments</b>        | <b>Assignments</b>  |                    | <b>12</b>   | <b>12</b>  |
|                           | Assignment-1        | NumericQuizzes     | 2           | 16.67      |
|                           | Assignment-2        | NumericQuizzes     | 2           | 16.67      |
|                           | Assignment-3        | NumericQuizzes     | 2           | 16.67      |
|                           | Assignment-4        | NumericQuizzes     | 2           | 16.67      |
|                           | Assignment-5        | NumericQuizzes     | 2           | 16.66      |
|                           | Assignment-6        | NumericQuizzes     | 2           | 16.66      |
| <b>Lab Pop-up Quizzes</b> | <b>Lab Quizes</b>   |                    | <b>3</b>    | <b>3</b>   |
|                           | Lab Quiz-1          | NumericQuizzes     | 0.5         | 16.66      |
|                           | Lab Quiz-2          | NumericQuizzes     | 0.5         | 16.66      |



| Category                      | Grade Item             | Type                  | Max. Points | Weight (%) |
|-------------------------------|------------------------|-----------------------|-------------|------------|
| <b>Lecture Pop-up Quizzes</b> | Lab Quiz-3             | NumericQuizzes        | 0.5         | 16.67      |
|                               | Lab Quiz-4             | NumericQuizzes        | 0.5         | 16.67      |
|                               | Lab Quiz-5             | NumericQuizzes        | 0.5         | 16.67      |
|                               | Lab Quiz-6             | NumericQuizzes        | 0.5         | 16.67      |
|                               | <b>Lecture Quizzes</b> |                       | <b>3</b>    | <b>3</b>   |
|                               | Lecture Quiz-1         | NumericQuizzes        | 0.5         | 16.66      |
|                               | Lecture Quiz-2         | NumericQuizzes        | 0.5         | 16.66      |
|                               | Lecture Quiz-3         | NumericQuizzes        | 0.5         | 16.67      |
|                               | Lecture Quiz-4         | NumericQuizzes        | 0.5         | 16.67      |
|                               | Lecture Quiz-5         | NumericQuizzes        | 0.5         | 16.67      |
|                               | Lecture Quiz-6         | NumericQuizzes        | 0.5         | 16.67      |
| <b>Midterm Test</b>           | <b>Mid Term Test</b>   | <b>NumericQuizzes</b> | <b>10.5</b> | <b>17</b>  |
| <b>Final Test</b>             | Final Test             | NumericQuizzes        | 50          | 25         |
| <b>Calculated Grades</b>      | Midterm Grade          | Calculated            | —           | —          |
|                               | Final Calculated Grade | —                     | 156.5       | 100        |

## Labs

- Late penalty is 25% per lab session.
- Labs 1-5 must be completed before Week 7.
- Bring your lab kit and PPE to every lab session.

## Assignments

- 6 Assignments worth 2% each, for a total of 12%.
- All are FOL quizzes, with at least one week open.

## Quizzes

- 6 Lecture Quizzes and 6 Lab Quizzes.
- Quizzes are worth 0.5% each, for a total of 6%.
- Quizzes are given randomly throughout the term as a “pop quiz”.
- Quizzes are given at the start of the lab session (10 minute log in window) or at the end of the lecture.
- **You must be in attendance to write the quizzes.**

## Tests

- During in-person lecture time (90 minutes). One midterm test worth 17%, one final test worth 25%.
- Both tests are cumulative and online, in-person.
- Closed-book with handwritten cheat sheet; details will be posted on FOL closer to the date.
- Kit calculators allowed.

## Other Grade Considerations

In order to pass, two conditions must be met: \* Overall passing grade (50%). \* Attendance and submission of the mandatory labs (labs 1 through 10)

## How to succeed

- Come to class and lab sessions Participate
- Complete examples given in class
- Do the homework and assigned reading
- Attend office hours

## Cheating and Plagiarism

Students must write their reports and assignments in their **own words**. Whenever students take an idea or a passage from another author, they must acknowledge their debt both by using quotation marks where appropriate and by proper referencing such as footnotes or citations. This includes non-published sources, such as work completed in previous years or a classmate's work. **This includes the use of "AI" Technologies.**

## FOL

- Lectures will be posted after class, as well as supplementary material.
- Assignments and lab manuals will be posted ahead of time. You will always have at least one week's notice.
- All announcements will be made through the FOL course.

## Accessibility

Please contact me if you require material in an alternate format or if any other arrangements can make this course more accessible to you.

Consider contacting Accessibility Services for specifics on accommodations that you may be eligible for. M-Fr 8:30-4:30 F2010, 519-452-4282, [accessibility@fanshawec.ca](mailto:accessibility@fanshawec.ca)

## **Fanshawe's Recording Policy**

It is forbidden for students to record without written permission from the instructor (and) as an approved accommodation. Recording without permission or sharing a recording will result in disciplinary action.

If a class is being recorded with permission: it will be announced, and alternate means to participation will be made available to protect students' privacy.



# Chapter 1

## Types of Manufacturing Processes

---

### 1.1 Learning Objectives

By the end of this chapter, you will be able to:

1. Identify and classify the seven main types of manufacturing processes
  2. Select appropriate manufacturing processes based on material, volume, and precision requirements
  3. Compare cost, speed, and quality trade-offs between process types
  4. Recognize manufacturing processes used in automotive, food, and defense industries
  5. Apply process selection criteria to real-world manufacturing scenarios
- 

### 1.2 Introduction to Manufacturing Processes

Manufacturing is defined as the conversion of raw materials into finished goods on a large scale using man and machine. **Manufacturing processes** are the specific methods used for this conversion. Based on product requirements, different types of manufacturing processes are selected to achieve the required output with optimal cost, quality, and efficiency.

### Seven Main Categories of Manufacturing Processes

Each category contains multiple specific processes for different applications

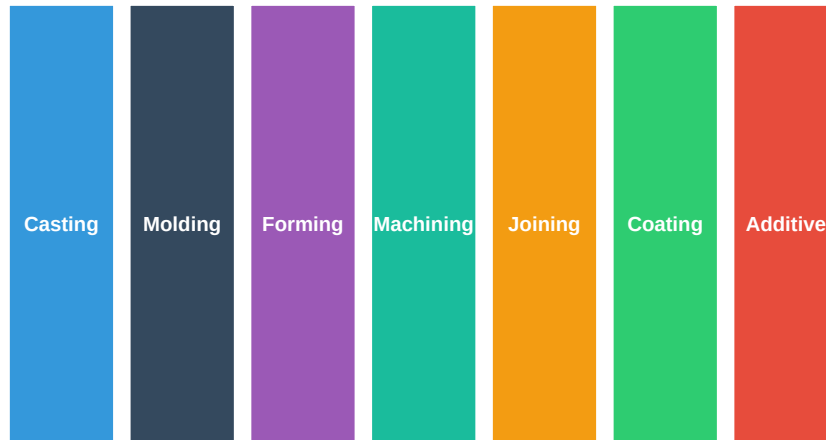


Figure 1.1: Overview of Manufacturing Process Categories

Why do we need different manufacturing processes?

Different products require different manufacturing approaches based on:

- **Material type:** Metals, plastics, ceramics, composites
- **Production volume:** One-off prototypes vs. millions of units
- **Geometric complexity:** Simple shapes vs. intricate details
- **Precision requirements:** Loose tolerances vs. micron-level accuracy
- **Cost constraints:** Budget limitations affect process selection
- **Lead time:** How quickly parts are needed

No single process is optimal for all situations!

A good video summarizing 6 of the 7 manufacturing processes is linked below (Coating/Plating is not covered):

**Manufacturing Processes Overview:** [https://www.youtube.com/watch?v=Um\\_g8sQ\\_p3Y](https://www.youtube.com/watch?v=Um_g8sQ_p3Y)

---

## 1.3 Process Selection Framework

Before examining each process type, it's important to understand how engineers select the right manufacturing process. The following visualization shows the key factors:

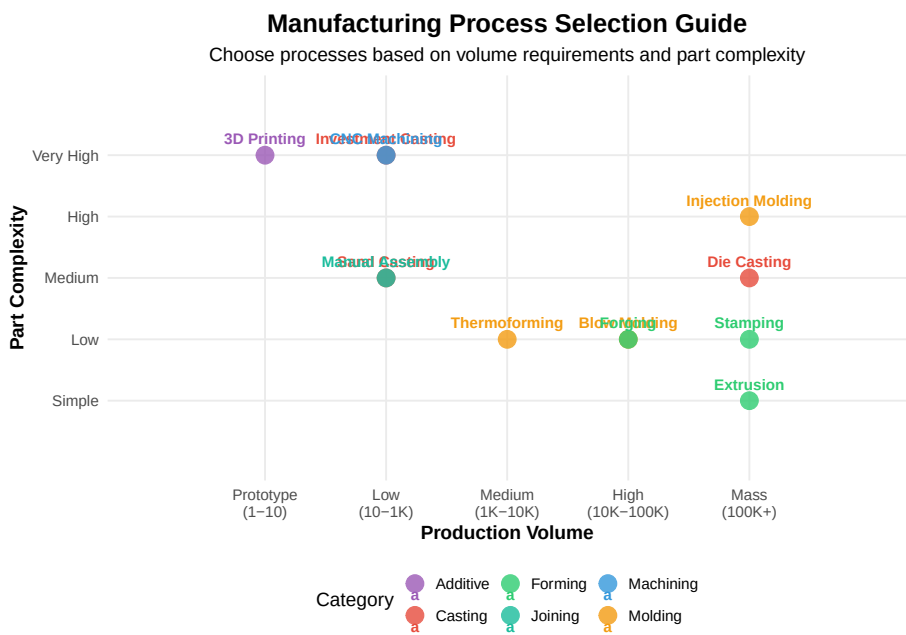


Figure 1.2: Process Selection Based on Volume and Complexity

Table 1.1: Manufacturing Process Comparison Summary

| Process Type | Typical Materials          | Production Volume   | Typical Tolerance | Tooling Cost | Unit Cost |
|--------------|----------------------------|---------------------|-------------------|--------------|-----------|
| Casting      | Metals, alloys             | Low to High         | ±0.5-2mm          | Medium-High  | Low       |
| Molding      | Plastics, rubber           | Medium to Very High | ±0.1-0.5mm        | High         | Very Low  |
| Forming      | Metals, sheet              | Medium to Very High | ±0.1-1mm          | Medium-High  | Low       |
| Machining    | All materials              | Low to Medium       | ±0.01-0.1mm       | Low          | High      |
| Joining      | All materials              | All volumes         | Varies            | Low          | Medium    |
| Coating      | All materials              | All volumes         | N/A               | Low          | Low       |
| Additive     | Plastics, metals, ceramics | Prototype to Low    | ±0.1-0.3mm        | None         | High      |

1.4 Casting Processes

**Casting** is a process of pouring liquid metal into a mold containing the hollow shape of the desired outcome. It uses sprue, gates, and runners to direct the molten metal flow.

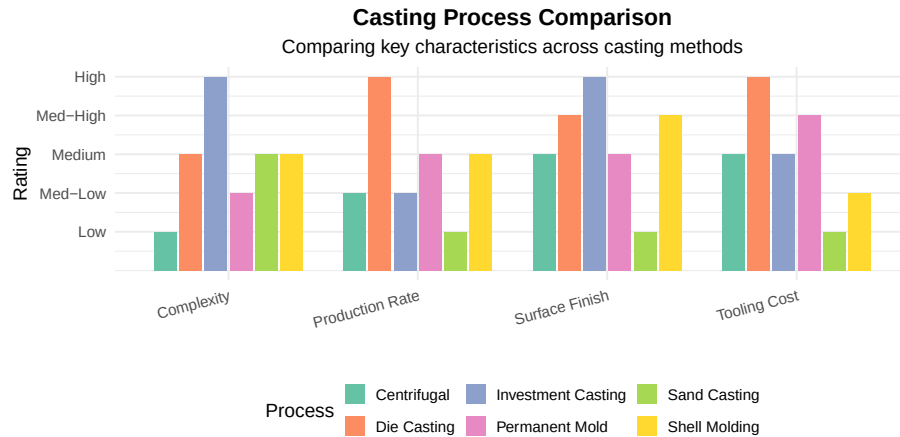


Figure 1.3: Comparison of Casting Processes

### 1.4.1 Sand Casting

Sand casting is a metal casting process that uses sand as the mold material. This process has a low production rate as the sand mold must be destroyed to remove the part. However, it is economical for low-volume production.

#### Process Steps:

1. Create a pattern (usually wood or metal)
2. Pack sand around the pattern to form the mold
3. Remove the pattern, leaving a cavity
4. Pour molten metal into the cavity
5. Allow solidification
6. Break the sand mold to remove the casting

**Advantages:** Low tooling cost, can produce very large parts, suitable for all metals

**Disadvantages:** Poor surface finish, loose tolerances, slow cycle time

**Examples:** Engine blocks, machine tool bases, cylinder heads, pump housings, large gears

Discussion: Why is sand casting still widely used despite its limitations?

Sand casting remains popular because:

1. **Low startup cost** - A wood pattern costs hundreds vs. steel dies costing tens of thousands
2. **Flexibility** - Easy to modify patterns for design changes
3. **Size capability** - Can cast parts weighing several tons
4. **Material versatility** - Works with nearly any metal



5. **Short lead time** - Parts can be produced in days vs. weeks for die casting

For prototypes and low-volume production (< 500 parts), sand casting is often the most economical choice even with its lower quality.

### 1.4.2 Die Casting

Die casting is used for producing metal parts where high precision is required. Molten metal is forced under high pressure into reusable metal dies, enabling high-volume production.

**Die Casting Process:** <https://www.youtube.com/watch?v=0oibUY8KUQM>  
**Process:**

1. Steel molds (dies) are machined to the part shape
2. Dies are mounted in a die casting machine
3. Molten metal is injected under high pressure (10-175 MPa)
4. Metal solidifies rapidly due to die cooling
5. Part is ejected and the cycle repeats

**Advantages:** Excellent dimensional accuracy, smooth surface finish, high production rates (up to 400 shots/hour)

**Disadvantages:** High die cost (\$50,000-\$500,000), limited to non-ferrous metals, porosity issues

**Examples:** Engine components, transmission housings, electronic enclosures, power tool housings

### 1.4.3 Investment Casting (Lost-Wax)

Investment casting produces complex, precision parts by creating a wax pattern that is “invested” (surrounded) with ceramic slurry, then melted out to leave a mold cavity.

**Investment Casting Process:** <https://www.youtube.com/watch?v=QtxVdC7pBQM>

**Advantages:** Excellent surface finish, tight tolerances ( $\pm 0.1\%$ ), complex geometries, any castable metal

**Disadvantages:** Expensive, slow cycle time, limited part size

**Examples:** Turbine blades, aerospace components, medical implants, jewelry, firearm components

Why do aerospace companies pay premium prices for investment casting?

Investment casting is essential for aerospace because:

1. **Complex internal passages** - Turbine blades have intricate cooling channels

2. **Superior metallurgy** - Controlled solidification produces better grain structure
3. **Near-net shape** - Minimizes machining of expensive superalloys
4. **Weight savings** - Can create hollow structures impossible with other methods
5. **Material options** - Works with titanium, Inconel, and other aerospace alloys

A single turbine blade might cost \$1,000-\$5,000, but there's no other process that can produce the required geometry and performance.

#### 1.4.4 Centrifugal Casting

Centrifugal casting spins molten metal in a rotating die, using centrifugal force to distribute the metal and float impurities to the inner surface.

**Applications:** Pipes, tubes, cylinder liners, bushings, rings

#### 1.4.5 Permanent Mold Casting

Uses reusable metal molds with gravity or low-pressure filling. Good for medium-volume production of aluminum and copper alloy parts.

**Applications:** Pistons, gear housings, wheels, pipe fittings

#### 1.4.6 Shell Molding

Uses resin-coated sand that forms a thin, strong shell around a heated pattern. Better dimensional accuracy than sand casting.

**Applications:** Gear housings, cylinder heads, connecting rods

---

### 1.5 Molding Processes

**Molding** uses a rigid frame to shape hot liquid or ductile raw material. It is predominantly used for plastics and rubber products.

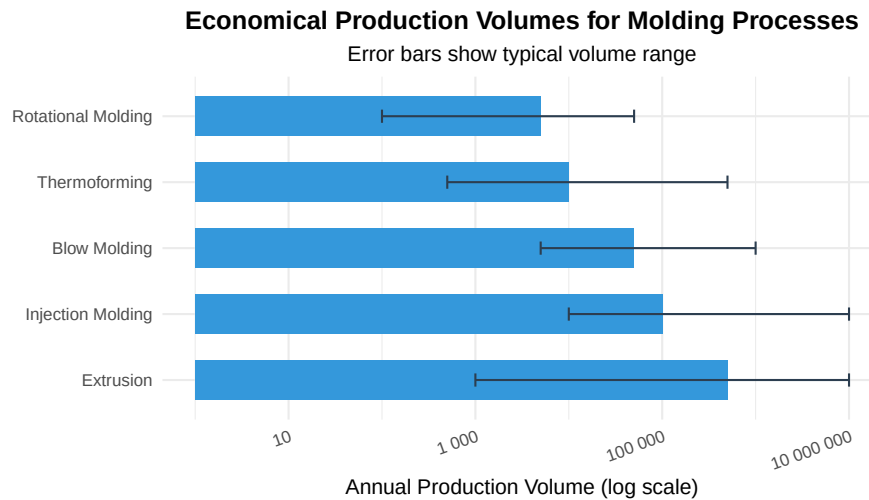


Figure 1.4: Molding Process Selection by Production Volume

### 1.5.1 Injection Molding

Injection molding is the most common process for manufacturing plastic parts. Plastic pellets are melted and injected under high pressure into a mold cavity.

**Injection Molding Process:** <https://www.youtube.com/watch?v=RMjtmr3CqA>

**Process Cycle:**

1. **Clamping** - Mold halves are clamped together
2. **Injection** - Molten plastic is injected into the cavity
3. **Cooling** - Part solidifies in the mold
4. **Ejection** - Part is pushed out by ejector pins

**Cycle times:** 10-60 seconds typical, as fast as 2 seconds for small parts

**Examples:** Automotive dashboards, phone cases, medical devices, food containers, toys

Live cell — try different volumes, mold cost, or cycle time.

```
#| label: injection-molding-cost
# Injection Molding Cost Analysis Example
# Comparing costs at different production volumes

mold_cost <- 50000 # Typical mold cost in dollars
material_cost_per_kg <- 2.50 # ABS plastic
part_weight_kg <- 0.1
cycle_time_sec <- 30
```

```

machine_rate_per_hour <- 75 # Machine + operator cost

# Calculate cost per part at different volumes
volumes <- c(1000, 10000, 50000, 100000, 500000)

cost_analysis <- data.frame(
  Volume = volumes,
  Mold_Per_Part = mold_cost / volumes,
  Material = part_weight_kg * material_cost_per_kg,
  Processing = (cycle_time_sec / 3600) * machine_rate_per_hour
)

cost_analysis$Total_Per_Part <- cost_analysis$Mold_Per_Part +
                                cost_analysis$Material +
                                cost_analysis$Processing

cat("Injection Molding Cost Breakdown:\n")
cat("=====\n")
for(i in 1:nrow(cost_analysis)) {
  cat(sprintf("At %s parts: %.2f per part (Tooling: %.2f, Material: %.2f, Processing: %.2f)\n",
    format(cost_analysis$Volume[i], big.mark = " "),
    cost_analysis$Total_Per_Part[i],
    cost_analysis$Mold_Per_Part[i],
    cost_analysis$Material[i],
    cost_analysis$Processing[i]))
}

```

Discussion: When does injection molding become cost-effective vs. 3D printing?

#### Break-even analysis:

- 3D printing: ~\$5-20 per part (no tooling, slow production)
- Injection molding: \$50,000 tooling + \$0.50-2 per part

**Break-even point:** Approximately 5,000-10,000 parts

**Rule of thumb:** - < 500 parts: 3D printing or machining - 500-5,000 parts: Consider soft tooling (aluminum molds) - > 5,000 parts: Steel injection molds become economical

### 1.5.2 Blow Molding

Blow molding creates hollow plastic parts by inflating a heated plastic tube (parison) inside a mold cavity.

**Types:** - **Extrusion blow molding** - Parison extruded, then blown - **Injection blow molding** - Preform injection molded, then blown - **Stretch blow molding** - Preform stretched and blown (PET bottles)

**Examples:** Bottles, containers, fuel tanks, automotive ducts, toys

### 1.5.3 Rotational Molding

Creates hollow parts by rotating a mold on two axes while heating, causing plastic powder to melt and coat the interior surface.

**Advantages:** Low tooling cost, no internal stresses, large hollow parts

**Examples:** Kayaks, playground equipment, tanks, coolers

### 1.5.4 Thermoforming

Heats a plastic sheet until pliable, then forms it over a mold using vacuum, pressure, or mechanical force.

**Examples:** Packaging (blister packs), disposable cups, refrigerator liners

### 1.5.5 Powder Metallurgy

Though involving metal, powder metallurgy uses molding principles. Metal powders are compacted in dies and sintered (heated below melting point) to fuse particles.

**Advantages:** Near-net shape, complex geometries, controlled porosity

**Examples:** Self-lubricating bearings, gears, filters, magnets

---

## 1.6 Forming Processes

**Forming** uses mechanical forces (compression, tension, shear) to shape material without removing any material. This preserves material grain structure and can improve mechanical properties.

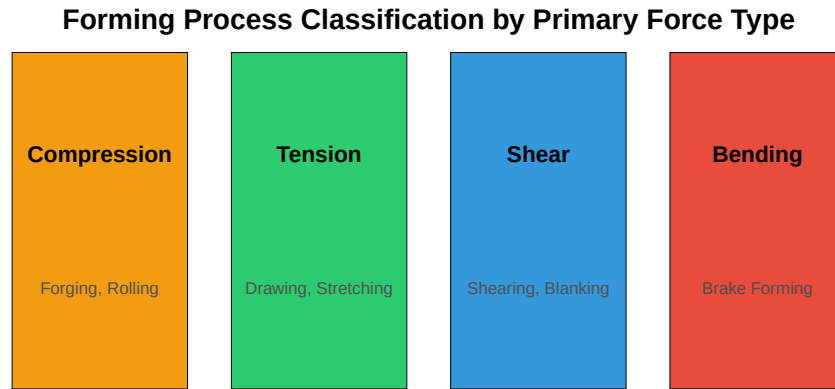


Figure 1.5: Types of Forming Forces

### 1.6.1 Forging

Forging uses compressive forces from hammers or presses to shape heated metal. Forged parts have superior strength due to aligned grain structure.

**Forging Process:** <https://www.youtube.com/watch?v=o4vQJnPwgiA>

**Types of Forging:** - **Open-die forging** - Simple shapes, large parts - **Closed-die (impression) forging** - Complex shapes, high volume - **Cold forging** - Room temperature, excellent finish - **Hot forging** - Heated material, easier deformation

**Examples:** Crankshafts, connecting rods, gears, aircraft structural components, hand tools

Why are aircraft landing gear forgings so expensive?

Aircraft landing gear forgings can cost \$100,000+ because:

1. **Material** - High-strength steel or titanium alloys
2. **Size** - Large parts require massive presses (50,000+ tons)
3. **Certification** - Extensive testing and documentation
4. **Grain flow** - Must be optimized for fatigue resistance
5. **NDT requirements** - 100% ultrasonic and magnetic particle inspection
6. **Low volume** - Only a few hundred per aircraft type per year

The forged grain structure provides 20-40% better fatigue life than equivalent castings or machined parts.

### 1.6.2 Stamping

Stamping uses dies and presses to form, cut, or shape sheet metal. Modern stamping lines can produce 1,000+ parts per hour.

**Operations include:** - **Blanking** - Cutting flat shapes from sheet - **Piercing** - Punching holes - **Drawing** - Forming cup shapes - **Bending** - Creating angles - **Coining** - Fine detail forming

**Examples:** Automotive body panels, appliance housings, electronic enclosures

### 1.6.3 Bending

Bending uses press brakes to create angular shapes in sheet metal.

**Key formula:**

$$\text{Bend Allowance} = \frac{\pi \times A \times (R + K \times T)}{180}$$

Where: A = bend angle, R = inside radius, T = material thickness, K = K-factor (typically 0.3-0.5)

### 1.6.4 Shearing

Shearing cuts sheet metal without chip formation using opposed cutting edges.

**Types:** Blanking, piercing, trimming, notching, slitting

### 1.6.5 Rolling

Reduces material thickness by passing through rotating rolls. Used for sheet, plate, and structural shapes.

### 1.6.6 Extrusion (Metal)

Forces material through a die to create constant cross-section profiles.

**Examples:** Aluminum window frames, heat sinks, structural shapes

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## 1.7 Machining Processes

**Machining** removes material using cutting tools to achieve precise dimensions and surface finishes. It is the most versatile manufacturing process but typically has higher per-part costs.

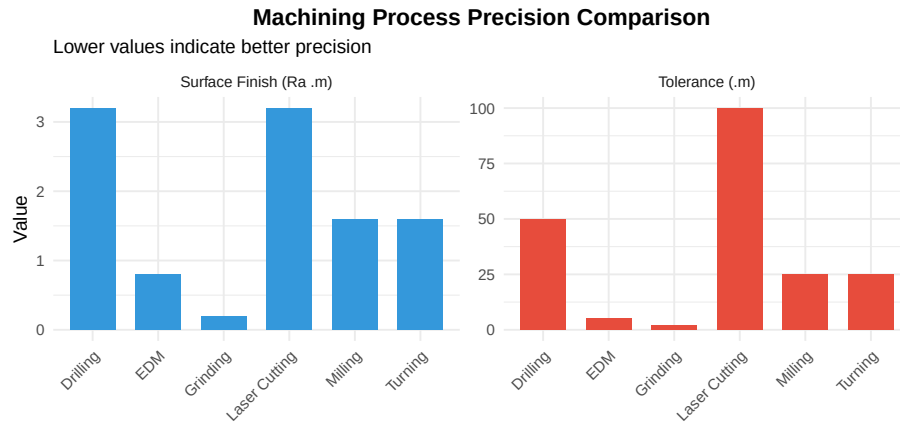


Figure 1.6: Machining Process Capabilities

### 1.7.1 Turning

Rotates the workpiece while a cutting tool moves to create cylindrical shapes.

**Performed on lathes or turning centers**

**Operations:** Facing, turning, boring, threading, grooving, parting

### 1.7.2 Milling

Rotates a cutting tool while moving the workpiece to remove material. The most versatile machining process.

**Types:** - **Peripheral milling** - Cutter axis parallel to surface - **Face milling** - Cutter axis perpendicular to surface - **End milling** - Creates slots, pockets, contours

### 1.7.3 Drilling

Creates cylindrical holes using rotating drill bits.

**Related operations:** Reaming (sizing), boring (enlarging), tapping (threading)

### 1.7.4 Grinding

Uses abrasive wheels for precision finishing. Achieves the finest tolerances and surface finishes.

**Types:** Surface grinding, cylindrical grinding, centerless grinding



1.7.5 Electrical Discharge Machining (EDM)

Uses electrical sparks to erode conductive materials. Essential for hardened materials and complex geometries.

**Applications:** Injection mold cavities, dies, aerospace components

When should you use EDM instead of conventional machining?

**Choose EDM when:**

- 1. **Hardened materials** - Can machine 60+ HRC steels that would destroy cutting tools
- 2. **Complex internal features** - Wire EDM cuts intricate shapes
- 3. **Tight corners** - Can achieve sharp internal corners impossible with rotary tools
- 4. **Fragile parts** - No cutting forces (non-contact process)
- 5. **Thin walls** - No deflection from tool pressure

**Limitations:** - Slow material removal (grams/hour vs. kg/hour for milling) - Only works on conductive materials - Recast layer may need removal - Higher cost per part

1.8 Joining Processes

**Joining** combines two or more components into a single assembly. Methods range from permanent (welding) to removable (fasteners).

Table 1.2: Joining Process Comparison

| Process              | Joint Strength | Dissimilar Metals | Automation | Typical Applications               |
|----------------------|----------------|-------------------|------------|------------------------------------|
| Arc Welding          | Excellent      | Limited           | Medium     | Structural steel, pressure vessels |
| Resistance Welding   | Excellent      | Limited           | High       | Auto body panels, wire mesh        |
| Laser Welding        | Excellent      | Limited           | High       | Electronics, medical devices       |
| Brazing              | Good           | Yes               | Medium     | Carbide tooling, HVAC              |
| Soldering            | Fair           | Yes               | High       | Electronics, plumbing              |
| Adhesive Bonding     | Good           | Yes               | High       | Composites, glass, plastics        |
| Mechanical Fastening | Excellent      | Yes               | High       | All industries                     |

1.8.1 Welding

Melts and fuses base materials together, often with added filler metal.

**Common Types:** - **MIG (GMAW)** - Gas Metal Arc Welding - fast, easy automation - **TIG (GTAW)** - Gas Tungsten Arc Welding - precise, clean welds

- **Stick (SMAW)** - Shielded Metal Arc Welding - versatile, field use - **Resistance** - Uses electrical resistance heating - **Laser/Electron Beam** - High precision, deep penetration

### 1.8.2 Brazing and Soldering

Joins metals using a filler metal with a lower melting point than the base materials. The base materials do not melt.

- **Brazing** - Filler melts above 450°C (840°F)
- **Soldering** - Filler melts below 450°C

### 1.8.3 Adhesive Bonding

Uses chemical adhesives to join materials. Essential for composites, glass, and dissimilar materials.

**Advantages:** Distributes stress, joins dissimilar materials, seals against fluids

**Disadvantages:** Surface preparation critical, limited temperature resistance, aging concerns

### 1.8.4 Mechanical Fastening

Uses bolts, screws, rivets, or clips. Allows disassembly for maintenance.

## 1.9 Surface Treatment and Coating

**Coating and surface treatment** processes protect parts from corrosion, wear, or environmental damage, and can improve appearance or electrical properties.

Table 1.3: Surface Treatment and Coating Processes

| Process                    | Typical Thickness | Base Material     | Primary Benefit       | Indus  |
|----------------------------|-------------------|-------------------|-----------------------|--------|
| <b>Electroplating</b>      | 1-50 m            | Conductive metals | Corrosion, appearance | Autom  |
| <b>Anodizing</b>           | 5-100 m           | Aluminum          | Corrosion, hardness   | Aerosp |
| <b>Powder Coating</b>      | 50-150 m          | Any metal         | Corrosion, appearance | Applia |
| <b>Painting</b>            | 25-75 m           | Any               | Appearance            | All    |
| <b>Hot Dip Galvanizing</b> | 45-85 m           | Steel             | Corrosion             | Const  |
| <b>PVD/CVD</b>             | 1-10 m            | Any               | Wear, hardness        | Toolin |
| <b>Thermal Spray</b>       | 100-500 m         | Any               | Wear, thermal barrier | Aerosp |

### 1.9.1 Electroplating

Deposits metal coating using electrical current. Common coatings include chrome, nickel, zinc, gold, and silver.

### 1.9.2 Anodizing

Electrochemically grows an oxide layer on aluminum. Provides corrosion resistance and can be dyed various colors.

### 1.9.3 Powder Coating

Applies dry powder electrostatically, then cures with heat. Produces thick, durable, attractive finishes.

### 1.9.4 Galvanizing

Coats steel with zinc for corrosion protection. Hot-dip galvanizing provides excellent outdoor durability.

## 1.10 Additive Manufacturing (3D Printing)

**Additive manufacturing** builds parts layer-by-layer from digital models. It's the opposite of machining (subtractive manufacturing).

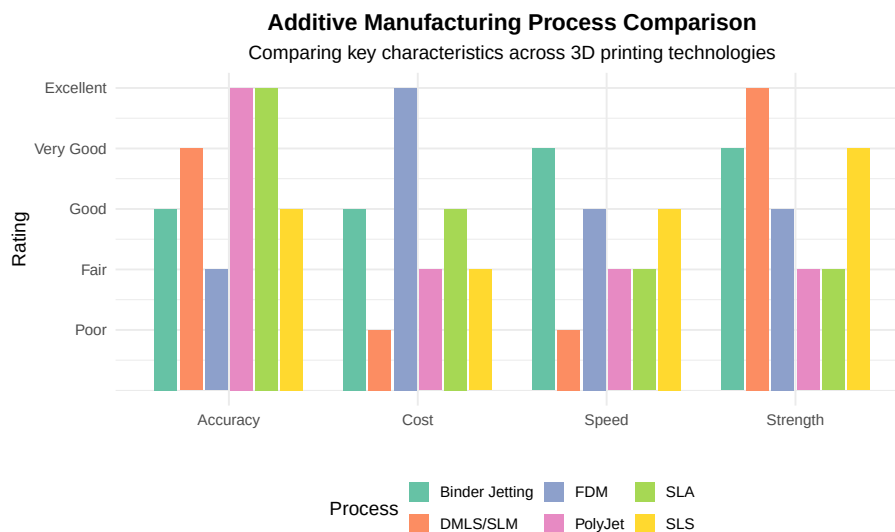


Figure 1.7: Additive Manufacturing Process Comparison

**How 3D Printing Works:** <https://www.youtube.com/watch?v=Vx0Z6LplaMU>

### 1.10.1 FDM (Fused Deposition Modeling)

Extrudes thermoplastic filament through a heated nozzle. Most common desktop 3D printing technology.

**Materials:** PLA, ABS, PETG, Nylon, TPU

### 1.10.2 SLA (Stereolithography)

Uses UV laser to cure liquid photopolymer resin. Excellent surface finish and detail.

### 1.10.3 SLS (Selective Laser Sintering)

Fuses powder material with a laser. Produces strong functional parts without support structures.

### 1.10.4 Metal 3D Printing (DMLS/SLM)

Fuses metal powder with high-power lasers. Used for aerospace, medical, and tooling applications.

When does metal 3D printing make sense vs. traditional manufacturing?

**Metal 3D printing is advantageous when:**

1. **Complex internal features** - Conformal cooling channels in molds
2. **Topology optimization** - Organic shapes that minimize weight
3. **Consolidation** - Combining multiple parts into one
4. **Low volume** - Too few parts to justify tooling
5. **Custom/one-off** - Medical implants, prototypes

**Traditional manufacturing is better when:** - High volumes (> 100-1000 parts depending on complexity) - Simple geometries - Large parts (3D print beds are limited) - Maximum strength required (forging still superior)

**Cost comparison:** - Simple bracket: \$5 machined vs. \$50+ 3D printed - Complex aerospace part: \$10,000 machined vs. \$3,000 3D printed

## 1.11 Industry Applications

Table 1.4: Manufacturing Process Applications by Industry

| Process | Automotive | Food Industry | Defense |
|---------|------------|---------------|---------|
|---------|------------|---------------|---------|

|                   |                                      |                              |                       |
|-------------------|--------------------------------------|------------------------------|-----------------------|
| Die Casting       | Engine blocks, transmission cases    | Equipment housings           | Aircraft structural   |
| Injection Molding | Dashboard, interior trim, connectors | Packaging, containers, caps  | Connectors, housing   |
| Forging           | Crankshafts, connecting rods, gears  | Mixer blades, pump impellers | Landing gear, turb    |
| Stamping          | Body panels, structural components   | Cans, trays, lids            | Aircraft skins, miss  |
| Machining         | Engine components, precision parts   | Valves, fittings             | Precision weapons c   |
| Welding           | Body structure, exhaust systems      | Tanks, piping, conveyors     | Aircraft structures,  |
| Powder Coating    | Wheels, chassis components           | Equipment frames             | Military vehicles, ec |
| 3D Printing       | Prototypes, custom fixtures          | Custom tooling, fixtures     | Turbine blades, bra   |

## 1.12 Sustainability in Manufacturing

Modern manufacturing must consider environmental impact. Here’s how different processes compare:

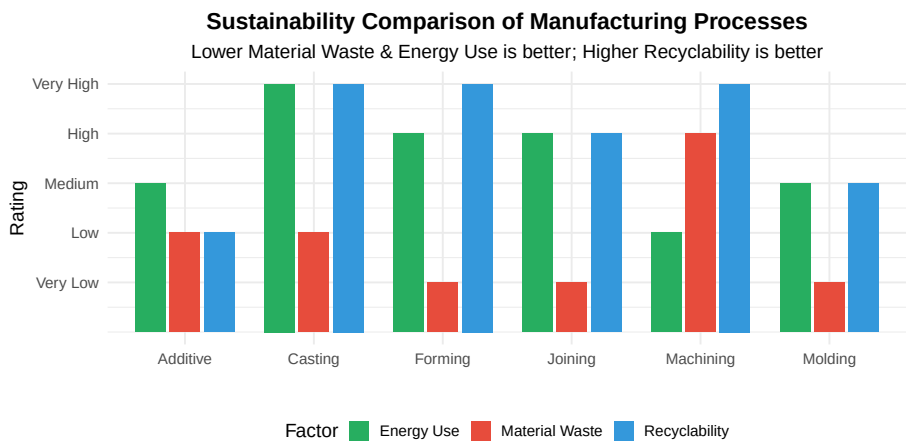


Figure 1.8: Environmental Considerations by Process Type

Discussion: How can manufacturers reduce environmental impact?

**Strategies for sustainable manufacturing:**

1. **Near-net-shape processes** - Casting, forging, and molding reduce machining waste
2. **Closed-loop recycling** - Reprocess scrap material on-site
3. **Energy-efficient equipment** - Modern machines use 30-50% less energy
4. **Additive manufacturing** - Only uses material where needed
5. **Water-based coatings** - Replace solvent-based paints
6. **Lean manufacturing** - Reduces overproduction and waste
7. **Life cycle assessment** - Consider total environmental impact

**Example:** Aluminum die casting recycles 95%+ of sprues and runners, while machining from billet may waste 80% of material.

---

## 1.13 Review Questions

Question 1: Process Identification

**A company needs to produce 50,000 aluminum housings per year. Each housing is a complex shape with internal fins for heat dissipation. The surface must be smooth and dimensions must be within  $\pm 0.3\text{mm}$ . Which process would you recommend?**

**Answer: Die Casting** is the best choice because: - High volume (50,000/year) justifies tooling cost - Aluminum is ideal for die casting - Complex internal features are possible -  $\pm 0.3\text{mm}$  tolerance is achievable - Good surface finish without secondary operations

Alternative: If volumes were lower ( $< 5,000$ ), investment casting might be considered.

Question 2: Cost Comparison

**Calculate the break-even point between 3D printing and injection molding for a small plastic part:** - 3D printing: \$8 per part (no tooling) - Injection molding: \$25,000 mold + \$0.50 per part

**Answer:** Let  $n$  = number of parts

Break-even:  $\$8n = \$25,000 + \$0.50n$

$\$7.50n = \$25,000$

$n = 3,333$  parts

**Below 3,333 parts:** 3D printing is cheaper **Above 3,333 parts:** Injection molding is cheaper

At 10,000 parts: - 3D printing: \$80,000 - Injection molding: \$30,000

Question 3: Material Selection

**Match each material/product to the most appropriate manufacturing process:**

- a) Stainless steel surgical instrument
- b) Large plastic kayak
- c) Titanium jet engine bracket
- d) Aluminum beverage can

**Answers:**

- a) **Investment casting** or **machining** - Complex shapes, high precision, excellent surface finish
- b) **Rotational molding** - Large hollow plastic part, low tooling cost
- c) **Additive manufacturing (DMLS)** or **forging** - Complex geometry, lightweight, strength critical
- d) **Deep drawing (stamping)** - Very high volume, thin-wall cylinder

Question 4: Process Selection Scenario

**You need to manufacture a new product with these requirements:** -

Quantity: 200 units for initial market test - Material: ABS plastic - Size: 150mm × 100mm × 50mm - Tolerance: ±0.5mm - Surface: Smooth, will be painted

**What process would you recommend and why?**

**Answer: 3D Printing (FDM or SLA)** would be most appropriate because:

1. **Low quantity (200)** - Injection molding tooling (\$30,000+) not justified
2. **Market test phase** - Design may change based on feedback
3. **Quick turnaround** - Parts in days vs. weeks for tooling
4. **Tolerances achievable** - ±0.5mm is within FDM capabilities
5. **Post-processing acceptable** - Parts will be painted anyway

**Cost comparison:** - FDM:  $200 \times \$15 = \$3,000$  - Injection molding:  $\$30,000 + (200 \times \$1) = \$30,200$

If the product succeeds and volume increases to 10,000+, transition to injection molding.

Question 5: Hybrid Manufacturing

**What is hybrid manufacturing and when is it used?**

**Answer: Hybrid manufacturing** combines additive and subtractive processes in a single machine or workflow.

**Common combinations:** - 3D print near-net shape → CNC machine critical surfaces - Cast blank → Machine precision features - Forge rough shape → Machine final dimensions

**Benefits:** - Reduced material waste vs. pure machining - Better precision than pure additive - Faster than pure machining for complex parts

**Example:** Aerospace companies 3D print titanium brackets, then machine mounting surfaces to achieve tight tolerances. This can reduce material waste by 90% compared to machining from billet.

---

## 1.14 Summary

Table 1.5: Manufacturing Processes Summary

| Process Category | Key Principle                  | Best For                                 |
|------------------|--------------------------------|------------------------------------------|
| <b>Casting</b>   | Pour liquid into mold          | Complex metal shapes, medium-high volume |
| <b>Molding</b>   | Shape heated/softened material | Plastic parts, very high volume          |
| <b>Forming</b>   | Deform solid with force        | Strong metal parts, high volume          |
| <b>Machining</b> | Remove material with tools     | Precision parts, any material            |
| <b>Joining</b>   | Combine multiple parts         | Assembly of components                   |
| <b>Coating</b>   | Apply protective layer         | Corrosion/wear protection                |
| <b>Additive</b>  | Build layer by layer           | Prototypes, complex geometry             |

### 1.14.1 Key Takeaways

1. **No single process is best for everything** - Selection depends on material, volume, complexity, and cost constraints
2. **Volume drives economics** - High tooling costs require high volumes to amortize
3. **Consider the complete supply chain** - Secondary operations, assembly, and finishing affect total cost
4. **Sustainability matters** - Modern manufacturing must minimize waste and energy use
5. **Hybrid approaches are growing** - Combining processes optimizes cost and performance

## 1.15 References

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3. DeGarmo, E.P., Black, J.T., & Kohser, R.A. (2019). *DeGarmo's Materials and Processes in Manufacturing* (13th ed.). Wiley.
4. Afzal, M. "Types of Manufacturing Processes." *Engineering Articles*.



## Chapter 2

# Process and Design

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### 2.1 Learning Objectives

By the end of this chapter, you will be able to:

1. Describe the stages of process plant project development from concept to operation
  2. Differentiate between fixed, programmable, and flexible automation systems
  3. Interpret Process Flow Diagrams (PFDs) and Piping & Instrumentation Diagrams (P&IDs)
  4. Explain the purpose and content of Front-End Engineering Design (FEED)
  5. Apply Design for Manufacturing (DFM) and Design for Assembly (DFA) principles
  6. Understand the role of HAZOP studies in process safety
  7. Identify key engineering deliverables for each project phase
- 

### 2.2 Introduction to Process Engineering

**Process engineering** is a specialized branch of engineering that focuses on the design, optimization, and implementation of manufacturing processes and production systems. Process engineers work at the intersection of design and production, ensuring that products can be manufactured efficiently, safely, and economically.

What is the role of a process engineer?

Process engineers are responsible for:

- **Designing production processes** - Selecting equipment, defining sequences, optimizing flow
- **Improving existing processes** - Reducing waste, increasing throughput, improving quality
- **Scaling up from lab to production** - Taking R&D concepts to full-scale manufacturing
- **Ensuring safety and compliance** - Meeting regulatory and environmental requirements
- **Supporting production** - Troubleshooting issues, training operators
- **Cost optimization** - Reducing manufacturing costs while maintaining quality

## 2.3 Types of Manufacturing Automation

Automation is essential for achieving high productivity, consistent quality, and competitive manufacturing costs. There are three fundamental types of automation, each suited to different production scenarios.

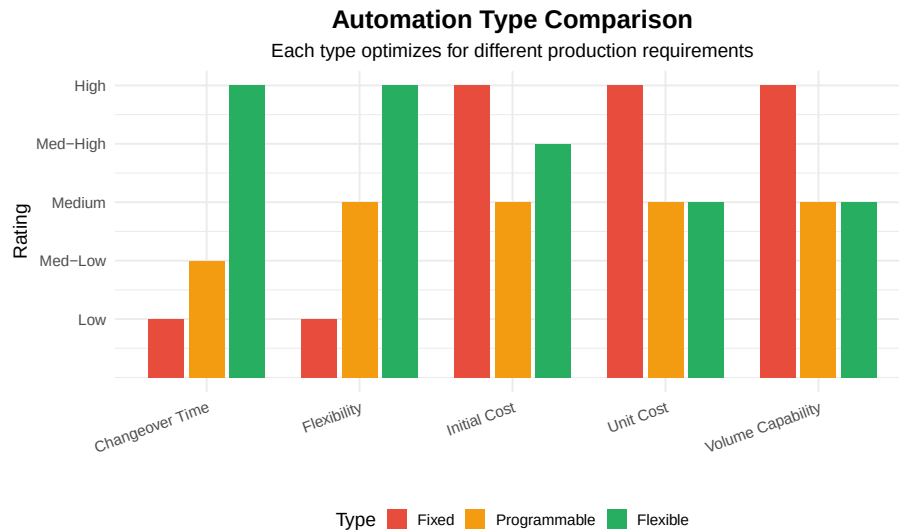


Figure 2.1: Comparison of Automation Types

### 2.3.1 Fixed Automation

**Fixed automation** (also called “hard automation”) uses specialized equipment designed for a specific product. The sequence of operations is determined by

the equipment configuration and cannot be easily changed.

**Characteristics:** - Very high production rates (thousands per hour) - Low per-unit cost at high volumes - High initial capital investment - Inflexible - dedicated to one product

**Examples:** - Automotive transfer lines for engine blocks - Beverage bottling lines - Chemical processing plants - Paper manufacturing

Live cell — try different equipment costs, production volumes, or operating costs.

```
#| label: fixed-automation-economics
# Fixed Automation Economic Analysis
# When does fixed automation make sense?

# Example: Automotive transfer line for cylinder heads
fixed_automation_cost <- 5000000 # Initial equipment cost
annual_production <- 500000      # Parts per year
fixed_cost_per_part <- 2.50      # Operating cost per part
equipment_life_years <- 10

# Calculate total cost per part including amortization
annual_equipment_cost <- fixed_automation_cost / equipment_life_years
total_annual_cost <- annual_equipment_cost + (fixed_cost_per_part * annual_production)
cost_per_part <- total_annual_cost / annual_production

cat("Fixed Automation Analysis:\n")
cat("=====\n")
cat(sprintf("Equipment cost: %s\n", format(fixed_automation_cost, big.mark = " ")))
cat(sprintf("Annual production: %s parts\n", format(annual_production, big.mark = " ")))
cat(sprintf("Operating cost per part: %.2f\n", fixed_cost_per_part))
cat(sprintf("Amortized equipment cost per part: %.2f\n", annual_equipment_cost/annual_production))
cat(sprintf("Total cost per part: %.2f\n", cost_per_part))
```

### 2.3.2 Programmable Automation

**Programmable automation** allows the production equipment to be reprogrammed for different products. It's suited for batch production where products change periodically.

**Characteristics:** - Medium production rates - Significant changeover time between products - Lower initial cost than fixed automation - Suitable for batch sizes of 100-10,000

**Examples:** - CNC machining centers - Industrial robots - PLC-controlled assembly machines - Batch chemical reactors

Discussion: When should you choose programmable over fixed automation?

**Choose programmable automation when:**

1. **Product variety** - You make multiple products on the same equipment
2. **Volume uncertainty** - Demand may shift between products
3. **Product evolution** - Frequent design changes expected
4. **Lower volumes** - Not enough volume to justify dedicated equipment
5. **Capital constraints** - Can't afford multiple dedicated lines

**Example scenario:** A machine shop that makes 20 different part numbers for aerospace customers would use CNC machines (programmable) rather than dedicated transfer lines (fixed) because: - Volumes per part number are low (100-1,000/year) - New part designs are introduced regularly - Flexibility to respond to customer changes is critical

**2.3.3 Flexible Automation**

**Flexible automation** can produce a variety of parts with minimal changeover time. It combines the flexibility of programmable automation with the productivity approaching fixed automation.

**Characteristics:** - Medium-high production rates - Very fast changeover (minutes, not hours) - High initial investment - Suitable for medium variety, medium volume

**Examples:** - Flexible Manufacturing Systems (FMS) - Robotic assembly cells - Automated guided vehicle (AGV) systems - Modern automotive final assembly lines

Table 2.1: Automation Type Selection Guide

| Production Characteristic   | Fixed Automation               | Programmable Automation       |
|-----------------------------|--------------------------------|-------------------------------|
| <b>Annual Volume</b>        | > 1,000,000 units              | 1,000 - 100,000 units         |
| <b>Product Variety</b>      | Single product                 | Several products in batches   |
| <b>Changeover Frequency</b> | Rarely (years)                 | Weekly to monthly             |
| <b>Initial Investment</b>   | Very High (\$5M+)              | Medium (\$500K-\$2M)          |
| <b>Operating Cost</b>       | Lowest per unit                | Medium per unit               |
| <b>Best Industries</b>      | Automotive, Beverage, Chemical | Job shops, Aerospace, Medical |

List industries where you can find various automated process systems

**By automation type:**

**Fixed Automation:** - Oil refineries - Chemical plants - Food processing (high-volume) - Beverage bottling - Automotive powertrain - Steel mills - Paper/pulp mills

**Programmable Automation:** - Aerospace manufacturing - Medical device production - Pharmaceutical (batch) - Job shops - Tool & die making - Defense contractors

**Flexible Automation:** - Electronics assembly - Automotive final assembly - Warehouse/distribution - Semiconductor fabrication - Consumer appliances - Furniture manufacturing

### 2.3.4 Automation Advantages and Disadvantages

Table 2.2: Advantages and Disadvantages of Manufacturing Automation

| Category      | Point                                                             |
|---------------|-------------------------------------------------------------------|
| Advantages    | Lowered Operating Costs - Reduced labor per unit                  |
| Advantages    | Improved Worker Safety - Removes humans from hazardous operations |
| Advantages    | Reduced Factory Lead Times - Faster, more predictable production  |
| Advantages    | Increased Production Output - 24/7 operation possible             |
| Advantages    | Smaller Environmental Footprint - Optimized resource usage        |
| Advantages    | Increased Productivity and Efficiency - Consistent cycle times    |
| Advantages    | Increased System Versatility - Modern systems adapt to changes    |
| Disadvantages | Higher Start-up Cost - Significant capital investment required    |
| Disadvantages | Higher Cost of Maintenance - Specialized technicians needed       |
| Disadvantages | Obsolescence/Depreciation - Technology changes rapidly            |
| Disadvantages | Workforce Displacement - Fewer direct labor jobs                  |
| Disadvantages | Not Economical for Small Scale - Minimum volumes required         |

## 2.4 Project Development Phases

Manufacturing projects follow a structured development process from initial concept to full operation. Understanding these phases is essential for process engineers.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use `linewidth` instead.

### Project Development Phases with Stage–Gate Reviews

*Stage–Gate Process: Each gate requires approval before proceeding*

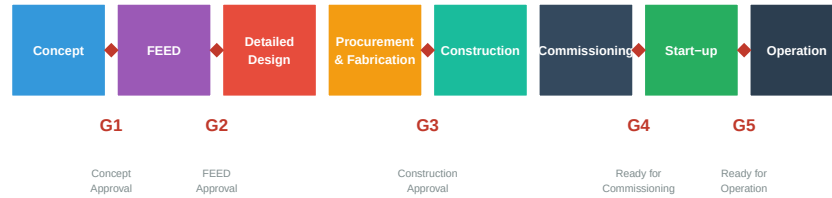


Figure 2.2: Project Development Phases and Gate Reviews

#### 2.4.1 Phase 1: Concept Development

The initial phase defines the project scope, evaluates feasibility, and develops preliminary cost estimates.

**Key Deliverables:** - Project charter and business case - Preliminary process flow diagrams - Order-of-magnitude cost estimate ( $\pm 30\text{-}50\%$ ) - Initial schedule - Risk assessment

#### 2.4.2 Phase 2: Front-End Engineering Design (FEED)

FEED (also called FEL - Front-End Loading) develops the project definition to a level sufficient for final investment decision.

**FEED Process Overview:** <https://www.youtube.com/watch?v=gSkoGFLDxF0>

**FEED Deliverables include:**

- Project organization chart
- Defined project scope
- Process Flow Diagrams (PFDs)
- Piping & Instrumentation Diagrams (P&IDs)
- Equipment datasheets and specifications
- Major equipment list
- 2D and 3D preliminary models
- Equipment layout and installation plan
- HAZOP and safety studies
- Automation strategy
- Budget estimate ( $\pm 10\text{-}15\%$ )
- Project timeline

- Fixed-bid quote capability

What are the steps of Process System Design and Engineering?

1. **Process Development**
  - Develop PFDs & P&IDs
  - Process simulation
  - Heat and mass balance
2. **Equipment Engineering**
  - 2D & 3D models
  - Equipment layout and installation plan
  - Skid design
  - Mechanical & structural design
3. **Systems Engineering**
  - Piping design
  - Electrical design
  - Instrumentation design
  - Control system architecture
4. **Installation Planning**
  - Lifts & installation planning
  - Construction sequencing
  - Tie-in planning

2.4.3 Phase 3: Detailed Engineering

Detailed engineering produces all documents needed for procurement and construction.

Table 2.3: Engineering Disciplines and Deliverables

| Discipline       | Key Deliverables                                                | Systems Provided        |
|------------------|-----------------------------------------------------------------|-------------------------|
| Process          | P&IDs, datasheets, process packages, operating procedures       | Process units, utility  |
| Mechanical       | Vessel drawings, heat exchanger specs, rotating equipment specs | Pressure vessels, tank  |
| Piping           | Isometrics, pipe specs, stress analysis, 3D models              | Process piping, utility |
| Electrical       | Single-line diagrams, load lists, cable schedules, lighting     | Power distribution, m   |
| I&C              | Logic diagrams, instrument index, control narratives, HMI specs | DCS/PLC, safety sys     |
| Civil/Structural | Foundation drawings, structural steel, buildings, roads         | Foundations, structur   |

Automation and Controls Engineering needs to include which tasks?

**Control System Design:** - Automation architecture design - Control panel fabrication drawings - PLC/DCS programming - HMI/SCADA design - Safety system design (SIS)

**Documentation:** - Control narratives - Logic diagrams (cause & effect) - Instrument datasheets - Loop diagrams - Cable schedules

**Integration:** - Controls integration services - Network architecture - Cybersecurity - Historian configuration - MES integration

#### 2.4.4 Phase 4-7: Construction, Commissioning, and Start-up

Table 2.4: Later Project Phases

| Phase                                | Key Activities                                                     |   |
|--------------------------------------|--------------------------------------------------------------------|---|
| <b>Procurement &amp; Fabrication</b> | Purchase equipment, fabricate skids, manufacture custom items      | I |
| <b>Construction</b>                  | Civil work, install equipment, piping, electrical, instrumentation | S |
| <b>Commissioning</b>                 | System checkout, loop testing, safety system validation            | V |
| <b>Start-up</b>                      | Initial operation, performance testing, operator training          | O |

What needs to be planned for Testing & Commissioning?

**Pre-Commissioning:** - Mechanical completion checks - Piping pressure tests  
- Electrical continuity tests - Instrument calibration

**Commissioning:** - Loop checking - Control system function tests - Safety system (SIS) validation - Utility systems start-up - Dry run testing

**Performance Testing:** - Factory Acceptance Tests (FAT) - Site Acceptance Tests (SAT) - Performance guarantee tests - Reliability testing

## 2.5 Process Documentation

### 2.5.1 Process Flow Diagrams (PFDs)

PFDs show the overall process flow, major equipment, and key operating conditions. They are the “big picture” view of the process.

Table 2.5: Process Flow Diagram (PFD) Content

| Element                     | What's Shown                                           | What's NOT Shown   |
|-----------------------------|--------------------------------------------------------|--------------------|
| <b>Major Equipment</b>      | Reactors, vessels, heat exchangers, pumps, compressors | Pipe sizes, valve  |
| <b>Process Streams</b>      | Main process flow lines with arrows                    | Minor bypasses,    |
| <b>Operating Conditions</b> | Temperature, pressure, flow rates at key points        | All intermediate   |
| <b>Control Loops</b>        | Major control loops only                               | Local instrument   |
| <b>Utility Streams</b>      | Steam, cooling water, nitrogen (simplified)            | Detailed utility c |
| <b>Stream Data</b>          | Stream numbers with heat/mass balance table            | All intermediate   |



### 2.5.2 Piping & Instrumentation Diagrams (P&IDs)

P&IDs are the detailed “roadmap” of the process, showing all equipment, piping, valves, and instrumentation.

**How to Read P&IDs:** <https://www.youtube.com/watch?v=j4EOTerfyTY>

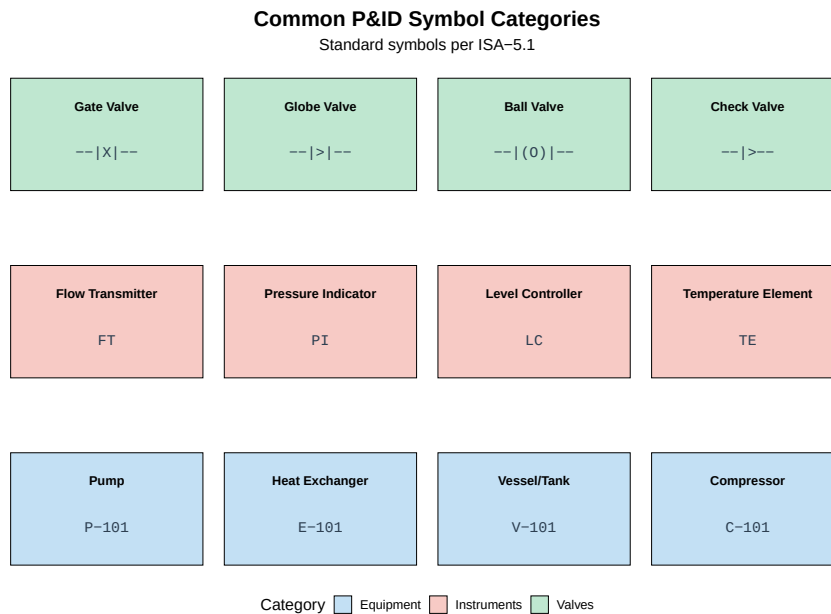


Figure 2.3: Common P&ID Symbols (Simplified)

Table 2.6: ISA Instrument Tag Structure

| First Letter | Measured Variable | Modifier Letters   | Example Tag                   |
|--------------|-------------------|--------------------|-------------------------------|
| <b>F</b>     | Flow              | T = Transmitter    | FT-101 (Flow Transmitter)     |
| <b>L</b>     | Level             | C = Controller     | LC-201 (Level Controller)     |
| <b>P</b>     | Pressure          | I = Indicator      | PI-301 (Pressure Indicator)   |
| <b>T</b>     | Temperature       | E = Element/Sensor | TE-401 (Temperature Element)  |
| <b>A</b>     | Analysis          | V = Valve          | AT-501 (Analyzer Transmitter) |
| <b>S</b>     | Speed             | A = Alarm          | SI-601 (Speed Indicator)      |

Quick Quiz: Decode these instrument tags

**Decode the following P&ID tags:**

1. **FIC-101** = ?
2. **TT-205** = ?

- 3. **PSV-301** = ?
- 4. **LCV-402** = ?

Answers:

- 1. **FIC-101** = Flow Indicating Controller (loop 101)
- 2. **TT-205** = Temperature Transmitter (loop 205)
- 3. **PSV-301** = Pressure Safety Valve (loop 301)
- 4. **LCV-402** = Level Control Valve (loop 402)

## 2.6 Design for Manufacturing (DFM) and Assembly (DFA)

**DFM** and **DFA** are methodologies that consider manufacturing and assembly constraints during product design, reducing cost and improving quality.

### 2.6.1 DFM Principles

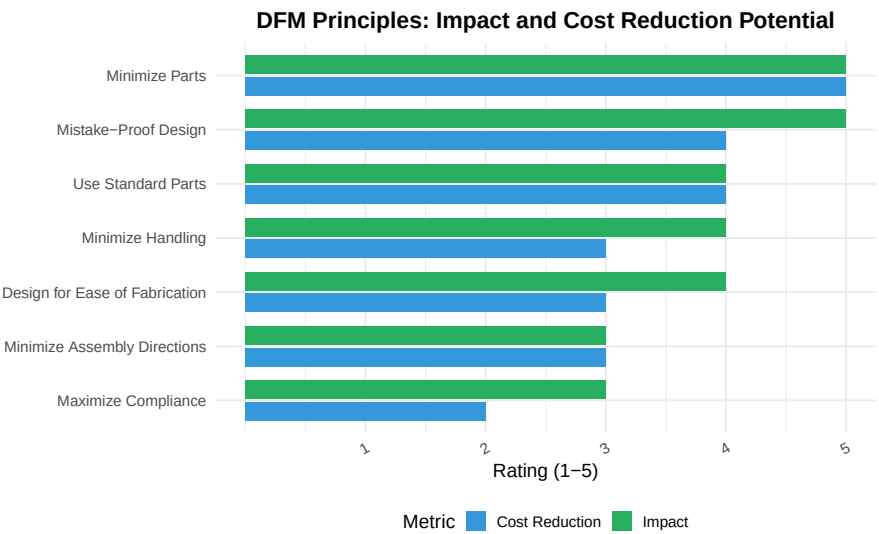


Figure 2.4: DFM Guidelines

Table 2.7: Design for Manufacturing Guidelines

| DFM Guideline | Why It Matters |
|---------------|----------------|
|---------------|----------------|

## 2.6. DESIGN FOR MANUFACTURING (DFM) AND ASSEMBLY (DFA) 37

|                                        |                                                                     |           |
|----------------------------------------|---------------------------------------------------------------------|-----------|
| <b>Minimize total number of parts</b>  | Fewer parts = fewer chances for failure, less inventory, lower cost | Combine   |
| <b>Use standard components</b>         | Standard parts are cheaper, faster to procure, easier to replace    | Use M6    |
| <b>Design for ease of fabrication</b>  | Complex features increase manufacturing cost and defect rates       | Avoid un  |
| <b>Design for ease of assembly</b>     | Reduces assembly time and errors                                    | Use snap  |
| <b>Minimize assembly directions</b>    | Multiple directions require repositioning, increasing time          | All parts |
| <b>Maximize compliance in assembly</b> | Tight tolerances increase cost; allow self-alignment where possible | Use tape  |
| <b>Design for ease of service</b>      | Products will need maintenance; design for technician access        | Locate s  |

### 2.6.2 DFA Analysis

DFA systematically evaluates assembly operations to identify improvement opportunities.

This one is live — try changing the part list or handling/insertion times and re-running.

```
#| label: dfa-example
# DFA Analysis Example: Simple Assembly
# Calculate assembly efficiency using Boothroyd-Dewhurst method

# Assembly data (simplified)
parts <- data.frame(
  Part = c("Base", "PCB", "Cover", "Screw 1", "Screw 2", "Screw 3", "Screw 4", "Label"),
  Handling_Time = c(2.5, 3.0, 2.5, 3.5, 3.5, 3.5, 3.5, 4.0), # seconds
  Insertion_Time = c(3.0, 4.0, 3.5, 6.0, 6.0, 6.0, 6.0, 2.0), # seconds
  Theoretically_Necessary = c(TRUE, TRUE, TRUE, FALSE, FALSE, FALSE, FALSE, FALSE)
)

# Calculations
total_parts <- nrow(parts)
total_time <- sum(parts$Handling_Time + parts$Insertion_Time)
min_parts <- sum(parts$Theoretically_Necessary)
ideal_time <- min_parts * 3 # 3 seconds is theoretical minimum per part

efficiency <- (ideal_time / total_time) * 100

cat("DFA Analysis Results:\n")
cat("=====\n")
cat(sprintf("Total parts: %d\n", total_parts))
cat(sprintf("Theoretically minimum parts: %d\n", min_parts))
cat(sprintf("Total assembly time: %.1f seconds\n", total_time))
cat(sprintf("Ideal assembly time: %.1f seconds\n", ideal_time))
cat(sprintf("Design Efficiency: %.1f%%\n", efficiency))
cat("\nOpportunity: Combine 4 screws into snap-fit design\n")
```

Case Study: Redesigning for DFM/DFA

**Original Design:** - 15 parts - 4 different screw types - Assembly time: 180 seconds - Multiple assembly directions - Special tools required

**Redesigned using DFM/DFA:** - 8 parts (47% reduction) - 1 standard screw type - Assembly time: 75 seconds (58% reduction) - Single assembly direction (top-down) - No special tools

**Cost Impact:** - Part cost reduced 35% - Assembly labor reduced 58% - Quality defects reduced 60% - Total product cost reduced 40%

## 2.7 HAZOP Studies

**HAZOP** (Hazard and Operability Study) is a systematic method to identify hazards and operability problems in process plants.

**HAZOP Study Overview:** <https://www.youtube.com/watch?v=6AqtX8oCpKI&t>

**HAZOP Study Overview:** <https://www.youtube.com/watch?v=ukn0RRt0ReA&list=PLeTiXOrLeXhI5IZKv3GeOUMOYpz5mG36S>

### 2.7.1 HAZOP Methodology

Table 2.8: HAZOP Guide Words and Examples

| Guide Word        | Meaning                            | Example with FLOW                       |
|-------------------|------------------------------------|-----------------------------------------|
| <b>NO/NOT</b>     | Complete negation of design intent | No flow - pump failure, blocked line    |
| <b>MORE</b>       | Quantitative increase              | More flow - control valve fails open    |
| <b>LESS</b>       | Quantitative decrease              | Less flow - partial blockage, pump wear |
| <b>AS WELL AS</b> | Qualitative modification/addition  | Flow + contamination - upstream leak    |
| <b>PART OF</b>    | Qualitative decrease               | Only part of intended flow path active  |
| <b>REVERSE</b>    | Logical opposite of intent         | Reverse flow - check valve failure      |
| <b>OTHER THAN</b> | Complete substitution              | Wrong material flowing                  |

### 2.7.2 HAZOP Example

| Node | Parameter | Guide Word | Deviation | Cause |
|------|-----------|------------|-----------|-------|
|------|-----------|------------|-----------|-------|

|                                  |             |      |                 |                               |
|----------------------------------|-------------|------|-----------------|-------------------------------|
| Feed Pump P-101 to Reactor R-101 | Flow        | No   | No Flow         | Pump failure, blocked suction |
| Feed Pump P-101 to Reactor R-101 | Flow        | More | More Flow       | Control valve fails open      |
| Feed Pump P-101 to Reactor R-101 | Pressure    | More | High Pressure   | Downstream blockage           |
| Feed Pump P-101 to Reactor R-101 | Temperature | Less | Low Temperature | Cooling water leak into feed  |

Discussion: Why is HAZOP done during FEED, not later?

HAZOP should be completed during FEED because:

1. **Cost of changes** - Changes during FEED cost 1x; during detailed design cost 10x; during construction cost 100x
2. **Design influence** - P&IDs are still being developed and can be modified
3. **Equipment selection** - Safety requirements may affect equipment sizing
4. **Budget accuracy** - Safety systems must be included in project budget
5. **Regulatory approval** - Many jurisdictions require HAZOP before construction permits

**Best practice:** Conduct preliminary HAZOP early in FEED, then detailed HAZOP when P&IDs are 60-80% complete.

## 2.8 Cost Estimation

Accurate cost estimation is critical for project approval and budget management.

Table 2.10: Cost Estimate Classification (AACE International)

| AACE Class     | Project Phase     | Engineering Complete | Accuracy Range | Purpose                |
|----------------|-------------------|----------------------|----------------|------------------------|
| <b>Class 5</b> | Concept Screening | 0-2%                 | -50% to +100%  | Screening alternatives |
| <b>Class 4</b> | Concept Study     | 1-15%                | -30% to +50%   | Feasibility            |
| <b>Class 3</b> | FEED/Budget       | 10-40%               | -20% to +30%   | Authorization          |
| <b>Class 2</b> | Detailed Design   | 30-75%               | -15% to +20%   | Control                |
| <b>Class 1</b> | Construction      | 65-100%              | -10% to +15%   | Bid/Tender             |

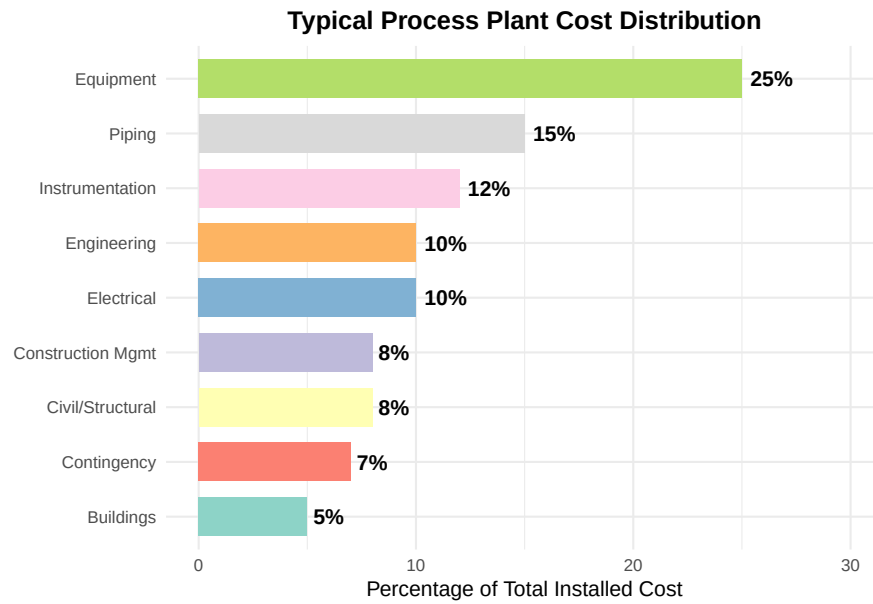


Figure 2.5: Typical Process Plant Cost Breakdown

## 2.9 Case Study: Automotive Paint Line Project

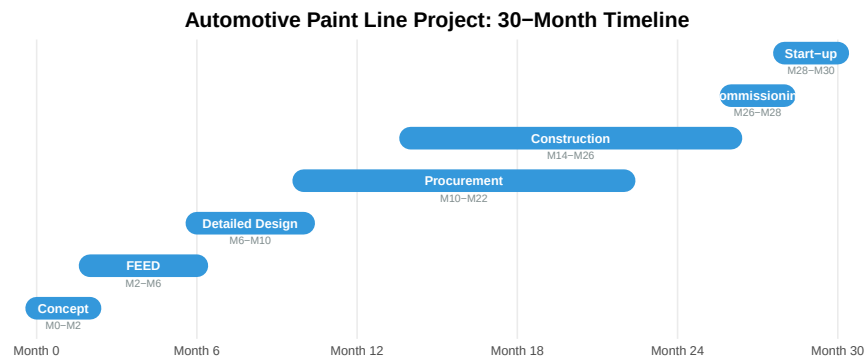


Figure 2.6: Automotive Paint Line Project Timeline

Case Study Details: Paint Line Engineering Challenges

**Project Overview:** - New automotive paint line for SUV body shop - Capacity: 60 jobs per hour (JPH) - Process: E-coat, primer, basecoat, clearcoat

**Key Engineering Challenges:**

- 1. Environmental Control** - Temperature:  $\pm 1^{\circ}\text{C}$  throughout paint booth - Humidity: 65%  $\pm 5\%$  RH - Air cleanliness: Class 10,000
  - 2. Automation Requirements** - 48 paint robots (8 per zone) - Conveyor system with 2m pitch - Zone tracking and recipe management
  - 3. Safety Systems** - Explosion-proof electrical classification - Fire suppression throughout - VOC monitoring and control
  - 4. Quality Systems** - Inline thickness measurement - Automated visual inspection - Recipe management and traceability
- Project Outcome:** - On-time completion - Under budget by 3% - Quality metrics exceeded targets - 99.2% first-pass yield achieved
- 

## 2.10 Review Questions

**Question 1: Automation Selection**

**A medical device company needs to produce 5,000 units per year of a surgical instrument. The design is mature and not expected to change. Volumes could grow to 50,000 units if successful. Which automation type should they initially choose?**

**Answer: Programmable automation** (CNC machining, robotic cells)

**Reasoning:** - Current volume (5,000/year) doesn't justify fixed automation - Flexibility allows for minor design refinements - Can scale up by adding shifts or machines - Lower initial investment preserves capital - If volume grows to 50,000, can justify dedicated equipment later

**Question 2: P&ID Interpretation**

**Decode the following instrument tags and explain their function:**

- a) FIC-201
- b) PSV-305
- c) TT-401
- d) LCV-150

**Answers:**

- a) **FIC-201** - Flow Indicating Controller (loop 201)
  - Measures flow, displays value, and controls a valve
- b) **PSV-305** - Pressure Safety Valve (loop 305)
  - Relieves pressure if it exceeds setpoint (safety device)
- c) **TT-401** - Temperature Transmitter (loop 401)
  - Converts temperature measurement to 4-20mA signal

- d) **LCV-150** - Level Control Valve (loop 150)
- Valve that is manipulated by a level controller

Question 3: Project Phases

**Why is FEED often considered the most critical project phase?**

**Answer:** FEED is critical because:

1. **Defines scope** - All subsequent work based on FEED deliverables
2. **Locks in 80% of cost** - Design decisions determine equipment and construction costs
3. **Lowest cost to change** - Changes cost 10-100x more in later phases
4. **Authorization basis** - Management approves project based on FEED estimate
5. **Risk identification** - HAZOP and risk studies identify safety requirements
6. **Contractor selection** - FEED package used for competitive bidding

**Poor FEED = project overruns, scope changes, and budget problems.**

Question 4: DFM Application

**A product currently uses 12 screws of 3 different sizes for assembly. Apply DFM principles to improve this design.**

**Answer:** Apply these DFM principles:

1. **Minimize parts:** Can some screws be eliminated by using integral snap-fits?
2. **Standardize:** Use one screw size instead of three
  - Reduces inventory complexity
  - Single tool for assembly
  - Lower procurement cost
3. **Design for ease:** Consider alternatives:
  - Snap-fit connections (no tools)
  - Twist-lock features
  - Press-fit pins

**Possible redesign:** - Reduce from 12 screws to 4 screws (for serviceability) - All same size (M3  $\times$  8) - Add 4 snap-fits for initial positioning - Result: 67% fewer fasteners, single tool, faster assembly

Question 5: HAZOP Analysis

**For a reactor feed pump, apply the HAZOP guide word “REVERSE” to the parameter “FLOW”. What could cause this? What are the consequences? What safeguards would you recommend?**

**Answer:**



**Deviation:** Reverse flow through feed pump

**Causes:** - Pump stops while discharge valve open - Check valve fails (stuck open or missing) - Parallel pump creates backflow - Reactor pressure exceeds pump shutoff head

**Consequences:** - Reactor contents backflow into feed tank - Contamination of feed material - Possible chemical reaction in feed system - Pump damage (running backward)

**Safeguards:** - Check valve on pump discharge - Automatic pump discharge block valve (closes on pump stop) - Low flow alarm and trip - Reactor pressure monitoring - Backflow-preventing orifice

## 2.11 Summary

Table 2.11: Chapter 2 Summary

| Topic                    | Key Takeaway                                                                        |
|--------------------------|-------------------------------------------------------------------------------------|
| <b>Automation Types</b>  | Fixed for high volume single product; Programmable for batch; Flexible for variety  |
| <b>Project Phases</b>    | Concept → FEED → Detailed Design → Construction → Commissioning → Start-up          |
| <b>PFDs vs P&amp;IDs</b> | PFD = big picture (equipment, flows); P&ID = detailed (all instruments, valves, pip |
| <b>DFM/DFA</b>           | Design for easy manufacturing and assembly reduces cost 20-40%                      |
| <b>HAZOP</b>             | Systematic hazard identification using guide words; do it during FEED               |
| <b>Cost Estimation</b>   | Estimate accuracy improves with engineering completion; AACE Class 1-5              |

### 2.11.1 Key Takeaways

1. **Automation selection** depends on production volume, variety, and changeover requirements
2. **FEED phase** defines 80% of project cost with only 10-15% of engineering effort
3. **P&IDs** are the critical document connecting design to construction
4. **DFM/DFA** should be applied early in design to maximize impact
5. **HAZOP** identifies hazards systematically and must be done before construction
6. **Cost estimates** improve in accuracy as engineering progresses

## 2.12 References

1. Towler, G., & Sinnott, R. (2021). *Chemical Engineering Design* (7th ed.). Elsevier.

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4. Boothroyd, G., Dewhurst, P., & Knight, W.A. (2011). *Product Design for Manufacture and Assembly* (3rd ed.). CRC Press.
5. CCPS. (2008). *Guidelines for Hazard Evaluation Procedures* (3rd ed.). Wiley.

## Chapter 3

# Integrated Manufacturing Systems

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### 3.1 Learning Objectives

By the end of this chapter, you will be able to:

1. Compare different types of production facility layouts and material handling methods
  2. Calculate Manufacturing Cycle Time (MCT) and Manufacturing Cycle Efficiency (MCE)
  3. Analyze production line performance including the impact of station failures
  4. Apply line balancing techniques to optimize workstation assignments
  5. Explain the concepts of takt time, cycle time, and their relationship
  6. Describe Flexible Manufacturing Systems (FMS) and their applications
  7. Understand modern concepts including digital twins and Industry 4.0
- 

### 3.2 Material Handling Methods & Systems

There are four common types of production facility layouts, each with distinct features and material handling requirements:

Table 3.1: Material Handling Methods by Plant Layout

---

| Layout Type                | Product Characteristics               | Material Handling                           |
|----------------------------|---------------------------------------|---------------------------------------------|
| <b>Fixed-Position</b>      | Large, heavy, or immobile products    | Cranes, hoists, fork lifts bring materials  |
| <b>Process (Job Shop)</b>  | High variety, low volume, custom work | Fork lifts, AGVs, carts move WIP            |
| <b>Cellular</b>            | Part families, medium variety/volume  | Conveyors within cells, carts between cells |
| <b>Product (Flow Line)</b> | Standardized products, high volume    | Conveyors, automated transfer systems       |

Discussion: How do you choose between layout types?

**Selection criteria:**

| Factor        | Favors Process Layout | Favors Product Layout |
|---------------|-----------------------|-----------------------|
| Volume        | Low (< 1,000/year)    | High (> 10,000/year)  |
| Variety       | High (many products)  | Low (1-2 products)    |
| Routing       | Variable by product   | Fixed, same for all   |
| Equipment     | General-purpose       | Specialized           |
| Worker skills | High, varied          | Specific, repetitive  |
| WIP inventory | High                  | Low                   |
| Lead time     | Long                  | Short                 |

**When in doubt:** If volume and variety are moderate, consider cellular manufacturing as a compromise.

### 3.3 Fundamentals of Production Lines

A **production line** consists of workstations arranged so that the product moves from one station to the next, with a portion of the total work performed at each location.

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i Please use ``linewidth`` instead.

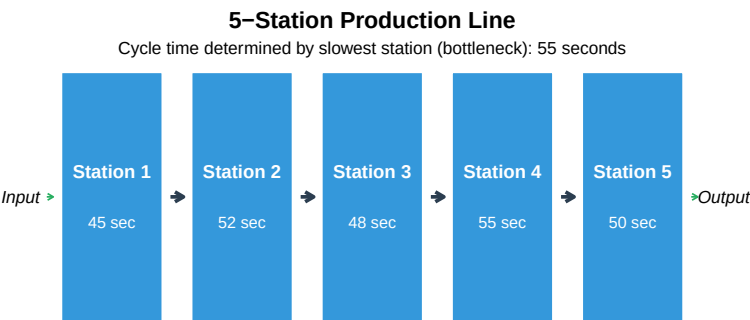


Figure 3.1: Production Line Concept

3.3.1 Methods of Work Transport

Table 3.2: Work Transport Methods

| Method                     | Description                                              | Advantages                         |
|----------------------------|----------------------------------------------------------|------------------------------------|
| Manual                     | Workers pass parts by hand or using simple carts         | Low cost, flexible, easy to change |
| Continuous                 | Conveyor moves at constant speed; work done while moving | Smooth flow, no transfer time      |
| Synchronous (Intermittent) | All units move simultaneously in discrete steps          | Fixed cycle time, easy to control  |
| Asynchronous               | Units move independently when released by worker         | Absorbs variation, no blocking     |

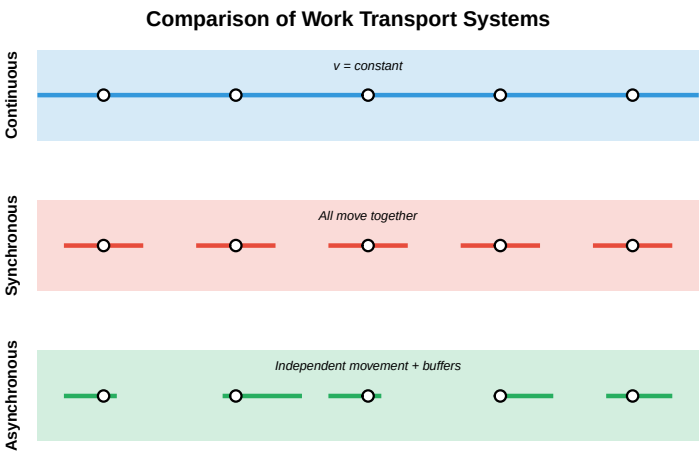


Figure 3.2: Work Transport System Types

What causes blocking and starving in production lines?

**Starving:** A station is ready to work but has no parts to process. - **Causes:** Upstream station is slower, upstream breakdown, material shortage

**Blocking:** A station finishes but cannot release the part. - **Causes:** Downstream station still working, downstream breakdown, no buffer space

**Solutions:** 1. **Buffers** - Storage between stations absorbs variation 2. **Line balancing** - Equalize work content across stations 3. **Parallel stations** - Add capacity at bottlenecks 4. **Preventive maintenance** - Reduce breakdowns

3.3.2 Work Cycle Types

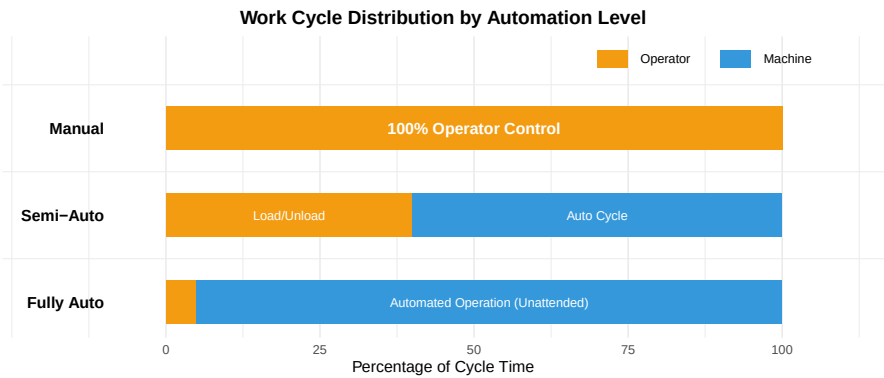


Figure 3.3: Manual, Semi-Automated, and Fully Automated Machine Cycles

3.4 Key Time Metrics

Understanding time metrics is essential for analyzing and improving production systems.

Table 3.3: Key Manufacturing

| Metric          | Definition                                  | Formula                                |
|-----------------|---------------------------------------------|----------------------------------------|
| Takt Time       | Available production time ÷ Customer demand | $T_{\text{takt}} = \frac{\text{text}}$ |
| Cycle Time      | Time between units coming off the line      | $T_c = \frac{60}{R_p}$                 |
| Lead Time       | Total time from order to delivery           | Order processing + Manuf               |
| Throughput Time | Time from raw material to finished goods    | Processing + Waiting + M               |

3.4.1 Takt Time vs. Cycle Time

Live cell — try different demand, shift length, or downtime figures.

```

#| label: takt-vs-cycle
# Takt Time Calculation Example
# Automotive assembly plant

# Given data
shift_length_min <- 480      # 8 hours = 480 minutes
breaks_min <- 30            # Two 15-minute breaks
planned_downtime_min <- 20  # Changeover, meetings
customer_demand <- 60       # Cars per shift

# Calculate available time
available_time <- shift_length_min - breaks_min - planned_downtime_min
cat("Available Production Time:", available_time, "minutes per shift\n")

# Calculate Takt Time
takt_time <- available_time / customer_demand
cat("Takt Time:", round(takt_time, 2), "minutes per car\n")
cat("      =", round(takt_time * 60, 1), "seconds per car\n\n")

# Compare to actual cycle time
actual_cycle_time <- 7.5 # minutes (current performance)
cat("Actual Cycle Time:", actual_cycle_time, "minutes per car\n")

if (actual_cycle_time <= takt_time) {
  cat("Status: MEETING DEMAND - Cycle time is within takt time\n")
  cat("Capacity margin:", round((1 - actual_cycle_time/takt_time) * 100, 1), "%\n")
} else {
  cat("Status: CANNOT MEET DEMAND - Cycle time exceeds takt time\n")
  cat("Shortfall:", round((actual_cycle_time/takt_time - 1) * 100, 1), "% over capacity\n")
}

```

Discussion: What happens when cycle time exceeds takt time?

**When Cycle Time > Takt Time, you cannot meet customer demand.**

**Options to fix this:**

1. **Reduce cycle time:**
  - Kaizen improvements at bottleneck
  - Better tools/fixtures
  - Automation of slow tasks
2. **Increase available time:**
  - Add overtime
  - Add shifts
  - Reduce breaks (carefully!)
3. **Add capacity:**
  - Parallel stations at bottleneck

- Additional production lines
  - Outsourcing
4. **Manage demand:**
- Negotiate delivery schedules
  - Build inventory in advance
  - Redirect to other facilities

**Key insight:** Takt time is the “heartbeat” of lean manufacturing - everything should be synchronized to it.

### 3.4.2 Manufacturing Cycle Time and Efficiency

This one is live — edit the time breakdown and re-run it.

```
#| label: mct-mce
# Manufacturing Cycle Time (MCT) and Efficiency (MCE) Example

# Given production data
production_time_min <- 480 # Total time equipment ran
units_produced <- 60      # Good units produced

# MCT Calculation
MCT <- production_time_min / units_produced
cat("Manufacturing Cycle Time (MCT):", MCT, "minutes per unit\n\n")

# MCE Calculation requires understanding of value-added time
# Typical breakdown of production time:
value_added_time <- 3.5 # Actual processing time (minutes)
inspection_time <- 0.5  # Quality checks
wait_time <- 2.5        # Queuing between operations
move_time <- 1.5        # Transport between stations

total_time_per_unit <- value_added_time + inspection_time + wait_time + move_time
cat("Time breakdown per unit:\n")
cat("  Value-added (processing):", value_added_time, "min\n")
cat("  Inspection:", inspection_time, "min\n")
cat("  Waiting:", wait_time, "min\n")
cat("  Moving:", move_time, "min\n")
cat("  Total:", total_time_per_unit, "min\n\n")

# MCE Calculation
MCE <- (value_added_time / total_time_per_unit) * 100
cat("Manufacturing Cycle Efficiency (MCE):", round(MCE, 1), "%\n")
cat("\nInterpretation: Only", round(MCE, 1), "% of time adds value.\n")
cat("Opportunity:", round(100 - MCE, 1), "% of time is waste that could be reduced.\n")
```



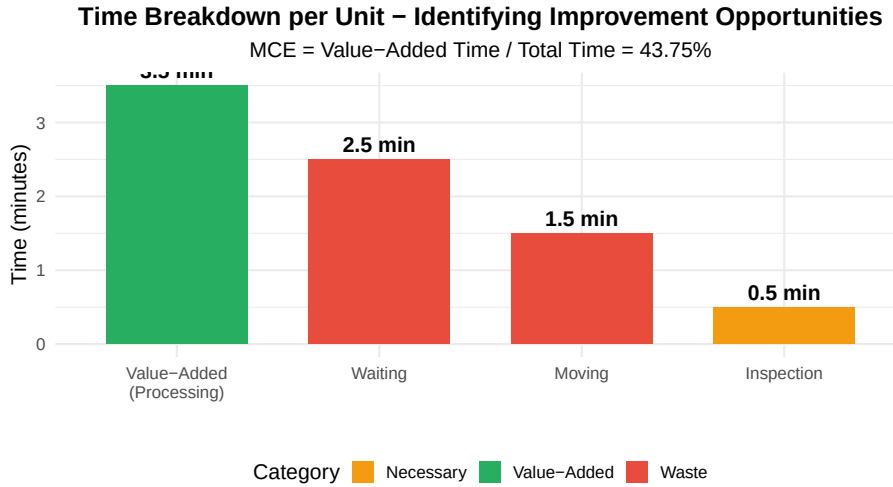


Figure 3.4: Manufacturing Cycle Efficiency - Where Does Time Go?

## 3.5 Manual Assembly Lines

Manual assembly lines use human workers at each station. The efficiency depends on how well work is balanced across stations.

### 3.5.1 Line Balancing

**Line balancing** assigns tasks to workstations to minimize idle time while meeting production rate requirements.

**Key formulas:**

$$T_c = \frac{\text{Available Time}}{\text{Required Output}} = \frac{60}{R_p}$$

$$TM = \frac{\sum t_i}{T_c} = \frac{T_{wc}}{T_c}$$

$$E_b = \frac{T_{wc}}{n \times T_c} \times 100\%$$

Where: -  $T_c$  = Cycle time (time available per unit) -  $R_p$  = Production rate (units per hour) -  $TM$  = Theoretical minimum stations -  $T_{wc}$  = Total work content (sum of all task times) -  $n$  = Actual number of stations -  $E_b$  = Line efficiency (balance efficiency)

### 3.5.2 Line Balancing Example: Electronics Assembly

Live cell — try different task times or a different desired output rate.

```
#| label: line-balance-example
# Line Balancing Example: Tablet Computer Assembly

# Task data
tasks <- data.frame(
  Task = c("A", "B", "C", "D", "E", "F", "G", "H"),
  Description = c("Install battery", "Mount display", "Connect cables",
    "Install motherboard", "Add memory", "Install camera",
    "Attach case back", "Final test"),
  Time_sec = c(30, 45, 20, 55, 15, 25, 35, 40),
  Predecessors = c("-", "A", "A", "B,C", "D", "D", "E,F", "G")
)

# Display task data
cat("Assembly Tasks:\n")
print(tasks[, c("Task", "Description", "Time_sec", "Predecessors")])

# Calculate metrics
total_work_content <- sum(tasks$Time_sec)
cat("\nTotal Work Content:", total_work_content, "seconds\n")

# Production requirements
desired_output <- 45 # units per hour
available_time <- 3600 # seconds per hour

# Step 1: Calculate cycle time
cycle_time <- available_time / desired_output
cat("\nStep 1 - Cycle Time:")
cat("\n Required output:", desired_output, "units/hour")
cat("\n Cycle time = 3600 /", desired_output, "=", cycle_time, "seconds\n")

# Step 2: Calculate theoretical minimum stations
TM <- total_work_content / cycle_time
cat("\nStep 2 - Theoretical Minimum Stations:")
cat("\n TM =", total_work_content, "/", cycle_time, "=", round(TM, 2))
cat("\n Minimum stations required:", ceiling(TM), "\n")

# Step 3: Assign tasks to stations (using largest task time rule)
cat("\nStep 3 - Station Assignments (respecting precedence):\n")
cat(" Station 1: A (30s) + C (20s) + B (45s) = 95s > 80s... EXCEEDS\n")
cat(" Let's try: A (30s) + C (20s) = 50s [30s idle]\n")
cat(" Station 2: B (45s) + E (15s) = 60s... wait, E needs D\n")
```

```

cat("  Station 2: B (45s) + D (55s) = 100s > 80s... EXCEEDS\n")
cat("  Station 2: B (45s) = 45s [35s idle]\n")
cat("  Station 3: D (55s) = 55s [25s idle]\n")
cat("  Station 4: E (15s) + F (25s) + G (35s) = 75s [5s idle]\n")
cat("  Station 5: H (40s) = 40s [40s idle]\n")

n_stations <- 5
efficiency <- (total_work_content / (n_stations * cycle_time)) * 100
balance_delay <- 100 - efficiency

cat("\nStep 4 - Results:")
cat("\n  Number of stations:", n_stations)
cat("\n  Line efficiency:", round(efficiency, 1), "%")
cat("\n  Balance delay (idle time):", round(balance_delay, 1), "%")
cat("\n\n  This is poor balance - consider redesigning tasks or adding parallel stations.")

```

### Precedence Diagram: Tablet Assembly

Arrows indicate task sequence requirements

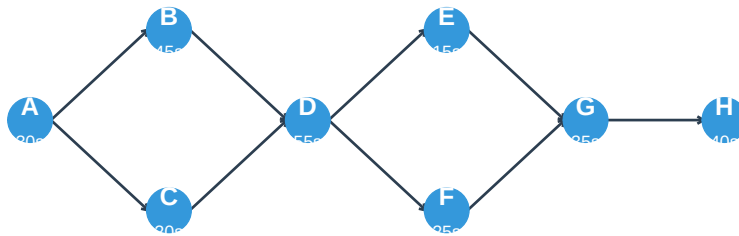


Figure 3.5: Precedence Diagram for Tablet Assembly

Try It: Can you find a better balance?

**Challenge:** Reassign tasks to improve efficiency above 80%.

**Hints:** - Can task A be split into two smaller tasks? - Could parallel workstations handle the bottleneck task D? - What if we allow 85 seconds per station (slower line, fewer stations)?

**One solution with parallel stations:** - Station 1: A + C = 50s - Station 2a: B = 45s (parallel) - Station 2b: B = 45s (parallel) - Station 3: D = 55s - Station 4: E + F = 40s - Station 5: G + H = 75s

With parallelism at Station 2, effective cycle time stays at 55s (Station 3 bot-

tleneck), but we can now achieve higher throughput.

---

## 3.6 Automated Production Lines

Automated lines reduce human intervention using mechanized transfer systems and automated workstations. They're used for high-volume production with well-defined work content.

### 3.6.1 Impact of Station Failures

The performance of automated lines is degraded by station failures. Even small failure probabilities can significantly reduce output.

**Actual Cycle Time:**

$$T_p = T_c + pT$$

**Actual Production Rate:**

$$R_p = \frac{60}{T_p} = \frac{60}{T_c + pT}$$

Where: -  $T_c$  = Ideal cycle time (minutes) -  $p$  = Probability of station failure per cycle -  $T$  = Average downtime per failure (minutes)

Live cell — try adding a scenario or changing the number of stations.

```
#| label: downtime-analysis
# Automated Line Performance Analysis

# Line parameters
ideal_cycle_time <- 1.0      # minutes per part (ideal)
n_stations <- 10             # Number of stations

# Scenario comparison
scenarios <- data.frame(
  Scenario = c("World Class", "Good", "Average", "Poor"),
  Failure_Prob = c(0.005, 0.02, 0.05, 0.10), # Per station per cycle
  Repair_Time = c(2, 5, 10, 15) # Minutes average
)

# Calculate line performance for each scenario
cat("Automated Line Performance Analysis\n")
cat("=====\n")
cat("Ideal cycle time:", ideal_cycle_time, "min/part\n")
cat("Number of stations:", n_stations, "\n")
```

```
cat("Ideal production rate:", 60/ideal_cycle_time, "parts/hour\n\n")

for(i in 1:nrow(scenarios)) {
  p <- scenarios$Failure_Prob[i]
  T <- scenarios$Repair_Time[i]

  # Line failure probability (any station)
  p_line <- 1 - (1 - p)^n_stations

  # Actual cycle time
  Tp <- ideal_cycle_time + p_line * T

  # Actual production rate
  Rp <- 60 / Tp

  # Efficiency
  efficiency <- (ideal_cycle_time / Tp) * 100

  cat(sprintf("%s Performance:\n", scenarios$Scenario[i]))
  cat(sprintf("  Station failure prob: %.1f%%\n", p * 100))
  cat(sprintf("  Line failure prob: %.1f%%\n", p_line * 100))
  cat(sprintf("  Average repair time: %d min\n", T))
  cat(sprintf("  Actual cycle time: %.2f min\n", Tp))
  cat(sprintf("  Actual production: %.1f parts/hour\n", Rp))
  cat(sprintf("  Line efficiency: %.1f%%\n\n", efficiency))
}
```

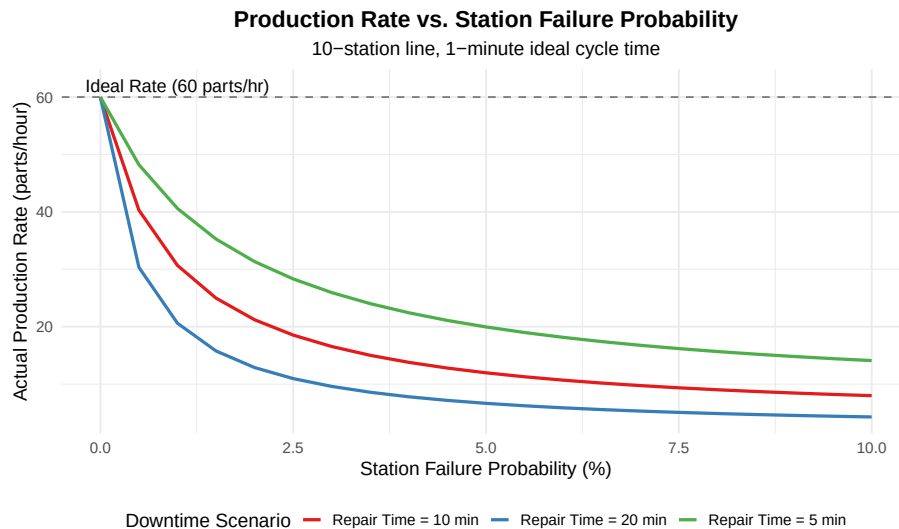


Figure 3.6: Impact of Station Failures on Production Rate

Discussion: Why is preventive maintenance critical for automated lines?

**The math shows why:** - At 5% failure probability with 10-minute repairs, you lose 25% of capacity - This costs more than the maintenance to prevent it

**Cost comparison (example):** - Lost production: 15 parts/hour  $\times$  \$50/part  $\times$  8 hours = \$6,000/day - Preventive maintenance: \$500/day - **ROI of PM: 12:1**

**Best practices:** 1. Track failure data by station 2. Prioritize PM on high-failure stations 3. Stock critical spare parts 4. Train operators on quick changeover/repair 5. Consider redundant stations for bottlenecks

### 3.6.2 Buffer Sizing

Buffers between stations decouple operations and improve overall line availability.

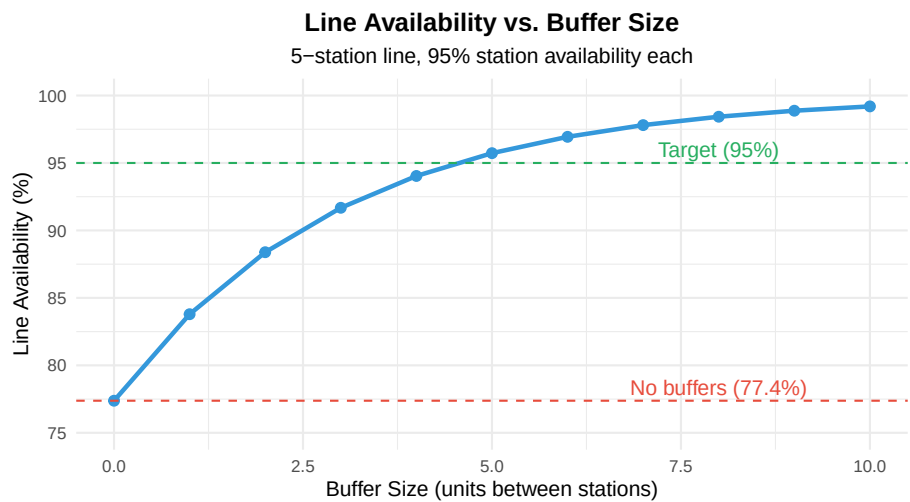


Figure 3.7: Effect of Buffers on Line Availability

### 3.7 Cellular Manufacturing

**Cellular manufacturing** applies Group Technology to organize production into cells that specialize in “families” of similar parts.

#### 3.7.1 Part Families and Machine Cells

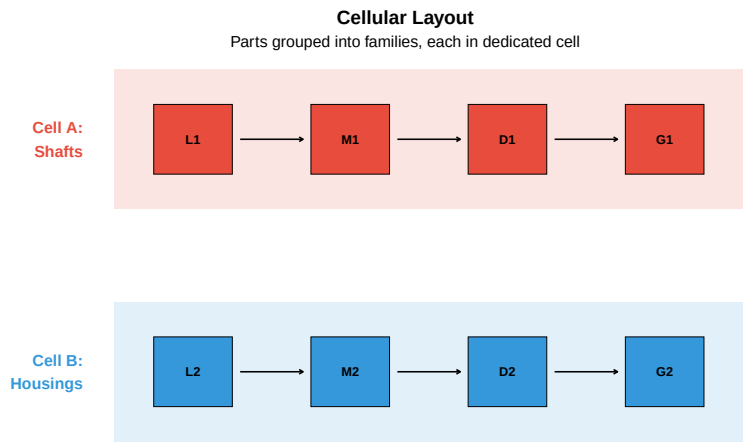


Figure 3.8: Cellular Manufacturing: From Process Layout to Cells

Table 3.4: Cellular Manufacturing Benefits vs. Traditional Process Layout

| Metric            | Traditional (Process)       | Cellular                  | Typical Improvement   |
|-------------------|-----------------------------|---------------------------|-----------------------|
| Setup time        | High - frequent changeovers | Low - dedicated to family | 50-90% reduction      |
| Work-in-process   | High - batches waiting      | Low - continuous flow     | 50-90% reduction      |
| Lead time         | Long - complex routing      | Short - U-cell flow       | 50-80% reduction      |
| Quality defects   | Higher - multiple handoffs  | Lower - cell ownership    | 30-50% reduction      |
| Material handling | High - long distances       | Low - within cell         | 40-70% reduction      |
| Floor space       | More - large aisles         | Less - compact cells      | 20-50% reduction      |
| Worker skills     | Specialized by machine      | Cross-trained on cell     | Increased flexibility |

### 3.8 Flexible Manufacturing Systems (FMS)

An **FMS** is a highly automated cell consisting of CNC machines interconnected by automated material handling, all controlled by a central computer.

#### 3.8.1 FMS Components

##### Flexible Manufacturing System (FMS) Layout

CNC machines connected by AGV with central computer control

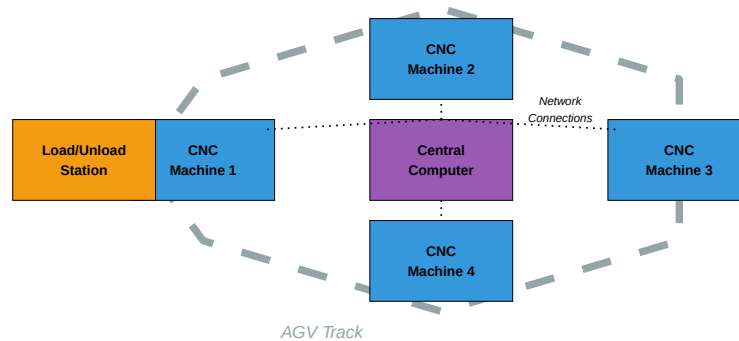


Figure 3.9: Flexible Manufacturing System Components

Table 3.5: FMS Characteristics and Applications

| Characteristic | FMS Capability |
|----------------|----------------|
|----------------|----------------|



|                   |                                                               |
|-------------------|---------------------------------------------------------------|
| Production Volume | Medium (2,000 - 100,000/year per part type)                   |
| Part Variety      | Medium (4-100 different part types)                           |
| Batch Size        | Small to medium (1-500 per batch)                             |
| Setup Time        | Minimal (automatic pallet/fixture changes)                    |
| Typical Parts     | Prismatic parts ( housings, brackets, covers)                 |
| Investment        | \$5M - \$50M+ depending on size                               |
| Best Suited For   | Aerospace, automotive components, job shops with repeat parts |

3.9 Modern Manufacturing Concepts

3.9.1 SMED (Single-Minute Exchange of Die)

SMED is a lean manufacturing technique to reduce changeover time to under 10 minutes (“single digit minutes”).

Table 3.6: SMED Methodology

| Step          | Description                                                                        | Example                     |
|---------------|------------------------------------------------------------------------------------|-----------------------------|
| 1. Separate   | Distinguish internal setup (machine stopped) from external setup (machine running) | Pre-staging                 |
| 2. Convert    | Convert as much internal setup to external setup as possible                       | Use quick change clamps     |
| 3. Streamline | Reduce time for remaining internal and external activities                         | Eliminate unnecessary steps |

Case Study: Injection Molding Changeover

**Before SMED:** - Changeover time: 90 minutes - Machine stopped for entire process - One operator working alone

**After SMED analysis:**

| Activity         | Before | Type     | After | Change                                |
|------------------|--------|----------|-------|---------------------------------------|
| Find tools       | 10 min | Internal | 0     | External - tools pre-staged           |
| Remove old mold  | 15 min | Internal | 5 min | Quick-release clamps                  |
| Get new mold     | 10 min | Internal | 0     | External - mold pre-staged at machine |
| Install new mold | 20 min | Internal | 8 min | Standardized mold heights             |

| Activity          | Before | Type     | After | Change                 |
|-------------------|--------|----------|-------|------------------------|
| Connect utilities | 10 min | Internal | 4 min | Quick-connect fittings |
| Adjust settings   | 15 min | Internal | 3 min | Pre-programmed recipes |
| First article     | 10 min | Internal | 5 min | Improved documentation |

**Results:** - Total changeover: 25 minutes (72% reduction) - Additional capacity: +15 changeovers/week possible - Smaller batch sizes now economical

### 3.9.2 AGV and AMR Systems

Table 3.7: AGV vs. AMR Comparison

| Feature                  | AGV (Automated Guided Vehicle)              | AMR (Autonomous Mobile Robot)             |
|--------------------------|---------------------------------------------|-------------------------------------------|
| <b>Navigation</b>        | Fixed path (wires, magnets, tape)           | Dynamic (SLAM, vision, sensors)           |
| <b>Flexibility</b>       | Low - follows predetermined routes          | High - calculates optimal routes          |
| <b>Infrastructure</b>    | Requires installation (floor modifications) | Minimal (maps environment)                |
| <b>Cost</b>              | Lower vehicle cost, higher infrastructure   | Higher vehicle cost, lower infrastructure |
| <b>Path Changes</b>      | Requires physical modification              | Software update only                      |
| <b>Obstacle Handling</b> | Stops and waits                             | Navigates around obstacles                |
| <b>Best Application</b>  | High-volume, repetitive routes              | Variable routes, dynamic environment      |

### 3.9.3 Digital Twin

A **digital twin** is a virtual representation of a physical manufacturing system that is updated in real-time with sensor data.

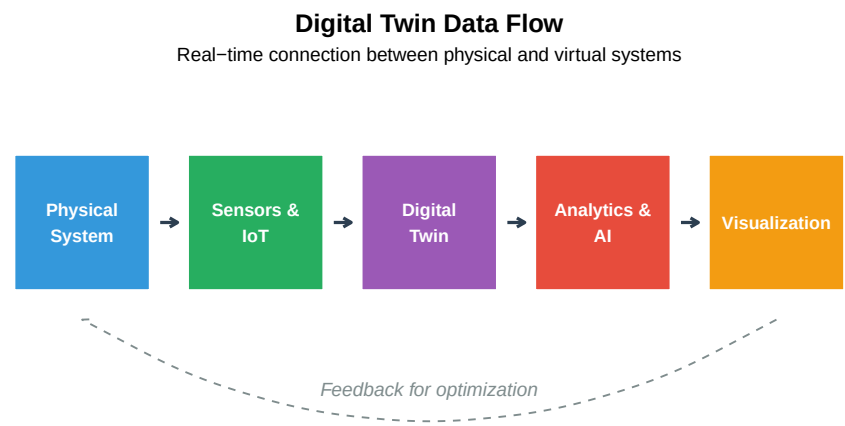


Figure 3.10: Digital Twin Concept

**Digital Twin Applications:** - Predictive maintenance (predict failures before they occur) - Process optimization (simulate changes before implementation) - Training (virtual commissioning and operator training) - Quality prediction (correlate parameters with outcomes)

3.10 Industry Applications

Table 3.8: Manufacturing System Applications by Industry

| System Type     | Automotive                                    | Food Processing                          |
|-----------------|-----------------------------------------------|------------------------------------------|
| Transfer Line   | Engine block machining, cylinder head lines   | Beverage bottling, canning lines         |
| Manual Assembly | Final vehicle assembly, interior trim         | Packaging, quality inspection            |
| FMS             | Transmission components, brake parts          | Multi-product bakery equipment           |
| Cellular        | Subassemblies, wiring harnesses               | Specialty product lines                  |
| AGV/AMR         | Body shop material delivery, parts sequencing | Ingredient delivery, packaging transport |

3.11 Review Questions

Question 1: Takt Time Calculation

A plant operates 8-hour shifts with 30 minutes of breaks. Daily demand is 400 units. Calculate the takt time.

**Solution:**

Available time = 8 hours  $\times$  60 min - 30 min = 450 minutes

Takt time = 450 min / 400 units = **1.125 minutes per unit** = 67.5 seconds

If actual cycle time is 60 seconds, the plant has  $(67.5 - 60)/67.5 = 11\%$  **capacity margin**.

Question 2: Line Efficiency

**A 6-station assembly line has a cycle time of 2 minutes. The task times at each station are: 1.8, 1.9, 2.0, 1.7, 1.6, and 1.5 minutes. Calculate the line efficiency.**

**Solution:**

Total work content = 1.8 + 1.9 + 2.0 + 1.7 + 1.6 + 1.5 = 10.5 minutes

Line efficiency =  $(10.5) / (6 \times 2.0) \times 100\% = 10.5/12 \times 100\% = 87.5\%$

Balance delay (idle time) =  $100\% - 87.5\% = 12.5\%$

Question 3: Downtime Impact

**An automated line has 8 stations, each with 3% failure probability per cycle and 8-minute average repair time. The ideal cycle time is 0.5 minutes. What is the actual production rate?**

**Solution:**

Line failure probability =  $1 - (1 - 0.03)^8 = 1 - 0.97^8 = 1 - 0.784 = 21.6\%$

Actual cycle time =  $0.5 + (0.216 \times 8) = 0.5 + 1.73 = 2.23$  **minutes**

Actual production rate =  $60 / 2.23 = 26.9$  **parts/hour**

(vs. ideal of 120 parts/hour - only 22% of ideal capacity!)

This shows why reliability is critical in automated systems.

Question 4: FMS Selection

**A company makes 15 different part numbers with annual volumes of 500-5,000 each. Parts are machined from aluminum and require milling, drilling, and tapping. Which system is most appropriate: dedicated transfer line, FMS, or CNC job shop?**

**Answer: FMS is most appropriate because:**

- **Volume range (500-5,000)** is ideal for FMS
- **Part variety (15 types)** requires flexibility
- **Similar operations** (milling, drilling, tapping) can share equipment
- **Material (aluminum)** machines well on CNC

Transfer line would require too much volume per part. Job shop would have high setup times and low efficiency for these volumes.

Question 5: SMED Application

**A stamping press has a 45-minute die change. Internal activities are 35 minutes, external are 10 minutes but currently done with machine stopped. How much can SMED reduce this?**

**Answer:**

**Step 1 - Separate:** Do the 10 minutes of external work while press runs. - New changeover: 35 minutes (30% reduction)

**Step 2 - Convert:** Analyze the 35 minutes: - If 15 minutes can be converted to external (pre-staging, etc.): New changeover = 20 minutes

**Step 3 - Streamline:** Quick-release clamps, standardization might save another 5 minutes. - Final changeover: **15 minutes (67% reduction)**

This enables smaller batch sizes and more frequent changeovers.

## 3.12 Summary

Table 3.9: Chapter 3 Summary

| Topic                         | Key Points                                                                              |
|-------------------------------|-----------------------------------------------------------------------------------------|
| <b>Production Lines</b>       | Manual, continuous, synchronous, or asynchronous transfer; each has trade-offs          |
| <b>Time Metrics</b>           | Takt time = demand pace; Cycle time = actual pace; MCE measures value-added %           |
| <b>Line Balancing</b>         | Assign tasks to minimize idle time; Efficiency = work content / (stations × cycle time) |
| <b>Automated Lines</b>        | Reliability is critical; Small failure probabilities cause large capacity losses        |
| <b>Cellular Manufacturing</b> | Group Technology groups part families; Reduces setup, WIP, lead time 50-90%             |
| <b>FMS</b>                    | Automated cells for medium variety/volume; Computer-controlled flexibility              |
| <b>Modern Concepts</b>        | SMED for quick changeover; AGV/AMR for material handling; Digital twins for opt         |

### 3.12.1 Key Takeaways

1. **Match system to product** - Volume and variety determine optimal manufacturing system
2. **Takt time sets the pace** - Everything should be designed to meet customer demand rate
3. **Balance matters** - Unbalanced lines waste capacity through idle time
4. **Reliability is critical** - Small failures compound to large losses in automated systems

5. **Flexibility has value** - FMS and cellular systems enable quick response to change
  6. **Technology evolves** - Digital twins and AMRs are transforming manufacturing
- 

### 3.13 References

1. Groover, M.P. (2020). *Automation, Production Systems, and Computer-Integrated Manufacturing* (5th ed.). Pearson.
2. Liker, J.K. (2004). *The Toyota Way*. McGraw-Hill.
3. Shingo, S. (1985). *A Revolution in Manufacturing: The SMED System*. Productivity Press.
4. Black, J.T. (1991). *The Design of the Factory with a Future*. McGraw-Hill.

## Chapter 4

# Plant Layout and Facility Design

---

### 4.1 Learning Objectives

By the end of this chapter, you will be able to:

- Identify and describe the four basic types of facility layouts
- Compare the advantages and disadvantages of each layout type
- Apply the steps involved in designing a process layout
- Calculate cycle time, theoretical minimum stations, and line efficiency for product layouts
- Understand the principles of warehouse and office layout design

Why is Plant Layout Important?

Plant layout directly impacts:

- **Productivity:** Efficient layouts minimize wasted movement and time
  - **Cost:** Poor layouts increase material handling costs by 20-50%
  - **Safety:** Well-designed layouts reduce accidents and injuries
  - **Flexibility:** The right layout allows adaptation to changing demands
  - **Employee Morale:** Good layouts improve working conditions
-

4.2 Types of Facility Layouts

There are **four basic layout types**, each suited to different production requirements:

Table 4.1: Overview of Facility Layout Types

| Layout Type            | Also Known As                 | Best For                        | Ex |
|------------------------|-------------------------------|---------------------------------|----|
| Process Layout         | Functional Layout, Job Shop   | High variety, low volume        | Ho |
| Product Layout         | Flow Line, Assembly Line      | Low variety, high volume        | Au |
| Hybrid/Cellular Layout | Group Technology, Cell Layout | Medium variety, medium volume   | El |
| Fixed-Position Layout  | Project Layout                | Very large or immobile products | Sh |

4.2.1 Visualizing the Layout Spectrum

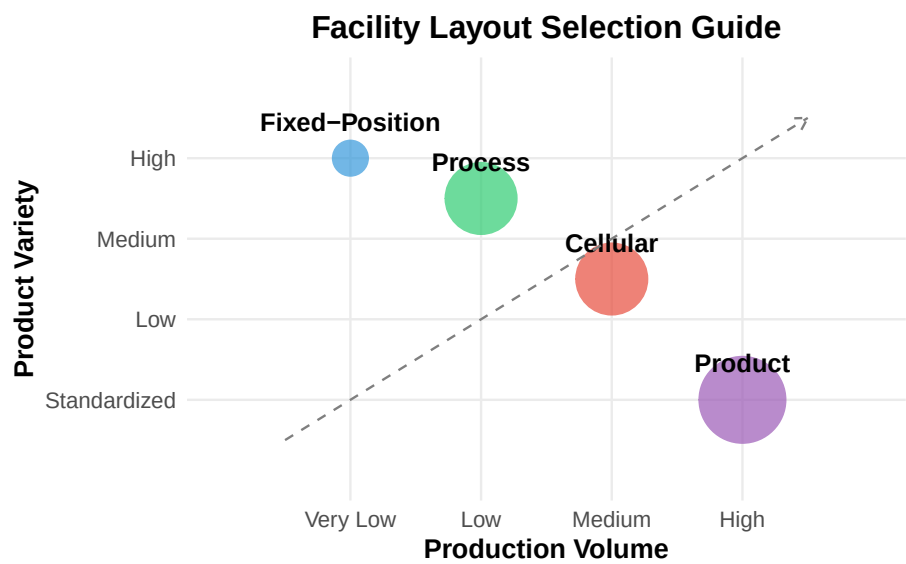


Figure 4.1: Product Variety vs. Production Volume for Different Layouts

4.3 Process Layout (Functional Layout)

In a **process layout**, similar resources (machines, equipment, workers with similar skills) are grouped together. Work flows between departments based on the specific requirements of each job.



Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use `linewidth` instead.

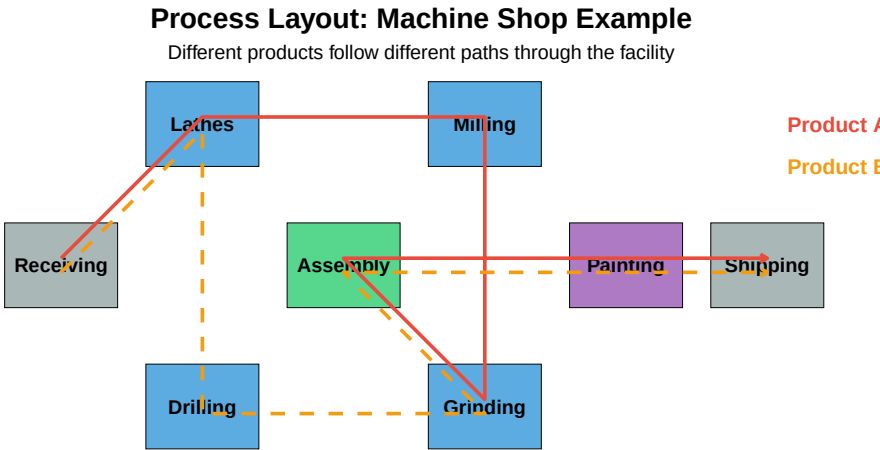


Figure 4.2: Example Process Layout - Machine Shop

### 4.3.1 Characteristics of Process Layouts

Table 4.2: Process Layout Characteristics

| Characteristic            | Description                    |
|---------------------------|--------------------------------|
| <b>Equipment</b>          | General-purpose machines       |
| <b>Labor</b>              | Skilled workers required       |
| <b>Flexibility</b>        | High - can handle variety      |
| <b>Processing Speed</b>   | Slower due to varied routing   |
| <b>Material Handling</b>  | Higher costs (complex paths)   |
| <b>Scheduling</b>         | Complex - different routes     |
| <b>Space Requirements</b> | Higher - WIP inventory storage |

Discussion Question: Can you identify a process layout in your daily life?

**Think about:** - A hospital emergency room (patients routed based on their needs)  
- A university campus (students move between buildings for different

classes) - A grocery store (customers choose their own path) - A machine shop (parts routed based on required operations)

**What makes these process layouts?** Similar functions are grouped together, and “products” (patients, students, customers) follow different paths based on their individual needs.

### 4.3.2 Real-World Examples

## 4.4 Product Layout (Assembly Line)

In a **product layout**, equipment and workstations are arranged in a line according to the sequence of operations needed to produce the product. Every unit follows the same path.

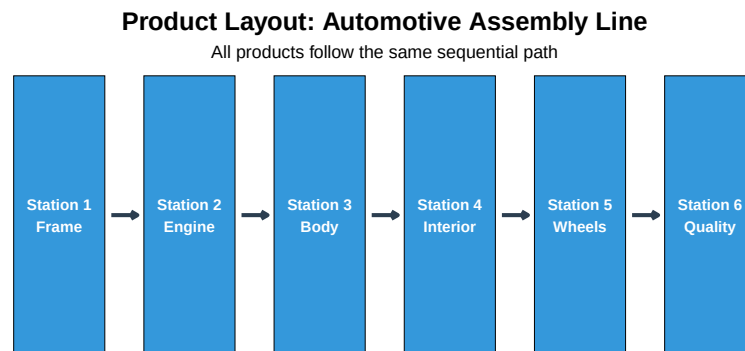


Figure 4.3: Product Layout - Assembly Line

### 4.4.1 Characteristics of Product Layouts

Table 4.3: Product Layout: Advantages vs. Disadvantages

| Advantage             | Disadvantage                |
|-----------------------|-----------------------------|
| High production rate  | High initial investment     |
| Low unit cost         | Low flexibility             |
| Low material handling | Line stops affect all       |
| Simple scheduling     | Monotonous work             |
| Less WIP inventory    | Requires balanced workloads |

Quick Quiz: Product Layout

**Question:** A car wash where vehicles move through a series of cleaning stations is an example of which layout type?

**Answer:** Product Layout - all cars follow the same path through the same sequence of operations.

## 4.5 Hybrid Layout (Cellular Manufacturing)

**Hybrid layouts** combine the flexibility of process layouts with the efficiency of product layouts. The most common type is the **cellular layout** using Group Technology.

### 4.5.1 Group Technology Concept

Group Technology (GT) identifies parts with similar characteristics and groups them into **part families**. Machines are then arranged into **cells** to process these families.

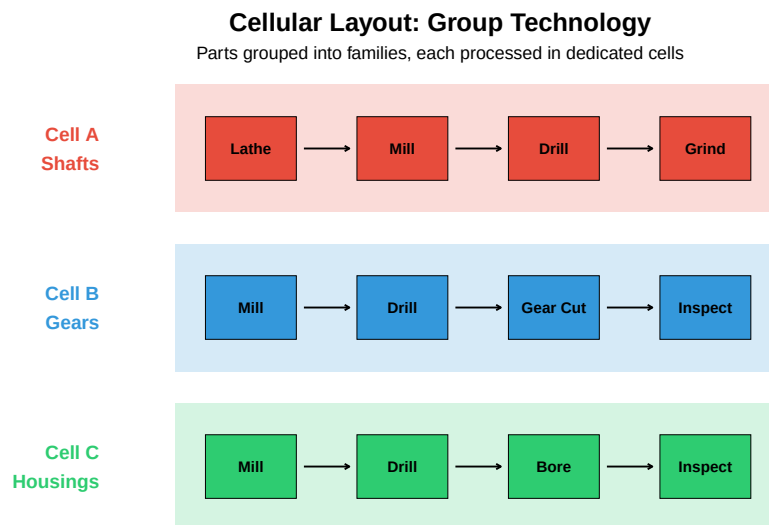


Figure 4.4: Cellular Layout with Manufacturing Cells

Discussion Question: Benefits of Cellular Manufacturing

**What are the main benefits of cellular manufacturing over traditional process layouts?**

1. **Reduced material handling** - Parts stay within the cell
2. **Shorter lead times** - No waiting between departments

3. **Less WIP inventory** - Smaller batches, faster flow
  4. **Improved quality** - Workers responsible for complete part
  5. **Team accountability** - Cell teams own their output
- 

## 4.6 Fixed-Position Layout

In a **fixed-position layout**, the product remains stationary while workers, equipment, and materials are brought to it. This is used when the product is too large or heavy to move.

### 4.6.1 Examples of Fixed-Position Layouts

- **Shipbuilding** - Ships built in dry docks
- **Aircraft manufacturing** - Large aircraft assembled in hangars
- **Construction** - Buildings, bridges, dams
- **Surgery** - Patient remains stationary

Challenge Question: Managing a Fixed-Position Layout

**What are the unique challenges of managing a fixed-position layout?**

Think about: - Scheduling deliveries of materials and equipment - Coordinating multiple contractors/teams - Limited space around the product - Weather and environmental factors (outdoor projects) - Ensuring safety with many workers in one area

---

## 4.7 Designing Process Layouts

Designing an effective process layout involves **three main steps**:

### 4.7.1 Step 1: Gather Information

You need to determine: - **Space requirements** for each department - **Available space** in the facility - **Closeness relationships** between departments

#### 4.7.1.1 From-To Matrix (Load Matrix)

A From-To matrix shows the number of trips or loads between departments:

Table 4.4: From-To Matrix: Loads per Day Between Departments

|           | Receiving | Machining | Assembly | Painting | Shipping |
|-----------|-----------|-----------|----------|----------|----------|
| Receiving | -         | 100       | 50       | 0        | 0        |

|                  |   |   |     |     |     |
|------------------|---|---|-----|-----|-----|
| <b>Machining</b> | 0 | - | 200 | 50  | 0   |
| <b>Assembly</b>  | 0 | 0 | -   | 100 | 60  |
| <b>Painting</b>  | 0 | 0 | 0   | -   | 150 |
| <b>Shipping</b>  | 0 | 0 | 0   | 0   | -   |

### 4.7.2 Step 2: Develop Block Plans

Use the **Load-Distance Model** to evaluate layout alternatives:

$$\text{Total Cost} = \sum_{i=1}^n \sum_{j=1}^n L_{ij} \times D_{ij} \times C$$

Where: -  $L_{ij}$  = Number of loads between departments  $i$  and  $j$  -  $D_{ij}$  = Distance between departments  $i$  and  $j$  -  $C$  = Cost per unit distance

### 4.7.3 Interactive Example: Comparing Two Layouts

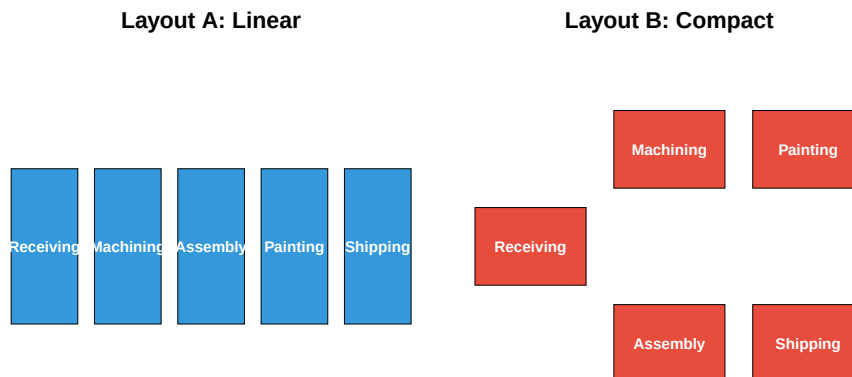


Figure 4.5: Comparing Two Process Layout Alternatives

Live cell — try different from-to volumes or layout distances.

```
#| label: load-distance-calc
# Calculate Load-Distance scores for both layouts
# From-To data: Receiving->Machining=100, Machining->Assembly=200,
#               Assembly->Painting=100, Painting->Shipping=150

# Layout A distances (linear): all adjacent = 1 unit
layout_A_score <- 100*1 + 200*1 + 100*1 + 150*1
cat("Layout A Total Load-Distance:", layout_A_score, "\n")
```

```
# Layout B distances (compact): most departments closer together
# Receiving to Machining = 1.12 (diagonal)
# Machining to Assembly = 1
# Assembly to Painting = 1.12 (diagonal)
# Painting to Shipping = 1
layout_B_score <- 100*1.12 + 200*1 + 100*1.12 + 150*1
cat("Layout B Total Load-Distance:", round(layout_B_score, 1), "\n")

cat("\nLayout A is better by:", round(layout_B_score - layout_A_score, 1), "units")
```

Why does the linear layout win in this case?

In this example, the linear layout (A) wins because the process flow is essentially sequential (Receiving → Machining → Assembly → Painting → Shipping).

When flow is **sequential**, a **product layout** (linear arrangement) is more efficient. Process layouts are better when there's **varied routing** between departments.

#### 4.7.4 Step 3: Develop Detailed Layout

Once the block plan is selected: - Determine exact sizes and shapes of departments - Plan aisle locations and widths - Consider utilities, safety exits, and accessibility - Use CAD software or 3D modeling for visualization

## 4.8 Designing Product Layouts (Line Balancing)

The key challenge in product layout design is **line balancing** - assigning tasks to workstations to minimize idle time while meeting production requirements.

### 4.8.1 Line Balancing Steps

1. Identify tasks and precedence relationships
2. Calculate required cycle time
3. Calculate theoretical minimum number of stations
4. Assign tasks to stations
5. Calculate efficiency

### 4.8.2 Key Formulas

Cycle Time:

$$C = \frac{\text{Available Production Time}}{\text{Desired Output}}$$

**Theoretical Minimum Stations:**

$$TM = \frac{\sum t_i}{C}$$

**Line Efficiency:**

$$\text{Efficiency} = \frac{\sum t_i}{n \times C} \times 100\%$$

Where: -  $C$  = Cycle time -  $\sum t_i$  = Total task time -  $n$  = Number of workstations

**4.8.3 Interactive Example: Pizza Assembly Line**

Table 4.5: Pizza Assembly Tasks

| Task | Description  | Time (sec) | Predecessors |
|------|--------------|------------|--------------|
| A    | Roll dough   | 40         | -            |
| B    | Spread sauce | 25         | A            |
| C    | Add cheese   | 20         | B            |
| D    | Add toppings | 35         | B            |
| E    | Season       | 15         | C, D         |
| F    | Box pizza    | 30         | E            |

**Precedence Diagram: Pizza Assembly**

Arrows show required sequence of operations

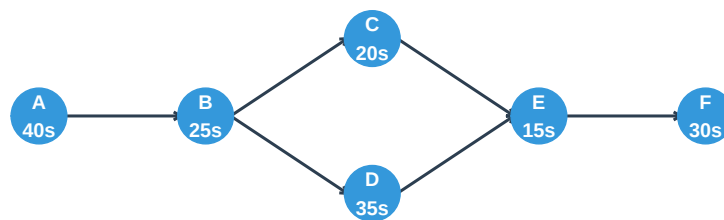


Figure 4.6: Precedence Diagram for Pizza Assembly

**4.8.4 Line Balancing Calculation**

This one is live — edit the numbers and re-run it to see how the results change.

```

#| label: line-balance-calc
# Given data
total_task_time <- 40 + 25 + 20 + 35 + 15 + 30 # seconds
desired_output <- 60 # pizzas per hour
available_time <- 3600 # seconds per hour

# Step 1: Calculate Cycle Time
cycle_time <- available_time / desired_output
cat("Cycle Time:", cycle_time, "seconds per pizza\n")

# Step 2: Calculate Theoretical Minimum Stations
TM <- total_task_time / cycle_time
cat("Theoretical Minimum Stations:", TM, "=", ceiling(TM), "stations\n")

# Step 3: Assign tasks to stations (one possible solution)
cat("\n--- Workstation Assignments ---\n")
cat("Station 1: A (40s) + B (25s) = 65s > 60s ... EXCEEDS!\n")
cat("Let's try again:\n")
cat("Station 1: A (40s) = 40s [Idle: 20s]\n")
cat("Station 2: B (25s) + D (35s) = 60s [Idle: 0s]\n")
cat("Station 3: C (20s) + E (15s) + F (30s) = 65s > 60s... EXCEEDS!\n")
cat("Revised:\n")
cat("Station 3: C (20s) + E (15s) = 35s [Idle: 25s]\n")
cat("Station 4: F (30s) = 30s [Idle: 30s]\n")

# With 4 stations
n_stations <- 4
efficiency <- (total_task_time / (n_stations * cycle_time)) * 100
cat("\n--- Results ---\n")
cat("Number of Stations:", n_stations, "\n")
cat("Line Efficiency:", round(efficiency, 1), "%\n")
cat("Balance Delay (Idle Time):", round(100 - efficiency, 1), "%\n")

```

Try It Yourself: Can you find a better balance?

**Challenge:** Can you assign tasks to only 3 stations while respecting the precedence constraints?

**Hint:** Consider these assignments: - Station 1: A (40s) + ? - Station 2: B (25s) + ? + ? - Station 3: ? + ?

Remember: Total time per station cannot exceed 60 seconds!

**Solution:** It's actually possible with careful planning: - Station 1: A (40s) = 40s - Station 2: B (25s) + C (20s) + E (15s) = 60s (but E needs C AND D complete!)

So 3 stations is NOT feasible due to precedence constraints. The minimum is 4



stations for this problem.

## 4.9 Warehouse Layout Design

Warehouse layouts are a special case of process layouts focused on **minimizing material handling**.

### 4.9.1 Key Principles

1. **Place high-volume items near the dock**
2. **Group items that are often picked together**
3. **Use ABC analysis** for storage assignment:
  - **A items** (20% of SKUs, 80% of picks): Prime locations
  - **B items** (30% of SKUs, 15% of picks): Secondary locations
  - **C items** (50% of SKUs, 5% of picks): Remote locations

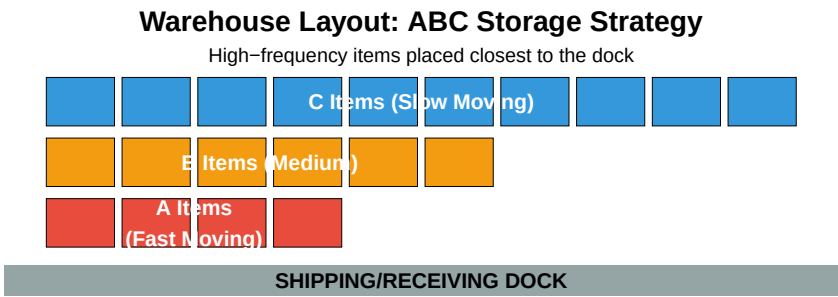


Figure 4.7: Warehouse Layout: ABC Storage Strategy

## 4.10 Office Layout Design

Office layouts balance **communication needs** with **privacy requirements**.

### 4.10.1 Two Main Approaches

Table 4.6: Office Layout Comparison

| Aspect               | Open Plan                  | Traditional/Closed        |
|----------------------|----------------------------|---------------------------|
| <b>Space</b>         | Shared workspaces          | Private offices           |
| <b>Walls</b>         | Partitions/dividers        | Permanent walls           |
| <b>Flexibility</b>   | High - easy to reconfigure | Low - expensive to change |
| <b>Communication</b> | Excellent                  | Limited                   |
| <b>Privacy</b>       | Low                        | High                      |
| <b>Cost</b>          | Lower                      | Higher                    |

### 4.10.2 Office Ergonomics Checklist

Click to expand: Office Ergonomics Best Practices

**Posture:** - Support the small of the back - Keep shoulders relaxed - Maintain neutral wrist position - Feet flat on floor or footrest

**Lighting:** - Position monitor to reduce glare - Adjust brightness for screen vs. paper work - Use task lighting when needed

**Workstation:** - Monitor at arm's length - Top of screen at or below eye level - Keyboard and mouse at elbow height

**Work Habits:** - Take breaks every 30-60 minutes - Vary tasks to prevent repetitive strain - Use keyboard shortcuts to reduce mouse use

### 4.10.3 The Future of Office Design

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## 4.11 Summary: Choosing the Right Layout

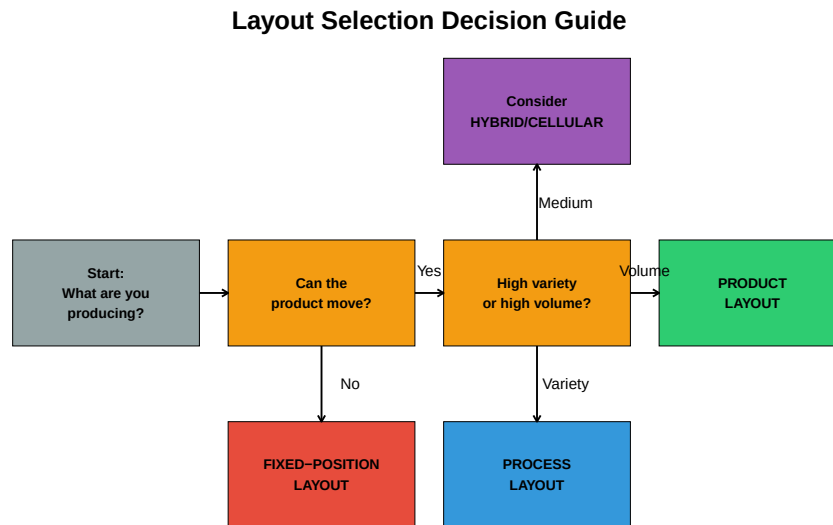


Figure 4.8: Layout Selection Decision Guide

### 4.11.1 Key Takeaways

1. **Process layouts** offer flexibility for varied products but have higher material handling costs
2. **Product layouts** are efficient for high-volume, standardized products
3. **Cellular layouts** provide a middle ground with benefits of both
4. **Fixed-position layouts** are necessary when products cannot be moved
5. **Line balancing** is critical for efficient product layouts
6. **Warehouse layouts** should minimize travel distance for high-frequency items

## 4.12 Industry Case Studies

### 4.12.1 Case Study 1: Food Processing Plant Layout

Food processing facilities have unique layout requirements due to food safety, hygiene, and regulatory compliance.

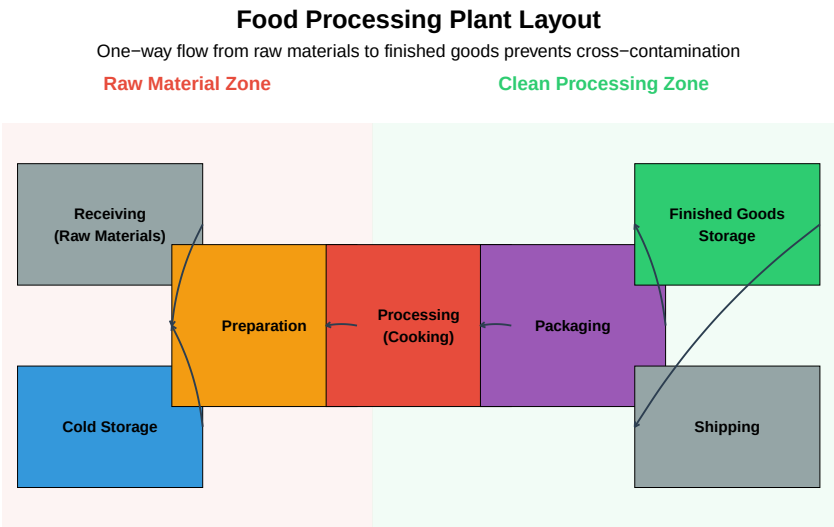


Figure 4.9: Food Processing Plant Layout Example

Key Design Principles for Food Processing:

Table 4.7: Food Processing Layout Design Principles

| Principle                  | Description                                                        | Regulation |
|----------------------------|--------------------------------------------------------------------|------------|
| <b>Linear Flow</b>         | Raw materials → Processing → Packaging (no backtracking)           | FSMA       |
| <b>Zone Separation</b>     | Physical barriers between raw and ready-to-eat areas               | HACCP      |
| <b>Cleanability</b>        | Smooth, non-porous surfaces; rounded corners; proper drainage      | 3-A Star   |
| <b>Temperature Control</b> | Maintain cold chain; separate refrigerated zones                   | FDA 21     |
| <b>Sanitary Design</b>     | Handwashing stations, foot baths, air curtains at zone transitions | USDA F     |
| <b>Allergen Control</b>    | Dedicated lines or thorough cleaning between allergen products     | FALCPA     |

Discussion Question: Cross-Contamination Prevention

**Question:** A food processing plant produces both peanut butter products and tree nut-free products. What layout considerations are essential?

Key Considerations:

1. **Physical Separation:** Completely separate production lines with physical barriers (walls, not just distance)
2. **Air Handling:** Separate HVAC systems to prevent airborne allergen transfer

3. **Personnel Flow:** Dedicated workers for each line, or strict gowning/cleaning procedures when crossing zones
4. **Scheduling:** If shared equipment is unavoidable, schedule allergen products last before deep cleaning
5. **Traffic Patterns:** Forklift and cart paths should not cross between allergen zones
6. **Verification:** Air sampling and surface swabs to verify effectiveness

#### 4.12.2 Case Study 2: Aerospace Manufacturing Facility

Aerospace facilities often use **fixed-position layouts** combined with **process layouts** for component fabrication.

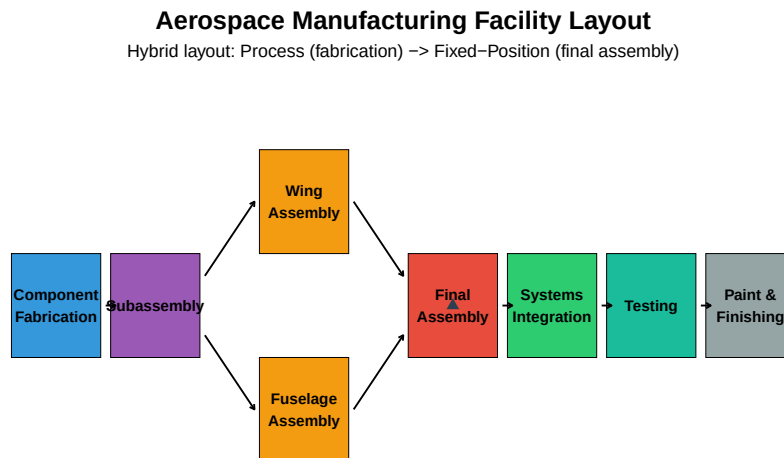


Figure 4.10: Aerospace Assembly Facility Layout

##### Aerospace Layout Characteristics:

- **Clean rooms** for precision components (hydraulics, avionics)
- **High-bay areas** for large assembly operations
- **Overhead cranes** for moving heavy components
- **Flow-line final assembly** (aircraft moves through stations) for high-volume commercial aircraft
- **Fixed-position** for military aircraft or low-volume production
- **Strict FOD (Foreign Object Debris)** control throughout

Video: Boeing 737 Assembly

Watch how Boeing uses a moving assembly line for high-volume aircraft production:

### 4.12.3 Case Study 3: Automotive Supplier - Hybrid Cell Layout

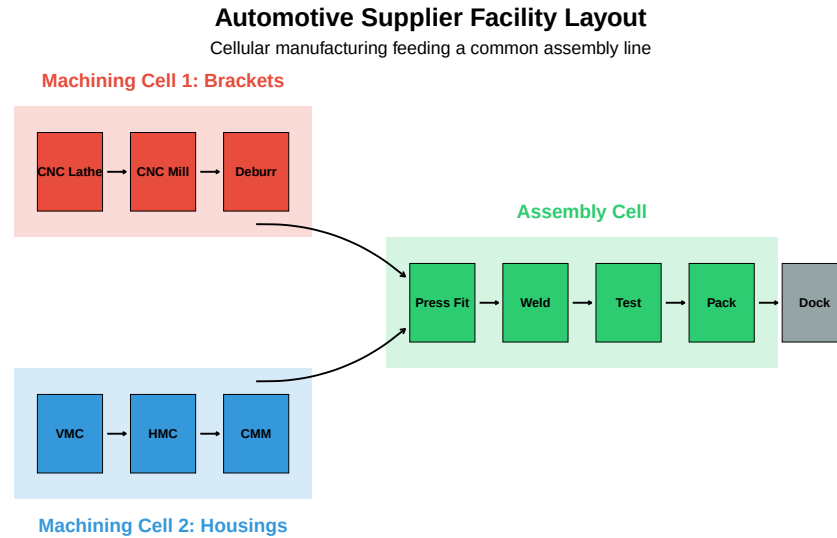


Figure 4.11: Automotive Supplier Facility with Cellular Layout

#### JIT Delivery Requirements:

Automotive suppliers must often deliver directly to the assembly line in **sequence** (JIS - Just-In-Sequence):

Table 4.8: JIT/JIS Requirements and Layout Implications

| Requirement              | Typical Standard                      | Layout Impact                            |
|--------------------------|---------------------------------------|------------------------------------------|
| <b>Delivery Window</b>   | ±30 minutes of scheduled time         | Dedicated shipping lane, staging area n  |
| <b>Sequence Accuracy</b> | 100% (0 sequence errors)              | Pick-to-light systems, visual managemen  |
| <b>Quality Level</b>     | 0 PPM target (Six Sigma)              | Inline inspection, mistake-proofing (pol |
| <b>Packaging</b>         | Returnable containers to line side    | Container cleaning/storage area needed   |
| <b>Traceability</b>      | Full lot/serial traceability required | Barcode/RFID scanning stations throu     |

## 4.13 Ergonomics in Manufacturing

Ergonomics (human factors engineering) is critical for plant layout design to ensure worker safety, reduce injuries, and improve productivity.

### 4.13.1 The NIOSH Lifting Equation

The National Institute for Occupational Safety and Health (NIOSH) developed an equation to evaluate manual lifting tasks:

$$RWL = LC \times HM \times VM \times DM \times AM \times FM \times CM$$

Where **RWL** = Recommended Weight Limit (in pounds or kg)

Table 4.9: NIOSH Lifting Equation Factors

| Factor    | Name                  | Formula                                   | Description                                    |
|-----------|-----------------------|-------------------------------------------|------------------------------------------------|
| <b>LC</b> | Load Constant         | 51 lb (23 kg)                             | Maximum weight under ideal conditions          |
| <b>HM</b> | Horizontal Multiplier | 10/H                                      | H = horizontal distance from spine to load     |
| <b>VM</b> | Vertical Multiplier   | $1 - 0.0075 \times V - 30 \times 10^{-4}$ | V = vertical height of hands at start (inches) |
| <b>DM</b> | Distance Multiplier   | $0.82 + 1.8/D$                            | D = vertical travel distance (inches)          |
| <b>AM</b> | Asymmetric Multiplier | $1 - 0.0032A$                             | A = angle of asymmetry (degrees)               |
| <b>FM</b> | Frequency Multiplier  | Table lookup                              | Based on lift frequency and duration           |
| <b>CM</b> | Coupling Multiplier   | Table lookup                              | Based on grip quality (good/fair/poor)         |

### 4.13.2 Lifting Index Calculation

$$LI = \frac{\text{Actual Weight}}{\text{RWL}}$$

Table 4.10: Lifting Index Interpretation

| Lifting Index       | Risk Level | Action Required                                   |
|---------------------|------------|---------------------------------------------------|
| LI 1.0              | Low        | Task is acceptable for most workers               |
| $1.0 < LI \leq 3.0$ | Moderate   | Task poses risk; engineering controls recommended |
| $LI > 3.0$          | High       | Task is unacceptable; immediate redesign required |

### 4.13.3 NIOSH Lifting Example

Also live — try changing the box weight, distances, or lift frequency below and re-running.

```
#| label: niosh-example-plant
# Example: Worker lifts 35 lb boxes from pallet to conveyor
# Conditions:
# - Horizontal distance (H) = 15 inches
# - Vertical height at start (V) = 10 inches
# - Vertical travel distance (D) = 20 inches
# - Asymmetry angle (A) = 30 degrees
```

```

# - Frequency = 4 lifts per minute, 2 hours
# - Coupling = Fair (handles present)

LC <- 51 # Load constant (lb)

# Calculate multipliers
HM <- 10 / 15 # Horizontal multiplier
VM <- 1 - 0.0075 * abs(10 - 30) # Vertical multiplier
DM <- 0.82 + (1.8 / 20) # Distance multiplier
AM <- 1 - 0.0032 * 30 # Asymmetric multiplier
FM <- 0.72 # From NIOSH table: 4 lifts/min, 1-2 hours
CM <- 0.95 # Fair coupling

cat("Multiplier Values:\n")
cat("  HM (Horizontal):", round(HM, 3), "\n")
cat("  VM (Vertical):", round(VM, 3), "\n")
cat("  DM (Distance):", round(DM, 3), "\n")
cat("  AM (Asymmetry):", round(AM, 3), "\n")
cat("  FM (Frequency):", FM, "\n")
cat("  CM (Coupling):", CM, "\n")

# Calculate RWL
RWL <- LC * HM * VM * DM * AM * FM * CM
cat("\nRecommended Weight Limit (RWL):", round(RWL, 1), "lb\n")

# Calculate Lifting Index
actual_weight <- 35
LI <- actual_weight / RWL
cat("Lifting Index (LI):", round(LI, 2), "\n")

if (LI <= 1.0) {
  cat("\nRisk Level: LOW - Task is acceptable\n")
} else if (LI <= 3.0) {
  cat("\nRisk Level: MODERATE - Consider engineering controls\n")
} else {
  cat("\nRisk Level: HIGH - Immediate redesign required\n")
}

```

Discussion Question: Reducing the Lifting Index

**Based on the example above, what layout or workstation changes could reduce the Lifting Index?**

**Solutions (in order of effectiveness):**

1. **Reduce horizontal distance (H):** Move conveyor closer to pallet position



- If  $H = 10''$ : HM improves from 0.67 to 1.0  $\rightarrow$  RWL increases by 50%
- 2. **Raise the starting height (V):** Use elevated pallet positions or pallet lifters
  - If  $V = 30''$  (optimal): VM improves from 0.85 to 1.0  $\rightarrow$  RWL increases by 18%
- 3. **Eliminate asymmetry (A):** Reposition conveyor directly in front of pallet
  - If  $A = 0^\circ$ : AM improves from 0.90 to 1.0  $\rightarrow$  RWL increases by 11%
- 4. **Reduce frequency:** Add a second worker or use automation
  - If 1 lift/min: FM improves from 0.72 to 0.94  $\rightarrow$  RWL increases by 31%
- 5. **Improve coupling:** Use boxes with good handles
  - If good coupling: CM improves from 0.95 to 1.0  $\rightarrow$  RWL increases by 5%

**Best solution:** Implement a scissor lift for the pallet (adjustable height) and move the conveyor closer. This addresses the largest multiplier reductions.

#### 4.13.4 Ergonomic Workstation Design

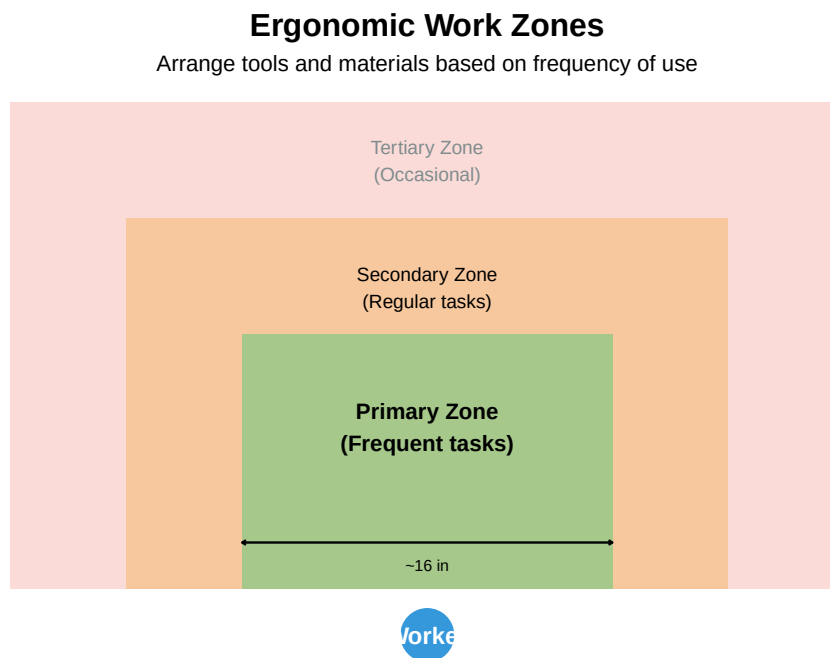


Figure 4.12: Ergonomic Work Zones for Standing Operations

**Layout Implications:**

- **Primary zone:** Most frequently used tools and parts
- **Secondary zone:** Regularly used items, occasional reaches
- **Tertiary zone:** Rarely needed items, should trigger redesign if frequent use required

## 4.14 Modern Layout Trends and Industry 4.0

### 4.14.1 Flexible Manufacturing Layouts

Modern facilities must adapt quickly to changing product demands. Key trends include:

Table 4.11: Modern Flexible Layout Trends

| Trend                   | Description                                                         | B  |
|-------------------------|---------------------------------------------------------------------|----|
| Modular Workstations    | Pre-built workstation modules that can be relocated in hours        | R  |
| Reconfigurable Cells    | Cells with standardized machine interfaces for quick changeover     | Sw |
| Mobile Equipment        | Equipment on wheels or air bearings for rapid repositioning         | El |
| Flexible Utilities      | Overhead utility drops and floor trenches allow equipment mobility  | N  |
| Digital Layout Planning | Digital twins enable virtual layout testing before physical changes | R  |

### 4.14.2 Industry 4.0 Layout Considerations

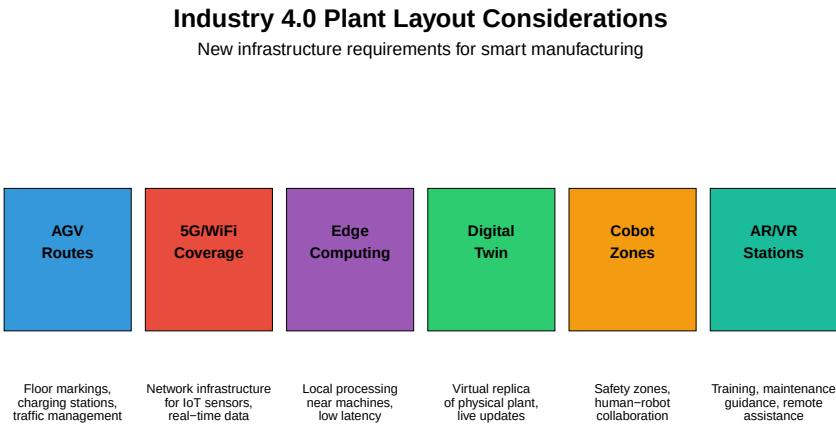


Figure 4.13: Industry 4.0 Enabled Plant Layout Elements

4.14.3 AGV and AMR Traffic Planning

Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs) require careful layout planning:

Table 4.12: AGV/AMR Layout Planning Guidelines

| Consideration         | Design Guidelines                                                                 |
|-----------------------|-----------------------------------------------------------------------------------|
| Route Design          | Unidirectional loops preferred; minimize intersections; avoid steep grades        |
| Charging Locations    | Opportunity charging at pickup/dropoff points; dedicated charging bays for breaks |
| Traffic Intersections | Yield zones, traffic signals, or elevation changes at crossings                   |
| Floor Requirements    | Smooth, level surfaces; magnetic tape or QR codes for navigation; no oil/water    |
| Safety Zones          | Minimum 1m clearance from pedestrian paths; light curtains at crossings           |
| Staging Areas         | Buffer areas for queue management; waiting positions clear of traffic             |

Video: AGV Systems in Modern Warehouses

4.15 Layout Simulation Software

Modern plant layout design relies heavily on simulation software to test alternatives before physical implementation.

4.15.1 Common Layout Planning Software

Table 4.13: Plant Layout Planning Software Comparison

| Software                     | Primary Use                  | Key Features                                                       |
|------------------------------|------------------------------|--------------------------------------------------------------------|
| AutoCAD Factory Design Suite | 2D/3D CAD layout design      | Extensive equipment libraries, interference checking, etc.         |
| FlexSim                      | 3D discrete event simulation | Drag-and-drop modeling, VR capability, real-time 3D rendering      |
| Arena                        | Process simulation           | Statistical analysis, what-if scenarios, bottleneck identification |
| Plant Simulation (Siemens)   | Digital factory simulation   | Full Siemens integration, AGV simulation, energy analysis          |
| SketchUp + Extensions        | Quick 3D visualization       | Low cost, easy learning curve, walkthrough capability              |

### 4.15.2 Simulation Benefits

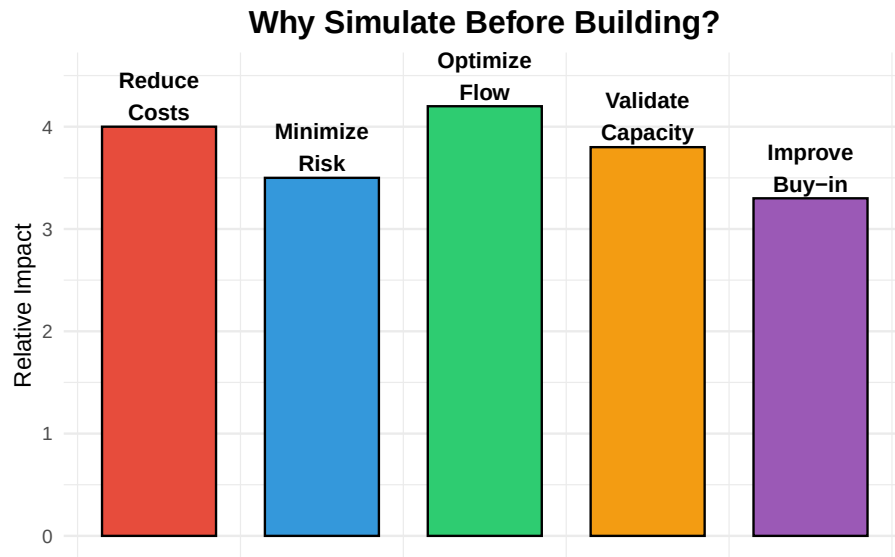


Figure 4.14: Benefits of Layout Simulation

#### ROI of Simulation:

- **10-20x** return on simulation investment typical
- **30-50%** reduction in layout change costs
- **Weeks to months** saved in ramp-up time
- **Stakeholder alignment** - visual communication of concepts

Video: Plant Simulation Demo

## 4.16 Practice Problems

### Problem 1: Layout Selection

**Scenario:** A company manufactures custom kitchen cabinets. Each order is unique, with different dimensions, materials, and finishes. Production volume is about 50 units per month.

**Question:** Which layout type would you recommend and why?

**Answer:** A **process layout** would be most appropriate because: - High product variety (custom orders) - Low production volume - Different operations needed for each order - Flexibility is more important than efficiency

## Problem 2: Line Balancing

**Given:** - Total task time: 240 seconds - Desired output: 30 units per hour - Available time: 3600 seconds per hour

**Calculate:** 1. Cycle time 2. Theoretical minimum number of stations 3. If you use 3 stations, what is the efficiency?

**Solution:** 1. Cycle time =  $3600 / 30 = 120$  seconds 2. TM =  $240 / 120 = 2$  stations 3. Efficiency =  $(240 / (3 \times 120)) \times 100 = 66.7\%$

## Problem 3: Warehouse Layout

**Scenario:** A warehouse has the following items: - Item X: 500 picks/day - Item Y: 100 picks/day - Item Z: 50 picks/day

**Question:** How should these items be positioned relative to the shipping dock?

**Answer:** Using ABC analysis: - Item X (high frequency): **Closest to dock** (A zone) - Item Y (medium frequency): **Middle distance** (B zone) - Item Z (low frequency): **Farthest from dock** (C zone)

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## 4.17 Review Questions

Question 1: What distinguishes a cellular layout from both process and product layouts?

**Answer:** A cellular layout combines characteristics of both:

- **From process layout:** Equipment variety within each cell, flexibility to handle part variations
- **From product layout:** Sequential flow within the cell, dedicated equipment for part families
- **Unique characteristics:**
  - Parts grouped into families using Group Technology
  - Cells are self-contained mini-factories
  - Workers often cross-trained to operate multiple machines
  - Reduced material handling compared to process layout
  - More flexibility than pure product layout

Question 2: Calculate the Lifting Index for the following scenario

**Scenario:** A worker lifts 25 kg boxes from floor level ( $V=0$ ) to a shelf at waist height. The horizontal distance is 40 cm, vertical travel is 75 cm, no asymmetry, frequency is 2 lifts per minute for 4 hours, and coupling is poor (no handles).

**Given multipliers:** - LC = 23 kg - HM = 25/H (H in cm) - VM = 1 - 0.003|V - 75| (V in cm) - DM = 0.82 + 4.5/D (D in cm) - AM = 1.0 (no asymmetry) - FM = 0.65 (from tables) - CM = 0.90 (poor coupling)

**Solution:**

$$HM = 25/40 = 0.625$$

$$VM = 1 - 0.003|0 - 75| = 1 - 0.225 = 0.775$$

$$DM = 0.82 + 4.5/75 = 0.82 + 0.06 = 0.88$$

$$AM = 1.0$$

$$FM = 0.65$$

$$CM = 0.90$$

$$RWL = 23 \times 0.625 \times 0.775 \times 0.88 \times 1.0 \times 0.65 \times 0.90$$

$$RWL = 23 \times 0.625 \times 0.775 \times 0.88 \times 0.65 \times 0.90$$

$$RWL = 5.7 \text{ kg}$$

$$LI = 25/5.7 = 4.4$$

**Risk Level:** HIGH (LI > 3.0) - Immediate redesign required

**Recommended actions:** Raise starting height (use pallet lifter), reduce horizontal distance, add handles to boxes, reduce frequency.

Question 3: List three Industry 4.0 technologies that impact plant layout design

**Answer:** (Any three of the following)

1. **AGVs/AMRs:** Require route planning, charging stations, floor markings, safety zones
2. **Collaborative robots (cobots):** Need safety zones, but smaller footprint than traditional robot cells
3. **Digital twins:** Enable virtual layout testing, reduce physical prototyping
4. **IoT sensors:** Require network infrastructure (5G/WiFi), edge computing locations
5. **AR/VR systems:** Training stations, remote assistance capabilities
6. **Flexible automation:** Modular equipment, quick-connect utilities

Question 4: What are the key differences between food processing plant layouts and general manufacturing?

**Answer:**

| Aspect          | Food Processing                     | General Manufacturing               |
|-----------------|-------------------------------------|-------------------------------------|
| Flow direction  | Strictly one-way (raw → finished)   | Can have backtracking               |
| Zone separation | Physical barriers between raw/clean | Logical separation often sufficient |

| Aspect           | Food Processing                         | General Manufacturing        |
|------------------|-----------------------------------------|------------------------------|
| Surfaces         | Sanitary design, smooth, cleanable      | Durability focused           |
| Temperature      | Multiple temperature zones (cold chain) | Generally ambient            |
| Regulatory       | HACCP, FSMA, FDA oversight              | OSHA, industry standards     |
| Allergen control | Dedicated lines or validated cleaning   | Not typically a concern      |
| Traceability     | Lot-level required by regulation        | Often batch-level sufficient |

Question 5: A warehouse has three product categories with the following pick frequencies. Design the storage zones.

**Data:** - Category A: 1200 picks/day, 500 SKUs - Category B: 600 picks/day, 1500 SKUs - Category C: 200 picks/day, 3000 SKUs

**Solution:**

Using ABC analysis based on pick frequency: - **Category A** (60% of picks, 10% of SKUs): Prime locations closest to shipping dock - **Category B** (30% of picks, 30% of SKUs): Secondary locations, moderate distance - **Category C** (10% of picks, 60% of SKUs): Remote locations, maximum distance from dock

**Additional considerations:** - Within each zone, use cube utilization for dense storage of slow movers - Consider pick path optimization (serpentine vs. return routing) - High-frequency items at ergonomic pick heights (waist level) - Reserve space near dock for cross-docking high-velocity items

4.18 Chapter Summary

Table 4.14: Chapter 4 Summary: Plant Layout and Fa

| Topic                                  | Key Concepts                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------------|
| <b>Layout Types</b>                    | Fixed-position, Process, Product, Hybrid/Cellular - selected based on volume and va |
| <b>Process Layout Design</b>           | From-To matrix, Load-Distance model, Block plans, Closeness relationships           |
| <b>Product Layout (Line Balancing)</b> | Cycle time, Theoretical minimum stations, Precedence diagrams, Line efficiency      |
| <b>Cellular Manufacturing</b>          | Group Technology, Part families, Self-contained cells, Reduced material handling    |
| <b>Warehouse Design</b>                | ABC analysis, High-frequency items near dock, Pick path optimization                |
| <b>Ergonomics</b>                      | NIOSH lifting equation, RWL, Lifting Index, Work zones, Workstation design          |

|                                           |                                                                                                                                              |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Modern Trends</b><br><b>Simulation</b> | Flexible layouts, Industry 4.0, AGV/AMR planning, Digital twins<br>FlexSim, Arena, Plant Simulation - virtual testing before physical change |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

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## 4.19 References

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## Chapter 5

# Lean Manufacturing

---

### 5.1 Learning Objectives

By the end of this chapter, you will be able to:

- Define Lean Manufacturing and explain its origins
- Identify and describe the 3 M's of Lean (Muda, Mura, Muri)
- Recognize and eliminate the seven types of waste
- Apply the 5S methodology to workplace organization
- Understand Just-in-Time (JIT) principles and Kanban systems
- Calculate Takt time and apply line balancing concepts
- Implement setup reduction techniques (SMED)

“Perfection is not attainable. But if we chase perfection, we can catch excellence.” — Vince Lombardi

---

### 5.2 Introduction to Lean Manufacturing

**Lean Manufacturing** is the systematic elimination of waste. The term can be traced to Jim Womack, Daniel Jones, and Daniel Roos' book *The Machine that Changed the World* (1991), which examined the Toyota Production System.

What does “Lean” actually mean?

The term “Lean” refers to cutting “fat” from production activities — eliminating anything that doesn't add value to the customer. Think of it like a lean athlete: no excess weight, maximum efficiency, peak performance.

5.2.1 Definition of Lean

Table 5.1: Lean Manufacturing Targets (Womack, Jones, Roos 1990)

| Metric                        | Lean Target         |
|-------------------------------|---------------------|
| Human effort in factory       | Half the hours      |
| Defects in finished product   | Half the defects    |
| Engineering effort            | One-third the hours |
| Factory space for same output | Half the space      |
| In-process inventories        | A tenth or less     |

5.3 Evolution of Manufacturing Systems

Manufacturing has evolved through three major paradigms. Understanding this evolution helps us appreciate why Lean emerged as the dominant philosophy.

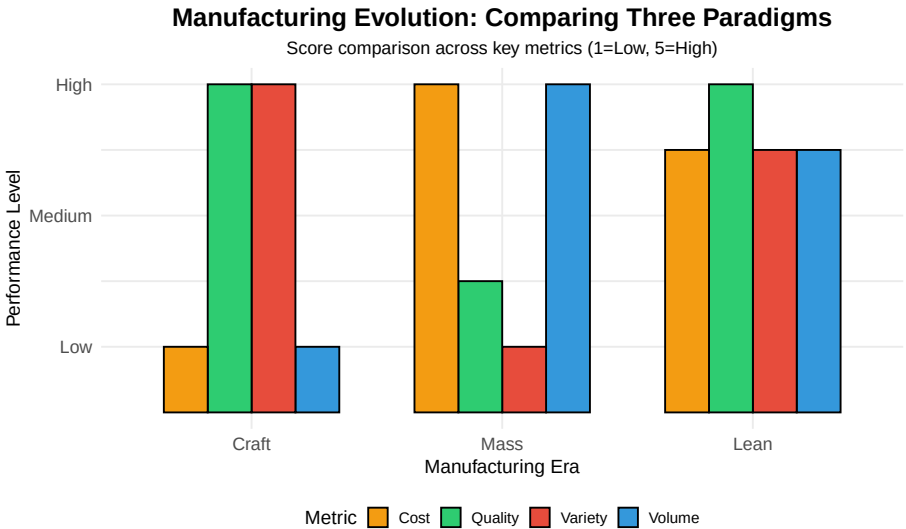


Figure 5.1: Evolution of Manufacturing Systems

5.3.1 Comparison of Manufacturing Systems

Table 5.2: Evolution of Manufacturing Systems

| Characteristic     | Craft Manufacturing                  | Mass Manufacturing           | Lean Manufacturing            |
|--------------------|--------------------------------------|------------------------------|-------------------------------|
| <b>Time Period</b> | Late 1800s                           | 1920s (Ford)                 | 1970s (Toyota)                |
| <b>Process</b>     | Built on blocks, workers walk around | Assembly line                | Cellular manufacturing        |
| <b>Workforce</b>   | Craftsmen with pride                 | Low skilled, simplistic jobs | Highly skilled, multi-tasking |
| <b>Components</b>  | Hand-crafted, hand-fitted            | Interchangeable parts        | Standardized parts            |
| <b>Quality</b>     | Excellent                            | Lower                        | Excellent                     |
| <b>Cost</b>        | Very expensive                       | Affordable                   | Low                           |
| <b>Volume</b>      | Few produced                         | Billions - identical         | Highly variable               |
| <b>Flexibility</b> | Very high                            | Very low                     | High                          |

Discussion: Why did Mass Manufacturing dominate for so long?

Mass manufacturing dominated because:

1. **Economies of scale** - Higher volume = lower unit cost
2. **Predictable demand** - Post-war boom created stable markets
3. **Limited competition** - Few global competitors
4. **Consumer acceptance** - “Any color as long as it’s black” (Henry Ford)

**What changed?** - Global competition (especially from Japan) - Customers demanded variety and quality - Product lifecycles shortened - Technology enabled flexibility

## 5.4 The Toyota Production System (TPS)

The most influential lean manufacturing model is the **Toyota Production System (TPS)**, developed by Taiichi Ohno and Shigeo Shingo at Toyota.

### 5.4.1 Historical Context

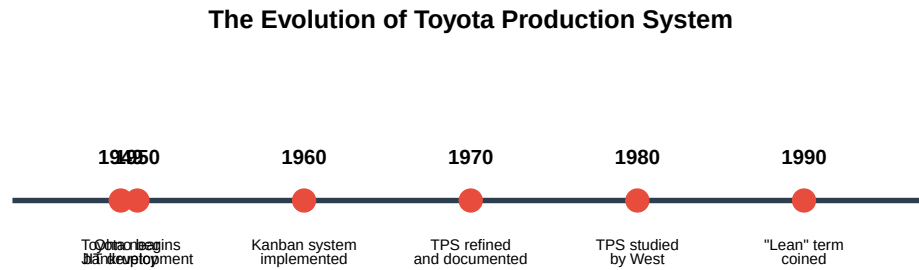


Figure 5.2: Toyota's Journey to Lean Excellence

**In 1949**, Toyota was on the brink of bankruptcy. Ford's car production was at least **8 times more efficient** than Toyota's.

**Kiichiro Toyoda's Challenge:** "To achieve the same rate of production as the United States in three years."

**Taiichi Ohno** accepted this challenge. Inspired by American supermarkets, he developed the Just-in-Time method.

### 5.4.2 Two Pillars of TPS

#### The House of Toyota Production System

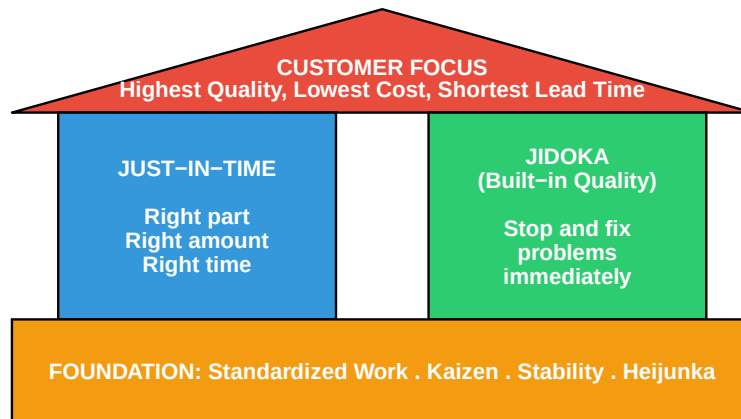


Figure 5.3: The Two Pillars of Toyota Production System

1. **Just-in-Time (JIT):** Produce only what is needed, when it is needed, in the amount needed.
  2. **Jidoka (Built-in Quality):** Automation with a human touch — stop production immediately when a defect is detected.
- 

## 5.5 The 3 M's of Lean

Lean manufacturing focuses on eliminating three types of inefficiency, known as the **3 M's**:

The 3 M's of Lean



Figure 5.4: The 3 M's of Lean Manufacturing

Table 5.3: The 3 M's Explained

| Japanese      | English                     | Description                                            |
|---------------|-----------------------------|--------------------------------------------------------|
| **Muda** (ムダ) | Waste                       | Activities that consume resources without adding value |
| **Mura** (ムラ) | Inconsistency/Unevenness    | Variation and lack of uniformity in processes          |
| **Muri** (ムリ) | Overburden/Unreasonableness | Overburdening people or equipment beyond capacity      |

5.6 The Seven Types of Waste (MUDA)

**Shigeo Shingo** identified seven categories of waste common to manufacturing. These are activities that add cost without adding value.

“Waste is everything that is not absolutely essential.” — Hiroyuki Hirano

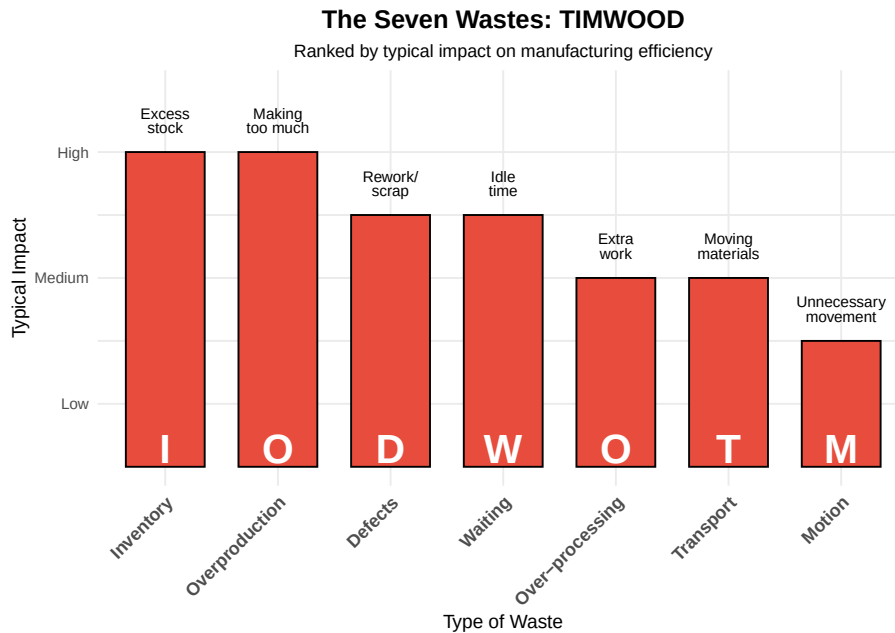


Figure 5.5: The Seven Types of Waste (TIMWOOD)

### 5.6.1 Detailed Breakdown of the Seven Wastes

Table 5.4: The Seven Wastes: Detailed

| # | Waste Type                 | Description                                                | Examples                              |
|---|----------------------------|------------------------------------------------------------|---------------------------------------|
| 1 | <b>**Overproduction**</b>  | Producing more than needed, earlier than needed            | Building to forecast                  |
| 2 | <b>**Inventory**</b>       | Excess raw materials, WIP, or finished goods               | Safety stock, excess stock            |
| 3 | <b>**Waiting**</b>         | Idle time waiting for materials, information, or equipment | Machine down, waiting for materials   |
| 4 | <b>**Transportation**</b>  | Unnecessary movement of materials between processes        | Long distances, multiple moves        |
| 5 | <b>**Over-processing**</b> | Doing more work than the customer requires                 | Tighter tolerances, unnecessary steps |
| 6 | <b>**Motion**</b>          | Unnecessary movement of people                             | Reaching, bending, carrying           |
| 7 | <b>**Defects**</b>         | Products that don't meet specifications                    | Scrap, rework, returns                |

Memory Aid: TIMWOOD

Use **TIMWOOD** to remember the seven wastes:

- **T**ransportation
- **I**nventory
- **M**otion
- **W**aiting

- **Overproduction**
- **Over-processing**
- **Defects**

Some people also use **DOWNTIME** (adding “Non-utilized talent” as an 8th waste).

Interactive Exercise: Identify the Waste

**Scenario 1:** A worker walks 50 feet to get a tool, uses it for 30 seconds, then walks back.

**Answer:** This is **Motion** waste. The solution would be to place tools at the point of use (5S).

**Scenario 2:** A machine produces 1000 parts, but only 800 are needed this week.

**Answer:** This is **Overproduction** waste. The solution would be to implement pull production based on actual demand.

**Scenario 3:** Parts sit in a queue for 3 days before being processed at the next station.

**Answer:** This is both **Inventory** and **Waiting** waste. The solution would be to balance the line and implement one-piece flow.

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## 5.7 The 5S Methodology

**5S** is a workplace organization methodology that creates a clean, efficient, and safe working environment.



The 5S Methodology for Workplace Organization

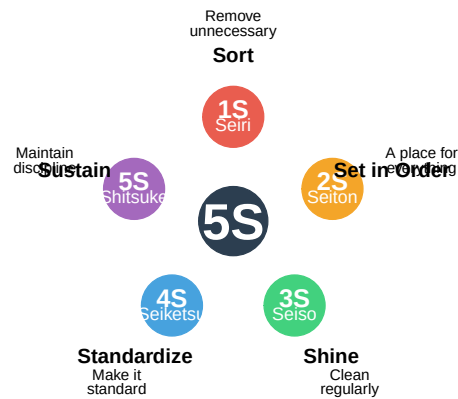


Figure 5.6: The 5S Methodology

Table 5.5: The 5S Steps Explained

| Step | Japanese     | English      | Action                                          | Key Question                    |
|------|--------------|--------------|-------------------------------------------------|---------------------------------|
| 1S   | **Seiri**    | Sort         | Remove unnecessary items from the workplace     | Do we need this?                |
| 2S   | **Seiton**   | Set in Order | Organize items so they are easy to find and use | Where does this belong?         |
| 3S   | **Seiso**    | Shine        | Clean the workplace regularly                   | Is everything clean?            |
| 4S   | **Seiketsu** | Standardize  | Create standards for maintaining 1S-3S          | How do we keep it this way?     |
| 5S   | **Shitsuke** | Sustain      | Build discipline to maintain standards          | Are we following our standards? |

5.7.1 Visual Factory

“The ability to understand the status of a production area in **5 minutes or less** by simple observation without use of computers or speaking to anyone.”

A well-implemented 5S creates a **visual factory** where problems are immediately apparent.

## 5.8 Just-in-Time (JIT) Production

### 5.8.1 The JIT Philosophy

“Deliver the right material, in the exact quantity, with perfect quality, in the right place, just before it is needed.” — Ohno and Shingo

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i Please use `linewidth` instead.

#### Just-in-Time: Delivering Value to the Customer



Figure 5.7: Just-in-Time: The Right Everything

### 5.8.2 Push vs. Pull Systems

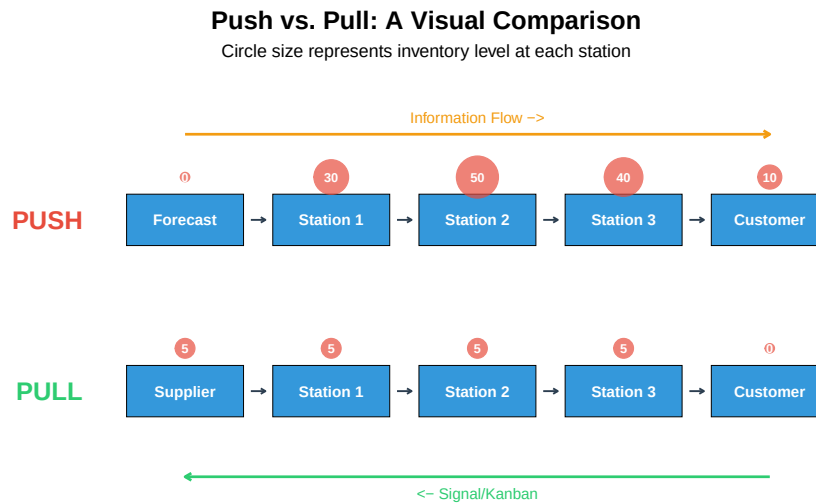


Figure 5.8: Push vs. Pull Production Systems

Table 5.6: Push vs. Pull System Comparison

| Aspect                         | Push System                      | Pull System                       |
|--------------------------------|----------------------------------|-----------------------------------|
| <b>Production trigger</b>      | Forecast/Schedule                | Actual demand/Kanban signal       |
| <b>Information flow</b>        | Forward (upstream to downstream) | Backward (downstream to upstream) |
| <b>Inventory levels</b>        | High (buffers at each station)   | Low (minimal buffers)             |
| <b>Flexibility</b>             | Low                              | High                              |
| <b>Problem visibility</b>      | Problems hidden by inventory     | Problems immediately visible      |
| <b>Customer responsiveness</b> | Slow                             | Fast                              |

### 5.8.3 Kanban System

**Kanban** ( ) is Japanese for “signboard” or “billboard.” It’s a scheduling system for lean and JIT production.

Sample Production Kanban Card

PRODUCTION KANBAN

|                |                        |                        |    |
|----------------|------------------------|------------------------|----|
| Part Number:   | 3278784                | <div><div></div></div> |    |
| Part Name:     | Securing Bracket       |                        |    |
| From Process:  | CNC Turning (CNC-      |                        |    |
| To Process:    | CNC Machining (CNC-M7) |                        |    |
| Container Qty: | 12                     | Container Type:        | A1 |

Figure 5.9: Example Kanban Card

5.9 Key Time Metrics

Understanding time metrics is essential for lean manufacturing analysis and improvement.

5.9.1 Definitions

Understanding Time Metrics in Manufacturing

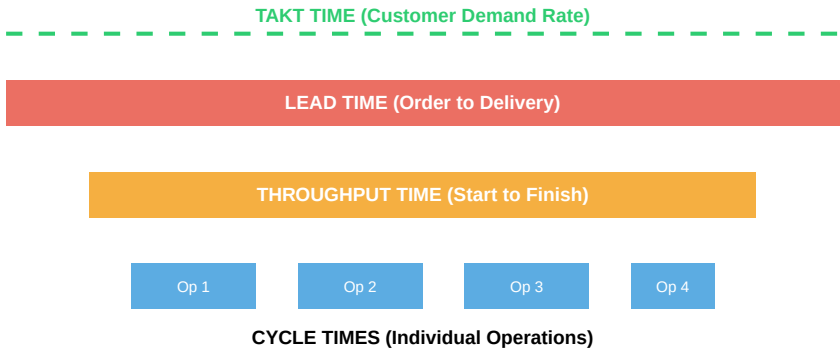


Figure 5.10: Relationship Between Time Metrics

### 5.9.2 Formulas and Calculations

**Takt Time** — The pace of production to meet customer demand:

$$\text{Takt Time} = \frac{\text{Available Production Time}}{\text{Customer Demand}}$$

**Cycle Time** — Time to complete one unit at a workstation:

$$\text{Cycle Time} = \frac{\text{Net Operating Time}}{\text{Units Produced}}$$

**Throughput Time** — Total time from start to finish:

$$\text{Throughput Time} = \text{Processing} + \text{Inspection} + \text{Queue} + \text{Move Time}$$

### 5.9.3 Interactive Example: Calculating Takt Time

Live cell — try different shift length, breaks, or customer demand.

```
#| label: takt-calculation
# Given data
available_time_per_day <- 8 * 60 # 8 hours = 480 minutes
breaks_and_meetings <- 60      # 60 minutes for breaks/meetings
customer_demand <- 240         # 240 units per day

# Calculate net available time
net_available_time <- available_time_per_day - breaks_and_meetings
cat("Net Available Time:", net_available_time, "minutes/day\n")

# Calculate Takt Time
takt_time <- net_available_time / customer_demand
cat("Takt Time:", takt_time, "minutes per unit\n")
cat("Takt Time:", takt_time * 60, "seconds per unit\n")

# Interpretation
cat("\n--- Interpretation ---\n")
cat("To meet customer demand, we must produce 1 unit every",
    takt_time, "minutes (", takt_time * 60, "seconds).\n")
```

Practice Problem: Calculate Your Takt Time

**Given:** - Shift length: 10 hours - Two 15-minute breaks and one 30-minute lunch - Customer requires 300 units per shift

**Calculate:** 1. Net available production time 2. Takt time in minutes 3. Takt time in seconds

**Solution:**

Net time =  $(10 \times 60) - (15 + 15 + 30) = 600 - 60 = 540$  minutes

Takt time =  $540 / 300 = 1.8$  minutes = 108 seconds per unit

## 5.10 Setup Reduction (SMED)

**SMED** (Single Minute Exchange of Dies) is a methodology developed by Shigeo Shingo to dramatically reduce changeover time.

### 5.10.1 The SMED Concept

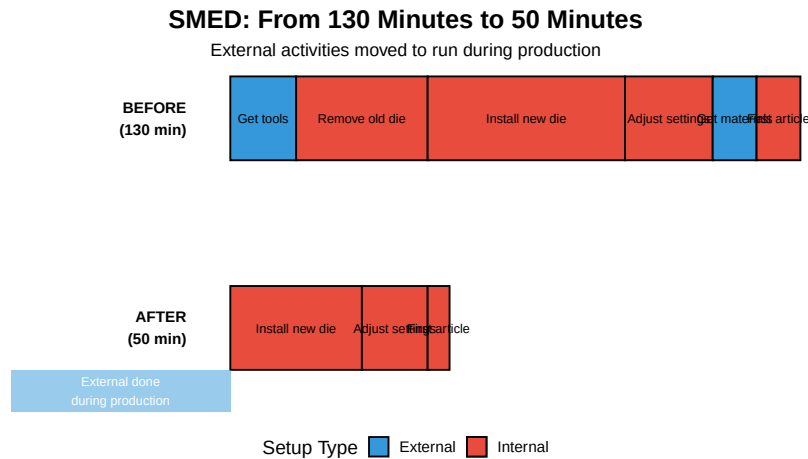


Figure 5.11: SMED: Internal vs External Setup

### 5.10.2 SMED Methodology

Table 5.7: The Four Steps of

| Step | Action                             | Description                                        |
|------|------------------------------------|----------------------------------------------------|
| 1    | **Document current process**       | Video the entire changeover, identify all steps    |
| 2    | **Separate internal and external** | Internal = machine must be stopped; External = c   |
| 3    | **Convert internal to external**   | Find ways to perform 'internal' tasks while machin |
| 4    | **Streamline all operations**      | Reduce time for remaining internal tasks through   |

Case Study: Press Setup Reduction

**Problem:** Presses were idle more than 50% of the time due to setup. - Average production run: 1-2 days - Setup time: 1-2 days

**Root Cause Analysis:** - Using standard bolt fasteners with open-end wrenches - No pre-staging of tools or dies - Sequential operations that could be parallel

**Solutions Implemented:**

| Change                    | Time Saved |
|---------------------------|------------|
| Socket sets + power tools | 10%        |
| Quick-release clamps      | 50%        |
| Pre-staging materials     | 20%        |
| Parallel operations       | 20%        |

**Result:** Setup time reduced from 1-2 days to 4-6 hours, with a target of under 1 hour.

## 5.11 Kaizen: Continuous Improvement

**Kaizen** ( ) means “change for the better” — the philosophy of continuous, incremental improvement.

### 5.11.1 Kaizen Principles

#### Plan–Do–Check–Act: The Engine of Kaizen



Figure 5.12: The Kaizen Continuous Improvement Cycle

### 5.11.2 Key Kaizen Concepts

- **Everyone participates** — from CEO to frontline workers
  - **Small, incremental changes** — not major overhauls
  - **Focus on process** — blame the process, not the person
  - **Data-driven decisions** — measure before and after
  - **Standardize improvements** — lock in gains
-



## 5.12 Value Stream Mapping (VSM)

**Value Stream Mapping** is a lean tool for analyzing the flow of materials and information required to bring a product to the customer.

### 5.12.1 VSM Symbols

Table 5.8: Key Metrics in Value Stream Maps

| Metric                    | Description                                   | Typical Location                  |
|---------------------------|-----------------------------------------------|-----------------------------------|
| **Process Time (PT)**     | Time to actually process one unit             | In process box                    |
| **Lead Time (LT)**        | Total time from raw material to finished good | Timeline at bottom                |
| **Changeover Time (C/O)** | Time to switch between products               | In process box                    |
| **Uptime**                | Percentage of time equipment is available     | In process box                    |
| **Takt Time**             | Required pace to meet demand                  | Noted on map                      |
| **WIP Inventory**         | Units waiting between processes               | Triangle symbol between processes |

## 5.13 Summary

Table 5.9: Lean Manufacturing: Key Concepts Summary

| Concept                | Definition                                                                      |
|------------------------|---------------------------------------------------------------------------------|
| **Lean Manufacturing** | Systematic elimination of waste                                                 |
| **3 M's**              | Muda (waste), Mura (inconsistency), Muri (overburden)                           |
| **7 Wastes**           | TIMWOOD: Transport, Inventory, Motion, Waiting, Overproduction, Over-processing |
| **5S**                 | Sort, Set in Order, Shine, Standardize, Sustain                                 |
| **JIT**                | Right part, right quantity, right quality, right place, right time              |
| **Kanban**             | Visual signal system for pull production                                        |
| **SMED**               | Single Minute Exchange of Dies - rapid changeover                               |
| **Kaizen**             | Continuous incremental improvement                                              |

## 5.14 Review Questions

Question 1: What are the seven types of waste, and which is often considered the most impactful?

The seven types of waste (TIMWOOD) are: 1. **T**ransportation 2. **I**nventory 3. **M**otion 4. **W**aiting 5. **O**verproduction 6. **O**ver-processing 7. **D**efects

**Inventory** is often considered the most impactful because it hides other problems, ties up capital, and can become obsolete.

Question 2: Explain the difference between Push and Pull production systems.

**Push System:** - Production based on forecasts - Materials pushed through the system - Results in high inventory - Problems hidden by buffer stock

**Pull System:** - Production triggered by actual demand - Materials pulled by downstream operations - Minimal inventory - Problems immediately visible

Question 3: Calculate the Takt time for a facility with 450 minutes of available time and demand of 150 units.

$$\text{Takt Time} = \frac{450 \text{ minutes}}{150 \text{ units}} = 3 \text{ minutes per unit}$$

This means we need to produce one unit every 3 minutes (180 seconds) to meet customer demand.

Question 4: What is the difference between internal and external setup in SMED?

**Internal Setup:** Activities that can ONLY be performed when the machine is stopped (e.g., physically removing the old die).

**External Setup:** Activities that CAN be performed while the machine is still running (e.g., getting tools, pre-staging the next die, preparing materials).

The key to SMED is converting as many internal activities to external as possible.

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## 5.15 References

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## Chapter 6

# Machine Guarding and Functional Safety

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### 6.1 Learning Objectives

By the end of this chapter, you will be able to:

- Apply the Hierarchy of Controls to manufacturing safety design
- Identify and specify appropriate physical guarding and interlock systems
- Select and position presence-sensing devices (light curtains, area scanners)
- Calculate safety distances for protective device placement
- Understand collaborative robot safety requirements
- Design emergency stop systems according to standards
- Differentiate between safety integrity levels (SIL) and performance levels (PL)

“Safety isn’t expensive, it’s priceless.” — Anonymous

---

### 6.2 Introduction: The Safety Hierarchy

In process engineering, safety is not merely a feature added at the end of a project; it is a **fundamental design constraint**. Before deploying safety devices, the engineer must apply the **Hierarchy of Controls** — a systematic approach to eliminating or reducing workplace hazards.

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i Please use ``linewidth`` instead.

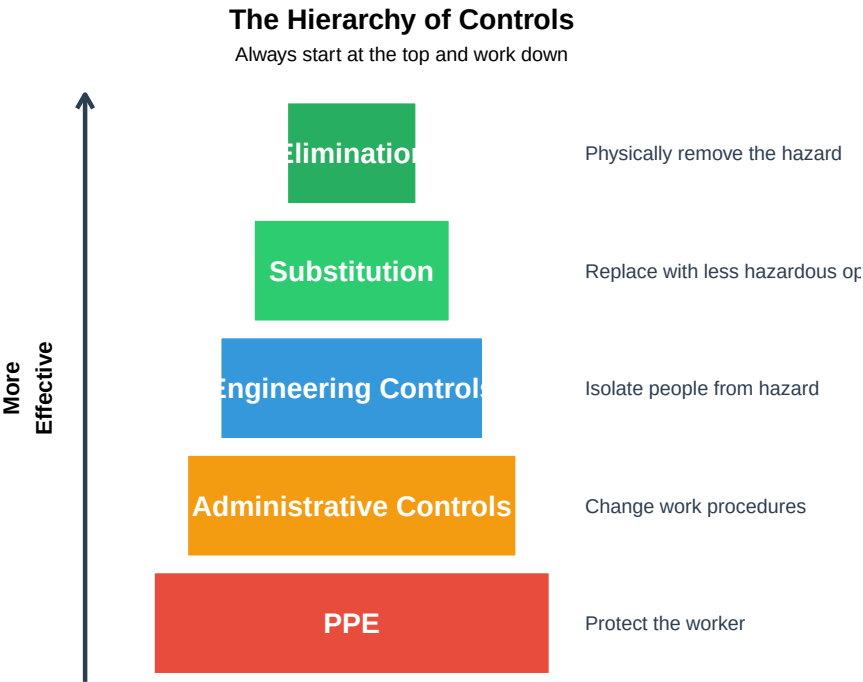


Figure 6.1: The Hierarchy of Controls (Most to Least Effective)

Table 6.1: Hierarchy of Controls: Detailed

| Priority | Control Level               | Description                                   | Example     |
|----------|-----------------------------|-----------------------------------------------|-------------|
| 1        | **Elimination**             | Physically remove the hazard entirely         | Elimination |
| 2        | **Substitution**            | Replace with something less hazardous         | Use low     |
| 3        | **Engineering Controls**    | Isolate people from the hazard through design | Install r   |
| 4        | **Administrative Controls** | Change the way people work                    | Safety p    |
| 5        | **PPE**                     | Protect the worker with equipment             | Safety g    |

Discussion: Why not just rely on PPE?

**PPE (Personal Protective Equipment)** is the least effective control because:

1. **It doesn't eliminate the hazard** — the danger still exists
2. **It depends on human compliance** — workers may forget or choose not to wear it
3. **It can fail** — equipment degrades, doesn't fit properly, or is damaged
4. **It's the last line of defense** — if PPE fails, the worker is exposed

**Key Principle:** The more we can design out hazards through engineering controls, the less we depend on human behavior to keep people safe.

“If you have to post a warning sign, you’ve already failed at design.”

### 6.3 Physical Guarding and Interlocks

Physical barriers are the most intuitive form of protection. They create a “hard” separation between the operator and moving parts, ensuring that humans cannot enter the danger zone during machine operation.

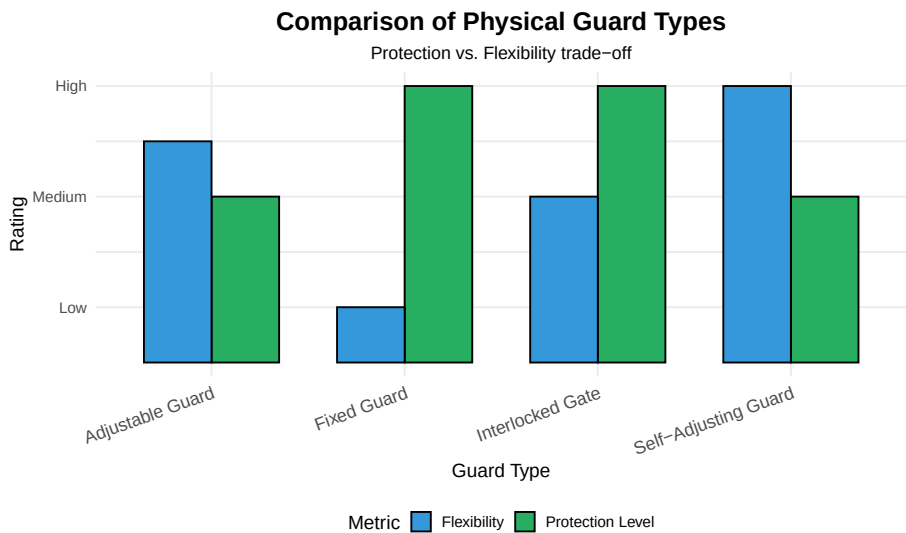


Figure 6.2: Types of Physical Guarding

#### 6.3.1 Fixed Guards

**Fixed guards** are permanent barriers that require tools to remove. They are the simplest and most reliable form of protection.

Table 6.2: Fixed Guards: Key Characteristics

| Characteristic   | Description                                            |
|------------------|--------------------------------------------------------|
| Construction     | Wire mesh, polycarbonate sheets, or solid metal panels |
| Removal          | Requires tools (screws, bolts) to remove               |
| Best Application | Areas that need infrequent access for maintenance only |
| Advantages       | Simple, reliable, low cost, no moving parts to fail    |

|                      |                                                                        |
|----------------------|------------------------------------------------------------------------|
| <b>Disadvantages</b> | Impedes visibility, slows maintenance, can be removed and not replaced |
|----------------------|------------------------------------------------------------------------|

### 6.3.2 Interlocked Guards

**Interlocked guards** are access points equipped with sensors. When the guard is opened, the safety controller initiates a stop.

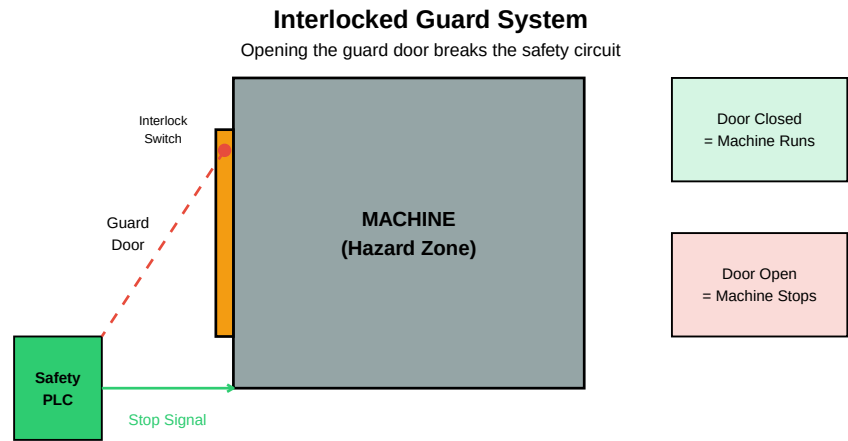


Figure 6.3: Interlocked Guard Operation Principle

### 6.3.3 Stop Categories

When an interlock is triggered, the machine must stop. The **stop category** determines how this happens:

Table 6.3: Machine Stop Categories (IEC 60204-1)

| Category              | Type              | Method                                      | Application                     |
|-----------------------|-------------------|---------------------------------------------|---------------------------------|
| <b>**Category 0**</b> | Uncontrolled Stop | Immediate removal of power to actuators     | Emergency stop                  |
| <b>**Category 1**</b> | Controlled Stop   | Controlled deceleration, then power removed | High-inertia loads              |
| <b>**Category 2**</b> | Controlled Stop   | Controlled stop with power maintained       | Process that must be maintained |

### 6.3.4 Trapped Key Systems (Lock-Out/Tag-Out)

**Trapped key systems** use a mechanical sequence to ensure that a machine cannot be energized unless all access keys are returned to the control panel.

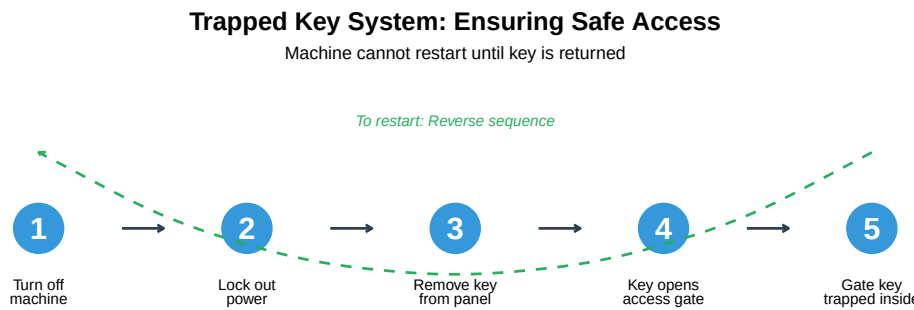


Figure 6.4: Trapped Key System Sequence

**Case Study: The Fatal Bypass**

**Incident:** A maintenance worker was killed when a robot arm unexpectedly activated while he was inside a robotic cell.

**What Happened:** - The guard interlock had been bypassed with a “defeat device” (a magnet taped to fool the sensor) - Workers had bypassed the interlock because it slowed down troubleshooting - The bypass had been in place for months without management knowledge

**Root Causes:** 1. Interlock caused inconvenience during frequent troubleshooting 2. No monitoring system to detect bypasses 3. Culture that tolerated “workarounds”

**Lessons Learned:** - Design interlocks that don’t impede normal operations - Use tamper-evident or tamper-resistant devices - Regular safety audits to detect bypasses - Zero-tolerance policy for defeating safety devices

**Never bypass a safety device, even temporarily.**

---

## 6.4 Presence-Sensing Devices

Presence-sensing devices allow for a “fence-less” environment, improving visibility and ergonomics while maintaining high safety levels. They detect when a person enters a hazardous zone and trigger a machine stop.

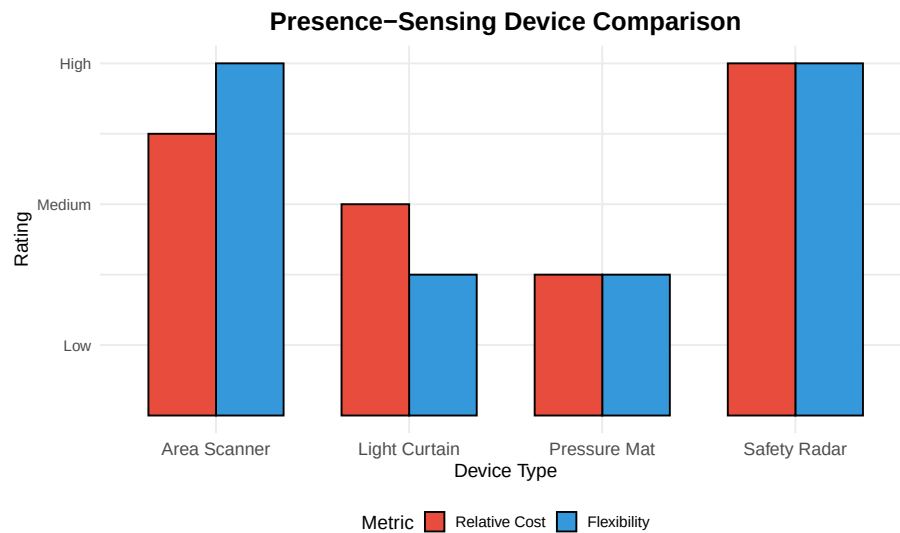


Figure 6.5: Types of Presence-Sensing Devices

### 6.4.1 Light Curtains

**Light curtains** use a transmitter and receiver to create an array of infrared beams. If any beam is broken, the safety circuit is interrupted.

#### Light Curtain: Detection Through Beam Interruption

Any object larger than the resolution will trigger a stop

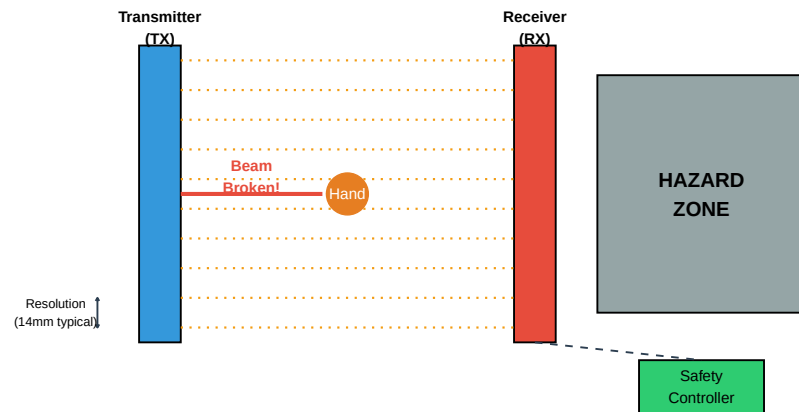


Figure 6.6: Light Curtain Operating Principle



Table 6.4: Light Curtain Specifications by Application

| Parameter     | Finger Detection (14mm)     | Hand Detection (30mm) | Body Detection (40mm) |
|---------------|-----------------------------|-----------------------|-----------------------|
| Resolution    | 14mm beam spacing           | 30mm beam spacing     | 40mm+ beam spacing    |
| Height        | 150-1800mm                  | 300-1800mm            | 900-2000mm            |
| Range         | 0.5-20m                     | 0.5-20m               | 0.5-30m               |
| Response Time | 5-15ms                      | 10-20ms               | 15-30ms               |
| Application   | Point of operation guarding | Perimeter guarding    | Area access detection |

6.4.2 Laser Area Scanners

Laser area scanners use LiDAR technology to monitor a 2D plane. They are highly flexible and allow for programmable zones.

Laser Area Scanner: Programmable Safety Zones  
270° scanning field with configurable warning and safety zones

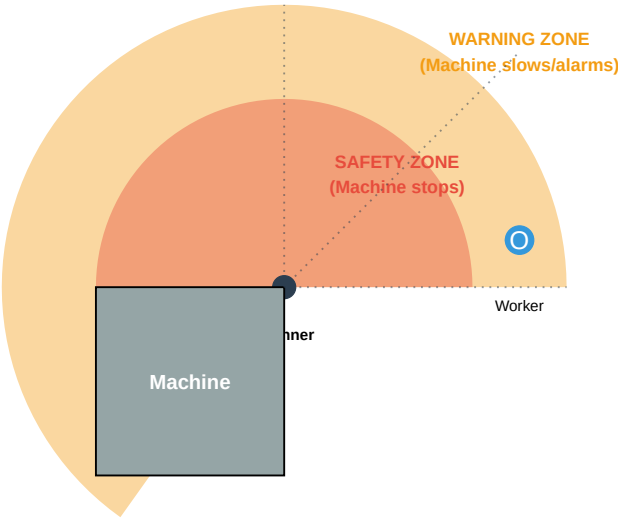


Figure 6.7: Laser Area Scanner Zones

Table 6.5: Light Curtains vs. Area Scanners

| Feature | Light Curtains | Area Scanners |
|---------|----------------|---------------|
|---------|----------------|---------------|

|                      |                                       |                                   |
|----------------------|---------------------------------------|-----------------------------------|
| Detection Type       | Vertical/Linear Barrier               | Horizontal/Area Coverage          |
| Flexibility          | Fixed installation                    | Programmable zones                |
| Zone Configuration   | Single detection plane                | Multiple warning + safety zones   |
| Cost                 | Generally lower (\$1,000-5,000)       | Higher (\$3,000-10,000)           |
| Best Use Case        | Clear entry/exit points, press brakes | Large open floors, AGV paths, fl  |
| Environmental Limits | Dust/debris can cause false trips     | More tolerant of dust, outdoor ra |

6.4.3 Pressure-Sensitive Safety Mats

**Pressure-sensitive mats** detect the presence of a person standing on them. They are used primarily in legacy systems or specific zones where optical sensors may be obscured.

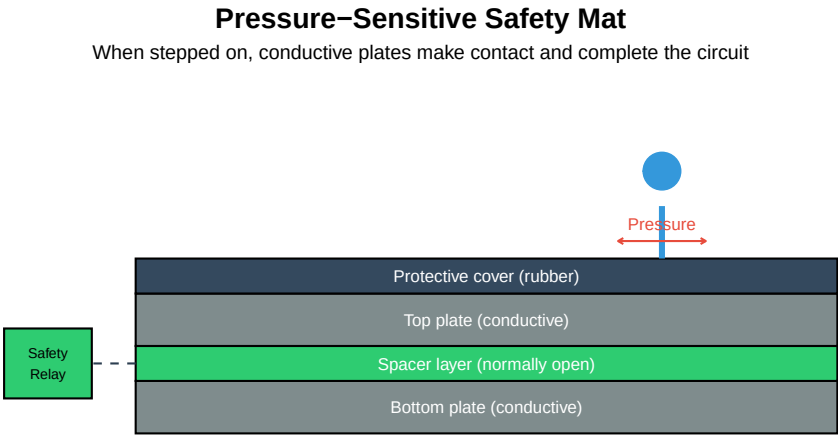


Figure 6.8: Pressure-Sensitive Safety Mat Construction

When to Use Pressure Mats vs. Optical Sensors

**Use Pressure Mats When:** - Environment has heavy dust, smoke, or steam (blinds optical sensors) - Detection area is well-defined and doesn't change - Workers need to stand in a specific location - Budget is limited

**Use Optical Sensors When:** - Flexibility in zone configuration is needed - Environment is relatively clean - Detection area is large or needs to change - Higher integrity level is required

**Hybrid Approach:** Many modern systems use both — area scanners for perimeter protection and pressure mats for specific danger zones near the machine.

## 6.5 Control-Actuated Safety Devices

These devices ensure that the operator’s body—specifically their hands—is in a predetermined **safe location** before the machine can cycle.

### 6.5.1 Two-Hand Controls

**Two-hand controls** are common in pressing or stamping operations. To initiate a cycle, the operator must press two buttons simultaneously.

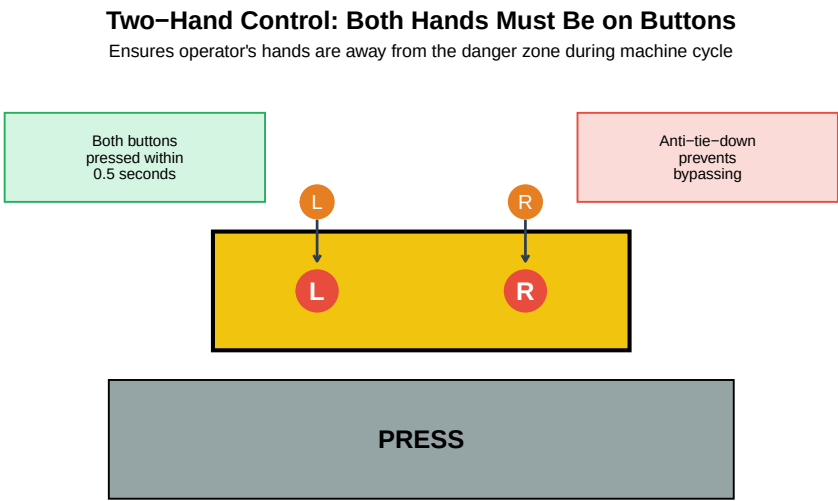


Figure 6.9: Two-Hand Control System

**Anti-Tie-Down Requirement:** Modern systems require both buttons to be pressed within **0.5 seconds** of each other. This prevents operators from taping one button down to work with one hand.

Table 6.6: Two-Hand Control Requirements

| Requirement            | Standard Value                          | Purpose                                   |
|------------------------|-----------------------------------------|-------------------------------------------|
| Synchronous activation | Within 0.5 seconds                      | Prevents one-hand operation               |
| Continuous pressure    | Must be held throughout cycle           | Ensures hands stay on controls            |
| Release before restart | Both must be released before next cycle | Prevents automatic cycling                |
| Button spacing         | Minimum 260mm (10 inches) apart         | Cannot press both with one hand/arm       |
| Guards around buttons  | Prevent accidental activation           | Prevents hip, elbow, or object activation |

### 6.5.2 Enable/Deadman Switches

**Enable switches** (also called deadman switches) are used during **Teach Mode** for robotics. The operator must hold a three-position trigger in the **middle** position.

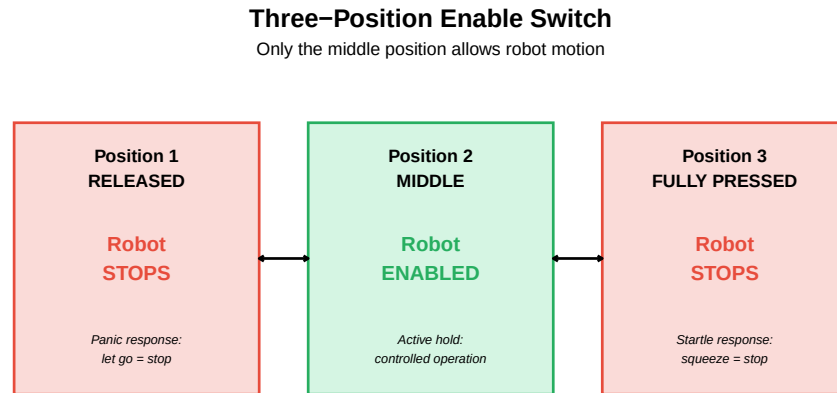


Figure 6.10: Three-Position Enable Switch Operation

Why Three Positions?

The three-position design accounts for **two natural human reactions** to danger:

1. **Panic (Release):** When frightened, people often release what they’re holding — robot stops.
2. **Startle (Squeeze):** When startled, people often grip tighter — robot stops.

Only a deliberate, controlled **middle hold** allows the robot to move. This is fundamental to safe teach pendant operation.

## 6.6 Collaborative Robotics (Cobots)

**Collaborative robots (cobots)** represent a shift from “separation” to “collaboration.” They rely on sophisticated internal sensors rather than external cages, allowing humans and robots to work in the same space.

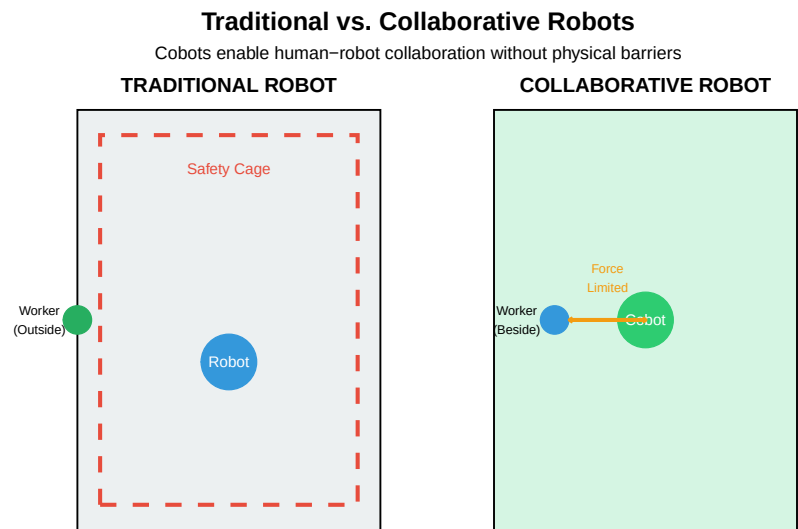


Figure 6.11: Traditional Robot vs. Collaborative Robot

6.6.1 Collaborative Operation Methods

ISO 10218 and ISO/TS 15066 define **four methods** for safe collaborative operation:

Table 6.7: Four Methods of Collaborative Robot Operation

| Method                                     | How It Works                                                | Sensors Used         |
|--------------------------------------------|-------------------------------------------------------------|----------------------|
| <b>**Safety-Rated Monitored Stop**</b>     | Robot stops when human enters workspace, resumes when clear | Area scanners        |
| <b>**Hand Guiding**</b>                    | Operator physically guides robot through motions            | Force/torque sensors |
| <b>**Speed and Separation Monitoring**</b> | Robot slows or stops based on human proximity               | Area scanners        |
| <b>**Power and Force Limiting**</b>        | Robot limits force/power to safe levels on contact          | Joint torque sensors |

6.6.2 Power and Force Limiting (PFL)

The most common cobot safety method is **Power and Force Limiting**. The robot detects a change in torque in its joints. If it bumps into a human, it stops before a regulated amount of force is exceeded.

Table 6.8: ISO/TS 15066 Biomechanical Limits (Quasi-static contact)

| Body Region    | Maximum Pressure (N/cm <sup>2</sup> ) | Maximum Force (N) | Risk Level |
|----------------|---------------------------------------|-------------------|------------|
| Skull/Forehead | 130                                   | 130               | Critical   |
| Face           | 65                                    | 65                | Critical   |
| Neck           | 145                                   | 150               | Critical   |
| Back/Shoulders | 210                                   | 210               | Moderate   |
| Chest          | 140                                   | 140               | Moderate   |
| Abdomen        | 110                                   | 110               | Moderate   |
| Hand/Fingers   | 300                                   | 140               | Lower      |

Discussion: Are Cobots Always Safe?

**Common Misconception:** “Cobots are safe, so we don’t need to do a risk assessment.”

**Reality:** Cobots are *potentially* safe, but a **risk assessment is always required**.

**Factors that can make a cobot unsafe:**

1. **End-of-arm tooling** — A soft gripper is different from a sharp blade
2. **Workpiece** — Carrying a heavy or sharp object increases risk
3. **Speed** — Even force-limited robots can cause injury at high speed
4. **Application** — Some tasks bring humans closer to hazards
5. **Environment** — Slippery floors, confined spaces, etc.

**Every collaborative application requires a documented risk assessment per ISO 12100.**

## 6.7 Safety Distance Calculation

A safety device cannot be placed arbitrarily. It must be far enough away that the machine comes to a **complete halt** before the human can reach the hazard.

### 6.7.1 The Safety Distance Formula

The minimum safety distance  $S$  is calculated as:

$$S = (K \times T) + C$$

Where:

- $K$ : The hand/body approach speed (standardized values)
- $T$ : The total stopping time of the system (device response + machine stop time)

- $C$ : The “penetration depth” (how far a hand can reach through before being detected)

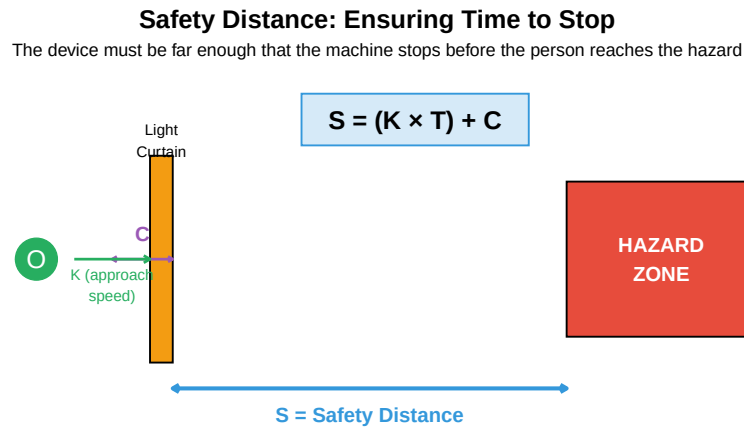


Figure 6.12: Safety Distance Components

### 6.7.2 Standard Values for $K$

Table 6.9: Standard Approach Speed Values (ISO 13855)

| Approach Type              | K Value (mm/s) | When to Use                          |
|----------------------------|----------------|--------------------------------------|
| Hand/Arm Approach (fast)   | 2000           | Close approach to point of operation |
| Hand/Arm Approach (normal) | 1600           | General approach to machines         |
| Walking Approach           | 1600           | Area access detection                |
| Standing Reach             | 0 (static)     | Worker already in position           |

### 6.7.3 Penetration Depth ( $C$ )

The penetration depth depends on the **resolution** of the sensing device:

Table 6.10: Penetration Depth Values Based on Resolution

| Resolution (mm) | C Value (mm) | Detection Type   |
|-----------------|--------------|------------------|
| 14              | 0            | Finger detection |
| >14 to 20       | 80           | Finger/Hand      |
| >20 to 30       | 130          | Hand detection   |
| >30 to 40       | 240          | Hand/Arm         |

&gt;40

850

Body detection

### 6.7.4 Example Calculation

Live cell — try different approach speed, response time, or resolution values.

```
#| label: safety-calc-example
# Given parameters
K <- 2000          # Approach speed (mm/s) - fast hand approach
T_device <- 0.020  # Light curtain response time (20 ms = 0.020 s)
T_machine <- 0.150 # Machine stopping time (150 ms = 0.150 s)
C <- 0            # Penetration depth for 14mm resolution

# Calculate total stopping time
T_total <- T_device + T_machine
cat("Total stopping time (T):", T_total * 1000, "ms\n")

# Calculate minimum safety distance
S <- (K * T_total) + C
cat("Minimum safety distance (S):", S, "mm\n")
cat("Minimum safety distance (S):", S / 25.4, "inches\n")

# Interpretation
cat("\n--- Interpretation ---\n")
cat("The light curtain must be at least", S, "mm from the hazard.\n")
cat("This ensures the machine stops before a person can reach the danger zone.\n")
```

Practice Problem: Calculate Safety Distance

**Given:** - Light curtain resolution: 30mm (hand detection) - Light curtain response time: 15ms - Machine stopping time: 200ms - Application: Operator hand approach

**Calculate the minimum safety distance.**

**Solution:**

```
K = 1600 mm/s (normal hand approach)
T = 0.015 + 0.200 = 0.215 seconds
C = 130 mm (for 30mm resolution)

S = (K × T) + C
S = (1600 × 0.215) + 130
S = 344 + 130
S = 474 mm (approximately 18.7 inches)
```

The light curtain must be installed at least **474mm** from the nearest hazard point.



## 6.8 Emergency Stop (E-Stop) Systems

The **Emergency Stop** is the “last resort” — it is not a safety device but a **complementary protective measure**. It must be available to stop the machine when all other measures have failed.

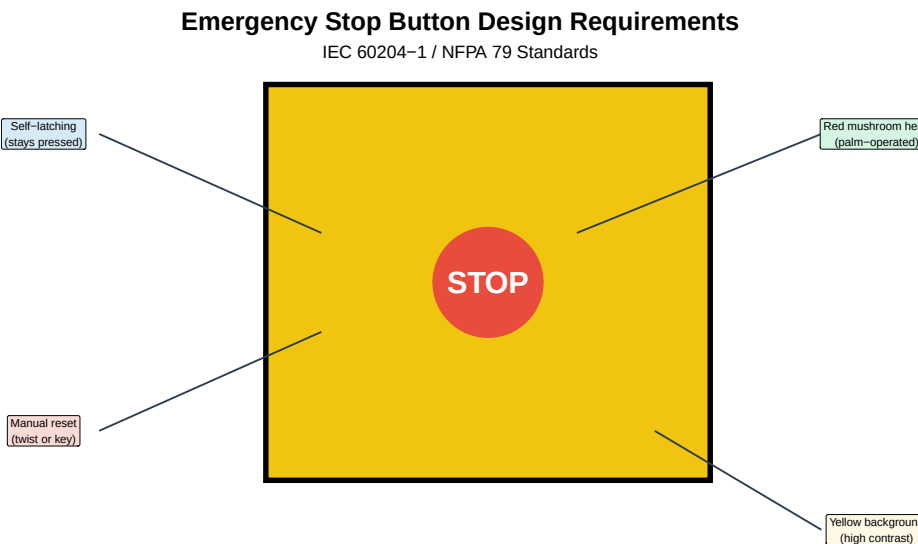


Figure 6.13: E-Stop Design Requirements

Table 6.11: E-Stop Requirements per IEC 60204-1

| Requirement   | Standard                                     |
|---------------|----------------------------------------------|
| Color         | Red (RAL 3000 or equivalent)                 |
| Shape         | Mushroom head (palm or fist operated)        |
| Background    | Yellow (high contrast)                       |
| Operation     | Self-latching (stays activated when pressed) |
| Reset         | Manual reset required (twist, pull, or key)  |
| Accessibility | Within easy reach of all operators           |
| Wiring        | Normally closed contacts (fail-safe)         |
| Function      | Category 0 or Category 1 stop                |

### 6.8.1 E-Stop Pull Cords

For long conveyors or machines where an operator cannot quickly reach a fixed button, **pull cords** provide continuous E-Stop coverage.

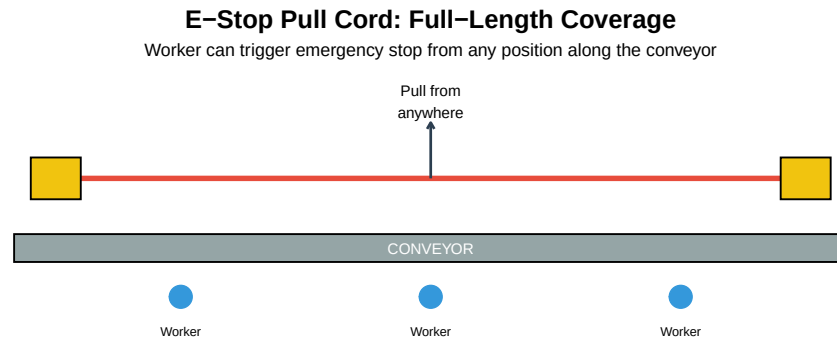


Figure 6.14: E-Stop Pull Cord System

#### Common E-Stop Mistakes

**Mistake 1: Using E-Stop for normal stopping** - E-Stops are for emergencies only - Normal stops should use a separate control - Frequent E-Stop use causes wear and desensitizes operators

**Mistake 2: E-Stop that doesn't stop everything** - All hazardous motion must stop - Auxiliary equipment (conveyors, feeders) often forgotten - Review entire system, not just the primary machine

**Mistake 3: Automatic restart after E-Stop reset** - Machine must NOT automatically restart when E-Stop is released - Separate "Start" action required - This prevents unexpected motion after reset

**Mistake 4: Hidden or blocked E-Stops** - E-Stops must be visible and accessible - Guard placement sometimes blocks access - Regular audits needed to ensure accessibility

## 6.9 Functional Safety Standards

Modern safety systems must meet specific **performance levels** to ensure reliability. Two main frameworks exist:

### 6.9.1 Safety Integrity Levels (SIL) - IEC 62061

Table 6.12: Safety Integrity Levels (PFH = Probability of Dangerous Failure per Hour)

| SIL Level    | PFH Range                             | Approximate Meaning                              | Typical Application       |
|--------------|---------------------------------------|--------------------------------------------------|---------------------------|
| <b>SIL 1</b> | 10 <sup>-6</sup> to <10 <sup>-5</sup> | Dangerous failure less than once per 11 years    | Low-risk tasks            |
| <b>SIL 2</b> | 10 <sup>-5</sup> to <10 <sup>-4</sup> | Dangerous failure less than once per 114 years   | Most industrial machinery |
| <b>SIL 3</b> | 10 <sup>-4</sup> to <10 <sup>-3</sup> | Dangerous failure less than once per 1,140 years | High-risk processes       |

### 6.9.2 Performance Levels (PL) - ISO 13849-1

Table 6.13: Performance Levels per ISO 13849-1

| Performance Level | PFH Range                               | Risk Reduction | Equivalent SIL |
|-------------------|-----------------------------------------|----------------|----------------|
| <b>PL a</b>       | 10 <sup>-6</sup> to <10 <sup>-5</sup>   | Lowest         | < SIL 1        |
| <b>PL b</b>       | 3×10 <sup>-6</sup> to <10 <sup>-5</sup> | Low            | SIL 1          |
| <b>PL c</b>       | 10 <sup>-5</sup> to <3×10 <sup>-5</sup> | Medium         | SIL 1          |
| <b>PL d</b>       | 10 <sup>-5</sup> to <10 <sup>-4</sup>   | High           | SIL 2          |
| <b>PL e</b>       | 10 <sup>-4</sup> to <10 <sup>-3</sup>   | Highest        | SIL 3          |

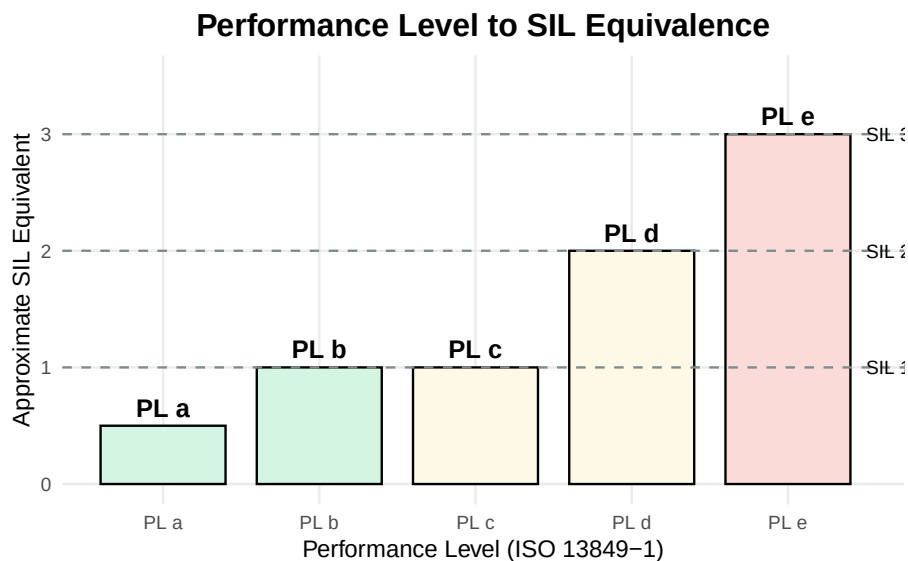


Figure 6.15: Relationship Between SIL and PL

Which Standard to Use: SIL or PL?

**Use ISO 13849-1 (Performance Levels) when:** - Designing safety functions for standard machinery - Using off-the-shelf safety components - Simpler systems with well-defined safety functions

**Use IEC 62061 (SIL) when:** - Complex programmable systems (safety PLCs) - Process industries - When customer or regulation specifies SIL

**Note:** Many modern systems use both standards, as they are harmonized and can be used together. The choice often depends on industry practice and customer requirements.

## 6.10 Risk Assessment Process

Before selecting any safety device, a **risk assessment** must be performed. This is required by ISO 12100.

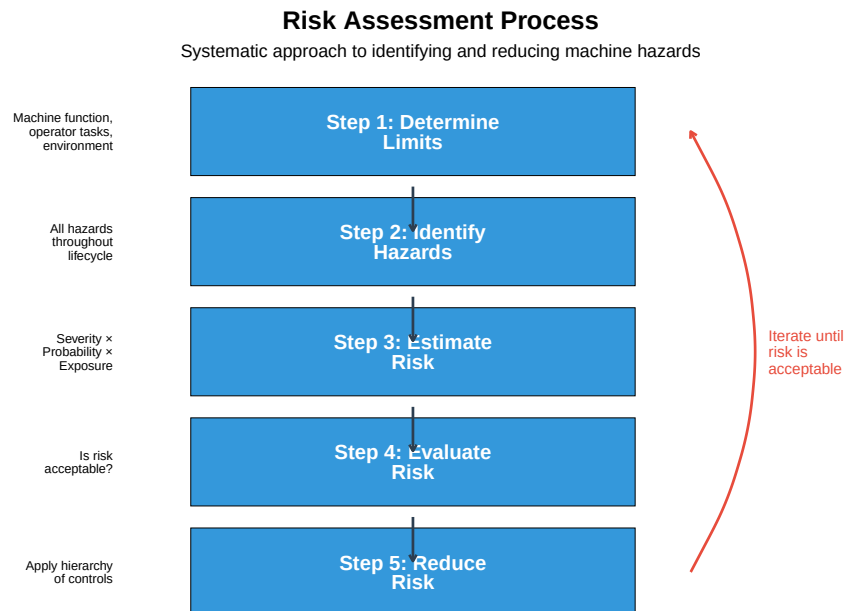


Figure 6.16: Risk Assessment Process (ISO 12100)

### 6.10.1 Risk Estimation

Risk is typically estimated using three factors:

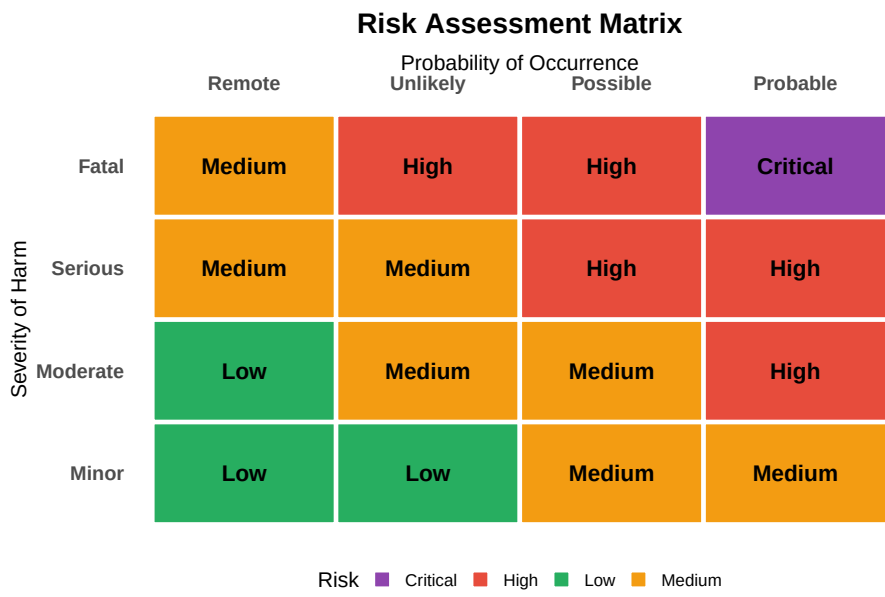


Figure 6.17: Risk Estimation Matrix

6.11 Summary

Table 6.14: Machine Guarding and Safety: Key Concepts Summary

| Topic                        | Key Points                                                      | Critic |
|------------------------------|-----------------------------------------------------------------|--------|
| **Hierarchy of Controls**    | Elimination > Substitution > Engineering > Administrative > PPE | Always |
| **Physical Guarding**        | Fixed guards, interlocked gates, trapped key systems            | Stop c |
| **Presence Sensing**         | Light curtains, area scanners, pressure mats                    | Resolu |
| **Control-Actuated Devices** | Two-hand controls, enable/deadman switches                      | Anti-t |
| **Collaborative Robots**     | PFL, speed/separation monitoring, hand guiding                  | Risk a |
| **Safety Distance**          | $S = (K \times T) + C$ formula for device placement             | Includ |
| **E-Stop Systems**           | Red mushroom on yellow, self-latching, manual reset             | E-Stop |
| **Functional Safety**        | SIL (IEC 62061) and PL (ISO 13849-1) standards                  | Requir |

## 6.12 Review Questions

Question 1: What are the five levels of the Hierarchy of Controls, and why is PPE the least effective?

The five levels are (from most to least effective):

1. **Elimination** — Physically remove the hazard
2. **Substitution** — Replace with something less hazardous
3. **Engineering Controls** — Isolate people from the hazard
4. **Administrative Controls** — Change how people work
5. **PPE** — Protect the worker with equipment

**PPE is least effective because:** - The hazard still exists - It depends on human compliance - Equipment can fail or be worn incorrectly - It's the last line of defense

Question 2: A light curtain has a response time of 12ms and resolution of 14mm. The machine takes 180ms to stop. Calculate the minimum safety distance for hand approach.

**Given:** -  $K = 2000$  mm/s (hand approach, use higher value for safety) -  $T = 0.012 + 0.180 = 0.192$  seconds -  $C = 0$  mm (14mm resolution)

**Calculation:**

$$S = (K \times T) + C$$

$$S = (2000 \times 0.192) + 0$$

$$S = 384 \text{ mm}$$

The light curtain must be installed at least **384mm** (approximately 15 inches) from the hazard.

Question 3: Explain why a three-position enable switch stops the robot in both position 1 (released) and position 3 (fully pressed).

The three-position design accounts for two natural human reactions to danger:

**Position 1 (Released):** - When frightened, people often release what they're holding (panic response) - Robot stops immediately

**Position 3 (Fully Pressed):** - When startled, people often grip tighter (startle response) - Robot stops immediately

**Position 2 (Middle):** - Only a deliberate, controlled hold allows operation - Requires conscious effort to maintain - This ensures the operator is alert and in control

Question 4: What are the four methods of collaborative robot operation according to ISO 10218?

1. **Safety-Rated Monitored Stop**

- Robot stops when human enters workspace
- Resumes when human leaves
- Uses area scanners or light curtains
- 2. **Hand Guiding**
  - Operator physically guides robot
  - Force/torque sensors enable safe teaching
- 3. **Speed and Separation Monitoring**
  - Robot speed varies based on human proximity
  - Uses external sensors to track human position
- 4. **Power and Force Limiting (PFL)**
  - Robot limits contact force to safe levels
  - Built-in joint torque sensors
  - Most common cobot method

Question 5: What are the requirements for an E-Stop button according to IEC 60204-1?

**Physical Requirements:** - **Color:** Red - **Shape:** Mushroom head (palm or fist operated) - **Background:** Yellow (high contrast)

**Functional Requirements:** - **Self-latching:** Stays activated when pressed - **Manual reset:** Requires deliberate action to release - **Accessibility:** Within easy reach of all operators - **Wiring:** Normally closed contacts (fail-safe) - **Function:** Category 0 or Category 1 stop - **No automatic restart:** Separate start action required after reset

## 6.13 References

- ISO 12100:2010 — Safety of machinery — General principles for design — Risk assessment and risk reduction
- ISO 13849-1:2015 — Safety of machinery — Safety-related parts of control systems
- ISO 13855:2010 — Safety of machinery — Positioning of safeguards with respect to approach speeds
- ISO 10218-1/2:2011 — Robots and robotic devices — Safety requirements for industrial robots
- ISO/TS 15066:2016 — Robots and robotic devices — Collaborative robots
- IEC 60204-1:2016 — Safety of machinery — Electrical equipment of machines
- IEC 62061:2021 — Safety of machinery — Functional safety of safety-related control systems
- OSHA 29 CFR 1910.212 — General requirements for all machines
- ANSI/RIA TR R15.306 — Task-based risk assessment methodology





## Chapter 7

# Ergonomics in Automated Manufacturing

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### 7.1 Learning Objectives

By the end of this chapter, you will be able to:

- Define ergonomics and explain its importance in manufacturing
- Identify the seven major ergonomic risk factors
- Recognize common musculoskeletal disorders (MSDs) and their causes
- Apply ergonomic design principles to workstation design
- Conduct a REBA (Rapid Entire Body Assessment) analysis
- Understand the role of AI and automation in ergonomic assessment
- Design workstations that optimize human performance and safety

“Ergonomics is the science of fitting the job to the worker, not forcing the worker to fit the job.” — International Ergonomics Association

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### 7.2 What is Ergonomics?

**Ergonomics** (from Greek *ergon* = work, *nomos* = laws) is the scientific discipline focused on understanding the interactions between humans and other elements of a system, with the goal of optimizing human well-being and overall system performance.

**The Three Domains of Ergonomics**  
Ergonomics considers the whole human experience at work

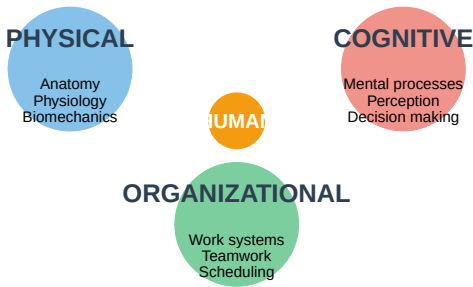


Figure 7.1: The Three Domains of Ergonomics

Table 7.1: The Three

| Domain                               | Focus                                         | Key Topics            |
|--------------------------------------|-----------------------------------------------|-----------------------|
| <b>**Physical Ergonomics**</b>       | Human body’s response to physical stress      | Posture, material han |
| <b>**Cognitive Ergonomics**</b>      | Mental processes and human-system interaction | Mental workload, dec  |
| <b>**Organizational Ergonomics**</b> | Optimization of sociotechnical systems        | Communication, team   |

### 7.2.1 Why Ergonomics Matters in Manufacturing

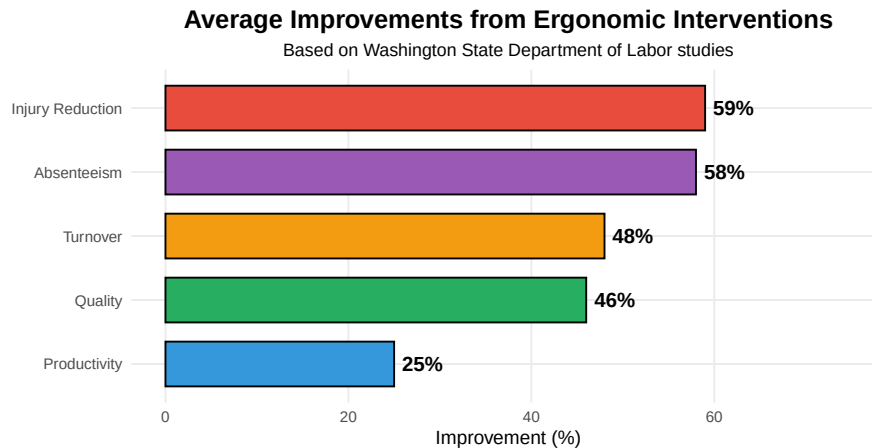


Figure 7.2: The Business Case for Ergonomics

#### The Cost of Ignoring Ergonomics

**Direct Costs:** - Workers' compensation claims - Medical expenses - Legal fees  
- Increased insurance premiums

**Indirect Costs (often 4-10x direct costs):** - Lost productivity - Training replacement workers - Reduced quality - Overtime to cover absent workers - Administrative time - Low morale

**Statistics:** - MSDs account for **33%** of all workplace injuries - Average MSD claim costs **\$15,000-\$20,000** - Lost workdays due to MSDs average **12 days** per case - Carpal tunnel surgery costs approximately **\$30,000** per case (including lost time)

## 7.3 The Seven Ergonomic Risk Factors

The **Initial Ergonomic Risk Assessment (INERA)** identifies seven primary risk factors that contribute to musculoskeletal disorders. Understanding these factors is the first step in prevention.

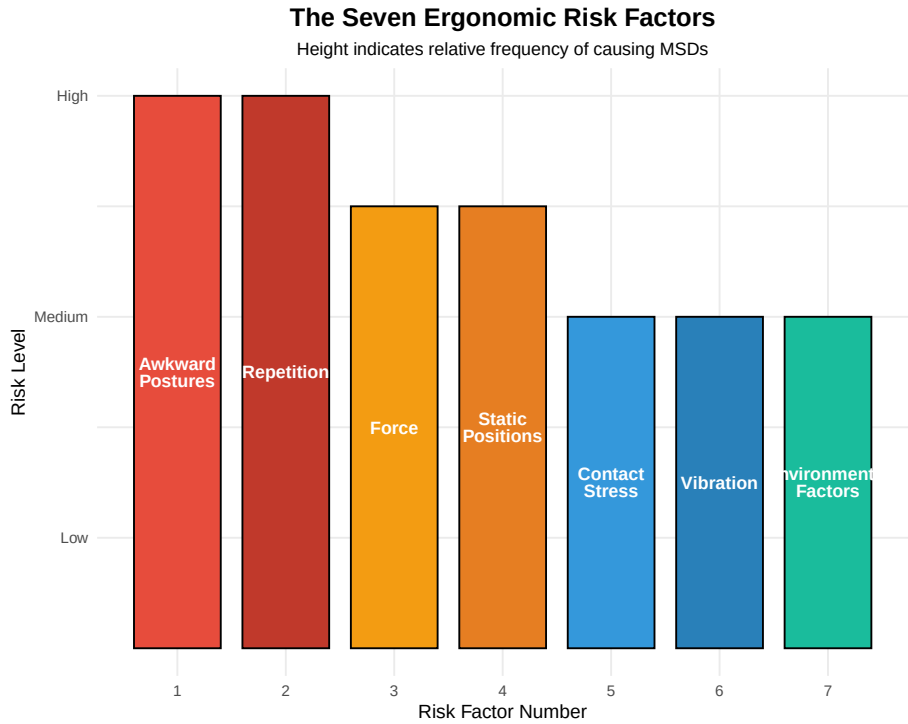


Figure 7.3: The Seven Ergonomic Risk Factors

Table 7.2: The Seven Ergonomic Risk Factors

| # | Risk Factor                      | Definition                                              |
|---|----------------------------------|---------------------------------------------------------|
| 1 | <b>**Awkward Postures**</b>      | Positions that deviate from neutral body alignment      |
| 2 | <b>**Repetition**</b>            | Performing the same motion repeatedly                   |
| 3 | <b>**Force**</b>                 | Amount of physical effort required to do a task         |
| 4 | <b>**Static Positions**</b>      | Maintaining the same position for extended periods      |
| 5 | <b>**Contact Stress**</b>        | Pressure from hard surfaces or edges on body tissues    |
| 6 | <b>**Vibration**</b>             | Oscillating movements transferred to the body           |
| 7 | <b>**Environmental Factors**</b> | Temperature, lighting, noise affecting work performance |

7.3.1 Risk Factor 1: Awkward Postures

**Neutral posture** is the position where joints are naturally aligned, minimizing stress on muscles, tendons, and skeletal system. Any deviation from neutral increases injury risk.

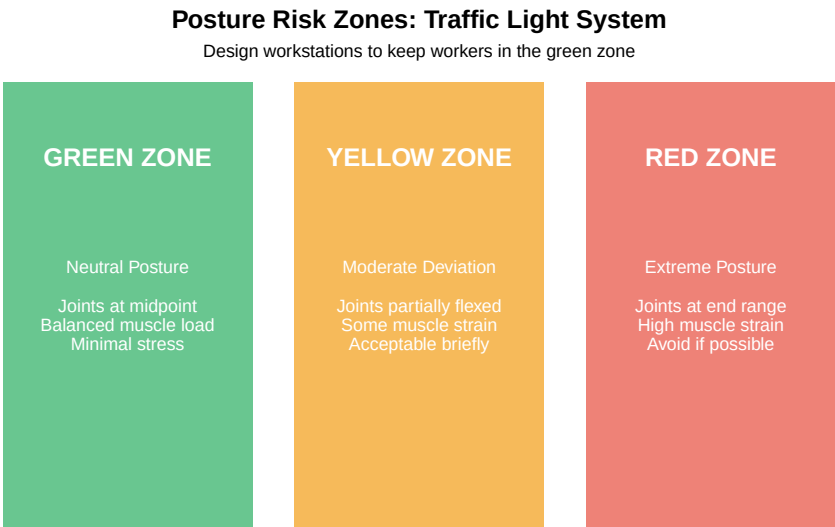


Figure 7.4: Posture Risk Zones

Table 7.3: Neutral vs. Awkward Postures by Body Part

| Body Part             | Neutral Position                          | Awkward Position                       | Risk Level |
|-----------------------|-------------------------------------------|----------------------------------------|------------|
| <b>**Head/Neck**</b>  | Balanced on spine, looking straight ahead | Bent forward >20°, tilted, twisted     | High       |
| <b>**Shoulders**</b>  | Relaxed, arms at sides                    | Raised, reaching overhead, behind body | High       |
| <b>**Elbows**</b>     | Close to body, bent 90-120°               | Fully extended or tightly bent         | High       |
| <b>**Wrists**</b>     | Straight, in line with forearm            | Bent up/down, twisted side-to-side     | High       |
| <b>**Back**</b>       | Natural S-curve maintained                | Bent forward, twisted, arched          | High       |
| <b>**Hips/Knees**</b> | Thighs parallel to floor, feet flat       | Squatting, kneeling, twisted           | High       |

7.3.2 Risk Factor 2: Repetition

**Repetitive motion** is one of the most significant risk factors in manufacturing. The risk increases with:

- **Frequency:** How often the motion occurs
- **Duration:** How long the task is performed
- **Recovery time:** Rest between repetitions

Table 7.4: Repetition Risk Thresholds

| Cycle Time   | Repetitions/Hour | Risk Level | Action Required |
|--------------|------------------|------------|-----------------|
| > 30 seconds | < 120            | Low        | Monitor         |

|               |         |          |                                  |
|---------------|---------|----------|----------------------------------|
| 15-30 seconds | 120-240 | Moderate | Investigate and consider changes |
| < 15 seconds  | > 240   | High     | Immediate intervention needed    |

### 7.3.3 Risk Factor 3: Force

The amount of **physical effort** required significantly impacts injury risk. Force requirements depend on:

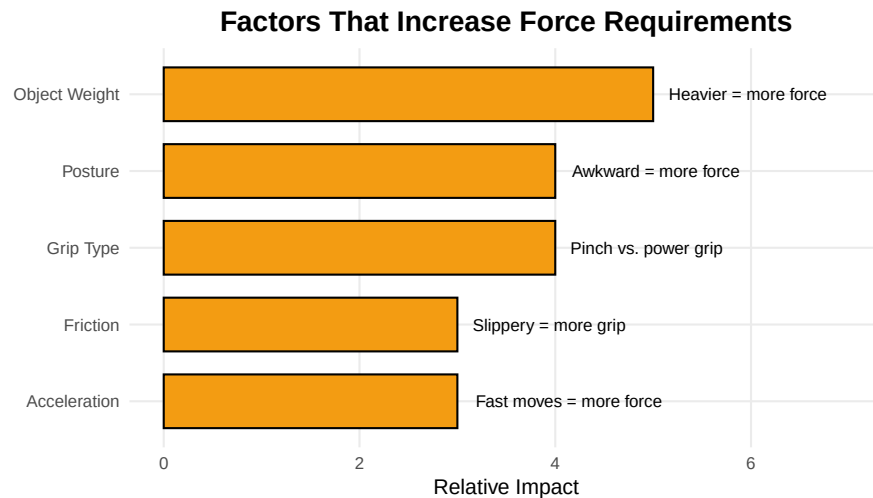


Figure 7.5: Factors Affecting Force Requirements

Interactive Exercise: Identify the Risk Factors

**Scenario:** An assembly line worker performs the following task 400 times per shift:

1. Reaches overhead to grab a part from a bin
2. Holds the part while inserting 4 screws with a power screwdriver
3. Places the completed assembly on a conveyor
4. The workstation has poor lighting and the floor is concrete

**Identify all risk factors present:**

**Answer:** 1. **Awkward Posture** - Reaching overhead 2. **Repetition** - 400 cycles per shift 3. **Force** - Gripping parts, controlling screwdriver 4. **Static Position** - Holding part during assembly 5. **Vibration** - Power screwdriver 6. **Contact Stress** - Standing on concrete (feet) 7. **Environmental** - Poor lighting

*This task has ALL SEVEN risk factors present!*

7.4 Musculoskeletal Disorders (MSDs)

**Musculoskeletal Disorders (MSDs)** are injuries and disorders affecting muscles, nerves, tendons, ligaments, joints, cartilage, and spinal discs. They are the most common and costly occupational health problem.

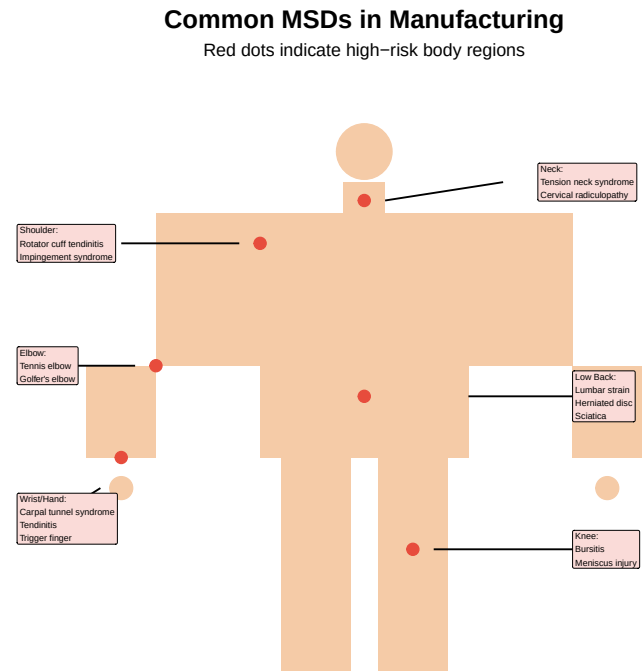


Figure 7.6: Common MSDs by Body Region

Table 7.5: Common Musculoskeletal Disorders in Manufacturing

| Disorder                          | Description                                           | Causes                                       |
|-----------------------------------|-------------------------------------------------------|----------------------------------------------|
| <b>**Carpal Tunnel Syndrome**</b> | Compression of median nerve in the wrist              | Repetitive wrist motions, forceful exertions |
| <b>**Tendinitis**</b>             | Inflammation of tendons from overuse                  | Repetitive motions, awkward postures         |
| <b>**Low Back Pain**</b>          | Strain or injury to muscles/discs of lower back       | Heavy lifting, bending, twisting             |
| <b>**Rotator Cuff Injury**</b>    | Damage to shoulder tendons                            | Overhead work, repetitive motions            |
| <b>**Epicondylitis**</b>          | Inflammation of elbow tendons (tennis/golfer's elbow) | Repetitive arm/wrist motions                 |
| <b>**Trigger Finger**</b>         | Finger gets stuck in bent position                    | Repetitive gripping, hand tools              |

### 7.4.1 The MSD Development Cycle

MSDs typically develop gradually through a predictable cycle:

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use `linewidth` instead.

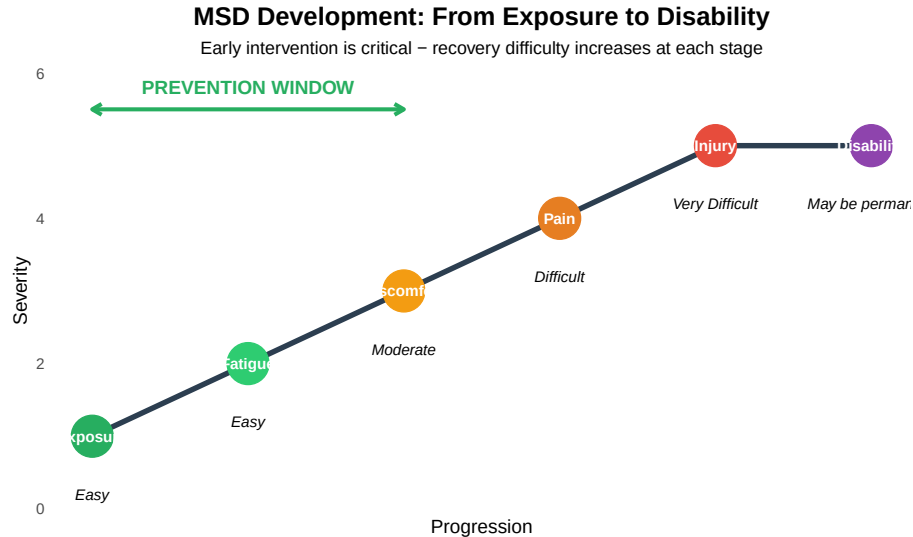


Figure 7.7: The MSD Development Cycle

Warning Signs: When to Take Action

**Early Warning Signs (Act Immediately):** - Persistent fatigue in specific body parts - Discomfort during or after work - Tightness or stiffness - Minor aching or soreness

**Serious Warning Signs (Seek Medical Attention):** - Persistent pain that doesn't go away with rest - Numbness or tingling - Loss of strength or grip - Swelling or inflammation - Reduced range of motion - Pain that wakes you at night

**Key Message:** Report symptoms early! The earlier the intervention, the better the outcome.

## 7.5 Preventing MSDs: Best Practices

Prevention is far more effective and less costly than treatment. Here are evidence-based strategies for MSD prevention:



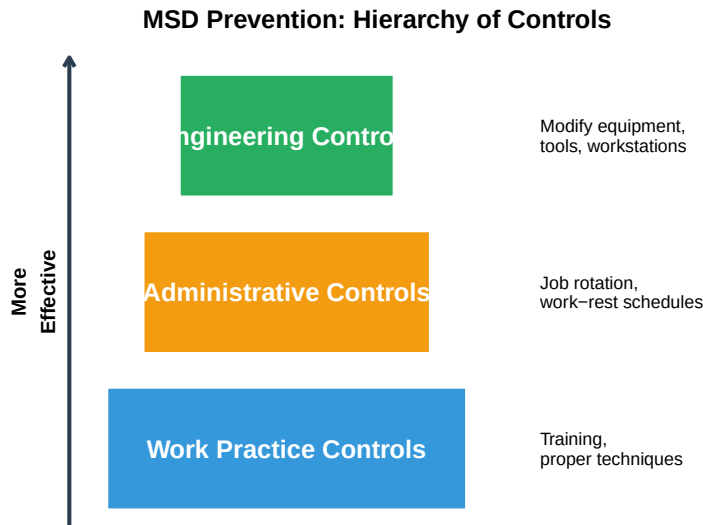


Figure 7.8: MSD Prevention Hierarchy

Table 7.6: MSD Prevention Strategies

| Strategy                            | Engineering Solutions                             | Worker Actions                 |
|-------------------------------------|---------------------------------------------------|--------------------------------|
| <b>**Maintain Neutral Posture**</b> | Adjustable workstations, proper tool selection    | Awareness of body position, s  |
| <b>**Reduce Repetition**</b>        | Automation, job rotation between stations         | Vary tasks when possible       |
| <b>**Minimize Force**</b>           | Mechanical assists, better handles, lighter parts | Use tools properly, ask for he |
| <b>**Avoid Static Positions**</b>   | Sit-stand workstations, adjustable fixtures       | Shift weight, change position  |
| <b>**Eliminate Contact Stress**</b> | Padding, rounded edges, anti-fatigue mats         | Use padding, wear appropriat   |
| <b>**Control Vibration**</b>        | Vibration-dampening tools, isolation mounts       | Limit exposure time, grip too  |
| <b>**Optimize Environment**</b>     | Proper lighting, climate control, noise reduction | Report issues, use available c |
| <b>**Allow Recovery**</b>           | Rest break areas, micro-break reminders           | Take breaks, report fatigue    |

7.5.1 Proper Lifting Technique

Improper lifting is a leading cause of back injuries. The NIOSH lifting equation provides guidelines, but proper technique is fundamental.

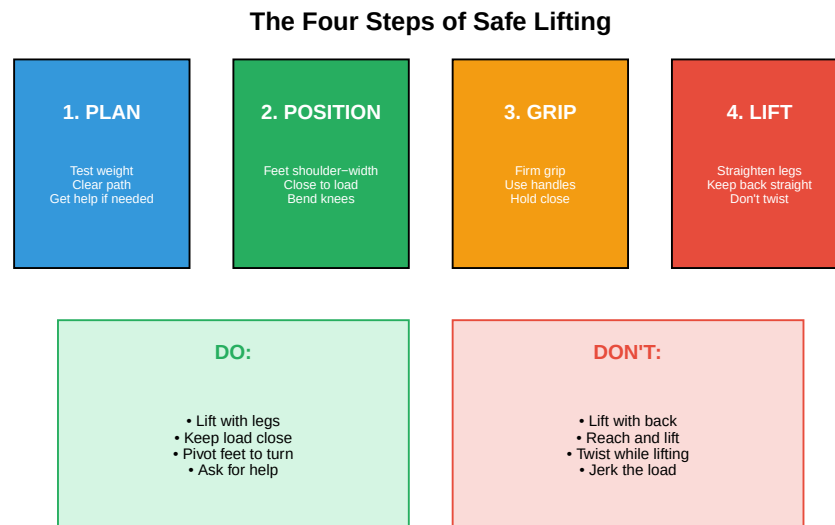


Figure 7.9: Proper Lifting Technique

## 7.6 Ergonomic Design Principles

Effective ergonomic design follows established principles that account for human capabilities and limitations.

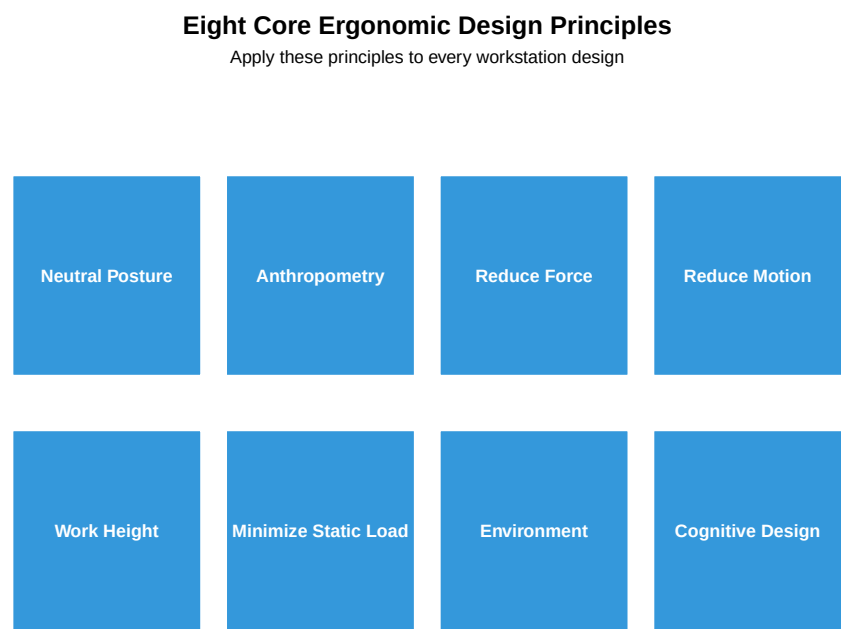


Figure 7.10: Ergonomic Design Principles

Table 7.7: Ergonomic Design Principles in Detail

| Principle                               | Description                                            | Implementation                                   |
|-----------------------------------------|--------------------------------------------------------|--------------------------------------------------|
| <b>**1. Neutral Posture**</b>           | Design to promote natural, balanced body positions     | Adjustable workstation                           |
| <b>**2. Accommodate Anthropometry**</b> | Accommodate different body sizes (5th-95th percentile) | Adjustable equipment                             |
| <b>**3. Reduce Excessive Force**</b>    | Minimize force requirements for all tasks              | Mechanical assists, better tool design           |
| <b>**4. Reduce Excessive Motion**</b>   | Reduce extreme motions, frequencies, and durations     | Bring work closer, eliminate unnecessary motions |
| <b>**5. Optimize Work Height**</b>      | Match work surface height to task requirements         | Precision work height adjustment                 |
| <b>**6. Minimize Static Load**</b>      | Avoid holding same position for extended periods       | Support arms, provide rest breaks                |
| <b>**7. Environmental Conditions**</b>  | Optimize lighting, temperature, noise, air quality     | Task lighting, climate control                   |
| <b>**8. Cognitive Support**</b>         | Design interfaces for easy understanding and use       | Clear displays, logical layout                   |

### 7.6.1 Anthropometry: Designing for Human Dimensions

**Anthropometry** is the scientific study of human body measurements. Workstations must accommodate the range of workers who will use them.

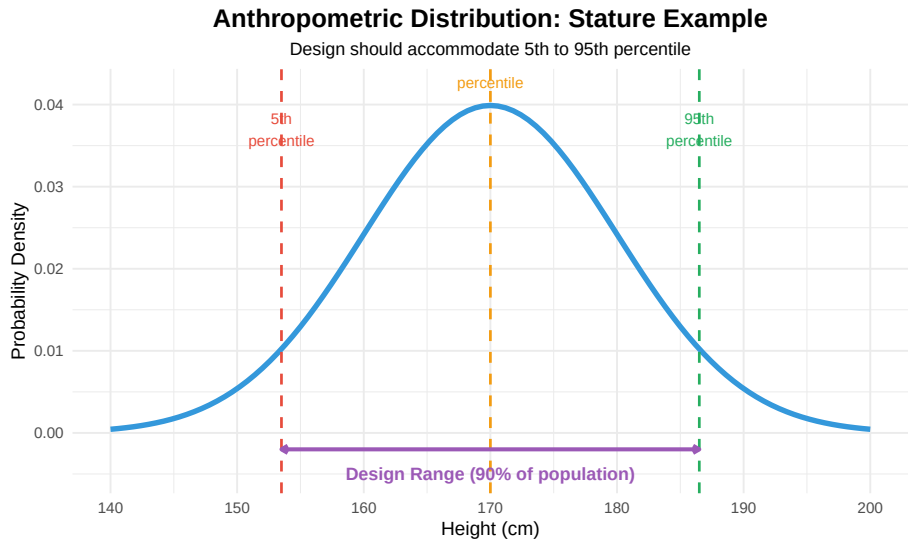


Figure 7.11: Anthropometric Design Considerations

Table 7.8: Key Anthropometric Measurements for Worksta

| Measurement               | 5th % Female (cm) | 50th % Combined (cm) | 95th |
|---------------------------|-------------------|----------------------|------|
| Stature (Standing Height) | 150               |                      | 170  |
| Eye Height (Standing)     | 138               |                      | 158  |
| Shoulder Height           | 121               |                      | 140  |
| Elbow Height              | 93                |                      | 104  |
| Knuckle Height            | 64                |                      | 74   |
| Sitting Height            | 79                |                      | 87   |
| Seated Eye Height         | 68                |                      | 76   |
| Seated Elbow Height       | 18                |                      | 24   |
| Thigh Clearance           | 12                |                      | 15   |
| Forward Reach             | 64                |                      | 73   |

### 7.6.2 Work Surface Height Guidelines

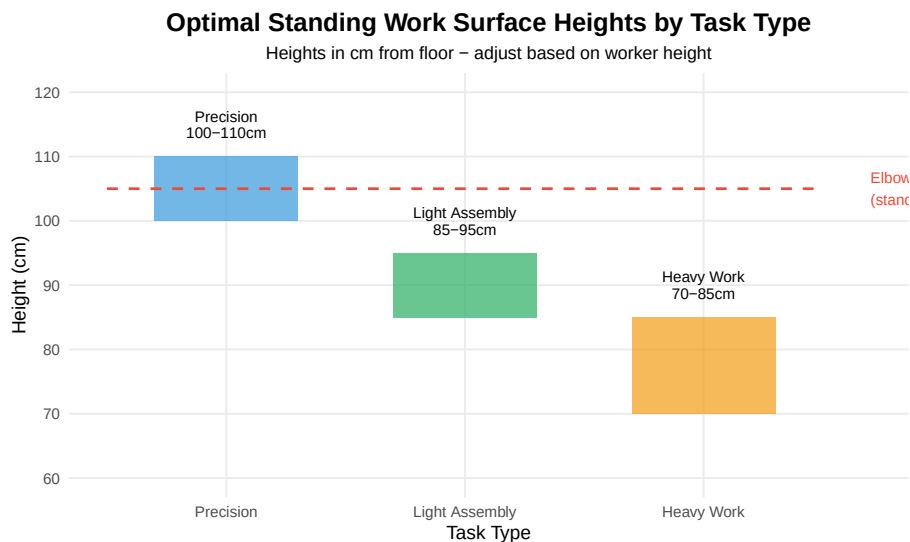


Figure 7.12: Recommended Work Surface Heights

Discussion: Why Different Heights for Different Tasks?

**Precision Work (100-110 cm):** - Requires close visual inspection - Hands need to be at or above eye level - Examples: Electronics assembly, inspection, watchmaking

**Light Assembly (85-95 cm):** - General manipulation tasks - Hands at about elbow height - Examples: Packaging, general assembly, sorting

**Heavy Work (70-85 cm):** - Requires pushing, pressing, or lifting - Lower height allows use of body weight - Examples: Packing boxes, pressing operations, heavy assembly

**Key Principle:** The work surface height should allow the worker to maintain **neutral posture** while performing the specific task.

## 7.7 Ergonomic Assessment Tools

Several standardized tools help assess ergonomic risk and guide interventions. Understanding these tools is essential for the process engineer.

Table 7.9: Common Ergonomic Assessment Tools

| Tool                              | Full Name                      | Focus Area                  | Best Used For                       |
|-----------------------------------|--------------------------------|-----------------------------|-------------------------------------|
| <b>**RULA**</b>                   | Rapid Upper Limb Assessment    | Upper body postures         | Sedentary/computer work             |
| <b>**REBA**</b>                   | Rapid Entire Body Assessment   | Whole body postures         | Manufacturing tasks                 |
| <b>**NIOSH Lifting Equation**</b> | NIOSH Lifting Equation         | Manual lifting tasks        | Lifting, lowering, pushing, pulling |
| <b>**OCRA**</b>                   | Occupational Repetitive Action | Repetitive upper limb tasks | Assembly line work                  |
| <b>**EAWS**</b>                   | Ergonomic Assessment Worksheet | Comprehensive workstation   | Complete work environment           |
| <b>**Snook Tables**</b>           | Snook Psychophysical Tables    | Manual material handling    | Push, pull, lift, carry             |

### 7.7.1 REBA: Rapid Entire Body Assessment

**REBA** is one of the most widely used ergonomic assessment tools in manufacturing. It evaluates the whole body posture and provides a risk score.

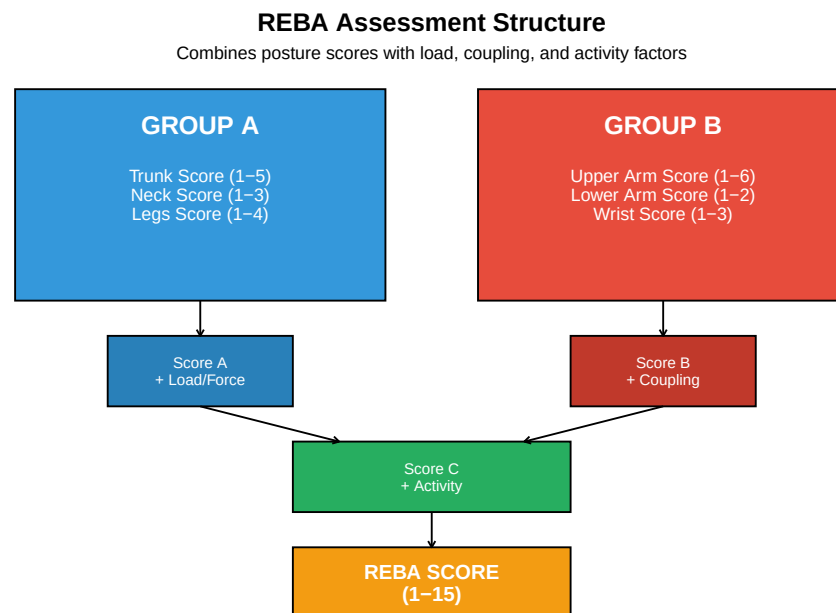


Figure 7.13: REBA Assessment Framework

### 7.7.2 REBA Scoring Tables

**Trunk Posture Scoring:**

Table 7.10: REBA Trunk Posture Scores

| Position                       | Score | Adjustment                    |
|--------------------------------|-------|-------------------------------|
| Upright (0°)                   | 1     | —                             |
| 0-20° flexion/extension        | 2     | +1 if twisted                 |
| 20-60° flexion, >20° extension | 3     | +1 if twisted                 |
| >60° flexion                   | 4     | +1 if twisted or side-bending |

**Neck Posture Scoring:**

Table 7.11: REBA Neck Posture Scores

| Position                  | Score | Adjustment                    |
|---------------------------|-------|-------------------------------|
| 0-20° flexion             | 1     | +1 if twisted or side-bending |
| >20° flexion or extension | 2     | +1 if twisted or side-bending |

**REBA Action Levels:**

Table 7.12: REBA Action Levels

| REBA Score | Risk Level | Action           |
|------------|------------|------------------|
| 1          | Negligible | None necessary   |
| 2-3        | Low        | May be necessary |
| 4-7        | Medium     | Necessary        |
| 8-10       | High       | Necessary soon   |
| 11-15      | Very High  | Necessary NOW    |

**7.7.3 REBA Employee Assessment Worksheet**

The REBA Employee Assessment Worksheet provides a systematic approach to evaluating whole-body posture risk. This worksheet is based on the original work by Hignett and McAtamney (2000) and is widely used in manufacturing ergonomic assessments.

**7.7.3.1 Group A: Neck, Trunk, and Leg Analysis****Step 1: Neck Posture Score**

Table 7.13: REBA Neck Posture Scoring

| Position      | Base Score | Adjustment                    | Max Score |
|---------------|------------|-------------------------------|-----------|
| 0-20° flexion | 1          | +1 if twisted or side-bending | 2         |

|              |   |                               |   |
|--------------|---|-------------------------------|---|
| >20° flexion | 2 | +1 if twisted or side-bending | 3 |
| In extension | 2 | +1 if twisted or side-bending | 3 |

**Step 2: Trunk Posture Score**

Table 7.14: REBA Trunk Posture Scoring

| Position                         | Base Score | Adjustment                    | Max Score |
|----------------------------------|------------|-------------------------------|-----------|
| Upright (0°)                     | 1          | —                             | 1         |
| 0-20° flexion or extension       | 2          | +1 if twisted or side-bending | 3         |
| 20-60° flexion or >20° extension | 3          | +1 if twisted or side-bending | 4         |
| >60° flexion                     | 4          | +1 if twisted or side-bending | 5         |

**Step 3: Legs Posture Score**

Table 7.15: REBA Legs Posture Scoring

| Position                                                      | Base Score | Knee Adjustment          |
|---------------------------------------------------------------|------------|--------------------------|
| Bilateral weight bearing, walking, or sitting                 | 1          | +1 if knees 30-60° flexi |
| Unilateral weight bearing, featherweight, or unstable posture | 2          | +1 if knees 30-60° flexi |

**Step 4: Look Up Table A Score**Table 7.16: REBA Table A: Posture Score A (Trunk  $\times$  Neck  $\times$  Legs)

| Trunk Score | Neck Score | Legs = 1 | Legs = 2 | Legs = 3 | Legs = 4 |
|-------------|------------|----------|----------|----------|----------|
| 1           | 1          | 1        | 1        | 3        | 4        |
| 1           | 2          | 2        | 2        | 3        | 4        |
| 1           | 3          | 2        | 3        | 4        | 5        |
| 2           | 1          | 2        | 3        | 4        | 5        |
| 2           | 2          | 3        | 4        | 5        | 6        |
| 2           | 3          | 4        | 5        | 6        | 7        |
| 3           | 1          | 3        | 4        | 5        | 6        |
| 3           | 2          | 4        | 5        | 6        | 7        |
| 3           | 3          | 5        | 6        | 7        | 8        |
| 4           | 1          | 4        | 5        | 6        | 7        |
| 4           | 2          | 5        | 6        | 7        | 8        |
| 4           | 3          | 6        | 7        | 8        | 9        |
| 5           | 1          | 6        | 7        | 8        | 9        |
| 5           | 2          | 7        | 8        | 9        | 9        |



|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 5 | 3 | 8 | 9 | 9 | 9 |
|---|---|---|---|---|---|

**Step 5: Add Load/Force Score**

Table 7.17: REBA Load/Force Scoring

| Load/Force          | Score | Shock Adjustment                       |
|---------------------|-------|----------------------------------------|
| < 5 kg (< 11 lbs)   | 0     | —                                      |
| 5-10 kg (11-22 lbs) | 1     | +1 if shock or rapid build-up of force |
| > 10 kg (> 22 lbs)  | 2     | +1 if shock or rapid build-up of force |

$$\text{Score A} = \text{Table A Score} + \text{Load/Force Score}$$

**7.7.3.2 Group B: Arm and Wrist Analysis****Step 6: Upper Arm Posture Score**

Table 7.18: REBA Upper Arm Posture Scoring

| Position                     | Base Score | Adjustments                                                    |
|------------------------------|------------|----------------------------------------------------------------|
| 20° extension to 20° flexion | 1          | +1 if shoulder raised; +1 if arm abducted; -1 if leaning/suppo |
| >20° extension               | 2          | +1 if shoulder raised; +1 if arm abducted; -1 if leaning/suppo |
| 20-45° flexion               | 2          | +1 if shoulder raised; +1 if arm abducted; -1 if leaning/suppo |
| 45-90° flexion               | 3          | +1 if shoulder raised; +1 if arm abducted; -1 if leaning/suppo |
| >90° flexion                 | 4          | +1 if shoulder raised; +1 if arm abducted; -1 if leaning/suppo |

**Step 7: Lower Arm Posture Score**

Table 7.19: REBA Lower Arm Posture Scoring

| Position                        | Score |
|---------------------------------|-------|
| 60-100° flexion (optimal range) | 1     |
| < 60° or > 100° flexion         | 2     |

**Step 8: Wrist Posture Score**

Table 7.20: REBA Wrist Posture Scoring

| Position | Base Score | Adjustment | Max Score |
|----------|------------|------------|-----------|
|----------|------------|------------|-----------|

|                                      |   |                                    |   |
|--------------------------------------|---|------------------------------------|---|
| 0-15° flexion or extension (neutral) | 1 | +1 if wrist is twisted or deviated | 2 |
| >15° flexion or extension            | 2 | +1 if wrist is twisted or deviated | 3 |

**Step 9: Look Up Table B Score**Table 7.21: REBA Table B: Posture Score B (Upper Arm  $\times$  Lower Arm  $\times$  Wrist)

| Upper Arm Score | Lower Arm Score | Wrist = 1 | Wrist = 2 | Wrist = 3 |
|-----------------|-----------------|-----------|-----------|-----------|
| 1               | 1               | 1         | 1         | 2         |
| 1               | 2               | 2         | 2         | 3         |
| 2               | 1               | 1         | 2         | 3         |
| 2               | 2               | 2         | 3         | 4         |
| 3               | 1               | 3         | 4         | 5         |
| 3               | 2               | 4         | 5         | 5         |
| 4               | 1               | 4         | 5         | 6         |
| 4               | 2               | 5         | 6         | 7         |
| 5               | 1               | 6         | 7         | 8         |
| 5               | 2               | 7         | 8         | 8         |
| 6               | 1               | 7         | 8         | 9         |
| 6               | 2               | 8         | 9         | 9         |

**Step 10: Add Coupling Score**

Table 7.22: REBA Coupling Score

| Coupling Quality | Description                                                            |
|------------------|------------------------------------------------------------------------|
| Good             | Well-fitting handle and mid-range power grip                           |
| Fair             | Hand hold acceptable but not ideal; coupling acceptable via other body |
| Poor             | Hand hold not acceptable although possible; no handles                 |
| Unacceptable     | Awkward, unsafe grip; no handles; coupling unacceptable using other bo |

$$\text{Score B} = \text{Table B Score} + \text{Coupling Score}$$

**7.7.3.3 Calculating the Final REBA Score****Step 11: Look Up Table C Score**

Table 7.23: REBA Table C: Combined Score (Score A  $\times$  Score B)

| Score A | B=1 | B=2 | B=3 | B=4 | B=5 | B=6 | B=7 | B=8 | B=9 | B=10 | B=11 | B=12 |
|---------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| 1       | 1   | 1   | 1   | 2   | 3   | 3   | 4   | 5   | 6   | 7    | 7    | 7    |
| 2       | 1   | 2   | 2   | 3   | 4   | 4   | 5   | 6   | 6   | 7    | 7    | 8    |
| 3       | 1   | 2   | 3   | 3   | 4   | 5   | 6   | 7   | 7   | 8    | 8    | 8    |
| 4       | 2   | 3   | 3   | 4   | 5   | 6   | 7   | 8   | 8   | 9    | 9    | 9    |
| 5       | 3   | 4   | 4   | 5   | 6   | 7   | 8   | 8   | 9   | 9    | 9    | 10   |
| 6       | 3   | 4   | 5   | 6   | 7   | 8   | 8   | 9   | 9   | 10   | 10   | 10   |
| 7       | 4   | 5   | 6   | 7   | 8   | 8   | 9   | 9   | 10  | 10   | 11   | 11   |
| 8       | 5   | 6   | 7   | 8   | 8   | 9   | 9   | 10  | 10  | 11   | 11   | 11   |
| 9       | 6   | 6   | 7   | 8   | 9   | 9   | 10  | 10  | 10  | 11   | 11   | 12   |
| 10      | 7   | 7   | 8   | 9   | 9   | 10  | 10  | 11  | 11  | 11   | 12   | 12   |
| 11      | 7   | 7   | 8   | 9   | 9   | 10  | 11  | 11  | 11  | 12   | 12   | 12   |
| 12      | 7   | 8   | 8   | 9   | 9   | 10  | 11  | 11  | 12  | 12   | 12   | 12   |

**Step 12: Add Activity Score**

Table 7.24: REBA Activity Score (Add +1 for each condition present)

| Activity Condition                                                        | Score |
|---------------------------------------------------------------------------|-------|
| One or more body parts are static (held > 1 minute)                       | +1    |
| Repeated small range actions (> 4 $\times$ per minute, excluding walking) | +1    |
| Action causes rapid large range changes in postures or unstable base      | +1    |

REBA Score = Table C Score + Activity Score

**7.7.3.4 REBA Score Interpretation**

Table 7.25: REBA Action Level Guidelines

| REBA Score | Risk Level | Action Required  | Priority                               |
|------------|------------|------------------|----------------------------------------|
| 1          | Negligible | None necessary   | Continue monitoring                    |
| 2-3        | Low        | May be necessary | Investigate further when convenient    |
| 4-7        | Medium     | Necessary        | Implement changes to reduce risk       |
| 8-10       | High       | Necessary soon   | Investigate and implement changes soon |
| 11-15      | Very High  | Necessary NOW    | Stop task and make immediate changes   |

### 7.7.4 REBA Example Calculation: Step-by-Step Walk-through

Let's work through a complete REBA assessment for a manufacturing worker performing a packaging task.

**Scenario:** A worker stands at a packaging station, bending forward to pick up items from a low bin and placing them into boxes on a conveyor. The task is performed continuously for 4-hour shifts.

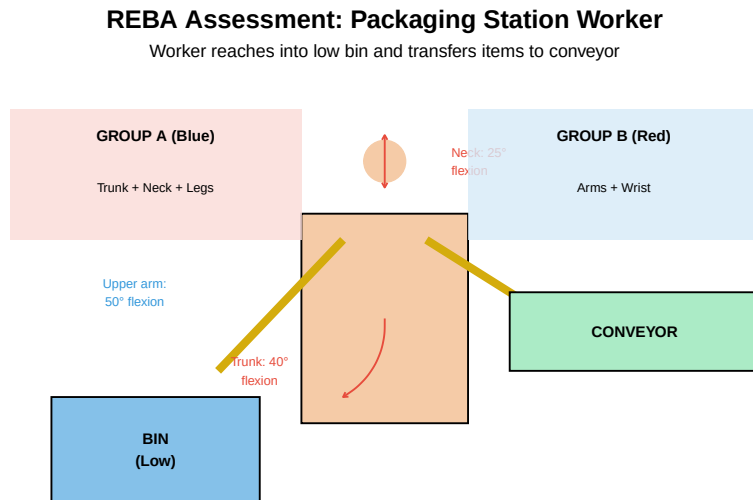


Figure 7.14: REBA Assessment Scenario: Packaging Worker

#### 7.7.4.1 Step-by-Step Assessment

Live cell — try different posture, load, or coupling scores.

```
#| label: reba-step-by-step
# =====
# REBA Assessment: Packaging Station Worker
# =====
# Worker bends forward to pick items from low bin and place on conveyor
# Items weigh approximately 7 kg each
# Task performed continuously with repetitive motions

cat("===== GROUP A: TRUNK, NECK, LEGS =====\n\n")

# Step 1: Neck Score
# Worker looks down into bin - approximately 25° flexion
neck_base <- 2      # >20° flexion
```

```

neck_twist <- 0      # No twist or side-bending
neck_score <- neck_base + neck_twist
cat("Step 1 - Neck Score:", neck_score, "\n")
cat("  Position: >20° flexion (looking down into bin)\n")
cat("  Base score: 2, Twist adjustment: 0\n\n")

# Step 2: Trunk Score
# Worker bends forward approximately 40° to reach bin
trunk_base <- 3      # 20-60° flexion
trunk_twist <- 1     # Slight twist when reaching
trunk_score <- trunk_base + trunk_twist
cat("Step 2 - Trunk Score:", trunk_score, "\n")
cat("  Position: 40° forward flexion with slight twist\n")
cat("  Base score: 3, Twist adjustment: +1\n\n")

# Step 3: Legs Score
# Worker stands with weight distributed evenly
legs_base <- 1       # Bilateral weight bearing
legs_knee <- 0       # Knees not flexed significantly
legs_score <- legs_base + legs_knee
cat("Step 3 - Legs Score:", legs_score, "\n")
cat("  Position: Standing, bilateral weight bearing\n")
cat("  Base score: 1, Knee adjustment: 0\n\n")

# Step 4: Table A Lookup
# Trunk = 4, Neck = 2, Legs = 1
# From Table A: Score = 5
table_a_score <- 5
cat("Step 4 - Table A Score:", table_a_score, "\n")
cat("  Lookup: Trunk=4, Neck=2, Legs=1 → Table A = 5\n\n")

# Step 5: Load/Force Score
# Items weigh approximately 7 kg
load_force <- 1      # 5-10 kg
shock_force <- 0     # No shock/rapid buildup
load_score <- load_force + shock_force
cat("Step 5 - Load/Force Score:", load_score, "\n")
cat("  Load: 7 kg (in 5-10 kg range) = 1\n\n")

# SCORE A CALCULATION
score_a <- table_a_score + load_score
cat(">>> SCORE A = Table A + Load/Force =", table_a_score, "+", load_score, "=", score_a, "\n\n")

cat("===== GROUP B: UPPER ARM, LOWER ARM, WRIST =====\n\n")

```

```

# Step 6: Upper Arm Score
# Arms reaching forward/down approximately 50° flexion
upper_arm_base <- 3 # 45-90° flexion
shoulder_raised <- 0 # Not raised
arm_abducted <- 0 # Not abducted
arm_support <- 0 # Not supported
upper_arm_score <- upper_arm_base + shoulder_raised + arm_abducted + arm_support
cat("Step 6 - Upper Arm Score:", upper_arm_score, "\n")
cat(" Position: 50° flexion reaching into bin\n")
cat(" Base score: 3, No adjustments needed\n\n")

# Step 7: Lower Arm Score
# Elbow bent approximately 70° (in optimal range)
lower_arm_score <- 1 # 60-100° flexion
cat("Step 7 - Lower Arm Score:", lower_arm_score, "\n")
cat(" Position: 70° elbow flexion (optimal range)\n\n")

# Step 8: Wrist Score
# Wrist in moderate flexion to grasp items
wrist_base <- 2 # >15° flexion
wrist_twist <- 0 # No significant twist
wrist_score <- wrist_base + wrist_twist
cat("Step 8 - Wrist Score:", wrist_score, "\n")
cat(" Position: ~20° wrist flexion when grasping\n")
cat(" Base score: 2, Twist adjustment: 0\n\n")

# Step 9: Table B Lookup
# Upper Arm = 3, Lower Arm = 1, Wrist = 2
# From Table B: Score = 4
table_b_score <- 4
cat("Step 9 - Table B Score:", table_b_score, "\n")
cat(" Lookup: Upper Arm=3, Lower Arm=1, Wrist=2 → Table B = 4\n\n")

# Step 10: Coupling Score
# Worker grasps boxes - acceptable grip but not ideal handles
coupling_score <- 1 # Fair coupling
cat("Step 10 - Coupling Score:", coupling_score, "\n")
cat(" Coupling: Fair - hand hold acceptable but not ideal\n\n")

# SCORE B CALCULATION
score_b <- table_b_score + coupling_score
cat(">>> SCORE B = Table B + Coupling =", table_b_score, "+", coupling_score, "=", score_b, "\n\n")

cat("===== FINAL REBA SCORE CALCULATION =====\n\n")

```

```

# Step 11: Table C Lookup
# Score A = 6, Score B = 5
# From Table C: Score = 7
table_c_score <- 7
cat("Step 11 - Table C Score:", table_c_score, "\n")
cat("  Lookup: Score A=6, Score B=5 → Table C = 7\n\n")

# Step 12: Activity Score
# Check all three conditions:
static_posture <- 0 # Body parts not held >1 minute
repeated_actions <- 1 # Repeated small range actions (continuous picking)
rapid_changes <- 0 # No rapid large range changes
activity_score <- static_posture + repeated_actions + rapid_changes
cat("Step 12 - Activity Score:", activity_score, "\n")
cat("  Static posture >1 min: No (+0)\n")
cat("  Repeated actions >4×/min: Yes (+1)\n")
cat("  Rapid large changes: No (+0)\n\n")

# FINAL REBA SCORE
reba_final <- table_c_score + activity_score
cat("                                \n")
cat("      FINAL REBA SCORE:", reba_final, "                                \n")
cat("                                \n\n")

# Interpretation
cat("INTERPRETATION:\n")
cat("                                \n")
if (reba_final == 1) {
  cat("Risk Level: NEGLIGIBLE\n")
  cat("Action: None necessary\n")
} else if (reba_final <= 3) {
  cat("Risk Level: LOW\n")
  cat("Action: May be necessary\n")
} else if (reba_final <= 7) {
  cat("Risk Level: MEDIUM\n")
  cat("Action: Necessary\n")
} else if (reba_final <= 10) {
  cat("Risk Level: HIGH\n")
  cat("Action: Necessary soon\n")
} else {
  cat("Risk Level: VERY HIGH\n")
  cat("Action: Necessary NOW - Stop task immediately\n")
}
cat("                                \n")

```

### 7.7.4.2 Recommendations for This Assessment

Based on the REBA score of **8 (HIGH risk)**, the following interventions should be implemented soon:

Table 7.26: Recommended Interventions Based on REBA Assessment

| Priority | Intervention           | Target Risk Factor | Expected Impact                |
|----------|------------------------|--------------------|--------------------------------|
| 1        | Raise bin height       | Trunk flexion      | Reduce trunk score from 4 to 2 |
| 2        | Add tilt mechanism     | Neck flexion       | Reduce neck score from 2 to 1  |
| 3        | Reduce load weight     | Load/force         | Reduce load score from 1 to 0  |
| 4        | Implement job rotation | Repetition         | Reduce activity score          |

### 7.7.5 REBA Example Calculation

Live cell — plug in your own posture scores and re-run.

```
#| label: reba-example
# REBA Assessment Example: Assembly Worker

# Group A Scores (Trunk, Neck, Legs)
trunk_score <- 3      # 20-60° forward bend
neck_score <- 2       # >20° flexion
legs_score <- 1       # Bilateral weight bearing

# Look up Table A score (trunk + neck + legs)
# For this combination: Score A = 4
table_a_score <- 4

# Load/Force Score
load_force <- 1       # 5-10 kg intermittent

# Score A with load
score_a <- table_a_score + load_force
cat("Score A (posture + load):", score_a, "\n")

# Group B Scores (Upper arm, Lower arm, Wrist)
upper_arm_score <- 3  # 45-90° flexion
lower_arm_score <- 2  # <60° or >100° flexion
wrist_score <- 2      # >15° flexion/extension

# Look up Table B score
# For this combination: Score B = 5
table_b_score <- 5

# Coupling Score
```



```

coupling <- 1          # Acceptable grip

# Score B with coupling
score_b <- table_b_score + coupling
cat("Score B (posture + coupling):", score_b, "\n")

# Look up Table C (Score A × Score B)
# For Score A = 5 and Score B = 6: Score C = 8
score_c <- 8

# Activity Score
activity <- 1          # One or more body parts static >1 min

# Final REBA Score
reba_final <- score_c + activity
cat("\n=== FINAL REBA SCORE:", reba_final, "===\n")

# Interpretation
cat("\nInterpretation:\n")
if (reba_final <= 1) {
  cat("Risk Level: Negligible - No action necessary\n")
} else if (reba_final <= 3) {
  cat("Risk Level: Low - Action may be necessary\n")
} else if (reba_final <= 7) {
  cat("Risk Level: Medium - Action necessary\n")
} else if (reba_final <= 10) {
  cat("Risk Level: High - Action necessary soon\n")
} else {
  cat("Risk Level: Very High - Action necessary NOW\n")
}

```

Practice Problem: REBA Assessment

**Scenario:** A worker is performing a task with the following postures:

- **Trunk:** 30° forward flexion, twisted
- **Neck:** 15° flexion
- **Legs:** Weight on one leg
- **Upper Arm:** 60° flexion, shoulder raised
- **Lower Arm:** 80° flexion
- **Wrist:** Neutral
- **Load:** 8 kg carried
- **Coupling:** Fair grip (handles exist but not ideal)
- **Activity:** Repeated small range actions

**Calculate the REBA score and determine the action level.**

**Solution:**

Trunk: 3 + 1 (twisted) = 4  
 Neck: 1 (0-20° flexion)  
 Legs: 2 (weight on one leg)  
 Table A = 5  
 Load/Force = 1 (5-10kg)  
 Score A = 6

Upper Arm: 3 + 1 (raised) = 4  
 Lower Arm: 1 (60-100°)  
 Wrist: 1 (neutral)  
 Table B = 4  
 Coupling = 1 (fair)  
 Score B = 5

Table C (Score A=6, Score B=5) = 8  
 Activity = +1 (repeated actions)  
 REBA Score = 9

Risk Level: HIGH - Action necessary soon

**7.7.6 NIOSH Lifting Equation**

The **NIOSH Lifting Equation** calculates a **Recommended Weight Limit (RWL)** for lifting tasks.

$$RWL = LC \times HM \times VM \times DM \times AM \times FM \times CM$$

Where:

Table 7.27: NIOSH Lifting Equation Factors

| Factor    | Name                  | Value                                 | Description                                             |
|-----------|-----------------------|---------------------------------------|---------------------------------------------------------|
| <b>LC</b> | Load Constant         | 23 kg (51 lb)                         | Maximum weight under ideal conditions                   |
| <b>HM</b> | Horizontal Multiplier | 25/H                                  | H = horizontal distance from midsagittal plane to hands |
| <b>VM</b> | Vertical Multiplier   | $1 - 0.0032V - 75 \times 10^{-6} V^2$ | V = vertical height of hands at lift origin             |
| <b>DM</b> | Distance Multiplier   | $0.82 + 4.5/D$                        | D = vertical travel distance (cm)                       |
| <b>AM</b> | Asymmetric Multiplier | $1 - 0.0032A$                         | A = angle of asymmetry (degrees)                        |
| <b>FM</b> | Frequency Multiplier  | Table lookup                          | Based on lift frequency and duration                    |
| <b>CM</b> | Coupling Multiplier   | Table lookup                          | Based on hand-load coupling quality                     |

**Lifting Index (LI)** indicates risk level:

$$LI = \frac{\text{Actual Weight}}{\text{RWL}}$$

Table 7.28: Lifting Index Interpretation

| Lifting Index | Risk      | Action                                         |
|---------------|-----------|------------------------------------------------|
| LI 1.0        | Low       | Task acceptable for most workers               |
| 1.0 < LI 2.0  | Moderate  | Some workers may be at risk - consider changes |
| 2.0 < LI 3.0  | High      | Many workers at risk - changes recommended     |
| LI > 3.0      | Very High | Unacceptable - immediate changes required      |

Live cell — change the distances, angle, or frequency and re-run.

```
#| label: niosh-example
# NIOSH Lifting Equation Example

# Task parameters
H <- 40    # Horizontal distance (cm)
V <- 30    # Vertical height at start (cm)
D <- 60    # Vertical travel distance (cm)
A <- 30    # Asymmetry angle (degrees)
F <- 2     # Lifts per minute
duration <- 2 # Hours of lifting

# Constants
LC <- 23   # Load constant (kg)

# Calculate multipliers
HM <- 25 / H
VM <- 1 - 0.003 * abs(V - 75)
DM <- 0.82 + 4.5 / D
AM <- 1 - 0.0032 * A

# Frequency multiplier (from table, for 2 lifts/min, 2 hours, V<75cm)
FM <- 0.84

# Coupling multiplier (assume "fair" coupling)
CM <- 0.95

cat("Multipliers:\n")
cat("HM =", round(HM, 3), "\n")
cat("VM =", round(VM, 3), "\n")
cat("DM =", round(DM, 3), "\n")
cat("AM =", round(AM, 3), "\n")
```

```

cat("FM =", FM, "\n")
cat("CM =", CM, "\n\n")

# Calculate RWL
RWL <- LC * HM * VM * DM * AM * FM * CM
cat("Recommended Weight Limit (RWL):", round(RWL, 1), "kg\n\n")

# Calculate Lifting Index for actual load
actual_weight <- 15 # kg
LI <- actual_weight / RWL
cat("Actual Weight:", actual_weight, "kg\n")
cat("Lifting Index (LI):", round(LI, 2), "\n\n")

# Interpretation
if (LI <= 1.0) {
  cat("Risk Level: LOW - Task acceptable\n")
} else if (LI <= 2.0) {
  cat("Risk Level: MODERATE - Consider changes\n")
} else if (LI <= 3.0) {
  cat("Risk Level: HIGH - Changes recommended\n")
} else {
  cat("Risk Level: VERY HIGH - Immediate changes required\n")
}

```

---

## 7.8 AI-Powered Ergonomic Assessment

Artificial intelligence is revolutionizing ergonomic assessment by enabling **real-time, automated analysis** of worker postures without disrupting operations.

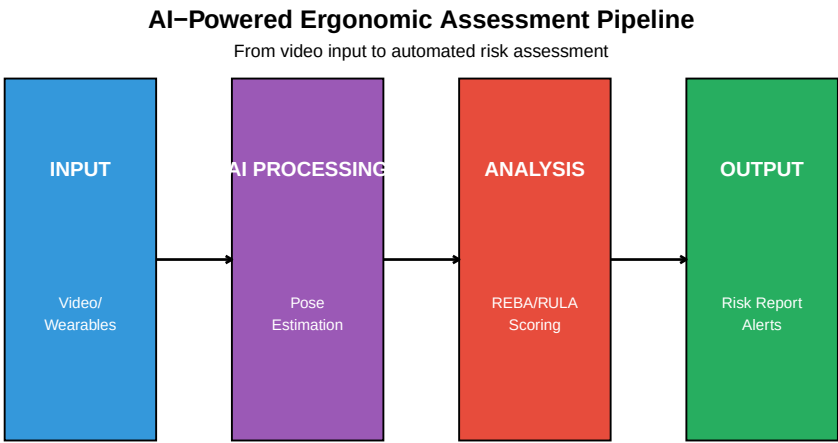


Figure 7.15: AI Ergonomics Assessment Pipeline

Table 7.29: AI-Powered Ergonomic Assessment Tools

| Tool                         | Technology                  | Key Features                                        |
|------------------------------|-----------------------------|-----------------------------------------------------|
| <b>**TuMeke Ergonomics**</b> | Computer vision + wearables | Real-time REBA scoring, multi-camera support        |
| <b>**VelocityEHS**</b>       | AI video analysis           | Automated job analysis, risk prioritization         |
| <b>**Protex AI**</b>         | Real-time video AI          | Privacy-preserving analysis, floor-wide monitoring  |
| <b>**SoterTask**</b>         | Wearable sensors + AI       | Wearable clips, real-time coaching, haptic feedback |
| <b>**Viso.ai**</b>           | Custom AI models            | Custom pose detection, integration APIs             |
| <b>**ErgoPlus Platform**</b> | Mobile app + AI             | Mobile assessment, improvement tracking             |

### 7.8.1 Benefits of AI Ergonomic Assessment

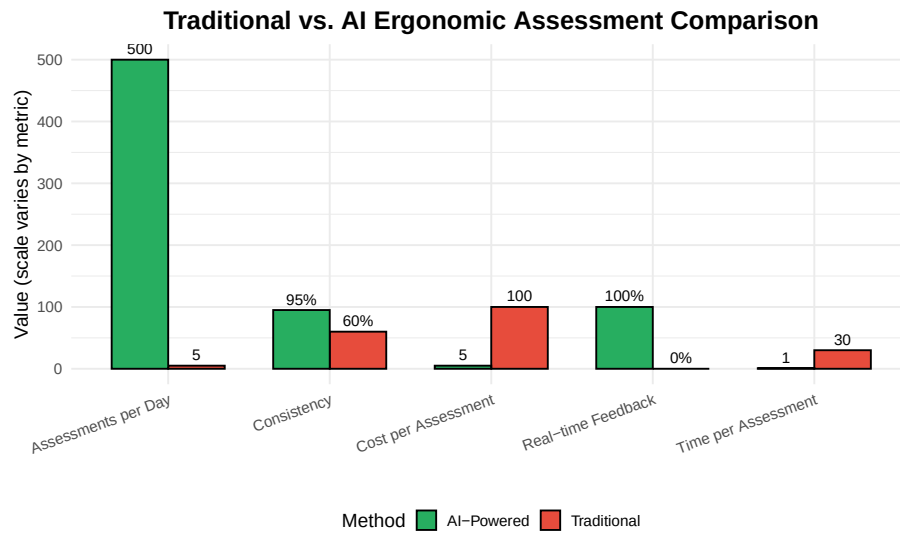


Figure 7.16: Traditional vs. AI Ergonomic Assessment

## 7.9 Ergonomics in Automated Manufacturing

Industrial automation inherently involves the mechanization of processes traditionally carried out by human operators. In the automation design phase, engineers must strike a balance between mechanization and the necessary human interaction with these systems.

### 7.9.1 Human-Automation Interaction Points

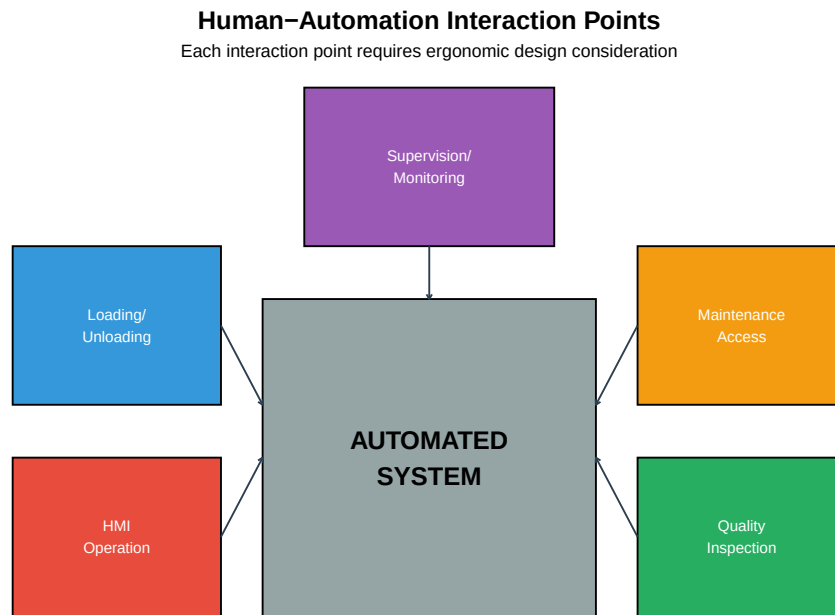


Figure 7.17: Ergonomic Considerations in Automated Systems

Table 7.30: Ergonomic Considerations for Automated Systems

| Interaction Point             | Ergonomic Challenges                                   | Design Solutions            |
|-------------------------------|--------------------------------------------------------|-----------------------------|
| <b>**Loading/Unloading**</b>  | Repetitive motion, lifting, reaching, awkward postures | Height-adjustable conveyors |
| <b>**HMI Operation**</b>      | Static posture, visual strain, cognitive load          | Adjustable screens, proper  |
| <b>**Maintenance**</b>        | Awkward access, confined spaces, tool use              | Adequate access space, tool |
| <b>**Quality Inspection**</b> | Static positions, visual strain, fine motor tasks      | Adjustable fixtures, magnif |
| <b>**Supervision**</b>        | Sedentary position, cognitive fatigue, vigilance       | Multiple monitors at eye le |

### 7.9.2 Collaborative Robot Ergonomics

Cobots present unique ergonomic opportunities and challenges:

Table 7.31: Ergonomic Considerations for Collaborative Robots

| Aspect                     | Opportunity                                         | Challenge                    |
|----------------------------|-----------------------------------------------------|------------------------------|
| <b>**Task Allocation**</b> | Robot handles heavy, repetitive, or precision tasks | Determining optimal task div |

|                               |                                                        |                                   |
|-------------------------------|--------------------------------------------------------|-----------------------------------|
| <b>**Workstation Layout**</b> | Shared workspace allows flexible positioning           | Ensuring adequate                 |
| <b>**Pacing**</b>             | Robot adapts to human pace, not vice versa             | Avoiding pressure                 |
| <b>**Force Assistance**</b>   | Robot provides powered assist for lifting              | Proper force limits               |
| <b>**Cognitive Load**</b>     | Robot handles routine tasks, human focuses on judgment | Maintaining situational awareness |

## 7.10 Employer Responsibilities

Employers have both legal and ethical obligations to protect workers from ergonomic hazards.

Table 7.32: Employer Ergonomics Program Responsibilities

| Responsibility                   | Actions                                                            |
|----------------------------------|--------------------------------------------------------------------|
| <b>**Management Commitment**</b> | Allocate resources, set goals, lead by example                     |
| <b>**Worker Involvement**</b>    | Include workers in assessments, design teams, solution development |
| <b>**Training**</b>              | Ergonomics awareness, proper techniques, symptom recognition       |
| <b>**Hazard Identification**</b> | Regular assessments, worker surveys, injury analysis               |
| <b>**Early Reporting**</b>       | Non-punitive reporting system, prompt response to symptoms         |
| <b>**Hazard Control**</b>        | Engineering controls, work organization, PPE as last resort        |
| <b>**Program Evaluation**</b>    | Track injuries, measure improvements, adjust programs              |

### OSHA Ergonomics Guidelines

While OSHA does not have a specific ergonomics standard, employers are still responsible under the **General Duty Clause** (Section 5(a)(1)) which requires employers to provide a workplace “free from recognized hazards.”

**OSHA’s Recommended Program Elements:** 1. Management leadership 2. Worker participation 3. Hazard identification and assessment 4. Hazard prevention and control 5. Education and training 6. Program evaluation

**Industry-Specific Guidelines:** - Poultry Processing - Nursing Homes - Retail Grocery Stores - Shipyards

Visit: OSHA Ergonomics eTool



## 7.11 Case Study: Ergonomics in Automated Assembly

**Company:** Automotive parts manufacturer

**Problem:** High rate of upper extremity MSDs in assembly department - 15 recordable injuries per year - Lost workdays averaging 20 days per case - Workers' comp costs exceeding \$300,000 annually

### Assessment Findings:

Table 7.33: Ergonomic Assessment Findings

| Workstation      | Primary Issues                        | REBA Score | Risk Level       |
|------------------|---------------------------------------|------------|------------------|
| Sub-assembly     | Overhead reaching, high repetition    | 9          | High             |
| Main line        | Forward bending, static standing      | 8          | High             |
| Inspection       | Awkward wrist postures, visual strain | 7          | Medium           |
| <b>Packaging</b> | <b>Lifting, carrying, repetition</b>  | <b>10</b>  | <b>Very High</b> |

### Interventions Implemented:

1. **Sub-assembly:**
  - Tilting fixtures to bring work closer
  - Parts bins repositioned to eliminate overhead reaching
  - Cobot added for repetitive insertion tasks
2. **Main line:**
  - Height-adjustable workstations installed
  - Anti-fatigue mats added
  - Job rotation implemented
3. **Inspection:**
  - Magnifying task lights installed
  - Angled fixtures to reduce wrist bending
  - Ergonomic seating provided
4. **Packaging:**
  - Vacuum lift assists installed
  - Conveyor heights adjusted
  - Case erectors automated

### Results After 18 Months:

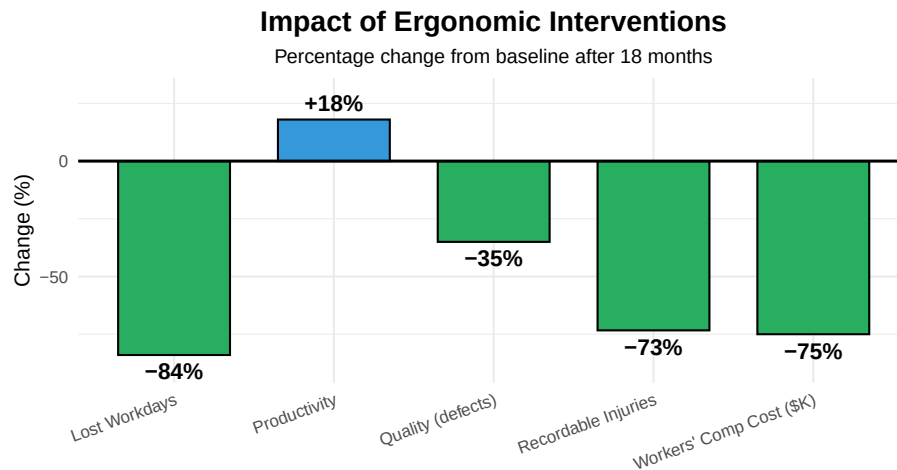


Figure 7.18: Ergonomic Intervention Results

**ROI Analysis:** (live cell — try different investment or savings figures)

```
#| label: roi-analysis
# Investment
equipment_cost <- 150000 # Lift assists, fixtures, cobots
training_cost <- 15000
total_investment <- equipment_cost + training_cost

# Annual Savings
injury_reduction <- (15 - 4) * 20000 # Avg cost per injury
productivity_gain <- 0.18 * 500000 # 18% improvement on labor cost
quality_improvement <- 0.35 * 50000 # 35% defect reduction

annual_savings <- injury_reduction + productivity_gain + quality_improvement

cat("Total Investment: $", format(total_investment, big.mark = " "), "\n")
cat("Annual Savings: $", format(annual_savings, big.mark = " "), "\n")
cat("Payback Period:", round(total_investment / annual_savings, 1), "years\n")
cat("5-Year ROI:", round((annual_savings * 5 - total_investment) / total_investment * 100, 1), "%\n")
```

## 7.12 Summary

Table 7.34: Ergonomics in Manufacturing: Key Concepts Summary

| Topic                            | Key Points                                                                           |
|----------------------------------|--------------------------------------------------------------------------------------|
| <b>**Ergonomics Definition**</b> | Fitting the job to the worker; physical, cognitive, organizational domains           |
| <b>**Risk Factors**</b>          | Posture, repetition, force, static positions, contact stress, vibration, environment |
| <b>**MSDs**</b>                  | Injuries to muscles, nerves, tendons; develop gradually from exposure to disability  |
| <b>**Prevention**</b>            | Engineering controls most effective; proper lifting, neutral postures, recovery time |
| <b>**Design Principles**</b>     | Neutral posture, anthropometry, work height optimization, environmental control      |
| <b>**Assessment Tools**</b>      | REBA for whole body, RULA for upper limb, NIOSH for lifting tasks                    |
| <b>**AI in Ergonomics**</b>      | Real-time monitoring, automated REBA/RULA scoring, continuous assessment             |
| <b>**Employer Role**</b>         | Management commitment, worker involvement, training, hazard control, evaluation      |

## 7.13 Review Questions

Question 1: What are the seven ergonomic risk factors, and which two are most commonly associated with carpal tunnel syndrome?

The seven ergonomic risk factors are: 1. Awkward Postures 2. Repetition 3. Force 4. Static Positions 5. Contact Stress 6. Vibration 7. Environmental Factors

**Carpal tunnel syndrome** is most commonly associated with: - **Repetition** - Repeated wrist movements compress the median nerve - **Force** - Forceful gripping increases pressure in the carpal tunnel - Also contributing: awkward wrist postures and vibration

Question 2: A worker performs a task 8 times per minute for a 4-hour shift. The cycle time is 7.5 seconds. Is this considered high repetition? Why?

**Analysis:** - Cycle time: 7.5 seconds ( $< 15$  seconds threshold) - Repetitions per hour:  $8 \times 60 = 480$  ( $> 240$  threshold)

**Yes, this is HIGH repetition** because: 1. Cycle time is less than 15 seconds 2. Repetitions exceed 240 per hour

**Risk Level:** High - Immediate intervention needed

**Recommended actions:** - Job rotation with different tasks - Automation of repetitive elements - Micro-breaks every 20-30 minutes - Workstation redesign to reduce cycle elements

Question 3: Explain why ergonomic design should accommodate the 5th to 95th percentile of the population rather than designing for the “average” person.

**Designing for the average excludes most people:** - The “average” person in ALL dimensions doesn’t exist - A person may be average height but have

long arms or short legs - Designing for only the 50th percentile fits only about 50% of users

**The 5th-95th percentile range:** - Accommodates 90% of the population - Addresses both small and large users - Provides adjustability rather than fixed dimensions

**Application rules:** - **Clearances** (overhead, legroom): Design for 95th percentile (largest) - **Reach distances:** Design for 5th percentile (smallest) - **Work surfaces:** Make adjustable to accommodate range

**Example:** A fixed workstation at 95cm height: - Too high for 5th percentile female (elbow at ~93cm) - Too low for 95th percentile male (elbow at ~117cm) - Adjustable range of 85-115cm accommodates both

Question 4: Calculate the NIOSH Lifting Index for the following task and determine if intervention is needed.

**Task Parameters:** - Horizontal distance (H): 35 cm - Vertical height (V): 80 cm - Travel distance (D): 50 cm - Asymmetry (A): 0° (symmetric lift) - Frequency: 4 lifts/minute for 1 hour - Coupling: Good (handles) - Actual load: 18 kg

**Solution:**

$$LC = 23 \text{ kg}$$

$$HM = 25/35 = 0.714$$

$$VM = 1 - 0.003|80-75| = 0.985$$

$$DM = 0.82 + 4.5/50 = 0.91$$

$$AM = 1 - 0.0032(0) = 1.0$$

$$FM = 0.75 \text{ (4/min, 1 hour, } V > 75\text{cm, from table)}$$

$$CM = 1.0 \text{ (good coupling)}$$

$$RWL = 23 \times 0.714 \times 0.985 \times 0.91 \times 1.0 \times 0.75 \times 1.0$$

$$RWL = 11.0 \text{ kg}$$

$$LI = 18 / 11.0 = 1.64$$

Risk Level: MODERATE ( $1.0 < LI < 2.0$ )

Action: Some workers may be at risk - consider changes

**Recommendations:** - Reduce horizontal distance (bring load closer) - Reduce lifting frequency - Use mechanical assist for loads >12 kg

Question 5: What are three advantages of AI-powered ergonomic assessment over traditional methods?

**1. Scale and Coverage:** - Traditional: 5-10 assessments per day by trained ergonomist - AI: Hundreds or thousands of assessments simultaneously - Can monitor entire facility continuously

**2. Consistency and Objectivity:** - Traditional: Inter-rater reliability issues; different assessors may score differently - AI: Consistent scoring algorithm applied uniformly - Removes subjective interpretation

**3. Real-time Feedback:** - Traditional: Assessment completed, report generated days/weeks later - AI: Immediate scoring and alerts - Can provide real-time coaching to workers

**Additional advantages:** - Lower cost per assessment - Non-disruptive (workers not aware of assessment) - Longitudinal tracking of posture patterns - Data for predictive analytics

## 7.14 Additional Resources

### 7.14.1 Videos

**Ergonomics Fundamentals:** - Introduction to Workplace Ergonomics - Safe Manual Handling Techniques

**Assessment Tools:** - REBA Assessment Tutorial - RULA Assessment Guide - NIOSH Lifting Equation Explained

**AI Ergonomics:** - TuMeke AI Ergonomics Demo - VelocityEHS Industrial Ergonomics

### 7.14.2 Online Tools

- ErgoPlus REBA Calculator
- NIOSH Lifting Equation Calculator
- Cornell University Ergonomics Web

## 7.15 References

- Berlin, C., & Adams, C. (2017). *Production Ergonomics: Designing Work Systems to Support Optimal Human Performance*. Ubiquity Press.
- Hignett, S., & McAtamney, L. (2000). Rapid Entire Body Assessment (REBA). *Applied Ergonomics*, 31(2), 201-205.
- McAtamney, L., & Corlett, E.N. (1993). RULA: A Survey Method for Investigation of Work-Related Upper Limb Disorders. *Applied Ergonomics*, 24(2), 91-99.
- NIOSH. (1994). *Applications Manual for the Revised NIOSH Lifting Equation*. DHHS (NIOSH) Publication No. 94-110.
- OSHA. (2000). *Ergonomics: The Study of Work*. OSHA 3125.

- Waters, T.R., Putz-Anderson, V., & Garg, A. (1993). Revised NIOSH Equation for the Design and Evaluation of Manual Lifting Tasks. *Ergonomics*, 36(7), 749-776.
  - ISO 11228-1:2021. Ergonomics — Manual handling — Part 1: Lifting, lowering and carrying.
  - ISO 11228-3:2007. Ergonomics — Manual handling — Part 3: Handling of low loads at high frequency.
-

## Chapter 8

# Industrial Robotics

---

### 8.1 Learning Objectives

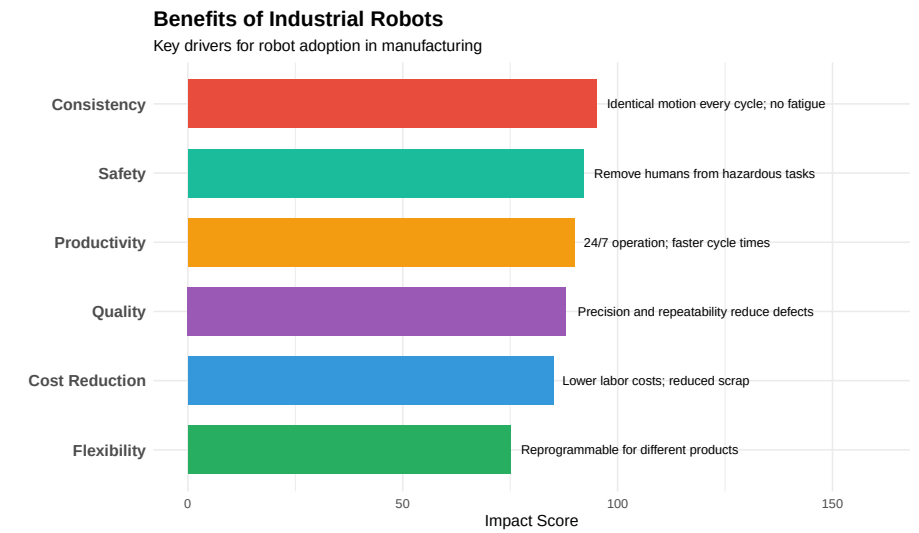
After completing this chapter, you will be able to:

1. Define industrial robots and explain their role in modern manufacturing
  2. Identify the main types of robot configurations and their applications
  3. Describe robot specifications including payload, reach, repeatability, and speed
  4. Explain the components of a robotic system (manipulator, controller, teach pendant, end effector)
  5. Understand robot coordinate systems and motion types
  6. Identify common robot applications in automotive, food, and defense industries
  7. Apply robot safety standards and safeguarding methods
  8. Describe collaborative robot (cobot) technology and applications
  9. Understand basic robot programming concepts
- 

### 8.2 Introduction to Industrial Robotics

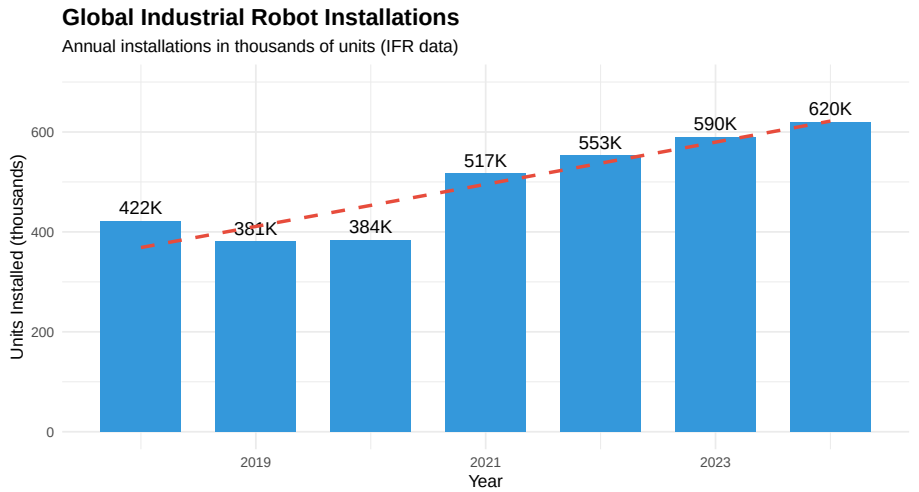
An **industrial robot** is defined by ISO 8373 as “an automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications.”

8.2.1 Why Use Robots?



8.2.2 Robot Market Growth

``geom_smooth()`` using formula = 'y ~ x'`



8.2.3 Industries Using Robots

Table 8.1: Industrial Robot Usage by Industry Sector

| Industry | Market Share (%) | Key Applications | Growth Rate (%) |
|----------|------------------|------------------|-----------------|
|----------|------------------|------------------|-----------------|

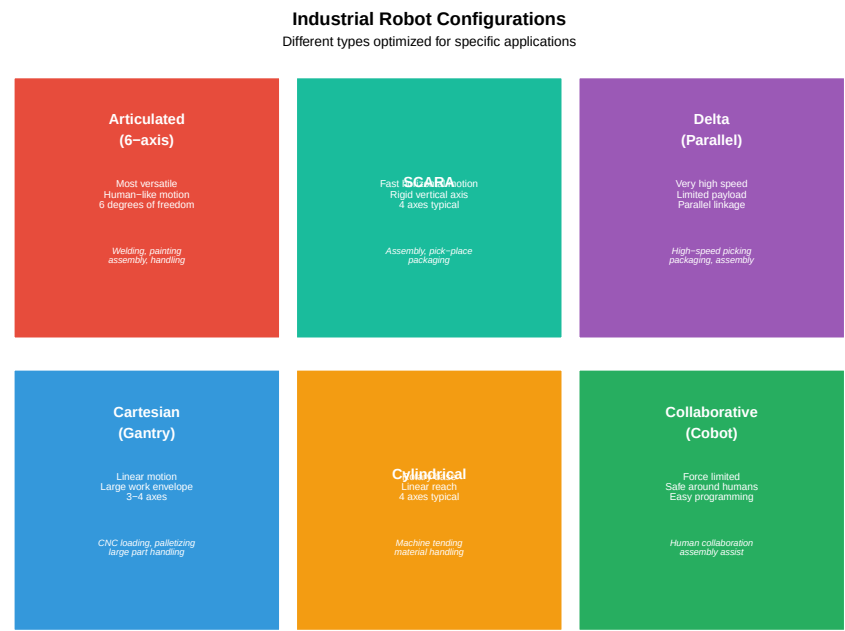


|                        |    |                                                  |          |
|------------------------|----|--------------------------------------------------|----------|
| Automotive             | 30 | Welding, painting, assembly, material handling   | Stable   |
| Electrical/Electronics | 25 | Assembly, testing, packaging, dispensing         | High     |
| Metal & Machinery      | 12 | Machine tending, welding, cutting                | Moderate |
| Food & Beverage        | 8  | Palletizing, packaging, pick-and-place           | High     |
| Plastics & Chemicals   | 5  | Injection molding, packaging, assembly           | Moderate |
| Aerospace/Defense      | 4  | Drilling, fastening, inspection, composite layup | High     |

8.3 Robot Configurations

Industrial robots come in various mechanical configurations, each suited to different applications.

8.3.1 Common Robot Types



8.3.2 Detailed Configuration Comparison

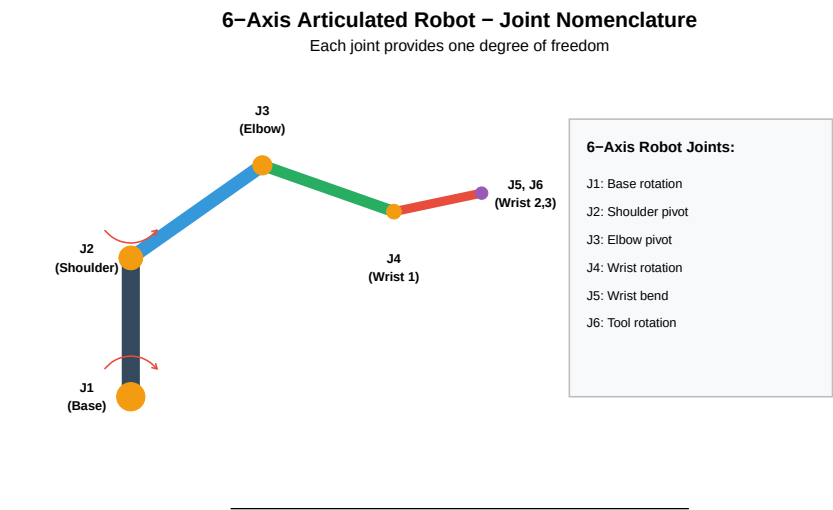
Table 8.2: Robot Configuration Comparison

| Configuration      | Axes | Payload Range | Reach   | Repeatability | Speed         | Cost          |
|--------------------|------|---------------|---------|---------------|---------------|---------------|
| 6-Axis Articulated | 6    | 3-2300 kg     | 0.5-4 m | ±0.02-0.1 mm  | Moderate-Fast | \$\$-\$\$\$\$ |

|                  |     |            |                 |                        |                |
|------------------|-----|------------|-----------------|------------------------|----------------|
| SCARA            | 4   | 1-20 kg    | 0.2-1.2 m       | $\pm 0.01$ - $0.05$ mm | Very Fast      |
| Delta/Parallel   | 3-4 | 0.5-12 kg  | 0.5-1.5 m       | $\pm 0.05$ - $0.1$ mm  | Extremely Fast |
| Cartesian/Gantry | 3-6 | 5-1000+ kg | Custom (meters) | $\pm 0.05$ - $0.5$ mm  | Moderate       |

8.3.3 Degrees of Freedom

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use `linewidth` instead.



8.4 Robot Specifications

Understanding robot specifications is essential for proper selection and application.

8.4.1 Key Specifications

Table 8.3: Key Robot Specifications

| Specification      | Definition                                            | Typical Values                            |
|--------------------|-------------------------------------------------------|-------------------------------------------|
| Payload            | Maximum mass the robot can handle at full speed       | 3-2300 kg (varies with speed)             |
| Reach              | Maximum distance from base to tool center point       | 500-4000 mm for articulated robots        |
| Repeatability      | Ability to return to same position repeatedly         | $\pm 0.01$ to $\pm 0.1$ mm                |
| Accuracy           | Ability to reach a commanded position exactly         | $\pm 0.1$ to $\pm 1.0$ mm                 |
| Maximum Speed      | Maximum velocity at tool center point or joint speeds | 1-12 m/s TCP; 100-300 deg/s joints        |
| Degrees of Freedom | Number of independent axes of motion                  | 4-7 axes                                  |
| Mounting           | How robot is installed (floor, ceiling, wall, angle)  | Floor (most common), Ceiling, Wall, Angle |

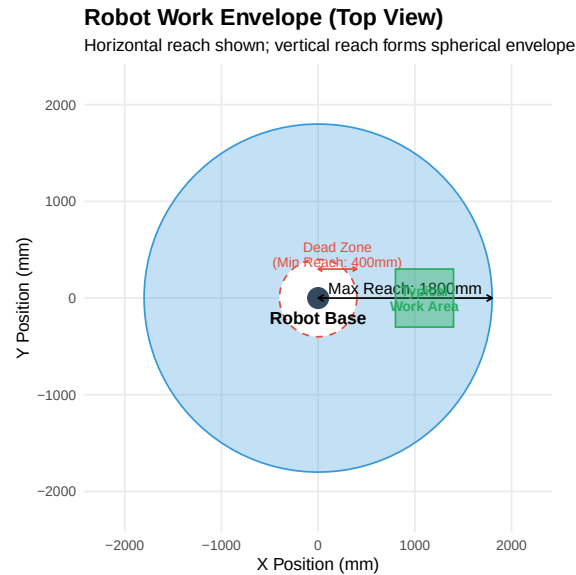
**IP Rating**

Ingress Protection rating for dust/water resistance

IP40 (standard) to IP67 (harsh)

**8.4.2 Work Envelope**

The **work envelope** (or workspace) is the three-dimensional space the robot can reach.

**8.4.3 Payload Considerations**

Live cell — try different payload, safety factor, or robot capacity values.

```
#| label: payload-calc
# Payload Calculation Example
robot_max_payload <- 10 # kg
gripper_weight <- 2.5 # kg
part_weight <- 5.0 # kg
safety_factor <- 1.2 # 20% safety margin

total_load <- gripper_weight + part_weight
required_capacity <- total_load * safety_factor
available_capacity <- robot_max_payload - gripper_weight

cat("Payload Analysis:\n")
cat("      \n")
cat("Robot maximum payload:", robot_max_payload, "kg\n")
cat("Gripper weight:", gripper_weight, "kg\n")
cat("Part weight:", part_weight, "kg\n")
```

```

cat("Total load:", total_load, "kg\n")
cat("With safety factor (", safety_factor, "x):", required_capacity, "kg\n")
cat("      \n")
cat("Available capacity for part:", available_capacity, "kg\n")

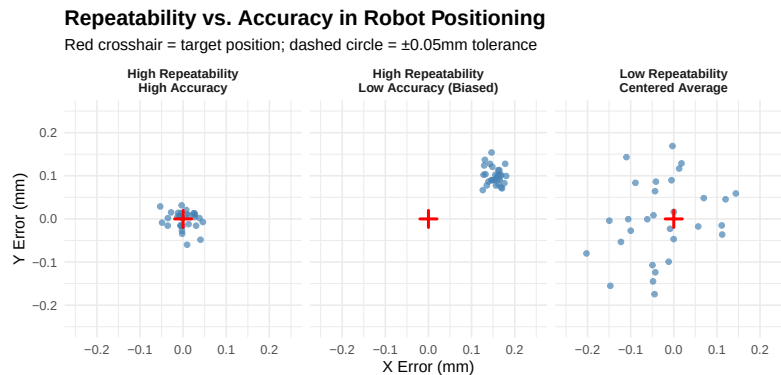
if(required_capacity <= robot_max_payload) {
  cat("Result: ACCEPTABLE - Robot can handle this load\n")
} else {
  cat("Result: EXCEEDED - Select larger robot or lighter gripper\n")
}

# Note about moment load
cat("\nNote: Also verify moment load (payload × distance from flange)\n")
cat("Moment at wrist:", part_weight, "kg ×", 0.3, "m =", part_weight * 0.3, "kg·m\n")

```

#### 8.4.4 Repeatability vs Accuracy

Warning in annotate("circle", x0 = 0, y0 = 0, r = 0.05, color = "red", linetype = "dashed"): Ignoring unknown parameters: `x0`, `y0`, and `r`



#### Understanding the Difference

**Repeatability:** - Ability to return to the SAME taught position - More important for most applications - Quoted spec (e.g.,  $\pm 0.05\text{mm}$ ) is typically 3 - Achieved through consistent mechanical design and servo control

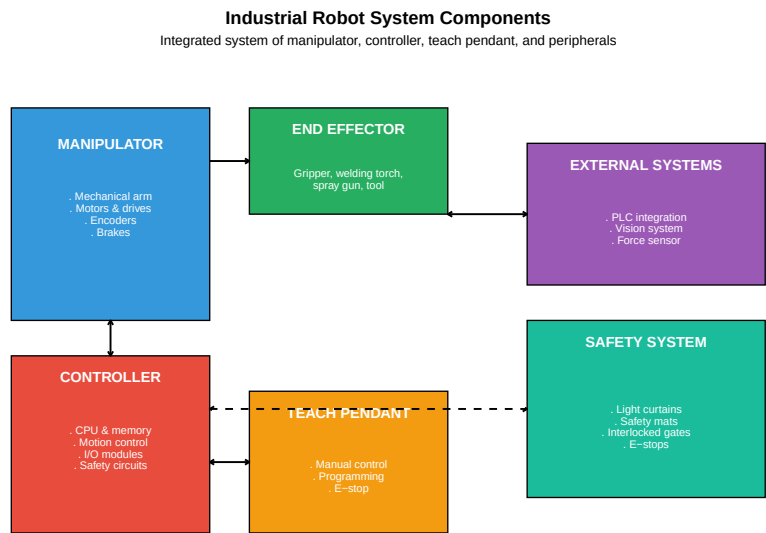
**Accuracy:** - Ability to reach a COMMANDED position - Important for offline programming - Affected by mechanical tolerances, deflection, calibration - Can be improved through calibration

**Why Repeatability Matters More:** In most applications, the robot is taught positions by jogging to them. As long as it returns to those positions consistently (repeatability), accuracy doesn't matter. Accuracy only matters when positions are calculated (offline programming) rather than taught.

## 8.5 Robot System Components

A complete robotic system consists of several integrated components.

### 8.5.1 System Architecture



### 8.5.2 End Effectors (EOAT)

**End of Arm Tooling (EOAT)** is the device attached to the robot's wrist to interact with parts.

Table 8.4: Common End Effector Types

| EOAT Type                  | Operating Principle                                   | Applications                      |
|----------------------------|-------------------------------------------------------|-----------------------------------|
| <b>Mechanical Gripper</b>  | Fingers actuated by pneumatic, electric, or hydraulic | General part handling; assembly   |
| <b>Vacuum Gripper</b>      | Suction cups powered by venturi or vacuum pump        | Flat surfaces; sheet material; bo |
| <b>Magnetic Gripper</b>    | Electromagnetic or permanent magnet                   | Ferrous metal parts; steel sheets |
| <b>Welding Torch</b>       | MIG, TIG, spot welding equipment                      | Automotive body, frames, comp     |
| <b>Spray Gun</b>           | Paint, adhesive, sealant dispensing                   | Automotive painting; sealing      |
| <b>Deburring Tool</b>      | Rotary tool for edge finishing                        | Casting, machining finishing      |
| <b>Force/Torque Sensor</b> | Measures forces and torques at tool                   | Assembly; polishing; insertion    |
| <b>Vision Camera</b>       | 2D or 3D imaging for guidance                         | Part location; inspection; guidan |

### 8.5.3 Gripper Selection Example

Live cell — try a different part mass, acceleration, or friction coefficient.

```
#| label: gripper-selection
# Gripper Force Calculation
part_mass <- 3.0          # kg
acceleration <- 15        # m/s2 (robot acceleration)
gravity <- 9.81           # m/s2
safety_factor <- 2.0     # Minimum safety factor
friction_coefficient <- 0.3 # Steel on rubber

# Calculate forces
weight_force <- part_mass * gravity
inertia_force <- part_mass * acceleration
total_force <- sqrt(weight_force^2 + inertia_force^2)

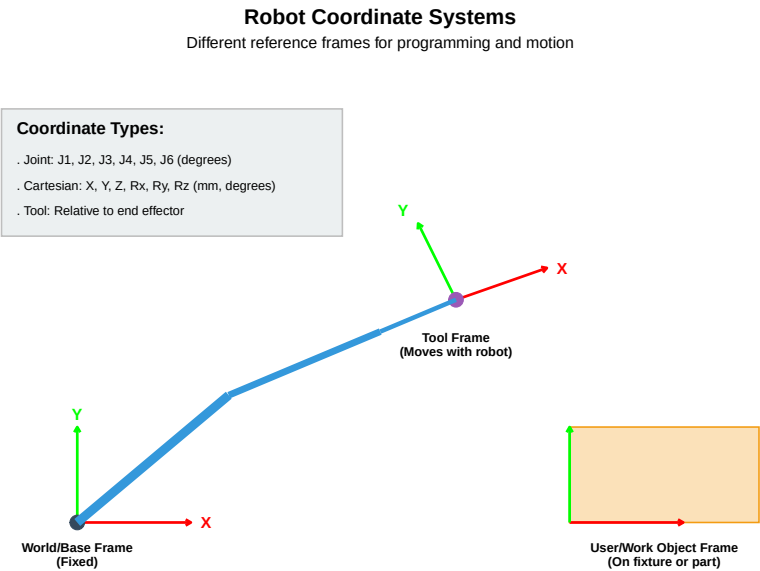
# Required grip force (friction gripper)
required_grip <- total_force / friction_coefficient * safety_factor

cat("Gripper Force Calculation:\n")
cat("      \n")
cat("Part mass:", part_mass, "kg\n")
cat("Weight force:", round(weight_force, 1), "N\n")
cat("Inertia force:", round(inertia_force, 1), "N\n")
cat("Combined force:", round(total_force, 1), "N\n")
cat("Friction coefficient:", friction_coefficient, "\n")
cat("Safety factor:", safety_factor, "\n")
cat("      \n")
cat("Required grip force:", round(required_grip, 1), "N\n")
cat("\nSelect gripper with grip force ", ceiling(required_grip/10)*10, "N\n")
```

## 8.6 Coordinate Systems and Motion

Understanding coordinate systems is essential for robot programming.

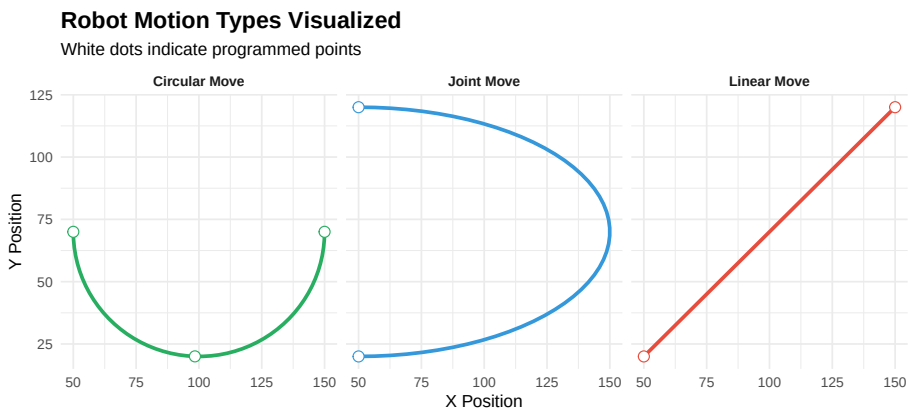
### 8.6.1 Coordinate Frames



### 8.6.2 Motion Types

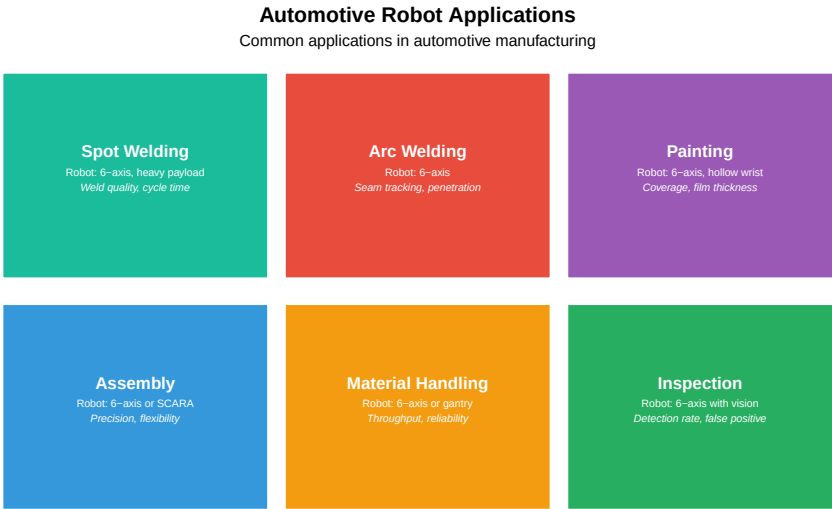
Table 8.5: Robot Motion Types

| Motion Type                  | Path Characteristic                                      | Speed Control               |
|------------------------------|----------------------------------------------------------|-----------------------------|
| <b>Joint Move (MoveJ)</b>    | Unpredictable path; each joint moves at constant speed   | Joint velocity (% or deg/s) |
| <b>Linear Move (MoveL)</b>   | Straight line from start to end; TCP follows linear path | TCP velocity (mm/s)         |
| <b>Circular Move (MoveC)</b> | Arc or circle; defined by start, via, and end points     | TCP velocity (mm/s)         |



## 8.7 Robot Applications

### 8.7.1 Automotive Applications



### 8.7.2 Food & Beverage Applications

Table 8.6: Food & Beverage Robot Applications

| Application        | Description                                  | Robot Type                |
|--------------------|----------------------------------------------|---------------------------|
| Palletizing        | Stack cases/bags onto pallets in patterns    | 4-axis palletizer, 6-axis |
| Case Packing       | Load products into cases/cartons             | Delta, SCARA, 6-axis      |
| Pick and Place     | Transfer products between conveyors/stations | Delta (high speed), SCARA |
| Cutting/Portioning | Protein portioning, cake cutting             | 6-axis with vision        |
| Decorating         | Applying icing, toppings, decorations        | 6-axis, delta             |
| Quality Inspection | Vision-based defect detection, grading       | 6-axis with vision        |

### 8.7.3 Aerospace/Defense Applications

Table 8.7: Aerospace/Defense Robot Applications

| Application          | Description                                      | Accuracy Required            |
|----------------------|--------------------------------------------------|------------------------------|
| Drilling & Fastening | Precision hole drilling; rivet/bolt installation | $\pm 0.05\text{mm}$ position |



|                   |                                                    |                         |          |
|-------------------|----------------------------------------------------|-------------------------|----------|
| Composite Layup   | Automated fiber placement (AFP); tape laying       | ±0.25mm fiber placement | Complex  |
| NDT/Inspection    | Ultrasonic, X-ray, visual inspection of structures | Full coverage           | Large pa |
| Surface Treatment | Painting, coating, surface prep for bonding        | Consistent thickness    | Large en |
| Assembly          | Large structure assembly; wing-to-fuselage         | ±0.5mm                  | Heavy p  |

8.8 Robot Safety

Robot safety is critical to protect workers from potential hazards.

8.8.1 Hazard Categories

Table 8.8: Robot Hazard Categories and Mitigations

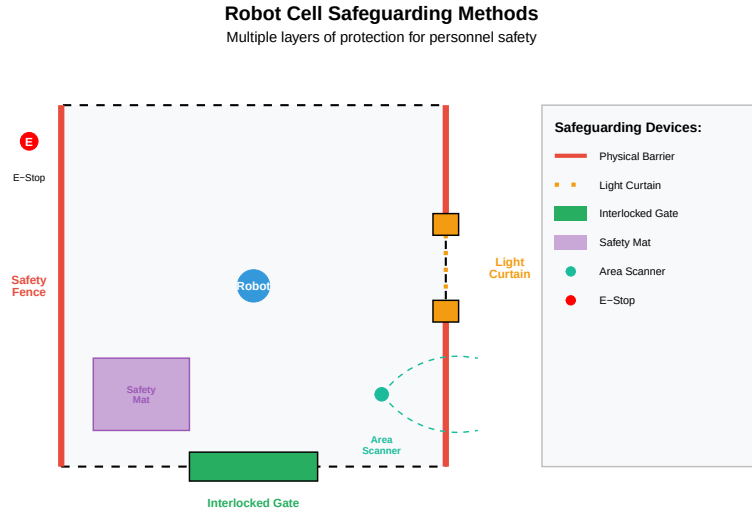
| Hazard   | Description                                            | Risk_Factors                                   |
|----------|--------------------------------------------------------|------------------------------------------------|
| Impact   | Robot strikes person with arm or tooling               | Speed, mass, sharp edges, unexpected motion    |
| Crushing | Person caught between robot and fixed object           | Limited escape space, high forces              |
| Trapping | Body part caught in articulating joints or mechanisms  | Pinch points at joints, insufficient clearance |
| Ejection | Part or tool flies off due to grip failure or breakage | Grip force, centrifugal force, tool condition  |
| Other    | Electrical, thermal, noise, radiation hazards          | Voltage, heat, welding arc, laser              |

8.8.2 Safety Standards

Table 8.9: Key Robot Safety Standards

| Standard          | Scope                                | Key Requirements                                       |
|-------------------|--------------------------------------|--------------------------------------------------------|
| ISO 10218-1:2011  | Robot manufacturer requirements      | Stop functions, speed limiting, singularity protection |
| ISO 10218-2:2011  | Robot system integrator requirements | Risk assessment, safeguarding, layout, validation      |
| ISO/TS 15066:2016 | Collaborative robot safety           | Force/pressure limits, speed/separation monitoring     |
| ANSI/RIA R15.06   | US equivalent to ISO 10218           | Safeguarding devices, risk assessment                  |
| ANSI/RIA R15.08   | Industrial mobile robots             | Navigation safety, personnel detection                 |

### 8.8.3 Safeguarding Methods



## 8.9 Collaborative Robots (Cobots)

**Collaborative robots** are designed to work safely alongside humans without traditional guarding.

### 8.9.1 Cobot Characteristics

Table 8.10: Traditional Robot vs. Collaborative Robot Comparison

|                        | Traditional Industrial Robot      | Collaborative Robot                  |
|------------------------|-----------------------------------|--------------------------------------|
| <b>Safety Approach</b> | Safeguarded cell; humans excluded | Inherently safe; force/speed limited |
| <b>Payload</b>         | Up to 2300 kg                     | Typically 3-35 kg                    |
| <b>Speed</b>           | Very high ( $>2$ m/s)             | Limited ( $<1.5$ m/s typically)      |
| <b>Reach</b>           | Up to 4 m                         | 500-1300 mm typical                  |
| <b>Programming</b>     | Specialized programmers           | Intuitive; hand guiding              |
| <b>Guarding</b>        | Required (fences, curtains)       | Often not required                   |
| <b>Investment</b>      | High (\$50K-500K+ with cell)      | Lower (\$25K-75K typical)            |
| <b>Applications</b>    | High-volume, dedicated tasks      | Flexible, shared workspace           |

### 8.9.2 Collaborative Operation Modes (ISO/TS 15066)

Table 8.11: ISO/TS 15066 Collabora

| Collaboration Mode                     | Description                                                          |
|----------------------------------------|----------------------------------------------------------------------|
| <b>Safety-Rated Monitored Stop</b>     | Robot stops when human enters collaborative zone; resumes when clear |
| <b>Hand Guiding</b>                    | Operator physically guides robot; robot follows input forces         |
| <b>Speed and Separation Monitoring</b> | Robot adjusts speed based on distance to human; stops if too close   |
| <b>Power and Force Limiting</b>        | Robot limits force/pressure on contact to safe levels                |

8.9.3 Force and Pressure Limits

ISO/TS 15066 specifies maximum contact forces and pressures for different body areas:

Table 8.12: ISO/TS 15066 Biomechanical Limits (Selected Values)

| Body Area                | Max Pressure (N/cm <sup>2</sup> ) | Quasi-Static Pressure | Max Force (N) | Quasi-Static Force |
|--------------------------|-----------------------------------|-----------------------|---------------|--------------------|
| <b>Skull/Forehead</b>    | 130                               | 130                   | 130           | 130                |
| <b>Face</b>              | 65                                | 65                    | 65            | 65                 |
| <b>Neck (front/back)</b> | 145                               | 145                   | 150           | 150                |
| <b>Chest</b>             | 140                               | 140                   | 140           | 140                |
| <b>Abdomen</b>           | 110                               | 110                   | 110           | 110                |
| <b>Hand/Finger</b>       | 280                               | 280                   | 140           | 140                |
| <b>Arm</b>               | 190                               | 190                   | 150           | 150                |
| <b>Leg</b>               | 220                               | 220                   | 220           | 220                |

8.10 Robot Programming Basics

8.10.1 Programming Methods

Table 8.13: Robot Programmi

| Method                           | Description                                                     | Advantages    |
|----------------------------------|-----------------------------------------------------------------|---------------|
| <b>Online/Teach Pendant</b>      | Use teach pendant to jog robot to positions and record          | Direct; prec  |
| <b>Lead-Through/Hand Guiding</b> | Physically move robot arm to desired positions                  | Intuitive; fa |
| <b>Offline Programming</b>       | Create programs on PC without robot; download later             | No producti   |
| <b>Simulation-Based</b>          | Program and test in virtual environment; transfer to real robot | Risk-free tes |

8.10.2 Basic Program Structure

Example Robot Program (ABB RAPID-style pseudocode):

```

PROGRAM Main()

    ! Initialize
    HomePosition()
    GripperOpen()

    ! Main cycle
    WHILE running DO

        ! Move to pick position
        MoveJ pApproach, v500, z50, tool1
        MoveL pPick, v100, fine, tool1

        ! Pick part
        GripperClose()
        WaitTime 0.2

        ! Move to place position
        MoveL pApproach, v200, z50, tool1
        MoveJ pPlaceApproach, v500, z50, tool1
        MoveL pPlace, v100, fine, tool1

        ! Place part
        GripperOpen()
        WaitTime 0.2

        ! Return
        MoveL pPlaceApproach, v200, z50, tool1

        ! Increment counter
        PartCount := PartCount + 1

    ENDWHILE

ENDPROGRAM

```

### 8.10.3 Key Programming Concepts

Table 8.14: Key Robot Programming Concepts

| Concept                   | Description                                                   | Example       |
|---------------------------|---------------------------------------------------------------|---------------|
| <b>Position/Target</b>    | Stored robot position (joint angles or Cartesian coordinates) | pHome, pPick  |
| <b>Motion Instruction</b> | Command to move robot (MoveJ, MoveL, MoveC)                   | MoveL pTarget |

|                     |                                                               |                        |
|---------------------|---------------------------------------------------------------|------------------------|
| <b>Speed Data</b>   | Velocity parameter (mm/s for TCP, % for joints)               | v100 = 100 mm/s, v50   |
| <b>Zone Data</b>    | Corner path blending (fine = stop at point, z10 = 10mm blend) | fine = stop, z5 = 5mm  |
| <b>Tool Data</b>    | Definition of tool center point relative to flange            | tool1 with TCP offset  |
| <b>Work Object</b>  | Definition of work coordinate system for part positions       | wobj_fixture with base |
| <b>I/O Commands</b> | Control external devices (SetDO, WaitDI, etc.)                | SetDO doGripper, 1; V  |

---

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## 8.11 Video Resources

### 8.11.1 Introduction to Industrial Robots

### 8.11.2 Collaborative Robots Explained

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## 8.12 Summary

Industrial robotics is a fundamental technology in modern manufacturing:

1. **Robot types** include articulated (most versatile), SCARA (fast assembly), delta (high-speed picking), and cartesian (large work areas)
  2. **Key specifications** include payload, reach, repeatability, speed, and degrees of freedom
  3. **System components** work together: manipulator, controller, teach pendant, end effector, and safety system
  4. **Coordinate systems** (world, tool, user) enable flexible programming and positioning
  5. **Applications** span automotive, food & beverage, and aerospace with industry-specific requirements
  6. **Safety** requires comprehensive risk assessment and appropriate safeguarding
  7. **Collaborative robots** enable human-robot cooperation with force/speed limiting
  8. **Programming** can be online (teach pendant), lead-through, or offline
- 

## 8.13 Review Questions

**Question 1:** Compare articulated (6-axis) robots, SCARA robots, and delta robots. For each, describe the typical applications and key advantages.

**Answer:**

| Robot Type                | Configuration                     | Key Advantages                                                      | Typical Applications                                            |
|---------------------------|-----------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>6-Axis Articulated</b> | 6 revolute joints; human-like arm | Maximum flexibility; can reach any orientation; large payload range | Welding, painting, assembly, material handling, machine tending |
| <b>SCARA</b>              | 4 axes; selective compliance      | Very fast horizontal motion; rigid vertical; compact; precise       | Pick-and-place, assembly, screw driving, packaging              |
| <b>Delta (Parallel)</b>   | 3-4 axes; parallel linkage        | Extremely fast (150+ picks/min); low moving mass                    | High-speed picking, packaging, sorting, light assembly          |

#### Detailed Comparison:

**6-Axis Articulated:** - Most versatile - can position tool at any angle - Wide payload range (3 kg to 2300+ kg) - Can reach around obstacles - Complex programming possible - *Best for:* Tasks requiring complex orientations, welding seams, painting

**SCARA:** - Inherently rigid in Z (vertical) - good for insertion - Very fast in X-Y plane - Lower cost than 6-axis - Compact footprint - *Best for:* Assembly operations, circuit board population, packaging

**Delta:** - Highest speed of any configuration - Low inertia (motors at base, not moving) - Limited payload (typically <15 kg) - Requires overhead mounting - *Best for:* High-speed sorting, packaging lines, food handling

**Question 2:** A robot has a maximum payload of 15 kg. The gripper weighs 4 kg and the part weighs 8 kg. Is this robot suitable? What other factors should be considered?

**Answer:**

#### Initial Assessment:

Total load = Gripper weight + Part weight

Total load = 4 kg + 8 kg = 12 kg

Robot capacity: 15 kg

Load: 12 kg

Remaining margin: 3 kg (20%)

#### Consideration with Safety Factor:

Typical safety factor: 1.2 - 1.5

Required capacity = 12 kg  $\times$  1.25 = 15 kg

This equals the robot's maximum capacity - marginal!

#### Additional Factors to Consider:

##### 1. Moment Load (Torque at Wrist)

- Distance from flange to center of gravity matters
- If part is held 300mm from flange:  $\text{Moment} = 8 \text{ kg} \times 0.3 \text{ m} = 2.4 \text{ kg} \cdot \text{m}$
- Robot specs include moment limits - must verify

##### 2. Inertia

- Fast motion requires acceleration
- Higher mass + longer distance = higher inertia
- May need to derate payload for high-speed applications

##### 3. Payload Derating with Reach

- Many robots derate payload at extended reach
- 15 kg at 500mm may become 10 kg at 1500mm
- Check payload diagram in robot specifications

##### 4. Orientation

- Payload may vary with wrist orientation
- Horizontal reach vs. vertical lift differ

##### 5. Acceleration Requirements

- Higher acceleration = higher effective load
- May need to reduce speed/acceleration

**Recommendation:** The 15 kg robot is marginal for this application. Consider:  
 - Select next size up (20 kg) for adequate margin - Use lighter gripper (3 kg or less)  
 - Reduce part handling distance from flange - Limit acceleration/speed if using this robot

**Question 3:** Explain the difference between repeatability and accuracy. Why is repeatability often more important than accuracy for robot applications?

**Answer:**

#### Definitions:

**Repeatability:** - Ability to return to the **same taught position** multiple times - Measures consistency/precision of motion - Quoted as  $\pm$  value (e.g.,  $\pm 0.05 \text{ mm}$  at 3 ) - Affected by: mechanical wear, servo control, thermal effects

**Accuracy:** - Ability to reach a **commanded position** exactly - Measures how close actual position is to theoretical position - Typically worse than repeatability (e.g.,  $\pm 0.5 \text{ mm}$ ) - Affected by: manufacturing tolerances, calibration, deflection, backlash

#### Why Repeatability is Usually More Important:

##### 1. Teaching Method

- Most robots are programmed by teaching (jogging to positions)

- Robot doesn't need to know where it "should" be - just return to where it was taught
- As long as it returns consistently (repeatability), accuracy doesn't matter

## 2. Process Requirements

- Welding: Need to follow same seam every time → repeatability
- Assembly: Need to insert part in same hole every time → repeatability
- Pick-place: Need to pick from same fixture position → repeatability

## 3. Compensation Possible

- Poor accuracy can be compensated by adjusting taught points
- Poor repeatability cannot be compensated

### When Accuracy Matters:

#### 1. Offline Programming

- Positions calculated from CAD data, not taught
- Robot must go where commanded, first time
- Common in aerospace (large parts, tight access)

#### 2. Multi-Robot Systems

- Robots must agree on positions
- Calibration to common frame requires accuracy

#### 3. Frequent Program Changes

- If programs are frequently changed without re-teaching
- Offline modifications need accurate execution

**Bottom Line:** For most industrial applications,  $\pm 0.05$  mm repeatability is far more valuable than  $\pm 0.5$  mm accuracy because taught positions are used.

**Question 4:** Describe the four collaborative operation modes defined in ISO/TS 15066. Give an example application for each.

**Answer:**

#### 1. Safety-Rated Monitored Stop

*Description:* Robot operates at normal speed when workspace is clear. When a person enters the collaborative zone, the robot stops and holds position. Robot resumes automatically when person exits.

*Technical Requirements:* - Safety-rated sensors (light curtains, scanners) - Safety-rated monitored stop function - Clearly defined collaborative zone

*Example Application:* Machine tending cell where operator occasionally loads raw material. Robot works at full speed during machining cycle but stops safely when operator enters to load/unload.

---

#### 2. Hand Guiding



*Description:* Operator physically contacts the robot through a hand guiding device and manually moves it. Robot follows operator's force input.

*Technical Requirements:* - Hand guiding device (force/torque sensing) - Emergency stop accessible at guiding device - Speed reduction while guiding - Safe torque control

*Example Application:* Teaching a paint path on an aerospace component. Operator guides robot along surface contour while robot records positions. Also used for "collaborative finishing" where operator guides robot holding a polishing tool.

### 3. Speed and Separation Monitoring

*Description:* Robot continuously monitors distance to the nearest person. Speed adjusts based on separation: full speed when far, reduced speed when closer, stop if minimum distance breached.

*Technical Requirements:* - Safety-rated distance monitoring (usually area scanners) - Safety-rated speed control - Multiple speed/distance zones defined - Real-time separation calculation

*Example Application:* Shared packaging area where robot palletizes cases while operator moves around the cell. Robot slows when operator approaches, stops if too close, and resumes at appropriate speed as operator moves away.

### 4. Power and Force Limiting

*Description:* Robot is designed so that forces and pressures from any contact are below injury thresholds. Contact is allowed because it cannot cause harm.

*Technical Requirements:* - Inherently safe design (compliant joints, low mass, rounded surfaces) - Force/torque sensing or current monitoring - Verified force limits per ISO/TS 15066 body area tables - May include padding/soft covers

*Example Application:* Assembly assist where robot holds a heavy part while human worker installs fasteners. Robot and human are in direct contact, with robot providing position support and human providing dexterity. Force is limited so accidental contact causes no injury.

**Question 5:** Calculate the required gripper force for a robot picking a 2 kg part with a friction coefficient of 0.4. The robot accelerates at 10 m/s<sup>2</sup>. Use a safety factor of 2.0.

**Answer:** (live — change the numbers to check your own version)

```
#| label: q5-answer
# Given values
mass <- 2.0           # kg
```

```

acceleration <- 10      # m/s2
gravity <- 9.81         # m/s2
friction_coef <- 0.4    # Coefficient of friction
safety_factor <- 2.0    # Safety factor

# Calculate forces
# Weight force (vertical)
F_gravity <- mass * gravity

# Inertia force (from acceleration)
F_inertia <- mass * acceleration

# Combined force (vector sum for worst case)
# Worst case: acceleration horizontal while holding against gravity
F_combined <- sqrt(F_gravity2 + F_inertia2)

# Required friction force to hold part
# F_friction =      × F_normal
# F_normal = Grip force
# F_friction must overcome F_combined

# Without safety factor:
F_grip_min <- F_combined / friction_coef

# With safety factor:
F_grip_required <- F_grip_min * safety_factor

cat("Gripper Force Calculation:\n")
cat("          \n\n")
cat("Input Parameters:\n")
cat("  Part mass:", mass, "kg\n")
cat("  Robot acceleration:", acceleration, "m/s2\n")
cat("  Friction coefficient:", friction_coef, "\n")
cat("  Safety factor:", safety_factor, "\n\n")

cat("Force Analysis:\n")
cat("  Gravity force: Fg = m × g =", mass, "×", gravity, "=",
    round(F_gravity, 1), "N\n")
cat("  Inertia force: Fi = m × a =", mass, "×", acceleration, "=",
    round(F_inertia, 1), "N\n")
cat("  Combined force: Fc = √(Fg2 + Fi2) =", round(F_combined, 1), "N\n\n")

cat("Grip Force Calculation:\n")
cat("  F_friction =      × F_grip  F_combined\n")
cat("  F_grip  F_combined /      =", round(F_combined, 1), "/", friction_coef,

```

```

    "=", round(F_grip_min, 1), "N\n")
cat("  With safety factor:", round(F_grip_min, 1), "x", safety_factor,
    "=", round(F_grip_required, 1), "N\n\n")

cat("                                \n")
cat("REQUIRED GRIPPER FORCE:", ceiling(F_grip_required), "N minimum\n")
cat("Select gripper rated for ", ceiling(F_grip_required/10)*10, "N\n")

```

**Question 6:** List and describe five safety devices used to protect personnel around industrial robots.

**Answer:**

### 1. Physical Barriers (Safety Fencing)

*Description:* Fixed guards or fencing that physically prevent access to the robot work envelope.

*Characteristics:* - Rigid construction (steel, aluminum, polycarbonate) - Must withstand expected impact forces - Minimum height (typically 1.8 m) - Properly anchored to floor - May include interlocked gates for authorized entry

*Application:* Primary safeguard for most traditional robot cells.

### 2. Light Curtains (AOPD - Active Opto-electronic Protective Device)

*Description:* Transmitter and receiver create a sensing field of infrared beams. Breaking any beam triggers a stop.

*Characteristics:* - Type 2 or Type 4 (safety-rated) - Resolution (14mm for finger, 40mm for hand/arm) - Muting capability for material transfer - Very fast response (<20ms)

*Application:* Access points where frequent entry needed; allows material flow while protecting personnel.

### 3. Safety Mats (Pressure-Sensitive Mats)

*Description:* Floor mats that detect weight/pressure and trigger stop when stepped on.

*Characteristics:* - Placed in hazardous zones - Withstand industrial environment (oil, debris) - Fast response - Self-monitoring for failures

*Application:* Areas where overhead detection isn't practical; around robot bases.

### 4. Area Scanners (Safety Laser Scanners)

*Description:* Laser scanner detects objects/personnel entering defined zones. Multiple warning and stop zones configurable.

*Characteristics:* - 270° field of view typical - Multiple programmable zones - Warning zone (slow robot) and stop zone (stop robot) - Type 3 safety rating - Detects at floor level

*Application:* Open cells requiring flexibility; mobile robot protection; speed/separation monitoring.

### 5. Interlocked Gates/Doors

*Description:* Access gates with safety switches that stop robot when opened.

*Characteristics:* - Mechanical or magnetic interlock switches - Trapped key systems for lockout - May include guard locking (prevent opening while robot moving) - Escape release from inside - Meets ISO 14119

*Application:* Authorized entry points for maintenance, setup, troubleshooting.

### Additional Devices:

- **E-Stop (Emergency Stop):** Palm-operated buttons to immediately stop robot
- **Two-Hand Controls:** Require both hands on buttons to initiate motion
- **Enabling Devices:** 3-position switches held by personnel in restricted area
- **Presence-Sensing Devices:** Capacitive, radar, or vision-based detection

**Question 7:** Explain the difference between MoveJ (Joint Move) and MoveL (Linear Move). When would you use each?

**Answer:**

### MoveJ (Joint Move)

*How It Works:* - Each joint moves from start to end angle independently - All joints start and stop together (coordinated) - Path of TCP (Tool Center Point) is unpredictable - curved - Robot calculates joint interpolation, not TCP path

*Characteristics:* - Fastest point-to-point motion - No singularity issues during move - Path may vary depending on starting configuration - Speed specified in % of max or joint velocity (°/s)

*When to Use:* - Moving between distant points through free space - Approach movements before precise positioning - Home position moves - Any time when path doesn't matter, only endpoints - Escaping from near-singularity positions

**MoveL (Linear Move)**

*How It Works:* - TCP follows a straight line from start to end - Robot calculates joint positions to maintain linear TCP path - Orientation is interpolated linearly as well - All axes coordinated to produce straight line

*Characteristics:* - Guaranteed straight path - Consistent, predictable motion - Slower than MoveJ for same endpoint distance - Can encounter singularities during move - Speed specified in mm/s at TCP

*When to Use:* - Process paths (welding, sealing, cutting) - Insertion/extraction operations - Approach to pick/place positions - Any time path must be straight - When moving close to obstacles

---

**Practical Example:**

Imagine picking a part from a fixture:

! Move from home to near pick position - path doesn't matter

MoveJ pApproach, v1000, z50, tool1 ! Joint move - fast

! Move down to pick - must be straight to avoid collision

MoveL pPick, v100, fine, tool1 ! Linear move - controlled

! Pick part, then retract straight up

GripperClose()

MoveL pApproach, v200, z10, tool1 ! Linear move - straight up

! Move to place area - through open space

MoveJ pPlaceApproach, v1000, z50, tool1 ! Joint move - fast

**Key Decision:** - If path matters → MoveL - If only destination matters → MoveJ

**Question 8:** A manufacturing engineer needs to select a robot for a palletizing application. The requirements are: 25 kg payload, 2.2 m horizontal reach, 10 cycles per minute, floor-mounted. Recommend a robot type and justify your selection.

**Answer:**

**Application Analysis:**

| Requirement | Value  | Implication                                  |
|-------------|--------|----------------------------------------------|
| Payload     | 25 kg  | Medium-heavy; rules out delta and most SCARA |
| Reach       | 2.2 m  | Large work envelope needed                   |
| Cycle rate  | 10/min | 6 seconds/cycle; moderate speed              |

| Requirement | Value       | Implication                                   |
|-------------|-------------|-----------------------------------------------|
| Mounting    | Floor       | Typical industrial installation               |
| Motion type | Palletizing | Mainly vertical and horizontal; some rotation |

### Robot Type Evaluation:

**1. 4-Axis Palletizing Robot (RECOMMENDED)** - Specifically designed for palletizing - 4 axes: base rotation, arm, elbow, wrist - High payload capability (25-700+ kg available) - Long reach available (up to 3.2m) - Fast for palletizing motions - Lower cost than 6-axis - Examples: FANUC M-410iC, KUKA KR QUANTEC PA, ABB IRB 460

*Verdict: Excellent match for this application*

**2. 6-Axis Articulated Robot** - Maximum flexibility - Can handle complex pallet patterns - Higher cost than 4-axis for same payload - May be over-specified for simple palletizing - Examples: FANUC M-710iC/50, ABB IRB 6700

*Verdict: Capable but potentially over-specified and more expensive*

**3. Cartesian/Gantry Robot** - Can cover very large areas - Simple motion control - Higher installation complexity - Less common for palletizing applications

*Verdict: Possible but not ideal for this payload/cycle rate*

### Recommended Selection:

**FANUC M-410iC/185** or similar 4-axis palletizer - Payload: 185 kg (well above 25 kg requirement with margin for gripper) - Reach: 3.143 m (exceeds 2.2 m requirement) - Repeatability:  $\pm 0.5$  mm - Axes: 4

**Justification:** 1. **Purpose-built:** Designed specifically for palletizing 2. **Adequate payload:** 185 kg allows for heavy gripper + multiple parts 3. **Reach:** 3.1 m provides margin for future needs 4. **Speed:** Optimized kinematics for palletizing motion profile 5. **Reliability:** Proven in thousands of palletizing applications 6. **Cost:** Lower than equivalent 6-axis robot 7. **Simplicity:** 4 axes easier to program and maintain than 6

**Additional Recommendations:** - Vacuum or mechanical gripper depending on product - Vision system for product detection if variety exists - Safety scanner for operator access during pallet changes - Interface to conveyor for product delivery

## 8.14 References

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6. Niku, S.B. (2020). *Introduction to Robotics: Analysis, Control, Applications* (3rd ed.). Wiley.
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9. ABB Robotics. (2023). *Technical Reference Manual - RAPID Instructions*. ABB.
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## Chapter 9

# Industrial Automation Fundamentals

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### 9.1 Learning Objectives

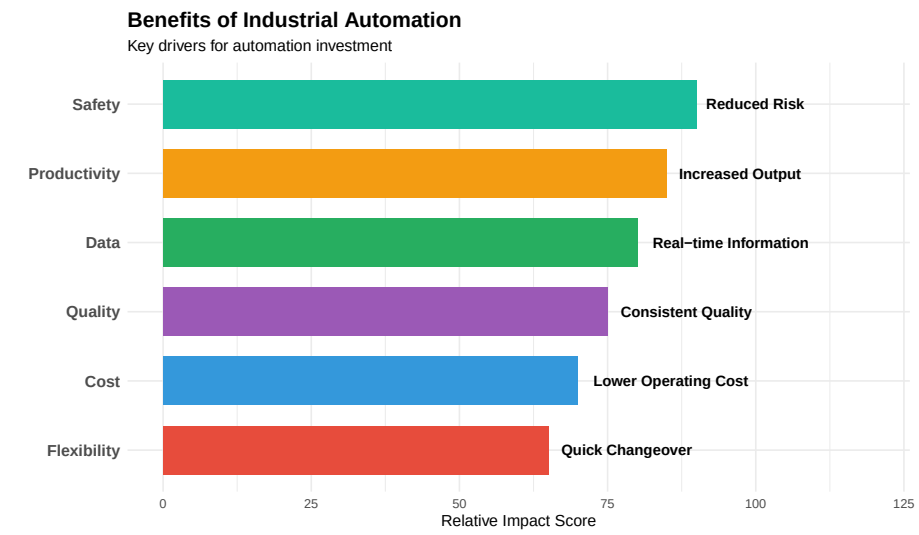
After completing this chapter, you will be able to:

1. Define industrial automation and explain its role in modern manufacturing
  2. Distinguish between fixed, programmable, and flexible automation
  3. Describe the architecture and function of Programmable Logic Controllers (PLCs)
  4. Identify common industrial sensors and their applications
  5. Explain motor control fundamentals including VFDs and servo systems
  6. Understand industrial communication protocols and networking
  7. Describe the role of HMI and SCADA in process monitoring
  8. Apply safety principles to automated systems
  9. Explain Industry 4.0 concepts and the Industrial Internet of Things (IIoT)
- 

### 9.2 Introduction to Industrial Automation

**Industrial automation** is the use of control systems, machinery, and information technologies to handle processes and machinery in an industry, replacing human intervention where possible to increase efficiency, quality, and safety.

9.2.1 Why Automate?



9.2.2 Automation in Key Industries

Table 9.1: Automation Applications by Industry

| Industry          | Application                  | Technology                | Benefit       |
|-------------------|------------------------------|---------------------------|---------------|
| Automotive        |                              |                           |               |
| Automotive        | Robotic welding and assembly | 6-axis robots, PLCs       | Consistent w  |
| Automotive        | Automated paint systems      | Conveyors, spray robots   | Uniform coat  |
| Automotive        | Vision inspection            | Machine vision, AI        | 100% inspect  |
| Food & Beverage   |                              |                           |               |
| Food & Beverage   | Filling and packaging lines  | Servo drives, sensors     | High speed,   |
| Food & Beverage   | Pasteurization control       | PLCs, temperature control | Food safety,  |
| Food & Beverage   | Sorting and grading          | Vision systems, conveyors | Quality sorti |
| Aerospace/Defense |                              |                           |               |
| Aerospace/Defense | CNC precision machining      | CNC, CMM integration      | Tight toleran |
| Aerospace/Defense | Automated testing systems    | Automated test equipment  | Comprehensi   |
| Aerospace/Defense | Clean room automation        | Robotics, HEPA systems    | Contaminati   |

9.3 Types of Automation

Industrial automation systems can be classified based on their flexibility and programming capability.

9.3.1 Automation Classification

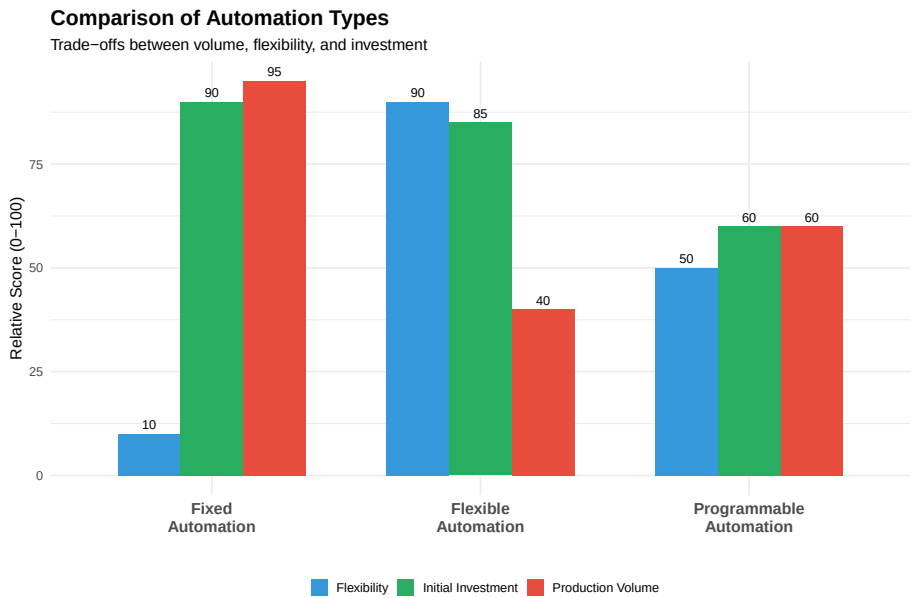
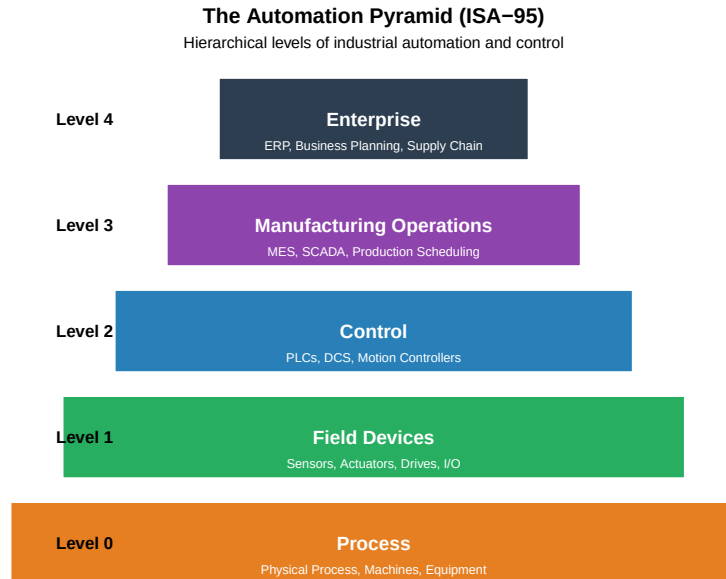


Table 9.2: Comparison of Automation Types

|                    | Fixed Automation                                   | Programmable Automation                              |
|--------------------|----------------------------------------------------|------------------------------------------------------|
| Definition         | Hard-wired, dedicated equipment for single product | Equipment can be reprogrammed for different products |
| Product Variety    | Single product or very similar variants            | Batches of different products                        |
| Production Volume  | Very high (millions of units)                      | Medium to high                                       |
| Changeover         | Difficult, expensive, time-consuming               | Requires reprogramming and setup                     |
| Initial Investment | Very high                                          | High                                                 |
| Unit Cost          | Very low per unit                                  | Medium                                               |
| Typical Equipment  | Transfer lines, dedicated assembly machines        | CNC machines, PLCs, industrial robots                |
| Best For           | Automotive components, fasteners, bottles          | Batch manufacturing, job shops                       |

### 9.3.2 The Automation Pyramid



#### Understanding the Pyramid Levels

**Level 0 - Process:** The actual physical equipment, machines, and processes being controlled. This includes conveyors, motors, valves, tanks, and the product being manufactured.

**Level 1 - Field Devices:** Sensors that measure process variables (temperature, pressure, flow, position) and actuators that affect the process (motors, valves, solenoids, drives).

**Level 2 - Control:** The “brain” of automation - PLCs, DCS, and motion controllers that execute control logic, process sensor inputs, and command actuators. Also includes HMI for operator interaction.

**Level 3 - Manufacturing Operations:** MES (Manufacturing Execution Systems), SCADA, production scheduling, quality management, and maintenance management. Bridges plant floor and business systems.

**Level 4 - Enterprise:** ERP systems, business intelligence, supply chain management, financial systems. Makes business decisions based on plant floor data.

**Key Principle:** Data flows up (process information), commands flow down (control directives).

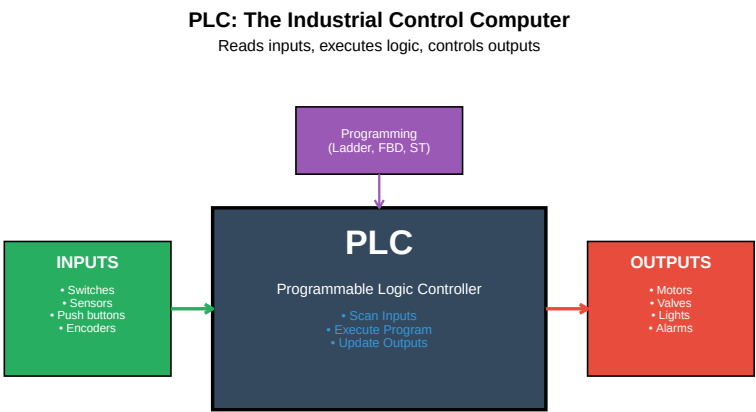
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## 9.4 Programmable Logic Controllers (PLCs)

The **PLC (Programmable Logic Controller)** is the workhorse of industrial automation, providing reliable, real-time control of machines and processes.

### 9.4.1 What is a PLC?

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use `linewidth` instead.



### 9.4.2 PLC Hardware Components

Table 9.3: PLC Hardware Components

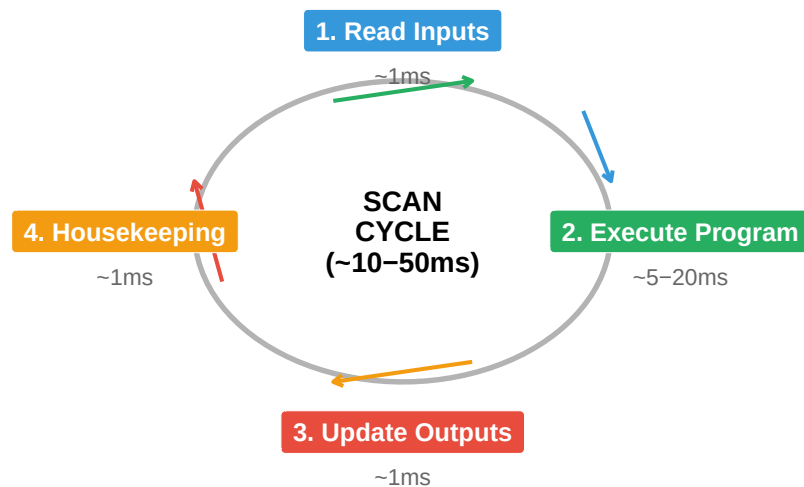
| Component                    | Function                                                                           |
|------------------------------|------------------------------------------------------------------------------------|
| <b>CPU (Processor)</b>       | Executes the control program, manages memory, coordinates all modules              |
| <b>Power Supply</b>          | Converts AC power to DC voltages required by PLC components                        |
| <b>Input Modules</b>         | Interface field devices (sensors, switches) to CPU; converts signals to digital    |
| <b>Output Modules</b>        | Interface CPU to field devices (motors, valves); converts digital to power signals |
| <b>Communication Modules</b> | Enable networking: Ethernet/IP, Profinet, Modbus, DeviceNet                        |
| <b>Programming Device</b>    | Laptop/PC with programming software for creating and downloading programs          |

### 9.4.3 The PLC Scan Cycle

PLCs operate in a continuous scan cycle:

## PLC Scan Cycle

Continuous loop executing at predictable intervals



Live cell — try different scan time components.

```
#| label: scan-time-calc
# Scan Time Calculation Example
input_scan_time <- 1.2      # ms
program_execution <- 15.5   # ms (depends on program size)
output_update <- 1.0        # ms
housekeeping <- 2.3         # ms

total_scan_time <- input_scan_time + program_execution + output_update + housekeeping

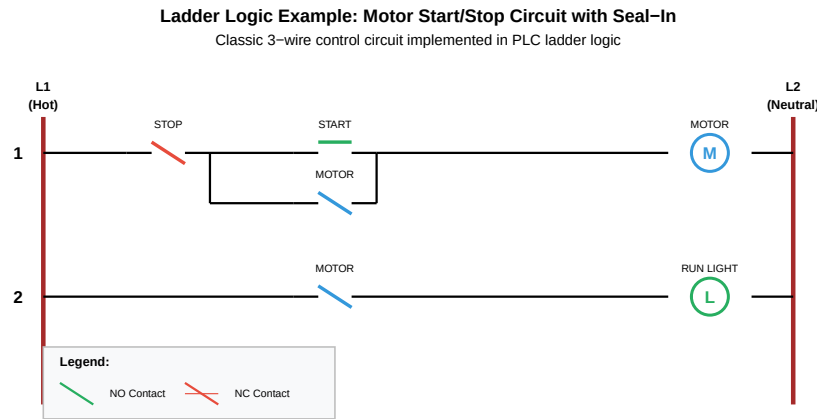
cat("PLC Scan Time Calculation:\n")
cat("      \n")
cat("Input Scan:      ", input_scan_time, "ms\n")
cat("Program Execution:", program_execution, "ms\n")
cat("Output Update:    ", output_update, "ms\n")
cat("Housekeeping:     ", housekeeping, "ms\n")
cat("      \n")
cat("Total Scan Time:  ", total_scan_time, "ms\n")
cat("\nScans per second: ", round(1000/total_scan_time, 0), "\n")
```

### 9.4.4 PLC Programming Languages (IEC 61131-3)

Table 9.4: IEC 61131-3 PLC Programming Language

| Language                        | Type      | Description                                                   |
|---------------------------------|-----------|---------------------------------------------------------------|
| Ladder Diagram (LD)             | Graphical | Resembles electrical relay circuits; most common in discrete  |
| Function Block Diagram (FBD)    | Graphical | Uses function blocks connected by lines; good for analog/pro  |
| Structured Text (ST)            | Textual   | High-level language similar to Pascal; powerful for complex c |
| Instruction List (IL)           | Textual   | Low-level assembly-like language; rarely used today           |
| Sequential Function Chart (SFC) | Graphical | Sequence/state-based programming; excellent for batch proce   |

9.4.5 Ladder Logic Example



How the Start/Stop Circuit Works

**Rung 1 - Motor Control:** 1. Power flows from L1 through the NC (Normally Closed) STOP button 2. If STOP is pressed, the circuit breaks and motor stops 3. START is NO (Normally Open) - pressing it allows power to flow to the MOTOR coil 4. When MOTOR energizes, its NO contact in the parallel branch closes 5. This “seals in” the circuit - motor stays running even after START is released 6. To stop, press STOP which breaks the seal-in circuit

**Rung 2 - Run Indicator:** 1. When MOTOR coil is energized, the MOTOR contact closes 2. This allows power to flow to the RUN LIGHT output 3. Light is ON whenever motor is running

**This is called a “3-wire control” circuit** - it provides: - Low voltage release protection (motor won’t restart after power failure) - Maintained contact operation (don’t need to hold START button)

## 9.5 Industrial Sensors

Sensors are the “eyes and ears” of automation systems, providing feedback about the process to the control system.

### 9.5.1 Sensor Classification

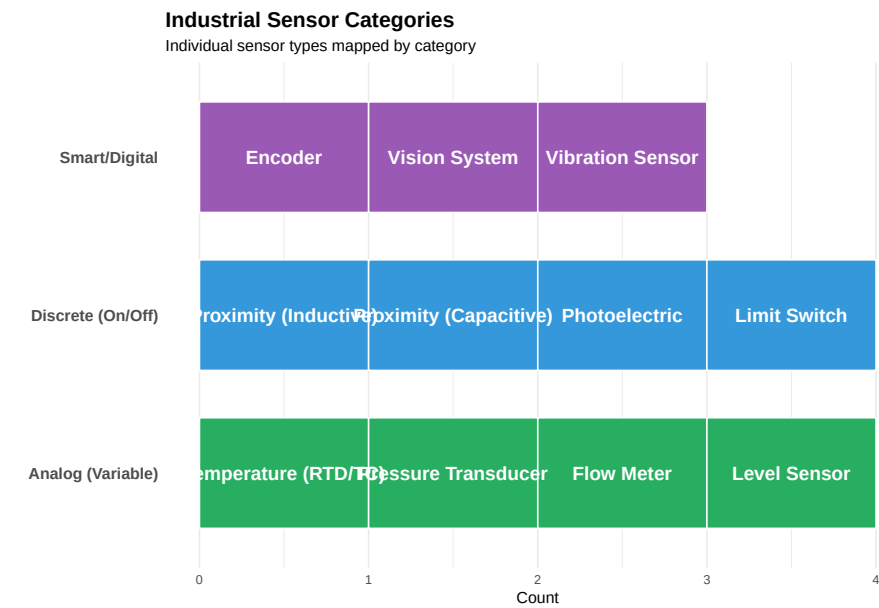
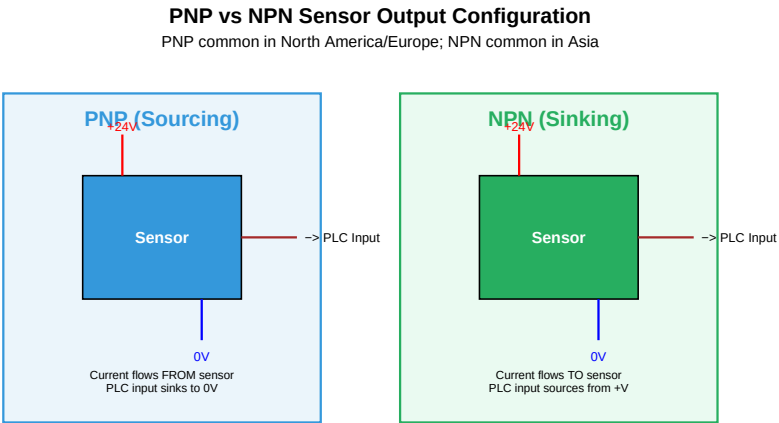


Table 9.5: Common Industrial Sensor Specifications

| Sensor               | Principle                                           | Range          | Adv |
|----------------------|-----------------------------------------------------|----------------|-----|
| Inductive Proximity  | Eddy current change in oscillating field            | 2-40mm typical | Dur |
| Capacitive Proximity | Capacitance change with target approach             | 2-25mm typical | Det |
| Photoelectric        | Light beam interrupted or reflected                 | 0.1-30m        | Lon |
| RTD (PT100)          | Resistance changes with temperature                 | -200 to 850°C  | Acc |
| Thermocouple         | Voltage generated at junction of dissimilar metals  | -200 to 2300°C | Wid |
| Pressure Transducer  | Diaphragm deflection converted to electrical signal | 0-10000 psi    | Acc |
| Incremental Encoder  | Optical/magnetic pulses per revolution              | 100-10000 PPR  | Sim |
| Absolute Encoder     | Unique code for each position                       | 12-25 bit      | Pos |



9.5.2 Sensor Wiring: PNP vs NPN



9.6 Actuators and Motor Control

Actuators convert control signals into physical motion or action.

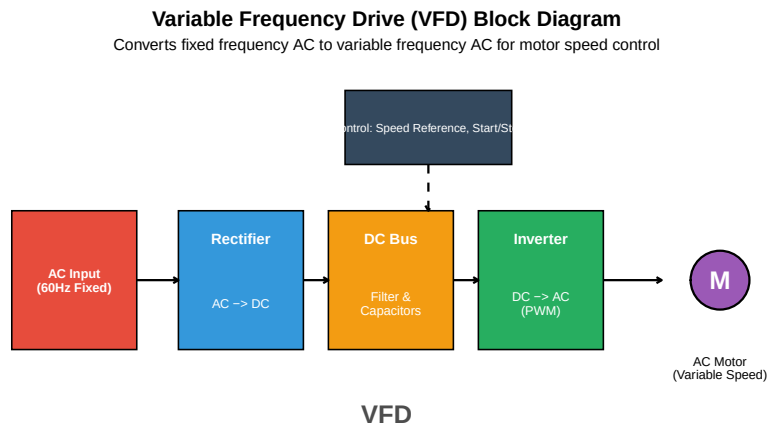
9.6.1 Types of Industrial Actuators

Table 9.6: Industrial Actuator Types and Applications

| Category         | Type               | Application                         | Control                              |
|------------------|--------------------|-------------------------------------|--------------------------------------|
| <b>Electric</b>  |                    |                                     |                                      |
| Electric Motors  | AC Induction Motor | Conveyors, pumps, fans, compressors | VFD for variable speed               |
| Electric Motors  | Servo Motor        | Precise positioning, robotics, CNC  | Servo drive with feedback            |
| Electric Motors  | Stepper Motor      | Indexing, low-speed positioning     | Stepper drive (open or closed loop)  |
| Electric Motors  | DC Motor           | Battery vehicles, legacy systems    | PWM, SCR drive                       |
| <b>Pneumatic</b> |                    |                                     |                                      |
| Pneumatic        | Cylinder           | Clamping, pushing, lifting          | Solenoid valves, proportional valves |
| Pneumatic        | Rotary Actuator    | Rotating grippers, indexing         | Solenoid valves                      |
| <b>Hydraulic</b> |                    |                                     |                                      |
| Hydraulic        | Cylinder           | Heavy lifting, presses              | Proportional/servo valves            |
| Hydraulic        | Motor              | Heavy machinery, mobile equipment   | Proportional/servo valves            |

9.6.2 Variable Frequency Drives (VFDs)

A **Variable Frequency Drive (VFD)** controls AC motor speed by varying the frequency and voltage of the power supplied to the motor.



Live cell — try different frequencies or motor parameters.

```

#| label: vfd-calculations
# VFD Speed and Torque Calculations

# Motor nameplate data
motor_hp <- 10
poles <- 4
rated_voltage <- 460 # V
rated_frequency <- 60 # Hz
rated_rpm <- 1750
slip_rpm <- (120 * rated_frequency / poles) - rated_rpm

# Synchronous speed at rated frequency
sync_speed_60 <- 120 * rated_frequency / poles
cat("Motor Data:\n")
cat("Synchronous speed at 60Hz:", sync_speed_60, "RPM\n")
cat("Rated speed:", rated_rpm, "RPM\n")
cat("Slip:", slip_rpm, "RPM (", round(slip_rpm/sync_speed_60*100, 1), "%)\n\n")

# Calculate speed at different frequencies
frequencies <- c(15, 30, 45, 60, 75)
speeds <- sapply(frequencies, function(f) {
  sync <- 120 * f / poles
  sync - slip_rpm # Assuming constant slip (approximation)
})

cat("Speed vs Frequency (V/Hz mode):\n")
cat("      \n")
for(i in seq_along(frequencies)) {
  cat(sprintf("%2d Hz: %4d RPM\n", frequencies[i], speeds[i]))
  
```

```
}

# V/Hz ratio
vhz_ratio <- rated_voltage / rated_frequency
cat("\nV/Hz Ratio:", round(vhz_ratio, 2), "V/Hz\n")
cat("At 30Hz, voltage should be:", 30 * vhz_ratio, "V\n")
```

9.6.3 VFD Benefits

Table 9.7: Variable Frequency Drive Benefits

| Benefit                   | Description                                                        | Typical |
|---------------------------|--------------------------------------------------------------------|---------|
| Energy Savings            | Match motor speed to load requirements; huge savings on fans/pumps | 20-50%  |
| Soft Start/Stop           | Ramp up/down gradually; eliminates inrush current (6-8x normal)    | Reduce  |
| Speed Control             | Precise speed control from 0-100%+ of base speed                   | N/A     |
| Process Control           | Maintain constant pressure, flow, or tension                       | Improv  |
| Reduced Mechanical Stress | Reduced wear on belts, gears, couplings from smooth acceleration   | Extend  |
| Power Factor              | VFD presents near-unity power factor to supply                     | Avoid F |

9.6.4 Servo Systems

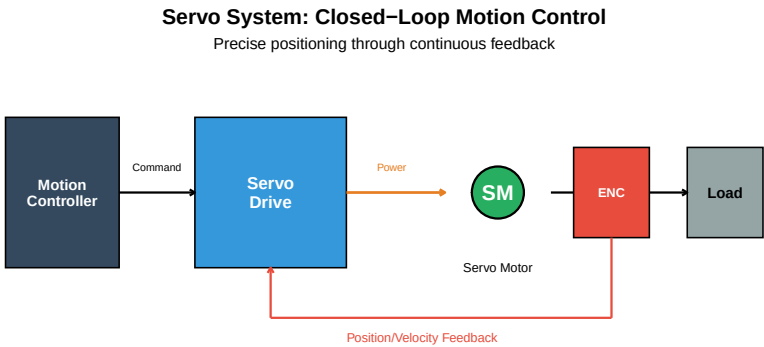


Table 9.8: VFD vs. Servo System Comparison

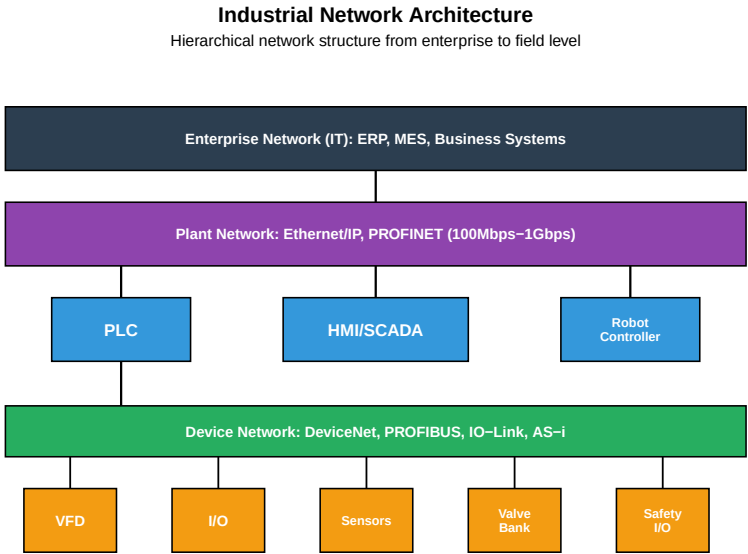
| Characteristic       | VFD + Induction Motor | Servo System                     |
|----------------------|-----------------------|----------------------------------|
| Control Type         | Open loop (usually)   | Closed loop (always)             |
| Feedback             | Optional encoder      | High-resolution encoder required |
| Positioning Accuracy | ±1-5% of speed        | ±0.01° or better                 |
| Dynamic Response     | Moderate (100-500ms)  | Fast (1-10ms)                    |
| Torque at Zero Speed | Limited (10-20%)      | 100% continuous                  |

|                     |                        |                          |
|---------------------|------------------------|--------------------------|
| Cost                | Lower (\$500-5000)     | Higher (\$2000-20000)    |
| Typical Application | Fans, pumps, conveyors | Robotics, CNC, packaging |

## 9.7 Industrial Communication Networks

Modern automation systems rely on communication networks to connect devices, share data, and enable coordinated control.

### 9.7.1 Network Architecture



### 9.7.2 Common Industrial Protocols

Table 9.9: Common Industrial Communication Protocols

| Protocol           | Developer         | Medium    | Speed         | Application                              |
|--------------------|-------------------|-----------|---------------|------------------------------------------|
| <b>Ethernet/IP</b> | ODVA (Rockwell)   | Ethernet  | 100Mbps/1Gbps | General automation, I/O                  |
| <b>PROFINET</b>    | PI (Siemens)      | Ethernet  | 100Mbps/1Gbps | Siemens ecosystem, motion control        |
| <b>Modbus TCP</b>  | Modicon/Schneider | Ethernet  | 100Mbps       | Simple, open, legacy systems             |
| <b>EtherCAT</b>    | Beckhoff          | Ethernet  | 100Mbps       | High-speed motion, precision positioning |
| <b>DeviceNet</b>   | ODVA              | CAN-based | 500kbps       | I/O, drives (legacy)                     |
| <b>PROFIBUS</b>    | PI (Siemens)      | RS-485    | 12Mbps        | I/O, drives (legacy)                     |

|                |                    |                |           |                         |
|----------------|--------------------|----------------|-----------|-------------------------|
| <b>IO-Link</b> | IO-Link Consortium | Point-to-point | 230.4kbps | Smart sensors           |
| <b>OPC UA</b>  | OPC Foundation     | Ethernet/Any   | Varies    | IT/OT integration, IIoT |

---

**Protocol Selection Guide**

**Choose Ethernet/IP if:** - Using Allen-Bradley/Rockwell equipment - Need general-purpose industrial Ethernet - Require seamless integration with IT networks

**Choose PROFINET if:** - Using Siemens equipment - Need deterministic communication for motion control - Require IRT (Isochronous Real-Time) performance

**Choose EtherCAT if:** - Need highest speed/lowest latency - High-precision motion control - Many axes of coordinated motion

**Choose Modbus TCP if:** - Simple, low-cost solution needed - Connecting legacy equipment - Open protocol preference (no licensing)

**Choose OPC UA if:** - Need IT/OT convergence - IIoT/Industry 4.0 implementation - Secure, platform-independent communication



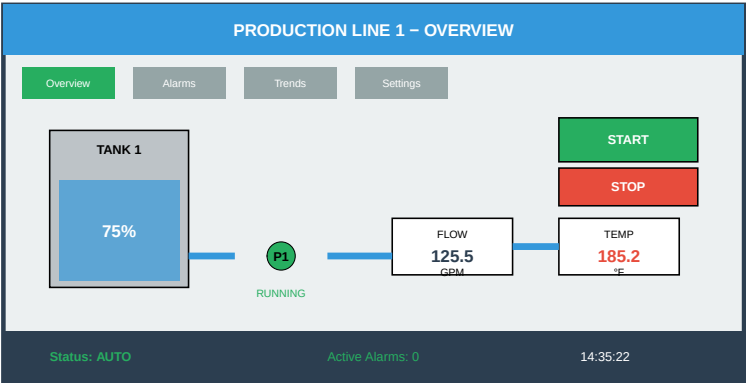
**9.8 Human-Machine Interface (HMI) and SCADA**

**9.8.1 HMI Overview**

An **HMI (Human-Machine Interface)** provides operators with a visual interface to monitor and control automated processes.

HMI Screen Example

Typical operator interface showing process status and controls



9.8.2 HMI Design Best Practices

Table 9.10: HMI Design Best Practices

| Principle                    | Best Practice                                                                |
|------------------------------|------------------------------------------------------------------------------|
| <b>Situational Awareness</b> | Show abnormal conditions prominently; operator should know status at a       |
| <b>Alarm Management</b>      | Prioritize alarms (critical/warning/info); avoid alarm floods; require ackno |
| <b>Navigation</b>            | Consistent layout; max 3 clicks to any screen; clear hierarchy               |
| <b>Color Usage</b>           | Use color sparingly for meaning; gray for normal; avoid red/green for criti  |
| <b>Data Display</b>          | Show trends, not just values; use appropriate precision; include units       |
| <b>Controls</b>              | Confirm destructive actions; use interlocks; provide feedback                |

9.8.3 SCADA Systems

**SCADA (Supervisory Control and Data Acquisition)** systems provide centralized monitoring and control across multiple locations or processes.

Table 9.11: SCADA System Components

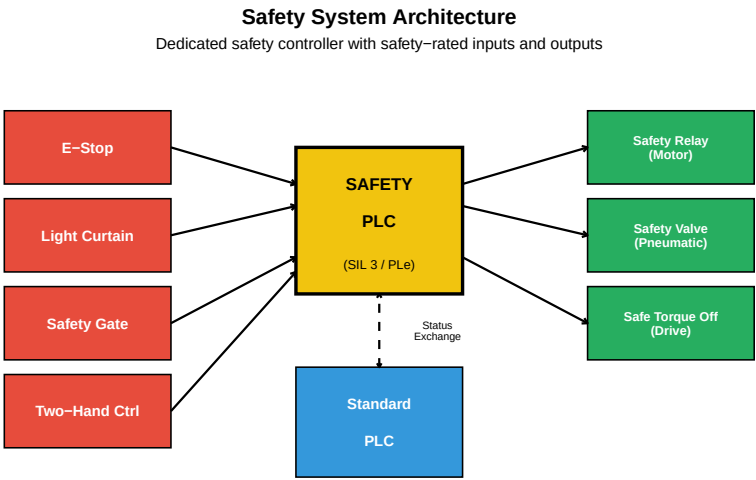
| Component                         | Function                                                       |
|-----------------------------------|----------------------------------------------------------------|
| <b>MTU (Master Terminal Unit)</b> | Central server running SCADA software; processes data, execute |
| <b>RTU (Remote Terminal Unit)</b> | Field device that collects data from sensors and sends to MTU  |
| <b>Communication Network</b>      | Links MTU to RTUs; can be radio, cellular, satellite, fiber    |
| <b>HMI/Workstations</b>           | Operator interface for monitoring and control                  |
| <b>Historian Database</b>         | Stores historical process data for trending and analysis       |

|              |                                                     |     |
|--------------|-----------------------------------------------------|-----|
| Alarm Server | Manages alarm generation, notification, and logging | Bui |
|--------------|-----------------------------------------------------|-----|

9.9 Safety in Automation

Safety is paramount in automated systems. Safety systems protect personnel and equipment from harm.

9.9.1 Safety System Architecture



9.9.2 Safety Integrity Levels (SIL)

Table 9.12: Safety Integrity Levels (IEC 61508)

| SIL Level    | PFD (avg)                             | Risk Reduction Factor | Risk Level                                        | Example |
|--------------|---------------------------------------|-----------------------|---------------------------------------------------|---------|
| <b>SIL 1</b> | 10 <sup>-2</sup> to <10 <sup>-1</sup> | 10-100                | Minor injury possible; first line of defense      | Warning |
| <b>SIL 2</b> | 10 <sup>-3</sup> to <10 <sup>-2</sup> | 100-1,000             | Serious injury possible; general industrial       | Machine |
| <b>SIL 3</b> | 10 <sup>-4</sup> to <10 <sup>-3</sup> | 1,000-10,000          | Death or severe injury; process industry standard | Emergen |
| <b>SIL 4</b> | 10 <sup>-5</sup> to <10 <sup>-4</sup> | 10,000-100,000        | Catastrophic; nuclear, aviation                   | Nuclear |

## 9.10 Industry 4.0 and IIoT

**Industry 4.0** represents the fourth industrial revolution, characterized by the integration of digital technologies into manufacturing.

### 9.10.1 Industry 4.0 Technologies



### 9.10.2 IIoT Architecture

Table 9.13: IIoT Architecture

| Layer             | Components                                                | Function         |
|-------------------|-----------------------------------------------------------|------------------|
| Edge/Device Layer | Sensors, actuators, PLCs, gateways, edge devices          | Collect data, lo |
| Network Layer     | Industrial Ethernet, Wi-Fi, 5G, LoRaWAN, MQTT             | Secure data tra  |
| Platform Layer    | Cloud/on-premise servers, databases, data lakes           | Store, process,  |
| Application Layer | Analytics dashboards, AI/ML models, business applications | Derive insights  |

### 9.10.3 Predictive Maintenance Example

```
# Simulating predictive maintenance with vibration data
set.seed(42)

# Generate 30 days of vibration readings
days <- 1:30
```



```

baseline_vibration <- 2.5 # mm/s RMS (normal)

# Simulate gradual bearing degradation
vibration <- baseline_vibration +
  0.05 * days + # Gradual increase
  0.5 * exp((days - 25)/5) * (days > 20) + # Accelerating failure
  rnorm(30, 0, 0.2) # Normal variation

# Alert thresholds
warning_threshold <- 4.0
alarm_threshold <- 6.0

# Find when thresholds crossed
warning_day <- min(which(vibration > warning_threshold))
alarm_day <- min(which(vibration > alarm_threshold))

```

Warning in min(which(vibration > alarm\_threshold)): no non-missing arguments to min; returning Inf

```

cat("Predictive Maintenance Analysis:\n")
cat("      \n")
cat("Baseline vibration:", baseline_vibration, "mm/s RMS\n")
cat("Warning threshold:", warning_threshold, "mm/s RMS\n")
cat("Alarm threshold:", alarm_threshold, "mm/s RMS\n\n")
cat("Warning triggered on day:", warning_day, "\n")
cat("Alarm triggered on day:", alarm_day, "\n")
cat("Days of warning before alarm:", alarm_day - warning_day, "\n")
cat("\nRecommendation: Schedule bearing replacement before day", alarm_day, "\n")

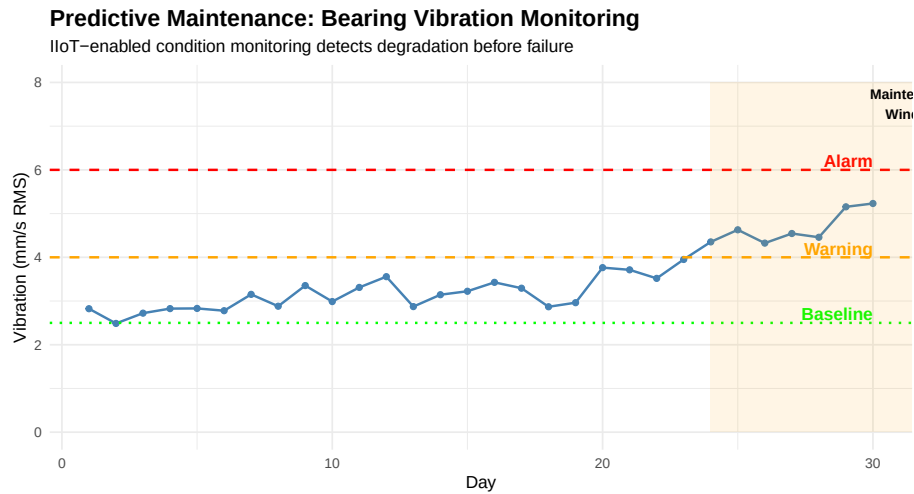
```

Predictive Maintenance Analysis:

Baseline vibration: 2.5 mm/s RMS  
Warning threshold: 4 mm/s RMS  
Alarm threshold: 6 mm/s RMS

Warning triggered on day: 24  
Alarm triggered on day: Inf  
Days of warning before alarm: Inf

Recommendation: Schedule bearing replacement before day Inf



## 9.11 Video Resources

### 9.11.1 Introduction to PLCs

### 9.11.2 Industrial Networks Explained

## 9.12 Summary

Industrial automation integrates multiple technologies to improve manufacturing efficiency, quality, and safety:

1. **Automation types** range from fixed (high-volume, single product) to flexible (low-volume, high variety)
2. **PLCs** are the workhorses of industrial control, using scan cycles to process inputs and control outputs
3. **Sensors** provide feedback on process conditions (discrete, analog, and smart)
4. **Motor control** through VFDs and servo systems enables precise speed and position control
5. **Industrial networks** connect devices using protocols like Ethernet/IP, PROFINET, and Modbus
6. **HMI and SCADA** provide operator interface for monitoring and control
7. **Safety systems** use dedicated controllers and devices to protect personnel
8. **Industry 4.0** brings IIoT, cloud computing, and AI to enable smart manufacturing

## 9.13 Review Questions

**Question 1:** Compare and contrast fixed, programmable, and flexible automation. Give an example application for each.

**Answer:**

| Aspect          | Fixed                | Programmable                  | Flexible                       |
|-----------------|----------------------|-------------------------------|--------------------------------|
| Product variety | Single product       | Batches of different products | High variety, mixed production |
| Volume          | Very high            | Medium-high                   | Low-medium                     |
| Changeover      | Difficult, expensive | Hours to days                 | Minutes                        |
| Investment      | Very high            | High                          | Very high                      |
| Flexibility     | None                 | Limited                       | High                           |

**Examples:**

1. **Fixed Automation:** Automotive transfer line producing engine blocks
  - Same machining operations repeated millions of times
  - Dedicated stations optimized for specific operations
  - Changing product would require complete retooling
2. **Programmable Automation:** CNC machining job shop
  - Different parts programmed and run in batches
  - Significant setup time between batches
  - Equipment reprogrammable but not instantaneous
3. **Flexible Automation:** Robotic welding cell in aerospace
  - Different assemblies run with quick changeover
  - Robot program changes automatically based on part ID
  - Same cell handles multiple product variants

**Question 2:** Explain the PLC scan cycle and why scan time is important.

**Answer:**

**The PLC Scan Cycle** consists of four main phases executed continuously:

1. **Input Scan** (~1ms): Read all physical inputs and store values in the input image table
2. **Program Execution** (~5-20ms): Execute the user program (ladder logic, etc.) using input image data
3. **Output Update** (~1ms): Write output image table values to physical outputs
4. **Housekeeping** (~1-3ms): Communications, diagnostics, self-testing

**Total scan time** = Sum of all phases, typically 10-50ms

**Why Scan Time Matters:**

1. **Response Time:** An input change isn't acted upon until the next scan completes. If scan time is 50ms, worst-case response time is 50ms.
2. **Safety:** Fast-moving machinery requires short scan times for safety functions. A 100ms scan time with a motor at 1800 RPM means 3 shaft revolutions between scans.
3. **Application Limits:**
  - Motion control may need <1ms scan times (special motion PLCs)
  - High-speed counting needs special modules
  - Safety PLCs have guaranteed maximum scan times
4. **Program Size Impact:** Larger programs take longer to execute, increasing scan time. Optimization may be needed for time-critical applications.
5. **Communication Delays:** Network updates typically happen once per scan, affecting data freshness.

**Question 3:** What are the key differences between PNP and NPN sensor outputs? When would you use each?

**Answer:**

**PNP (Sourcing):** - Sensor output sources current TO the load - When activated, output connects to +V - Load is connected between sensor output and 0V - Current flows: +V → Sensor → Output → Load → 0V - Common in North America and Europe

**NPN (Sinking):** - Sensor output sinks current FROM the load - When activated, output connects to 0V (ground) - Load is connected between sensor output and +V - Current flows: +V → Load → Output → Sensor → 0V - Common in Asia (Japan)

**Selection Criteria:**

1. **PLC Input Type:**
  - Sinking PLC inputs work with PNP (sourcing) sensors
  - Sourcing PLC inputs work with NPN (sinking) sensors
2. **Regional Standards:**
  - North America: PNP is standard
  - Europe: PNP is standard
  - Japan/Asia: NPN is common
3. **Safety Considerations:**
  - PNP: A ground fault can cause false ON signal
  - NPN: A ground fault typically causes safe-off condition
  - For safety applications, consider sensor design carefully
4. **Existing Infrastructure:**

- Match new sensors to existing system type
- Converting between types requires different PLC input cards

**Question 4:** A VFD is controlling a 10 HP, 4-pole motor rated at 1750 RPM and 460V at 60Hz. Calculate the motor speed at 30Hz and 45Hz. What voltage should the VFD supply at these frequencies?

**Answer:** (live — change the numbers to check your own version)

```
#| label: q4-answer
# Motor parameters
poles <- 4
rated_voltage <- 460 # V
rated_frequency <- 60 # Hz
rated_rpm <- 1750

# Calculate synchronous speed and slip at rated conditions
sync_speed_60 <- 120 * rated_frequency / poles
slip_rpm <- sync_speed_60 - rated_rpm
slip_percent <- (slip_rpm / sync_speed_60) * 100

cat("Motor Analysis:\n")
cat("Synchronous speed at 60Hz:", sync_speed_60, "RPM\n")
cat("Rated speed:", rated_rpm, "RPM\n")
cat("Slip:", slip_rpm, "RPM (", round(slip_percent, 1), "%)\n\n")

# V/Hz ratio
vHz_ratio <- rated_voltage / rated_frequency
cat("V/Hz Ratio:", round(vHz_ratio, 2), "V/Hz\n\n")

# Calculate at 30Hz
freq_30 <- 30
sync_30 <- 120 * freq_30 / poles
speed_30 <- sync_30 - slip_rpm # Assuming constant slip in RPM
voltage_30 <- freq_30 * vHz_ratio

cat("At 30Hz:\n")
cat("  Synchronous speed:", sync_30, "RPM\n")
cat("  Motor speed (approx):", speed_30, "RPM\n")
cat("  VFD output voltage:", voltage_30, "V\n\n")

# Calculate at 45Hz
freq_45 <- 45
sync_45 <- 120 * freq_45 / poles
speed_45 <- sync_45 - slip_rpm
voltage_45 <- freq_45 * vHz_ratio
```

```
cat("At 45Hz:\n")
cat("  Synchronous speed:", sync_45, "RPM\n")
cat("  Motor speed (approx):", speed_45, "RPM\n")
cat("  VFD output voltage:", voltage_45, "V\n")
```

**Note:** The calculation assumes constant slip in RPM, which is an approximation. Actual slip varies somewhat with load and frequency. The V/Hz ratio is maintained constant to provide constant torque capability below base speed.

**Question 5:** Compare Ethernet/IP and PROFINET. When would you choose each?

**Answer:**

| Feature              | Ethernet/IP                                             | PROFINET                                   |
|----------------------|---------------------------------------------------------|--------------------------------------------|
| <b>Developer</b>     | ODVA (Rockwell Automation)                              | PROFIBUS International (Siemens)           |
| <b>Base Protocol</b> | CIP over TCP/UDP                                        | Based on standard Ethernet                 |
| <b>Real-time</b>     | CIP Motion for motion control                           | IRT (Isochronous Real-Time)                |
| <b>Determinism</b>   | Standard: non-deterministic;<br>CIP Sync: deterministic | RT: soft real-time;<br>IRT: hard real-time |
| <b>Typical Cycle</b> | 2-10ms (standard); <1ms (CIP Motion)                    | 1-10ms (RT); <1ms (IRT)                    |
| <b>Ecosystem</b>     | Allen-Bradley, DeviceNet-heritage                       | Siemens, PROFIBUS-heritage                 |

**Choose Ethernet/IP when:** - Using Allen-Bradley/Rockwell PLCs - Integrating with DeviceNet legacy systems - Standard industrial Ethernet needs - North American installations (common) - Need CIP protocol compatibility

**Choose PROFINET when:** - Using Siemens PLCs (S7-1200, S7-1500) - Migrating from PROFIBUS installations - Need IRT for high-performance motion - European installations (common) - Require I-Device functionality

**General Guidance:** - Match the protocol to your PLC vendor's ecosystem - Both are capable industrial Ethernet solutions - Consider existing infrastructure and expertise - For motion, evaluate specific timing requirements

**Question 6:** What is the difference between HMI and SCADA? Where would each be used?

**Answer:**

**HMI (Human-Machine Interface):** - Local operator interface for a single machine or process - Typically a touchscreen panel mounted on or near

equipment - Communicates directly with one or few PLCs - Provides real-time control and monitoring - Limited data storage (trends, alarms) - Examples: Allen-Bradley PanelView, Siemens Comfort Panel

**SCADA (Supervisory Control and Data Acquisition):** - Centralized system monitoring multiple processes/locations - Server-based architecture with multiple client workstations - Communicates with many PLCs/RTUs across large distances - Supervisory control - high-level commands, not direct control - Extensive historical data collection and analysis - Examples: Wonderware, Ignition, FactoryTalk View SE

**Where Each is Used:**

| Application           | HMI        | SCADA         |
|-----------------------|------------|---------------|
| Single CNC machine    |            |               |
| Packaging line        |            |               |
| Entire factory floor  |            |               |
| Water treatment plant | (local)    | (central)     |
| Oil pipeline network  |            |               |
| Building automation   | (per zone) | (campus-wide) |
| Power grid            |            |               |

**Key Distinction:** HMI is typically embedded/local for one process; SCADA is a system supervising many processes across the enterprise.

**Question 7:** Explain what SIL 3 means and give an example of a SIL 3 application.

**Answer:**

**SIL 3 (Safety Integrity Level 3)** is defined in IEC 61508 and represents:

- **PFD (Probability of Failure on Demand):**  $10^{-4}$  to  $10^{-3}$  (0.01% to 0.1%)
- **Risk Reduction Factor:** 1,000 to 10,000
- **Availability:** 99.9% to 99.99%

**What SIL 3 Means:** - The safety function will fail to operate when needed less than 1 in 1,000 demands - Requires redundant architecture (typically 2oo3 or 1oo2D) - Requires certified safety-rated components - Requires rigorous design, testing, and validation processes - Requires regular proof testing and maintenance

**SIL 3 Requirements:** - Hardware fault tolerance  $\geq 1$  (single fault won't cause dangerous failure) - Safe Failure Fraction  $> 90\%$  (for Type A components) - Systematic capability rating of SC3 - Extensive documentation and lifecycle management

**SIL 3 Application Examples:**

1. **Emergency Shutdown System (ESD)** in oil refinery
  - Detects dangerous conditions (high pressure, fire)
  - Shuts down process and isolates fuel sources
  - Failure could result in explosion and fatalities
2. **Burner Management System (BMS)**
  - Controls furnace/boiler ignition sequence
  - Monitors flame presence and fuel/air ratio
  - Prevents furnace explosions
3. **High Integrity Pressure Protection System (HIPPS)**
  - Prevents overpressure in pipelines
  - Isolates high-pressure source before relief valve capacity exceeded
  - Protects against catastrophic pipeline rupture

**Question 8:** Describe three benefits of implementing IIoT in a manufacturing facility and give a specific example of each.

**Answer:**

### 1. Predictive Maintenance

*Benefit:* Detect equipment degradation before failure, enabling planned maintenance during scheduled downtime rather than emergency repairs.

*Example:* Vibration sensors on a conveyor gearbox transmit data to cloud analytics. Machine learning algorithms detect bearing wear signature 3 weeks before expected failure. Maintenance schedules replacement during next planned shutdown, avoiding \$50,000 in lost production from unplanned downtime.

### 2. Real-Time Production Visibility

*Benefit:* Instant access to production metrics, OEE, and quality data from anywhere, enabling faster decision-making.

*Example:* Dashboard shows real-time OEE for all production lines on plant manager's tablet. Sudden drop in Line 3 performance triggers alert. Investigation reveals material feed issue; corrected within 30 minutes instead of waiting for end-of-shift report.

### 3. Energy Optimization

*Benefit:* Monitor energy consumption at machine level, identify waste, and optimize based on production schedule.

*Example:* Power meters on each machine report to central system. Analysis reveals HVAC in unused areas runs at full capacity during weekends. Automated scheduling reduces weekend HVAC operation, saving 15% on energy bills (\$30,000/year).

**Additional Benefits:**

4. **Quality Traceability** - Every part traced through production with complete process data



5. **Remote Monitoring** - Engineers can diagnose issues without traveling to site
  6. **Supply Chain Integration** - Real-time inventory and production data shared with suppliers
- 

## 9.14 References

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## Chapter 10

# Statistical Process Control (SPC)

---

### 10.1 Learning Objectives

By the end of this chapter, you will be able to:

- Explain the purpose and benefits of Statistical Process Control
- Distinguish between common cause and special cause variation
- Select the appropriate control chart for different data types
- Construct and interpret  $\bar{X}$ -bar/R,  $\bar{X}$ -bar/S, and Individual-MR charts
- Construct and interpret attribute control charts (p, np, c, u)
- Apply control chart interpretation rules (Western Electric rules)
- Calculate and interpret process capability indices ( $C_p$ ,  $C_{pk}$ ,  $P_p$ ,  $P_{pk}$ )
- Respond appropriately to out-of-control signals

“In God we trust; all others must bring data.” — W. Edwards Deming

---

### 10.2 Introduction to SPC

#### 10.2.1 What is Statistical Process Control?

**Statistical Process Control (SPC)** is a method of quality control that uses statistical methods to monitor and control a process. It helps ensure that the process operates at its full potential to produce conforming product.

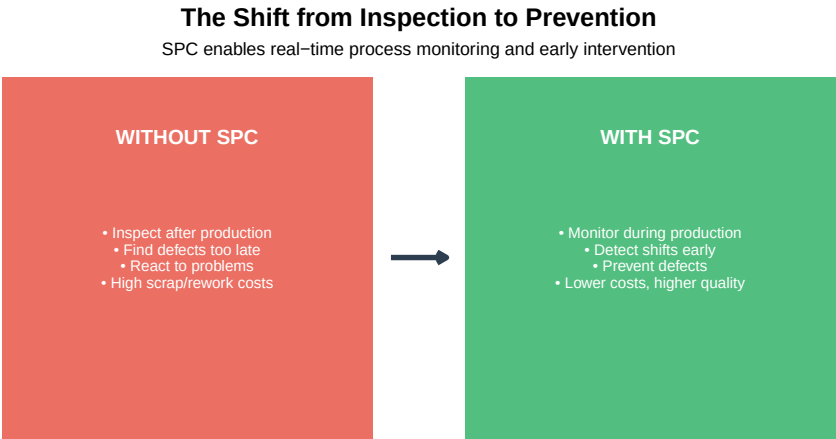


Figure 10.1: The SPC Philosophy: Prevention Through Monitoring

10.2.2 Brief History of SPC

Table 10.1: Evolution of Statistical Process Control

| Year      | Development                                          | Significance                              |
|-----------|------------------------------------------------------|-------------------------------------------|
| 1920s     | Walter Shewhart develops control charts at Bell Labs | Birth of modern quality control           |
| 1930s-40s | SPC used in WWII military production                 | Proved effectiveness in manufacturing     |
| 1950s     | Deming teaches SPC to Japanese industry              | Foundation of Japanese quality management |
| 1980s     | US automotive industry adopts SPC                    | Response to Japanese competition          |
| 1990s     | SPC becomes requirement in ISO/QS-9000               | Industry-wide standardization             |
| 2000s+    | Real-time SPC, automated data collection             | Integration with Industry 4.0             |

### 10.2.3 Why SPC Matters

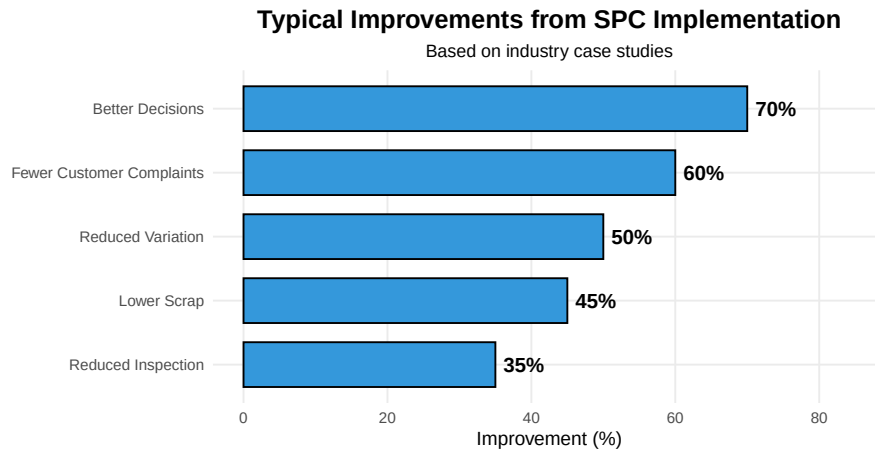


Figure 10.2: Benefits of SPC Implementation

## 10.3 Understanding Variation

**All processes exhibit variation.** No two products are ever exactly identical. The key insight of SPC is that variation comes from two fundamentally different sources, and we must respond to them differently.

10.3.1 Common Cause vs. Special Cause Variation

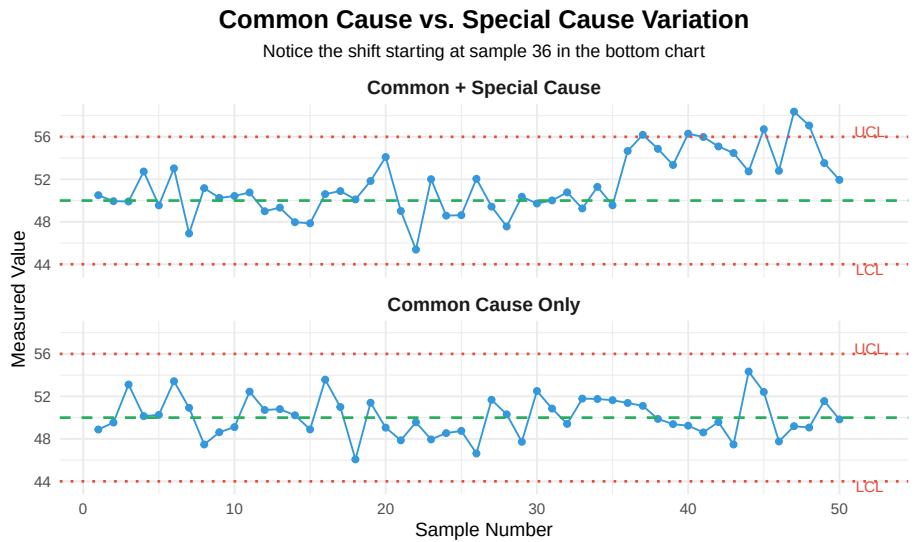


Figure 10.3: Two Types of Variation

Table 10.2: Common Cause vs. Special Cause Variation

| Characteristic             | Common Cause Variation                                     | Special Cause Variation       |
|----------------------------|------------------------------------------------------------|-------------------------------|
| <b>**Other Names**</b>     | Random, chance, noise, inherent                            | Assignable                    |
| <b>**Source**</b>          | Built into the process; many small factors                 | Specific, identifiable        |
| <b>**Pattern**</b>         | Random, stable, predictable pattern                        | Non-random                    |
| <b>**Predictability**</b>  | Statistically predictable within limits                    | Unpredictable                 |
| <b>**Examples**</b>        | Machine vibration, material variation, ambient temperature | Tool breakage, operator error |
| <b>**Action Required**</b> | Improve the system (management decision)                   | Find and eliminate the cause  |
| <b>**Who Acts**</b>        | Management — requires system change                        | Operators                     |

10.3.2 The Two Mistakes in Process Control

Understanding variation prevents two costly mistakes:

### Two Mistakes That Waste Resources and Reduce Quality

Control charts help us avoid both mistakes



Figure 10.4: The Two Fundamental Mistakes

The Funnel Experiment: Why Tampering Makes Things Worse

**Deming's Funnel Experiment** demonstrates why adjusting a stable process increases variation:

**Setup:** Drop a ball through a funnel onto a target. The ball lands near (but not exactly on) the target due to common cause variation.

**Rule 1 (Correct):** Leave the funnel alone. Variation remains constant.

**Rule 2 (Tampering):** After each drop, move the funnel to compensate for the last error. - Result: Variation DOUBLES!

**Rule 3 (More Tampering):** Move funnel to compensate relative to the target. - Result: Variation increases without bound!

**Lesson:** Adjusting a stable process based on individual deviations (common cause) actually makes the process worse. Only adjust when special causes are identified.

**Real-World Example:** An operator notices a part is 0.02mm below target and adjusts the machine. The next part is 0.03mm above target (the adjustment plus normal variation). They adjust again... and the process oscillates with increasing variation.

### 10.3.3 When is a Process “In Control”?

A process is **in statistical control** when:

1. Only common cause variation is present
2. All points fall within control limits
3. No non-random patterns exist

4. The process is stable and predictable

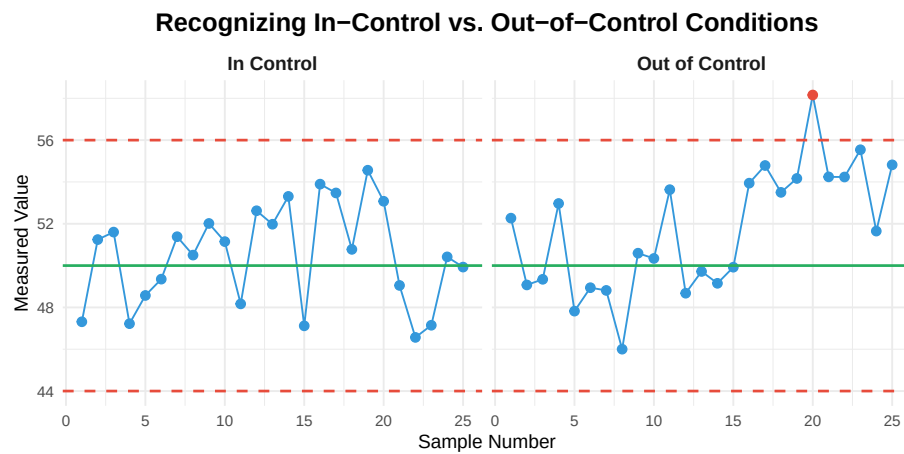


Figure 10.5: In-Control vs. Out-of-Control Process

## 10.4 Control Charts: The Foundation of SPC

### 10.4.1 What is a Control Chart?

A **control chart** is a graph that shows process data over time with statistically determined control limits. It provides a visual method for distinguishing between common and special cause variation.



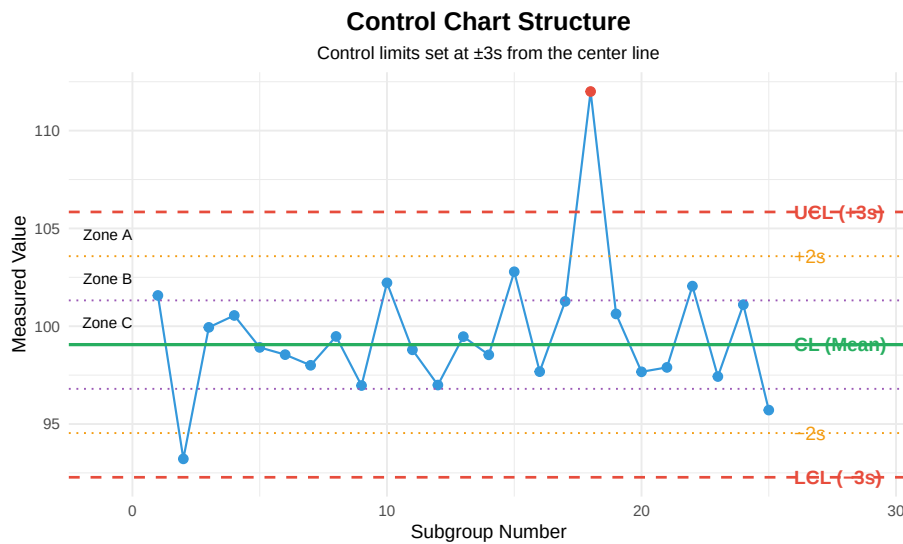


Figure 10.6: Anatomy of a Control Chart

### 10.4.2 Control Chart Components

Table 10.3: Control Chart Components

| Component                            | Description                                                          | Purpose                   |
|--------------------------------------|----------------------------------------------------------------------|---------------------------|
| <b>**Center Line (CL)**</b>          | The process average; target for the process                          | Shows where process       |
| <b>**Upper Control Limit (UCL)**</b> | Upper boundary of expected variation (+3 )                           | Signals if process shift  |
| <b>**Lower Control Limit (LCL)**</b> | Lower boundary of expected variation (-3 )                           | Signals if process shift  |
| <b>**Data Points**</b>               | Individual or subgroup statistics plotted over time                  | Visual representation     |
| <b>**Zones A, B, C**</b>             | Regions between $\pm 1$ , $\pm 2$ , and $\pm 3$ for pattern analysis | Used for detecting trends |

### 10.4.3 Why $\pm 3$ Sigma?

Control limits are set at  **$\pm 3$  standard deviations** from the center line. This is not arbitrary:

- For a normal distribution, 99.73% of data falls within  $\pm 3$
- Only 0.27% (about 3 in 1000) will fall outside by chance
- This balances the risk of both mistakes (false alarms vs. missed signals)

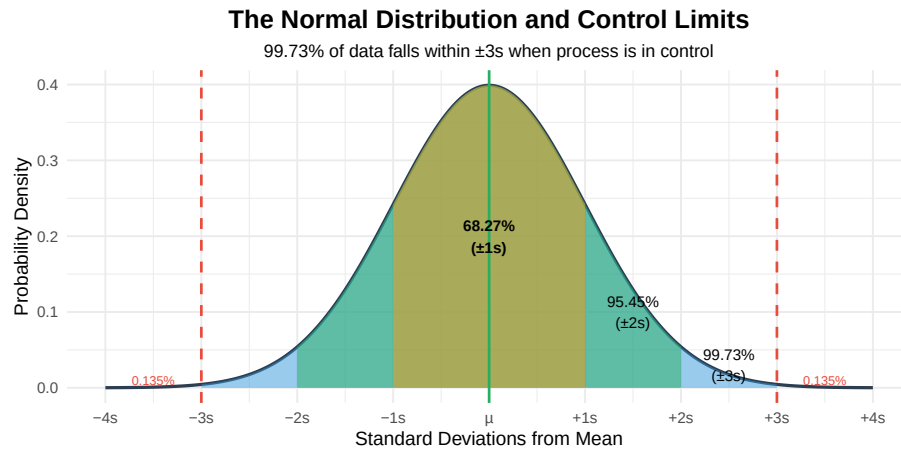


Figure 10.7: Normal Distribution and Control Limits

#### 10.4.4 Choosing the Right Control Chart

The type of control chart depends on the **type of data** being collected:

##### Selecting the Right Control Chart

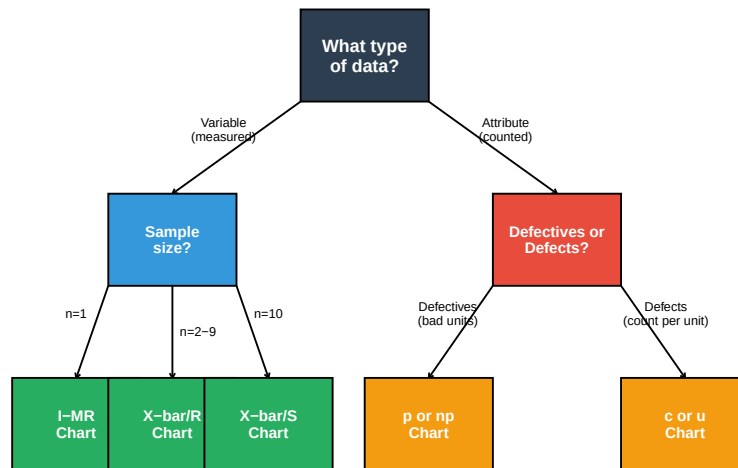


Figure 10.8: Control Chart Selection Flowchart

Table 10.4: Control Chart Summary

| Chart Type           | Data Type | Sample Size          | What It Charts                     | Typical    |
|----------------------|-----------|----------------------|------------------------------------|------------|
| <b>**X-bar / R**</b> | Variable  | 2-9 per subgroup     | Subgroup mean and range            | Production |
| <b>**X-bar / S**</b> | Variable  | 10 per subgroup      | Subgroup mean and std dev          | Large sub  |
| <b>**I-MR**</b>      | Variable  | 1 (individuals)      | Individual values and moving range | Continuo   |
| <b>**p chart**</b>   | Attribute | Variable or constant | Proportion defective               | Pass/fail  |
| <b>**np chart**</b>  | Attribute | Constant             | Number defective                   | Pass/fail  |
| <b>**c chart**</b>   | Attribute | Constant             | Number of defects                  | Defects p  |
| <b>**u chart**</b>   | Attribute | Variable             | Defects per unit                   | Defects p  |

Question: Variable vs. Attribute Data - What's the Difference?

**Variable Data (Measured):** - Can take any value within a range - Measured on a continuous scale - Examples: length, weight, temperature, voltage, pressure - More information per measurement - Smaller sample sizes needed

**Attribute Data (Counted):** - Discrete categories (good/bad, pass/fail) - Counted, not measured - Examples: number of defects, number of rejects, yes/no - Less information per observation - Larger sample sizes needed

**When to Use Each:** - Use variable data when you CAN measure (more powerful) - Use attribute data when you can only classify or count - Sometimes you convert variable to attribute (e.g., pass/fail based on tolerance)

## 10.5 Variable Control Charts

### 10.5.1 X-bar and R Chart

The **X-bar and R chart** is the most common control chart for variable data. It tracks both the process average (X-bar) and the process spread (Range).

#### 10.5.1.1 Constructing an X-bar/R Chart

**Step 1:** Collect data in subgroups (typically 4-5 samples)

```
# Example: Measuring shaft diameter (mm)
# 25 subgroups of 5 measurements each

set.seed(100)
n_subgroups <- 25
n_samples <- 5
```

```
# Generate data
data <- matrix(rnorm(n_subgroups * n_samples, mean = 25.00, sd = 0.05),
              nrow = n_subgroups, ncol = n_samples)

# Add a special cause (shift) at subgroup 18
data[18, ] <- data[18, ] + 0.12

# Calculate statistics for each subgroup
xbar <- apply(data, 1, mean)
R <- apply(data, 1, function(x) max(x) - min(x))

cat("First 5 subgroups:\n")
print(round(data[1:5, ], 4))
cat("\nX-bar values:", round(xbar[1:5], 4))
cat("\nRange values:", round(R[1:5], 4))
```

First 5 subgroups:

|      | [,1]    | [,2]    | [,3]    | [,4]    | [,5]    |
|------|---------|---------|---------|---------|---------|
| [1,] | 24.9749 | 24.9781 | 24.9776 | 25.0006 | 24.9834 |
| [2,] | 25.0066 | 24.9640 | 24.9131 | 24.9456 | 25.0682 |
| [3,] | 24.9961 | 25.0115 | 25.0089 | 25.0135 | 24.9765 |
| [4,] | 25.0443 | 24.9421 | 25.0949 | 25.0504 | 25.0421 |
| [5,] | 25.0058 | 25.0124 | 24.8864 | 24.8963 | 24.9271 |

X-bar values: 24.9829 24.9795 25.0013 25.0348 24.9456

Range values: 0.0257 0.1551 0.037 0.1528 0.126

### Step 2: Calculate control limits

```
# Control chart constants for n=5
A2 <- 0.577
D3 <- 0      # D3 = 0 for n < 7
D4 <- 2.114

# Calculate grand mean and average range
X_double_bar <- mean(xbar)
R_bar <- mean(R)

cat("Grand Mean (X-double-bar):", round(X_double_bar, 4), "mm\n")
cat("Average Range (R-bar):", round(R_bar, 4), "mm\n\n")

# X-bar chart limits
UCL_xbar <- X_double_bar + A2 * R_bar
LCL_xbar <- X_double_bar - A2 * R_bar

cat("X-bar Chart:\n")
```

```
cat("  UCL =", round(UCL_xbar, 4), "mm\n")
cat("  CL  =", round(X_double_bar, 4), "mm\n")
cat("  LCL =", round(LCL_xbar, 4), "mm\n\n")

# R chart limits
UCL_R <- D4 * R_bar
LCL_R <- D3 * R_bar

cat("R Chart:\n")
cat("  UCL =", round(UCL_R, 4), "mm\n")
cat("  CL  =", round(R_bar, 4), "mm\n")
cat("  LCL =", round(LCL_R, 4), "mm\n")
```

Grand Mean (X-double-bar): 25.004 mm  
Average Range (R-bar): 0.1023 mm

X-bar Chart:

UCL = 25.0631 mm  
CL = 25.004 mm  
LCL = 24.945 mm

R Chart:

UCL = 0.2164 mm  
CL = 0.1023 mm  
LCL = 0 mm

**Step 3:** Plot the charts

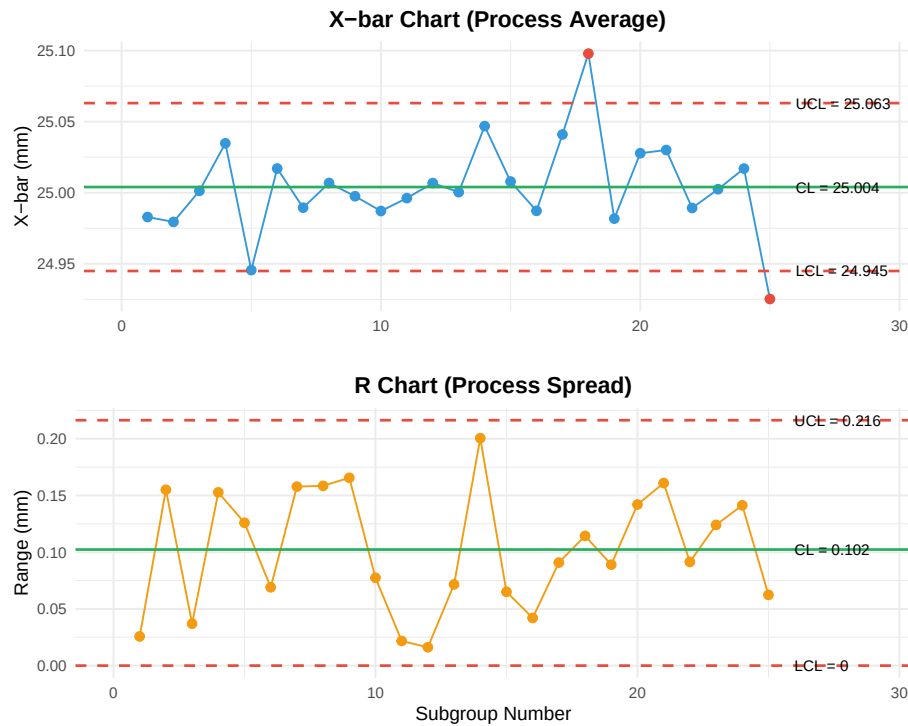


Figure 10.9: X-bar and R Control Charts

**Interpretation:** Subgroup 18 shows a point above the UCL on the X-bar chart, indicating a special cause. Investigation reveals the cause and corrective action is taken.

### 10.5.2 Control Chart Constants

Table 10.5: Control Chart Constants for X-bar/R Charts

| n | A2    | D3    | D4    | d2    |
|---|-------|-------|-------|-------|
| 2 | 1.880 | 0.000 | 3.267 | 1.128 |
| 3 | 1.023 | 0.000 | 2.574 | 1.693 |
| 4 | 0.729 | 0.000 | 2.282 | 2.059 |
| 5 | 0.577 | 0.000 | 2.114 | 2.326 |
| 6 | 0.483 | 0.000 | 2.004 | 2.534 |
| 7 | 0.419 | 0.076 | 1.924 | 2.704 |
| 8 | 0.373 | 0.136 | 1.864 | 2.847 |
| 9 | 0.337 | 0.184 | 1.816 | 2.970 |

|    |       |       |       |       |
|----|-------|-------|-------|-------|
| 10 | 0.308 | 0.223 | 1.777 | 3.078 |
|----|-------|-------|-------|-------|

10.5.3 Individual and Moving Range (I-MR) Chart

When sample size is  $n = 1$  (one measurement per time period), use the **I-MR chart**.

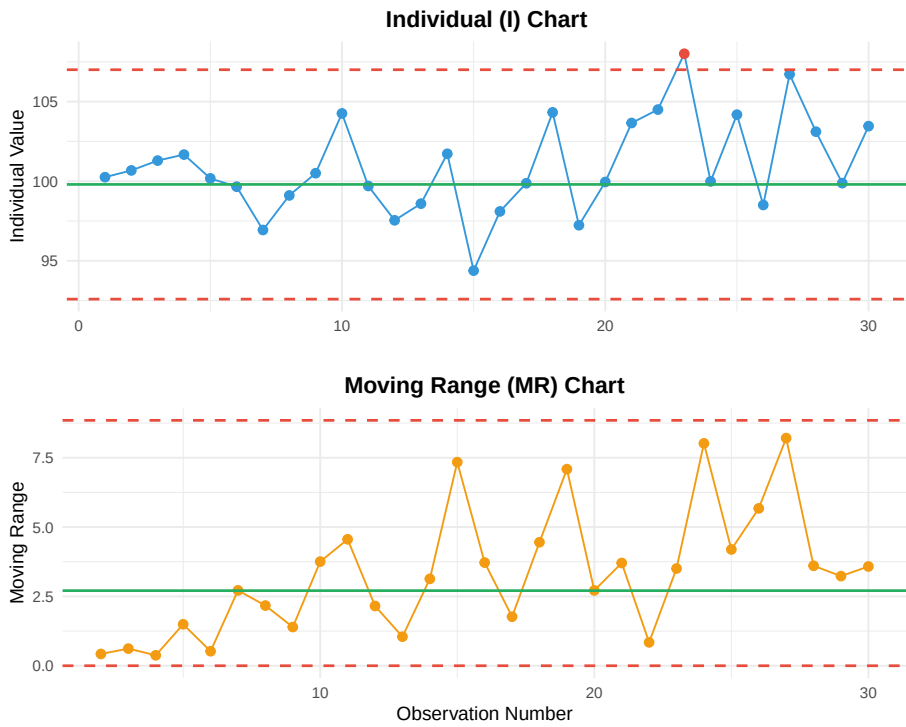


Figure 10.10: Individual and Moving Range Chart

**I-MR Chart Formulas:**

| Chart    | Center Line | UCL                      | LCL                      |
|----------|-------------|--------------------------|--------------------------|
| I Chart  | $\bar{X}$   | $\bar{X} + 2.66\bar{MR}$ | $\bar{X} - 2.66\bar{MR}$ |
| MR Chart | $\bar{MR}$  | $3.267 \times \bar{MR}$  | 0                        |

## 10.6 Attribute Control Charts

### 10.6.1 p Chart (Proportion Defective)

The **p chart** tracks the proportion of defective units when sample sizes may vary.

```
# Example: Daily inspection data
# Varying sample sizes (different production volumes)

set.seed(300)
days <- 20
sample_sizes <- sample(80:120, days, replace = TRUE)
defectives <- rbinom(days, sample_sizes, prob = 0.05)

# Add a special cause on day 15
defectives[15] <- rbinom(1, sample_sizes[15], prob = 0.12)

# Calculate proportion
p <- defectives / sample_sizes

# Calculate p-bar (weighted average)
p_bar <- sum(defectives) / sum(sample_sizes)

cat("p-bar (average proportion defective):", round(p_bar, 4), "\n\n")

# Calculate control limits (vary with sample size)
UCL_p <- p_bar + 3 * sqrt(p_bar * (1 - p_bar) / sample_sizes)
LCL_p <- pmax(0, p_bar - 3 * sqrt(p_bar * (1 - p_bar) / sample_sizes))

cat("Sample of control limits (varying with n):\n")
cat("Day 1: UCL =", round(UCL_p[1], 4), ", LCL =", round(LCL_p[1], 4), "(n =", sample_sizes[1], ")\n")
cat("Day 5: UCL =", round(UCL_p[5], 4), ", LCL =", round(LCL_p[5], 4), "(n =", sample_sizes[5], ")\n")
```

p-bar (average proportion defective): 0.0493

Sample of control limits (varying with n):

Day 1: UCL = 0.1167 , LCL = 0 (n = 93 )

Day 5: UCL = 0.1143 , LCL = 0 (n = 100 )



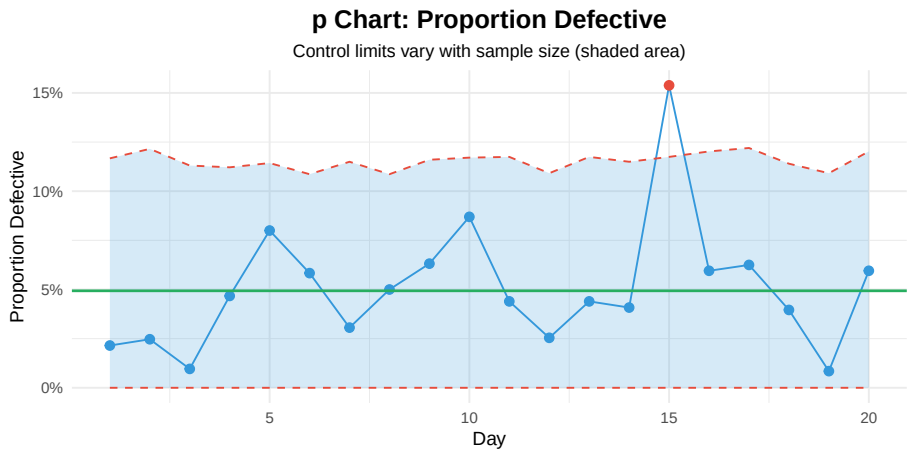


Figure 10.11: p Chart for Proportion Defective

10.6.2 c Chart (Count of Defects)

The **c chart** tracks the number of defects when the sample size (area of opportunity) is constant.

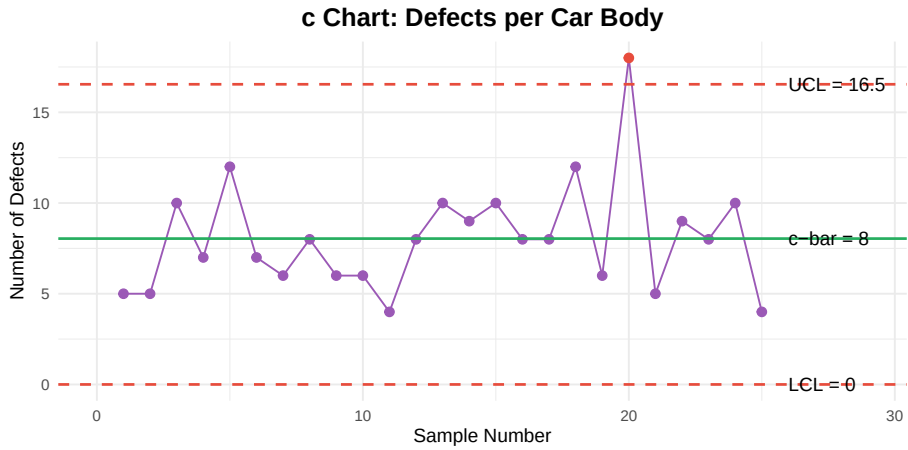


Figure 10.12: c Chart for Defects per Unit

Table 10.6: Attribute Control C

| Chart       | Center Line                       | UCL                                              |
|-------------|-----------------------------------|--------------------------------------------------|
| **p chart** | $\bar{p} = \frac{\sum d}{\sum n}$ | $\bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$ |

|                     |                                   |                                          |
|---------------------|-----------------------------------|------------------------------------------|
| <b>**np chart**</b> | $\bar{np} = \frac{\sum d}{k}$     | $\bar{np} + 3\sqrt{\bar{np}(1-\bar{p})}$ |
| <b>**c chart**</b>  | $\bar{c} = \frac{\sum c}{k}$      | $\bar{c} + 3\sqrt{\bar{c}}$              |
| <b>**u chart**</b>  | $\bar{u} = \frac{\sum c}{\sum n}$ | $\bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$    |

## 10.7 Interpreting Control Charts

### 10.7.1 Out-of-Control Signals (Western Electric Rules)

A point doesn't have to be outside the control limits to indicate a problem. The **Western Electric Rules** identify non-random patterns:

Table 10.7: Western Electric Rules for Control Chart Interpretation

| Rule   | Pattern                               | Description                                            |
|--------|---------------------------------------|--------------------------------------------------------|
| Rule 1 | **Point beyond control limit**        | One point above UCL or below LCL                       |
| Rule 2 | **Zone A: 2 of 3 consecutive points** | 2 out of 3 consecutive points in Zone A ( $>2\sigma$ ) |
| Rule 3 | **Zone B: 4 of 5 consecutive points** | 4 out of 5 consecutive points in Zone B ( $>1\sigma$ ) |
| Rule 4 | **8 consecutive points on one side**  | 8 consecutive points above or below the center line    |

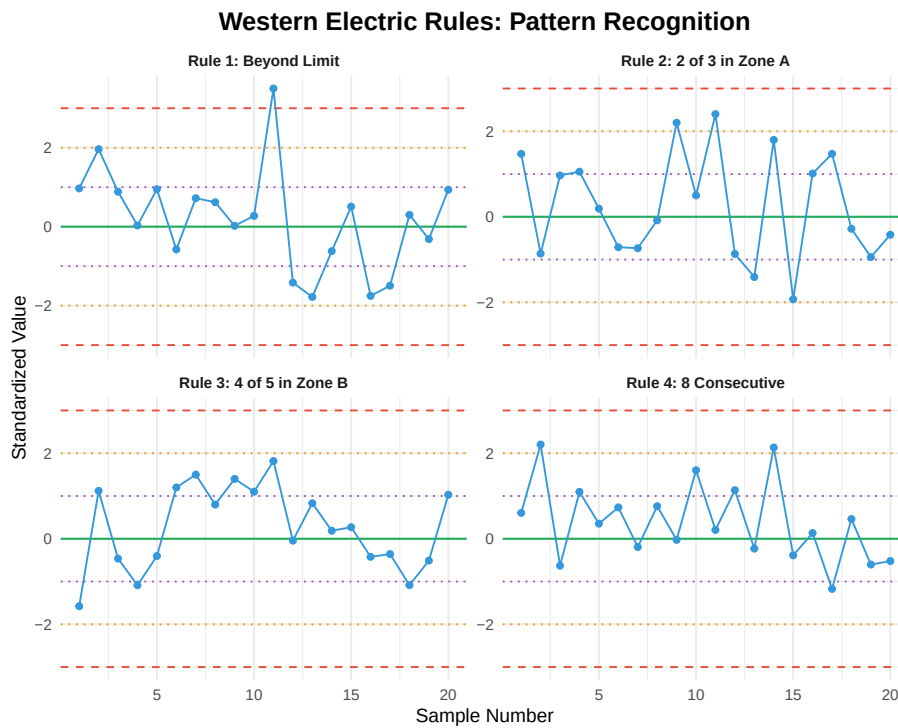


Figure 10.13: Visual Examples of Western Electric Rules

### 10.7.2 Additional Patterns to Watch For

Table 10.8: Additional Control Chart Patterns

| Pattern                   | Description                                                    | Possible Causes                     |
|---------------------------|----------------------------------------------------------------|-------------------------------------|
| <b>**Trend**</b>          | 6 or more consecutive points steadily increasing or decreasing | Tool wear, temperature              |
| <b>**Cycle**</b>          | Repeating high-low pattern                                     | Shift changes, batch effect         |
| <b>**Stratification**</b> | Points consistently near center line (too little variation)    | Incorrect limits, data manipulation |
| <b>**Mixture**</b>        | Points avoiding center line (bimodal distribution)             | Two machines, two operators         |

Interactive Exercise: Identify the Out-of-Control Condition

**Look at this sequence of points (standardized values):**

0.5, 0.8, 1.2, 0.9, 1.4, 1.6, 1.3, 1.8, -0.2, 0.3

**Questions:**

1. Are any points beyond  $\pm 3$  ?
2. Is there a Rule 4 violation (8 consecutive on one side)?

3. What pattern do you see?

**Answers:**

- 1. No — all points are within  $\pm 3$
- 2. **YES** — Points 1-8 are all positive (above center line)
- 3. This is a **Rule 4 violation**: 8 consecutive points above the center line indicates a likely process shift upward

**Action:** Investigate what changed — new material lot? Different operator? Equipment adjustment?

## 10.8 Process Capability

### 10.8.1 What is Process Capability?

**Process capability** compares the process performance to the specification limits. It answers: “Can this process consistently produce parts within tolerance?”

Warning: Using ``size`` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use ``linewidth`` instead.

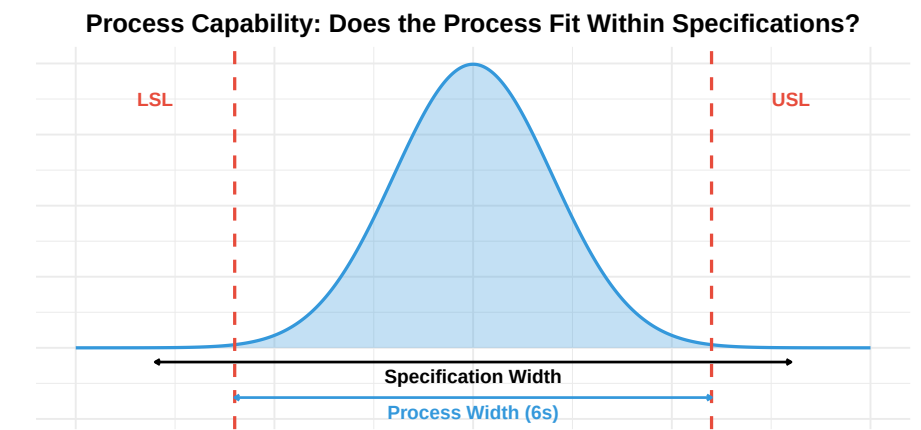


Figure 10.14: Process Capability: Comparing Process to Specifications

### 10.8.2 Capability Indices

Table 10.9: Process Capability

| Index | Formula |
|-------|---------|
|-------|---------|

|                |                                                                                 |           |
|----------------|---------------------------------------------------------------------------------|-----------|
| <b>**Cp**</b>  | $\frac{USL - LSL}{6\sigma}$                                                     | Potential |
| <b>**Cpk**</b> | $\min\left(\frac{USL - \bar{X}}{3\sigma}, \frac{\bar{X} - LSL}{3\sigma}\right)$ | Actual ca |
| <b>**Pp**</b>  | $\frac{USL - LSL}{6s}$                                                          | Overall p |
| <b>**Ppk**</b> | $\min\left(\frac{USL - \bar{X}}{3s}, \frac{\bar{X} - LSL}{3s}\right)$           | Overall p |

10.8.3 Cp vs. Cpk: Why Both Matter

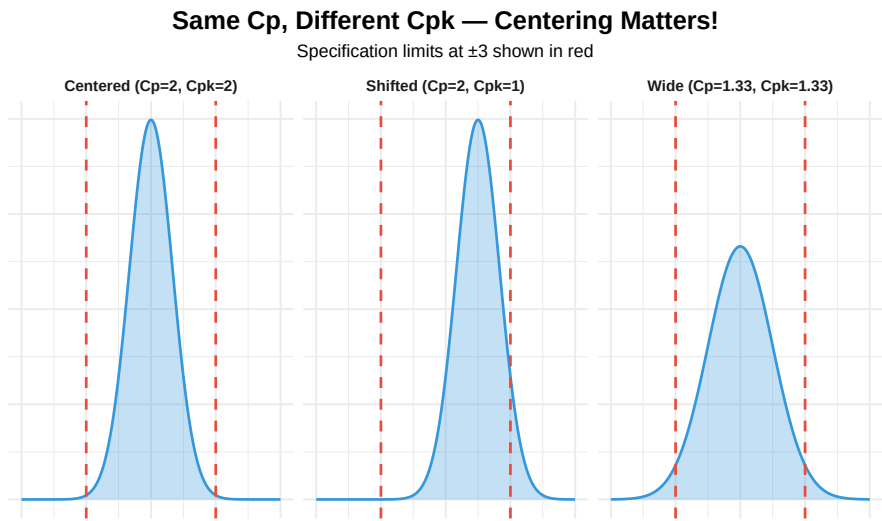


Figure 10.15: Cp vs. Cpk: The Importance of Centering

10.8.4 Capability Example Calculation

```
# Example: Shaft diameter specification 25.00 ± 0.15 mm
# Process data from 25 subgroups of 5

USL <- 25.15
LSL <- 24.85
Target <- 25.00

# Using earlier X-bar/R data
X_bar_cap <- mean(xbar)
R_bar_cap <- mean(R)

# Estimate sigma using R-bar/d2 (d2 = 2.326 for n=5)
d2 <- 2.326
sigma_within <- R_bar_cap / d2
```

```

cat("Process Statistics:\n")
cat("  Mean (X-bar):", round(X_bar_cap, 4), "mm\n")
cat("  Sigma (estimated):", round(sigma_within, 4), "mm\n\n")

# Calculate Cp
Cp <- (USL - LSL) / (6 * sigma_within)
cat("Cp =", round(Cp, 2), "\n")

# Calculate Cpk
Cpu <- (USL - X_bar_cap) / (3 * sigma_within)
Cpl <- (X_bar_cap - LSL) / (3 * sigma_within)
Cpk <- min(Cpu, Cpl)

cat("Cpu =", round(Cpu, 2), "\n")
cat("Cpl =", round(Cpl, 2), "\n")
cat("Cpk =", round(Cpk, 2), "\n\n")

# Interpretation
cat("Interpretation:\n")
if (Cpk >= 1.33) {
  cat("Process is CAPABLE (Cpk >= 1.33)\n")
} else if (Cpk >= 1.00) {
  cat("Process is marginally CAPABLE (1.00 <= Cpk < 1.33)\n")
} else {
  cat("Process is NOT CAPABLE (Cpk < 1.00)\n")
}

```

```

Process Statistics:
  Mean (X-bar): 25.004 mm
  Sigma (estimated): 0.044 mm

```

```

Cp = 1.14
Cpu = 1.11
Cpl = 1.17
Cpk = 1.11

```

```

Interpretation:
Process is marginally CAPABLE (1.00 <= Cpk < 1.33)

```

### 10.8.5 Capability Index Interpretation

Table 10.10: Capability Index Interpretation Guide

| Cpk Value | Sigma Level | PPM Defective | Interpretation                     |
|-----------|-------------|---------------|------------------------------------|
| < 1.00    | < 3         | > 2,700       | Not capable — improvement required |

|             |       |             |                                        |              |
|-------------|-------|-------------|----------------------------------------|--------------|
| 1.00 - 1.33 | 3 - 4 | 2,700 - 64  | Marginally capable — monitor closely   | Consumer p   |
| 1.33 - 1.67 | 4 - 5 | 64 - 0.6    | Capable — meets typical requirements   | General ma   |
| 1.67 - 2.00 | 5 - 6 | 0.6 - 0.002 | Good capability — automotive/aerospace | Automotive   |
| > 2.00      | > 6   | < 0.002     | Excellent — Six Sigma level            | Aerospace, r |

Question: When to use Cp/Cpk vs. Pp/Ppk?

**Cp/Cpk (Process Capability):** - Uses within-subgroup variation (estimated from  $\bar{R}/d_2$ ) - Represents the POTENTIAL capability if process is in control - Use for ongoing process monitoring - Requires process to be in statistical control

**Pp/Ppk (Process Performance):** - Uses overall standard deviation ( $s$ ) of all data - Represents ACTUAL performance including all variation - Use for initial capability studies or when process stability is unknown - Includes both common and special cause variation

**Rule of Thumb:** - If  $Pp \geq Cp$  and  $Ppk \geq Cpk \rightarrow$  Process is stable - If  $Pp < Cp$  or  $Ppk < Cpk \rightarrow$  Process has special cause variation (not in control)

## 10.9 Responding to Out-of-Control Signals

### 10.9.1 The Response Process

#### Out-of-Control Response Procedure

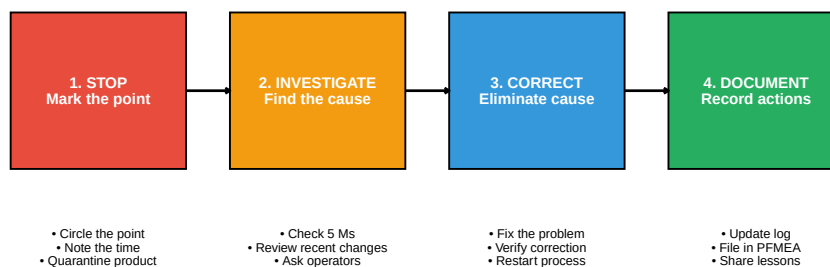


Figure 10.16: Responding to Out-of-Control Conditions

### 10.9.2 Investigation Checklist (5 Ms + E)

Table 10.11: Investigation Checklist: 5 Ms + E

| Category               | Questions to Ask                                                 |
|------------------------|------------------------------------------------------------------|
| <b>**Man**</b>         | Different operator? New employee? Fatigue? Training issue?       |
| <b>**Machine**</b>     | Equipment malfunction? Tool wear? Maintenance due? Setup change? |
| <b>**Material**</b>    | New lot? Different supplier? Out-of-spec material?               |
| <b>**Method**</b>      | Procedure changed? Parameter drift? Wrong program?               |
| <b>**Measurement**</b> | Gauge drift? Calibration due? Different inspector?               |
| <b>**Environment**</b> | Temperature change? Humidity? Vibration? Contamination?          |

## 10.10 SPC Implementation

### 10.10.1 Steps to Implement SPC

Table 10.12: SPC Implementation Steps

| Step | Action                                      | Key Considerations                                |
|------|---------------------------------------------|---------------------------------------------------|
| 1    | Select the process and characteristic       | Start with critical/problem processes             |
| 2    | Determine measurement system adequacy (MSA) | Gauge R&R must be acceptable (< 10%)              |
| 3    | Collect initial data (25+ subgroups)        | Process must be running normally                  |
| 4    | Calculate trial control limits              | Limits based on actual process, not specification |
| 5    | Identify and remove special causes          | Investigate each out-of-control point             |
| 6    | Recalculate limits with stable data         | These become the standard limits                  |
| 7    | Implement ongoing monitoring                | Train operators, establish reaction plan          |
| 8    | Review and improve continuously             | Update limits when process improves               |

### 10.10.2 Common SPC Mistakes

Table 10.13: Common SPC Mistakes and Why They Fail

| Mistake                                                 | Why It's Wrong                                                     |
|---------------------------------------------------------|--------------------------------------------------------------------|
| <b>**Using specification limits as control limits**</b> | Control limits are based on process performance, not specification |
| <b>**Changing limits too frequently**</b>               | Limits should only change when process fundamentally changes       |
| <b>**Ignoring the R chart**</b>                         | R chart often signals problems before X-bar chart                  |
| <b>**Over-adjusting (tampering)**</b>                   | Adjusting for common cause variation increases variation           |
| <b>**Not acting on signals**</b>                        | Defeats the purpose of SPC                                         |



## 10.11 Summary

Table 10.14: SPC Key Concepts Summary

| Topic                       | Key Points                                                                               |
|-----------------------------|------------------------------------------------------------------------------------------|
| <b>**Variation**</b>        | Common cause (inherent) vs. special cause (assignable); respond differently              |
| <b>**Control Charts**</b>   | Graph with UCL/LCL at $\pm 3$ ; distinguishes between variation types                    |
| <b>**Variable Charts**</b>  | X-bar/R (n=2-9), X-bar/S (n 10), I-MR (n=1)                                              |
| <b>**Attribute Charts**</b> | p (proportion), np (count defectives), c (defects), u (defects/unit)                     |
| <b>**Interpretation**</b>   | Western Electric rules: beyond limits, 2 of 3 in Zone A, 4 of 5 in Zone B, 8 consecutive |
| <b>**Capability**</b>       | Cp (potential), Cpk (actual); 1.33 is capable; 1.67 for automotive                       |

## 10.12 Review Questions

### 10.12.1 Conceptual Questions

Question 1: What is the difference between common cause and special cause variation?

**Common Cause Variation:** - Inherent to the process - Always present - Random and predictable - Many small factors - System-level improvement required - Examples: machine vibration, material variation, ambient temperature

**Special Cause Variation:** - External to normal process - Intermittent or new - Non-random patterns - Specific assignable cause - Local action to eliminate - Examples: tool breakage, new operator, bad material lot

**Key Point:** Only act on special causes. Attempting to adjust for common cause variation (tampering) makes the process worse.

Question 2: Why are control limits set at  $\pm 3$  sigma rather than  $\pm 2$  sigma?

**At  $\pm 3$ :** - 99.73% of data falls within limits (when in control) - Only 0.27% false alarm rate (about 3 per 1000 points) - Good balance between detecting real problems and avoiding false alarms

**If we used  $\pm 2$ :** - 95.45% of data within limits - 4.55% false alarm rate (about 45 per 1000 points!) - Too many false alarms  $\rightarrow$  operators ignore the chart

**If we used  $\pm 4$ :** - 99.99% within limits - Almost no false alarms - But we'd miss real problems (low sensitivity)

**The  $\pm 3$  choice balances:** - Risk of false alarm (treating common cause as special) - Risk of missed signal (treating special cause as common)

Question 3: When would you use an I-MR chart instead of an X-bar/R chart?

**Use I-MR Chart When:**

1. **Sample size is  $n = 1$**  — only one measurement per time period
2. **Destructive testing** — can't afford multiple samples
3. **Expensive measurement** — time or cost prohibitive
4. **Continuous process** — each reading is independent (batch process)
5. **Long intervals** — measurements far apart in time
6. **Homogeneous batches** — within-batch variation is negligible

**Examples:** - Daily chemical batch analysis - Weekly calibration check - Each lot incoming inspection (one sample per lot) - Continuous process temperature readings

**Caution:** I-MR charts are less sensitive than X-bar charts because averaging in subgroups reduces noise.

**10.12.2 Calculation Questions**

Question 4: Calculate control limits for an X-bar/R chart given the following data.

**Given:** - 20 subgroups of  $n = 4$  measurements - Grand mean (X-double-bar) = 50.25 - Average range (R-bar) = 2.40 - Constants for  $n = 4$ :  $A_2 = 0.729$ ,  $D_3 = 0$ ,  $D_4 = 2.282$

**Calculate UCL and LCL for both charts.**

**Solution:**

**X-bar Chart:**

$$UCL = \bar{\bar{X}} + A_2 \times \bar{R}$$

$$UCL = 50.25 + 0.729 \times 2.40 = 50.25 + 1.75 = 52.00$$

$$LCL = \bar{\bar{X}} - A_2 \times \bar{R}$$

$$LCL = 50.25 - 0.729 \times 2.40 = 50.25 - 1.75 = 48.50$$

$$CL = 50.25$$

**R Chart:**

$$UCL = D_4 \times \bar{R} = 2.282 \times 2.40 = 5.48$$

$$LCL = D_3 \times \bar{R} = 0 \times 2.40 = 0$$

$$CL = 2.40$$

Question 5: Calculate  $C_p$  and  $C_{pk}$  for a process with the following data.

**Given:** - Specification:  $100 \pm 5$  ( $USL = 105$ ,  $LSL = 95$ ) - Process mean = 102  
- Process standard deviation ( $\sigma$ ) = 1.5

**Solution:**

**Cp (Potential Capability):**

$$C_p = (USL - LSL) / (6 \sigma)$$

$$C_p = (105 - 95) / (6 \times 1.5)$$

$$C_p = 10 / 9 = 1.11$$

**Cpk (Actual Capability):**

$$C_{pu} = (USL - \bar{X}) / (3 \sigma) = (105 - 102) / (3 \times 1.5) = 3 / 4.5 = 0.67$$

$$C_{pl} = (\bar{X} - LSL) / (3 \sigma) = (102 - 95) / (3 \times 1.5) = 7 / 4.5 = 1.56$$

$$C_{pk} = \min(C_{pu}, C_{pl}) = \min(0.67, 1.56) = 0.67$$

**Interpretation:** -  $C_p = 1.11$  suggests process spread could fit within specs IF centered -  $C_{pk} = 0.67$  shows process is NOT capable because it's shifted toward USL - Action: Center the process closer to 100 (target)

Question 6: A p chart has  $\bar{p} = 0.04$  and sample size  $n = 200$ . Calculate control limits.

**Solution:**

**Center Line:**

$$CL = \bar{p} = 0.04 \text{ (4\% defective)}$$

**Upper Control Limit:**

$$UCL = \bar{p} + 3 \times \sqrt{\bar{p} \times (1 - \bar{p}) / n}$$

$$UCL = 0.04 + 3 \times \sqrt{(0.04 \times 0.96) / 200}$$

$$UCL = 0.04 + 3 \times \sqrt{0.000192}$$

$$UCL = 0.04 + 3 \times 0.01386$$

$$UCL = 0.04 + 0.0416 = 0.0816 \text{ (8.16\%)}$$

**Lower Control Limit:**

$$LCL = \bar{p} - 3 \times \sqrt{\bar{p} \times (1 - \bar{p}) / n}$$

$$LCL = 0.04 - 0.0416 = -0.0016$$

Since LCL cannot be negative:  $LCL = 0$

**Final Limits:** -  $UCL = 8.16\%$  -  $CL = 4.00\%$  -  $LCL = 0\%$

### 10.12.3 Application Questions

Question 7: You observe 8 consecutive points above the center line but all within control limits. What does this indicate?

**This is a Rule 4 violation (Western Electric Rules).**



**Rule:** If R chart is out of control, fix that FIRST before interpreting the X-bar chart.

---

## 10.13 References

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  - ASTM E2587. Standard Practice for Use of Control Charts in Statistical Process Control.
-



## Chapter 11

# Measurement Systems Analysis

---

### 11.1 Learning Objectives

After completing this chapter, you will be able to:

1. Explain the importance of measurement system quality in process control
  2. Identify the components of measurement system variation
  3. Define and calculate accuracy, bias, linearity, stability, repeatability, and reproducibility
  4. Conduct and interpret a Gauge R&R study using both the Range and ANOVA methods
  5. Determine measurement system acceptability using %GRR and ndc criteria
  6. Apply attribute measurement system analysis for pass/fail decisions
  7. Recognize common sources of measurement error and implement improvements
  8. Understand the relationship between MSA and SPC effectiveness
- 

### 11.2 Introduction to Measurement Systems Analysis

**Measurement Systems Analysis (MSA)** is the science of evaluating the quality of measurement processes. Before we can trust our data for process

control, quality decisions, or capability studies, we must first verify that our measurement system is capable of providing reliable results.

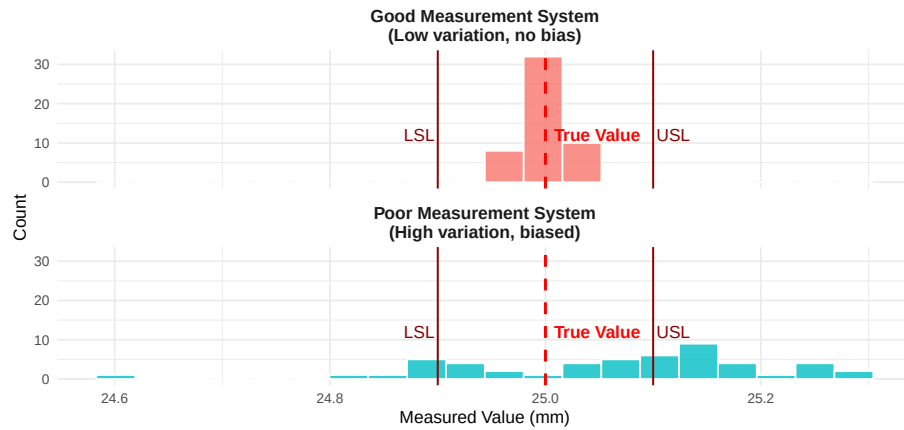
### 11.2.1 Why MSA Matters

“You can’t improve what you can’t measure - but you also can’t improve what you measure incorrectly.”

Consider this scenario:

#### The Same Parts Measured by Two Different Systems

Poor measurement systems lead to incorrect accept/reject decisions



### 11.2.2 The Cost of Poor Measurement

Table 11.1: Business Impact of Poor Measurement System

| Impact                               | Description                                                    |
|--------------------------------------|----------------------------------------------------------------|
| <b>False Accepts (Type II Error)</b> | Bad parts passed as good; reach customer                       |
| <b>False Rejects (Type I Error)</b>  | Good parts rejected as bad; scrapped or reworked               |
| <b>Process Adjustment Errors</b>     | Adjusting process based on measurement noise, not real changes |
| <b>Capability Misassessment</b>      | Cp/Cpk calculations are wrong; true capability unknown         |
| <b>Customer Complaints</b>           | Inconsistent quality due to measurement-based decisions        |

### 11.2.3 MSA in Industry Standards

MSA is required by multiple quality standards:

- **IATF 16949** (Automotive): Requires MSA for all measurement systems referenced in control plans
- **AS9100** (Aerospace): Requires demonstrated measurement capability
- **ISO 9001**: Requires monitoring and measurement resources to be suitable



- **FDA 21 CFR Part 820** (Medical Devices): Requires validated measurement equipment

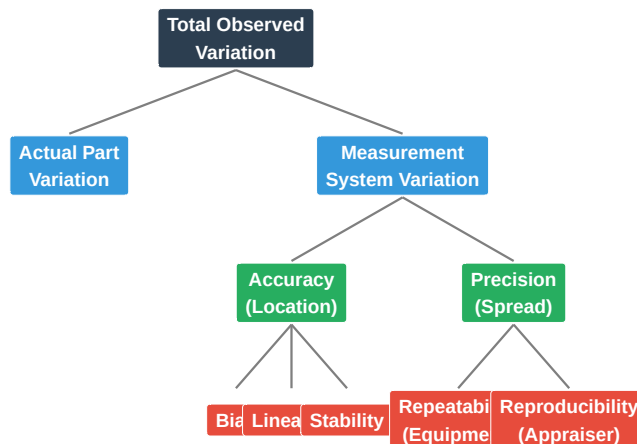
## 11.3 Components of Measurement Variation

Total measurement variation comes from multiple sources. Understanding these components is essential for improvement.

### 11.3.1 The Measurement System Model

#### Components of Measurement System Variation

Total variation = Part variation + Measurement system variation



### 11.3.2 Accuracy vs. Precision

These are often confused but are fundamentally different concepts:

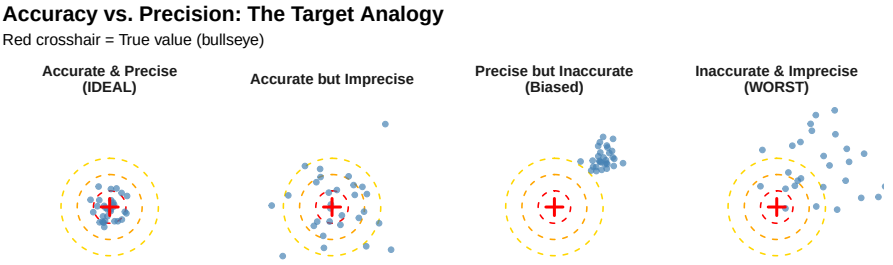


Table 11.2: Accuracy v

|                  | Definition                                             | Affected By                     |
|------------------|--------------------------------------------------------|---------------------------------|
| <b>Accuracy</b>  | How close measurements are to the true value (average) | Calibration, bias, linearity    |
| <b>Precision</b> | How close measurements are to each other (spread)      | Repeatability, reproducibility, |

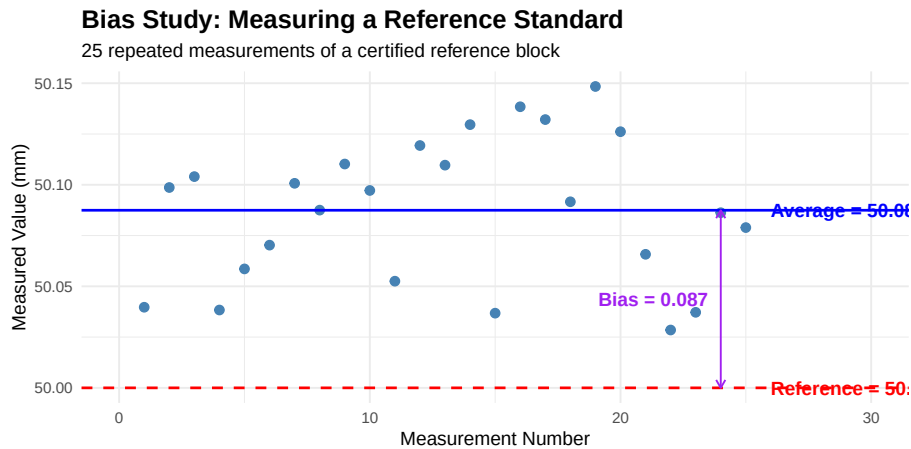
## 11.4 Accuracy Studies

Accuracy studies evaluate how well the measurement system indicates the true value.

### 11.4.1 Bias

**Bias** is the systematic error - the difference between the average of measurements and the true (reference) value.

$$\text{Bias} = \bar{x}_{\text{observed}} - x_{\text{reference}}$$



Live cell — try different reference and measured values.

```
#| label: bias-calculation
# Bias study calculation
reference_value <- 50.000 # Certified reference standard
measurements <- c(50.08, 50.07, 50.09, 50.08, 50.06, 50.10, 50.07, 50.08,
                  50.09, 50.08, 50.07, 50.11, 50.08, 50.09, 50.07)

# Calculate bias
average_measured <- mean(measurements)
bias <- average_measured - reference_value

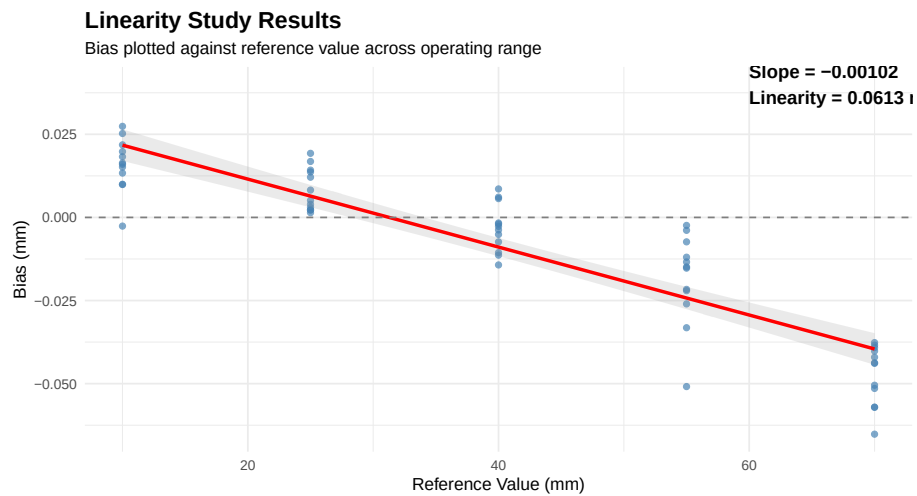
cat("Reference Value:", reference_value, "mm\n")
cat("Average Measured:", round(average_measured, 4), "mm\n")
cat("Bias:", round(bias, 4), "mm\n")
cat("Bias as % of Tolerance (±0.10):", round(abs(bias) / 0.10 * 100, 1), "%\n")

# Statistical test for significance
t_test <- t.test(measurements, mu = reference_value)
cat("\nt-test p-value:", round(t_test$p.value, 4))
if(t_test$p.value < 0.05) {
  cat(" - Bias is statistically significant\n")
} else {
  cat(" - Bias is not statistically significant\n")
}
```

### 11.4.2 Linearity

**Linearity** measures whether bias changes across the measurement range. A gauge might be accurate at one end of its range but biased at the other.

```
`geom_smooth()` using formula = 'y ~ x'
```



Live cell — try a different set of reference points and biases.

```
#| label: linearity-calc
# Linearity calculation
reference <- c(10, 25, 40, 55, 70)
avg_bias <- c(0.020, 0.008, -0.002, -0.018, -0.048)

# Fit linear regression
model <- lm(avg_bias ~ reference)

# Calculate linearity
slope <- coef(model)[2]
range_used <- max(reference) - min(reference)
linearity <- abs(slope) * range_used

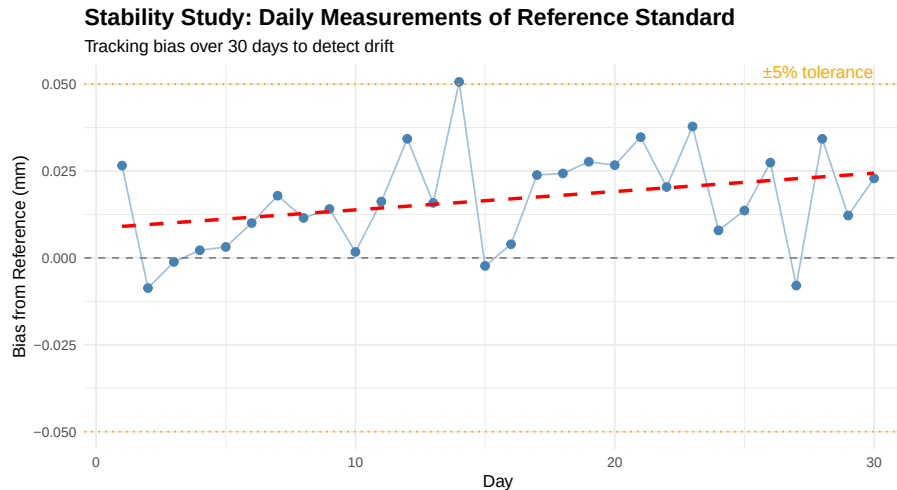
cat("Regression equation: Bias =", round(coef(model)[1], 4), "+",
    round(slope, 5), "× Reference\n")
cat("Linearity (slope × range):", round(linearity, 4), "mm\n")
cat("As % of tolerance (±0.10):", round(linearity / 0.10 * 100, 1), "%\n")

# R-squared
cat("R² =", round(summary(model)$r.squared, 3),
    "- indicates", ifelse(summary(model)$r.squared > 0.7, "significant", "minor"),
    "linearity issue\n")
```

### 11.4.3 Stability

**Stability** (also called drift) measures whether the measurement system's accuracy changes over time.

```
`geom_smooth()` using formula = 'y ~ x'
```



#### 11.4.4 Stability Control Chart

For ongoing stability monitoring, use a control chart. Live cell — try different measurements or a different historical sigma.

```
#| label: stability-control-chart
# Stability monitoring with control chart
reference <- 25.000
measurements <- c(25.012, 25.008, 25.015, 25.018, 25.022, 25.019, 25.025,
                  25.028, 25.031, 25.026, 25.033, 25.038, 25.035, 25.041, 25.039)
days <- 1:15

# Calculate control limits (based on historical sigma)
historical_sigma <- 0.010
x_bar <- mean(measurements[1:5]) # Baseline from first 5 days
UCL <- x_bar + 3 * historical_sigma
LCL <- x_bar - 3 * historical_sigma

cat("Baseline Average:", round(x_bar, 4), "\n")
cat("UCL:", round(UCL, 4), "\n")
cat("LCL:", round(LCL, 4), "\n")

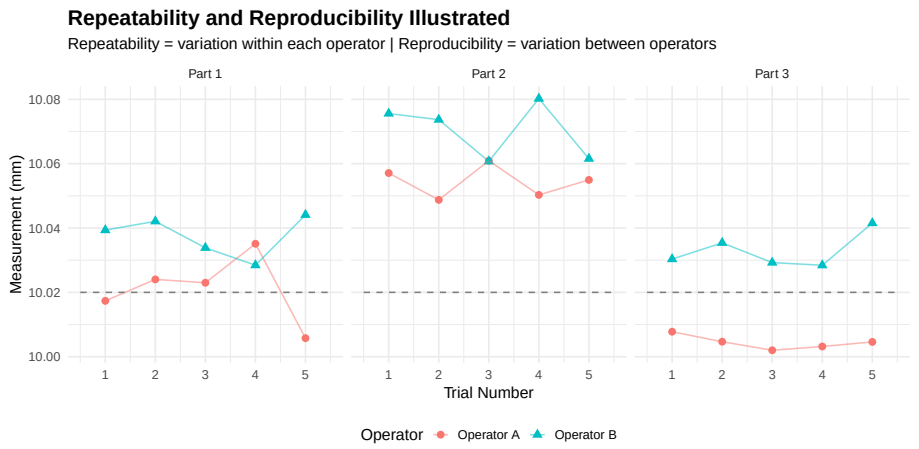
# Check for out-of-control
out_of_control <- which(measurements > UCL | measurements < LCL)
if(length(out_of_control) > 0) {
  cat("\nOut-of-control points on days:", out_of_control, "\n")
  cat("Drift detected - recalibration recommended\n")
} else {
```

```
cat("\nNo out-of-control points - measurement system stable\n")
}
```

## 11.5 Precision Studies: Gauge R&R

**Gauge R&R** (Repeatability and Reproducibility) is the most common MSA study. It quantifies precision variation.

### 11.5.1 Repeatability vs. Reproducibility



Table

| Component                   | Also Known As                                   | Definition               |
|-----------------------------|-------------------------------------------------|--------------------------|
| <b>Repeatability (EV)</b>   | Equipment Variation, Within-operator variation  | Variation when same op   |
| <b>Reproducibility (AV)</b> | Appraiser Variation, Between-operator variation | Variation when different |
| <b>Gauge R&amp;R (GRR)</b>  | Total measurement system variation              | Combined repeatability   |

### 11.5.2 Gauge R&R Study Design

Table 11.4: Standard Gauge R&R Study Design Parameters

| Parameter                  | Typical_Value | Notes                                    |
|----------------------------|---------------|------------------------------------------|
| <b>Number of Parts</b>     | 10            | Minimum 5, 10 recommended for statisti   |
| <b>Number of Operators</b> | 3             | Minimum 2, 3 is standard for detecting c |
| <b>Number of Trials</b>    | 3             | Minimum 2, 3 is standard for repeatabili |

|                           |                              |                                                     |
|---------------------------|------------------------------|-----------------------------------------------------|
| <b>Total Measurements</b> | $10 \times 3 \times 3 = 90$  | More measurements = better precision in estimates   |
| <b>Part Selection</b>     | Cover the full process range | Parts should represent typical production variation |
| <b>Randomization</b>      | Randomize measurement order  | Blind measurements when possible; prevents bias     |

---

### 11.5.3 Range Method (Short Form)

The Range Method is a quick approximation useful for initial screening.

Live cell — try different measurement data for the two operators.

```
#| label: range-method
# Range Method Gauge R&R Calculation
# Two operators, five parts, two trials each

# Data: Measurements by each operator (2 trials per part)
operator_a <- matrix(c(
  0.85, 0.87, # Part 1
  0.92, 0.91, # Part 2
  0.78, 0.80, # Part 3
  0.95, 0.96, # Part 4
  0.88, 0.85 # Part 5
), ncol = 2, byrow = TRUE)

operator_b <- matrix(c(
  0.86, 0.88, # Part 1
  0.93, 0.92, # Part 2
  0.80, 0.79, # Part 3
  0.94, 0.97, # Part 4
  0.87, 0.88 # Part 5
), ncol = 2, byrow = TRUE)

# Step 1: Calculate ranges for each part-operator combination
range_a <- apply(operator_a, 1, function(x) max(x) - min(x))
range_b <- apply(operator_b, 1, function(x) max(x) - min(x))

# Step 2: Average range
R_bar <- mean(c(range_a, range_b))

# Step 3: Calculate Repeatability (EV)
# d2 for 2 trials = 1.128
d2_trials <- 1.128
EV <- R_bar / d2_trials

# Step 4: Calculate part averages by operator
avg_a <- rowMeans(operator_a)
avg_b <- rowMeans(operator_b)
```

```

# Step 5: Range of operator averages
X_diff <- abs(mean(avg_a) - mean(avg_b))

# d2 for 2 operators = 1.128
d2_operators <- 1.128
n_parts <- 5
n_trials <- 2

# Step 6: Reproducibility (AV)
AV_squared <- (X_diff / d2_operators)^2 - (EV^2 / (n_parts * n_trials))
AV <- sqrt(max(0, AV_squared))

# Step 7: Gauge R&R
GRR <- sqrt(EV^2 + AV^2)

# Step 8: Express as % of tolerance
tolerance <- 0.30 # Example tolerance of ±0.15 = 0.30 total

cat("=== Range Method Gauge R&R Results ===\n\n")
cat("Average Range (R-bar):", round(R_bar, 4), "\n")
cat("Repeatability (EV):", round(EV, 4), "\n")
cat("Reproducibility (AV):", round(AV, 4), "\n")
cat("Gauge R&R (GRR):", round(GRR, 4), "\n\n")
cat("%EV (of tolerance):", round(EV / tolerance * 100 * 5.15, 1), "%\n")
cat("%AV (of tolerance):", round(AV / tolerance * 100 * 5.15, 1), "%\n")
cat("%GRR (of tolerance):", round(GRR / tolerance * 100 * 5.15, 1), "%\n")

```

#### 11.5.4 ANOVA Method (Full Study)

**ANOVA** (Analysis of Variance) is a statistical technique that partitions the total observed variation in a dataset into distinct components, each attributable to a specific source. In the context of Gauge R&R, ANOVA separates the total measurement variation into contributions from:

- **Parts** (the actual differences between items being measured)
- **Operators** (systematic differences between people taking measurements)
- **Part × Operator interaction** (whether certain operators measure certain parts differently)
- **Repeatability** (random error when the same operator measures the same part multiple times)

##### 11.5.4.1 Why Use ANOVA for Gauge R&R?

The fundamental question in any measurement system analysis is: *“How much of the variation I observe comes from my measurement system versus the ac-*



*tual differences between parts?”* ANOVA answers this question by statistically decomposing variance—it calculates how much each source contributes to what we observe.

Consider the alternative: if we simply looked at the spread of all measurements, we wouldn’t know whether a wide spread indicates highly variable parts (good—our gauge can detect differences) or a poor measurement system (bad—we’re adding noise). ANOVA untangles these effects mathematically.

#### 11.5.4.2 Why ANOVA is the Preferred Method

While the Range Method provides a quick estimate, ANOVA offers several critical advantages for Gauge R&R studies:

| Aspect                      | Range Method                              | ANOVA Method                                   |
|-----------------------------|-------------------------------------------|------------------------------------------------|
| <b>Interaction effects</b>  | Cannot detect                             | Identifies Part $\times$ Operator interactions |
| <b>Statistical rigor</b>    | Approximation using range-based estimates | Exact variance component estimation            |
| <b>Hypothesis testing</b>   | Not available                             | Provides F-tests and p-values for each source  |
| <b>Accuracy</b>             | Tends to overestimate GRR                 | More precise variance estimates                |
| <b>Information richness</b> | Basic EV/AV breakdown                     | Full decomposition with confidence intervals   |
| <b>Industry acceptance</b>  | Screening only                            | Required for formal MSA studies (AIAG)         |

The ability to detect **interaction effects** is particularly important. If Operator A consistently measures Part 3 higher than Operator B does (but they agree on other parts), this indicates a training issue or technique difference that the Range Method would miss entirely. ANOVA flags this interaction, enabling targeted corrective action.

#### 11.5.4.3 How ANOVA Works in Gauge R&R

ANOVA uses the **F-statistic** to test whether variation between groups (e.g., between operators) is significantly larger than variation within groups (repeatability). The mathematical basis is:

$$F = \frac{\text{Mean Square Between Groups}}{\text{Mean Square Within Groups}} = \frac{MS_{\text{factor}}}{MS_{\text{error}}}$$

A large F-value indicates that the factor (Part, Operator, or their interaction) contributes meaningfully to the observed variation. The p-value tells us the probability of seeing such an F-value if the factor had no real effect.

From the ANOVA table's Mean Squares, we extract **variance components** ( $\sigma^2$ ) for each source, then combine them to calculate the standard metrics: %EV (repeatability), %AV (reproducibility), %GRR, and the number of distinct categories (ndc).

```
# Full Gauge R&R using ANOVA method
# 10 parts, 3 operators, 3 trials

set.seed(42)

# Generate realistic GRR data
n_parts <- 10
n_operators <- 3
n_trials <- 3

# Part true values (most of the variation)
part_effects <- rnorm(n_parts, 0, 0.05)

# Operator effects (some variation)
operator_effects <- c(-0.008, 0.003, 0.005)

# Create dataset
grr_data <- expand.grid(
  Part = 1:n_parts,
  Operator = paste0("Op", 1:n_operators),
  Trial = 1:n_trials
)

# Add measurements
grr_data$Measurement <- 10 +
  part_effects[grr_data$Part] +
  operator_effects[as.numeric(factor(grr_data$Operator))] +
  rnorm(nrow(grr_data), 0, 0.006) # Repeatability error

# Convert to factors
grr_data$Part <- factor(grr_data$Part)
grr_data$Operator <- factor(grr_data$Operator)

# Fit ANOVA model
anova_model <- aov(Measurement ~ Part + Operator + Part:Operator, data = grr_data)

# Extract variance components
anova_table <- anova(anova_model)
print(anova_table)
```

Analysis of Variance Table

Response: Measurement

|               | Df | Sum Sq   | Mean Sq   | F value  | Pr(>F)        |
|---------------|----|----------|-----------|----------|---------------|
| Part          | 9  | 0.136635 | 0.0151816 | 343.6163 | < 2.2e-16 *** |
| Operator      | 2  | 0.003430 | 0.0017152 | 38.8223  | 1.52e-11 ***  |
| Part:Operator | 18 | 0.000430 | 0.0000239 | 0.5401   | 0.9263        |
| Residuals     | 60 | 0.002651 | 0.0000442 |          |               |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

```
# Extract Mean Squares from ANOVA
MS_part <- anova_table["Part", "Mean Sq"]
MS_operator <- anova_table["Operator", "Mean Sq"]
MS_interaction <- anova_table["Part:Operator", "Mean Sq"]
MS_error <- anova_table["Residuals", "Mean Sq"]

# Calculate variance components
n_p <- 10 # number of parts
n_o <- 3 # number of operators
n_r <- 3 # number of trials

# Repeatability variance
var_repeatability <- MS_error

# Interaction variance
var_interaction <- max(0, (MS_interaction - MS_error) / n_r)

# Operator variance
var_operator <- max(0, (MS_operator - MS_interaction) / (n_p * n_r))

# Part variance
var_part <- max(0, (MS_part - MS_interaction) / (n_o * n_r))

# Reproducibility = Operator + Interaction
var_reproducibility <- var_operator + var_interaction

# Total Gauge R&R
var_GRR <- var_repeatability + var_reproducibility

# Total variation
var_total <- var_part + var_GRR

# Convert to standard deviations
sigma_repeatability <- sqrt(var_repeatability)
sigma_reproducibility <- sqrt(var_reproducibility)
sigma_GRR <- sqrt(var_GRR)
```

```

sigma_part <- sqrt(var_part)
sigma_total <- sqrt(var_total)

# Calculate study variation (5.15 sigma for 99% of distribution)
SV_repeatability <- 5.15 * sigma_repeatability
SV_reproducibility <- 5.15 * sigma_reproducibility
SV_GRR <- 5.15 * sigma_GRR
SV_part <- 5.15 * sigma_part
SV_total <- 5.15 * sigma_total

cat("\n=== ANOVA Gauge R&R Results ===\n\n")
cat("Variance Components:\n")
cat("  Repeatability:", round(var_repeatability, 8), "\n")
cat("  Reproducibility:", round(var_reproducibility, 8), "\n")
cat("  Part-to-Part:", round(var_part, 8), "\n")
cat("  Total:", round(var_total, 8), "\n")

cat("\nStudy Variation (5.15):\n")
cat("  Repeatability:", round(SV_repeatability, 5), "\n")
cat("  Reproducibility:", round(SV_reproducibility, 5), "\n")
cat("  Gauge R&R:", round(SV_GRR, 5), "\n")
cat("  Part-to-Part:", round(SV_part, 5), "\n")
cat("  Total:", round(SV_total, 5), "\n")

```

```
=== ANOVA Gauge R&R Results ===
```

```
Variance Components:
```

```

  Repeatability: 4.418e-05
  Reproducibility: 5.638e-05
  Part-to-Part: 0.0016842
  Total: 0.00178476

```

```
Study Variation (5.15):
```

```

  Repeatability: 0.03423
  Reproducibility: 0.03867
  Gauge R&R: 0.05164
  Part-to-Part: 0.21135
  Total: 0.21757

```

### 11.5.5 Calculating %GRR and Acceptance Criteria

```

# Calculate %GRR using both methods
tolerance <- 0.30 # Total tolerance

```

```

# Method 1: % of Tolerance (%GRR_Tolerance)
pct_GRR_tol <- (SV_GRR / tolerance) * 100

# Method 2: % of Total Variation (%GRR_TV)
pct_GRR_tv <- (sigma_GRR / sigma_total) * 100

# Number of Distinct Categories (ndc)
ndc <- 1.41 * (sigma_part / sigma_GRR)

cat("=== Gauge R&R Acceptance Criteria ===\n\n")
cat("%GRR (of Tolerance):", round(pct_GRR_tol, 1), "%\n")
cat("%GRR (of Total Variation):", round(pct_GRR_tv, 1), "%\n")
cat("Number of Distinct Categories (ndc):", round(ndc, 1), "%\n\n")

# Acceptance criteria
cat("AIAG Acceptance Guidelines:\n")
cat("      \n")
if(pct_GRR_tol < 10) {
  cat("%GRR < 10%: ACCEPTABLE - Measurement system is acceptable\n")
} else if(pct_GRR_tol < 30) {
  cat("%GRR 10-30%: MARGINAL - May be acceptable based on application\n")
} else {
  cat("%GRR > 30%: UNACCEPTABLE - Measurement system needs improvement\n")
}

if(ndc >= 5) {
  cat("ndc 5: ACCEPTABLE - Adequate discrimination\n")
} else if(ndc >= 2) {
  cat("ndc 2-4: MARGINAL - Limited discrimination ability\n")
} else {
  cat("ndc < 2: UNACCEPTABLE - Cannot distinguish between parts\n")
}

```

=== Gauge R&R Acceptance Criteria ===

%GRR (of Tolerance): 17.2 %

%GRR (of Total Variation): 23.7 %

Number of Distinct Categories (ndc): 5.8

AIAG Acceptance Guidelines:

%GRR 10-30%: MARGINAL - May be acceptable based on application

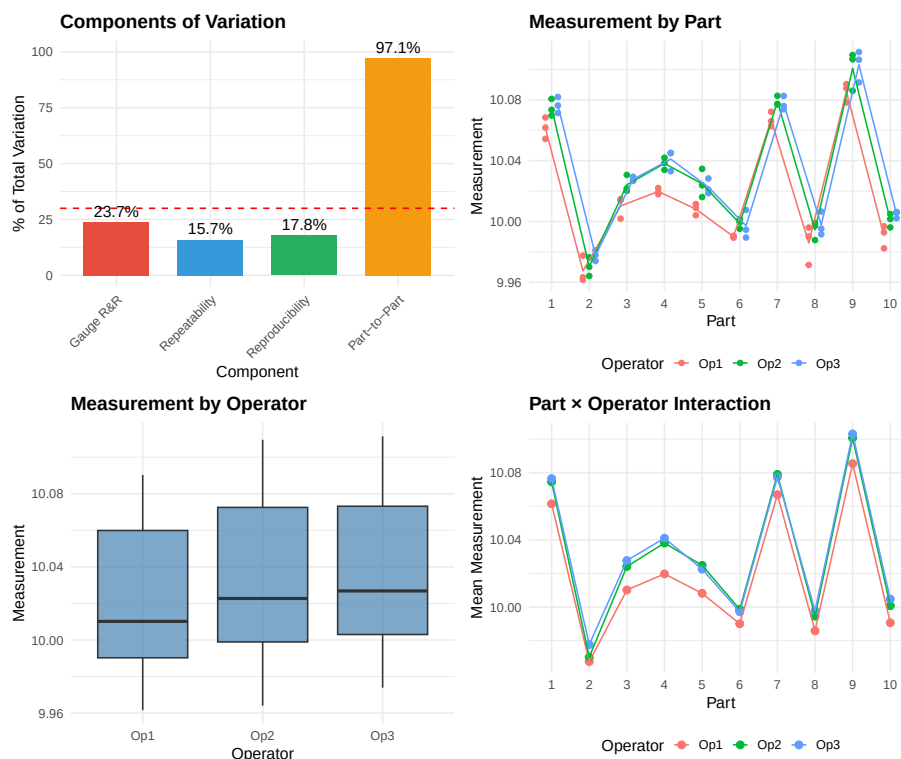
ndc 5: ACCEPTABLE - Adequate discrimination

### 11.5.6 Gauge R&R Acceptance Criteria Summary

Table 11.5: AIAG Gauge R&amp;R Acceptance Guidelines

| Criterion            | Status       | Interpretation                                                      |
|----------------------|--------------|---------------------------------------------------------------------|
| <b>%GRR &lt; 10%</b> | Acceptable   | Measurement system is acceptable for process control and capability |
| <b>%GRR 10-30%</b>   | Marginal     | May be acceptable depending on importance, cost of gauge, repair    |
| <b>%GRR &gt; 30%</b> | Unacceptable | Measurement system is not acceptable; action required               |
| <b>ndc = 5</b>       | Acceptable   | Gauge can adequately distinguish between parts                      |
| <b>ndc = 2-4</b>     | Marginal     | Limited ability to distinguish; may be OK for simple pass/fail      |
| <b>ndc &lt; 2</b>    | Unacceptable | Gauge cannot distinguish between parts; almost useless for control  |

### 11.5.7 Visual Analysis of Gauge R&R



## 11.6 Attribute Measurement Systems

For pass/fail or categorical measurements (visual inspection, go/no-go gauges), we use **Attribute MSA** or **Attribute Agreement Analysis**.

### 11.6.1 Kappa Statistic

When evaluating attribute measurement systems (pass/fail, accept/reject, visual inspection), you might think that **percent agreement** would be sufficient—if two inspectors agree 90% of the time, isn’t that good enough? The problem is that some agreement happens purely by chance, and percent agreement doesn’t account for this.

#### 11.6.1.1 Why Percent Agreement Can Be Misleading

Consider a visual inspection where 95% of parts are good. If an inspector simply guesses “Accept” for every part (without even looking), they would be “correct” 95% of the time. Two such inspectors would agree 90.5% of the time by pure chance:

$$P_e = P(\text{both say Accept}) + P(\text{both say Reject}) = 0.95^2 + 0.05^2 = 0.905$$

This is why we need **Cohen’s Kappa** ( $\kappa$ )—it measures agreement *beyond what we’d expect from chance alone*.

#### 11.6.1.2 The Kappa Formula

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

Where:

- $P_o$  = Observed proportion of agreement (how often they actually agree)
- $P_e$  = Expected proportion of agreement by chance (how often they’d agree if guessing randomly based on their individual tendencies)
- $1 - P_e$  = Maximum possible agreement beyond chance

**Interpreting the formula:** The numerator ( $P_o - P_e$ ) represents *actual agreement minus chance agreement*—the “real” agreement attributable to consistent measurement. The denominator ( $1 - P_e$ ) normalizes this to a 0–1 scale by dividing by the maximum possible improvement over chance.

| Kappa Value  | Meaning                                                         |
|--------------|-----------------------------------------------------------------|
| $\kappa = 1$ | Perfect agreement (everyone agrees, and not by chance)          |
| $\kappa = 0$ | Agreement equals chance—no better than random guessing          |
| $\kappa < 0$ | Agreement is <i>worse</i> than chance (systematic disagreement) |

### 11.6.1.3 When to Use Kappa

Use Kappa when evaluating:

- **Inspector vs. Reference:** Does the operator match the known correct answer? (Most important for MSA)
- **Inspector vs. Inspector:** Do two operators classify the same parts the same way?
- **Inspector vs. Self (Repeatability):** Does the same operator give the same answer when re-inspecting the same part?

For a robust attribute MSA, you should calculate Kappa for all three comparisons. High agreement with other operators but low agreement with the reference standard indicates *consistent but incorrect* inspection—a training problem.

Table 11.6: Kappa Statistic Interpretation Guide

| Kappa              | Interpretation             | Action                           |
|--------------------|----------------------------|----------------------------------|
| $< 0$              | Less than chance agreement | Major issues - do not use        |
| <b>0.00 - 0.20</b> | Slight agreement           | Major improvement needed         |
| <b>0.21 - 0.40</b> | Fair agreement             | Significant improvement needed   |
| <b>0.41 - 0.60</b> | Moderate agreement         | Some improvement needed          |
| <b>0.61 - 0.80</b> | Substantial agreement      | Acceptable for most applications |
| <b>0.81 - 1.00</b> | Almost perfect agreement   | Excellent - acceptable           |

### 11.6.2 Attribute Agreement Study Example

```
# Attribute Agreement Analysis
# 3 operators inspect 30 samples, 2 trials each
# Reference standard (known good/bad) available

set.seed(123)
n_samples <- 30
n_operators <- 3
n_trials <- 2

# Reference classification (10 bad, 20 good)
reference <- c(rep("Reject", 10), rep("Accept", 20))

# Simulate operator decisions (some disagreement)
simulate_decision <- function(ref, accuracy) {
  sapply(ref, function(r) {
    if(runif(1) < accuracy) r else ifelse(r == "Accept", "Reject", "Accept")
  })
}
```



```

# Operators with different accuracy levels
op1_t1 <- simulate_decision(reference, 0.92)
op1_t2 <- simulate_decision(reference, 0.92)
op2_t1 <- simulate_decision(reference, 0.88)
op2_t2 <- simulate_decision(reference, 0.88)
op3_t1 <- simulate_decision(reference, 0.95)
op3_t2 <- simulate_decision(reference, 0.95)

# Calculate agreement metrics

# 1. Within-operator agreement (repeatability)
within_op1 <- mean(op1_t1 == op1_t2)
within_op2 <- mean(op2_t1 == op2_t2)
within_op3 <- mean(op3_t1 == op3_t2)

# 2. Each operator vs. reference
op1_vs_ref <- mean(op1_t1 == reference & op1_t2 == reference)
op2_vs_ref <- mean(op2_t1 == reference & op2_t2 == reference)
op3_vs_ref <- mean(op3_t1 == reference & op3_t2 == reference)

# 3. All operators agree (both trials)
all_agree_t1 <- mean(op1_t1 == op2_t1 & op2_t1 == op3_t1)
all_agree_both <- mean(op1_t1 == op1_t2 & op1_t1 == op2_t1 & op2_t1 == op2_t2 &
                      op2_t1 == op3_t1 & op3_t1 == op3_t2)

cat("=== Attribute Agreement Analysis ===\n\n")
cat("Within-Operator Agreement (Repeatability):\n")
cat("  Operator 1:", round(within_op1 * 100, 1), "%\n")
cat("  Operator 2:", round(within_op2 * 100, 1), "%\n")
cat("  Operator 3:", round(within_op3 * 100, 1), "%\n\n")

cat("Each Operator vs. Reference (both trials correct):\n")
cat("  Operator 1:", round(op1_vs_ref * 100, 1), "%\n")
cat("  Operator 2:", round(op2_vs_ref * 100, 1), "%\n")
cat("  Operator 3:", round(op3_vs_ref * 100, 1), "%\n\n")

cat("All Operators Agreement:\n")
cat("  All agree (one trial):", round(all_agree_t1 * 100, 1), "%\n")
cat("  All agree (both trials):", round(all_agree_both * 100, 1), "%\n")

```

=== Attribute Agreement Analysis ===

Within-Operator Agreement (Repeatability):

Operator 1: 83.3 %

Operator 2: 76.7 %

Operator 3: 93.3 %

Each Operator vs. Reference (both trials correct):

Operator 1: 83.3 %

Operator 2: 73.3 %

Operator 3: 93.3 %

All Operators Agreement:

All agree (one trial): 70 %

All agree (both trials): 53.3 %

### 11.6.3 Calculating Kappa

```
# Calculate Kappa for Operator 1 vs Reference
# Create confusion matrix for trial 1
actual <- factor(reference, levels = c("Accept", "Reject"))
predicted <- factor(op1_t1, levels = c("Accept", "Reject"))

confusion <- table(Predicted = predicted, Actual = actual)
print(confusion)

# Calculate proportions
n <- sum(confusion)
p_o <- sum(diag(confusion)) / n # Observed agreement

# Expected agreement by chance
p_accept <- sum(confusion[1,]) / n * sum(confusion[,1]) / n
p_reject <- sum(confusion[2,]) / n * sum(confusion[,2]) / n
p_e <- p_accept + p_reject

# Kappa
kappa <- (p_o - p_e) / (1 - p_e)

cat("\nKappa Calculation (Operator 1 vs Reference):\n")
cat("Observed agreement (Po):", round(p_o, 3), "\n")
cat("Expected agreement (Pe):", round(p_e, 3), "\n")
cat("Kappa:", round(kappa, 3), "\n")

# Interpretation
if(kappa >= 0.81) {
  cat("Interpretation: Almost perfect agreement\n")
} else if(kappa >= 0.61) {
  cat("Interpretation: Substantial agreement\n")
} else if(kappa >= 0.41) {
  cat("Interpretation: Moderate agreement\n")
}
```

```

} else {
  cat("Interpretation: Fair or less agreement - needs improvement\n")
}

```

|           | Actual |        |
|-----------|--------|--------|
| Predicted | Accept | Reject |
| Accept    | 17     | 1      |
| Reject    | 3      | 9      |

Kappa Calculation (Operator 1 vs Reference):

Observed agreement (Po): 0.867

Expected agreement (Pe): 0.533

Kappa: 0.714

Interpretation: Substantial agreement

#### 11.6.4 Attribute MSA Best Practices

Table 11.7: Best Practices for Attribute MSA Studies

| Practice                           | Description                                                               |
|------------------------------------|---------------------------------------------------------------------------|
| <b>Use boundary samples</b>        | Include samples at the accept/reject boundary where decisions are hardest |
| <b>Include clear accept/reject</b> | Include obvious accept and obvious reject samples as controls             |
| <b>Blind testing</b>               | Operators should not know which samples are repeated or reference         |
| <b>Multiple trials</b>             | Minimum 2-3 trials per operator to assess within-operator repeatability   |
| <b>Reference standard</b>          | Must have known correct answers to assess accuracy, not just agreement    |
| <b>Adequate sample size</b>        | Minimum 30-50 samples recommended for statistical validity                |

## 11.7 Measurement System Resolution

**Resolution** (also called discrimination) is the smallest increment a measurement system can detect.

### 11.7.1 Rule of Ten

The **Rule of Ten** states that measurement resolution should be at least 1/10 of: - The tolerance, OR - The process variation (6 )

Whichever is smaller.

Live cell — try different tolerance, process spread, or gauge resolution values.

```

#| label: resolution-check
# Resolution adequacy check

```

```

tolerance <- 0.100 # mm total tolerance
process_6sigma <- 0.080 # 6 sigma of process
gauge_resolution <- 0.010 # mm (smallest increment)

# Check against tolerance
ratio_tolerance <- tolerance / gauge_resolution
# Check against process
ratio_process <- process_6sigma / gauge_resolution

cat("=== Resolution Adequacy Check ===\n\n")
cat("Tolerance:", tolerance, "mm\n")
cat("Process 6 :", process_6sigma, "mm\n")
cat("Gauge Resolution:", gauge_resolution, "mm\n\n")

cat("Tolerance / Resolution:", ratio_tolerance, ":1")
if(ratio_tolerance >= 10) cat(" - ADEQUATE\n") else cat(" - INADEQUATE\n")

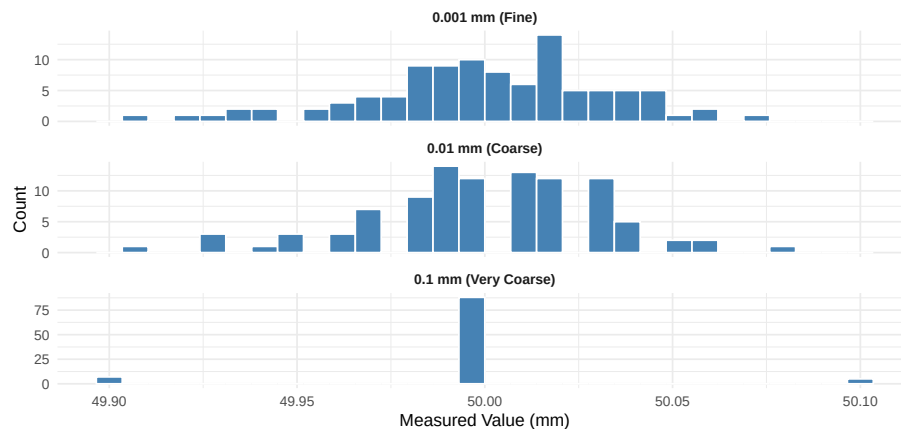
cat("Process 6 / Resolution:", ratio_process, ":1")
if(ratio_process >= 10) cat(" - ADEQUATE\n") else cat(" - INADEQUATE\n")

```

### 11.7.2 Resolution Impact on Gauge R&R

#### Effect of Gauge Resolution on Measurement Distribution

Same parts measured with different resolution gauges



11.8 Common Sources of Measurement Error

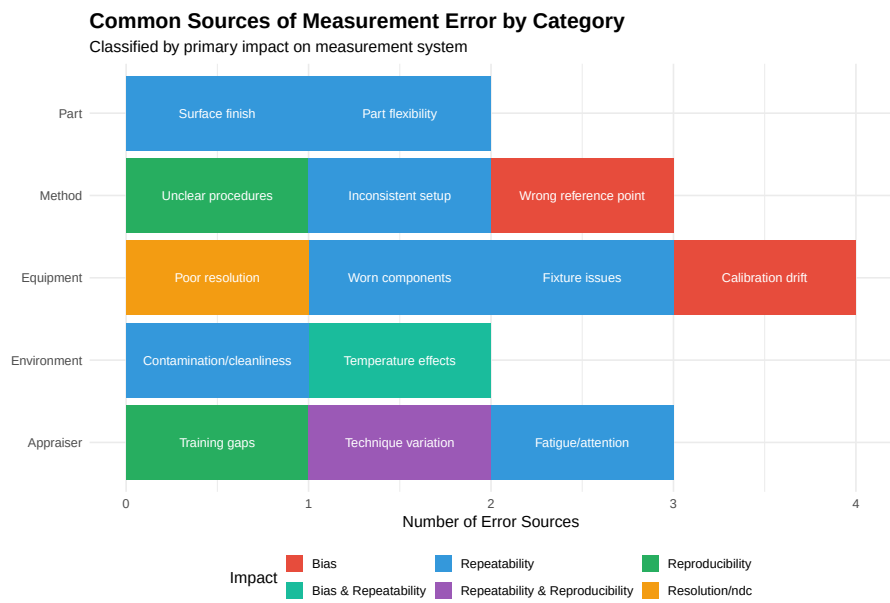


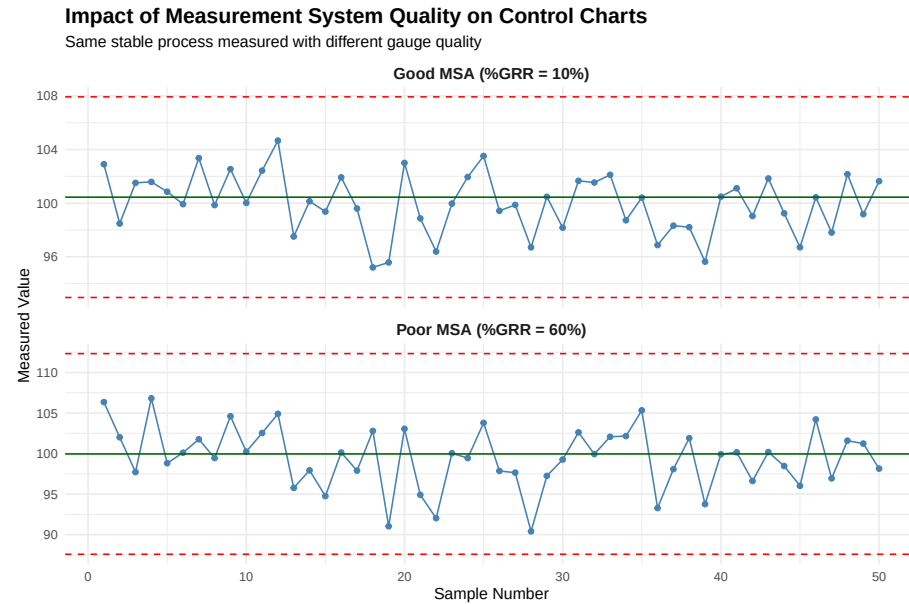
Table 11.8: MSA Issues and Improvement Actions

| Issue Identified          | Likely Causes                                                           | Improvement Actions     |
|---------------------------|-------------------------------------------------------------------------|-------------------------|
| High Repeatability (EV)   | Gauge condition, fixture, resolution, part variation during measurement | Maintain                |
| High Reproducibility (AV) | Training, technique, procedure clarity                                  | Standardize             |
| Significant Bias          | Calibration, master accuracy, technique                                 | Recalibrate             |
| Poor Linearity            | Gauge mechanism, calibration across range                               | Repair/Replace          |
| Drift/Stability Issues    | Environmental effects, gauge wear, master deterioration                 | Environment Control     |
| Low ndc                   | Resolution, excessive gauge variation relative to part variation        | Higher Resolution Gauge |

11.9 MSA and SPC Relationship

Poor measurement systems directly impact SPC effectiveness.

### 11.9.1 Effect on Control Charts



### 11.9.2 Effect on Capability Studies

Live cell — try different tolerance/sigma values or GRR percentages.

```
#| label: msa-capability-effect
# True process parameters
true_mean <- 100
true_sigma <- 2.0
USL <- 106
LSL <- 94
tolerance <- USL - LSL

# True capability
true_Cp <- tolerance / (6 * true_sigma)
true_Cpk <- min((USL - true_mean), (true_mean - LSL)) / (3 * true_sigma)

cat("=== True Process Capability ===\n")
cat("True :", true_sigma, "\n")
cat("True Cp:", round(true_Cp, 2), "\n")
cat("True Cpk:", round(true_Cpk, 2), "\n\n")

# With good measurement system (10% GRR)
grr_good <- 0.10 * tolerance / 5.15
observed_sigma_good <- sqrt(true_sigma^2 + grr_good^2)
```

```
observed_Cp_good <- tolerance / (6 * observed_sigma_good)
observed_Cpk_good <- min((USL - true_mean), (true_mean - LSL)) / (3 * observed_sigma_good)

cat("=== With Good MSA (10% GRR) ===\n")
cat("Observed  :", round(observed_sigma_good, 3), "\n")
cat("Observed Cp:", round(observed_Cp_good, 2), "\n")
cat("Observed Cpk:", round(observed_Cpk_good, 2), "\n\n")

# With poor measurement system (50% GRR)
grr_poor <- 0.50 * tolerance / 5.15
observed_sigma_poor <- sqrt(true_sigma^2 + grr_poor^2)
observed_Cp_poor <- tolerance / (6 * observed_sigma_poor)
observed_Cpk_poor <- min((USL - true_mean), (true_mean - LSL)) / (3 * observed_sigma_poor)

cat("=== With Poor MSA (50% GRR) ===\n")
cat("Observed  :", round(observed_sigma_poor, 3), "\n")
cat("Observed Cp:", round(observed_Cp_poor, 2), "\n")
cat("Observed Cpk:", round(observed_Cpk_poor, 2), "\n\n")

cat("Poor MSA makes a capable process (Cpk=1.0) appear incapable (Cpk=",
    round(observed_Cpk_poor, 2), ")\n")
```

---

## 11.10 Conducting an MSA Study: Step-by-Step

### Gauge R&R Study Process Flow

Steps for conducting a complete MSA study



### 11.10.1 Data Collection Sheet Template

Table 11.9: Gauge R&R Data Collection Sheet (Partial Example)

| Part | Operator | Trial | Measurement |
|------|----------|-------|-------------|
| 1    | A        | 1     | _____       |
| 1    | A        | 2     | _____       |
| 1    | B        | 1     | _____       |
| 1    | B        | 2     | _____       |
| 1    | C        | 1     | _____       |
| 1    | C        | 2     | _____       |
| 2    | A        | 1     | _____       |
| 2    | A        | 2     | _____       |
| 2    | B        | 1     | _____       |
| 2    | B        | 2     | _____       |
| 2    | C        | 1     | _____       |



---

2 C                      2 \_\_\_\_\_

---

## 11.11 Video Resources

### 11.11.1 Understanding Gauge R&R

### 11.11.2 MSA in Practice

---

## 11.12 Summary

Measurement Systems Analysis ensures that your data can be trusted for process control and decision-making:

1. **Before SPC, do MSA** - Garbage in, garbage out; verify measurement quality first
2. **Accuracy components** - Bias, linearity, and stability affect location (average)
3. **Precision components** - Repeatability (within operator) and reproducibility (between operators)
4. **Key metrics** - %GRR < 30% and  $ndc \geq 5$  for acceptable systems
5. **Resolution matters** - 10:1 ratio vs. tolerance or process variation
6. **Attribute MSA** - Use Kappa for pass/fail decisions
7. **Continuous improvement** - Regular verification and recertification

“The measurement system is part of the process. Fix the measurement system before trying to fix the process.”

---

## 11.13 Review Questions

**Question 1:** A gauge R&R study yields %GRR = 45% and  $ndc = 2$ . What does this mean and what should be done?

**Answer:**

This measurement system is **unacceptable**:

- **%GRR = 45%** exceeds the 30% maximum threshold, meaning 45% of the observed variation is due to the measurement system, not the parts
- **$ndc = 2$**  indicates the gauge can only distinguish between 2 categories of parts, which is insufficient for process control (minimum 5 required)

**Actions required:** 1. Investigate root causes of high variation: - Check gauge calibration and condition - Evaluate operator technique and training - Review measurement procedure for ambiguity - Assess fixture and part positioning - Check gauge resolution adequacy

2. Prioritize improvement efforts:

- If repeatability (EV) is high: Focus on gauge condition, fixture, technique consistency
- If reproducibility (AV) is high: Focus on training, procedure clarity, technique standardization

3. Re-run study after improvements

4. If improvements insufficient, consider upgrading to a higher-capability gauge

5. Until resolved, this gauge should not be used for process control or capability studies

**Question 2:** What is the difference between repeatability and reproducibility? Give an example of each source of error.

**Answer:**

**Repeatability (Equipment Variation - EV):** - Variation when the **same operator** measures the **same part** multiple times with the **same gauge** - Also called “within-operator” variation - Represents the inherent precision of the gauge and measurement technique

*Examples of repeatability errors:* - Gauge resolution limitations - Gauge mechanism play/backlash - Inconsistent part positioning in fixture - Variations in technique even by same operator - Environmental micro-changes during measurement

**Reproducibility (Appraiser Variation - AV):** - Variation when **different operators** measure the **same parts** - Also called “between-operator” variation - Represents differences in how operators apply the measurement method

*Examples of reproducibility errors:* - Different technique between operators (pressure applied, angle, etc.) - Different interpretation of measurement procedure - Different reading of analog scales - Training differences - Physical differences (eyesight, hand steadiness)

**Question 3:** Calculate the bias and determine if it is acceptable. Reference value = 25.000 mm, Tolerance =  $\pm 0.050$  mm, Measured values: 25.012, 25.015, 25.008, 25.011, 25.014, 25.009, 25.013, 25.010, 25.012, 25.011

**Answer:**

Live cell — check your own version by changing the numbers.

```

#| label: q3-answer
reference <- 25.000
tolerance_total <- 0.100 # ±0.050 = 0.100 total
measurements <- c(25.012, 25.015, 25.008, 25.011, 25.014,
                  25.009, 25.013, 25.010, 25.012, 25.011)

# Calculate bias
mean_measured <- mean(measurements)
bias <- mean_measured - reference

cat("Reference value:", reference, "mm\n")
cat("Mean measured:", round(mean_measured, 4), "mm\n")
cat("Bias:", round(bias, 4), "mm\n\n")

# Express as % of tolerance
bias_pct <- abs(bias) / tolerance_total * 100
cat("Bias as % of tolerance:", round(bias_pct, 1), "%\n\n")

# T-test for significance
t_result <- t.test(measurements, mu = reference)
cat("T-test p-value:", round(t_result$p.value, 6), "\n\n")

# Acceptability
cat("Acceptability Assessment:\n")
if(bias_pct < 10) {
  cat("- Bias < 10% of tolerance: ACCEPTABLE\n")
} else if(bias_pct < 25) {
  cat("- Bias 10-25% of tolerance: MARGINAL\n")
} else {
  cat("- Bias > 25% of tolerance: UNACCEPTABLE\n")
}

if(t_result$p.value < 0.05) {
  cat("- Bias is statistically significant (p < 0.05)\n")
  cat("- Consider recalibration or technique adjustment\n")
}

```

The bias of 0.0115 mm (11.5% of tolerance) is **marginally acceptable** and statistically significant. Recalibration should be considered.

**Question 4:** Why is it important to conduct MSA before capability studies ( $C_p$ ,  $C_{pk}$ )?

**Answer:**

MSA must precede capability studies because measurement error directly inflates observed variation:

**Mathematical Relationship:**

$$\sigma_{observed}^2 = \sigma_{process}^2 + \sigma_{measurement}^2$$

**Impacts on Capability:**

1. **Understated Cp/Cpk:** Observed sigma is always larger than true process sigma, so calculated capability will be lower than actual capability
2. **False Alarms:** A capable process may appear incapable due to measurement noise, leading to unnecessary process adjustments or investment
3. **Missed Issues:** High measurement variation masks actual process variation; you can't see problems through the "fog" of gauge error
4. **Bad Decisions:** Incorrect capability data leads to:
  - Wrong process acceptance decisions
  - Inappropriate control limits
  - Misguided improvement investments
  - Customer quality issues

**Example:** - True process: Cpk = 1.5 (very capable) - With 50% GRR: Observed Cpk = 0.9 (appears incapable) - Result: Unnecessary process "improvement" efforts

**Rule of Thumb:** If %GRR > 30%, any capability study is essentially meaningless because too much of the observed variation is measurement noise, not real process variation.

**Question 5:** A visual inspection has the following results: Kappa vs. reference = 0.72, within-operator agreement = 88%, between-operator agreement = 75%. Evaluate this attribute measurement system.

**Answer:**

**Results Interpretation:**

1. **Kappa vs. Reference = 0.72**
  - Falls in "Substantial agreement" range (0.61-0.80)
  - Operators are correctly classifying parts most of the time
  - Acceptable for many applications, but improvement possible
2. **Within-Operator Agreement = 88%**
  - Operators are fairly consistent with themselves
  - 12% of the time, same operator gives different result on same part
  - Should target >90% for critical inspections
3. **Between-Operator Agreement = 75%**
  - Operators disagree on 25% of parts
  - This is a significant reproducibility issue
  - Major source of inconsistency in product quality

**Assessment: MARGINAL - Improvement Needed****Recommended Actions:**

1. **Standardize criteria:** Create clear visual standards with boundary samples
  - Photos of accept/reject borderline cases
  - Written descriptions of defect criteria
2. **Training:**
  - Calibration session with all operators and reference samples
  - Identify operators with lower agreement for targeted training
3. **Improve conditions:**
  - Check lighting adequacy and consistency
  - Ensure proper viewing distance and angle
  - Reduce fatigue with appropriate break schedules
4. **Consider automation:** For critical inspections, automated vision systems may provide better consistency
5. **Re-study after improvements** to verify effectiveness

**Question 6:** What is the “Rule of Ten” for gauge resolution, and how would you apply it?

**Answer:**

**The Rule of Ten:** The gauge resolution should be at least **1/10 of the tolerance** or **1/10 of the process variation (6 $\sigma$ )**, whichever is smaller.

$$\text{Resolution} \leq \frac{\min(\text{Tolerance}, 6\sigma)}{10}$$

**Application Example:**

Given: - Part tolerance:  $\pm 0.05$  mm (total = 0.10 mm) - Process 6 : 0.08 mm - Available gauges: 0.01 mm resolution, 0.001 mm resolution

**Calculation:**

Required resolution     $\min(0.10, 0.08) / 10$   
 Required resolution     $0.08 / 10$   
 Required resolution    0.008 mm

**Assessment:** - 0.01 mm gauge:  $0.01 > 0.008 \rightarrow$  **INADEQUATE** - 0.001 mm gauge:  $0.001 < 0.008 \rightarrow$  **ADEQUATE**

**Why It Matters:** 1. If resolution is too coarse, gauge cannot detect small part differences 2. Results in low ndc (number of distinct categories) 3. Control charts will show “stair-step” patterns 4. Capability studies will be inaccurate

**Practical Tips:** - For SPC applications, 10:1 is minimum; 20:1 is preferred - For capability studies, 10:1 is acceptable - For simple pass/fail gauging, 5:1 may be acceptable

**Question 7:** Given the following ANOVA table from a Gauge R&R study, calculate the variance components and %GRR.

| Source        | DF | SS     | MS      |
|---------------|----|--------|---------|
| Part          | 9  | 0.2850 | 0.03167 |
| Operator      | 2  | 0.0025 | 0.00125 |
| Part×Operator | 18 | 0.0090 | 0.00050 |
| Repeatability | 60 | 0.0180 | 0.00030 |
| Total         | 89 | 0.3145 |         |

Study design: 10 parts, 3 operators, 3 trials

**Answer:**

Live cell — check your own version by changing the ANOVA table values.

```
#| label: q7-answer
# Given values from ANOVA table
MS_part <- 0.03167
MS_operator <- 0.00125
MS_interaction <- 0.00050
MS_error <- 0.00030

n_parts <- 10
n_operators <- 3
n_trials <- 3

# Variance components
var_repeatability <- MS_error
cat("Var(Repeatability) =", var_repeatability, "\n")

var_interaction <- max(0, (MS_interaction - MS_error) / n_trials)
cat("Var(Interaction) =", round(var_interaction, 6), "\n")

var_operator <- max(0, (MS_operator - MS_interaction) / (n_parts * n_trials))
cat("Var(Operator) =", round(var_operator, 6), "\n")

var_part <- max(0, (MS_part - MS_interaction) / (n_operators * n_trials))
cat("Var(Part) =", round(var_part, 6), "\n\n")

# Reproducibility = Operator + Interaction
var_reproducibility <- var_operator + var_interaction
```

```

cat("Var(Reproducibility) =", round(var_reproducibility, 6), "\n")

# Total Gauge R&R
var_GRR <- var_repeatability + var_reproducibility
cat("Var(GRR) =", round(var_GRR, 6), "\n")

# Total variation
var_total <- var_part + var_GRR
cat("Var(Total) =", round(var_total, 6), "\n\n")

# Calculate %GRR (of total variation)
sigma_GRR <- sqrt(var_GRR)
sigma_total <- sqrt(var_total)
pct_GRR <- (sigma_GRR / sigma_total) * 100

cat("%GRR (of Total Variation) =", round(pct_GRR, 1), "%\n\n")

# ndc
sigma_part <- sqrt(var_part)
ndc <- 1.41 * (sigma_part / sigma_GRR)
cat("ndc =", round(ndc, 1), "\n\n")

if(pct_GRR < 10) {
  cat("Assessment: ACCEPTABLE (< 10%)\n")
} else if(pct_GRR < 30) {
  cat("Assessment: MARGINAL (10-30%)\n")
} else {
  cat("Assessment: UNACCEPTABLE (> 30%)\n")
}

```

**Question 8:** What actions would you take if a Gauge R&R study showed high repeatability (EV) but low reproducibility (AV)?

**Answer:**

When **repeatability is high** (main contributor to GRR) but **reproducibility is low**, the measurement variation is coming from the gauge/equipment/technique rather than differences between operators.

**Root Cause Investigation - Focus Areas:**

1. **Gauge Condition**
  - Worn measuring surfaces or contacts
  - Loose components or mechanism play
  - Damaged or dirty probe/sensor
2. **Gauge Resolution**
  - Resolution may be inadequate for the tolerance

- Check 10:1 rule compliance
- 3. **Fixture/Fixturing**
  - Part not held consistently
  - Fixture wear or damage
  - Part not seating properly
- 4. **Measurement Technique**
  - Inconsistent contact force
  - Varying measurement location on part
  - Speed of measurement varies
- 5. **Part Characteristics**
  - Part surface finish affecting readings
  - Part flexibility/deflection during measurement
  - Part temperature variation
- 6. **Environment**
  - Vibration affecting readings
  - Temperature instability
  - Contamination

**Improvement Actions:**

| Priority | Action                                                           |
|----------|------------------------------------------------------------------|
| 1        | Verify gauge calibration and condition; repair/replace if needed |
| 2        | Check and improve fixture; ensure consistent part positioning    |
| 3        | Evaluate and potentially upgrade gauge resolution                |
| 4        | Standardize technique (force, speed, location)                   |
| 5        | Control environmental factors                                    |
| 6        | Re-run study to verify improvement                               |

**Note:** Low AV in this case is actually good - it means operators are consistent with each other. The problem is the measurement equipment itself.

## 11.14 References

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## Chapter 12

# Total Productive Maintenance (TPM)

---

### 12.1 Learning Objectives

By the end of this chapter, you will be able to:

- Define Total Productive Maintenance and explain its objectives
- Describe the 8 pillars of TPM and their interconnections
- Calculate Overall Equipment Effectiveness (OEE) and its components
- Identify and categorize the Six Big Losses
- Implement autonomous maintenance activities
- Develop preventive and predictive maintenance strategies
- Apply TPM principles to improve equipment reliability and productivity

“Take care of your equipment, and it will take care of your production.” — Seiichi Nakajima, Father of TPM

---

### 12.2 Introduction to TPM

#### 12.2.1 What is Total Productive Maintenance?

**Total Productive Maintenance (TPM)** is a holistic approach to equipment maintenance that strives to achieve perfect production with no breakdowns, no defects, and no accidents. It involves everyone in the organization, from operators to top management.

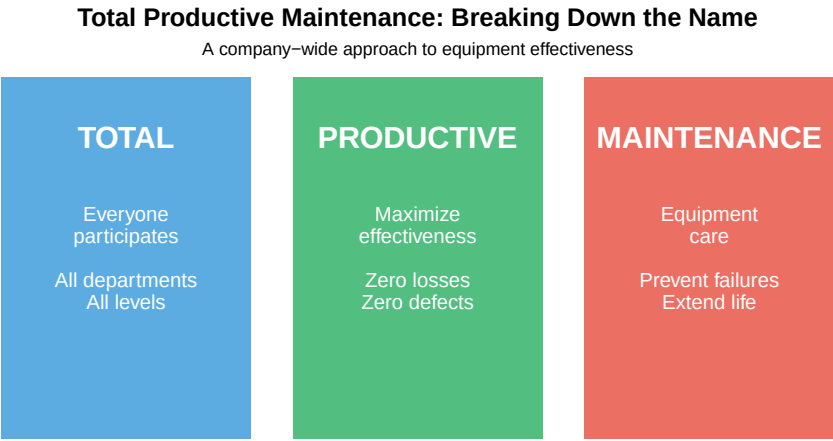


Figure 12.1: The TPM Philosophy

12.2.2 History of TPM

Table 12.1: Evolution of Maintenance Philosophy

| Era           | Maintenance Approach   | Description                                            |
|---------------|------------------------|--------------------------------------------------------|
| Pre-1950s     | Breakdown Maintenance  | Fix it when it breaks; reactive only                   |
| 1950s         | Preventive Maintenance | Scheduled maintenance to prevent failures              |
| 1960s         | Productive Maintenance | Maintenance focused on reliability and maintainability |
| 1970s         | TPM Developed          | Nakajima develops TPM at Nippondenso (Toyota supplier) |
| 1980s         | TPM Spreads Globally   | Adopted by automotive, electronics, process industries |
| 1990s-Present | TPM + Industry 4.0     | Integration with predictive analytics, IoT sensors, AI |

12.2.3 The Goals of TPM

TPM aims to achieve **zero** targets:

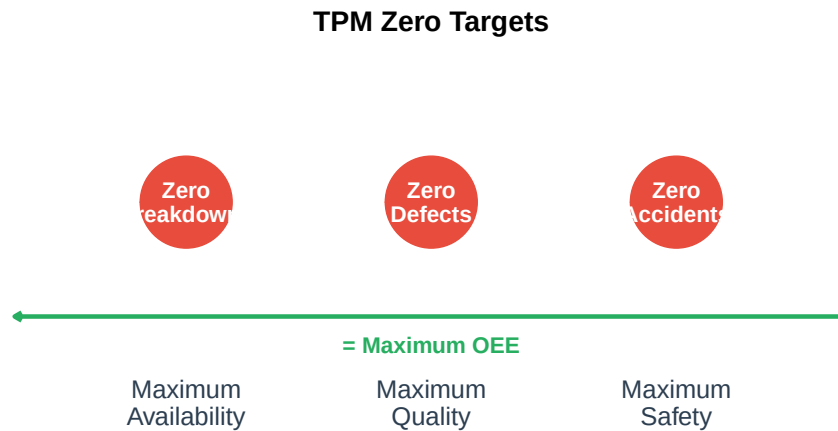


Figure 12.2: TPM Zero Targets

### 12.2.4 Traditional vs. TPM Approach

Table 12.2: Traditional Maintenance vs. TPM

| Aspect                                | Traditional Maintenance               | TPM Approach                       |
|---------------------------------------|---------------------------------------|------------------------------------|
| <b>**Equipment ownership**</b>        | 'Maintenance owns the machines'       | 'We all own the machines'          |
| <b>**Maintenance responsibility**</b> | Maintenance department only           | Everyone (operators + maintenance) |
| <b>**Operator role**</b>              | Operate; call maintenance when broken | Operate, clean, inspect, maintain  |
| <b>**Maintenance focus**</b>          | Fix breakdowns quickly                | Prevent breakdowns; extend life    |
| <b>**Problem solving**</b>            | Reactive; fire-fighting               | Proactive; root cause elimination  |
| <b>**Goal**</b>                       | Minimize maintenance cost             | Maximize equipment effectiveness   |

Question: Why involve operators in maintenance?

**Operators are the first line of defense because:**

1. **They know the equipment:** Operators work with machines every day and notice subtle changes
2. **Early detection:** Small problems are detected before becoming major failures
3. **Ownership mindset:** People take better care of things they “own”
4. **Reduced response time:** Operators can address minor issues immediately
5. **Freed maintenance resources:** Technicians can focus on complex tasks

**Example:** An operator notices a slight vibration change in a motor. Under traditional maintenance, they might ignore it. Under TPM, they report it, potentially preventing a \$50,000 breakdown.

“No one knows a machine better than the person who operates it every day.”

## 12.3 The Eight Pillars of TPM

TPM is built on **eight pillars**, each addressing a specific aspect of equipment management and organizational culture.

### The House of TPM: Eight Pillars Built on 5S Foundation

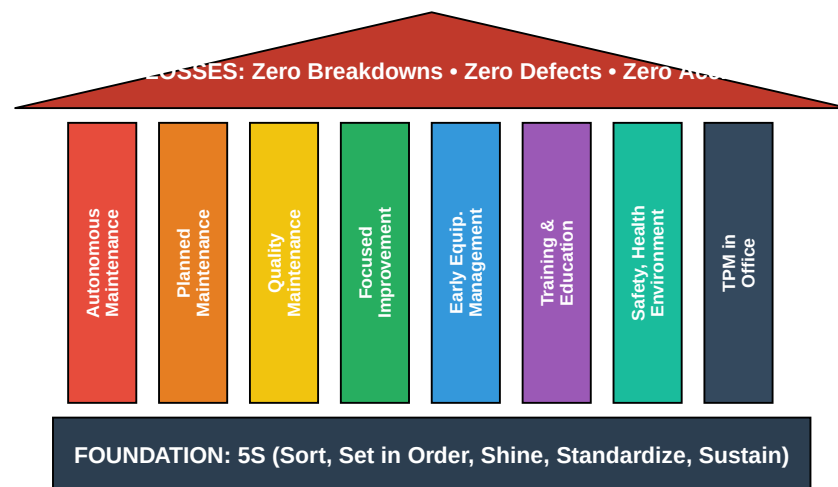


Figure 12.3: The Eight Pillars of TPM

Table 12.3: The Eight Pillars of TPM

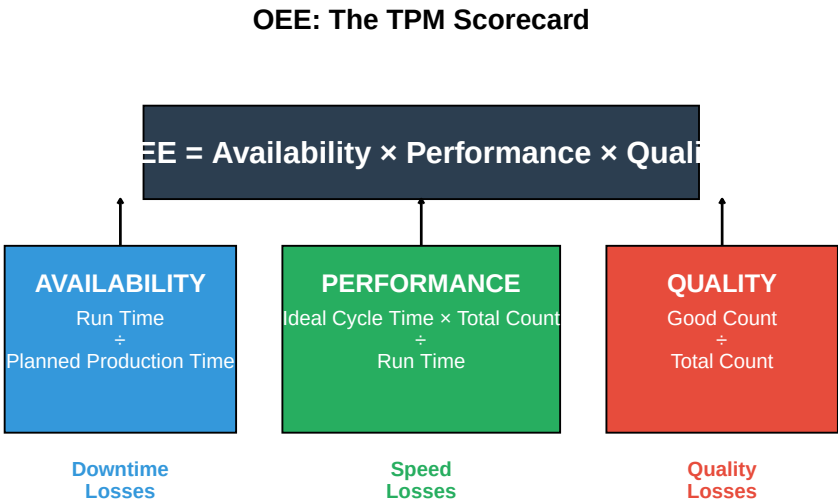
| # | Pillar                     | Purpose                                |
|---|----------------------------|----------------------------------------|
| 1 | **Autonomous Maintenance** | Operators maintain their own equipment |
| 2 | **Planned Maintenance**    | Scheduled maintenance by specialists   |
| 3 | **Quality Maintenance**    | Zero defects through equipment care    |

|   |                                        |                                            |
|---|----------------------------------------|--------------------------------------------|
| 4 | <b>**Focused Improvement**</b>         | Eliminate losses through kaizen activities |
| 5 | <b>**Early Equipment Management**</b>  | Design equipment for maintainability       |
| 6 | <b>**Training &amp; Education**</b>    | Develop skilled operators and technicians  |
| 7 | <b>**Safety, Health, Environment**</b> | Zero accidents, safe workplace             |
| 8 | <b>**TPM in Office**</b>               | Apply TPM principles to administrative pr  |

## 12.4 Overall Equipment Effectiveness (OEE)

### 12.4.1 What is OEE?

**Overall Equipment Effectiveness (OEE)** is the primary metric of TPM. It measures how effectively equipment is being used by combining three factors: Availability, Performance, and Quality.



|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| <b>**Performance**</b> | $\frac{\text{Ideal Cycle Time}}{\text{Run Time}} \times \text{Total Count}$ |
| <b>**Quality**</b>     | $\frac{\text{Good Count}}{\text{Total Count}} \times 100\%$                 |
| <b>**OEE**</b>         | $\text{Availability} \times \text{Performance} \times \text{Quality}$       |

### 12.4.3 OEE Calculation Example

Live cell — try different downtime, production, or defect numbers.

```
#| label: oee-calculation
# OEE Calculation Example: Injection Molding Machine

cat("=== SHIFT DATA ===\n")
# Time data
shift_length <- 480 # minutes (8 hours)
planned_downtime <- 30 # breaks, planned maintenance
planned_production_time <- shift_length - planned_downtime
cat("Planned Production Time:", planned_production_time, "minutes\n")

# Downtime losses
breakdown_time <- 25 # equipment failure
changeover_time <- 35 # mold changes
adjustment_time <- 10 # setup adjustments
total_downtime <- breakdown_time + changeover_time + adjustment_time
cat("Unplanned Downtime:", total_downtime, "minutes\n")

run_time <- planned_production_time - total_downtime
cat("Run Time:", run_time, "minutes\n\n")

# Production data
ideal_cycle_time <- 0.5 # minutes per part (design spec)
total_parts <- 700 # parts produced
good_parts <- 665 # parts passing inspection
defects <- total_parts - good_parts

cat("=== PRODUCTION DATA ===\n")
cat("Total Parts Produced:", total_parts, "\n")
cat("Good Parts:", good_parts, "\n")
cat("Defects:", defects, "\n\n")

# Calculate OEE Components
cat("=== OEE CALCULATION ===\n\n")

# Availability
availability <- (run_time / planned_production_time) * 100
cat("AVAILABILITY:\n")
cat(" = Run Time / Planned Production Time\n")
```



```

cat("  =", run_time, "/", planned_production_time, "\n")
cat("  =", round(availability, 1), "%\n\n")

# Performance
# Actual output vs. theoretical output during run time
theoretical_output <- run_time / ideal_cycle_time
performance <- (total_parts / theoretical_output) * 100
cat("PERFORMANCE:\n")
cat("  Theoretical Output =", theoretical_output, "parts\n")
cat("  Actual Output =", total_parts, "parts\n")
cat("  = Actual / Theoretical\n")
cat("  =", round(performance, 1), "%\n\n")

# Quality
quality <- (good_parts / total_parts) * 100
cat("QUALITY:\n")
cat("  = Good Parts / Total Parts\n")
cat("  =", good_parts, "/", total_parts, "\n")
cat("  =", round(quality, 1), "%\n\n")

# Overall OEE
oeo <- (availability/100) * (performance/100) * (quality/100) * 100
cat("=== OVERALL OEE ===\n")
cat("OEE =", round(availability, 1), "% ×", round(performance, 1), "% ×", round(quality, 1), "%\n")
cat("OEE =", round(oeo, 1), "%\n")

```

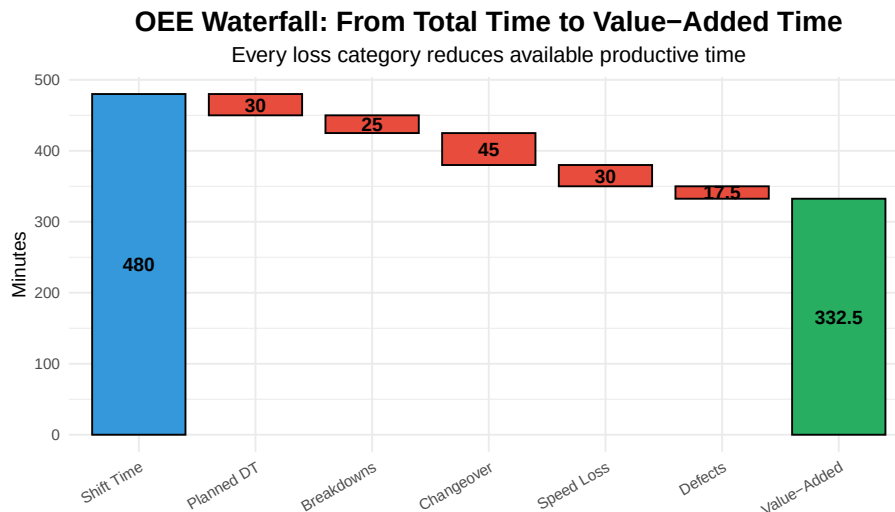


Figure 12.5: OEE Waterfall Chart: Where Time Goes

### 12.4.4 OEE Benchmarks

Table 12.5: OEE Benchmarks

| OEE Level | Classification | Interpretation                              | Component Ta    |
|-----------|----------------|---------------------------------------------|-----------------|
| < 65%     | Unacceptable   | Significant improvement opportunities exist | —               |
| 65% - 75% | Typical        | Average for discrete manufacturing          | A: 90%, P: 85%, |
| 75% - 85% | Good           | Well-managed operations                     | A: 92%, P: 90%, |
| > 85%     | World Class    | Best-in-class performance                   | A: 95%, P: 95%  |

Interactive Exercise: Calculate OEE

Given the following shift data, calculate OEE:

- Shift length: 480 minutes
- Planned breaks: 40 minutes
- Equipment breakdown: 30 minutes
- Setup/changeover: 20 minutes
- Ideal cycle time: 1 minute/part
- Total parts produced: 350
- Rejected parts: 14

Work through the calculation:

**Solution:**

Planned Production Time = 480 - 40 = 440 minutes

Run Time = 440 - 30 - 20 = 390 minutes

Availability = 390 / 440 = 88.6%

Theoretical Output = 390 / 1 = 390 parts

Performance = 350 / 390 = 89.7%

Quality = (350 - 14) / 350 = 336 / 350 = 96.0%

OEE = 88.6% × 89.7% × 96.0% = 76.3%

**Interpretation:** This is “Good” performance, above the typical range. The biggest opportunity is in Performance (speed losses).

## 12.5 The Six Big Losses

TPM categorizes all equipment losses into **six categories**, which map directly to the OEE components.

The Six Big Losses: Targets for TPM Improvement

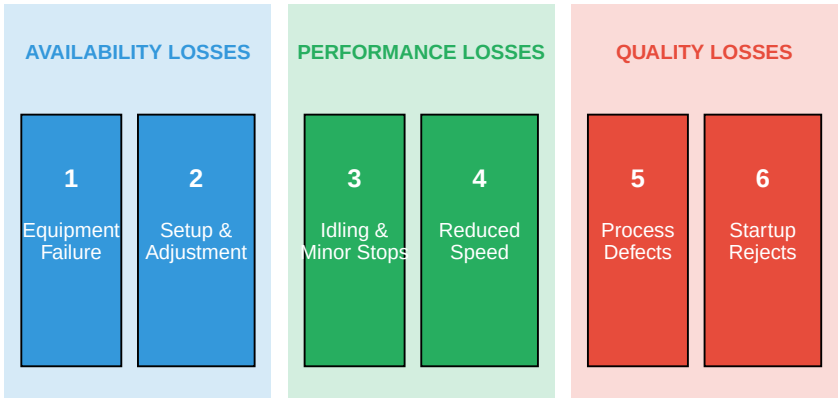


Figure 12.6: The Six Big Losses

Table 12.6: The Six Big Losses

| # | Loss Category            | OEE Component | Description                      |
|---|--------------------------|---------------|----------------------------------|
| 1 | **Equipment Failure**    | Availability  | Unplanned equipment breakdown    |
| 2 | **Setup & Adjustment**   | Availability  | Time for changeover, tool change |
| 3 | **Idling & Minor Stops** | Performance   | Brief stoppages (jams, blockage  |
| 4 | **Reduced Speed**        | Performance   | Running slower than design spe   |
| 5 | **Process Defects**      | Quality       | Defects produced during norma    |
| 6 | **Startup Rejects**      | Quality       | Defects during warmup, startu    |

### 12.5.1 Typical Loss Distribution

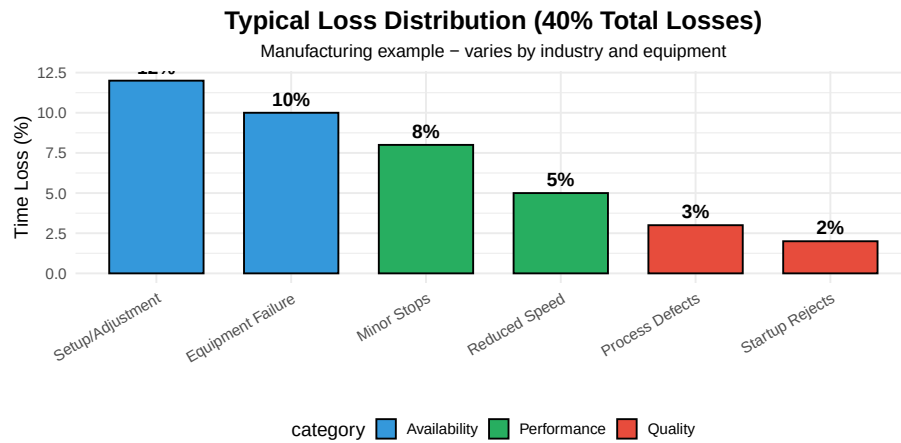


Figure 12.7: Typical Distribution of the Six Big Losses

---

## 12.6 Autonomous Maintenance

**Autonomous Maintenance (AM)** is the first and most visible pillar of TPM. It transfers basic maintenance tasks from maintenance specialists to operators.

### 12.6.1 The Seven Steps of Autonomous Maintenance



Figure 12.8: Seven Steps of Autonomous Maintenance

Table 12.7: Autonomous Maintenance Steps in Detail

| Step | Name                                 | Activities                                             |
|------|--------------------------------------|--------------------------------------------------------|
| 1    | **Initial Cleaning**                 | Deep clean equipment; identify abnormalities           |
| 2    | **Eliminate Contamination Sources**  | Find and fix sources of dirt, leaks, contamination     |
| 3    | **Cleaning & Lubrication Standards** | Create standards for cleaning, lubricating, inspection |
| 4    | **General Inspection**               | Train operators on equipment function; expand skills   |
| 5    | **Autonomous Inspection**            | Operators perform scheduled inspections independently  |
| 6    | **Organization & Tidiness**          | Standardize workplace; visual management               |
| 7    | **Full Autonomous Maintenance**      | Continuous improvement; operators propose improvements |

### 12.6.2 The AM Tag System

During autonomous maintenance activities, operators identify abnormalities using **tags**:

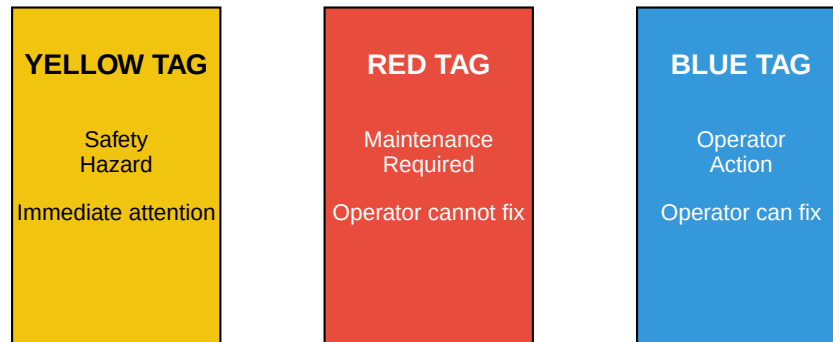
**AM Tag System: Identifying and Tracking Abnormalities**

Figure 12.9: AM Tag System

**12.6.3 Example: AM Cleaning and Inspection Checklist**

Table 12.8: Sample AM Inspection Checklist

| # | Check Item                | Method            | Standard                       | Frequency  |
|---|---------------------------|-------------------|--------------------------------|------------|
| 1 | General cleanliness       | Visual            | No dust, debris, contamination | Daily      |
| 2 | Oil/lubricant levels      | Visual/gauge      | Between min-max marks          | Daily      |
| 3 | Loose bolts/fasteners     | Touch/wrench      | Hand tight + 1/4 turn          | Weekly     |
| 4 | Unusual sounds            | Listen            | Normal operating sound         | Each start |
| 5 | Vibration                 | Touch             | Minimal, no unusual patterns   | Each start |
| 6 | Temperature               | Touch/thermometer | < 60°C (or per spec)           | Hourly     |
| 7 | Leaks (oil, air, coolant) | Visual            | No visible leaks               | Daily      |
| 8 | Safety guards in place    | Visual            | All guards secure              | Each shift |

**12.7 Planned Maintenance**

**Planned Maintenance** is the second pillar, focusing on scheduled maintenance activities performed by maintenance specialists.

12.7.1 Types of Maintenance

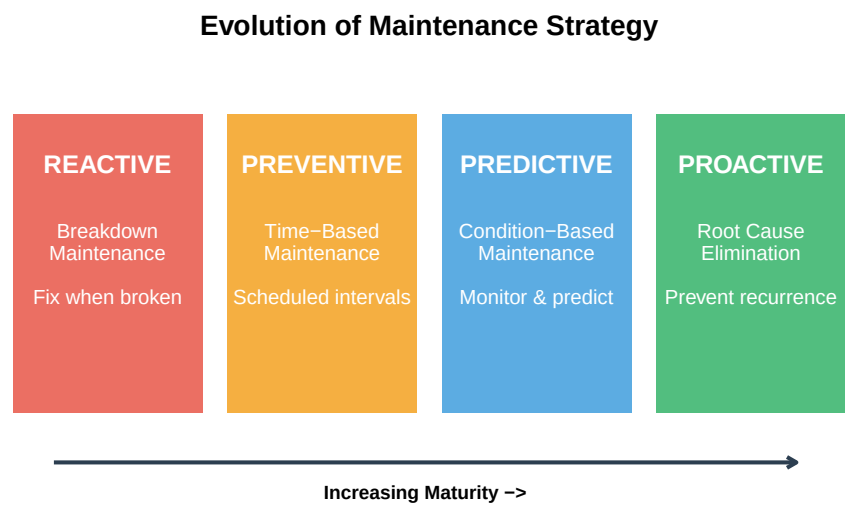


Figure 12.10: Types of Maintenance Activities

Table 12.9: Maintenance Strategy Comparison

| Type           | Approach                     | Advantages                               | Disadvantages                   |
|----------------|------------------------------|------------------------------------------|---------------------------------|
| **Breakdown**  | Run to failure               | No upfront cost                          | Unplanned downtime              |
| **Preventive** | Time/cycle-based replacement | Planned downtime, parts availability     | May replace good components     |
| **Predictive** | Condition monitoring         | Optimal component life, targeted actions | Monitoring equipment cost       |
| **Proactive**  | Root cause elimination       | Prevents future failures                 | Requires analysis and expertise |

12.7.2 Predictive Maintenance Technologies

Table 12.10: Predictive Maintenance Technologies

| Technology                 | What It Detects                                             | Equipment Applications                   |
|----------------------------|-------------------------------------------------------------|------------------------------------------|
| **Vibration Analysis**     | Imbalance, misalignment, bearing wear, looseness            | Rotating equipment (motors, pumps, fans) |
| **Thermography**           | Hot spots, electrical issues, friction, insulation problems | Electrical panels, motors, bearings      |
| **Oil Analysis**           | Contamination, wear particles, lubricant degradation        | Gearboxes, hydraulics, engines           |
| **Ultrasound**             | Leaks, bearing defects, electrical discharge                | Steam traps, valves, bearings            |
| **Motor Current Analysis** | Motor winding issues, rotor bar problems, power quality     | Electric motors, drives, generators      |

### 12.7.3 The P-F Curve

The **P-F Curve** illustrates the concept of detecting potential failure (P) before functional failure (F).

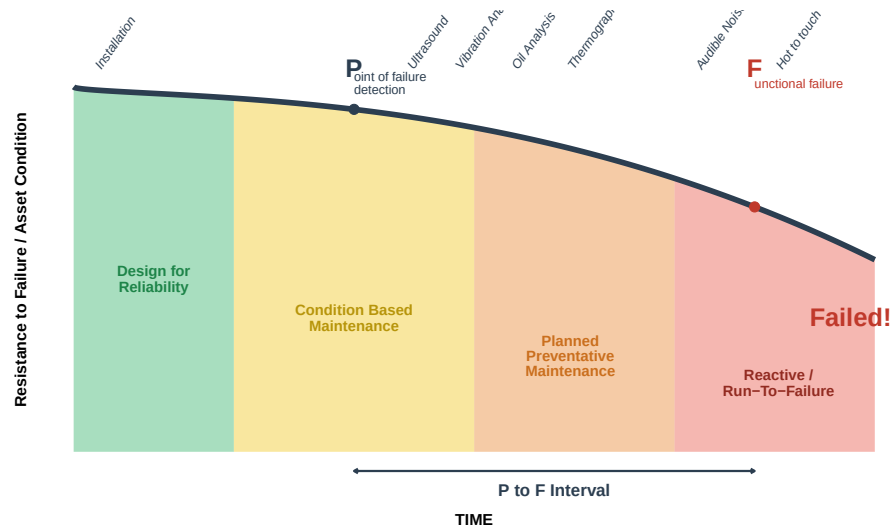


Figure 12.11: The P-F Curve: Detecting Failure Early

Question: Why is predictive maintenance better than preventive maintenance?

**Preventive Maintenance (Time-Based):** - Replaces parts at fixed intervals regardless of condition - May replace parts still in good condition (waste) - May miss failures that occur before scheduled replacement - Based on average life, not individual component condition

**Predictive Maintenance (Condition-Based):** - Monitors actual equipment condition - Replaces parts when condition indicates need - Maximizes component life - Provides warning time to plan repairs

**Example:** - Preventive: Replace bearing every 6 months (may last 9 months or fail at 4 months) - Predictive: Monitor vibration; replace when trending indicates 2-3 weeks to failure - Result: Optimal component life, planned downtime, no unexpected failures

**Cost Comparison:** - Reactive repair: \$10,000 (including secondary damage, lost production) - Preventive replacement: \$3,000 (may waste \$1,000 in parts life) - Predictive replacement: \$2,500 (optimal timing, minimal waste)



## 12.8 Focused Improvement (Kobetsu Kaizen)

**Focused Improvement** uses cross-functional teams to eliminate specific losses identified through OEE analysis.

### 12.8.1 The Kaizen Process

#### Focused Improvement (Kobetsu Kaizen) Process

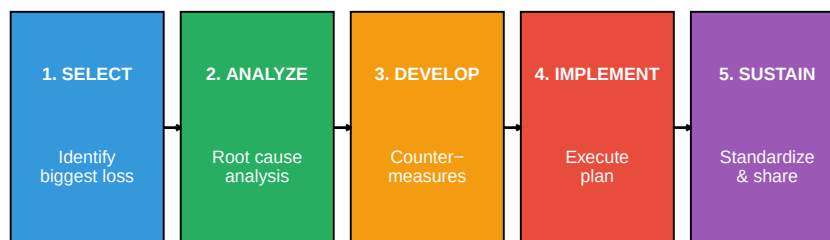


Figure 12.12: Focused Improvement Process

#### 12.8.1.1 Understanding Kaizen

**Kaizen** ( ) is a Japanese term meaning “change for better” or **continuous improvement**. Unlike large-scale transformation projects, Kaizen focuses on small, incremental improvements made consistently over time. The philosophy is built on the belief that every process can be improved, and everyone—from operators to managers—should participate in identifying and implementing improvements.

##### Core Principles of Kaizen:

- **Good processes bring good results** — Focus on improving the process, not blaming people
- **Go see for yourself (Gemba)** — Understand the actual situation at the workplace
- **Speak with data** — Base decisions on facts and measurements
- **Take action to contain and correct root causes** — Don’t just treat symptoms
- **Work as a team** — Cross-functional collaboration produces better solutions
- **Kaizen is everybody’s business** — Every employee can contribute improvements

### 12.8.1.2 The Five Steps in Detail

#### Step 1: SELECT — Identify the Biggest Loss

The first step uses data to identify which loss has the greatest impact on OEE. This typically involves:

- Reviewing OEE data and breakdown logs
- Creating Pareto charts to visualize loss magnitude
- Selecting a loss that is significant, measurable, and within the team's ability to address
- Forming a cross-functional team with members who understand the equipment and process

*Key question: “Which single problem, if solved, would give us the biggest improvement?”*

#### Step 2: ANALYZE — Root Cause Analysis

Once a loss is selected, the team investigates its root cause using structured problem-solving tools:

- **Why-Why Analysis (5 Whys):** Repeatedly asking “why” to drill down from symptoms to root causes
- **Fishbone Diagram (Ishikawa):** Categorizing potential causes by Machine, Method, Material, Man, Measurement, Environment
- **PM Analysis:** A systematic method for chronic losses that examines physical mechanisms
- **Data collection:** Gathering facts at the Gemba (actual workplace)

*Key principle: Never assume you know the cause—verify with evidence.*

#### Step 3: DEVELOP — Create Countermeasures

Based on root cause analysis, the team develops countermeasures—actions designed to eliminate or reduce the root cause:

- Brainstorm multiple potential solutions
- Evaluate options based on effectiveness, cost, and implementation time
- Create an action plan with clear responsibilities and deadlines
- Consider both immediate fixes and permanent solutions
- Apply error-proofing (Poka-Yoke) where possible

*Important: Countermeasures address root causes, not just symptoms.*

#### Step 4: IMPLEMENT — Execute the Plan

The team puts countermeasures into action:

- Follow the action plan systematically
- Make changes during planned downtime when possible
- Document all modifications made
- Train affected personnel on new procedures

- Monitor results immediately after implementation
- Be prepared to adjust if initial results are not as expected

*Tip: Start with small-scale trials before full implementation when possible.*

### Step 5: SUSTAIN — Standardize and Share

The final step ensures improvements are maintained and shared across the organization:

- **Standardize:** Update procedures, checklists, and work instructions to reflect the new method
- **Visualize:** Create visual controls that make the standard easy to follow
- **Train:** Ensure all shifts and new employees learn the improved method
- **Verify:** Include checks in AM routines or audits to confirm the standard is being followed
- **Share:** Communicate successful improvements to other areas that might benefit (Yokoten—horizontal deployment)

*Kaizen is never finished—today’s standard is tomorrow’s baseline for further improvement.*

#### 12.8.1.3 Kaizen Event vs. Daily Kaizen

| Aspect          | Kaizen Event (Blitz)                | Daily Kaizen                              |
|-----------------|-------------------------------------|-------------------------------------------|
| <b>Duration</b> | 3-5 days intensive                  | Ongoing, small improvements               |
| <b>Scope</b>    | Focused on one specific problem     | Any improvement opportunity               |
| <b>Team</b>     | Dedicated cross-functional team     | Individual or small group                 |
| <b>Impact</b>   | Significant, measurable improvement | Cumulative effect over time               |
| <b>Example</b>  | Reduce changeover time by 50%       | Reorganize tool storage for easier access |

Both approaches are valuable—Kaizen events tackle significant losses while daily Kaizen builds a culture of continuous improvement.

#### 12.8.2 Loss Elimination Tools

Effective loss elimination requires systematic tools that help teams identify problems, find root causes, and implement lasting solutions. The following tools are essential for Focused Improvement activities.

### 12.8.2.1 Pareto Analysis

**Pareto Analysis** applies the 80/20 rule: roughly 80% of losses come from 20% of causes. By creating a Pareto chart, teams can visualize which losses have the greatest impact and prioritize their improvement efforts accordingly.

**How to use:**

1. Collect data on all losses (downtime events, defects, speed reductions)
2. Categorize losses by type or cause
3. Calculate the frequency or duration of each category
4. Rank categories from highest to lowest
5. Create a bar chart with a cumulative percentage line
6. Focus improvement efforts on the “vital few” categories that account for most losses

### 12.8.2.2 Why-Why Analysis (5 Whys)

**Why-Why Analysis** is a simple but powerful technique for drilling down from a symptom to its root cause by repeatedly asking “Why?” The goal is to go beyond surface-level explanations to find the fundamental cause that, if addressed, will prevent recurrence.

**Example:**

| Level             | Question                      | Answer                                             |
|-------------------|-------------------------------|----------------------------------------------------|
| Problem           | Machine stopped               | Overload tripped                                   |
| Why 1?            | Why did the overload trip?    | Motor drew excessive current                       |
| Why 2?            | Why excessive current?        | Bearing was seized                                 |
| Why 3?            | Why was bearing seized?       | Insufficient lubrication                           |
| Why 4?            | Why insufficient lubrication? | Lubrication schedule not followed                  |
| Why 5?            | Why not followed?             | No visual indicator when lubrication is due        |
| <b>Root Cause</b> |                               | Missing visual management for lubrication schedule |

**Tips for effective Why-Why Analysis:**

- Stay focused on one causal chain at a time
- Base answers on facts, not assumptions—verify at the Gemba
- Don’t stop at “human error”—ask why the error was possible
- The number 5 is a guideline, not a rule—continue until you reach a controllable root cause

### 12.8.2.3 PM Analysis

**PM Analysis** (Physical Mechanism Analysis or Phenomenon-Mechanism Analysis) is a systematic method for analyzing **chronic losses**—problems that occur repeatedly despite multiple repair attempts. It examines the physical principles behind equipment failures.

**The PM Analysis Process:**

1. **Clarify the phenomenon:** Describe exactly what happens, when, and under what conditions
2. **Conduct physical analysis:** Understand the physical principles involved (friction, heat, vibration, etc.)
3. **Identify conditions that produce the phenomenon:** List all conditions required for the problem to occur
4. **Investigate equipment, materials, and methods:** Examine each condition against the 4Ms (Machine, Material, Method, Man)
5. **Plan and implement countermeasures:** Address the conditions that allow the problem to occur
6. **Verify results:** Confirm the chronic loss has been eliminated

*PM Analysis is particularly valuable when Why-Why Analysis doesn't resolve a recurring problem.*

### 12.8.2.4 SMED (Single-Minute Exchange of Die)

**SMED** is a systematic approach to reducing changeover or setup time—the time between the last good part of one product and the first good part of the next. The goal is to complete changeovers in under 10 minutes (single-digit minutes).

**The SMED Process:**

**SMED: Reducing Changeover Time**

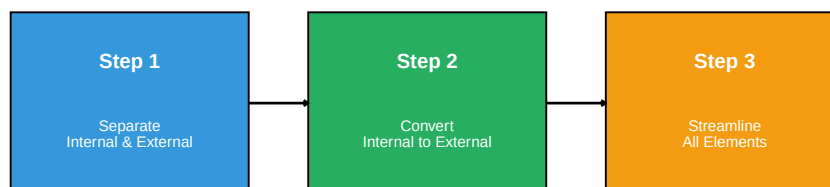


Figure 12.13: SMED Methodology

**Key Concepts:**

- **Internal setup:** Tasks that can only be done when the machine is stopped (changing the die itself)
- **External setup:** Tasks that can be done while the machine is running (preparing tools, preheating)

#### Typical SMED Improvements:

| Technique                | Example                                               |
|--------------------------|-------------------------------------------------------|
| Prepare in advance       | Stage tools and materials before stopping the machine |
| Use quick-release clamps | Replace bolts with cam locks or hydraulic clamps      |
| Eliminate adjustments    | Use fixtures, stops, and standardized settings        |
| Parallel operations      | Two people working simultaneously on different tasks  |
| Standardize components   | Use common bolt sizes, connection types               |

#### 12.8.2.5 Poka-Yoke (Error-Proofing)

**Poka-Yoke** ( ) means “mistake-proofing” or “error-proofing.” These are devices or methods that either prevent errors from occurring or detect them immediately when they do occur.

#### Types of Poka-Yoke:

| Type              | Function                           | Example                                         |
|-------------------|------------------------------------|-------------------------------------------------|
| <b>Prevention</b> | Makes errors impossible            | Asymmetric connector that only fits one way     |
| <b>Detection</b>  | Identifies errors immediately      | Sensor that stops the line if a part is missing |
| <b>Warning</b>    | Alerts operator to potential error | Light or buzzer when a step is skipped          |

#### Poka-Yoke Design Principles:

1. **Design out the possibility of error** — The best Poka-Yoke makes the error impossible
2. **Detect at the source** — Catch errors where they occur, not downstream
3. **Stop and fix immediately** — Don’t pass defects to the next process
4. **Keep it simple** — Low-cost, reliable solutions are better than complex ones

#### Examples in Manufacturing:

- **Guide pins** that prevent a part from being loaded incorrectly
- **Counters** that verify the correct number of components were installed
- **Checklists with physical verification** (e.g., removing a flag for each completed step)
- **Limit switches** that confirm proper positioning before the cycle starts
- **Color coding** to match parts with their correct locations

Question: How do you decide which tool to use?

#### Tool Selection Guide:

- **Pareto Analysis:** Use first to identify which loss to focus on
- **Why-Why Analysis:** Use for sporadic failures with clear cause-effect relationships
- **PM Analysis:** Use for chronic losses that keep recurring despite repairs
- **SMED:** Use specifically for setup/changeover losses (Loss #2)
- **Poka-Yoke:** Use to prevent defects and human errors (Losses #5, #6)

Often, multiple tools are used together—Pareto to select the problem, Why-Why or PM Analysis to find the root cause, and Poka-Yoke to prevent recurrence.

#### 12.8.2.6 Summary of Loss Elimination Tools

Table 12.11: Tools for Loss Elimination

| Tool                        | Purpose                               | Application                              |
|-----------------------------|---------------------------------------|------------------------------------------|
| <b>**Pareto Analysis**</b>  | Identify biggest losses to target     | Prioritize improvement efforts           |
| <b>**Why-Why Analysis**</b> | Find root cause of failures           | Breakdown analysis, defect investigation |
| <b>**PM Analysis**</b>      | Analyze chronic losses systematically | Complex, recurring problems              |
| <b>**SMED**</b>             | Reduce setup/changeover time          | Availability improvement (Loss #2)       |
| <b>**Poka-Yoke**</b>        | Error-proof the process               | Quality improvement (Losses #5, #6)      |

## 12.9 TPM Implementation

### 12.9.1 The 12 Steps of TPM Implementation

Table 12.12: 12 Steps of TPM Implementation

| Phase                  | Step | Activity                                    | Duration   |
|------------------------|------|---------------------------------------------|------------|
| <b>**Preparation**</b> | 1    | Announce TPM introduction by top management | 1 month    |
| <b>**Preparation**</b> | 2    | Education and campaign to introduce TPM     | 3-6 months |
| <b>**Preparation**</b> | 3    | Create TPM promotion organization           | 1-2 months |

|                           |    |                                                  |              |
|---------------------------|----|--------------------------------------------------|--------------|
| <b>**Preparation**</b>    | 4  | Establish basic TPM policies and goals           | 1-2 months   |
| <b>**Preparation**</b>    | 5  | Formulate master plan for TPM development        | 1-2 months   |
| <b>**Implementation**</b> | 6  | Hold TPM kick-off                                | 1 day        |
| <b>**Implementation**</b> | 7  | Improve effectiveness of each piece of equipment | 6-12 months  |
| <b>**Implementation**</b> | 8  | Develop autonomous maintenance program           | 12-24 months |
| <b>**Implementation**</b> | 9  | Develop planned maintenance program              | 12-24 months |
| <b>**Implementation**</b> | 10 | Conduct training to improve skills               | Ongoing      |
| <b>**Stabilization**</b>  | 11 | Develop early equipment management program       | 12-24 months |
| <b>**Stabilization**</b>  | 12 | Perfect TPM implementation and raise levels      | Ongoing      |

### 12.9.2 Success Factors

Table 12.13: TPM Success Factors

| Factor                               | Why Critical                                     |
|--------------------------------------|--------------------------------------------------|
| <b>**Top Management Commitment**</b> | Resources, culture change, sustained focus       |
| <b>**Training and Education**</b>    | Skills to identify problems, perform maintenance |
| <b>**Small Group Activities**</b>    | Operator involvement, team problem-solving       |
| <b>**Visual Management**</b>         | Make abnormalities obvious, track progress       |
| <b>**Patience and Persistence**</b>  | TPM takes 3-5 years to fully implement           |

## 12.10 Case Study: TPM Implementation

### 12.10.1 Background

**Company:** Automotive component manufacturer

**Equipment:** CNC machining center

**Initial OEE:** 52%

### 12.10.2 Problem Analysis

```
# Initial State
cat("=== INITIAL OEE BREAKDOWN ===\n\n")

availability_initial <- 78 # %
performance_initial <- 82 # %
quality_initial <- 81 # %

oee_initial <- (availability_initial/100) * (performance_initial/100) * (quality_initial/100)
```



```
cat("Availability:", availability_initial, "%\n")
cat("Performance:", performance_initial, "%\n")
cat("Quality:", quality_initial, "%\n")
cat("OEE:", round(oe_initial, 1), "%\n\n")
```

```
# Loss breakdown
cat("=== TOP LOSSES IDENTIFIED ===\n")
cat("1. Equipment breakdowns: 8% of time\n")
cat("2. Setup/changeover: 7% of time\n")
cat("3. Minor stops: 12% of time\n")
cat("4. Defects: 4% of production\n")
```

=== INITIAL OEE BREAKDOWN ===

Availability: 78 %  
 Performance: 82 %  
 Quality: 81 %  
 OEE: 51.8 %

=== TOP LOSSES IDENTIFIED ===

1. Equipment breakdowns: 8% of time  
 2. Setup/changeover: 7% of time  
 3. Minor stops: 12% of time  
 4. Defects: 4% of production

### 12.10.3 Improvements Implemented

| Loss        | Root Cause                | Countermeasure                 | Result          |
|-------------|---------------------------|--------------------------------|-----------------|
| Breakdowns  | Lack of basic maintenance | Autonomous maintenance program | 8% → 3%         |
| Setup time  | No standard procedure     | SMED implementation            | 35 min → 12 min |
| Minor stops | Chip buildup in sensors   | Improved chip management       | 12% → 4%        |
| Defects     | Tool wear not detected    | In-process gauging             | 4% → 1.5%       |

### 12.10.4 Results After 18 Months

```
# After TPM Implementation
cat("=== FINAL OEE BREAKDOWN ===\n\n")

availability_final <- 91 # %
```

```

performance_final <- 94      # %
quality_final <- 98.5      # %

oeo_final <- (availability_final/100) * (performance_final/100) * (quality_final/100)
cat("Availability:", availability_final, "% (was", availability_initial, "%)\n")
cat("Performance:", performance_final, "% (was", performance_initial, "%)\n")
cat("Quality:", quality_final, "% (was", quality_initial, "%)\n")
cat("OEE:", round(oeo_final, 1), "% (was", round(oeo_initial, 1), "%)\n\n")

improvement <- oeo_final - oeo_initial
cat("=== IMPROVEMENT ===\n")
cat("OEE Improvement:", round(improvement, 1), "percentage points\n")
cat("Relative Improvement:", round((improvement/oeo_initial)*100, 0), "%\n")

```

=== FINAL OEE BREAKDOWN ===

Availability: 91 % (was 78 %)  
 Performance: 94 % (was 82 %)  
 Quality: 98.5 % (was 81 %)  
 OEE: 84.3 % (was 51.8 %)

=== IMPROVEMENT ===

OEE Improvement: 32.4 percentage points  
 Relative Improvement: 63 %

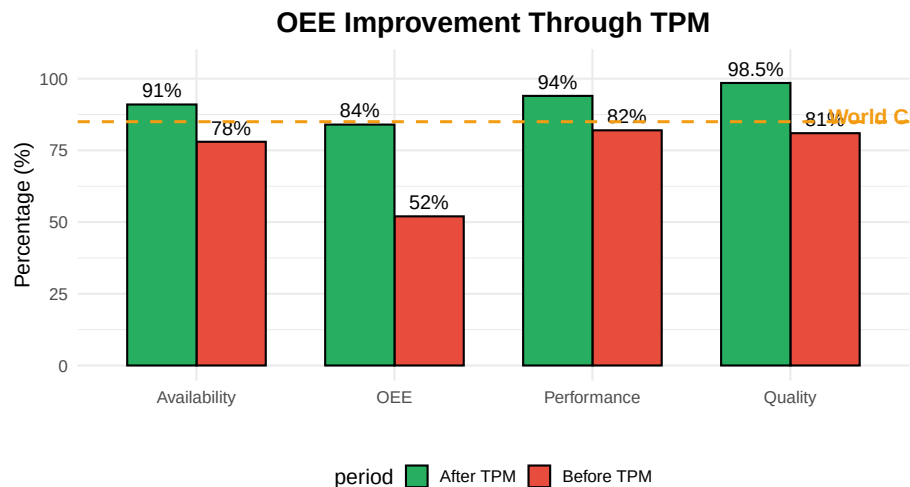


Figure 12.14: OEE Improvement: Before and After TPM

## 12.11 Summary

Table 12.14: TPM Key Concepts Summary

| Topic                      | Key Points                                                                                    |
|----------------------------|-----------------------------------------------------------------------------------------------|
| **TPM Definition**         | Total involvement, productive focus, maintenance excellence; zero breakdowns/defects          |
| **Eight Pillars**          | AM, PM, Quality, Focused Improvement, Early Equipment, Training, Safety, Office               |
| **OEE**                    | OEE = Availability $\times$ Performance $\times$ Quality; World class > 85%                   |
| **Six Big Losses**         | Equipment failure, setup, minor stops, speed loss, defects, startup rejects                   |
| **Autonomous Maintenance** | Operators maintain own equipment; 7 steps from cleaning to full ownership                     |
| **Planned Maintenance**    | Preventive $\rightarrow$ Predictive $\rightarrow$ Proactive; use P-F curve for optimal timing |

## 12.12 Review Questions

### 12.12.1 Conceptual Questions

Question 1: What does “Total” mean in Total Productive Maintenance?

“Total” has three dimensions:

1. **Total participation:** Everyone participates, from operators to top management
2. **Total equipment effectiveness:** Aim for maximum OEE (zero losses)
3. **Total system:** Covers the entire equipment lifecycle, from design to disposal

**Key insight:** TPM is not just a maintenance program — it’s a company-wide philosophy where everyone takes responsibility for equipment care.

Question 2: Why is OEE calculated as a product rather than a sum?

**OEE uses multiplication because losses are sequential and compound:**

1. **Availability** determines how much time is available
2. **Performance** determines how much of that time is productive
3. **Quality** determines how much of that production is good

**Example:** - If each factor is 90%:  $OEE = 0.9 \times 0.9 \times 0.9 = 72.9\%$  (not 270% or 90%) - This reflects reality: you can only achieve quality on parts that were produced (performance) during time the machine ran (availability)

**Why this matters:** - Small improvements in all three factors compound - 5% improvement in each:  $0.85 \times 0.85 \times 0.85 = 61\% \rightarrow 0.90 \times 0.90 \times 0.90 = 73\%$   
- Focus on the weakest factor first for biggest impact

Question 3: What is the difference between autonomous maintenance and planned maintenance?

| Aspect        | Autonomous Maintenance                 | Planned Maintenance        |
|---------------|----------------------------------------|----------------------------|
| <b>Who</b>    | Operators                              | Maintenance technicians    |
| <b>What</b>   | Basic care (clean, inspect, lubricate) | Complex repairs, overhauls |
| <b>When</b>   | Daily, during operation                | Scheduled intervals        |
| <b>Skills</b> | Basic equipment knowledge              | Technical expertise        |
| <b>Goal</b>   | Prevent deterioration                  | Restore/improve equipment  |

**They complement each other:** - AM catches problems early (first line of defense) - PM provides specialized skills when needed - Together they achieve zero breakdowns

### 12.12.2 Calculation Questions

Question 4: Calculate OEE given the following data.

**Given:** - Shift length: 480 minutes - Breaks: 30 minutes - Changeover: 45 minutes - Breakdown: 20 minutes - Ideal cycle time: 30 seconds (0.5 min) - Total parts produced: 580 - Rejected parts: 23

**Solution:**

Planned Production Time = 480 - 30 = 450 minutes

Run Time = 450 - 45 - 20 = 385 minutes

Availability = 385 / 450 = 85.6%

Theoretical Output = 385 / 0.5 = 770 parts

Performance = 580 / 770 = 75.3%

Quality = (580 - 23) / 580 = 557 / 580 = 96.0%

OEE = 85.6% × 75.3% × 96.0% = 61.9%

**Analysis:** Performance is the biggest opportunity for improvement (75.3% vs. targets of 95%). Investigate minor stops and speed losses.

Question 5: A machine has OEE of 65% with Availability 90%, Performance 80%, Quality 90%. Which factor should be prioritized?

**Analysis:**

Current state: - Availability: 90% (Gap to world class: 5%) - Performance: 80% (Gap to world class: 15%) - Quality: 90% (Gap to world class: 9%)

**Performance should be prioritized because:**

1. **Biggest gap:** 15 percentage points below world class (95%)
2. **Biggest OEE impact:** Improving performance 10%  $\rightarrow$   $OEE = 90\% \times 90\% \times 90\% = 72.9\%$
3. **Often overlooked:** Speed losses and minor stops are frequently accepted as normal

**Investigation areas for performance:** - Minor stops (jams, blockages, sensor trips) - Reduced speed operation - Idling and waiting - Operator pace

Question 6: If current OEE is 60% and target is 85%, what improvement is needed in each component assuming equal improvement?

**Solution:**

Current:  $60\% = A \times P \times Q$

If we improve each equally, we need: - Target:  $85\% = A' \times P' \times Q'$  - If  $A' = P' = Q' = x$ , then  $x^3 = 0.85$  -  $x = \sqrt[3]{0.85} = 0.947 = 94.7\%$

**Each component needs to reach approximately 95%**

**Check:**  $0.947 \times 0.947 \times 0.947 = 0.849 = 84.9\%$

**If current factors are A=80%, P=85%, Q=88%:** - Availability needs: +15 points (80%  $\rightarrow$  95%) - Performance needs: +10 points (85%  $\rightarrow$  95%) - Quality needs: +7 points (88%  $\rightarrow$  95%)

**Prioritize by gap size:** Availability first, then Performance, then Quality.

### 12.12.3 Application Questions

Question 7: An operator notices a small oil leak on a machine. Under traditional maintenance vs. TPM, what happens?

**Traditional Maintenance Approach:** 1. Operator continues running machine 2. May or may not report to maintenance 3. If reported, added to backlog 4. Maintenance addresses when convenient (days/weeks) 5. Leak worsens, potential bearing damage 6. Eventually leads to breakdown

**TPM Approach:** 1. Operator notices during routine inspection 2. Tags the abnormality immediately 3. Determines if operator can fix (blue tag) or needs maintenance (red tag) 4. If simple: operator tightens fitting or adds sealant 5. If complex: maintenance prioritizes based on severity 6. Root cause investigated to prevent recurrence 7. Finding shared in team meeting

**Key Differences:** - **Ownership:** Operator takes responsibility vs. "not my job" - **Speed:** Immediate action vs. delayed response - **Prevention:** Fixed before failure vs. reactive repair - **Cost:** \$10 seal vs. \$5,000 bearing replacement + downtime

Question 8: List five items an operator should check during daily autonomous maintenance.

**Five Essential Daily AM Checks:**

1. **Cleanliness:** Remove chips, debris, dust; clean safety windows and sensors
  - *Why:* Contamination causes wear, sensor malfunctions, quality issues
2. **Lubrication:** Check oil levels, grease points, coolant level
  - *Why:* Inadequate lubrication causes friction, heat, accelerated wear
3. **Fasteners:** Check for loose bolts, guards, covers
  - *Why:* Loose fasteners cause vibration, safety hazards, misalignment
4. **Leaks:** Look for oil, coolant, air, hydraulic leaks
  - *Why:* Leaks indicate seal failure, waste resources, create hazards
5. **Abnormal sounds/vibration:** Listen and feel during startup
  - *Why:* Changes indicate developing problems (bearings, alignment)

**Bonus items:** - Safety guards in place - Emergency stops functional - Pressure/temperature gauges in normal range - No unusual smells (burning, chemical)

Question 9: Explain how SMED relates to OEE improvement.

**SMED (Single Minute Exchange of Die) directly improves Availability:**

**Loss #2: Setup and Adjustment** is an availability loss captured in OEE.

**How SMED helps:**

1. **Reduces changeover time:**
  - Before SMED: 45-minute changeover
  - After SMED: 10-minute changeover
  - Gain: 35 minutes per changeover
2. **Impact on OEE:**
  - If 2 changeovers per shift: 70 minutes saved
  - On 450-minute shift: Availability improves 15%+
3. **Secondary benefits:**
  - More changeovers become practical
  - Smaller batch sizes possible
  - Flexibility increased
  - Less WIP inventory

**SMED Process:** 1. Separate internal (machine stopped) from external (while running) setup 2. Convert internal to external where possible 3. Streamline remaining internal operations 4. Practice and standardize

**Example:** Die change on press - External: Stage next die, prepare tools, pre-heat - Internal: Remove old die, install new die, adjust - Target: All possible work done while machine runs

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## 12.13 References

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## Chapter 13

# Quality Management Systems

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### 13.1 Learning Objectives

After completing this chapter, you will be able to:

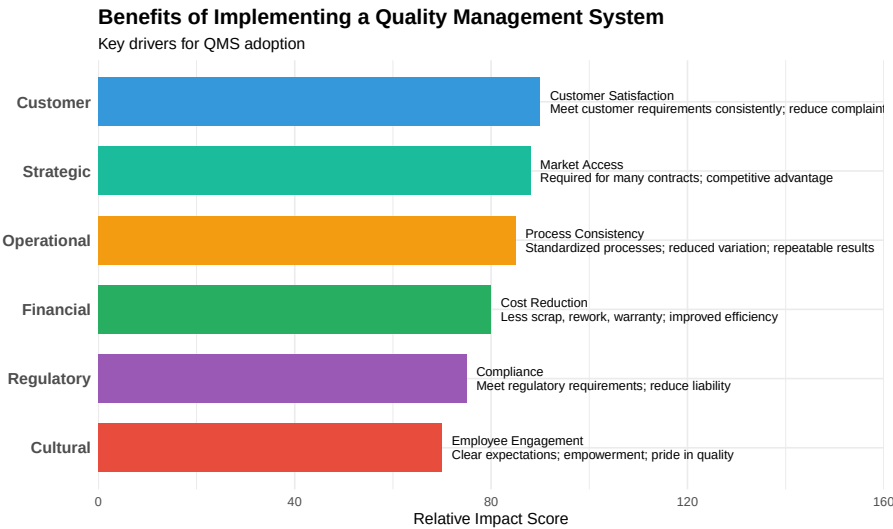
1. Define quality management systems and explain their purpose in manufacturing
  2. Describe the structure and requirements of ISO 9001:2015
  3. Identify industry-specific quality standards (IATF 16949, AS9100, ISO 22000)
  4. Explain the documentation hierarchy and control requirements
  5. Conduct internal audits using systematic methodologies
  6. Implement effective corrective and preventive action (CAPA) processes
  7. Understand management review and continuous improvement requirements
  8. Integrate quality tools (SPC, PFMEA, MSA) within a QMS framework
- 

### 13.2 Introduction to Quality Management Systems

A **Quality Management System (QMS)** is a formalized system that documents processes, procedures, and responsibilities for achieving quality policies and objectives. It coordinates and directs an organization's activities to meet

customer and regulatory requirements while continuously improving effectiveness.

13.2.1 Why Implement a QMS?



13.2.2 The Evolution of Quality Management

Table 13.1: Evolution of Quality Management

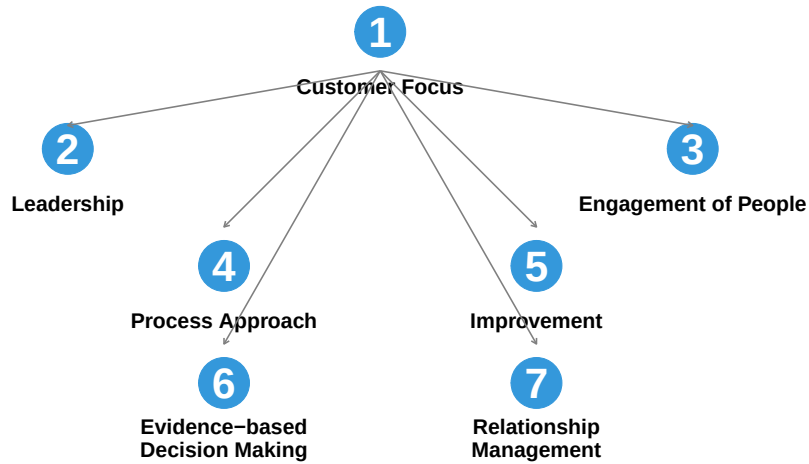
| Era           | Approach                    | Focus                                                        |
|---------------|-----------------------------|--------------------------------------------------------------|
| Pre-1900s     | Craftsmanship               | Individual skill; master craftsmen                           |
| 1900-1940s    | Inspection                  | Sort good from bad; end-of-line inspection                   |
| 1940-1960s    | Statistical Quality Control | Control charts; sampling; Shewhart, Deming                   |
| 1960-1980s    | Quality Assurance           | Prevention; systems approach; documented procedures          |
| 1980-2000s    | Total Quality Management    | Company-wide quality; customer focus; continuous improvement |
| 2000s-Present | Integrated Management       | Risk-based thinking; integration with business strategy      |

13.2.3 Quality Management Principles

ISO 9000 defines seven quality management principles that form the foundation of all QMS standards:

**Seven Quality Management Principles (ISO 9000:2015)**

Customer Focus is the primary principle; others support it



### 13.3 ISO 9001:2015 Structure

**ISO 9001:2015** is the international standard for quality management systems. It uses the High-Level Structure (HLS) common to all ISO management system standards.

### 13.3.1 The Ten Clauses

#### ISO 9001:2015 Structure – The Ten Clauses

Clauses 4–10 contain the auditable requirements

|    |                             |                                                         |
|----|-----------------------------|---------------------------------------------------------|
| 1  | Scope                       | Defines applicability of the standard                   |
| 2  | Normative References        | References to ISO 9000:2015                             |
| 3  | Terms and Definitions       | Terms used in the standard                              |
| 4  | Context of the Organization | Understanding the organization and its context          |
| 5  | Leadership                  | Top management commitment and policy                    |
| 6  | Planning                    | Actions to address risks and opportunities              |
| 7  | Support                     | Resources, competence, awareness, communication         |
| 8  | Operation                   | Operational planning and control                        |
| 9  | Performance Evaluation      | Monitoring, measurement, analysis, evaluation           |
| 10 | Improvement                 | Nonconformity, corrective action, continual improvement |

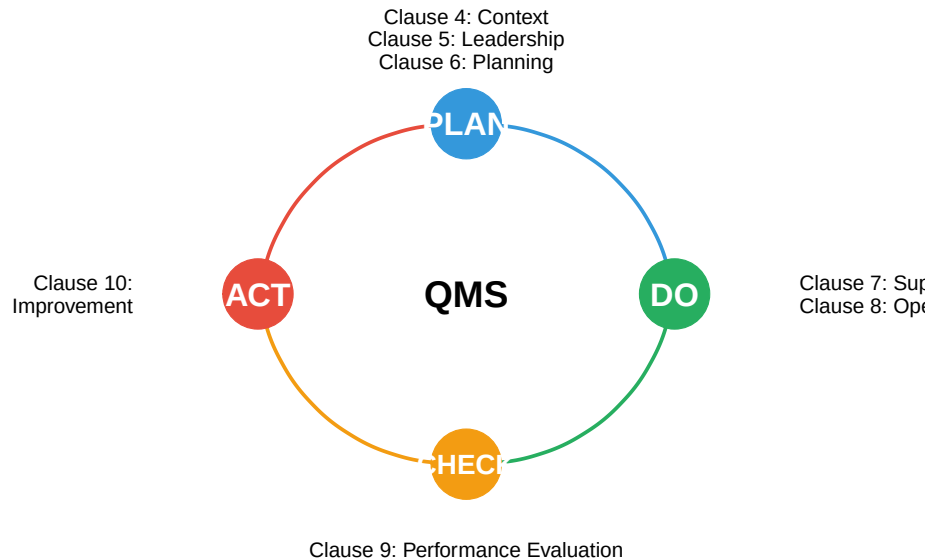
### 13.3.2 The Process Approach and PDCA

ISO 9001:2015 emphasizes the **process approach** combined with the **Plan-Do-Check-Act (PDCA)** cycle:

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i Please use `linewidth` instead.

### PDCA Cycle and ISO 9001:2015 Clauses

The continuous improvement cycle embedded in the standard



### 13.3.3 Risk-Based Thinking

ISO 9001:2015 introduced **risk-based thinking** throughout the QMS:

Table 13.2: Risk-Based Thinking in ISO 9001:2015

| Clause | Requirement                                | Risk_Application                                                 |
|--------|--------------------------------------------|------------------------------------------------------------------|
| 4.1    | Context of the Organization                | Identify internal/external issues that could affect QMS outcomes |
| 4.2    | Interested Parties                         | Understand needs/expectations that could impact quality          |
| 6.1    | Actions to Address Risks and Opportunities | Plan actions to address risks and opportunities; integrate       |
| 8.1    | Operational Planning and Control           | Control processes considering risks identified in planning       |
| 9.1    | Monitoring and Measurement                 | Monitor effectiveness of actions taken to address risks          |
| 10.2   | Nonconformity and Corrective Action        | Analyze nonconformities; update risks/opportunities as needed    |

#### Understanding Risk-Based Thinking

**Risk-based thinking** doesn't require formal risk management (like FMEA for every process), but rather:

1. **Consider risks** when designing and implementing the QMS
2. **Preventive action** is built into the system, not a separate activity

3. **Proportional response** - more rigorous controls for higher-risk processes
4. **Opportunities** are also considered (not just negative risks)

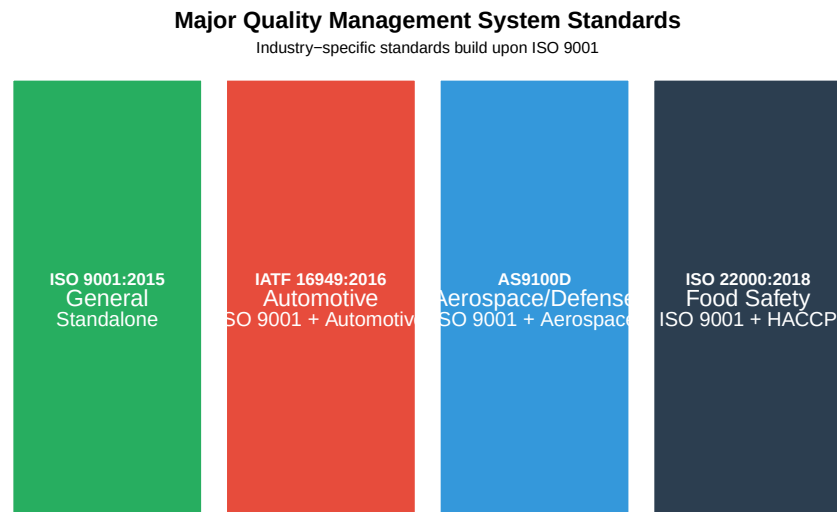
**Example:** A food manufacturer identifies “allergen cross-contamination” as a significant risk. Risk-based thinking leads them to: - Dedicated production lines for allergen-containing products - Enhanced cleaning verification procedures - Staff training on allergen awareness - Increased inspection frequency after changeovers

The extent of controls is proportional to the risk level.

## 13.4 Industry-Specific Quality Standards

Different industries have developed standards that build upon ISO 9001 with additional sector-specific requirements.

### 13.4.1 Comparison of Major Standards



### 13.4.2 IATF 16949:2016 (Automotive)

**IATF 16949** is the automotive quality standard, required by most major OEMs (Ford, GM, Toyota, VW, etc.).

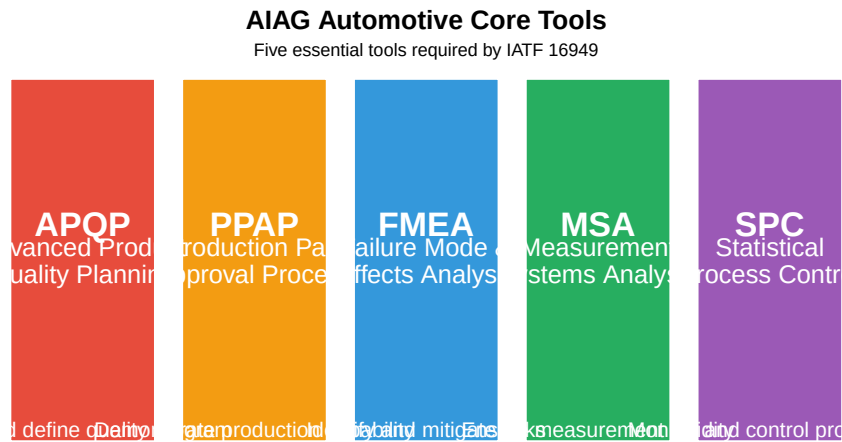
Table 13.3: IATF 16949 Additional Requirements Beyond ISO 9001

| Requirement Area | ISO 9001 | IATF 16949 Addition |
|------------------|----------|---------------------|
|------------------|----------|---------------------|

|                           |                        |                                                               |
|---------------------------|------------------------|---------------------------------------------------------------|
| Product Safety            | Implied                | Explicit requirements for product safety processes            |
| APQP/PPAP                 | Not specified          | APQP phases required; PPAP submission mandatory               |
| Control Plans             | Not specified          | Control plans required for all parts; specific format         |
| MSA                       | General requirement    | MSA studies required per AIAG MSA manual                      |
| SPC                       | General requirement    | SPC for all special characteristics                           |
| Supplier Quality          | Basic purchasing       | Supplier development; second-party audits; IATF certification |
| Warranty Management       | Complaint handling     | NTF analysis; warranty data analysis; field return analysis   |
| Manufacturing Feasibility | Review of requirements | Documented manufacturing feasibility review                   |

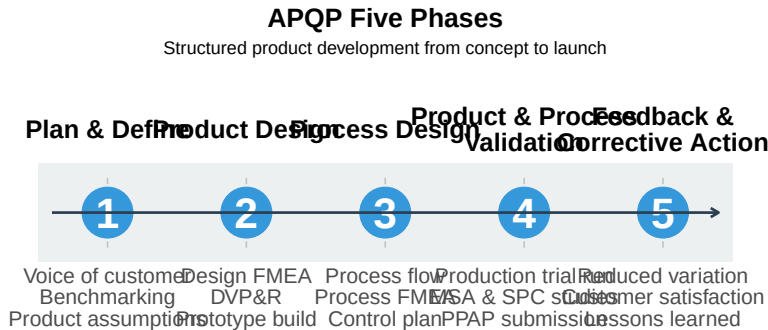
13.4.3 Automotive Core Tools

IATF 16949 requires the use of **five core tools**:



13.4.3.1 APQP: Advanced Product Quality Planning

**APQP** is a structured framework for developing products that ensures customer requirements are met through careful planning and defined milestones.



**Example: New Brake Caliper Development**

A Tier 1 supplier developing a new brake caliper for an OEM would follow APQP:

- **Phase 1:** Gather customer requirements (stopping distance, weight limits, environmental conditions), benchmark competitor products
- **Phase 2:** Design the caliper, create Design FMEA (what if piston seal fails?), build prototypes for testing
- **Phase 3:** Design manufacturing process, create Process FMEA, develop control plan for machining and assembly
- **Phase 4:** Run production trial, validate  $C_{pk} > 1.67$  for critical dimensions, submit PPAP package
- **Phase 5:** Monitor warranty returns, implement kaizen improvements, document lessons for next program

**13.4.3.2 PPAP: Production Part Approval Process**

**PPAP** provides evidence that the supplier understands customer requirements and can consistently produce conforming parts during actual production runs.

Table 13.4: PPAP 18 Elements Required for Submission

| Element                                      | Description                                                       |
|----------------------------------------------|-------------------------------------------------------------------|
| <b>1. Design Records</b>                     | Part drawings, CAD files, specifications                          |
| <b>2. Engineering Change Documents</b>       | Documentation of any design changes since original approval       |
| <b>3. Customer Engineering Approval</b>      | Evidence of customer sign-off on design                           |
| <b>4. Design FMEA</b>                        | Risk analysis of product design failure modes                     |
| <b>5. Process Flow Diagram</b>               | Visual representation of manufacturing steps                      |
| <b>6. Process FMEA</b>                       | Risk analysis of process failure modes                            |
| <b>7. Control Plan</b>                       | Document describing controls for each process step                |
| <b>8. MSA Studies</b>                        | Gage R&R studies proving measurement system adequacy              |
| <b>9. Dimensional Results</b>                | Actual measurements vs. specifications for all dimensions         |
| <b>10. Material/Performance Test Results</b> | Test results for material properties and performance requirements |
| <b>11. Initial Process Studies (SPC)</b>     | $C_{pk}/P_{pk}$ studies demonstrating process capability          |
| <b>12. Qualified Lab Documentation</b>       | Scope and accreditation of testing laboratories                   |
| <b>13. Appearance Approval Report</b>        | Required for parts with color, grain, or surface requirements     |
| <b>14. Sample Production Parts</b>           | Parts from production run for customer evaluation                 |
| <b>15. Master Sample</b>                     | Retained reference sample for future comparison                   |
| <b>16. Checking Aids</b>                     | Fixtures, gages, or templates used to inspect parts               |
| <b>17. Customer-Specific Requirements</b>    | OEM-specific requirements (Ford, GM, Toyota, etc.)                |
| <b>18. Part Submission Warrant (PSW)</b>     | Summary document with supplier certification                      |

**PPAP Submission Levels:**



| Level | Requirements                                   | When Used                          |
|-------|------------------------------------------------|------------------------------------|
| 1     | PSW only                                       | Low-risk parts, trusted suppliers  |
| 2     | PSW + samples + limited data                   | Standard submission                |
| 3     | PSW + samples + complete data                  | Default level; new parts/suppliers |
| 4     | PSW + other per customer direction             | Customer-specified requirements    |
| 5     | PSW + samples + complete data at supplier site | Full documentation review on-site  |

#### Example: Fuel Injector PPAP Submission

A supplier submitting PPAP for a new fuel injector would include:

- **Dimensional results:** 100% inspection of 300 parts showing all 47 dimensions in spec
- **MSA study:** Gage R&R < 10% for critical spray angle measurement
- **Process capability:** Cpk = 1.89 for nozzle hole diameter (spec requires Cpk = 1.67)
- **Material certs:** Steel chemistry and hardness test results
- **Performance tests:** Flow rate, spray pattern, durability (1 million cycles)
- **PSW:** Signed warrant certifying all requirements met

#### 13.4.3.3 FMEA: Failure Mode and Effects Analysis

**FMEA** is a systematic method for identifying potential failures, their causes, and effects, then prioritizing actions to reduce risk. Two types are used in automotive:

- **Design FMEA (DFMEA):** Analyzes potential failures in product design
- **Process FMEA (PFMEA):** Analyzes potential failures in manufacturing processes

Table 13.5: Process FMEA Structure and Example

| Column                        | Description                                                | Example                     |
|-------------------------------|------------------------------------------------------------|-----------------------------|
| <b>Process Step</b>           | The operation being analyzed                               | Torque fastener to 25 Nm    |
| <b>Potential Failure Mode</b> | How the process could fail to meet requirements            | Under-torque / Over-torque  |
| <b>Potential Effect(s)</b>    | Impact on customer if failure occurs                       | Joint loosens in service; w |
| <b>Severity (S)</b>           | 1-10 rating of effect seriousness (10 = safety/regulatory) | 10 (safety critical)        |
| <b>Potential Cause(s)</b>     | Root cause of the failure mode                             | Worn tool, wrong program    |
| <b>Occurrence (O)</b>         | 1-10 rating of cause likelihood (10 = very frequent)       | 4 (occasional occurrence)   |

|                            |                                                       |                          |
|----------------------------|-------------------------------------------------------|--------------------------|
| <b>Current Controls</b>    | Existing prevention or detection controls             | Torque monit             |
| <b>Detection (D)</b>       | 1-10 rating of detection ability (10 = cannot detect) | 3 (high detect           |
| <b>RPN</b>                 | $S \times O \times D$ = Risk Priority Number          | $10 \times 4 \times 3 =$ |
| <b>Recommended Actions</b> | Actions to reduce S, O, or D                          | Add DC torq              |

Action Priority Based on RPN:

RPN-Based Action Priority

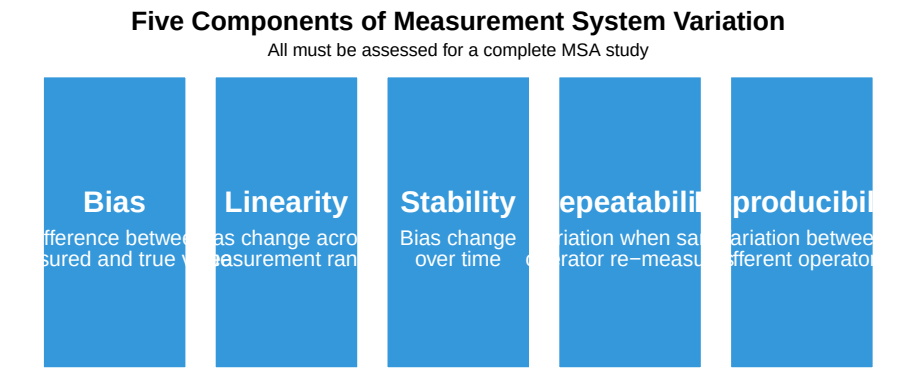


Example: Engine Block Machining PFMEA

| Step          | Failure Mode            | Effect                       | S | Cause             | O | Control                      | D | RPN | Action                                             |
|---------------|-------------------------|------------------------------|---|-------------------|---|------------------------------|---|-----|----------------------------------------------------|
| Bore cylinder | Oversize bore           | Piston slap, oil consumption | 7 | Tool wear         | 5 | SPC chart, tool life monitor | 4 | 140 | Add in-process bore gage                           |
| Hone surface  | Insufficient crosshatch | Poor ring seal, oil burning  | 8 | Wrong hone stones | 3 | Visual inspection            | 6 | 144 | Add surface roughness measurement                  |
| Wash block    | Residual chips          | Engine seizure               | 9 | Blocked nozzle    | 4 | Cleanliness spec check       | 5 | 180 | Add particle counter, automated flush verification |

13.4.3.4 MSA: Measurement Systems Analysis

**MSA** ensures that measurement systems provide reliable data for decision-making. Poor measurement systems can cause good parts to be rejected (false rejects) or bad parts to be accepted (false accepts).



**Gage R&R Study Acceptance Criteria:**

| %GRR   | Interpretation | Action                                                                    |
|--------|----------------|---------------------------------------------------------------------------|
| < 10%  | Excellent      | Acceptable for all applications                                           |
| 10-30% | Acceptable     | May be acceptable depending on application, cost of gage, cost of repairs |
| > 30%  | Unacceptable   | Measurement system must be improved; cannot make reliable decisions       |

**Number of Distinct Categories (ndc):** Should be 5 to adequately discriminate between parts.

**Example: Gage R&R Study for Shaft Diameter**

A supplier conducts a Gage R&R study for a micrometer used to measure shaft diameter (spec: 25.00 ± 0.05 mm):

**Study Setup:** - 3 operators measure 10 parts, 3 times each = 90 measurements  
- Parts selected to represent the range of production variation

**Results:** | Source | Variance | % Contribution | |:—|:—|:—| |



**Process Capability Indices:**

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min\left(\frac{USL - \bar{x}}{3\sigma}, \frac{\bar{x} - LSL}{3\sigma}\right)$$

| Cpk Value   | Interpretation                    | PPM Defective |
|-------------|-----------------------------------|---------------|
| < 1.00      | Not capable                       | > 2,700       |
| 1.00 - 1.33 | Marginally capable                | 64 - 2,700    |
| 1.33 - 1.67 | Capable                           | 0.6 - 64      |
| 1.67        | Highly capable (IATF requirement) | < 0.6         |

**Example: SPC for Injection Molded Connector**

A supplier monitors critical dimensions on an injection molded electrical connector:

**Process Setup:** - Critical dimension: Pin spacing =  $2.54 \pm 0.08$  mm - Sample 5 parts every 30 minutes - X-bar and R charts maintained at the press

**Results over one shift:** - Process mean (X-double-bar) = 2.539 mm - Average range (R-bar) = 0.024 mm - Estimated  $\sigma = R\text{-bar} / d_2 = 0.024 / 2.326 = 0.0103$  mm

**Capability:** -  $C_p = (2.62 - 2.46) / (6 \times 0.0103) = 2.59$  -  $C_{pk} = \min[(2.62 - 2.539)/(3 \times 0.0103), (2.539 - 2.46)/(3 \times 0.0103)] = \min[2.62, 2.56] = \mathbf{2.56}$

**Conclusion:**  $C_{pk} = 2.56$  exceeds IATF requirement of 1.67. Process is highly capable.

**13.4.4 AS9100D (Aerospace/Defense)**

**AS9100D** adds aerospace-specific requirements to ISO 9001:

Table 13.6: AS9100D Key Additional Requirements

| Area                                | Requirement                                                           |
|-------------------------------------|-----------------------------------------------------------------------|
| <b>Configuration Management</b>     | Control product configuration throughout lifecycle; change management |
| <b>First Article Inspection</b>     | FAI per AS9102; documented verification of first production parts     |
| <b>Counterfeit Parts Prevention</b> | Controls to prevent counterfeit parts entering supply chain           |
| <b>Special Processes</b>            | NADCAP accreditation for special processes (welding, heat treat, NDT) |
| <b>Product Safety</b>               | Product safety requirements; reporting of unsafe conditions           |
| <b>Risk Management</b>              | Explicit risk management process (often using AS/NZS 4360 or similar) |

|               |                                                  |
|---------------|--------------------------------------------------|
| Human Factors | Consideration of human factors in process design |
|---------------|--------------------------------------------------|

13.4.5 ISO 22000:2018 (Food Safety)

ISO 22000 integrates quality management with HACCP (Hazard Analysis Critical Control Points):

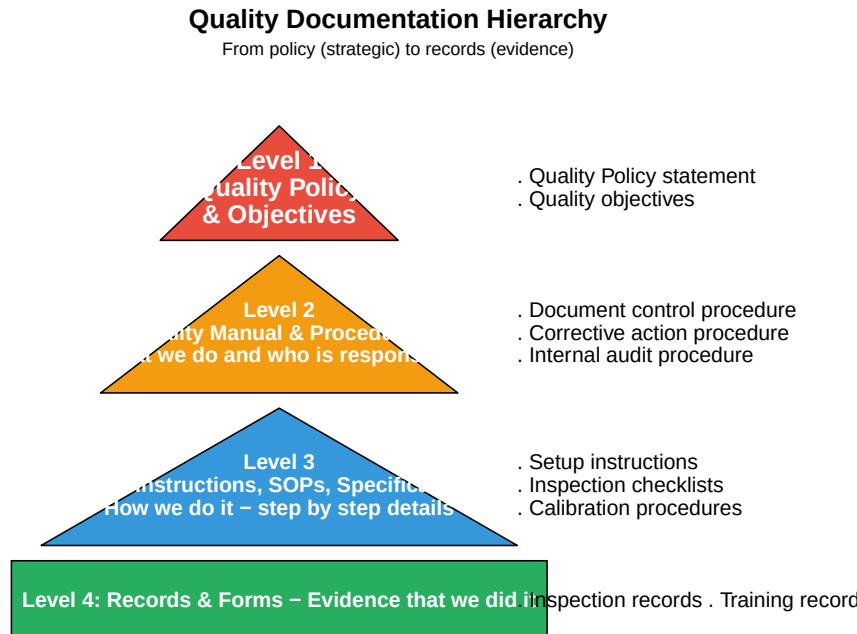
Table 13.7: ISO 22000 Food Safety M

| Element                      | Description                                                             |
|------------------------------|-------------------------------------------------------------------------|
| Prerequisite Programs (PRPs) | Basic hygiene conditions: cleaning, pest control, personnel hygiene     |
| Operational PRPs (OPRPs)     | PRPs essential to control specific identified hazards                   |
| HACCP Plan                   | Critical Control Points (CCPs) for significant hazards; critical limits |
| Traceability                 | One-step-back, one-step-forward traceability; recall procedures         |
| Emergency Preparedness       | Procedures for food safety emergencies; recall/withdrawal               |
| Validation and Verification  | Validate control measures; verify HACCP plan effectiveness              |

13.5 Documentation Requirements

A QMS requires documented information to ensure consistency, provide evidence, and enable improvement.

### 13.5.1 Documentation Hierarchy



### 13.5.2 Document Control Requirements

Table 13.8: Document Control Requirements (ISO 9001 Clause 7.5)

| Requirement                | Description                                                     | Implementation                 |
|----------------------------|-----------------------------------------------------------------|--------------------------------|
| <b>Approval</b>            | Documents reviewed and approved for adequacy before issue       | Approval signatures            |
| <b>Review &amp; Update</b> | Reviewed, updated as necessary, and re-approved                 | Periodic review schedule       |
| <b>Identification</b>      | Identified with title, date, revision, author                   | Document numbering             |
| <b>Availability</b>        | Relevant versions available at points of use                    | Controlled copies available    |
| <b>Protection</b>          | Protected from loss, damage, unauthorized changes               | Backup systems; access control |
| <b>Obsolete Control</b>    | Obsolete documents identified and prevented from unintended use | Stamp 'OBSOLETE'               |
| <b>External Documents</b>  | External documents identified and distribution controlled       | List of external documents     |
| <b>Record Retention</b>    | Records retained for specified periods; protected; retrievable  | Retention matrix; storage      |

### 13.5.3 Documented Information Required by ISO 9001:2015

Table 13.9: Documented Information Required by ISO 9001:2015

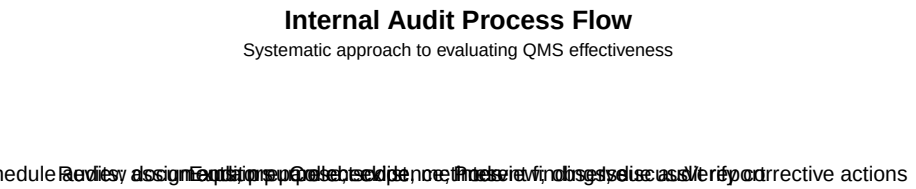
| Type                                   | Clause  | Requirement                                      |
|----------------------------------------|---------|--------------------------------------------------|
| <b>Documents (shall be maintained)</b> |         |                                                  |
| Documents (shall be maintained)        | 4.3     | Scope of the QMS                                 |
| Documents (shall be maintained)        | 5.2     | Quality policy                                   |
| Documents (shall be maintained)        | 6.2     | Quality objectives                               |
| Documents (shall be maintained)        | 7.1.5   | Monitoring and measuring resources (calibration) |
| Documents (shall be maintained)        | 7.2     | Evidence of competence                           |
| Documents (shall be maintained)        | 8.1     | Evidence of conformity to processes              |
| Documents (shall be maintained)        | 8.5.1   | Controlled conditions for production             |
| <b>Records (shall be retained)</b>     |         |                                                  |
| Records (shall be retained)            | 7.1.5.1 | Calibration/verification results                 |
| Records (shall be retained)            | 7.2     | Training records                                 |
| Records (shall be retained)            | 8.2.3.2 | Results of review of requirements                |
| Records (shall be retained)            | 8.3.3   | Design and development inputs                    |
| Records (shall be retained)            | 8.3.4   | Design and development controls                  |
| Records (shall be retained)            | 8.3.5   | Design and development outputs                   |
| Records (shall be retained)            | 8.3.6   | Design and development changes                   |
| Records (shall be retained)            | 8.5.2   | Traceability requirements                        |
| Records (shall be retained)            | 8.6     | Release of products/services                     |
| Records (shall be retained)            | 9.1.1   | Results of monitoring and measurement            |

## 13.6 Internal Auditing

**Internal audits** are systematic, independent evaluations of the QMS to determine if it conforms to requirements and is effectively implemented.



13.6.1 The Audit Process



13.6.2 Audit Checklist Example

Table 13.10: Sample Audit Checklist: Clause 7.1.5 Monitoring and Resources

| Clause  | Requirement                                                            | Question                          |
|---------|------------------------------------------------------------------------|-----------------------------------|
| 7.1.5   | Monitoring and measuring resources suitable for the type of activities | How do you determine what me      |
| 7.1.5   | Resources maintained to ensure continued fitness for purpose           | What maintenance is performed     |
| 7.1.5   | Documented information on fitness for purpose retained                 | Can you show me calibration re    |
| 7.1.5   | Measuring equipment identified to determine status                     | How can I tell if this gauge is c |
| 7.1.5.2 | Measurement traceability when required                                 | What standards are these gauge    |

13.6.3 Types of Audit Findings

Table

| Finding Type                | Definition                                                                   |
|-----------------------------|------------------------------------------------------------------------------|
| Major Nonconformity         | Absence or total breakdown of a system; significant risk to product/customer |
| Minor Nonconformity         | Single lapse or isolated incident; system exists but not fully effective     |
| Observation                 | Weakness that could lead to nonconformity if not addressed                   |
| Opportunity for Improvement | Suggestion for enhancement; not a nonconformity                              |

13.6.4 Auditor Competence

Table 13.12: Internal Auditor Competency Requirements

| Competency                    | Description                                                  | How to Develop             |
|-------------------------------|--------------------------------------------------------------|----------------------------|
| <b>Knowledge of Standards</b> | Understanding of ISO 9001 and applicable industry standards  | Training, self-study       |
| <b>Audit Techniques</b>       | Planning, interviewing, evidence collection, reporting       | Audit experience, training |
| <b>Industry Knowledge</b>     | Understanding of processes, products, and industry practices | Work experience, training  |
| <b>Communication Skills</b>   | Active listening, clear questioning, professional writing    | Practice, training         |
| <b>Objectivity</b>            | Independence from area being audited; impartial              | Audit experience, training |
| <b>Professional Judgment</b>  | Ability to evaluate significance of findings                 | Experience, training       |

## 13.7 Corrective and Preventive Action (CAPA)

**Corrective action** eliminates the cause of a detected nonconformity to prevent recurrence. **Preventive action** eliminates the cause of a potential nonconformity to prevent occurrence.

### 13.7.1 The CAPA Process

#### CAPA Process Flow

Eight-step process for effective corrective and preventive action



#### 13.7.1.1 Step 1: Identification

The CAPA process begins when a nonconformity or potential problem is identified. Sources include:

Table 13.13: Common CAPA Initiation Sources

| Source                     | Description                                            | Priority   |
|----------------------------|--------------------------------------------------------|------------|
| <b>Customer Complaints</b> | Product returns, warranty claims, customer feedback    | High       |
| <b>Internal Audits</b>     | Nonconformities found during scheduled internal audits | Medium     |
| <b>Process Monitoring</b>  | SPC out-of-control conditions, trend analysis          | Medium     |
| <b>Inspection/Testing</b>  | Failed inspections, test failures, rejected lots       | High       |
| <b>Supplier Issues</b>     | Incoming inspection failures, supplier audit findings  | Medium     |
| <b>Employee Reports</b>    | Near-misses, safety concerns, improvement suggestions  | Low-Medium |
| <b>Management Review</b>   | Issues identified during management review meetings    | Medium     |
| <b>External Audits</b>     | Findings from certification body or customer audits    | High       |

### 13.7.1.2 Step 2: Evaluation

Evaluate the significance of the issue to determine the appropriate response level:

#### Evaluation Criteria:

| Factor                 | Questions to Consider                                 |
|------------------------|-------------------------------------------------------|
| <b>Safety</b>          | Could this harm people or the environment?            |
| <b>Customer Impact</b> | How many customers affected? Severity of impact?      |
| <b>Regulatory</b>      | Does this violate regulations or standards?           |
| <b>Financial</b>       | What is the cost of the problem? Potential liability? |
| <b>Frequency</b>       | Is this a one-time event or recurring pattern?        |
| <b>Detection</b>       | How was it found? Could others exist undetected?      |

**Containment Actions** - Immediate actions to limit the impact while the investigation proceeds:

- Quarantine suspect product
- Sort and inspect inventory
- Notify affected customers
- Implement temporary controls
- Stop shipments if necessary

#### Example: Evaluation of Dimensional Nonconformity

A batch of machined shafts is found with diameter 0.02 mm oversize:

- **Safety:** Low - not a safety-critical dimension
- **Customer Impact:** Medium - may cause assembly issues
- **Regulatory:** None
- **Financial:** 500 parts  $\times$  \$15/part = \$7,500 potential scrap

- **Containment:** Quarantine batch, 100% inspection of last 3 lots, notify customer

**Decision:** Proceed with formal CAPA (cost exceeds \$5,000 threshold)

13.7.1.3 Step 3: Investigation

Gather all relevant data and evidence to understand what happened:

**Investigation Activities:**

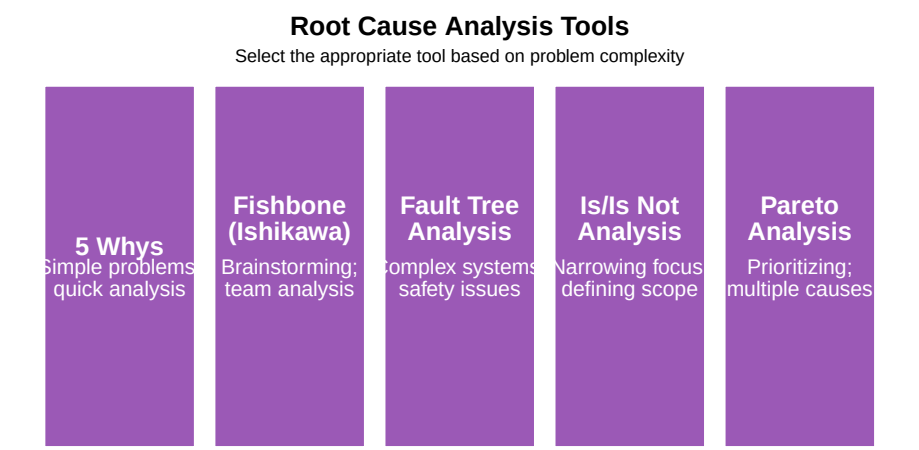
- **Review documentation:** Work instructions, control plans, inspection records
- **Examine physical evidence:** Failed parts, raw materials, tooling
- **Interview personnel:** Operators, inspectors, supervisors
- **Analyze data:** SPC charts, measurement data, process parameters
- **Timeline reconstruction:** When did the problem start? What changed?

Table 13.14: Data Collection for CAPA Investigation

| Data Type                 | Examples                                               | Collection Method    |
|---------------------------|--------------------------------------------------------|----------------------|
| <b>Process Data</b>       | Temperatures, pressures, speeds, feeds, cycle times    | Data historians, SPC |
| <b>Material Data</b>      | Lot numbers, certificates, incoming inspection results | Traceability system, |
| <b>Equipment Data</b>     | Maintenance records, calibration status, tool wear     | CMMS, calibration    |
| <b>Personnel Data</b>     | Training records, shift assignments, experience level  | HR system, training  |
| <b>Environmental Data</b> | Temperature, humidity, contamination levels            | Environmental moni   |

13.7.1.4 Step 4: Root Cause Analysis

Determine the fundamental cause(s) of the problem using structured analysis tools:



**The 5 Whys Example:**

| Level             | Question                                                    | Answer                                    |
|-------------------|-------------------------------------------------------------|-------------------------------------------|
| Problem           | Shaft diameter is oversize                                  | -                                         |
| Why 1?            | Why is the diameter oversize?                               | Tool is worn                              |
| Why 2?            | Why is the tool worn?                                       | Tool wasn't changed at scheduled interval |
| Why 3?            | Why wasn't it changed?                                      | Operator didn't know it was due           |
| Why 4?            | Why didn't operator know?                                   | Tool life counter was reset incorrectly   |
| Why 5?            | Why was it reset incorrectly?                               | No verification step in setup procedure   |
| <b>Root Cause</b> | <b>Setup procedure lacks tool counter verification step</b> |                                           |

**The 6M Categories (Fishbone Diagram):**

- **Man** (People): Training, skills, fatigue, attention
- **Machine**: Equipment condition, capability, maintenance
- **Material**: Raw material quality, specifications, storage
- **Method**: Procedures, work instructions, standards
- **Measurement**: Gages, calibration, measurement technique
- **Mother Nature** (Environment): Temperature, humidity, contamination

**13.7.1.5 Step 5: Action Plan**

Define specific corrective and/or preventive actions to address the root cause:

Table 13.15: Types of CAPA Actions

| Action_Type              | Definition                                       | Example                                                     |
|--------------------------|--------------------------------------------------|-------------------------------------------------------------|
| <b>Correction</b>        | Fix the immediate problem (symptom treatment)    | Rework or scrap the nonconforming part                      |
| <b>Corrective Action</b> | Eliminate the root cause to prevent recurrence   | Revise setup procedure to include tool counter verification |
| <b>Preventive Action</b> | Eliminate potential causes before problems occur | Add tool counter verification to ALL new tool setups        |

**SMART Action Planning:**

- **Specific**: Clearly defined action (not vague)
- **Measurable**: Can verify completion and effectiveness
- **Assignable**: Single owner responsible
- **Realistic**: Achievable with available resources

- **Time-bound:** Due date specified

#### **Example: SMART Corrective Action**

**Poor:** “Improve the setup procedure”

**SMART:** “Quality Engineer (J. Smith) will revise Work Instruction WI-2345 to add Step 4.7: ‘Verify tool life counter matches Tool Setup Sheet before starting production. Record counter reading on Setup Checklist Form F-102.’ Revision to be completed by April 15, training completed by April 22.”

#### **13.7.1.6 Step 6: Implementation**

Execute the action plan with clear ownership and tracking:

##### **Implementation Requirements:**

- Assign responsible person for each action
- Provide necessary resources (time, budget, authority)
- Communicate changes to affected personnel
- Update documentation (procedures, work instructions, forms)
- Train personnel on new requirements
- Implement changes in a controlled manner

##### **Change Control Considerations:**

| Change Type       | Approval Required                            | Documentation                              |
|-------------------|----------------------------------------------|--------------------------------------------|
| Document revision | Document owner, Quality                      | Revision history, training records         |
| Process change    | Process owner, Engineering, Quality          | Change request, validation if needed       |
| Equipment change  | Maintenance, Engineering, Quality            | Equipment qualification, capability study  |
| Product change    | Engineering, Quality, Customer (if required) | ECN, customer approval, PPAP if applicable |

#### **13.7.1.7 Step 7: Verification**

Verify that actions were completed AND effective:

**Completion Verification:** - Was the action actually implemented? - Is documentation updated and approved? - Are personnel trained? - Is the change being followed?

**Effectiveness Verification:** - Has the problem stopped recurring? - Do metrics show improvement? - Are there any unintended consequences?

Table 13.16: CAPA Effectiveness Verification Methods

| Method                      | Description                                                                 | When_to_U   |
|-----------------------------|-----------------------------------------------------------------------------|-------------|
| <b>Monitoring Period</b>    | Track for recurrence over defined period (e.g., 90 days, 3 production runs) | All CAPAs - |
| <b>Statistical Analysis</b> | Compare defect rates before/after; statistical significance testing         | When baseli |
| <b>Audit Verification</b>   | Include in next internal audit; verify implementation                       | Process/doc |
| <b>Customer Feedback</b>    | Monitor complaints related to this issue                                    | Customer-re |
| <b>Process Capability</b>   | Cpk study to confirm process improvement                                    | Manufacturi |

### 13.7.1.8 Step 8: Closure

Complete documentation and share lessons learned:

#### Closure Requirements:

- All actions verified complete
- Effectiveness confirmed (no recurrence)
- Documentation updated and approved
- CAPA record completed with all evidence
- Management approval for closure

#### Lessons Learned:

- What was the root cause?
- Could this happen elsewhere? (Horizontal deployment)
- Should procedures/training be updated system-wide?
- Are there similar products/processes at risk?
- What can we do to prevent similar issues?

#### Example: Horizontal Deployment

The shaft diameter CAPA revealed that setup procedures lacked tool counter verification.

**Horizontal deployment:** Quality reviewed ALL 47 CNC machine setup procedures and found 12 others with the same gap. All procedures were revised, and the finding was shared at the monthly quality meeting as a lesson learned.

### 13.7.2 CAPA Effectiveness Criteria

This worked example is fixed as a reference — read through it once, then try the *editable* example further on in this chapter for yourself.

```
# CAPA Effectiveness Tracking
set.seed(42)

# Simulate CAPA data for a year
months <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
```

```

capa_data <- data.frame(
  Month = factor(months, levels = months),
  Total_CAPAs = c(12, 15, 18, 14, 11, 13, 10, 9, 8, 7, 9, 6),
  Closed_On_Time = c(8, 10, 12, 11, 9, 11, 9, 8, 7, 6, 8, 5),
  Effective = c(7, 9, 10, 10, 8, 10, 8, 8, 7, 6, 8, 5),
  Recurrence = c(2, 1, 3, 1, 0, 1, 0, 0, 1, 0, 0, 0)
)

# Calculate metrics
capa_data$On_Time_Rate <- round(capa_data$Closed_On_Time / capa_data$Total_CAPAs * 100
capa_data$Effectiveness_Rate <- round(capa_data$Effective / capa_data$Closed_On_Time *

cat("CAPA Performance Metrics Summary:\n")
cat("\n")
cat("Total CAPAs initiated:", sum(capa_data$Total_CAPAs), "\n")
cat("Closed on time:", sum(capa_data$Closed_On_Time), "(",
  round(sum(capa_data$Closed_On_Time)/sum(capa_data$Total_CAPAs)*100, 1), "%)\n")
cat("Verified effective:", sum(capa_data$Effective), "(",
  round(sum(capa_data$Effective)/sum(capa_data$Closed_On_Time)*100, 1), "%)\n")
cat("Recurrences:", sum(capa_data$Recurrence), "\n")

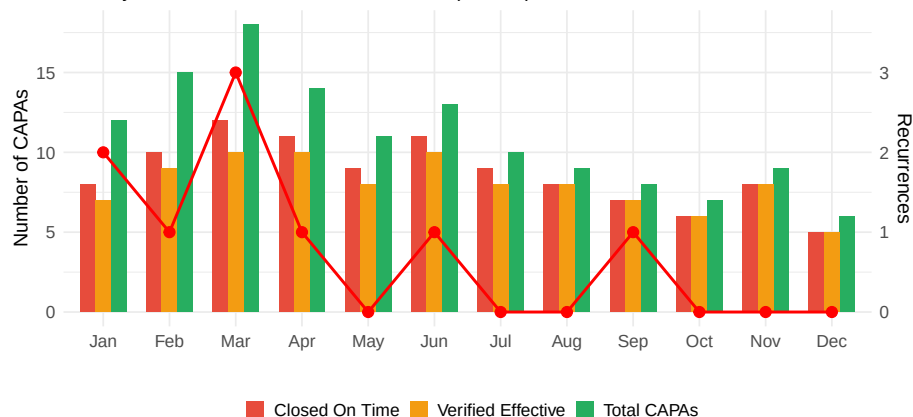
```

CAPA Performance Metrics Summary:

Total CAPAs initiated: 132  
 Closed on time: 104 ( 78.8 %)  
 Verified effective: 96 ( 92.3 %)  
 Recurrences: 9

### CAPA Performance Tracking

Monthly CAPA metrics with recurrence trend (red line)





## 13.8 Management Review

**Management review** is a formal evaluation of the QMS by top management to ensure its continuing suitability, adequacy, and effectiveness.

### 13.8.1 Management Review Inputs

Table 13.17: Management Review

| Required Input                   | Details to Review                                                                           |
|----------------------------------|---------------------------------------------------------------------------------------------|
| <b>Previous Review Actions</b>   | Status of actions from previous management reviews                                          |
| <b>Changes in Issues/Context</b> | Changes in external/internal issues; interested party requirements                          |
| <b>Quality Performance</b>       | Customer satisfaction; quality objectives achievement; process performance; nonconformities |
| <b>Resource Adequacy</b>         | Adequacy of resources for QMS maintenance and improvement                                   |
| <b>Risk/Opportunity Actions</b>  | Effectiveness of actions taken to address risks and opportunities                           |
| <b>Improvement Opportunities</b> | Opportunities for improvement identified from various sources                               |

### 13.8.2 Management Review Outputs

Table 13.18: Management Review Outputs (ISO 9001:2015)

| Required Output              | ISO Requirement                                            | Examples                                       |
|------------------------------|------------------------------------------------------------|------------------------------------------------|
| <b>Improvement Decisions</b> | Decisions and actions related to improvement opportunities | Launch kaizen project                          |
| <b>QMS Changes</b>           | Any need for changes to the QMS                            | Revise control plan based on customer feedback |
| <b>Resource Needs</b>        | Any resource needs                                         | Hire additional quality engineers              |

13.8.3 Sample Management Review Dashboard



13.9 Continuous Improvement

**Continuous improvement** is a fundamental principle of quality management - the ongoing effort to improve products, services, and processes.

13.9.1 Improvement Methodologies

Table 13.19: Continuous Improvement

| Methodology               | Origin          | Best_For                                        | Step  |
|---------------------------|-----------------|-------------------------------------------------|-------|
| <b>PDCA</b>               | Deming/Shewhart | General improvement cycle; process changes      | Plan  |
| <b>DMAIC</b>              | Six Sigma       | Data-driven problem solving; reducing variation | Defin |
| <b>Kaizen</b>             | Toyota/Lean     | Small, incremental improvements; team-based     | Iden  |
| <b>A3 Problem Solving</b> | Toyota          | Visual problem solving; root cause analysis     | Back  |
| <b>8D</b>                 | Ford            | Customer complaints; complex problems           | D0-I  |

### 13.9.2 Key Performance Indicators (KPIs)

Table 13.20: Common Quality Management KPIs

| Category                | KPI                       | Formula                                                                | Ta  |
|-------------------------|---------------------------|------------------------------------------------------------------------|-----|
| <b>Quality Metrics</b>  |                           |                                                                        |     |
| Quality                 | First Pass Yield (FPY)    | $(\text{Good units first time} / \text{Total units}) \times 100\%$     | 98  |
| Quality                 | Customer Complaints (PPM) | $(\text{Complaints} / \text{Units shipped}) \times 1,000,000$          | <1  |
| Quality                 | Internal Reject Rate      | $(\text{Rejected units} / \text{Total inspected}) \times 100\%$        | <1  |
| Quality                 | Cost of Quality           | Prevention + Appraisal + Internal Failure + External Failure           | <3  |
| <b>Delivery Metrics</b> |                           |                                                                        |     |
| Delivery                | On-Time Delivery (OTD)    | $(\text{Orders delivered on time} / \text{Total orders}) \times 100\%$ | 98  |
| Delivery                | Lead Time                 | Order date to delivery date                                            | Inc |
| Delivery                | Order Fulfillment Rate    | $(\text{Complete orders shipped} / \text{Total orders}) \times 100\%$  | 99  |
| <b>Cost Metrics</b>     |                           |                                                                        |     |
| Cost                    | Scrap Rate                | $(\text{Scrap value} / \text{Total production value}) \times 100\%$    | <2  |
| Cost                    | Rework Cost               | Labor + Material for rework                                            | Mi  |
| Cost                    | Warranty Cost             | Warranty claims cost / Revenue                                         | <0  |
| <b>Safety Metrics</b>   |                           |                                                                        |     |
| Safety                  | Recordable Incident Rate  | $(\text{Recordable incidents} \times 200,000) / \text{Hours worked}$   | <1  |
| Safety                  | Near Miss Reports         | Count of near miss reports                                             | En  |

### 13.9.3 Cost of Quality (COQ)

```
# Cost of Quality Analysis
coq_data <- data.frame(
  Category = c("Prevention", "Appraisal", "Internal Failure", "External Failure"),
  Description = c("Quality planning, training, process control",
                 "Inspection, testing, audits, calibration",
                 "Scrap, rework, reinspection, downtime",
                 "Warranty, returns, complaints, recalls"),
  Cost = c(50000, 75000, 120000, 180000)
)

total_coq <- sum(coq_data$Cost)
revenue <- 1000000 # $10M revenue

coq_data$Percent_of_COQ <- round(coq_data$Cost / total_coq * 100, 1)
coq_data$Percent_of_Revenue <- round(coq_data$Cost / revenue * 100, 2)

cat("Cost of Quality Analysis:\n")
cat("      \n")
print(coq_data[, c("Category", "Cost", "Percent_of_COQ")])
cat("      \n")
```

```

cat("Total Cost of Quality: $", format(total_coq, big.mark = " "), "\n")
cat("COQ as % of Revenue:", round(total_coq/revenue*100, 1), "%\n\n")

# Analysis
conformance <- coq_data$Cost[1] + coq_data$Cost[2]
nonconformance <- coq_data$Cost[3] + coq_data$Cost[4]

cat("Conformance Costs (Prevention + Appraisal): $", format(conformance, big.mark = " "),
    "(", round(conformance/total_coq*100, 1), "%)\n")
cat("Nonconformance Costs (Failures): $", format(nonconformance, big.mark = " "),
    "(", round(nonconformance/total_coq*100, 1), "%)\n")
cat("\nRecommendation: Increase prevention spending to reduce failure costs\n")

```

Cost of Quality Analysis:

|   | Category         | Cost   | Percent_of_COQ |
|---|------------------|--------|----------------|
| 1 | Prevention       | 50000  | 11.8           |
| 2 | Appraisal        | 75000  | 17.6           |
| 3 | Internal Failure | 120000 | 28.2           |
| 4 | External Failure | 180000 | 42.4           |

Total Cost of Quality: \$ 425 000

COQ as % of Revenue: 4.2 %

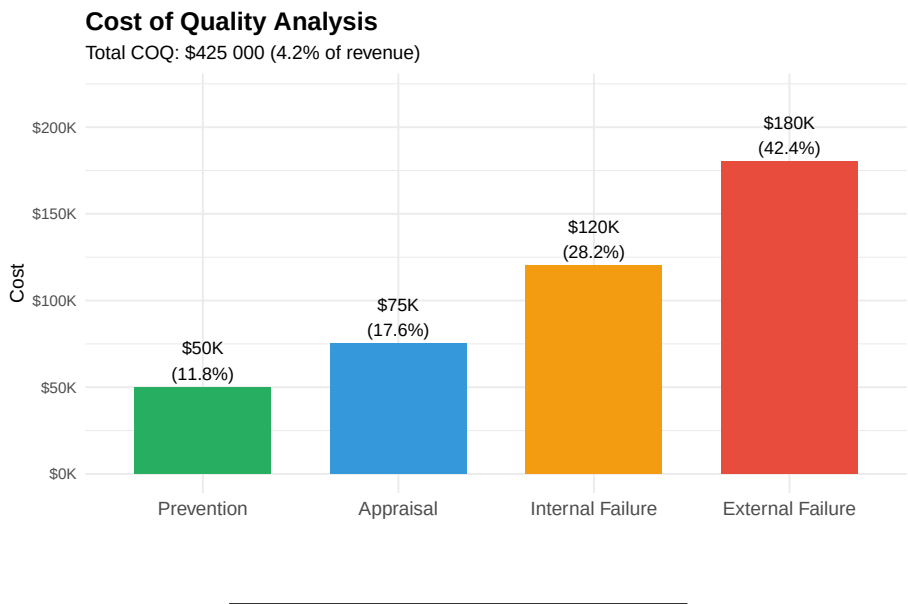
Conformance Costs (Prevention + Appraisal): \$ 125 000 ( 29.4 %)

Nonconformance Costs (Failures): \$ 3e+05 ( 70.6 %)

Recommendation: Increase prevention spending to reduce failure costs

Warning: Removed 1 row containing missing values or values outside the scale range  
(`geom\_text()`).

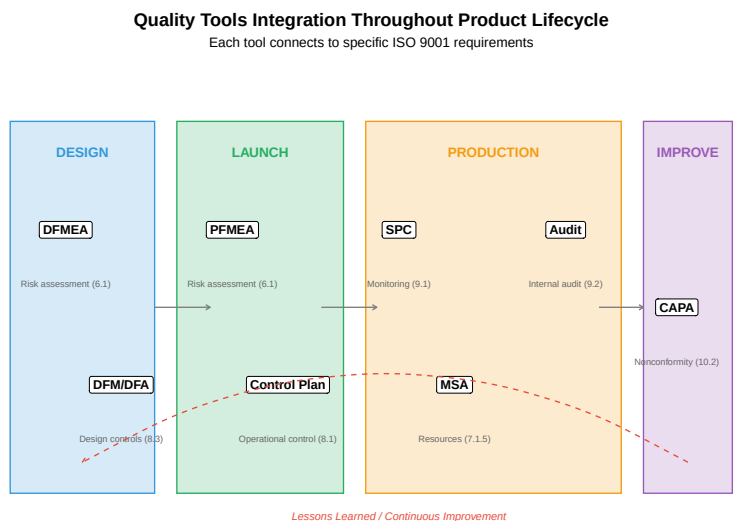
Removed 1 row containing missing values or values outside the scale range  
(`geom\_text()`).



## 13.10 Integration with Quality Tools

A mature QMS integrates various quality tools throughout the organization.

### 13.10.1 Quality Tools Integration Map



13.10.2 Linking Tools to QMS Requirements

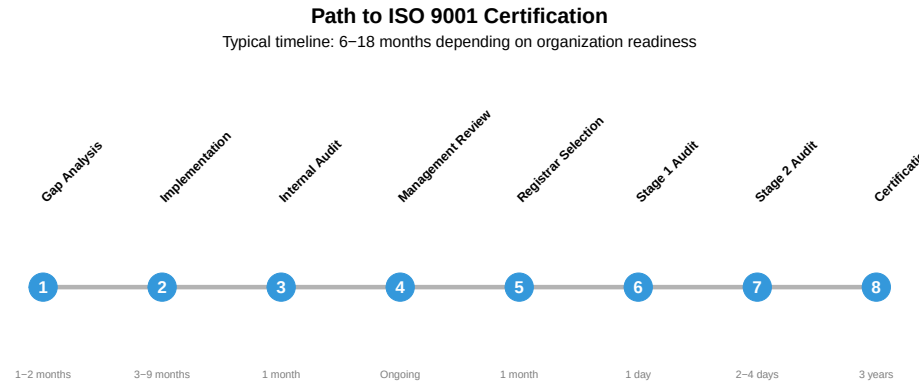
Table 13.21: Quality Tools and QMS Integration

| Quality Tool      | ISO 9001 Clause | Purpose                                               |
|-------------------|-----------------|-------------------------------------------------------|
| PFMEA             | 6.1, 8.1        | Identify and mitigate process risks before production |
| Control Plan      | 8.1, 8.5.1      | Define controls for special characteristics           |
| SPC               | 9.1.1           | Monitor process performance; detect special causes    |
| MSA               | 7.1.5           | Ensure measurement system is adequate                 |
| Internal Audit    | 9.2             | Verify QMS conformance and effectiveness              |
| CAPA              | 10.2            | Eliminate causes of nonconformities                   |
| Management Review | 9.3             | Evaluate QMS suitability and effectiveness            |

13.11 Certification Process

Organizations can seek **third-party certification** to demonstrate QMS conformance to customers and stakeholders.

13.11.1 The Certification Journey



13.11.2 Certification Audit Types

Table 13.22: Third-Party Certification Audit Types

| Audit Type | Purpose | Focus Areas |
|------------|---------|-------------|
|------------|---------|-------------|

|                                |                                                        |                                        |
|--------------------------------|--------------------------------------------------------|----------------------------------------|
| <b>Stage 1 (Documentation)</b> | Review QMS documentation; verify readiness for Stage 2 | Quality manual, procedures             |
| <b>Stage 2 (Certification)</b> | Evaluate implementation and effectiveness of QMS       | All clauses; process effectiveness     |
| <b>Surveillance</b>            | Verify continued conformance (annual)                  | Selected processes; preventive actions |
| <b>Recertification</b>         | Full re-audit at end of 3-year cycle                   | Complete review similar to Stage 2     |

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## 13.12 Video Resources

### 13.12.1 ISO 9001:2015 Overview

### 13.12.2 Internal Auditing Best Practices

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## 13.13 Summary

A Quality Management System provides the framework for consistently meeting customer requirements and improving organizational performance:

1. **ISO 9001:2015** provides the foundational requirements using risk-based thinking and PDCA
2. **Industry standards** (IATF 16949, AS9100, ISO 22000) add sector-specific requirements
3. **Documentation** establishes the hierarchy from policy to procedures to records
4. **Internal audits** verify conformance and identify improvement opportunities
5. **CAPA** eliminates root causes of nonconformities and prevents recurrence
6. **Management review** ensures QMS suitability and drives strategic improvement
7. **Continuous improvement** is embedded through KPIs, COQ analysis, and improvement methodologies
8. **Quality tools** (PFMEA, SPC, MSA) integrate throughout the product lifecycle

“Quality is not an act, it is a habit.” — Aristotle

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## 13.14 Review Questions

**Question 1:** Explain the seven quality management principles and how they relate to each other.

**Answer:**

The seven quality management principles form an interconnected foundation:

1. **Customer Focus** (Central principle)
  - Understanding and meeting customer needs is the primary purpose
  - All other principles support this goal
2. **Leadership**
  - Leaders establish unity of purpose and direction
  - Create conditions for people to achieve quality objectives
  - Enables all other principles
3. **Engagement of People**
  - Competent, empowered people at all levels
  - Essential for effective processes and customer focus
4. **Process Approach**
  - Activities managed as interrelated processes
  - Enables consistent, predictable results
5. **Improvement**
  - Ongoing focus on enhancing performance
  - Essential for maintaining customer satisfaction
6. **Evidence-based Decision Making**
  - Decisions based on analysis of data
  - Reduces uncertainty; improves decision quality
7. **Relationship Management**
  - Managing relationships with interested parties (suppliers, partners)
  - Optimizes their impact on performance

**Relationships:** - Customer Focus drives the need for all others - Leadership enables and supports all principles - Engagement of People is essential for Process Approach and Improvement - Evidence-based Decision Making supports Improvement - Relationship Management extends quality beyond the organization

**Question 2:** Compare ISO 9001, IATF 16949, and AS9100. When would each be required?

**Answer:**

| Standard               | Base                      | Industry          | Additional Focus                                |
|------------------------|---------------------------|-------------------|-------------------------------------------------|
| <b>ISO 9001:2015</b>   | Standalone                | General           | Quality fundamentals                            |
| <b>IATF 16949:2016</b> | ISO 9001<br>+ auto-motive | Automotive        | PPAP, APQP, Core Tools                          |
| <b>AS9100D</b>         | ISO 9001<br>+ aerospace   | Aerospace/Defense | Configuration mgmt, FAI, counterfeit prevention |

**When Required:**



**ISO 9001:** - Any organization wanting to demonstrate quality capability - Customer or contract requirement for QMS certification - Basis for all other standards - Suitable for service, manufacturing, any industry

**IATF 16949:** - Suppliers to automotive OEMs (Ford, GM, Toyota, VW, etc.) - Required for production parts/materials - Required throughout automotive supply chain - Includes Customer-Specific Requirements (CSRs)

**AS9100:** - Suppliers to aerospace/defense contractors - Boeing, Airbus, Lockheed Martin, military programs - Required for flight-critical components - OASIS database registration required

**Key Differences:** - IATF 16949: Emphasizes PPAP, APQP, manufacturing process control - AS9100: Emphasizes traceability, configuration control, first article inspection - Both build on ISO 9001 with sector-specific additions

**Question 3:** Describe the documentation hierarchy in a QMS. Give an example of each level.

**Answer:**

**Level 1: Quality Policy and Objectives** - Strategic direction for quality - Signed by top management - Communicated to all employees - *Example:* “ABC Company is committed to delivering defect-free products that meet customer requirements through continuous improvement of our processes and people.”

**Level 2: Quality Manual and Procedures** - Describes WHAT is done and WHO is responsible - System-level documents - *Example:* Document Control Procedure (QP-001) - Purpose: Control creation, approval, distribution of documents - Scope: All QMS documents - Responsibilities: Document Control Coordinator, Department Managers - Process flow for document changes

**Level 3: Work Instructions and Specifications** - Describes HOW tasks are performed - Detailed step-by-step instructions - *Example:* Work Instruction for CNC Lathe Setup (WI-MAC-015) - Step 1: Verify program number matches work order - Step 2: Install correct tooling per setup sheet - Step 3: Set tool offsets using touch-off procedure - Step 4: Run first piece and verify dimensions

**Level 4: Records and Forms** - Evidence that activities were performed - Completed forms, inspection results, logs - *Example:* First Article Inspection Report (FM-QC-008) - Part number, revision, date - Dimensional measurements vs. specifications - Inspector signature - Disposition (Accept/Reject)

**Question 4:** What are the key elements of an effective internal audit program?

**Answer:**

**1. Audit Program Planning** - Annual audit schedule covering all QMS processes - Risk-based frequency (higher risk = more frequent) - Documented audit program procedure - Resource allocation (trained auditors, time)

**2. Auditor Competence** - Trained in audit techniques - Knowledge of standards (ISO 9001, industry-specific) - Understanding of processes being audited - Independence from area being audited

**3. Audit Preparation** - Review of relevant documents (procedures, previous audits) - Prepared checklist linked to requirements - Notification to auditee - Defined scope and objectives

**4. Audit Execution** - Opening meeting (purpose, scope, schedule) - Evidence collection (documents, records, interviews, observation) - Objective evaluation against criteria - Note taking and evidence documentation

**5. Reporting** - Clear finding statements (nonconformities, observations) - Reference to requirement violated - Objective evidence cited - Closing meeting to present findings

**6. Follow-up** - Corrective action requests issued - Root cause analysis required - Verification of corrective action implementation - Verification of effectiveness

**7. Continuous Improvement of Audit Program** - Evaluate audit program effectiveness - Update based on organizational changes - Calibrate auditors periodically - Incorporate lessons learned

**Question 5:** Explain the difference between corrective action and preventive action. Provide an example of each.

**Answer:**

**Corrective Action:** - Eliminates the cause of a *detected* nonconformity - Reactive - responds to something that has already happened - Goal: Prevent recurrence of the specific problem

*Example:* - **Nonconformity:** Customer received shipment with wrong part numbers (3 incidents in past month) - **Root Cause:** Shipping labels printed from outdated part number list - **Corrective Action:** 1. Update master part number list 2. Link shipping system directly to ERP for current data 3. Add verification step comparing label to packing list - **Verification:** No recurrence after 3 months

**Preventive Action:** - Eliminates the cause of a *potential* nonconformity - Proactive - addresses something that hasn't happened yet - Goal: Prevent occurrence before it happens

*Example:* - **Potential Nonconformity:** New CNC machine may produce out-of-spec parts due to operator unfamiliarity - **Risk Identified Through:** PFMEA analysis during new equipment installation - **Preventive Action:** 1. Develop comprehensive training program before machine goes live 2. Create detailed setup procedures with photos 3. Implement first-article inspection for first 30 days 4. Assign experienced mentor to new operators - **Verification:** Track first-pass yield during initial production period

**Key Difference:** - Corrective = Fix what went wrong - Preventive = Stop what might go wrong

**Note:** ISO 9001:2015 doesn't explicitly require "preventive action" as a separate process. Instead, risk-based thinking throughout the QMS serves the preventive function.

**Question 6:** Calculate and analyze the Cost of Quality for the following data. What recommendations would you make?

| Category              | Cost      |
|-----------------------|-----------|
| Quality planning      | \$35,000  |
| Training              | \$25,000  |
| Incoming inspection   | \$40,000  |
| In-process inspection | \$55,000  |
| Final inspection      | \$30,000  |
| Scrap                 | \$85,000  |
| Rework                | \$65,000  |
| Warranty claims       | \$120,000 |
| Customer returns      | \$45,000  |

Annual revenue: \$8,000,000

**Answer:** (live — change the cost figures to check your own version)

```
#| label: q6-answer
# Cost of Quality Analysis
prevention <- 35000 + 25000 # Planning + Training
appraisal <- 40000 + 55000 + 30000 # Inspections
internal_failure <- 85000 + 65000 # Scrap + Rework
external_failure <- 120000 + 45000 # Warranty + Returns

total_coq <- prevention + appraisal + internal_failure + external_failure
revenue <- 8000000

cat("Cost of Quality Breakdown:\n")
cat("      \n")
cat("Prevention:      $", format(prevention, big.mark = " "),
    "(", round(prevention/total_coq*100, 1), "%)\n")
cat("Appraisal:      $", format(appraisal, big.mark = " "),
    "(", round(appraisal/total_coq*100, 1), "%)\n")
cat("Internal Failure: $", format(internal_failure, big.mark = " "),
    "(", round(internal_failure/total_coq*100, 1), "%)\n")
cat("External Failure: $", format(external_failure, big.mark = " "),
    "(", round(external_failure/total_coq*100, 1), "%)\n")
cat("      \n")
```

```

cat("Total COQ:      $", format(total_coq, big.mark = " "), "\n")
cat("COQ % of Revenue:", round(total_coq/revenue*100, 1), "%\n\n")

conformance <- prevention + appraisal
nonconformance <- internal_failure + external_failure

cat("Conformance Costs:  $", format(conformance, big.mark = " "),
    "(", round(conformance/total_coq*100, 1), "%)\n")
cat("Nonconformance Costs: $", format(nonconformance, big.mark = " "),
    "(", round(nonconformance/total_coq*100, 1), "%)\n")

```

**Analysis:** - COQ at 7.5% of revenue is HIGH (target <3%) - External failure costs are the largest category (27.5%) - Nonconformance costs (60.8%) greatly exceed conformance (39.2%) - Prevention is only 10% of COQ - very low

**Recommendations:** 1. **Increase Prevention Investment:** - Implement SPC on critical processes to reduce defects - Enhance operator training on quality standards - Apply PFMEA to identify and control failure modes

2. **Reduce External Failures:**

- Analyze warranty data for root causes
- Improve final inspection effectiveness
- Implement containment for known issues

3. **Optimize Appraisal:**

- Use SPC to reduce inspection where processes are capable
- Implement risk-based inspection (more for critical, less for proven)

4. **Target COQ Reduction:**

- Short-term: Reduce external failures by 50% (\$82,500 savings)
- Medium-term: Reduce internal failures by 30% (\$45,000 savings)
- Long-term: Achieve COQ <3% of revenue (\$240,000)

**Question 7:** What is risk-based thinking in ISO 9001:2015? How is it different from formal risk management?

**Answer:**

**Risk-Based Thinking in ISO 9001:2015:**

Risk-based thinking is a way of considering risks and opportunities when designing, implementing, and operating the QMS. It:

- Applies throughout the standard (not a separate clause)
- Is proportional to potential impact on conformity
- Makes preventive action part of routine planning
- Considers both risks (negative) and opportunities (positive)

**Where Risk-Based Thinking Applies:**

| Clause                 | Application                                         |
|------------------------|-----------------------------------------------------|
| 4.1 Context            | Identify external/internal issues affecting QMS     |
| 4.2 Interested Parties | Understand requirements that could impact quality   |
| 6.1 Planning           | Plan actions to address risks and opportunities     |
| 8.1 Operations         | Consider risks in operational planning              |
| 9.1 Performance        | Monitor effectiveness of risk actions               |
| 10.2 Corrective Action | Update risks/opportunities based on nonconformities |

#### How It Differs from Formal Risk Management:

| Aspect         | Risk-Based Thinking              | Formal Risk Management           |
|----------------|----------------------------------|----------------------------------|
| Requirement    | Implicit throughout ISO 9001     | Explicit (ISO 31000)             |
| Documentation  | No formal risk register required | Documented risk register         |
| Methodology    | No specific method required      | Defined methodology (FMEA, etc.) |
| Quantification | Not required                     | Often quantitative               |
| Scope          | QMS processes                    | Enterprise-wide                  |

#### Practical Implementation:

Organizations should: 1. Consider risks when designing processes 2. Build controls proportional to risk level 3. Monitor and review effectiveness 4. Update as context changes

ISO 9001 does NOT require: - Formal risk management process - Risk register - Specific risk assessment methodology - Documented risk assessments (unless organization decides to)

However, many organizations do use tools like PFMEA to fulfill risk-based thinking requirements, especially in automotive (IATF 16949 requires it).

**Question 8:** Describe what management review is and what should be included in the inputs and outputs.

**Answer:**

#### What is Management Review?

Management review is a formal, periodic evaluation of the QMS by top management to: - Ensure continuing suitability (appropriate for the organization) - Ensure adequacy (sufficient to meet requirements) - Ensure effectiveness (achieving

intended results) - Ensure alignment with strategic direction - Identify improvement opportunities

**Frequency:** At least annually; many organizations conduct quarterly reviews

**Required Inputs (ISO 9001 Clause 9.3.2):**

| Input                               | What to Review                                                                                                       |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Status of previous actions          | Actions from prior reviews - completed? effective?                                                                   |
| Changes in external/internal issues | Market changes, regulations, technology, organizational changes                                                      |
| QMS performance and effectiveness   | Customer satisfaction, quality objectives, process performance, nonconformities, audit results, supplier performance |
| Resource adequacy                   | People, infrastructure, environment, knowledge                                                                       |
| Effectiveness of risk actions       | Did actions address risks/opportunities?                                                                             |
| Improvement opportunities           | Suggestions, benchmarking, new technologies                                                                          |

**Required Outputs (ISO 9001 Clause 9.3.3):**

| Output                | Examples                                                          |
|-----------------------|-------------------------------------------------------------------|
| Improvement decisions | Launch specific improvement projects, address recurring issues    |
| QMS changes           | Revise quality objectives, update processes, modify documentation |
| Resource needs        | Budget approvals, hiring, equipment purchases, training           |

**Best Practices:** - Use a standard agenda aligned with required inputs - Prepare data in advance (dashboards, trend charts) - Document decisions and action items with owners and due dates - Follow up on action completion - Keep minutes as quality records

**Evidence Required:** - Meeting minutes or management review report - Attendance record (top management participation) - Action items with status tracking - Data/reports presented

## 13.15 References

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## Chapter 14

# Process Failure Mode and Effects Analysis (PFMEA)

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### 14.1 Learning Objectives

By the end of this chapter, you will be able to:

- Explain the purpose and benefits of PFMEA in manufacturing
- Distinguish between DFMEA (Design) and PFMEA (Process)
- Apply the 10-step PFMEA methodology to a manufacturing process
- Correctly rate Severity, Occurrence, and Detection using standardized scales
- Calculate Risk Priority Numbers (RPN) and prioritize actions
- Develop recommended actions to reduce process risks
- Create and maintain PFMEA documentation as a living document

“An ounce of prevention is worth a pound of cure.” — Benjamin Franklin

---

### 14.2 Introduction to PFMEA

#### 14.2.1 What is PFMEA?

**Process Failure Mode and Effects Analysis (PFMEA)** is a systematic, proactive method for evaluating a manufacturing or business process to identify where and how it might fail, and to assess the relative impact of different failures.

The goal is to identify and prioritize actions that will prevent defects before they occur.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use `linewidth` instead.

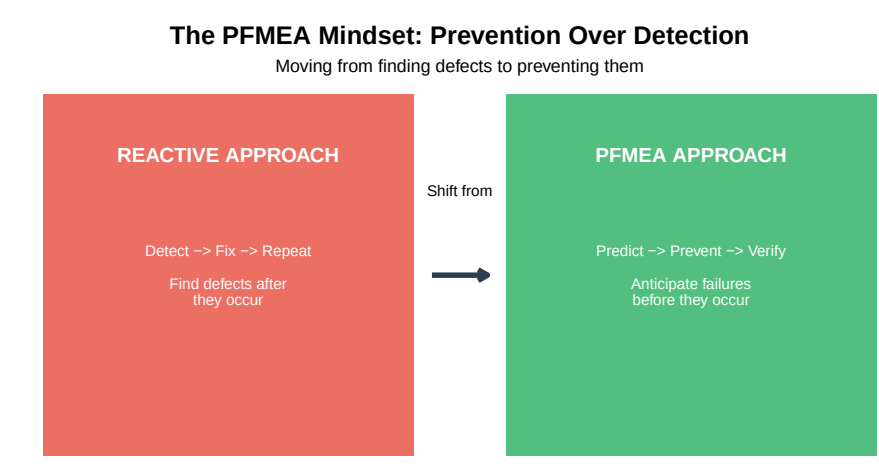


Figure 14.1: PFMEA: A Proactive Quality Tool

14.2.2 Brief History of FMEA

Table 14.1: Evolution of FMEA

| Year  | Development                                         | Significance                                |
|-------|-----------------------------------------------------|---------------------------------------------|
| 1949  | US Military develops FMEA (MIL-P-1629)              | First formal FMEA method                    |
| 1960s | NASA adopts FMEA for Apollo program                 | Proven reliability in life-critical systems |
| 1970s | Automotive industry begins using FMEA               | Ford, GM, Chrysler adopt FMEA               |
| 1980s | AIAG publishes first FMEA manual                    | Industry-wide consistency                   |
| 1990s | QS-9000 requires FMEA for automotive suppliers      | FMEA becomes mandatory                      |
| 2000s | FMEA expands to healthcare, service industries      | Methodology proves universal                |
| 2019  | AIAG-VDA FMEA Handbook published (current standard) | Harmonized global standard                  |

14.2.3 DFMEA vs. PFMEA

Two main types of FMEA are used in product development:

### Two Types of FMEA in Product Development

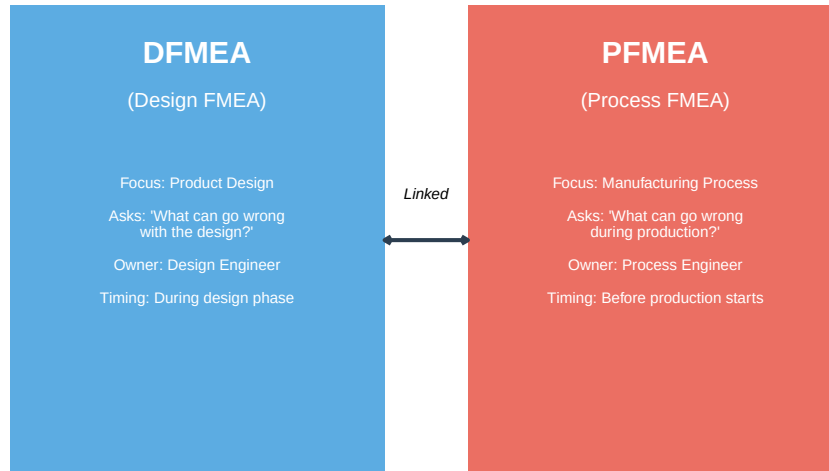


Figure 14.2: DFMEA vs PFMEA: Different Focus, Same Methodology

Table 14.2: DFMEA vs PFMEA Comparison

| Aspect                          | DFMEA (Design)                     | PFMEA (Process)                            |
|---------------------------------|------------------------------------|--------------------------------------------|
| <b>**Primary Question**</b>     | Will the design meet requirements? | Will the process produce conforming parts? |
| <b>**Failure Mode Example**</b> | Shaft diameter too small           | Shaft turned undersize                     |
| <b>**Effect Example**</b>       | Premature bearing failure          | Part fails incoming inspection             |
| <b>**Cause Example**</b>        | Inadequate stress analysis         | Tool wear not monitored                    |
| <b>**Control Example**</b>      | FEA simulation, prototype testing  | In-process gauging, SPC                    |
| <b>**Responsible Team**</b>     | Design engineering, R&D            | Manufacturing, quality, production         |

Question: When should PFMEA be conducted?

#### Ideal Timing for PFMEA:

1. **New Process Development:** Before production begins, during process design
2. **New Product Introduction:** When a new product will use existing processes
3. **Process Changes:** Whenever significant changes are made to equipment, methods, or materials
4. **Quality Issues:** When field failures or customer complaints indicate process problems
5. **Regular Review:** Periodically (annually) to ensure continued relevance

**Key Principle:** The earlier PFMEA is conducted, the lower the cost of implementing improvements. Changes during design cost 10x less than changes during production, and 100x less than changes after customer delivery.

#### 14.2.4 Why PFMEA Matters

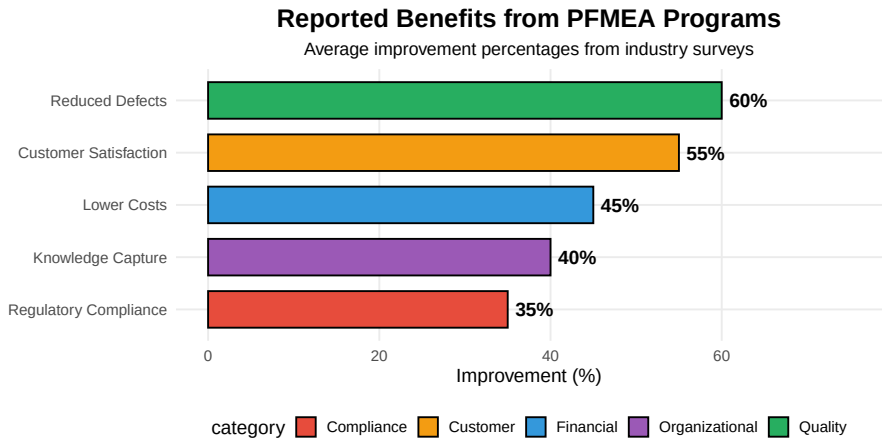


Figure 14.3: Benefits of PFMEA Implementation

### 14.3 PFMEA Fundamentals

#### 14.3.1 Core Objectives

The fundamental objective of PFMEA is **prevention over detection** — identifying and eliminating potential failures before they occur, rather than finding defects after production.

Table 14.3: Core Objectives of PFMEA

| Objective                         | Description                                                      | Outcom   |
|-----------------------------------|------------------------------------------------------------------|----------|
| <b>**Identify Failure Modes**</b> | Systematically identify all ways the process can fail            | Comprel  |
| <b>**Assess Risk**</b>            | Evaluate severity, likelihood, and detectability of each failure | Quantifi |
| <b>**Prioritize Actions**</b>     | Focus resources on highest-risk failure modes                    | Action p |
| <b>**Document Knowledge**</b>     | Capture institutional knowledge in a structured format           | Living d |

|                                  |                                                       |                      |
|----------------------------------|-------------------------------------------------------|----------------------|
| <b>**Drive<br/>Improvement**</b> | Implement and verify effectiveness of countermeasures | Measurable reduction |
|----------------------------------|-------------------------------------------------------|----------------------|

14.3.2 The PFMEA Team

PFMEA is most effective when conducted by a **cross-functional team**. No single person has all the knowledge needed to identify every potential failure.

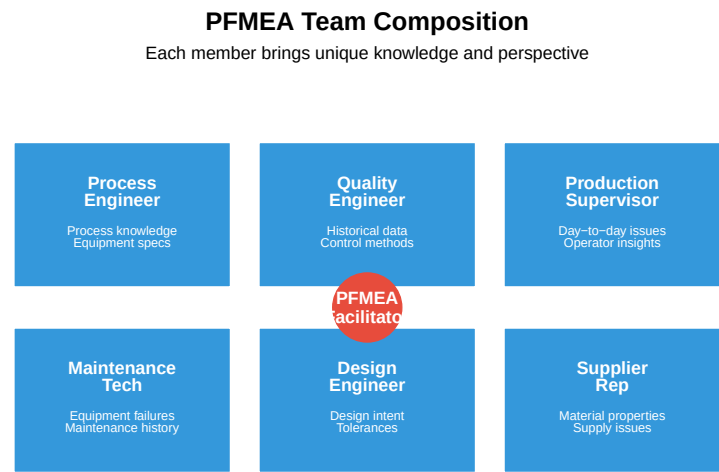


Figure 14.4: PFMEA Cross-Functional Team

Question: Why is a cross-functional team essential?

**A cross-functional team is essential because:**

1. **Diverse Knowledge:** No single person understands all aspects of the process
2. **Different Perspectives:** Quality sees defects, production sees constraints, maintenance sees failures
3. **Buy-in:** People support what they help create
4. **Historical Knowledge:** Operators often know about problems never formally documented
5. **System Thinking:** Process interactions are visible when multiple functions participate

**Common Mistake:** Having one engineer complete PFMEA alone as a “paperwork exercise.” This misses critical failure modes and produces an ineffective document.

**Best Practice:** Minimum 4-6 team members, including at least one person with hands-on process experience.

### 14.3.3 Key Terminology

Understanding PFMEA terminology is essential for effective analysis:

Table 14.4: PFMEA Key Terminology

| Term                            | Definition                                                           |
|---------------------------------|----------------------------------------------------------------------|
| <b>**Process Function**</b>     | What the process step is supposed to do                              |
| <b>**Failure Mode**</b>         | How the process can fail to perform its function                     |
| <b>**Failure Effect**</b>       | The consequence of the failure on the customer or next operation     |
| <b>**Severity (S)**</b>         | Rating (1-10) of how serious the failure effect is                   |
| <b>**Failure Cause**</b>        | Why the failure mode might occur                                     |
| <b>**Occurrence (O)**</b>       | Rating (1-10) of how likely the cause is to happen                   |
| <b>**Current Controls**</b>     | What currently prevents the cause or detects the failure             |
| <b>**Detection (D)**</b>        | Rating (1-10) of how likely current controls will detect the failure |
| <b>**RPN**</b>                  | Risk Priority Number = $S \times O \times D$                         |
| <b>**Action Priority (AP)**</b> | Alternative prioritization method (AIAG-VDA)                         |

### 14.3.4 The PFMEA Form

The PFMEA is documented in a structured form. Here is a simplified template:

Table 14.5: PFMEA Form Template (Simplified)

| Process Step | Function   | Failure Mode   | Effect                | S | Cause      | O | Current Controls  |
|--------------|------------|----------------|-----------------------|---|------------|---|-------------------|
| Example:     | Drill hole | Hole undersize | Assembly interference | 7 | Drill wear | 5 | Visual inspection |

## 14.4 The PFMEA Methodology: 10 Steps

PFMEA follows a systematic 10-step methodology. Each step builds on the previous one.

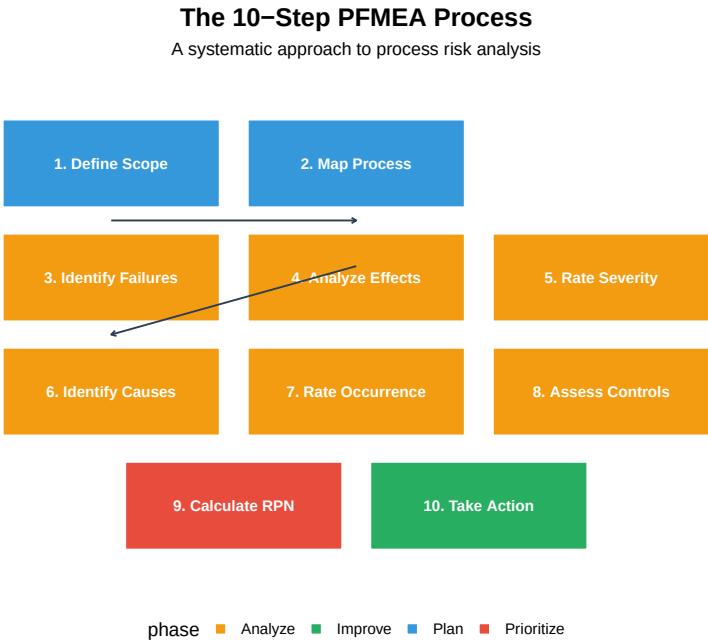


Figure 14.5: The 10-Step PFMEA Methodology

14.4.1 Step 1: Define Scope and Process Boundaries

Before starting the analysis, clearly define what is included and excluded.

Table 14.6: Step 1: Defining PFMEA Scope

| Element                        | Description                                 | Example: Injection Molding                       |
|--------------------------------|---------------------------------------------|--------------------------------------------------|
| <b>**Process Name**</b>        | Clear name identifying the process          | Plastic Part Injection Molding                   |
| <b>**Start Boundary**</b>      | Where does our analysis begin?              | Raw material loading into hopper                 |
| <b>**End Boundary**</b>        | Where does our analysis end?                | Part removed from mold, ready for packaging      |
| <b>**Included Operations**</b> | All operations within scope                 | Material prep, injection, cooling, ejection      |
| <b>**Excluded Operations**</b> | Operations analyzed separately or by others | Mold design (covered by DFMEA), part inspection  |
| <b>**Key Assumptions**</b>     | Conditions assumed to be true               | Material meets incoming spec, mold is maintained |

14.4.2 Step 2: Process Mapping and Identifying Functions

Create a detailed process flow diagram and identify the function of each step.

### Process Flow Diagram with Functions

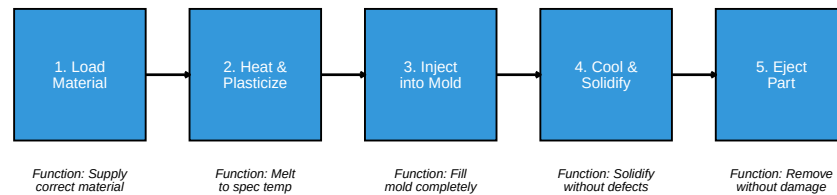


Figure 14.6: Example: Injection Molding Process Flow

### 14.4.3 Step 3: Identifying Potential Failure Modes

For each process function, ask: **“How can this step fail to perform its intended function?”**

Table 14.7: Step 3: Identifying Failure Modes

| Process Function                  | Potential Failure Mode       | Category             |
|-----------------------------------|------------------------------|----------------------|
| Load correct material             | Wrong material loaded        | Wrong item           |
| Load correct material             | Material contaminated        | Degraded item        |
| Load correct material             | Insufficient material        | Missing item         |
| Melt to specification temperature | Temperature too low          | Below specification  |
| Melt to specification temperature | Temperature too high         | Above specification  |
| Fill mold completely              | Short shot (incomplete fill) | Incomplete operation |
| Fill mold completely              | Flash (overfill)             | Excessive operation  |
| Fill mold completely              | Air entrapment               | Unintended result    |

**Common failure mode categories:** - **No operation** (step doesn’t happen)  
 - **Partial operation** (incomplete) - **Degraded operation** (below standard)  
 - **Excessive operation** (too much) - **Wrong operation** (incorrect action) -  
**Intermittent operation** (inconsistent)

Interactive Exercise: Identify Failure Modes

**Process Step:** Heat treat steel parts at 850°C for 2 hours

**Function:** Achieve required hardness (58-62 HRC)

**Identify at least 5 potential failure modes:**

**Answers:** 1. Temperature too low → Parts soft (below hardness spec) 2. Temperature too high → Parts brittle/cracked 3. Time too short → Incomplete



hardening 4. Time too long → Over-hardening, distortion 5. Uneven heating → Inconsistent hardness across part 6. Wrong atmosphere → Surface decarburization 7. Quench delay too long → Soft spots

#### 14.4.4 Step 4: Analyzing Failure Effects

For each failure mode, determine: “**What is the consequence if this failure occurs?**”

Consider effects at multiple levels: 1. **Local effect:** Impact on immediate operation 2. **Next operation effect:** Impact on downstream process 3. **End customer effect:** Impact on final user

Table 14.8: Step 4: Analyzing Effects at Multiple Levels

| Failure Mode | Effect Level   | Effect Description                                |
|--------------|----------------|---------------------------------------------------|
| Short shot   | Local          | Part is incomplete/missing features               |
| Short shot   | Next Operation | Part rejected at inspection, line stoppage        |
| Short shot   | End Customer   | If shipped: product malfunction, safety hazard    |
| Flash        | Local          | Excess material on part edges                     |
| Flash        | Next Operation | Secondary trimming required, increased cycle time |
| Flash        | End Customer   | If shipped: poor fit, appearance defect           |

#### 14.4.5 Step 5: Determining Severity Ratings

**Severity (S)** rates the seriousness of the effect on a scale of 1-10.

Table 14.9: Severity Rating Scale (S)

| Rating | Effect                      | Criteria                                                 |
|--------|-----------------------------|----------------------------------------------------------|
| 10     | Hazardous - without warning | Safety hazard; noncompliance with regulations; n         |
| 9      | Hazardous - with warning    | Safety hazard; noncompliance with regulations; v         |
| 8      | Very High                   | Product inoperable; loss of primary function             |
| 7      | High                        | Product operable but at reduced performance              |
| 6      | Moderate                    | Product operable; comfort/convenience affected; noticea  |
| 5      | Low                         | Product operable; reduced comfort; noticed by average c  |
| 4      | Very Low                    | Product operable; slightly reduced comfort; noticed by c |
| 3      | Minor                       | Fit/finish nonconforming; noticed by discriminating cus  |
| 2      | Very Minor                  | Fit/finish nonconforming; noticed only by trained obser  |
| 1      | None                        | No discernible effect                                    |

**Important:** Severity is determined by the **effect**, not the failure mode. Severity can only be reduced by changing the design — process changes typically cannot reduce severity.

### 14.4.6 Step 6: Identifying Potential Causes

For each failure mode, identify all possible causes: “**Why might this failure occur?**”

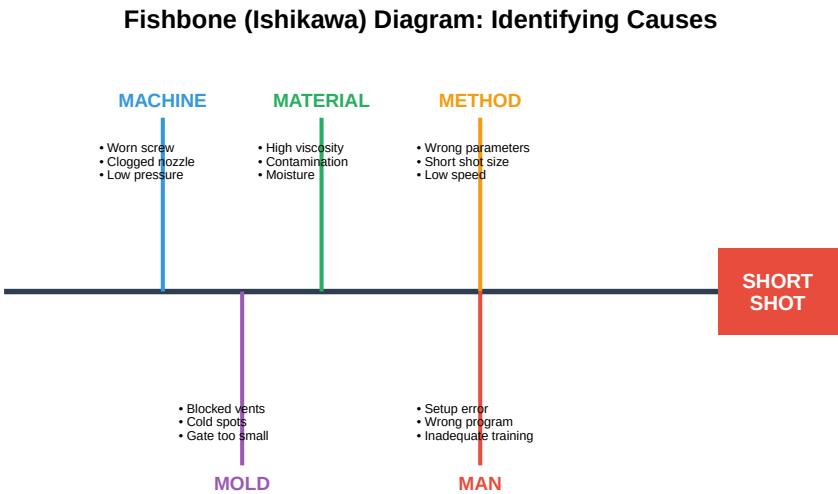


Figure 14.7: Fishbone Diagram: Causes of Short Shot

### 14.4.7 Step 7: Rating Occurrence

**Occurrence (O)** rates the likelihood that the cause will occur on a scale of 1-10.

Table 14.10: Occurrence Rating Scale (O)

| Rating | Likelihood | Criteria                        | Approximate Rate        | Cpk    |
|--------|------------|---------------------------------|-------------------------|--------|
| 10     | Very High  | Failure is almost inevitable    | 100 per 1000 ( 10%)     | < 0.33 |
| 9      | Very High  | Failures occur very frequently  | 50 per 1000 (5%)        | 0.33   |
| 8      | High       | Failures occur frequently       | 20 per 1000 (2%)        | 0.51   |
| 7      | High       | Failures occur on regular basis | 10 per 1000 (1%)        | 0.67   |
| 6      | Moderate   | Occasional failures             | 5 per 1000 (0.5%)       | 0.83   |
| 5      | Moderate   | Occasional failures expected    | 2 per 1000 (0.2%)       | 1.00   |
| 4      | Low        | Relatively few failures         | 1 per 1000 (0.1%)       | 1.17   |
| 3      | Low        | Few failures                    | 0.5 per 1000 (500 ppm)  | 1.33   |
| 2      | Remote     | Failure unlikely                | 0.1 per 1000 (100 ppm)  | 1.50   |
| 1      | Remote     | Failure nearly impossible       | 0.01 per 1000 ( 10 ppm) | 1.67   |

### 14.4.8 Step 8: Current Controls and Detection Ratings

Identify current controls and rate their ability to detect the failure or cause.

#### Types of Process Controls:

Table 14.11: Types of Process Controls

| Control Type                   | Purpose                                          | Examples                               |
|--------------------------------|--------------------------------------------------|----------------------------------------|
| <b>**Prevention Controls**</b> | Prevent the cause from occurring                 | Error-proofing (poka-yoke), interlocks |
| <b>**Detection Controls**</b>  | Detect the failure mode or cause after it occurs | Inspection, testing, gauging, SPC      |

#### Detection Rating Scale:

Table 14.12: Detection Rating Scale (D)

| Rating | Detection            | Criteria                                             | Example Controls                 |
|--------|----------------------|------------------------------------------------------|----------------------------------|
| 10     | Absolute Uncertainty | No current control; cannot detect or is not analyzed | No inspection                    |
| 9      | Very Remote          | Control will probably not detect                     | Indirect measurement             |
| 8      | Remote               | Control has poor chance of detection                 | Visual inspection                |
| 7      | Very Low             | Control has low chance of detection                  | Manual 100% inspection           |
| 6      | Low                  | Control may detect                                   | Variable gauging                 |
| 5      | Moderate             | Control has moderate chance of detection             | SPC charting                     |
| 4      | Moderately High      | Control has moderately high chance of detection      | In-station gauging               |
| 3      | High                 | Control has high chance of detection                 | Automated 100% inspection        |
| 2      | Very High            | Control almost certain to detect                     | Multiple independent inspections |
| 1      | Almost Certain       | Control will detect; error-proofed                   | Error-proofing                   |

**Important:** Detection ratings are often **harder to reduce** than occurrence ratings. Adding inspection doesn't prevent defects — it only catches them. Prevention is always better than detection.

Question: Why are detection ratings often harder to improve?

#### Detection ratings are harder to reduce because:

1. **Detection doesn't prevent:** The defect has already been made; you're just trying to find it
2. **Human inspection is unreliable:** Typical effectiveness is only 80-85% for visual inspection
3. **100% inspection is expensive:** Automated inspection requires significant investment
4. **Some defects are hard to detect:** Internal defects, intermittent problems, latent failures

5. **Sampling has limitations:** Even good sampling plans miss some defects

**Better approach:** Focus on reducing occurrence through: - Error-proofing (poka-yoke) - Process capability improvement - Preventive maintenance - Better training

### 14.4.9 Step 9: Calculating RPN and Prioritization

The **Risk Priority Number (RPN)** combines all three ratings:

$$\text{RPN} = \text{Severity (S)} \times \text{Occurrence (O)} \times \text{Detection (D)}$$

Live cell — try different Severity, Occurrence, and Detection ratings.

```
#| label: rpn-calculation
# Example RPN Calculations

# Failure Mode 1: Short shot
S1 <- 7    # Severity: Product inoperable (assembly)
O1 <- 5    # Occurrence: Occasional
D1 <- 6    # Detection: Moderate (visual inspection)

RPN1 <- S1 * O1 * D1
cat("Short shot RPN:", RPN1, "\n")

# Failure Mode 2: Flash
S2 <- 4    # Severity: Appearance issue
O2 <- 6    # Occurrence: Regular
D2 <- 3    # Detection: High (easy to see)

RPN2 <- S2 * O2 * D2
cat("Flash RPN:", RPN2, "\n")

# Failure Mode 3: Contamination
S3 <- 9    # Severity: Safety concern (medical device)
O3 <- 3    # Occurrence: Low
D3 <- 7    # Detection: Low (internal contamination)

RPN3 <- S3 * O3 * D3
cat("Contamination RPN:", RPN3, "\n")

# Comparison
cat("\n--- Priority Order by RPN ---\n")
cat("1. Short shot (RPN =", RPN1, ")\n")
cat("2. Contamination (RPN =", RPN3, ")\n")
cat("3. Flash (RPN =", RPN2, ")\n")
```

**RPN Action Thresholds (Traditional):**

Table 14.13: RPN Action Thresholds

| RPN Range | Risk Level  | Action Required                              |
|-----------|-------------|----------------------------------------------|
| 1-50      | Low         | Monitor; no immediate action                 |
| 51-100    | Moderate    | Consider action if resources available       |
| 101-200   | Significant | Action required within planning cycle        |
| 201-500   | High        | Immediate action required                    |
| 501-1000  | Critical    | Stop production; immediate corrective action |

Interactive Exercise: Calculate RPN

**Calculate the RPN for each failure mode and rank them:**

| Failure Mode             | Severity | Occurrence | Detection | RPN |
|--------------------------|----------|------------|-----------|-----|
| A: Dimension out of spec | 6        | 4          | 5         | ?   |
| B: Surface scratch       | 3        | 7          | 2         | ?   |
| C: Missing component     | 8        | 2          | 8         | ?   |
| D: Wrong material        | 9        | 3          | 6         | ?   |

**Answers:** - A:  $6 \times 4 \times 5 = 120$  - B:  $3 \times 7 \times 2 = 42$  - C:  $8 \times 2 \times 8 = 128$  - D:  $9 \times 3 \times 6 = 162$

**Priority Order:** D (162) > C (128) > A (120) > B (42)

**Question:** Should we always address the highest RPN first? See limitations discussion below.

**14.4.10 Limitations of RPN**

RPN has known limitations that must be understood:

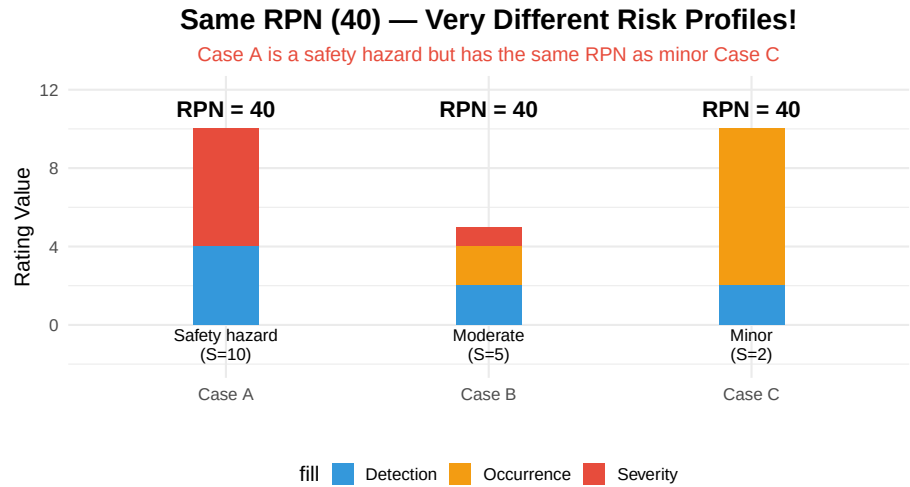


Figure 14.8: RPN Limitation Example

**Key RPN Limitations:**

1. **Same RPN, different risk:**  $10 \times 1 \times 4 = 40$ , but  $2 \times 10 \times 2 = 40$
2. **Severity often overlooked:** High severity should always trigger action
3. **Arbitrary thresholds:** Why is 100 the magic number?
4. **Non-linear scale:** Difference between 2 and 3 is not same as 8 and 9

**14.4.11 Action Priority (AP) Method — AIAG-VDA Alternative**

The 2019 AIAG-VDA FMEA Handbook introduced the **Action Priority (AP)** method as an alternative to RPN.

Table 14.14: Action Priority (AP) Method

| Priority          | Criteria                                        | Action Required             |
|-------------------|-------------------------------------------------|-----------------------------|
| <b>**HIGH**</b>   | Severity = 9 or 10 (any O, any D)               | Must take action to improve |
| <b>**HIGH**</b>   | Severity = 8 AND Occurrence 4                   | Must take action            |
| <b>**MEDIUM**</b> | Severity = 7 AND Occurrence 5                   | Should take action          |
| <b>**MEDIUM**</b> | Severity = 5-6 AND Occurrence 6 AND Detection 5 | Should take action          |
| <b>**LOW**</b>    | All other combinations                          | May take action; document   |

**14.4.12 Step 10: Recommended Actions and Follow-Up**

For high-priority items, develop specific recommended actions.

Table 14.15: Step 10: Recommended Actions with T

| Failure Mode | Current RPN | Recommended Action                                   | Responsible      | T |
|--------------|-------------|------------------------------------------------------|------------------|---|
| Short shot   | <b>210</b>  | Install shot size monitoring with automatic reject   | Process Engineer | 2 |
| Short shot   | <b>210</b>  | Implement preventive maintenance for nozzle cleaning | Maintenance      | 2 |
| Short shot   | <b>210</b>  | Add mold temperature monitoring with alarm           | Quality Engineer | 2 |

## 14.5 Complete PFMEA Example: Injection Molding

Let's work through a complete PFMEA for an injection molding process step by step.

Table 14.16: Complete PFMEA Example: Injection Molding Process

| Step      | Function              | Failure Mode   | Effect                       | S | Cause                    |
|-----------|-----------------------|----------------|------------------------------|---|--------------------------|
| 3. Inject | Fill mold completely  | Short shot     | Part scrapped, line stoppage | 7 | Insufficient shot size   |
| 3. Inject | Fill mold completely  | Flash          | Secondary trim required      | 4 | Excessive pressure       |
| 3. Inject | Fill mold completely  | Air entrapment | Weak area, visual defect     | 6 | Blocked vent             |
| 4. Cool   | Solidify uniformly    | Warping        | Dimensional rejection        | 7 | Uneven cooling           |
| 4. Cool   | Solidify uniformly    | Sink marks     | Visual defect                | 5 | Thick section, short p   |
| 5. Eject  | Remove without damage | Stuck in mold  | Part damage, mold damage     | 8 | Insufficient draft angle |

Live cell — try different S/O/D ratings for each improvement option.

```
#| label: rpn-improvement
# Improvement Analysis for Short Shot (Highest RPN = 210)

cat("=== CURRENT STATE ===\n")
S_current <- 7 # Severity (cannot change - inherent to failure)
O_current <- 5 # Occurrence
D_current <- 6 # Detection
RPN_current <- S_current * O_current * D_current
cat("Current RPN:", RPN_current, "(S=7, O=5, D=6)\n\n")

cat("=== IMPROVEMENT OPTIONS ===\n\n")

# Option 1: Improve Detection
cat("Option 1: Add automatic shot weight monitoring\n")
D_option1 <- 2 # Automated detection
RPN_option1 <- S_current * O_current * D_option1
```

```
cat("New RPN:", RPN_option1, "(S=7, O=5, D=2)\n")
cat("Reduction:", round((1 - RPN_option1/RPN_current)*100), "%\n\n")

# Option 2: Improve Occurrence (Prevention)
cat("Option 2: Add screw position monitoring with feedback control\n")
O_option2 <- 2 # Much lower occurrence
RPN_option2 <- S_current * O_option2 * D_current
cat("New RPN:", RPN_option2, "(S=7, O=2, D=6)\n")
cat("Reduction:", round((1 - RPN_option2/RPN_current)*100), "%\n\n")

# Option 3: Both improvements
cat("Option 3: Both improvements combined\n")
RPN_option3 <- S_current * O_option2 * D_option1
cat("New RPN:", RPN_option3, "(S=7, O=2, D=2)\n")
cat("Reduction:", round((1 - RPN_option3/RPN_current)*100), "%\n\n")
```

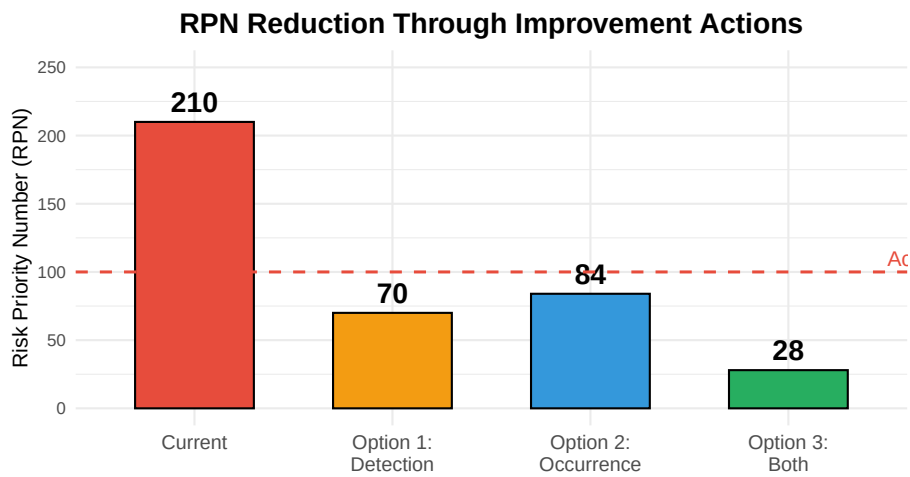


Figure 14.9: RPN Improvement Comparison

## 14.6 Industry-Specific PFMEA Examples

### 14.6.1 Example: Welding Process

Table 14.17: PFMEA Example: Welding Process



| Failure Mode      | Effect                                | S | Cause                               | O |
|-------------------|---------------------------------------|---|-------------------------------------|---|
| Incomplete fusion | Weak joint, potential crack           | 9 | Low heat input, contamination       | 4 |
| Porosity          | Leak path, strength reduction         | 8 | Shielding gas issue, moisture       | 5 |
| Undercut          | Stress concentration, fatigue failure | 7 | Excessive current, wrong angle      | 4 |
| Spatter           | Appearance, cleanup required          | 3 | Wrong parameters, contaminated wire | 6 |
| Distortion        | Dimensional non-conformance           | 6 | Excessive heat, improper sequence   | 4 |

### 14.6.2 Example: Heat Treatment Process

Table 14.18: PFMEA Example: Heat Treatment Process

| Failure Mode            | Effect                        | S  | Cause                        | O | Current |
|-------------------------|-------------------------------|----|------------------------------|---|---------|
| Under-hardened          | Premature wear, field failure | 8  | Low temp, short time         | 4 | Hardne  |
| Over-hardened/Brittle   | Part cracks under load        | 9  | High temp, long time         | 3 | Hardne  |
| Surface decarburization | Soft surface, wear            | 7  | Wrong atmosphere             | 4 | Surface |
| Distortion/Warpage      | Dimensional rejection         | 6  | Uneven heating, rapid quench | 5 | CMM s   |
| Quench cracking         | Scrap, safety hazard          | 10 | Thermal shock, design        | 2 | Visual  |

### 14.6.3 Example: Food Processing (Pasteurization)

Table 14.19: PFMEA Example: Pasteurization Process

| Failure Mode          | Effect                         | S  | Cause                              | O | C |
|-----------------------|--------------------------------|----|------------------------------------|---|---|
| Under-processing      | Pathogen survival, food safety | 10 | Equipment malfunction, calibration | 2 | C |
| Over-processing       | Nutrient loss, off-flavor      | 5  | Control system error               | 3 | 7 |
| Temperature deviation | Inconsistent kill, shelf life  | 8  | Sensor drift, flow variation       | 3 | C |
| Time deviation        | Process variation              | 6  | Timer malfunction                  | 3 | 7 |
| Cross-contamination   | Product recall, illness        | 10 | Gasket leak, CIP failure           | 2 | C |

Interactive Exercise: Create a PFMEA

**Create a simple PFMEA for a coffee brewing process in a commercial setting.**

**Process Steps:** 1. Fill water reservoir 2. Load coffee grounds 3. Brew coffee 4. Dispense into cup

**Identify at least 4 failure modes with S, O, D ratings and RPN.**

**Example Answer:**

| Step                  | Failure Mode           | Effect                  | S | Cause                           | O | Control         | D | RPN |
|-----------------------|------------------------|-------------------------|---|---------------------------------|---|-----------------|---|-----|
| 1.<br>Fill<br>water   | Wrong water<br>level   | Weak/stro-<br>ng coffee | 4 | No<br>level<br>indica-<br>tor   | 5 | Visual<br>check | 5 | 100 |
| 2.<br>Load<br>grounds | Wrong<br>amount        | Weak/stro-<br>ng coffee | 4 | No<br>mea-<br>suring            | 6 | Visual<br>check | 6 | 144 |
| 3.<br>Brew            | Temperature<br>too low | Under-<br>extracted     | 6 | Heating<br>ele-<br>ment<br>worn | 3 | Temp<br>display | 4 | 72  |
| 4.<br>Dis-<br>pense   | Overflow               | Burn<br>hazard,<br>mess | 8 | Cup<br>sensor<br>failure        | 2 | Cup<br>sensor   | 4 | 64  |

**Priority:** Wrong amount (144) > Water level (100) > Temperature (72) > Overflow (64)

## 14.7 Integration with Other Quality Tools

PFMEA doesn't exist in isolation — it connects to other quality tools and systems.

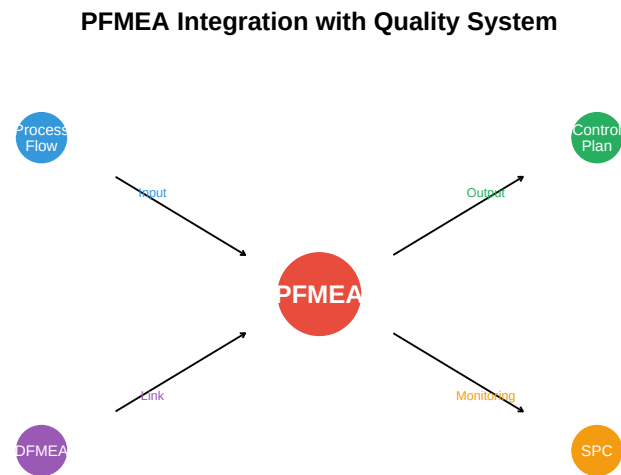


Figure 14.10: PFMEA Integration with Quality Tools

**14.7.1 From PFMEA to Control Plan**

The PFMEA directly feeds the **Control Plan** — a document that specifies what controls are needed at each process step.

Table 14.20: PFMEA to Control Plan Linkage

| From PFMEA      | To Control Plan                       |
|-----------------|---------------------------------------|
| Process Step    | Operation/Step                        |
| Failure Mode    | Characteristic to Control             |
| Severity        | Classification (Critical/Significant) |
| Cause           | Process Parameter                     |
| Current Control | Control Method                        |
| Detection       | Reaction Plan                         |

**14.8 Common Pitfalls and Best Practices**

**14.8.1 Common Pitfalls to Avoid**

Table 14.21: Common PFMEA Pitfalls and Solutions

| Pitfall                         | Problem                                                   | Solution  |
|---------------------------------|-----------------------------------------------------------|-----------|
| <b>**Paperwork Exercise**</b>   | PFMEA done only to satisfy customer requirement           | Conduct   |
| <b>**One-Person Show**</b>      | Engineer completes alone without team input               | Require c |
| <b>**Optimistic Detection**</b> | Overestimating effectiveness of current controls          | Validate  |
| <b>**Static Document**</b>      | Never updated after process changes                       | Review a  |
| <b>**RPN Obsession**</b>        | Focusing only on RPN numbers, missing high-severity items | Also cons |
| <b>**Vague Failure Modes**</b>  | Failure modes too general to be actionable                | Be specif |

### 14.8.2 Best Practices

Table 14.22: PFMEA Best Practices

| Practice                       | Description                                                 |
|--------------------------------|-------------------------------------------------------------|
| <b>**Start Early**</b>         | Begin PFMEA during process design, not after problems occur |
| <b>**Include Operators**</b>   | Operators know failures that never get documented           |
| <b>**Use Historical Data**</b> | Use warranty data, scrap reports, customer complaints       |
| <b>**Be Specific**</b>         | Specific failure modes lead to specific actions             |
| <b>**Follow Up**</b>           | Verify that actions were implemented and effective          |
| <b>**Living Document**</b>     | Review and update regularly (minimum annually)              |

## 14.9 Case Study: Preventing a Major Quality Issue

### 14.9.1 Background

**Company:** Automotive brake component manufacturer

**Product:** Brake caliper mounting bracket

**Situation:** New manufacturing process being implemented for a high-volume contract

### 14.9.2 PFMEA Analysis

During the PFMEA session, the team identified a critical failure mode:

Table 14.23: Case Study: Critical Failure Mode Identified

| Element         | Analysis                                         |
|-----------------|--------------------------------------------------|
| Process Step    | Drill mounting holes                             |
| Function        | Create holes at 45.0 $\pm$ 0.1mm spacing         |
| Failure Mode    | Hole spacing out of tolerance                    |
| Effect          | Caliper misalignment, brake performance affected |
| <b>Severity</b> | <b>9 (safety-related)</b>                        |
| Cause           | Fixture wear, thermal expansion                  |
| Occurrence      | 4 (occasional)                                   |
| Current Control | CMM check 5 per shift                            |
| Detection       | 5 (moderate sampling)                            |
| <b>RPN</b>      | <b>180</b>                                       |

### 14.9.3 Actions Taken

Based on the PFMEA, the team implemented:

1. **In-process gauging:** Automated measurement after every part (D: 5  $\rightarrow$  2)
2. **Fixture monitoring:** Temperature compensation and wear tracking (O: 4  $\rightarrow$  2)
3. **Enhanced fixture design:** Carbide wear surfaces (O: 4  $\rightarrow$  2)

Live cell — try different before/after ratings.

```

#| label: case-study-improvement
# Before improvements
S_before <- 9
O_before <- 4
D_before <- 5
RPN_before <- S_before * O_before * D_before
cat("Before RPN:", RPN_before, "\n")

# After improvements
S_after <- 9    # Severity unchanged (inherent)
O_after <- 2    # Reduced through fixture improvements
D_after <- 2    # Reduced through 100% gauging
RPN_after <- S_after * O_after * D_after
cat("After RPN:", RPN_after, "\n")
cat("Reduction:", round((1 - RPN_after/RPN_before)*100), "%\n")

```

### 14.9.4 Results

- **Zero field failures** related to hole spacing after 18 months of production
- **Estimated avoided cost:** \$2.4 million in potential recall and warranty

- **Investment:** \$85,000 for gauging and fixture upgrades
- **ROI:** 28:1

**Key Lesson:** The PFMEA investment of \$85,000 prevented a potential \$2.4 million problem. This is the power of proactive risk management.

## 14.10 Summary

Table 14.24: PFMEA Key Concepts Summary

| Topic                      | Key Points                                                                    |
|----------------------------|-------------------------------------------------------------------------------|
| <b>**PFMEA Purpose**</b>   | Prevent defects before they occur; proactive vs. reactive quality             |
| <b>**Key Ratings**</b>     | Severity (effect), Occurrence (likelihood), Detection (control effectiveness) |
| <b>**RPN Calculation**</b> | $RPN = S \times O \times D$ ; Range 1-1000; Higher = More risk                |
| <b>**Prioritization**</b>  | Address high RPN first, but always act on high severity (9-10)                |
| <b>**Action Priority**</b> | AIAG-VDA method: High/Medium/Low based on S-O-D combination                   |
| <b>**Best Practice**</b>   | Cross-functional team, living document, follow-up on actions                  |

## 14.11 Review Questions

### 14.11.1 Conceptual Questions

Question 1: What is the difference between a failure mode and a failure effect?

**Failure Mode:** How the process fails to perform its intended function - Example: “Hole drilled undersize” - Example: “Temperature too low” - Example: “Missing component”

**Failure Effect:** The consequence of the failure mode - Example: “Part rejected at assembly” (effect of undersize hole) - Example: “Product not properly cured” (effect of low temperature) - Example: “Product doesn’t function” (effect of missing component)

**Key Relationship:** One failure mode can have multiple effects at different levels (local, downstream, customer).

Question 2: Why should PFMEA be conducted by a cross-functional team?

**Reasons for Cross-Functional Team:**

1. **Diverse expertise:** No single person understands all aspects
2. **Multiple perspectives:** Quality sees defects differently than production

3. **Historical knowledge:** Operators know undocumented problems
4. **Buy-in:** People support what they help create
5. **System thinking:** Process interactions become visible
6. **Better solutions:** Different functions propose different improvements

**Typical team composition:** - Process Engineer (owner) - Quality Engineer - Production Supervisor/Operator - Maintenance Technician - Design Engineer - Supplier Representative (when applicable)

Question 3: Explain why detection ratings are often harder to reduce than occurrence ratings.

**Detection is harder to reduce because:**

1. **Detection doesn't prevent:** Defect already exists; just trying to find it
2. **Human inspection is unreliable:** 80-85% effectiveness typical
3. **100% inspection is expensive:** Requires automation investment
4. **Some defects are hidden:** Internal defects, intermittent failures
5. **Sampling limitations:** Cannot catch all defects with sampling

**Better approach:** Focus on occurrence reduction: - Error-proofing (poka-yoke) - Process capability improvement - Preventive maintenance - Better training

**Example:** Reducing drill wear (occurrence) is more effective than adding more inspection (detection).

Question 4: At what point in process development should PFMEA ideally be conducted?

**Ideal Timing:** During process design, before production begins

**Why early is better:** - Changes cost 1x during design - Changes cost 10x during production - Changes cost 100x after customer delivery

**PFMEA Timing Points:** 1. **New process development:** Before equipment is ordered 2. **New product introduction:** During process planning 3. **Process changes:** Before implementing changes 4. **Quality issues:** When problems indicate process risk 5. **Regular review:** Annually at minimum

**Key Principle:** The earlier PFMEA is conducted, the more opportunity exists to design out failures rather than inspect them out.

### 14.11.2 Calculation Questions

Question 5: Given S=8, O=6, D=4, calculate RPN. If improved controls reduce detection to 2, what is the new RPN?

**Initial Calculation:**

$$\text{RPN} = S \times O \times D$$

$$\text{RPN} = 8 \times 6 \times 4 = 192$$

**After Detection Improvement:**

New RPN =  $8 \times 6 \times 2 = 96$

**Analysis:** - Original RPN: 192 (Significant risk, action required) - New RPN: 96 (Moderate risk) - Reduction: 50%

**Note:** Severity (8) remains unchanged because it's determined by the effect, not by process controls. Only occurrence and detection can be reduced through process improvements.

Question 6: Two failure modes have RPNs of 120 and 150. Should the higher RPN always be addressed first?

**Answer:** Not necessarily!

**Consider the following scenarios:**

**Scenario A:** - Failure Mode 1: RPN = 150 (S=3, O=10, D=5) — Minor cosmetic issue - Failure Mode 2: RPN = 120 (S=10, O=3, D=4) — Safety hazard

**Priority:** Address Failure Mode 2 first despite lower RPN because: - Severity 10 indicates safety hazard - High-severity items should always be addressed regardless of RPN - The AP (Action Priority) method would classify this as HIGH priority

**When to prioritize lower RPN:** - Higher severity (especially 9-10) - Regulatory or safety implications - Customer-critical characteristic - Easier/faster to implement

**Key Lesson:** RPN is a prioritization tool, not the only decision criterion. Use engineering judgment.

### 14.11.3 Application Questions

Question 7: For a metal stamping operation, identify at least five potential failure modes and their likely effects.

**Metal Stamping PFMEA:**

| # | Failure Mode             | Effect                                    |
|---|--------------------------|-------------------------------------------|
| 1 | Part not fully formed    | Dimensional rejection, functional failure |
| 2 | Burrs on edges           | Injury hazard, assembly interference      |
| 3 | Cracks/splits            | Structural failure, scrap                 |
| 4 | Wrong material thickness | Dimensional/strength issues               |
| 5 | Mislocated features      | Assembly issues, rejection                |



| # | Failure Mode                | Effect                                        |
|---|-----------------------------|-----------------------------------------------|
| 6 | Surface scratches           | Appearance rejection,<br>corrosion initiation |
| 7 | Springback out of tolerance | Dimensional rejection                         |
| 8 | Die mark/impression         | Appearance defect                             |

**Potential Causes for “Part not fully formed”:** - Insufficient press tonnage  
- Die wear - Material too hard - Lubrication insufficient - Press speed too fast

Question 8: A process has RPN of 240 (S=8, O=6, D=5). Propose three improvement strategies and calculate resulting RPN.

**Current State:** S=8, O=6, D=5, RPN=240

**Strategy 1: Improve Detection** - Add automated inspection (100% gauging)  
- New D = 2 - New RPN =  $8 \times 6 \times 2 = 96$  (60% reduction)

**Strategy 2: Improve Occurrence** - Add error-proofing (poka-yoke) to prevent cause - New O = 2 - New RPN =  $8 \times 2 \times 5 = 80$  (67% reduction)

**Strategy 3: Combined Approach** - Error-proofing ( $O \rightarrow 3$ ) + Better detection ( $D \rightarrow 3$ ) - New RPN =  $8 \times 3 \times 3 = 72$  (70% reduction)

**Recommendation:** Strategy 2 (error-proofing) is typically preferred because:  
- Prevents defects from occurring - More sustainable long-term - Reduces inspection costs - Better for customer confidence

#### 14.11.4 Critical Thinking Questions

Question 9: Discuss the limitations of using RPN as the sole prioritization method.

##### RPN Limitations:

- Equal weighting problem:**
  - $10 \times 1 \times 4 = 40$  (safety hazard, rare)
  - $2 \times 10 \times 2 = 40$  (minor, frequent)
  - Same RPN, very different risk!
- Severity often overlooked:**
  - High-severity items (9-10) should always be addressed
  - RPN can hide safety-critical issues
- Non-linear scales:**
  - Difference between S=2 and S=3 is not same as S=8 and S=9
  - Multiplication assumes linear relationship
- Arbitrary thresholds:**
  - Why is RPN=100 the “action” threshold?
  - No scientific basis for specific cutoffs
- Gaming the system:**

- Teams may manipulate ratings to avoid action
- Subjectivity in ratings

**Better Approaches:** - Action Priority (AP) method - Always act on S = 9 regardless of RPN - Consider multiple criteria (cost, feasibility, timing) - Use risk matrix visualization

Question 10: How would you handle a situation where team members disagree on severity ratings?

#### Approach to Rating Disagreements:

- 1. Clarify the failure effect:** - Ensure everyone understands the same effect  
- Be specific about which customer sees which impact
- 2. Reference the rating criteria:** - Use the standardized scale consistently  
- Point to specific criteria, not opinions
- 3. Consider worst-case scenario:** - PFMEA should consider worst reasonable case - “Could this effect happen?” vs “Will it happen?”
- 4. Gather data:** - Review historical data if available - What have similar failures caused in the past?
- 5. When in doubt, go higher:** - Conservative approach is appropriate for safety - Can always reduce later with evidence
- 6. Document the discussion:** - Record the rationale for the rating - Note if there was disagreement and why decision was made
- 7. Get external input:** - Bring in customer perspective if needed - Consult subject matter experts

**Final Rule:** If a safety or regulatory issue is possible, severity should be 9 or 10 regardless of probability.

---

## 14.12 References

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-



## Chapter 15

# Troubleshooting and Root Cause Analysis

### 15.1 Learning Objectives

After completing this chapter, you will be able to:

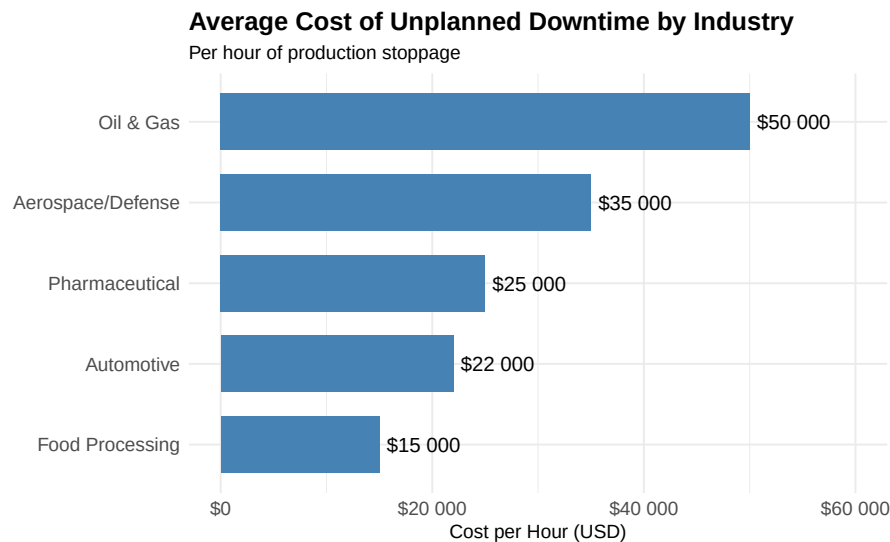
1. Apply systematic troubleshooting methodologies to diagnose equipment and process failures
  2. Distinguish between symptoms, immediate causes, and root causes
  3. Use the 5 Whys technique to drill down to fundamental causes
  4. Construct and analyze Ishikawa (fishbone) diagrams
  5. Apply Fault Tree Analysis (FTA) to complex failure scenarios
  6. Use Pareto analysis to prioritize problem-solving efforts
  7. Implement the 8D problem-solving methodology
  8. Document findings and implement effective corrective actions
  9. Prevent problem recurrence through systemic improvements
- 

### 15.2 Introduction to Troubleshooting

**Troubleshooting** is the systematic process of identifying, diagnosing, and resolving problems in equipment, processes, or systems. For electromechanical technicians, effective troubleshooting is perhaps the most valuable skill you can develop.

#### 15.2.1 The Cost of Downtime

Equipment failures and process problems have significant financial impacts:



### 15.2.2 Reactive vs. Proactive Problem Solving

Table 15.1: Comparison of Problem-Solving Approaches

| Aspect               | Reactive Approach              | Proactive Approach         |
|----------------------|--------------------------------|----------------------------|
| <b>Timing</b>        | After failure occurs           | Before or early in failure |
| <b>Focus</b>         | Quick fix to restore operation | Root cause elimination     |
| <b>Stress Level</b>  | High - production pressure     | Lower - planned approach   |
| <b>Cost</b>          | High - emergency repairs       | Lower - scheduled work     |
| <b>Learning</b>      | Limited - rush to fix          | High - thorough analysis   |
| <b>Documentation</b> | Often skipped                  | Systematic and complete    |

**Think About It:** Why do organizations often default to reactive troubleshooting?

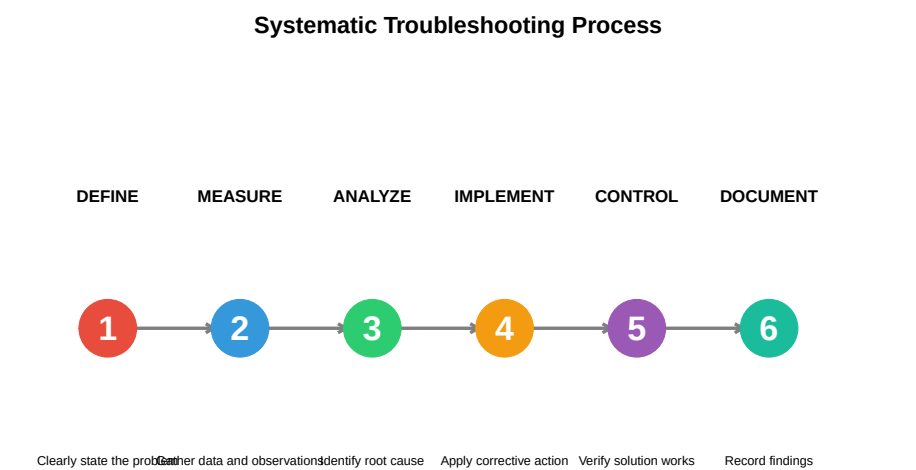
Common reasons include: - **Time pressure:** Production demands make it tempting to apply quick fixes - **Lack of training:** Technicians may not know RCA methodologies - **Reward systems:** Organizations may reward “firefighters” who quickly restore production - **Missing data:** Without good records, patterns are hard to identify - **Culture:** “We’ve always done it this way” mentality

The irony is that reactive troubleshooting actually costs more time in the long run due to recurring problems.

## 15.3 Systematic Troubleshooting Methodology

Effective troubleshooting follows a structured approach rather than random trial-and-error.

### 15.3.1 The DMAIC Troubleshooting Framework



### 15.3.2 Step 1: Define the Problem

A well-defined problem is half-solved. Use the **5W2H** method:

Table 15.2: 5W2H Problem Definition Framework

| Question        | What to Ask                                                        | Example Response                       |
|-----------------|--------------------------------------------------------------------|----------------------------------------|
| <b>WHAT</b>     | What is the problem? What is happening vs. what should happen?     | Motor overheating on conveyor          |
| <b>WHERE</b>    | Where does the problem occur? Location, machine, station?          | Discharge end of conveyor              |
| <b>WHEN</b>     | When did it start? When does it occur? Continuous or intermittent? | Started Monday, occurs continuously    |
| <b>WHO</b>      | Who discovered it? Who is affected? Who was operating?             | Operator noticed, maintenance notified |
| <b>WHY</b>      | Why is this a problem? What is the impact?                         | Risk of motor failure, line stoppage   |
| <b>HOW</b>      | How was the problem detected? How is it manifesting?               | Detected by high-temperature sensor    |
| <b>HOW MUCH</b> | How often? How many units affected? What is the magnitude?         | Occurs every day, 100% of units        |

### 15.3.3 Problem Statement Template

**Good Problem Statement Format:** “[WHAT] is occurring at [WHERE] since [WHEN], affecting [WHO/WHAT], with a magnitude of [HOW MUCH].”

**Example:** > “Motor overheating (85°C vs. normal 70°C) is occurring at Conveyor Line 3 discharge motor since Monday morning, affecting packaging department production, with the condition occurring every shift after approximately 2 hours of operation.”

**Practice:** Write a problem statement for this scenario

**Scenario:** An injection molding machine is producing parts with short shots (incomplete fill). The problem was noticed by the quality inspector during the night shift. About 15% of parts are affected. The issue seems worse when running the larger 500g parts compared to the smaller 200g parts.

**Sample Problem Statement:** “Short shots (incomplete cavity fill) are occurring on Injection Molding Machine #4 since night shift on Tuesday, affecting 15% of parts produced, with the defect rate increasing significantly when running 500g parts compared to 200g parts.”

### 15.3.4 Step 2: Gather Data and Observations

Before making any changes, collect information systematically:

Table 15.3: Systematic Data Gathering Checklist

| Data Source               | What to Check                                              | Tools/Methods           |
|---------------------------|------------------------------------------------------------|-------------------------|
| <b>Visual Inspection</b>  | Physical condition, wear, contamination, alignment, damage | Flashlight, mirror      |
| <b>Operator Interview</b> | What changed? When did it start? Any unusual events?       | Open-ended questions    |
| <b>Machine Parameters</b> | Temperatures, pressures, speeds, currents, positions       | HMI screens, gauges     |
| <b>Historical Records</b> | Maintenance logs, previous repairs, similar issues         | CMMS, logbooks          |
| <b>Product/Output</b>     | Defect patterns, measurements, reject rates                | Calipers, gauges        |
| <b>Environmental</b>      | Temperature, humidity, vibration, power quality            | Thermometer, hygrometer |

### 15.3.5 The “Change Analysis” Approach

Many problems are caused by changes. Ask:

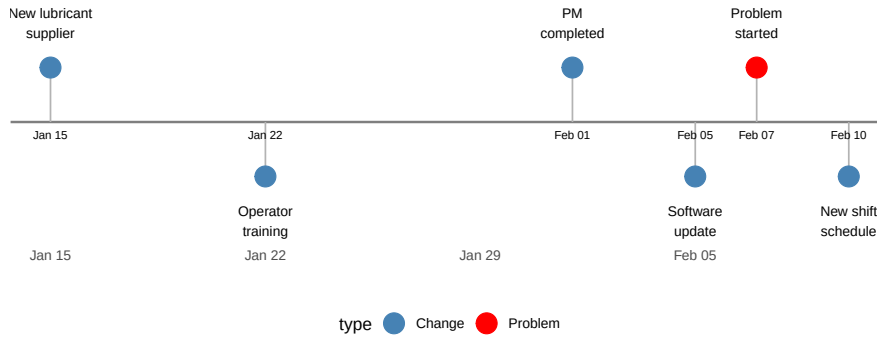
- What changed recently?
- New materials, suppliers, or batches?
- New operators or different shifts?
- Maintenance performed?
- Environmental changes (weather, season)?
- Software updates or parameter changes?



- New products or recipes running?

### Change Timeline Analysis

Map changes against problem occurrence to identify correlations



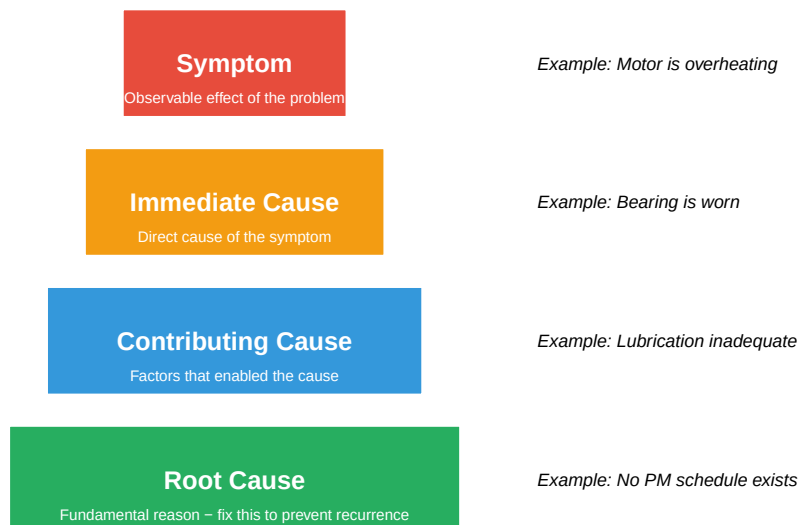
## 15.4 Understanding Root Cause vs. Symptoms

A critical concept in troubleshooting is distinguishing between different levels of causation.

### 15.4.1 The Causation Hierarchy

#### Causation Hierarchy

Dig deeper to find the root cause – don't stop at symptoms



15.4.2 Why Finding Root Cause Matters

Table 15.4: Comparison of Fix

| Fix Level       | Action Taken                                 | Immediate Result      |
|-----------------|----------------------------------------------|-----------------------|
| Symptom         | Add cooling fan to motor                     | Motor runs cooler tem |
| Immediate Cause | Replace worn bearing                         | Motor works for a whi |
| Root Cause      | Implement PM schedule with lubrication tasks | Problem eliminated    |

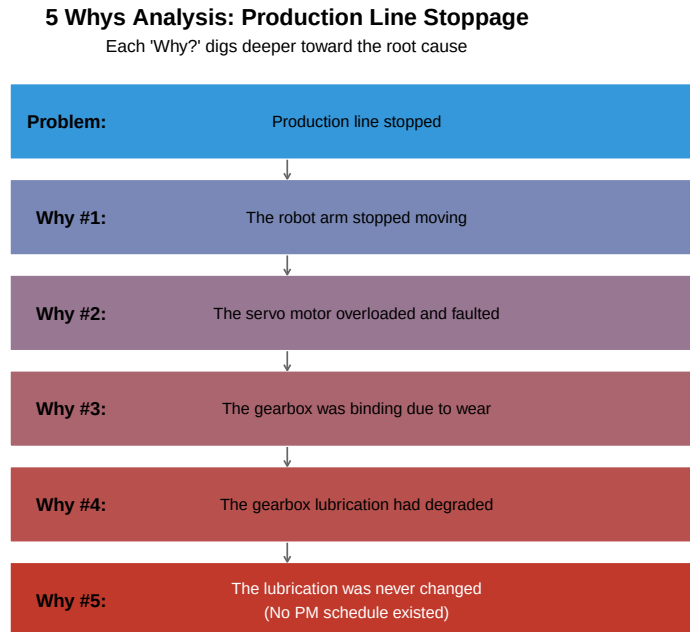
15.5 The 5 Whys Technique

The **5 Whys** is a simple but powerful technique developed by Sakichi Toyoda and used within Toyota’s manufacturing methods.

15.5.1 How It Works

1. State the problem
2. Ask “Why?” to identify the cause
3. Ask “Why?” again about that cause
4. Continue asking “Why?” until you reach the root cause
5. Typically requires 5 iterations (sometimes more, sometimes less)

### 15.5.2 5 Whys Example: Production Line Stoppage



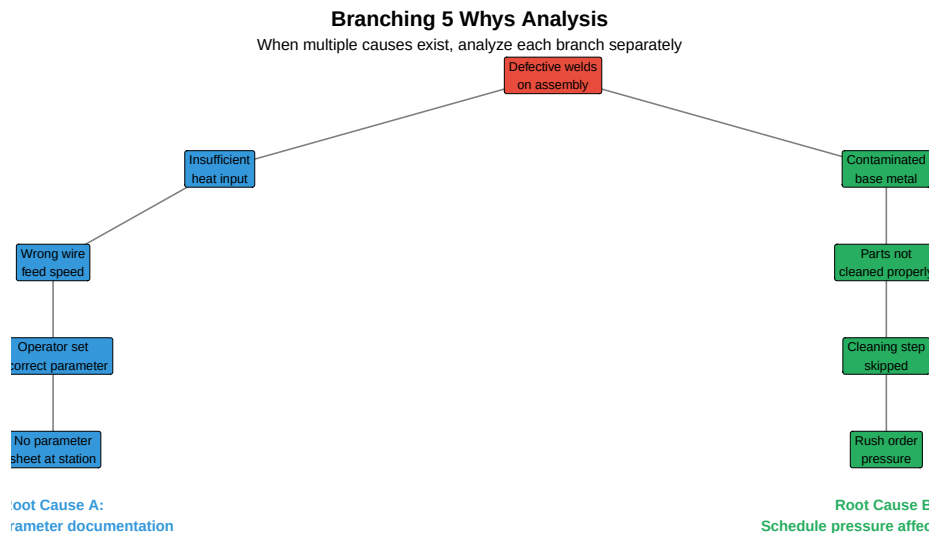
### 15.5.3 5 Whys Template

Table 15.5: 5 Whys Analysis Template

| Step                     | Description                                                      | Typical Cause Type |
|--------------------------|------------------------------------------------------------------|--------------------|
| <b>Problem Statement</b> | Clear, specific description of what happened                     | Symptom            |
| <b>Why #1</b>            | First-level cause (usually technical/physical)                   | Physical           |
| <b>Why #2</b>            | Why did that first cause occur?                                  | Physical/Human     |
| <b>Why #3</b>            | Why did the second cause occur?                                  | Human/Process      |
| <b>Why #4</b>            | Why did the third cause occur?                                   | Process/System     |
| <b>Why #5</b>            | Why did the fourth cause occur?                                  | System/Culture     |
| <b>Root Cause</b>        | <b>The fundamental cause that, if fixed, prevents recurrence</b> | <b>Root</b>        |

### 15.5.4 Multiple Branches in 5 Whys

Sometimes a problem has multiple causes. When you encounter this, branch your analysis:



**Practice Exercise:** Perform a 5 Whys Analysis

**Scenario:** A CNC machine produced 50 parts that were all 0.5mm undersize on a critical diameter.

Try to complete the 5 Whys analysis before looking at the sample answer below.

---

#### Sample Analysis:

1. **Problem:** CNC produced 50 undersize parts
2. **Why #1:** The tool offset was set incorrectly
3. **Why #2:** The operator entered the wrong offset value
4. **Why #3:** The operator misread the setup sheet
5. **Why #4:** The setup sheet had poor formatting and small print
6. **Why #5:** No standard format exists for setup documentation

**Root Cause:** Lack of standardized, clear setup documentation

**Corrective Actions:** - Create standardized setup sheet template with large, clear fonts - Add visual verification step (photo of correct setup) - Implement first-piece inspection requirement before production run

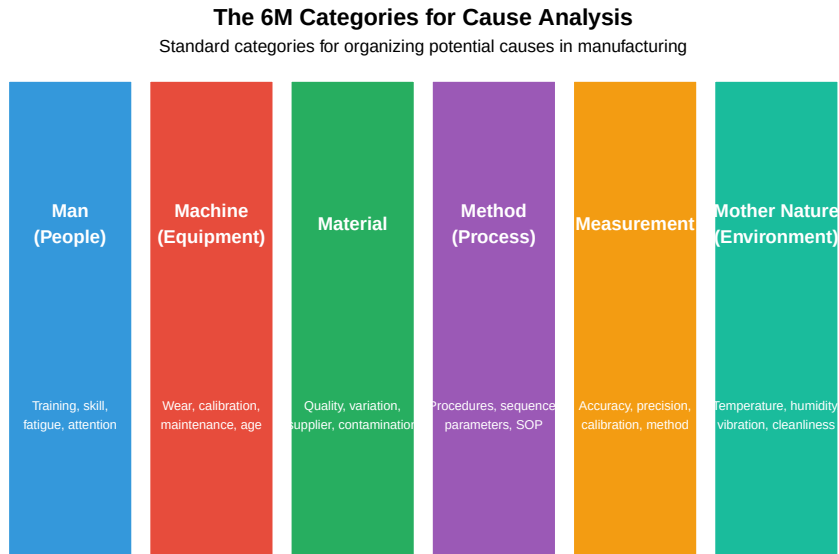
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## 15.6 Ishikawa (Fishbone) Diagram

The **Ishikawa diagram**, also called a **fishbone diagram** or **cause-and-effect diagram**, was developed by Kaoru Ishikawa in the 1960s. It provides a structured way to brainstorm and categorize potential causes.

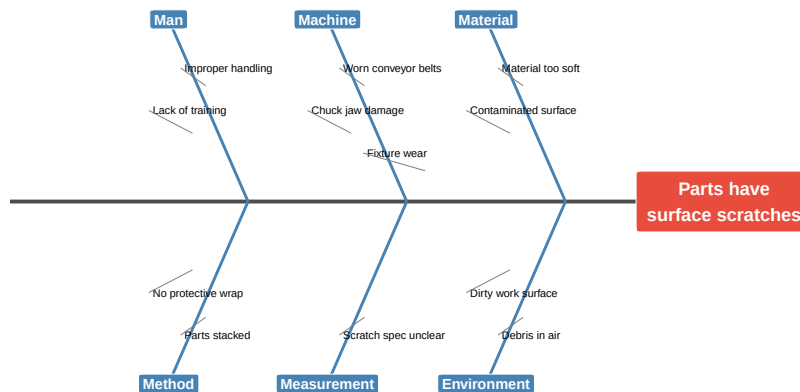
### 15.6.1 The 6M Categories

Manufacturing problems are typically organized using the **6M** categories:



### 15.6.2 Constructing a Fishbone Diagram

**Fishbone Diagram: Parts Have Surface Scratches**  
Brainstorming potential causes organized by 6M categories



### 15.6.3 Fishbone Diagram Best Practices

Table 15.6: Best Practices for Fishbone Diagram Development

| Guideline                                 | Description                                                               |
|-------------------------------------------|---------------------------------------------------------------------------|
| <b>Team Approach</b>                      | Include operators, engineers, maintenance, quality - diverse perspectives |
| <b>No Evaluation During Brainstorming</b> | Capture all ideas first; evaluate feasibility later                       |
| <b>Use Specific Language</b>              | Not 'machine problem' but 'bearing wear on spindle motor'.                |
| <b>Aim for 3-5 Causes per Category</b>    | Some categories may have more; don't force causes into categories         |
| <b>Circle the Most Likely</b>             | After brainstorming, identify 2-3 most probable causes to investigate     |
| <b>Verify with Data</b>                   | Don't assume - test hypotheses with data before implementing fixes        |

#### 15.6.4 Video: How to Create a Fishbone Diagram

### 15.7 Fault Tree Analysis (FTA)

**Fault Tree Analysis** is a top-down, deductive analysis method that uses Boolean logic to analyze undesired events. Originally developed for the aerospace industry, FTA is particularly useful for complex systems with multiple potential failure paths.

#### 15.7.1 FTA Symbols

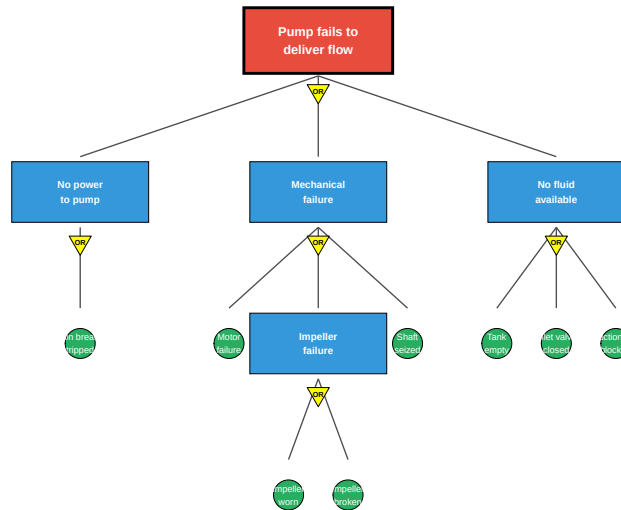
Table 15.7: Fault Tree Analysis Symbols

| Symbol           | Name               | Meaning                                                         |
|------------------|--------------------|-----------------------------------------------------------------|
| <b>Rectangle</b> | Intermediate Event | A fault event that results from combination of other events     |
| <b>Circle</b>    | Basic Event        | A basic failure that cannot be broken down further              |
| <b>Diamond</b>   | Undeveloped Event  | Event not analyzed further (outside scope or insufficient data) |
| <b>House</b>     | Normal Event       | An event expected to occur during normal operation              |
| <b>AND Gate</b>  | AND Gate           | Output occurs only if ALL inputs occur                          |
| <b>OR Gate</b>   | OR Gate            | Output occurs if ANY input occurs                               |

### 15.7.2 FTA Example: Pump Fails to Deliver Flow

#### Fault Tree Analysis: Pump Fails to Deliver Flow

OR gates mean any single path can cause the top event



### 15.7.3 Calculating Probability with FTA

FTA allows quantitative risk analysis by calculating the probability of the top event:

**OR Gate:**  $P(\text{output}) = 1 - [(1 - P_A) \times (1 - P_B) \times \dots]$  *Simplified:*  $P(\text{output}) = P_A + P_B$  (when probabilities are small)

**AND Gate:**  $P(\text{output}) = P_A \times P_B \times \dots$

Live cell — try different basic-event probabilities.

```

#| label: fta-calculation
# Example: Calculate top event probability
# Basic event probabilities (annual failure rates)
P_breaker_trip <- 0.02
P_motor_fail <- 0.01
P_impeller_worn <- 0.03
P_impeller_break <- 0.005
P_shaft_seize <- 0.008
P_tank_empty <- 0.05
P_valve_closed <- 0.01
P_line_blocked <- 0.02
  
```

```
# Impeller failure (OR gate)
P_impeller_fail <- 1 - (1 - P_impeller_worn) * (1 - P_impeller_break)
cat("P(Impeller failure):", round(P_impeller_fail, 4), "\n")

# Mechanical failure (OR gate)
P_mechanical <- 1 - (1 - P_motor_fail) * (1 - P_impeller_fail) * (1 - P_shaft_seize)
cat("P(Mechanical failure):", round(P_mechanical, 4), "\n")

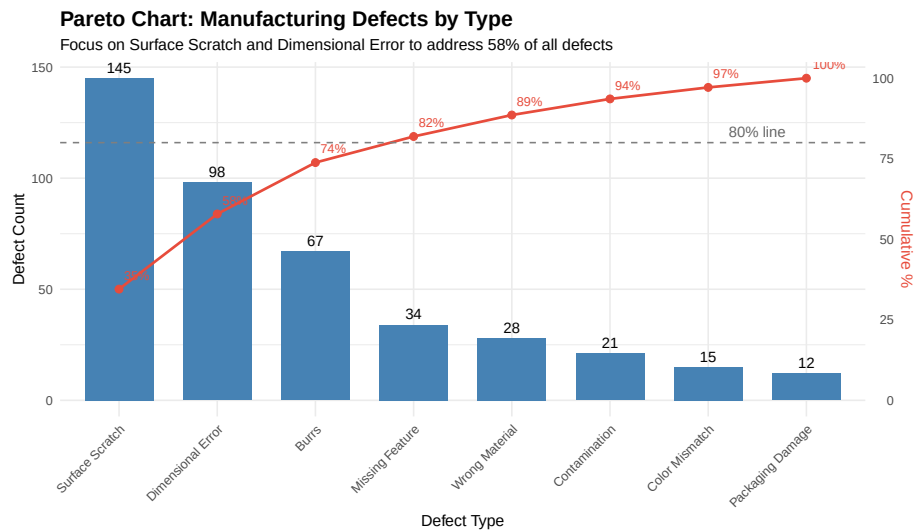
# No fluid available (OR gate)
P_no_fluid <- 1 - (1 - P_tank_empty) * (1 - P_valve_closed) * (1 - P_line_blocked)
cat("P(No fluid available):", round(P_no_fluid, 4), "\n")

# Top event: Pump fails (OR gate)
P_top <- 1 - (1 - P_breaker_trip) * (1 - P_mechanical) * (1 - P_no_fluid)
cat("P(Pump fails to deliver flow):", round(P_top, 4), "\n")
cat("This represents a", round(P_top * 100, 1), "% annual probability of failure")
```

## 15.8 Pareto Analysis

**Pareto Analysis** is based on the Pareto Principle (80/20 rule): roughly 80% of effects come from 20% of causes. This helps prioritize problem-solving efforts.

### 15.8.1 Creating a Pareto Chart





## 15.8.2 Interpreting Pareto Analysis

Live cell — try different defect counts.

```
#| label: pareto-interpretation
# Pareto analysis interpretation
defects <- data.frame(
  Defect = c("Surface Scratch", "Dimensional Error", "Burrs",
             "Missing Feature", "Wrong Material", "Contamination",
             "Color Mismatch", "Packaging Damage"),
  Count = c(145, 98, 67, 34, 28, 21, 15, 12)
)

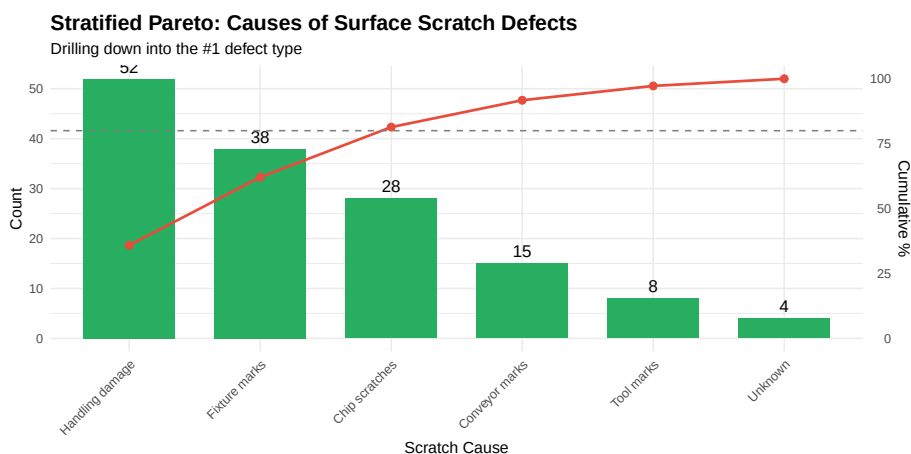
defects <- defects[order(-defects$Count),]
defects$Cumulative_Pct <- cumsum(defects$Count) / sum(defects$Count) * 100

# Identify the "vital few" (those that contribute to 80%)
vital_few <- defects[defects$Cumulative_Pct <= 80 |
  c(TRUE, defects$Cumulative_Pct[-nrow(defects)] < 80),]

cat("Total defects:", sum(defects$Count), "\n")
cat("\nVital Few (contributing to ~80% of defects):\n")
print(vital_few[, c("Defect", "Count", "Cumulative_Pct")])
cat("\nFocusing on these", nrow(vital_few), "defect types addresses",
    round(max(vital_few$Cumulative_Pct), 1), "% of all defects")
```

## 15.8.3 Stratified Pareto Analysis

Sometimes you need to break down categories further:



## 15.9 Is/Is Not Analysis

**Is/Is Not Analysis** is a structured comparison technique that helps narrow down the scope and identify distinguishing factors of a problem.

### 15.9.1 Is/Is Not Matrix

Table 15.8: Is/Is Not Analysis: Motor Overheating Problem

| Dimension     | Question                           | IS                         | IS NOT                 |
|---------------|------------------------------------|----------------------------|------------------------|
| <b>WHAT</b>   | What object has the problem?       | Motor on Line 3            | Motors on Lines 1, 2   |
| <b>WHAT</b>   | What is the defect/symptom?        | Overheating (85°C)         | Noise, vibration, or f |
| <b>WHERE</b>  | Where is the problem observed?     | Packaging area, Station 4  | Any other station      |
| <b>WHERE</b>  | Where on the object is the defect? | Motor housing, bearing end | Drive end, junction b  |
| <b>WHEN</b>   | When was it first observed?        | Monday morning shift       | Before weekend (Fric   |
| <b>WHEN</b>   | When in the process/cycle?         | After 2 hours of operation | Cold start or end of   |
| <b>EXTENT</b> | How many units affected?           | 100% - every shift         | Intermittent           |
| <b>EXTENT</b> | Is it trending up/down/stable?     | Stable - same every day    | Getting worse over t   |

### 15.9.2 Key Insights from Is/Is Not Analysis

From the example above, we can identify:

1. **Line 3 specific** - Not a general problem across all lines
2. **Thermal, not mechanical** - Overheating but no noise/vibration yet
3. **After warm-up** - Heat accumulation issue
4. **Started after weekend** - Something changed

**Investigation focus:** What changed over the weekend? Check maintenance logs, any work done on Line 3, any environmental changes.

---

## 15.10 The 8D Problem-Solving Methodology

The **8D (Eight Disciplines)** methodology is a comprehensive problem-solving approach developed by Ford Motor Company. It's widely used in automotive and other industries.

| The 8D Problem–Solving Methodology                                          |                    |                                                        |
|-----------------------------------------------------------------------------|--------------------|--------------------------------------------------------|
| A structured approach for complex problems requiring cross–functional teams |                    |                                                        |
| D0                                                                          | Plan               | Plan the 8D approach; determine if 8D is appropriate   |
| D1                                                                          | Team               | Establish a cross–functional team with required skills |
| D2                                                                          | Problem            | Define and describe the problem in measurable terms    |
| D3                                                                          | Containment        | Implement temporary actions to protect customer        |
| D4                                                                          | Root Cause         | Identify and verify root cause(s) using RCA tools      |
| D5                                                                          | Corrective Actions | Develop and verify permanent corrective actions        |
| D6                                                                          | Implementation     | Implement and validate permanent corrective actions    |
| D7                                                                          | Prevention         | Prevent recurrence; update systems and procedures      |
| D8                                                                          | Closure            | Recognize team; document learnings; close out          |

15.10.1 Detailed 8D Steps

Table 15.9: 8D Methodology Detailed Breakdown

| Discipline             | Key Activities                                                         | Deliverable                 |
|------------------------|------------------------------------------------------------------------|-----------------------------|
| D0: Plan               | Emergency response actions, initial assessment, determine if 8D needed | 8D initiation               |
| D1: Team               | Select team leader, identify required expertise, define roles          | Team charter                |
| D2: Problem            | 5W2H, Is/Is Not, quantify impact, identify affected parts/lots         | Problem statement           |
| D3: Containment        | Sort suspect material, increase inspection, alert downstream           | Containment actions         |
| D4: Root Cause         | 5 Whys, Fishbone, FTA, verify cause by testing                         | Verified root cause         |
| D5: Corrective Actions | Brainstorm solutions, verify effectiveness before full rollout         | Verified corrective actions |
| D6: Implementation     | Execute action plan, validate results, monitor KPIs                    | Implementation              |
| D7: Prevention         | Update FMEAs, procedures, training, share across organization          | Updated procedures          |
| D8: Closure            | Final documentation, lessons learned, team recognition                 | 8D report, team recognition |

15.10.2 D3: Containment Actions

Containment is crucial to protect the customer while you investigate:

Table 15.10: Types of Containment Actions

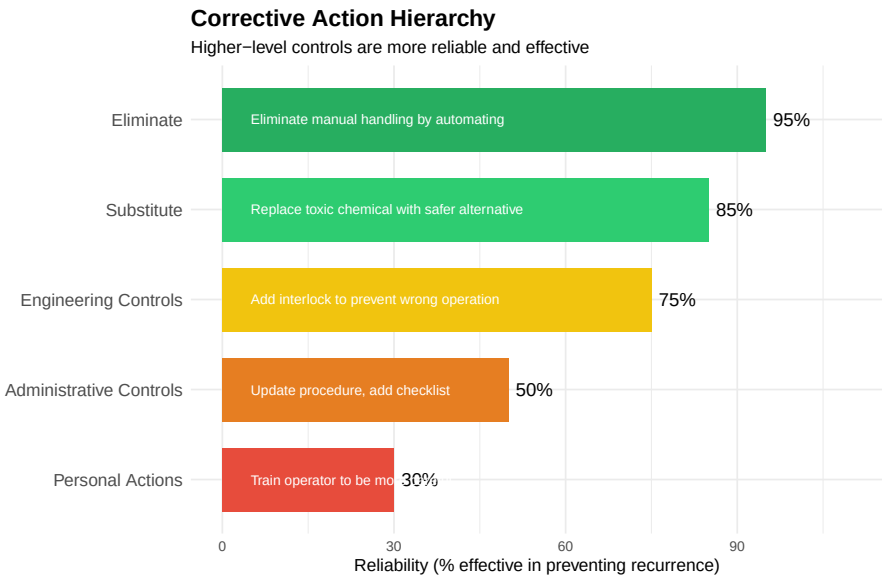
| Type           | Description                                                    | Example                |
|----------------|----------------------------------------------------------------|------------------------|
| Sort & Inspect | 100% inspection of suspect inventory to separate good from bad | Inspect all parts from |

|                              |                                                             |            |
|------------------------------|-------------------------------------------------------------|------------|
| <b>Rework/Repair</b>         | Repair nonconforming units if possible and economical       | Re-machine |
| <b>Scrap</b>                 | Dispose of units that cannot be repaired                    | Dispose    |
| <b>Hold/Quarantine</b>       | Prevent movement of suspect material until disposition      | Tag and    |
| <b>Increase Monitoring</b>   | Add inspections, increase sampling, tighten control limits  | Add 100    |
| <b>Supplier Notification</b> | Notify suppliers if incoming material is suspected          | Request    |
| <b>Customer Alert</b>        | Alert customers to check their inventory if product shipped | Contact    |

## 15.11 Corrective Action Implementation

Effective corrective actions address the root cause and prevent recurrence.

### 15.11.1 Types of Corrective Actions



### 15.11.2 SMART Corrective Actions

Corrective actions should be **SMART**:

Table 15.11: SMART Corrective Action Criteria

|          | Meaning    | Description                             | Poor Example                 | Good Example        |
|----------|------------|-----------------------------------------|------------------------------|---------------------|
| <b>S</b> | Specific   | Clearly defined action, not vague       | Improve quality              | Install process     |
| <b>M</b> | Measurable | Can verify completion and effectiveness | Train operators              | Train all operators |
| <b>A</b> | Achievable | Realistic with available resources      | Redesign entire product line | Add inspection      |

|          |            |                                   |                       |                      |
|----------|------------|-----------------------------------|-----------------------|----------------------|
| <b>R</b> | Relevant   | Directly addresses the root cause | Repaint the floor     | Add check fixture to |
| <b>T</b> | Time-bound | Has a target completion date      | Do this sometime soon | Complete by March 1  |

## 15.12 Verification and Validation

Before closing out a problem, verify that your actions worked.

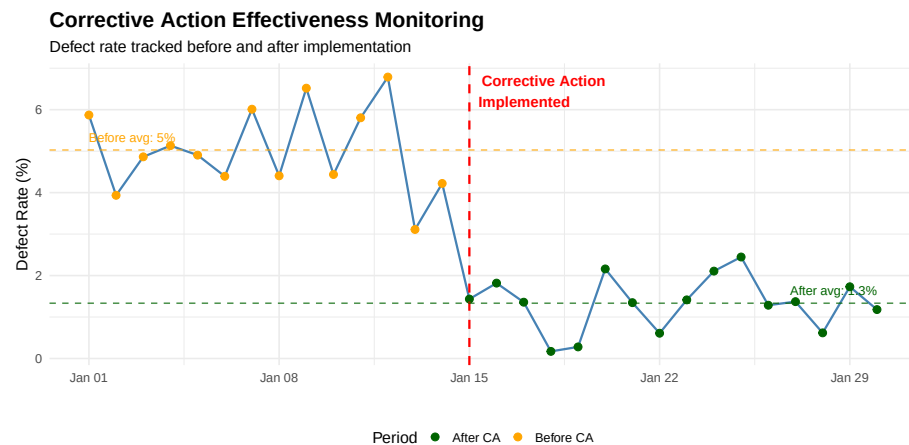
### 15.12.1 Verification vs. Validation

Table 15.12: Verification vs. Validation of Corrective Actions

| Aspect            | Verification                           | Validation                                      |
|-------------------|----------------------------------------|-------------------------------------------------|
| <b>Definition</b> | Did we implement the action correctly? | Did the action solve the problem?               |
| <b>Focus</b>      | Process compliance                     | Results and effectiveness                       |
| <b>Question</b>   | Did we build the fix right?            | Did we build the right fix?                     |
| <b>Timing</b>     | During/after implementation            | After implementation, during monitoring         |
| <b>Methods</b>    | Audits, inspections, checklists        | Data analysis, trend charts, capability studies |

### 15.12.2 Effectiveness Monitoring

Warning in `scale_x_date()`: A <numeric> value was passed to a Date scale.  
 i The value was converted to a <Date> object.



```
# Calculate improvement
before_avg <- mean(defect_rate[1:14])
after_avg <- mean(defect_rate[15:30])
improvement <- (before_avg - after_avg) / before_avg * 100
```

```
cat("Before corrective action: ", round(before_avg, 2), "% defect rate\n")
cat("After corrective action:   ", round(after_avg, 2), "% defect rate\n")
cat("Improvement:              ", round(improvement, 1), "% reduction\n")
```

```
Before corrective action: 5.03 % defect rate
After corrective action:  1.33 % defect rate
Improvement:              73.5 % reduction
```

---

### 15.13 Case Study: Automotive Welding Defects

Let's work through a comprehensive troubleshooting case using multiple RCA tools.

#### 15.13.1 The Problem

A Tier-1 automotive supplier is experiencing weld porosity defects on a critical structural component. Customer complaints have increased, and the plant has received a quality alert.

#### 15.13.2 Initial Problem Definition (5W2H)

Table 15.13: Case Study Problem Definition

| Question        | Details                                                          |
|-----------------|------------------------------------------------------------------|
| <b>WHAT</b>     | Porosity (gas pockets) in MIG welds on crossmember subassembly   |
| <b>WHERE</b>    | Robot welding Cell 4, occurring at weld joint W-23 (corner weld) |
| <b>WHEN</b>     | Started approximately 2 weeks ago, occurs on all shifts          |
| <b>WHO</b>      | Detected by customer during their incoming inspection            |
| <b>WHY</b>      | Customer rejecting parts; risk of production line stop           |
| <b>HOW</b>      | Visible voids on X-ray inspection; some visible to naked eye     |
| <b>HOW MUCH</b> | Estimated 8% of parts affected (up from baseline of 0.5%)        |

#### 15.13.3 5 Whys Analysis

Table 15.14: 5 Whys Analysis: Weld Porosity

| Level          | Finding                              |
|----------------|--------------------------------------|
| <b>Problem</b> | Porosity in welds at joint W-23      |
| <b>Why #1</b>  | Gas entrapment during solidification |
| <b>Why #2</b>  | Inadequate shielding gas coverage    |

**Why #3** Gas flow rate dropped below specification

**Why #4** Flow meter was not calibrated; showing higher than actual

**Why #5** No calibration schedule existed for welding gas flow meters

#### 15.13.4 Fishbone Diagram Findings

After brainstorming with the cross-functional team:

Table 15.15: Fishbone Analysis Results with Verification

| Category           | Potential Cause                           | Likelihood | Verification Result      |
|--------------------|-------------------------------------------|------------|--------------------------|
| <b>Machine</b>     | Gas flow meter out of calibration         | HIGH       | YES - Root cause         |
| <b>Machine</b>     | Worn contact tip affecting arc stability  | Medium     | No - tips OK             |
| <b>Material</b>    | Different wire lot from new supplier      | Medium     | No - tested OK           |
| <b>Material</b>    | Base metal surface contamination          | Low        | No - clean parts         |
| <b>Method</b>      | Robot path too fast for corner weld       | Medium     | Partial contributor      |
| <b>Method</b>      | Gas pre-flow time insufficient            | HIGH       | YES - Contributing       |
| <b>Man</b>         | Recent operator turnover, less experience | Low        | No - all shifts affected |
| <b>Environment</b> | Seasonal humidity changes                 | Low        | No - controlled          |

#### 15.13.5 8D Report Summary

Table 15.16: 8D Report: Weld Porosity Case Study

| D#        | Discipline                | Summary                                                                               |
|-----------|---------------------------|---------------------------------------------------------------------------------------|
| <b>D1</b> | <b>Team</b>               | Team: Weld Engineer (lead), Quality, Maintenance, Production Supervisor, Supplier     |
| <b>D2</b> | <b>Problem</b>            | 8% porosity rate at W-23 joint since Jan 15; customer quality alert received Jan 28   |
| <b>D3</b> | <b>Containment</b>        | 100% X-ray inspection of all crossmembers; sort and quarantine 3 days production      |
| <b>D4</b> | <b>Root Cause</b>         | Primary: Flow meter out of calibration (reading 25 CFH, actual 18 CFH). Contributor   |
| <b>D5</b> | <b>Corrective Actions</b> | 1. Recalibrate flow meter immediately; 2. Increase pre-flow time to 0.5s; 3. Add flow |
| <b>D6</b> | <b>Implementation</b>     | Actions completed Feb 5; validation testing shows 0.1% porosity rate (below 0.5% t    |
| <b>D7</b> | <b>Prevention</b>         | Updated PM schedule to include all gas flow meters quarterly; added to PFMEA w        |
| <b>D8</b> | <b>Closure</b>            | Customer satisfied with response; 8D closed Feb 15; cost avoidance: \$45,000 (avoided |

### 15.14 Documentation and Knowledge Management

Proper documentation ensures learnings are captured and problems don't recur.

15.14.1 Key Documentation Elements

Table 15.17: RCA Documentation Checklist

| Element                     | Contents                                                         | Benefit  |
|-----------------------------|------------------------------------------------------------------|----------|
| <b>Problem Description</b>  | 5W2H, problem statement, impact/cost                             | Clear re |
| <b>Timeline</b>             | When discovered, key investigation dates, implementation dates   | Underst  |
| <b>Data Collected</b>       | Measurements, photos, test results, operator interviews          | Evidenc  |
| <b>Root Cause Analysis</b>  | 5 Whys, fishbone diagrams, FTA as applicable                     | Docume   |
| <b>Corrective Actions</b>   | Actions taken, responsible parties, target/actual dates          | Account  |
| <b>Verification Results</b> | Before/after data, statistical comparison, effectiveness %       | Proof o  |
| <b>Lessons Learned</b>      | What we'd do differently, recommendations for similar situations | Continu  |
| <b>System Updates</b>       | PFMEA updates, procedure changes, training records               | Prevent  |

15.14.2 Creating a Troubleshooting Knowledge Base

Build organizational memory by:

1. **Standardized templates** - Use consistent formats for all RCA documents
2. **Searchable database** - Store in CMMS, SharePoint, or dedicated quality system
3. **Categories and tags** - Equipment type, failure mode, root cause category
4. **Cross-references** - Link related issues and solutions
5. **Regular reviews** - Periodic analysis of trends and patterns

15.15 Troubleshooting Tools Quick Reference

Table 15.18: RCA Tool Selection Guide

| Tool                       | Best For                                          | When to Use               |
|----------------------------|---------------------------------------------------|---------------------------|
| <b>5 Whys</b>              | Simple problems with linear cause chains          | Quick investigations, sta |
| <b>Fishbone Diagram</b>    | Brainstorming multiple potential causes           | Team problem-solving s    |
| <b>Fault Tree Analysis</b> | Complex systems with multiple failure paths       | Safety-critical systems,  |
| <b>Pareto Chart</b>        | Prioritizing which problems to tackle first       | When facing multiple p    |
| <b>Is/Is Not</b>           | Narrowing down scope and identifying distinctions | When problem is hard t    |
| <b>8D</b>                  | Customer complaints requiring formal response     | Formal quality requirem   |
| <b>Change Analysis</b>     | Problems that started suddenly                    | When timing of problem    |



## 15.16 Video Resources

### 15.16.1 Understanding 5 Whys

### 15.16.2 8D Problem Solving

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## 15.17 Summary

Effective troubleshooting and root cause analysis require:

1. **Systematic approach** - Don't guess; follow a structured methodology
2. **Clear problem definition** - Use 5W2H to clearly define what you're solving
3. **Data-driven analysis** - Collect facts before jumping to conclusions
4. **Root cause focus** - Don't stop at symptoms; dig to fundamental causes
5. **Multiple tools** - Use 5 Whys, Fishbone, FTA, Pareto as appropriate
6. **Effective corrective actions** - Aim for elimination, not just mitigation
7. **Verification** - Confirm your fix actually works with data
8. **Documentation** - Capture learnings to prevent recurrence

Remember: The goal isn't just to fix today's problem - it's to prevent tomorrow's.

---

## 15.18 Review Questions

**Question 1:** What is the difference between a symptom, immediate cause, and root cause? Provide an example of each for a machine that is producing parts with excessive surface roughness.

**Answer:**

- **Symptom:** The observable effect of the problem
  - *Example: Parts have Ra surface roughness of 3.2  $\mu$ m when specification is 1.6  $\mu$ m*
- **Immediate cause:** The direct, technical cause of the symptom
  - *Example: Cutting tool is worn beyond its effective life*
- **Root cause:** The fundamental reason that, if eliminated, prevents recurrence
  - *Example: No tool life monitoring system exists; operators judge tool changes by intuition rather than data*

Fixing the root cause (implementing tool life tracking) prevents the problem from recurring, whereas just replacing the tool (addressing immediate cause) means it will happen again.

**Question 2:** A food processing plant is experiencing frequent jams on their packaging line. Perform a 5 Whys analysis given the following information: The jam occurs at the carton loading station, started last week, happens 4-5 times per shift, and the maintenance team found the timing of the pusher mechanism is off.

**Answer:**

1. **Problem:** Carton loading station jams 4-5 times per shift
2. **Why #1:** Cartons are not in position when pusher activates → Timing mismatch
3. **Why #2:** Pusher timing is off by approximately 0.3 seconds
4. **Why #3:** Timing adjustment was changed during last week's maintenance
5. **Why #4:** Technician didn't have correct specification for timing setting
6. **Why #5:** Maintenance procedure doesn't include timing specifications

**Root Cause:** Maintenance procedures lack critical parameter specifications

**Corrective Actions:** - Update maintenance procedure with correct timing specification ( $180\text{ms} \pm 10\text{ms}$ ) - Restore timing to correct setting immediately - Review other procedures for missing specifications - Add parameter verification checklist to PM activities

**Question 3:** Create a fishbone diagram outline (list the potential causes under each 6M category) for the problem "CNC machine producing parts 0.05mm oversize."

**Answer:**

**MAN (People)** - Operator entered wrong offset - Incorrect setup procedure followed - Inadequate training on new program - Fatigue/distraction during setup

**MACHINE (Equipment)** - Spindle thermal growth - Ballscrew wear/backlash - Tool holder runout - Axis positioning error

**MATERIAL** - Material hardness variation - Different lot from supplier - Material springback - Incorrect material grade

**METHOD (Process)** - Wrong tool offset entered - Program error in G-code - Incorrect cutting parameters - Missing roughing pass

**MEASUREMENT** - Gauge out of calibration - Wrong measurement technique - Temperature affecting measurement - Incorrect datum reference

**MOTHER NATURE (Environment)** - Shop temperature variation - Coolant temperature change - Vibration from nearby equipment - Humidity affecting gauges

**Question 4:** Given the following defect data from a sheet metal stamping operation, create a Pareto analysis and identify the vital few defects to focus

on.

| Defect Type      | Count |
|------------------|-------|
| Scratches        | 89    |
| Burrs            | 156   |
| Wrinkles         | 45    |
| Splits/Cracks    | 28    |
| Dimensional      | 112   |
| Surface Dents    | 67    |
| Missing Features | 18    |

**Answer:** (live — try your own defect counts)

```
#| label: pareto-practice
# Create and analyze the data
defects <- data.frame(
  Type = c("Burrs", "Dimensional", "Scratches", "Surface Dents",
           "Wrinkles", "Splits/Cracks", "Missing Features"),
  Count = c(156, 112, 89, 67, 45, 28, 18)
)

# Sort by count descending
defects <- defects[order(-defects$Count),]

# Calculate percentages
defects$Percent <- defects$Count / sum(defects$Count) * 100
defects$Cumulative <- cumsum(defects$Percent)

# Display results
print(defects)

# Identify vital few
vital_few <- defects$Type[defects$Cumulative <= 80 |
                          c(TRUE, defects$Cumulative[-nrow(defects)] < 80)]
cat("\nVital Few (contributing to ~80%):", paste(vital_few, collapse = ", "))
```

The **vital few** are Burrs, Dimensional errors, and Scratches - addressing these three defect types would address approximately 69% of all defects. Adding Surface Dents would bring the total to 82%.

**Question 5:** What are the key differences between verification and validation in the context of corrective actions?

**Answer:**

| Aspect         | Verification                                       | Validation                                            |
|----------------|----------------------------------------------------|-------------------------------------------------------|
| Question asked | Did we implement the action correctly?             | Did the action solve the problem?                     |
| Focus          | Process compliance                                 | Results/outcomes                                      |
| Timing         | During/immediately after implementation            | Ongoing monitoring after implementation               |
| Methods        | Audits, checklists, inspections                    | Data analysis, trend charts, SPC                      |
| Example        | Confirm new sensor was installed per specification | Measure defect rate over 30 days to confirm reduction |

**Verification** confirms the action was done right. **Validation** confirms we did the right action.

Both are required - an action can be implemented perfectly (verified) but still not solve the problem (not validated).

**Question 6:** An aerospace manufacturer has a requirement to use the 8D methodology for customer complaints. What happens in D3 (Containment) and why is it critical?

**Answer:**

**D3 Containment Actions** protect the customer while the root cause investigation continues. Key activities include:

1. **Immediate actions:**

- Quarantine suspect inventory (tagged, segregated)
- 100% inspection of suspect lots
- Increase monitoring at subsequent operations
- Sort parts at customer location if already shipped

2. **Documentation:**

- Identify affected lot numbers, date codes, serial numbers
- Quantify the scope of potential exposure
- Document all containment actions taken

3. **Communication:**

- Notify customer of actions being taken
- Alert downstream operations
- Inform suppliers if incoming material involved

**Why critical:** - Prevents additional defective product from reaching the customer - Limits the scope of the problem (cost, liability) - Demonstrates responsiveness to customer - Buys time for proper root cause investigation - In aerospace, safety-of-flight concerns make containment essential

Without effective containment, the problem continues while you investigate, increasing customer impact and costs exponentially.

**Question 7:** Why might a team's 5 Whys analysis stop too early? What are signs that you haven't reached the true root cause?

**Answer:**

**Reasons teams stop too early:** 1. **Blame arrives** - Analysis ends at "operator error" without asking why the error occurred 2. **Comfort zone** - Team stops when they reach a cause they can easily fix 3. **Time pressure** - Rush to implement a fix, skipping deeper analysis 4. **Technical limit** - Team lacks expertise to go deeper into certain areas 5. **Defensiveness** - Reaching causes that implicate management or systems creates resistance

**Signs you haven't reached root cause:** 1. **Recurrence** - If the fix is "more training" or "be more careful," problem will return 2. **Human-based answer** - Root causes should typically be systems/processes, not individuals 3. **No prevention mechanism** - If fixing this cause doesn't prevent future occurrences, keep asking why 4. **Can still ask "why?"** - If there's a logical next level to investigate 5. **Administrative fix only** - If the solution is just a procedure or checklist with no engineering controls

**Example of stopping too early:** - Stopping at: "Operator set wrong parameter" - Should continue: Why? → No parameter sheet at workstation → Why? → No standard work documentation exists → Root cause: Missing standardized work process

The true root cause typically relates to systems, processes, or management decisions rather than individual actions.

**Question 8:** Describe how you would use Pareto analysis in a stratified manner to drill down into a quality problem.

**Answer:**

**Stratified Pareto Analysis** involves creating multiple levels of Pareto charts, progressively drilling down into the vital few categories.

**Example: High scrap rate at a machining shop**

**Level 1: Scrap by defect type** 1. Create Pareto of all defect types 2. Result: "Dimensional errors" is 45% of all scrap - investigate this first

**Level 2: Dimensional errors by machine** 1. Create Pareto of dimensional errors broken down by machine 2. Result: Machine #4 accounts for 60% of dimensional errors

**Level 3: Machine #4 dimensional errors by feature** 1. Create Pareto of which dimensions are failing on Machine #4 2. Result: Bore diameter is 70% of Machine #4 dimensional issues

**Level 4: Bore diameter issues by shift/operator** 1. Create Pareto by shift 2. Result: Evenly distributed - not operator-dependent (machine issue confirmed)

**Final Analysis:** Investigate boring operation on Machine #4 specifically - likely tool wear, spindle issue, or program error.

The stratified approach efficiently narrows focus from “we have a scrap problem” to “boring operation on Machine #4 has a systematic error.”

**Question 9:** A defense contractor needs to analyze a safety-critical failure. Why might Fault Tree Analysis (FTA) be preferred over simpler methods like 5 Whys?

**Answer:**

**FTA advantages for safety-critical analysis:**

1. **Handles complexity:**
  - Safety-critical systems often have multiple redundancies and complex failure paths
  - FTA can model combinations of events that must occur together (AND gates) or alternatively (OR gates)
  - 5 Whys follows single linear path; FTA captures parallel paths
2. **Quantitative capability:**
  - FTA allows probability calculations for top event
  - Can identify minimum cut sets (smallest combinations that cause failure)
  - Supports reliability and safety integrity level (SIL) calculations
  - Defense/aerospace require quantitative risk assessment
3. **Documentation requirements:**
  - Regulatory bodies (FAA, DoD) often require FTA for critical systems
  - Provides auditable, traceable analysis
  - Standard symbols and methodology
4. **Identifies hidden failures:**
  - Can reveal single points of failure
  - Shows where redundancy is (or isn't) effective
  - Identifies common cause failures that defeat redundancy
5. **Design improvement:**
  - Visual representation helps identify where to add safeguards
  - Sensitivity analysis shows which basic events most affect risk
  - Supports design reviews and safety cases

**Example:** A missile guidance system failure - 5 Whys might identify one cause path - FTA would show all possible failure paths, calculate overall failure probability, and identify that two specific sensor failures AND a software error all occurring together could cause the top event, even though each alone is unlikely.

---

## 15.19 References

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## Chapter 16

# Simulation Modeling with AnyLogic

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### 16.1 Learning Objectives

After completing this chapter, you will be able to:

1. Explain what simulation modeling is and why it is essential for manufacturing system design
  2. Describe the three simulation paradigms available in AnyLogic (discrete-event, agent-based, and system dynamics)
  3. Identify the key components and libraries in AnyLogic for manufacturing simulation
  4. Build a basic production line simulation using the Process Modeling Library
  5. Model warehouse picking and order fulfillment operations
  6. Analyze simulation results to optimize throughput, reduce bottlenecks, and improve efficiency
  7. Apply simulation to validate design decisions before physical implementation
  8. Use built-in AnyLogic examples to accelerate model development
- 

### 16.2 Introduction to Simulation Modeling

**Before You Read:** Imagine you are designing a new automotive assembly line that will cost \$50 million to build. How would you test whether your design

will meet production targets *before* spending any money on construction?

In manufacturing, making the wrong design decision can be incredibly expensive. A misplaced workstation, an undersized conveyor, or an incorrect staffing level can result in millions of dollars in lost productivity. **Simulation modeling** allows engineers to build virtual representations of manufacturing systems and test them under various conditions—all without spending a single dollar on physical equipment.

### 16.2.1 What is Simulation?

**Simulation** is the process of creating a computer model that mimics the behavior of a real-world system over time. Unlike static calculations or spreadsheet analysis, simulation captures the *dynamic* nature of manufacturing systems:

- Products arrive at random intervals
- Machines break down unexpectedly
- Workers take breaks
- Quality defects occur randomly
- Rush orders disrupt schedules

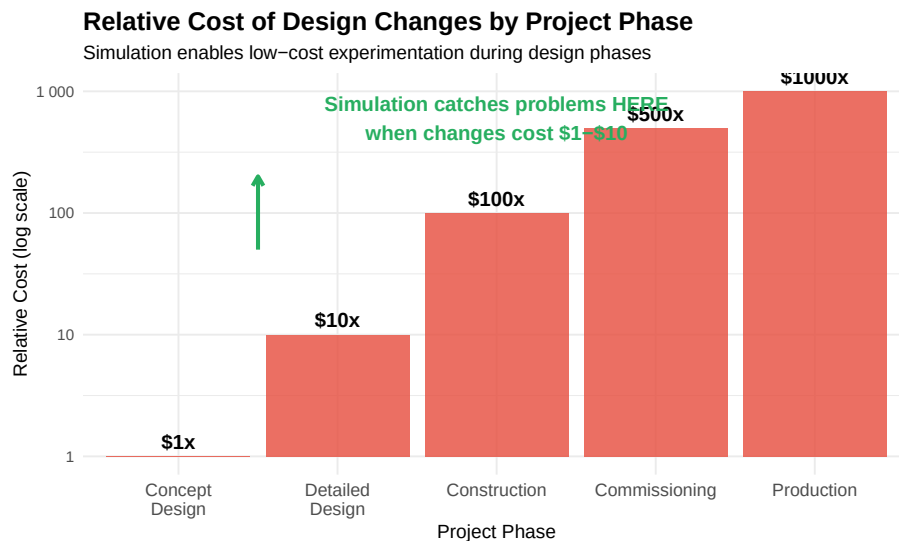


Figure 16.1: The value of simulation in reducing project risk

**Industry Application:** Mercedes-Benz engineers used AnyLogic simulation to optimize their Sprinter van production line, achieving a **5% efficiency gain** by identifying and eliminating bottlenecks before any physical changes were made. This saved millions in potential rework costs.

### 16.2.2 Why AnyLogic?

**AnyLogic** is one of the world’s leading simulation platforms, used by companies including Tesla, BMW, Amazon, Ford, and General Electric. It stands out from other simulation software for several key reasons:

Table 16.1: AnyLogic compared to typical simulation software

| Feature                 | AnyLogic                  | Other Software         |
|-------------------------|---------------------------|------------------------|
| Multi-method modeling   | Yes - all three paradigms | Usually single method  |
| Manufacturing libraries | Material Handling Library | Varies                 |
| 3D visualization        | Built-in 3D               | Often requires plugins |
| Cloud-based simulation  | AnyLogic Cloud            | Limited                |
| Academic version        | Free for students         | Varies                 |
| Agent-based modeling    | Industry-leading          | Limited support        |

## 16.3 The Three Simulation Paradigms

AnyLogic is unique because it supports **three different simulation approaches** within a single platform. Understanding when to use each paradigm is essential for effective modeling.

### 16.3.1 Discrete-Event Simulation (DES)

**Discrete-event simulation** models systems as a sequence of events that occur at specific points in time. Between events, nothing changes. This is the most common approach for manufacturing simulation.

**Key Concept: Discrete-Event Simulation** In DES, entities (products, orders, customers) flow through a series of processes (workstations, queues, resources). The simulation “jumps” from event to event, making it computationally efficient.

**Best for:** - Assembly lines and production systems - Warehouse operations and logistics - Service processes and queuing systems - Call centers and order processing

### 16.3.2 Agent-Based Modeling (ABM)

**Agent-based modeling** simulates systems by defining individual agents (workers, robots, vehicles) that have their own behaviors, goals, and decision rules. Agents interact with each other and their environment, and complex system behavior *emerges* from these interactions.

## Discrete-Event Simulation: Entities Flow Through Processes

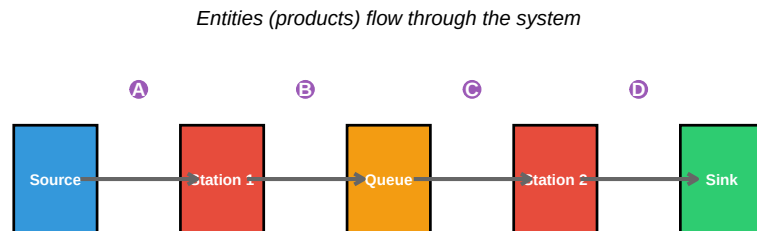


Figure 16.2: Discrete-event simulation flow diagram

**Best for:** - Autonomous mobile robots (AMRs) and AGVs - Pedestrian and crowd simulation - Market and consumer behavior - Complex adaptive systems

**Industry Application:** Tesla uses agent-based simulation in AnyLogic to optimize the deployment of autonomous mobile robots in their manufacturing facilities. Each robot is modeled as an agent with its own task allocation logic, collision avoidance, and charging behavior.

### 16.3.3 System Dynamics (SD)

**System dynamics** models systems using stocks, flows, and feedback loops. It focuses on aggregate behavior over time rather than individual entities.

**Best for:** - Strategic planning and capacity modeling - Supply chain dynamics - Policy analysis - Long-term trends and forecasting

### 16.3.4 Choosing the Right Paradigm

Warning: The ``label.size`` argument of ``geom_label()`` is deprecated as of ggplot2 3.5.0. Please use the ``linewidth`` argument instead.

**Quick Check:** A company wants to simulate their new automated warehouse with 50 mobile robots that must navigate, avoid collisions, and coordinate pick-ups. Which paradigm would you recommend and why?

## 16.4 The AnyLogic Interface and Libraries

### 16.4.1 The AnyLogic Workspace

When you open AnyLogic, you'll see several key areas:

## Selecting the Appropriate Simulation Paradigm

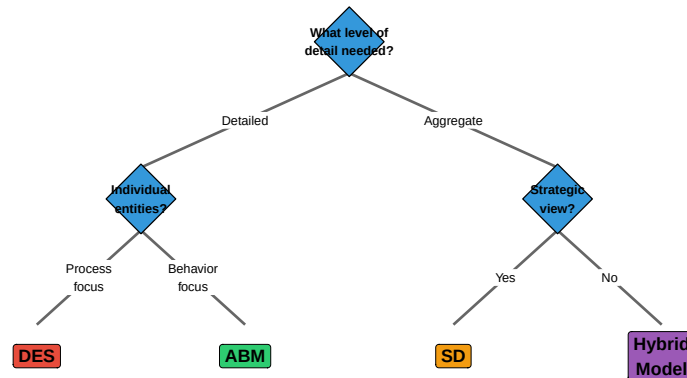


Figure 16.3: Decision framework for selecting simulation paradigm

1. **Model Browser** (left panel) - Shows all elements in your model
2. **Graphical Editor** (center) - Where you build your model visually
3. **Palette** (right panel) - Contains libraries and elements you can drag into your model
4. **Properties** (bottom) - Configure selected elements

## 16.4.2 Essential Libraries for Manufacturing

AnyLogic includes specialized libraries that dramatically accelerate model development:

Table 16.2: Key AnyLogic libraries for manufacturing simulation

| Library                          | Primary Use                              | Key Elements                          |
|----------------------------------|------------------------------------------|---------------------------------------|
| <b>Process Modeling Library</b>  | Discrete-event simulation of any process | Source, Queue, Delay, Service, Sink   |
| <b>Material Handling Library</b> | Conveyors, cranes, AGVs, forklifts       | Conveyor, Station, Transporter, Crane |
| <b>Pedestrian Library</b>        | Worker movement and ergonomics           | PedSource, PedService, PedSink        |
| <b>Road Traffic Library</b>      | Truck logistics and yard management      | CarSource, Road, Intersection         |
| <b>Rail Library</b>              | Rail-based transport systems             | RailNetwork, Train, RailYard          |
| <b>Fluid Library</b>             | Bulk material and liquid processing      | Tank, Pipe, Valve, FlowRate           |

### 16.4.3 The Process Modeling Library

The **Process Modeling Library** is the foundation for most manufacturing simulations. It provides building blocks for creating flowchart-based models:

#### Process Modeling Library Elements

Drag and connect these elements to build your simulation



Figure 16.4: Key Process Modeling Library elements

**Hands-On Activity:** Open AnyLogic and create a new model. Drag a Source, Queue, Delay, and Sink from the Process Modeling Library. Connect them in sequence. Run the simulation and observe entities flowing through.

## 16.5 Production Line Simulation

Now let's apply these concepts to design and analyze an automotive production line. We'll walk through a complete example using AnyLogic's built-in examples as a starting point.

### 16.5.1 Case Study: Automotive Door Panel Assembly Line

**The Scenario:** You are designing a door panel assembly line for an automotive supplier. The line must:

- Process **480 door panels per 8-hour shift** (1 panel per minute)
- Include 5 workstations: stamping, welding, painting, assembly, and inspection

- Meet quality target of **< 2% defect rate**
- Maintain **85% overall equipment effectiveness (OEE)**

Your task is to simulate this line to verify it meets these requirements before construction begins.

### 16.5.2 Step 1: Calculate Required Takt Time

Before building the simulation, we need to calculate the **takt time**—the maximum time allowed per unit to meet demand.

$$Takt\ Time = \frac{Available\ Production\ Time}{Customer\ Demand} \quad (16.1)$$

For our example:

$$Takt\ Time = \frac{8\ hours \times 60\ min/hour}{480\ panels} = 1\ minute/panel$$

This means each workstation must complete its operation in **less than 1 minute** to avoid becoming a bottleneck.

### 16.5.3 Step 2: Define Process Parameters

Table 16.3: Process parameters for door panel assembly line

| Station           | Mean Time (sec) | Std Dev (sec) | Defect Rate (%) | Availability (%) |
|-------------------|-----------------|---------------|-----------------|------------------|
| <b>Stamping</b>   | 45              | 5             | 0.5             | 95               |
| <b>Welding</b>    | 52              | 8             | 0.8             | 92               |
| <b>Painting</b>   | 48              | 6             | 0.6             | 90               |
| <b>Assembly</b>   | 55              | 7             | 0.7             | 94               |
| <b>Inspection</b> | 30              | 4             | 0.0             | 98               |

### 16.5.4 Step 3: Build the Simulation Model

In AnyLogic, we would create this model by:

1. Dragging a **Source** block to generate door panels at the required rate
2. Adding **Service** blocks for each workstation (with appropriate process times)
3. Connecting **Queue** blocks between stations to handle waiting
4. Adding **SelectOutput** blocks after each station to route defective parts
5. Placing a **Sink** at the end to collect finished panels

Production Line Model Structure in AnyLogic

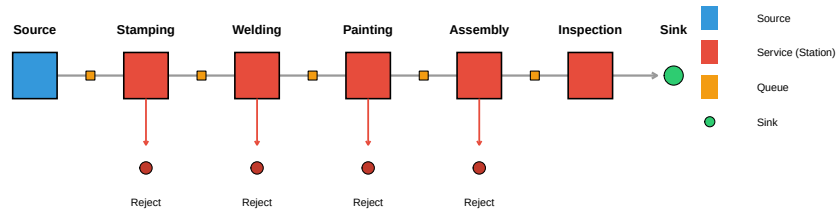


Figure 16.5: AnyLogic production line model structure

### 16.5.5 Step 4: Simulate and Analyze Results

Let's simulate what the output might look like when we run this model:

```
# Simulate production line results
set.seed(123)

# Generate hourly production data
hours <- 1:8
good_parts <- c(56, 58, 55, 57, 59, 54, 58, 57)
defects <- c(1, 2, 2, 1, 1, 3, 1, 2)
target <- rep(60, 8)

production_df <- data.frame(
  Hour = hours,
  Good = good_parts,
  Defects = defects,
  Target = target
)

# Generate queue length data over time
time_points <- seq(0, 480, by = 5)
queue_data <- data.frame(
  Time = rep(time_points, 5),
  Queue = rep(c("Q1", "Q2", "Q3", "Q4", "Q5"), each = length(time_points)),
  Length = c(
    pmax(0, rnorm(length(time_points), 2, 1)), # Q1
    pmax(0, rnorm(length(time_points), 5, 2)), # Q2 - bottleneck
    pmax(0, rnorm(length(time_points), 3, 1.5)), # Q3
    pmax(0, rnorm(length(time_points), 4, 1.8)), # Q4
    pmax(0, rnorm(length(time_points), 1, 0.5)) # Q5
  )
)
```



```

# Create multi-panel plot
p1 <- ggplot(production_df, aes(x = Hour)) +
  geom_col(aes(y = Good), fill = "#2ECC71", alpha = 0.8) +
  geom_col(aes(y = -Defects), fill = "#E74C3C", alpha = 0.8) +
  geom_line(aes(y = Target), color = "#3498DB", linewidth = 1.2, linetype = "dashed") +
  geom_hline(yintercept = 0, color = "black") +
  annotate("text", x = 8.5, y = 60, label = "Target", color = "#3498DB", fontface = "bold") +
  labs(
    title = "Hourly Production Output",
    subtitle = "Green = Good parts, Red = Defects, Dashed = Target",
    x = "Hour",
    y = "Units"
  ) +
  theme_minimal()

p2 <- ggplot(queue_data, aes(x = Time, y = Length, color = Queue)) +
  geom_line(alpha = 0.7) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "red") +
  annotate("text", x = 450, y = 6, label = "Warning\nThreshold",
    color = "red", size = 3 * diagram_text_scale) +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Queue Lengths Over Shift",
    subtitle = "Q2 (after Stamping) shows potential bottleneck",
    x = "Time (minutes)",
    y = "Queue Length"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")

gridExtra::grid.arrange(p1, p2, nrow = 2)

```

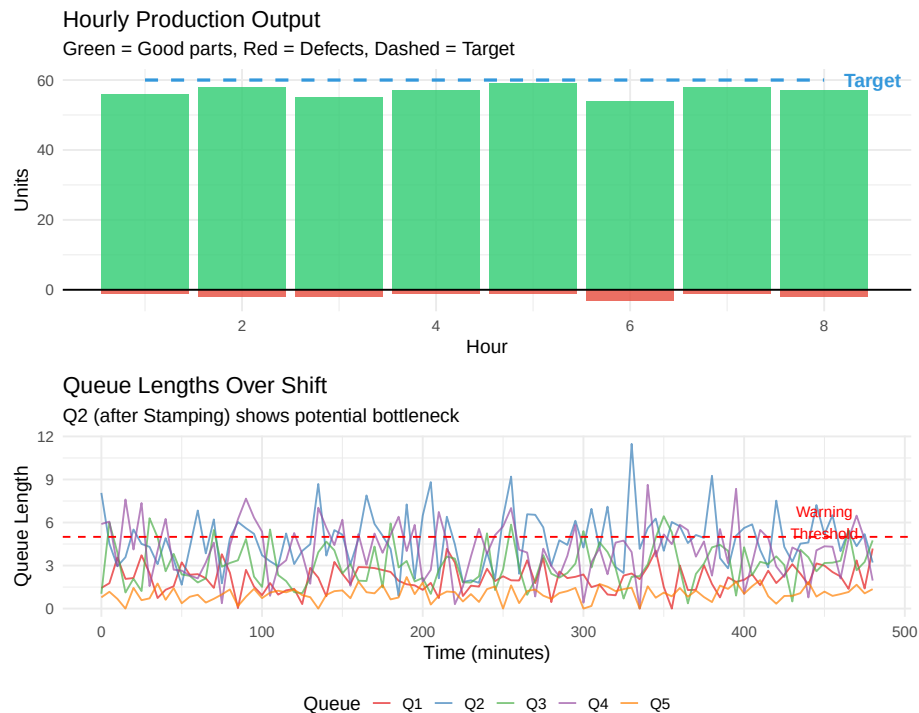


Figure 16.6: Simulated production line performance over 8-hour shift

### 16.5.6 Interpreting the Results

From our simulation, we can observe:

1. **Production Target:** The line produces approximately 454 good parts per shift ( $454/480 = 94.6\%$  of target)
2. **Defect Rate:** 13 defects / 467 total = 2.8% (exceeds our 2% target)
3. **Bottleneck Identified:** Queue Q2 (after Stamping) consistently has the longest queue, indicating the **Welding station** is the bottleneck

#### Warning

**Critical Finding:** The Welding station with a mean process time of 52 seconds and 8% variability is creating a bottleneck. Even though 52 seconds is under the 60-second takt time, the variability combined with 92% availability causes cumulative delays.

### 16.5.7 Step 5: Test Improvement Scenarios

The power of simulation is testing “what-if” scenarios. Let’s compare different improvement options:

```
# Create scenario comparison data
scenarios <- data.frame(
  Scenario = factor(c("Current State", "Add Buffer", "Reduce Variability",
    "Add Parallel Station", "Combined Approach"),
    levels = c("Current State", "Add Buffer", "Reduce Variability",
      "Add Parallel Station", "Combined Approach")),
  Throughput = c(454, 462, 471, 485, 498),
  Investment = c(0, 15, 50, 200, 180),
  ROI = c(0, 53, 34, 15.5, 24.4)
)

# Throughput comparison
ggplot(scenarios, aes(x = Scenario, y = Throughput, fill = Scenario)) +
  geom_col() +
  geom_hline(yintercept = 480, linetype = "dashed", color = "red", linewidth = 1) +
  geom_text(aes(label = Throughput), vjust = -0.5, fontface = "bold") +
  annotate("text", x = 5, y = 485, label = "Target: 480", color = "red") +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Throughput by Improvement Scenario",
    subtitle = "Simulation results for 8-hour shift",
    x = "",
    y = "Parts per Shift"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 30, hjust = 1)
  )
```

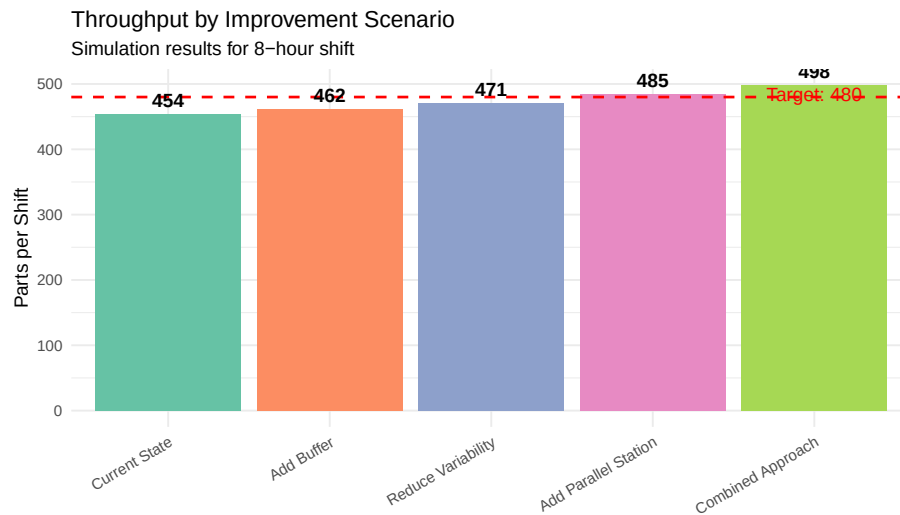


Figure 16.7: Comparison of improvement scenarios

**Group Discussion:** Based on these results, which improvement option would you recommend? Consider not just throughput but also implementation cost, timeline, and risk. Discuss the trade-offs with your team.

## 16.6 Warehouse Simulation: Picking and Order Fulfillment

Modern warehouses face immense pressure to fulfill orders quickly and accurately. Simulation helps optimize picking strategies, staffing levels, and automation investments.

### 16.6.1 The Warehouse Challenge

**Before You Read:** If you had to fill 1,000 customer orders per day from a warehouse with 50,000 different products (SKUs), how would you organize the picking process? What factors would affect efficiency?

Warehouse operations involve complex interactions between:

- **Order arrivals** (often unpredictable)
- **Picker movement** (walking is typically 50% of picker time)
- **Storage locations** (where products are stored affects travel time)
- **Picking strategies** (single-order vs. batch picking)

- **Material handling equipment** (carts, forklifts, conveyor systems)
- **Packing and shipping** (labor-intensive final steps)

### 16.6.2 Case Study: E-Commerce Distribution Center

**The Scenario:** An e-commerce company is designing a new distribution center to handle:

- **13,000 order lines per day** (an order line = one product in one order)
- **750 picking cartons** processed daily
- Peak periods with **3x normal order volume**
- Delivery promise of **same-day shipping** for orders before 2 PM

They need to decide: What picking algorithm will minimize fulfillment time while meeting labor cost targets?

### 16.6.3 Picking Strategies Compared

AnyLogic allows us to simulate and compare different picking strategies:

Table 16.4: Warehouse picking strategies

| Strategy                    | Description                                                | Best For                             |
|-----------------------------|------------------------------------------------------------|--------------------------------------|
| <b>Single-Order Picking</b> | One picker completes one order at a time                   | Small operations, high accuracy      |
| <b>Batch Picking</b>        | Picker collects items for multiple orders simultaneously   | High-volume with clustered orders    |
| <b>Zone Picking</b>         | Warehouse divided into zones; each picker handles one zone | Very large warehouses                |
| <b>Wave Picking</b>         | Orders grouped into waves released at intervals            | Managing workload throughout the day |
| <b>Cluster Picking</b>      | Multiple orders picked into compartmentalized cart         | Medium volume, diverse product range |

### 16.6.4 Simulating Picker Movement

One of the key factors in warehouse efficiency is picker travel distance. AnyLogic can model this using the **Pedestrian Library** combined with the **Process Modeling Library**.

```
# Simulate picker travel data for different storage strategies
set.seed(456)

# Random storage vs. velocity-based storage
n_picks <- 500
random_storage <- rnorm(n_picks, mean = 180, sd = 40) # feet per pick
velocity_storage <- rnorm(n_picks, mean = 95, sd = 25) # feet per pick
abc_storage <- rnorm(n_picks, mean = 120, sd = 30) # feet per pick

travel_data <- data.frame(
  Strategy = rep(c("Random Storage", "ABC Analysis", "Velocity-Based"), each = n_picks),
  Distance = c(random_storage, abc_storage, velocity_storage)
```

```

)

ggplot(travel_data, aes(x = Strategy, y = Distance, fill = Strategy)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = mean(velocity_storage), linetype = "dashed", color = "#2ECC71") +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 4, color = "red") +
  annotate("text", x = 3.35, y = mean(velocity_storage),
           label = paste0("Best: ", round(mean(velocity_storage)), " ft"),
           color = "#2ECC71", size = 4 * diagram_text_scale, fontface = "bold") +
  scale_fill_manual(values = c("#E74C3C", "#F39C12", "#2ECC71")) +
  labs(
    title = "Picker Travel Distance by Storage Strategy",
    subtitle = "Red diamond = mean; Velocity-based storage reduces travel by 47%",
    x = "Storage Assignment Strategy",
    y = "Travel Distance per Pick (feet)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

```

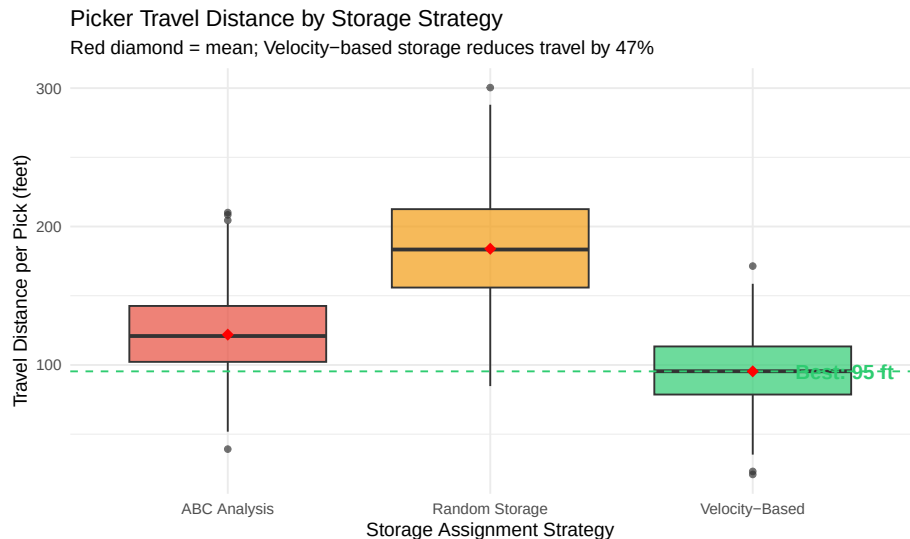


Figure 16.8: Impact of storage assignment on picker travel distance

**Industry Application:** Kuehne+Nagel used AnyLogic simulation to design a new warehouse processing 13,000 order lines per day. By testing different picking algorithms in simulation, they identified the optimal approach before building the physical facility, avoiding costly redesigns.

### 16.6.5 Autonomous Mobile Robots (AMRs) in Warehouses

Modern warehouses increasingly use **autonomous mobile robots** to reduce picker travel time. Instead of workers walking to products, robots bring shelving units to stationary pick stations.

```
# Simulate AMR performance
amr_counts <- c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50)
fulfillment_times <- c(45, 28, 18, 13, 10.5, 9, 8.2, 7.8, 7.6, 7.5)
utilization <- c(98, 95, 88, 78, 68, 58, 52, 48, 45, 43)

amr_data <- data.frame(
  AMRs = amr_counts,
  Time = fulfillment_times,
  Utilization = utilization
)

# Dual y-axis plot
coeff <- 0.5

ggplot(amr_data, aes(x = AMRs)) +
  geom_line(aes(y = Time), color = "#3498DB", linewidth = 1.5) +
  geom_point(aes(y = Time), color = "#3498DB", size = 3) +
  geom_line(aes(y = Utilization * coeff), color = "#E74C3C", linewidth = 1.5, linetype = "dashed") +
  geom_point(aes(y = Utilization * coeff), color = "#E74C3C", size = 3) +
  geom_vline(xintercept = 25, linetype = "dotted", color = "#2ECC71", linewidth = 1.2) +
  annotate("text", x = 27, y = 40, label = "Optimal\nPoint", color = "#2ECC71", fontface = "bold")
  scale_y_continuous(
    name = "Fulfillment Time (minutes)",
    sec.axis = sec_axis(~ . / coeff, name = "Robot Utilization (%)")
  ) +
  labs(
    title = "Finding the Optimal Number of Autonomous Mobile Robots",
    subtitle = "Blue = Fulfillment time (lower is better); Red = Robot utilization (higher is bet",
    x = "Number of AMRs"
  ) +
  theme_minimal() +
  theme(
    axis.title.y.left = element_text(color = "#3498DB"),
    axis.title.y.right = element_text(color = "#E74C3C")
  )
)
```

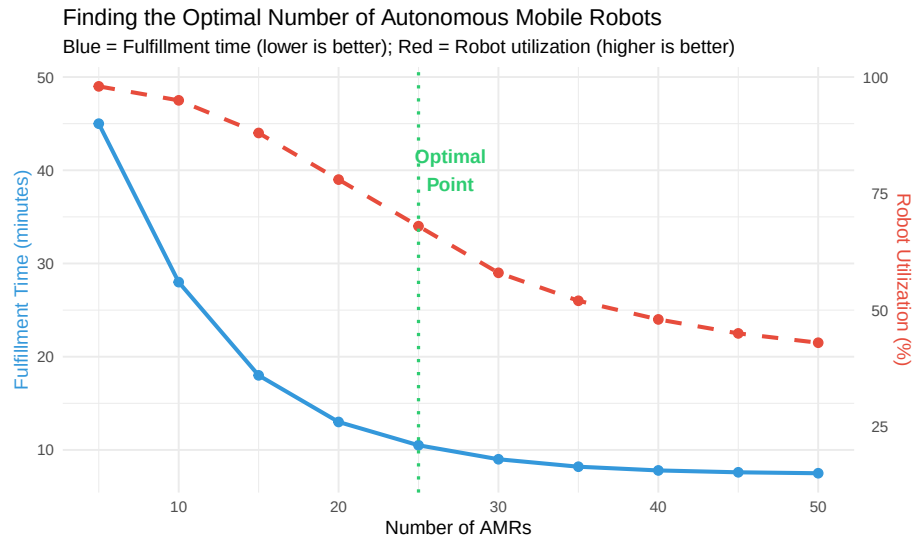


Figure 16.9: Simulation results: order fulfillment time vs. number of AMRs

This simulation reveals a classic trade-off: adding more robots reduces order fulfillment time but also decreases utilization (robots spend more time idle). The **optimal point** balances these factors based on business requirements.

**Quick Check:** 1. Why does adding more robots eventually show diminishing returns? 2. If robots cost \$50,000 each and labor costs \$25/hour, how would you calculate the breakeven point? 3. What other factors should be considered beyond raw cost?

## 16.7 Built-in AnyLogic Examples

AnyLogic includes numerous pre-built example models that serve as excellent learning resources and starting points for your own projects. Here are the key examples relevant to manufacturing and warehouse operations:

### 16.7.1 Manufacturing Examples

Table 16.5: Key manufacturing example models in AnyLogic

| Example Name         | Location                              | What You'll Learn           |
|----------------------|---------------------------------------|-----------------------------|
| <b>Job Shop</b>      | Help > Example Models > Manufacturing | Routing jobs through a shop |
| <b>Assembly Line</b> | Help > Example Models > Manufacturing | Line balancing              |



|                                      |                                           |                             |
|--------------------------------------|-------------------------------------------|-----------------------------|
| <b>Flexible Manufacturing System</b> | Help > Example Models > Manufacturing     | Flexible routing, machine s |
| <b>Conveyor System</b>               | Help > Example Models > Material Handling | Conveyor design, merging,   |
| <b>AGV System</b>                    | Help > Example Models > Material Handling | Vehicle dispatching, path p |

16.7.2 Warehouse and Logistics Examples

Table 16.6: Key warehouse example models in AnyLogic

| Example Name               | Location                                  | What You’ll Learn                         |
|----------------------------|-------------------------------------------|-------------------------------------------|
| <b>Warehouse</b>           | Help > Example Models > Logistics         | Storage assignment, inventory managem     |
| <b>Distribution Center</b> | Help > Example Models > Logistics         | Receiving, put-away, picking, shipping in |
| <b>Order Picking</b>       | Help > Example Models > Logistics         | Picking route optimization, batch vs. sin |
| <b>AS/RS System</b>        | Help > Example Models > Material Handling | Automated storage retrieval, crane sequ   |
| <b>Cross-Docking</b>       | Help > Example Models > Logistics         | Minimal storage, direct transfer operatio |

**Hands-On Activity:** Open AnyLogic and navigate to Help > Example Models. Open the “Assembly Line” example and run it. Then: 1. Identify the bottleneck station (highest queue length) 2. Modify the process time for that station 3. Observe how throughput changes 4. Document your findings

16.8 Building Your First Model: Step-by-Step Tutorial

Let’s walk through building a simple but complete production line model from scratch.

16.8.1 Tutorial: Three-Station Production Line

**Objective:** Model a production line with three workstations and analyze throughput.

16.8.1.1 Step 1: Create a New Model

- 1. Open AnyLogic
- 2. File > New > Model
- 3. Name it “ProductionLine\_Tutorial”
- 4. Select “Discrete Event” as the model type

16.8.1.2 Step 2: Add the Process Flow

From the **Process Modeling Library**, drag these elements onto the canvas in order:

1. **Source** - Generates parts entering the system
2. **Queue** - Holds parts waiting for Station 1
3. **Delay** - Represents Station 1 processing time
4. **Queue** - Holds parts waiting for Station 2
5. **Service** - Represents Station 2 (requires a worker resource)
6. **Queue** - Holds parts waiting for Station 3
7. **Delay** - Represents Station 3 processing time
8. **Sink** - Removes completed parts from the system

### 16.8.1.3 Step 3: Connect the Elements

Click on the output port of each element and drag to the input port of the next element. The flow should look like:

Source → Queue → Delay → Queue → Service → Queue → Delay → Sink

### 16.8.1.4 Step 4: Configure Parameters

Click each element and set these properties:

**Source:** - Arrivals defined by: Inter-arrival time - Inter-arrival time: `exponential(1)` (1 part per minute on average)

**Delays (Stations 1 and 3):** - Delay time: `triangular(0.5, 0.8, 1.2)` (minutes)

**Service (Station 2):** - Delay time: `triangular(0.6, 0.9, 1.5)` (minutes) - Requires a Resource Pool (add one and set capacity to 1)

**Queues:** - Maximum capacity: 10

### 16.8.1.5 Step 5: Add Statistics Collection

Add these statistics to track performance:

1. **TimeMeasureStart** - After Source
2. **TimeMeasureEnd** - Before Sink
3. **Dataset** - To log throughput over time

### 16.8.1.6 Step 6: Run and Analyze

```
# Simulate expected results
set.seed(789)
time <- seq(0, 480, by = 1)

# Generate cumulative throughput
throughput <- cumsum(rpois(length(time), lambda = 0.85))

# Generate queue lengths
```

```

q1 <- pmax(0, rnorm(length(time), 1.5, 0.8))
q2 <- pmax(0, rnorm(length(time), 3.2, 1.2)) # Bottleneck
q3 <- pmax(0, rnorm(length(time), 0.8, 0.5))

tutorial_data <- data.frame(
  Time = time,
  Throughput = throughput,
  Q1 = q1,
  Q2 = q2,
  Q3 = q3
)

p1 <- ggplot(tutorial_data, aes(x = Time, y = Throughput)) +
  geom_line(color = "#2ECC71", linewidth = 1) +
  geom_hline(yintercept = 400, linetype = "dashed", color = "red") +
  annotate("text", x = 50, y = 420, label = "Target: 400", color = "red") +
  labs(title = "Cumulative Throughput", x = "Time (min)", y = "Parts Completed") +
  theme_minimal()

tutorial_long <- tutorial_data %>%
  tidyr::pivot_longer(cols = c(Q1, Q2, Q3), names_to = "Queue", values_to = "Length")

p2 <- ggplot(tutorial_long, aes(x = Time, y = Length, color = Queue)) +
  geom_line(alpha = 0.6) +
  scale_color_manual(values = c("#3498DB", "#E74C3C", "#F39C12"),
                    labels = c("Before Station 1", "Before Station 2", "Before Station 3")) +
  labs(title = "Queue Lengths", x = "Time (min)", y = "Parts Waiting") +
  theme_minimal() +
  theme(legend.position = "bottom")

gridExtra::grid.arrange(p1, p2, ncol = 2)

```

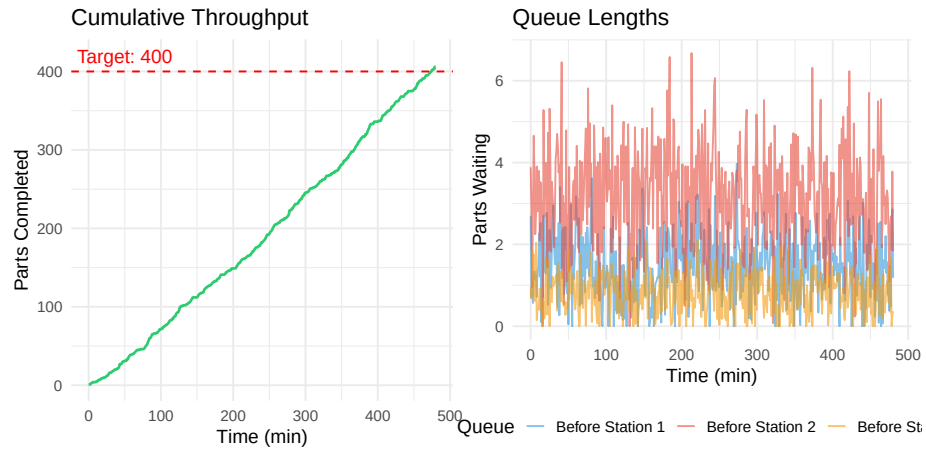


Figure 16.10: Expected results from tutorial model

**Verify Your Model:** After running, you should observe: - Approximately 400-420 parts completed in 480 minutes - Queue 2 (before Station 2) has the longest average length - The system is stable (queues don't grow without bound)

## 16.9 Advanced Topics

### 16.9.1 3D Visualization

AnyLogic includes powerful 3D visualization capabilities that help communicate simulation results to stakeholders who may not understand abstract flowcharts.

Benefits of 3D visualization: - Easier validation with plant managers (“Does this look like our facility?”) - Executive presentations and funding requests - Training material for new employees - Virtual commissioning before physical construction

### 16.9.2 Integration with Real Data

Production simulations become more valuable when connected to real data:

- **Historical data import:** Load actual order data, machine logs, or production records
- **Real-time dashboards:** Connect to databases for live monitoring
- **Digital twin operation:** Mirror physical operations in real-time

### 16.9.3 Optimization Experiments

AnyLogic includes an **Optimization** experiment type that automatically searches for the best parameter values:

```
# Simulate optimization process
iterations <- 1:50
staff_levels <- c(
  rep(5, 5), rep(7, 5), rep(6, 5), rep(8, 5), rep(7, 5),
  rep(6, 10), rep(7, 5), rep(7, 5), rep(7, 5)
)
throughput <- c(
  rnorm(5, 380, 10), rnorm(5, 445, 8), rnorm(5, 420, 9), rnorm(5, 450, 7), rnorm(5, 448, 7),
  rnorm(10, 425, 8), rnorm(5, 450, 6), rnorm(5, 452, 5), rnorm(5, 455, 4)
)

opt_data <- data.frame(
  Iteration = iterations,
  Staff = staff_levels,
  Throughput = throughput
)

opt_data$Best <- cummax(opt_data$Throughput)

ggplot(opt_data, aes(x = Iteration)) +
  geom_point(aes(y = Throughput), alpha = 0.5, color = "#3498DB") +
  geom_line(aes(y = Best), color = "#E74C3C", linewidth = 1.2) +
  geom_hline(yintercept = 455, linetype = "dashed", color = "#2ECC71") +
  annotate("text", x = 45, y = 460, label = "Optimal: 7 staff, 455 parts",
    color = "#2ECC71", fontface = "bold") +
  labs(
    title = "Optimization Experiment: Finding Optimal Staffing",
    subtitle = "Blue points = individual runs; Red line = best found so far",
    x = "Iteration",
    y = "Throughput (parts/shift)"
  ) +
  theme_minimal()
```

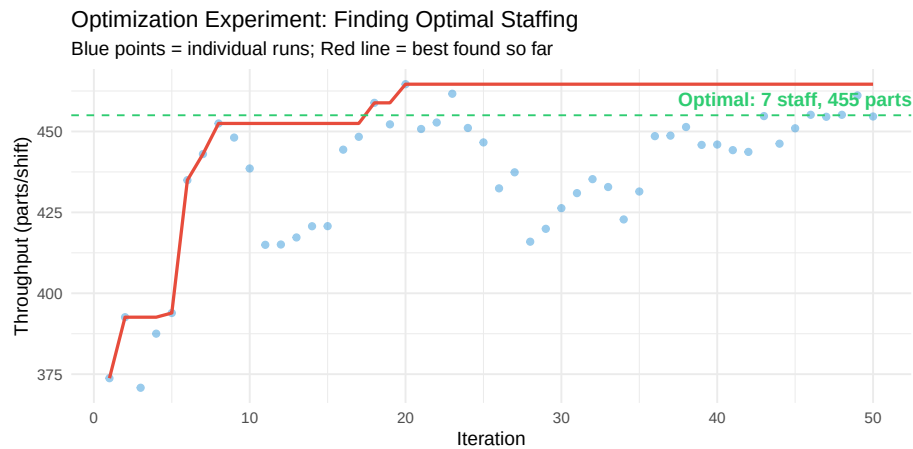


Figure 16.11: Optimization search finding the best staffing level

## 16.10 Key Takeaways

### What We Learned:

1. **Simulation enables risk-free experimentation** - Test design changes in virtual environments before spending money on physical implementation
2. **AnyLogic supports three paradigms** - Discrete-event (for process flows), agent-based (for autonomous entities), and system dynamics (for strategic planning)
3. **Manufacturing simulation identifies bottlenecks** - Queue analysis reveals which stations limit throughput
4. **Warehouse simulation optimizes picking** - Compare strategies like batch picking, zone picking, and AMR deployment before implementation
5. **Built-in examples accelerate learning** - Start from AnyLogic's example models to understand best practices
6. **The value is in the “what-if”** - Simulation's true power is comparing scenarios to find optimal solutions

**Key Equations:** - Takt Time: Equation Equation 16.1

**Before Moving On:** - Can you explain when to use discrete-event vs. agent-based simulation? - Can you build a basic three-station production line in AnyLogic? - Can you identify how simulation could help in your workplace or industry?

---

## 16.11 Practice Problems

### 16.11.1 Problem 1: Takt Time Calculation

A refrigerator assembly plant operates two 8-hour shifts per day with a 30-minute lunch break per shift. Customer demand is 720 units per day.

- Calculate the takt time in seconds
- If the bottleneck station has a cycle time of 62 seconds, will they meet demand? Why or why not?
- Suggest two approaches to resolve the issue

### 16.11.2 Problem 2: Simulation Model Design

You are asked to simulate a paint shop with: - 3 spray booths operating in parallel - Conveyor system bringing parts from preparation area - Curing oven with 30-minute cycle time - Quality inspection with 5% rejection rate

- Sketch the simulation model structure
- List which AnyLogic libraries you would use
- Identify potential bottlenecks to investigate

### 16.11.3 Problem 3: Warehouse Picking Analysis

A warehouse uses single-order picking with 5 pickers. Each picker walks an average of 3 miles per 8-hour shift. The company is considering batch picking with specialized carts.

- If batch picking reduces travel by 35%, how many miles would be saved per picker per shift?
- If workers walk at 3 mph and pick items at a rate of 2 per minute while at a location, what percentage of time is currently spent walking?
- What information would you need to simulate this scenario in AnyLogic?

### 16.11.4 Problem 4: AMR Fleet Sizing

Using the graph in Figure 16.9 as a reference:

- If the company requires fulfillment time under 12 minutes, what is the minimum number of AMRs needed?
- What trade-off occurs when moving from 25 to 50 AMRs?
- If each AMR costs \$50,000 and operates for 5 years, what is the daily cost per AMR?

**Challenge Problem:** Design a simulation experiment to determine the optimal number of pickers and AMRs for a warehouse with variable order volumes.

The solution should: - Handle daily order volumes ranging from 5,000 to 20,000 lines - Meet a 2-hour fulfillment target - Minimize total labor and equipment costs

---

## 16.12 Further Resources

### 16.12.1 AnyLogic Official Resources

- **AnyLogic University:** Free online courses covering all simulation paradigms
- **Example Models:** Help > Example Models in AnyLogic software
- **Cloud Platform:** [cloud.anylogic.com](https://cloud.anylogic.com) for running simulations online

### 16.12.2 Academic Resources

- Borshchev, A. (2013). *The Big Book of Simulation Modeling*. AnyLogic North America.
- Law, A.M. (2015). *Simulation Modeling and Analysis* (5th ed.). McGraw-Hill.

### 16.12.3 Industry Case Studies

- Mercedes-Benz Sprinter production optimization
- Kuehne+Nagel warehouse picking algorithm selection
- Tesla autonomous mobile robot deployment

#### Tip

**Getting Started:** Download the free Personal Learning Edition of AnyLogic from [anylogic.com](https://anylogic.com). It includes all features and example models—the only limitation is model size.



## Chapter 17

# Working Effectively with AI

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### 17.1 Learning Objectives

After completing this chapter, you will be able to:

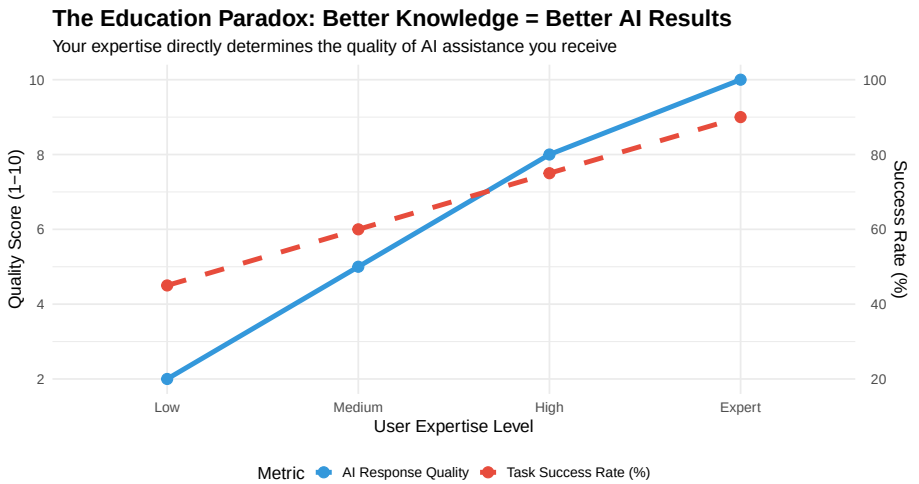
1. Understand AI's role in modern engineering work and its limitations
  2. Apply the four steps of effective AI use: Define, Prompt, Iterate, Validate
  3. Apply prompt engineering principles to get better AI responses
  4. Distinguish between poor, adequate, and excellent prompts
  5. Select appropriate AI tools for different tasks (chat interfaces, NotebookLM, Claude Code, browser extensions)
  6. Use iterative AI collaboration to solve complex engineering problems
  7. Critically evaluate AI-generated content for accuracy and safety
  8. Identify tasks suited for AI assistance versus human expertise
  9. Apply ethical considerations when using AI in engineering contexts
- 

### 17.2 Introduction: AI in the Modern Workplace

Artificial Intelligence has fundamentally changed how engineering work is performed. According to research, AI is predominantly used for **coding and technical work** (34% of professional use), with complex tasks showing potential **12x speedups** compared to traditional methods.

However, there's a critical paradox: **more complex tasks have lower success rates** (61-66% versus 70% for simpler tasks). This means engineers must develop skills to effectively collaborate with AI, not just delegate to it.

17.2.1 The Education Paradox



**Key Insight:** There is a nearly perfect correlation between prompt sophistication and response quality. You need strong foundational knowledge to get quality AI assistance. “AI will do it for me” is a trap.

Discussion: Why does expertise improve AI results?

The “education paradox” reveals a fundamental truth about AI collaboration:

- **Better questions lead to better answers:** Experts know what to ask and how to frame problems
- **Domain knowledge enables validation:** You can only verify AI output if you understand the domain
- **Iterative refinement requires expertise:** Knowing when to push back or ask follow-up questions
- **Critical evaluation is essential:** AI can sound confident while being wrong

**Think about it:** If you don’t understand the fundamentals of motor selection, how would you know if an AI recommendation is correct or dangerously undersized?

17.2.2 AI Impact by Industry

Table 17.1: AI Impact Across Engineering Industries

| Industry        | Common AI Applications                                       | Potential Impact |
|-----------------|--------------------------------------------------------------|------------------|
| Automotive      | Simulation modeling, control systems, predictive maintenance | 10-12x           |
| Food Processing | Process optimization, quality control, equipment diagnostics | 8-10x            |

|                              |                                                                |        |
|------------------------------|----------------------------------------------------------------|--------|
| <b>Defense</b>               | System design, modeling, compliance documentation              | 10-12x |
| <b>Mining</b>                | Equipment diagnostics, predictive maintenance, safety analysis | 8-10x  |
| <b>General Manufacturing</b> | Documentation, troubleshooting guides, process analysis        | 6-8x   |

---

Why do automotive and defense require “Critical” human oversight?

**Safety-critical applications** require mandatory human validation because:

1. **Regulatory requirements:** ISO 26262 (automotive) and MIL-STDs (defense) require documented human review
2. **Liability:** Engineers are legally responsible for designs, not AI tools
3. **Failure consequences:** Errors can cause injury, death, or mission failure
4. **AI limitations:** Current AI cannot reliably reason about edge cases and failure modes
5. **Standards compliance:** AI cannot certify compliance with safety standards

**Best practice:** Use AI for initial research and drafting, but always validate with human experts and formal analysis methods (FMEA, FTA, HAZOP).

---

## 17.3 Prompt Engineering Fundamentals

**Prompt engineering** is the skill of crafting inputs to AI systems that produce useful, accurate, and relevant outputs. In engineering contexts, this skill is becoming as important as technical writing.

Before you read: What makes a good question?

Think about the difference between these two questions:

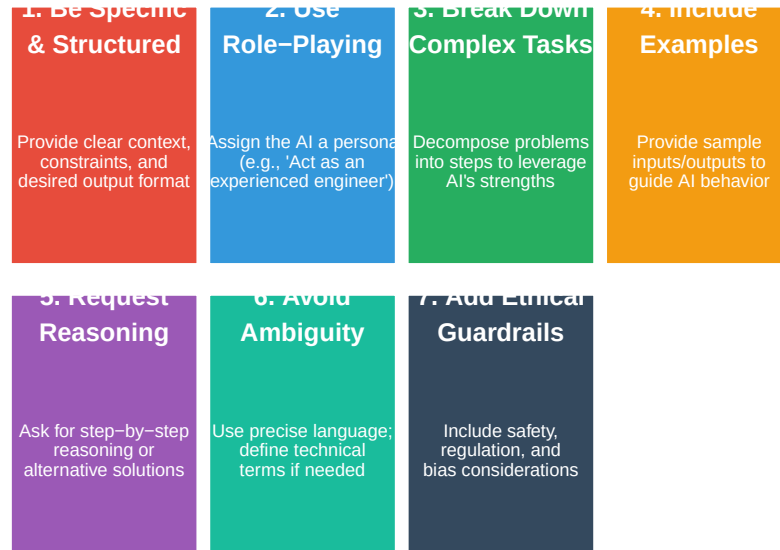
1. “Can you help me with my motor problem?”
2. “I have a 3-phase 460V 50HP motor that trips its overload after 15 minutes of running. Ambient temperature is 35°C. What diagnostic steps should I follow?”

Which question would get better help from a human expert? The same principle applies to AI. Before reading about prompt engineering, consider what information you would need to give a colleague to get useful help.

### 17.3.1 The Seven Rules of Effective Prompts

#### The Seven Rules of Effective Engineering Prompts

Master these principles to dramatically improve AI collaboration



### 17.4 Prompt Quality Comparison: Bad, Medium, and Good

The following examples demonstrate the dramatic difference that prompt quality makes. Each example shows the same task approached three ways.

#### 17.4.1 Example 1: Motor Selection

| Quality Level | Prompt                                                                                |
|---------------|---------------------------------------------------------------------------------------|
| <b>Bad</b>    | Help me pick a motor.                                                                 |
| <b>Medium</b> | I need to select a motor for a conveyor system. The conveyor moves boxes that weigh   |
| <b>Good</b>   | Act as a senior electromechanical engineer. I need to select an AC induction motor fo |

#### 17.4.2 Example 2: Troubleshooting a PLC System

| Quality Level | Prompt                                                                                           |
|---------------|--------------------------------------------------------------------------------------------------|
| <b>Bad</b>    | My PLC isn't working. Fix it.                                                                    |
| <b>Medium</b> | I have an Allen-Bradley PLC that's showing a fault. The output for a motor starter isn't turning |
| <b>Good</b>   | Act as an experienced automation technician troubleshooting a control system issue.\n\nSystem:   |

17.4.3 Example 3: Safety Analysis

| Quality Level | Prompt                                                                                             |
|---------------|----------------------------------------------------------------------------------------------------|
| <b>Bad</b>    | Is this machine safe?                                                                              |
| <b>Medium</b> | I have a robotic welding cell. What safety features does it need?                                  |
| <b>Good</b>   | Act as a certified machine safety engineer (TUV or equivalent).\n\nI need to perform a risk assess |

**Warning:** For safety-critical applications, AI should NEVER be the final authority. Always validate AI-generated safety analyses with certified professionals and applicable standards.

17.5 Anatomy of an Excellent Engineering Prompt

Let's break down the components of a well-structured prompt:

Anatomy of an Excellent Engineering Prompt

Each component serves a specific purpose in guiding AI response quality

|               |                                                                                 |                                              |
|---------------|---------------------------------------------------------------------------------|----------------------------------------------|
| Assignment    | "Act as a senior electromechanical engineer specializing in automotive systems" | Give expert perspective for response         |
| Context       | "I am designing a regenerative braking system for an electric vehicle"          | Establish the problem domain and application |
| Specification | "Vehicle mass: 2000kg, max speed: 150km/h, battery: 400V nominal"               | Provide concrete parameters for calculations |
| Constraint    | "Must comply with ISO 26262 ASIL-B, Budget under \$500/unit"                    | Define boundaries and requirements           |
| Deliverable   | "Provide: 1) Energy calculations, 2) Component selection, 3) Control strategy"  | Specify what you need from the AI            |
| Format        | "Format as a technical design document with diagrams described"                 | Ensure output is immediately usable          |
| Standards     | "Reference ISO 26262 and SAE J2954 where applicable"                            | Anchor response in industry requirements     |

17.6 Interactive Exercise: Rate These Prompts

For each prompt below, identify what’s missing and how you would improve it.

17.6.1 Exercise 1: Pump Selection

Original Prompt: “What pump should I use for water?”

Table 17.5: Analysis: What’s Missing from This Prompt?

| Missing Element           | Why It’s Important                                          |
|---------------------------|-------------------------------------------------------------|
| Flow rate requirement     | Cannot size pump without knowing GPM or L/min needed        |
| Head/pressure requirement | Total dynamic head determines pump curve selection          |
| Fluid properties          | Temperature, viscosity, solids content affect pump type     |
| Operating environment     | Indoor/outdoor, hazardous area, food-grade requirements     |
| Power supply              | Single/three phase, voltage determines motor options        |
| Budget/efficiency goals   | Premium efficiency vs. standard, capital vs. operating cost |

**Your Task:** Rewrite this prompt to include all necessary information for a food processing application pumping tomato paste at 80°C.

17.6.2 Exercise 2: Identify the Quality Level

Rate each prompt as Bad, Medium, or Good and justify your rating:

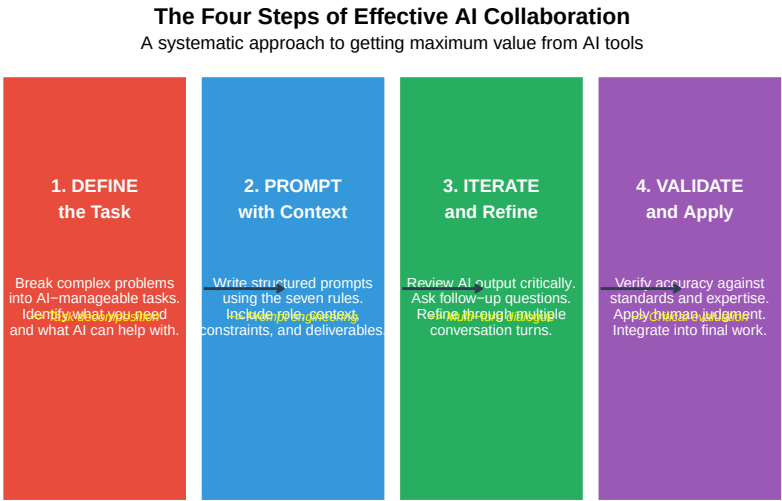
Table 17.6: Exe

| # | Prompt                                                                                                      |
|---|-------------------------------------------------------------------------------------------------------------|
| 1 | How do I wire a motor?                                                                                      |
| 2 | Explain the difference between VFD and soft starter for a 50HP motor application in a cement plant with hig |
| 3 | Write code for Arduino.                                                                                     |
| 4 | Design a sensor network for predictive maintenance on a CNC machine. Include vibration, temperature, and c  |
| 5 | Help with my project.                                                                                       |

17.7 The Four Steps of Effective AI Use

The research shows that effectively working with AI follows a consistent four-step process. Master these steps to maximize the value of AI collaboration.

Warning: Using ``size`` aesthetic for lines was deprecated in ggplot2 3.4.0.  
i Please use ``linewidth`` instead.



17.7.1 Step 1: Define the Task

Before engaging with AI, clearly define what you need:

- **What is the end goal?** A calculation, a design, troubleshooting guidance?

- **What constraints apply?** Budget, standards, materials, timeline?
- **What do you already know?** Your expertise shapes the conversation
- **Can AI actually help with this?** Some tasks require physical inspection or human judgment

**Key Insight:** The better you understand the problem, the better AI can help. Vague understanding leads to vague prompts and useless responses.

### 17.7.2 Step 2: Prompt with Context

Apply the seven rules of effective prompting:

1. Assign a role to establish expertise level
2. Provide complete context and background
3. Specify constraints and requirements
4. List clear deliverables
5. Request step-by-step reasoning
6. Define the output format
7. Include relevant standards

### 17.7.3 Step 3: Iterate and Refine

Rarely is the first response perfect. Plan for 3-6 turns of conversation:

- Review critically: What's useful? What's missing? What's wrong?
- Ask targeted follow-up questions
- Request expansions on specific areas
- Challenge assumptions and ask for alternatives

### 17.7.4 Step 4: Validate and Apply

AI output is a **starting point**, not a final answer:

- Verify calculations and units
- Cross-check against standards and codes
- Apply your engineering judgment
- Document what came from AI assistance
- Take responsibility for the final work product

Case Study: When validation caught an AI error

**Scenario:** An engineering student used AI to calculate the required conveyor belt width for a food processing line handling 500 kg/hr of bulk product.

**AI Response:** Recommended a 300mm belt based on throughput calculations.

**Validation Issue:** The AI correctly calculated volumetric flow but failed to consider: - Minimum belt width for the discharge chute (400mm required) - Product spread factor for irregular materials - Safety factor for surge capacity



**Actual Requirement:** 500mm belt per facility standards.

**Lesson:** AI calculations may be mathematically correct but miss practical engineering constraints. Always validate against physical requirements, codes, and site-specific standards.

Discussion: When should you NOT use AI?

AI is not appropriate when:

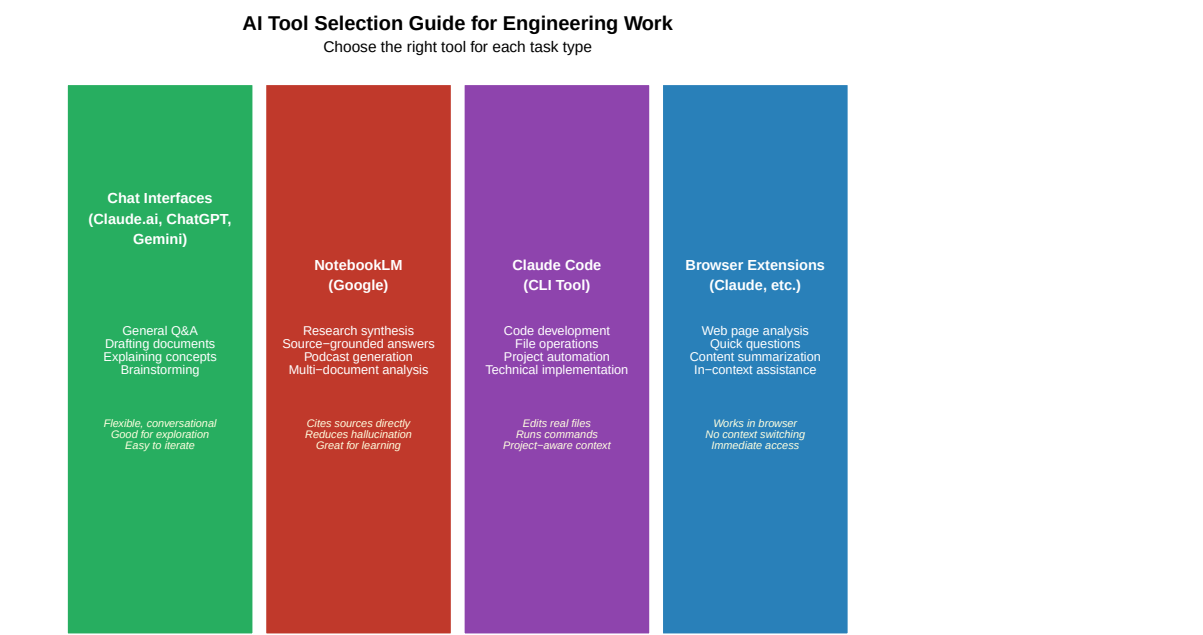
1. **Physical inspection is required:** AI cannot see, touch, or measure actual equipment
2. **Real-time safety decisions:** Lockout/tagout procedures, emergency response
3. **Legal certification:** Stamping drawings, signing off on safety assessments
4. **Novel situations:** First-of-a-kind designs with no training data
5. **Classified/proprietary work:** Confidential designs should not be uploaded

**Rule of thumb:** If a human expert would need to be physically present, AI cannot substitute.

---

## 17.8 AI Tools and Their Use Cases

Different AI tools serve different purposes. Understanding when to use each tool maximizes productivity.



17.8.1 Chat Interfaces (Claude.ai, ChatGPT, Gemini)

Web-based conversational AI platforms are the most versatile starting point for AI collaboration.

Table 17.7: Common Engineering Use Cases

| Use Case                 | Description                                                                         |
|--------------------------|-------------------------------------------------------------------------------------|
| Technical Explanations   | Explain complex concepts, standards, or technical principles in accessible language |
| Document Drafting        | Draft reports, specifications, procedures, or technical memos                       |
| Calculation Verification | Double-check hand calculations; request step-by-step solutions                      |
| Brainstorming Solutions  | Generate alternative approaches to engineering problems                             |
| Code Generation          | Write PLC ladder logic concepts, Python scripts, or MATLAB code                     |
| Standard Interpretation  | Interpret requirements from ISO, OSHA, NEC, or other standards                      |

Best Practices for Chat Interfaces:

- Start new conversations for new topics (avoid context pollution)
- Use the “regenerate” feature to see alternative responses
- Upload images of diagrams, schematics, or equipment for visual analysis
- Copy important responses to notes (conversations can be lost)

### 17.8.2 NotebookLM (Google)

NotebookLM is specifically designed for **source-grounded research**. Upload documents, and the AI answers questions based only on your sources.

Table 17.8: NotebookLM Features for Engineering

| Feature                  | Description                                                                |
|--------------------------|----------------------------------------------------------------------------|
| Source-Based Answers     | Answers come directly from your uploaded documents, reducing hallucination |
| Citation Links           | Every answer includes clickable citations to the exact source passage      |
| Audio Overview           | Generates AI podcast-style audio summaries of your documents for learning  |
| Multi-Document Synthesis | Combines information from multiple PDFs, websites, or text files           |
| Note Generation          | Creates structured notes and summaries from source materials               |

Example NotebookLM Workflow:

1. Upload: ISO 12100, ISO 13849-1, equipment manual PDF
2. Ask: “What risk assessment steps does ISO 12100 require?”
3. Review: AI provides answer with direct citations to document sections
4. Generate: Create an audio overview to reinforce learning

**When to Use NotebookLM:** When you need answers grounded in specific documents rather than general AI knowledge. Ideal for standards compliance, equipment documentation, and research synthesis.

### 17.8.3 Claude Code (Command Line Tool)

Claude Code is a **terminal-based AI assistant** that can read, write, and execute code in your actual project environment.

Table 17.9: Claude Code Capabilities for Technical Work

| Capability        | Description                                          | Example                                           |
|-------------------|------------------------------------------------------|---------------------------------------------------|
| File Operations   | Read, write, and edit files directly in your project | Edit src/motor_control.py to add a new function   |
| Code Development  | Generate, refactor, and debug code with full context | Write a Python script to parse CSV data           |
| Command Execution | Run build commands, tests, and system tools          | Run pytest and fix any failing tests              |
| Project Context   | Understands your entire codebase structure           | Find all files that reference the 'config' module |
| Git Integration   | Create commits, branches, and pull requests          | Commit changes with descriptive messages          |

Engineering Applications:

- **PLC Code Analysis:** Review and understand legacy ladder logic or structured text
- **Data Processing:** Create scripts to analyze production data, SPC charts, or sensor logs

- **Documentation Generation:** Auto-generate code documentation or API references
- **Test Automation:** Write unit tests for engineering calculations

**Example Session:**

```
> claude "Analyze the motor_control.py file and suggest improvements for  
        handling encoder signal noise"
```

Claude Code reads the file, understands the context, and provides specific recommendations with code changes that can be applied directly.

17.8.4 Browser Extensions (Claude Extension, etc.)

Browser extensions provide **instant AI assistance** without leaving your current workflow.

Table 17.10: Browser Extension Use Cases

| Use Case                       | How It Helps                                              | Example                                                                |
|--------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------|
| <b>Webpage Summarization</b>   | Summarize long technical articles or datasheets instantly | On a 20-page document                                                  |
| <b>Technical Documentation</b> | Ask questions about the documentation page you’re reading | On All about Circuits                                                  |
| <b>Email Composition</b>       | Draft professional responses to technical inquiries       | ‘Help me write an email to the client about the delay in the project.’ |
| <b>Research Assistance</b>     | Explain unfamiliar terms or concepts on any webpage       | Highlighting key points in a research paper                            |
| <b>Quick Calculations</b>      | Get quick engineering calculations without switching apps | ‘Convert 1000 mm to inches.’                                           |

**Best Practices:**

- Use for quick queries and summaries
- For complex problems, move to a full chat interface
- Be cautious with confidential information on web pages
- Extensions work best for reading comprehension tasks

17.8.5 Choosing the Right Tool

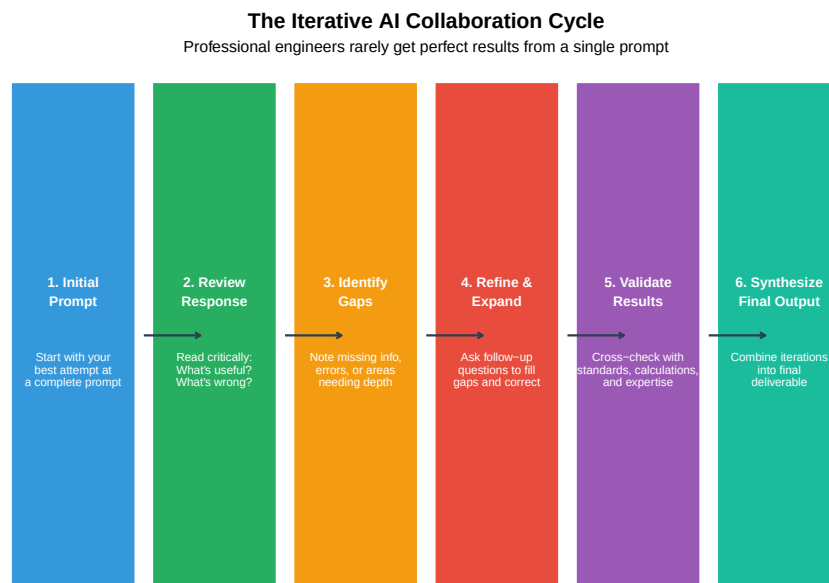
Table 17.11: Tool Selection Quick Reference

| Situation                              | Best Tool                         | Why                                                |
|----------------------------------------|-----------------------------------|----------------------------------------------------|
| Need to understand a technical concept | <a href="#">Chat Interface</a>    | Flexible conversation, can ask follow-up questions |
| Writing code for a project             | <a href="#">Claude Code</a>       | Can directly modify files and run code             |
| Researching from specific documents    | <a href="#">NotebookLM</a>        | Answers grounded in your uploaded documents        |
| Quick question about a webpage         | <a href="#">Browser Extension</a> | No context switching, immediate answers            |
| Drafting technical documentation       | <a href="#">Chat Interface</a>    | Good at structured document generation             |
| Analyzing production data              | <a href="#">Claude Code</a>       | Can write and execute analysis scripts             |

## 17.9 The Iterative Approach: Multi-Turn AI Collaboration

Complex engineering problems are rarely solved in a single prompt. The iterative approach mirrors how you would work with a human expert.

### 17.9.1 The Iteration Cycle



### 17.9.2 Worked Example: Designing a Conveyor Control System

#### Turn 1 - Initial Prompt:

Act as an automation engineer. Design a basic control system for a conveyor belt in a packaging line. The conveyor is 15m long and transports boxes to a palletizing station.

**AI Response Summary:** Generic overview with basic motor control, start/stop logic, and simple sensors.

#### Turn 2 - Adding Specificity:

Good start. Now expand with these details:

- Add accumulation zones (3 zones, 5m each) with photo-eye sensors
- Include a reject station at the 10m mark for overweight boxes
- Integrate with an upstream case sealer (need handshake signals)
- Use Allen-Bradley CompactLogix as the controller

Show the I/O list and basic ladder logic structure.

**AI Response Summary:** Detailed I/O list, zone control logic, sensor placement, and communication requirements.

**Turn 3 - Safety and Edge Cases:**

Now address:

1. What happens if the downstream palletizer faults?
2. Add E-stop integration per ISO 13849 PLd requirements
3. How do we handle box jams in each zone?
4. What's the startup sequence after E-stop reset?

Include safety relay wiring diagram description.

**AI Response Summary:** Complete safety integration, fault handling, jam detection, and recovery procedures.

**Turn 4 - Final Validation:**

Review your complete design and:

1. List any assumptions you made that I should verify
2. Identify potential failure modes you haven't addressed
3. What information would you need to finalize this for implementation?
4. Are there any code standards (ISA, IEC) I should reference?

**AI Response Summary:** Comprehensive assumption list, remaining questions, and standards references for documentation.

**Key Point:** This four-turn conversation produced a design that would have required a much longer single prompt to achieve, and the iterative approach allowed for course correction along the way.

Try it yourself: Practice iterative prompting

**Your Challenge:** Use the four-turn approach to design a temperature monitoring system for a food processing facility.

**Turn 1:** Define the basic requirements (number of zones, temperature ranges, alerting needs)

**Turn 2:** Add specifics (sensor types, communication protocol, integration with existing systems)

**Turn 3:** Address failure modes and safety (what happens if a sensor fails? How are critical temperature excursions handled?)

**Turn 4:** Request validation questions and identify what you need to verify before implementation

**Compare your results:** Did the iterative approach give you a more complete design than your initial prompt would have?

## 17.10 Practice Topics for Iterative AI Sessions

Use these topics to practice the iterative approach. Each topic is designed to require multiple turns to fully develop.

### 17.10.1 Beginner Level

Table 17.12: Beginner Practice Topics

| Topic                                           | Iteration Hint                                                         |
|-------------------------------------------------|------------------------------------------------------------------------|
| <b>Motor Starter Selection</b>                  | Start with application, then add environment, protection requirements  |
| <b>Sensor Selection for Proximity Detection</b> | Begin with what you're detecting, then material, distance, environment |
| <b>Basic PLC Program for Traffic Light</b>      | Define sequence first, then timing, then inputs/outputs, then safety   |
| <b>Belt Conveyor Speed Calculation</b>          | Start with throughput requirement, then add physical constraints       |

### 17.10.2 Intermediate Level

Table 17.13: Intermediate Practice Topics

| Topic                                           | Iteration Hint                                                         |
|-------------------------------------------------|------------------------------------------------------------------------|
| <b>VFD Parameter Setup for Pump Application</b> | Define motor and load first, then control requirements, then safety    |
| <b>PFMEA for Automated Assembly Station</b>     | Describe process steps, then potential failures, then iterate          |
| <b>Pneumatic Circuit Design with Safety</b>     | Start with actuator requirements, then add safety, then valves         |
| <b>HMI Screen Layout for Batch Process</b>      | Define operator tasks first, then alarms, then navigation, then safety |

### 17.10.3 Advanced Level

Table 17.14: Advanced Practice Topics

| Topic                                                   | Iteration Hint                                               |
|---------------------------------------------------------|--------------------------------------------------------------|
| <b>Predictive Maintenance System Architecture</b>       | Define assets and failure modes, then sensors, then data     |
| <b>Robot Cell Integration with Vision System</b>        | Robot selection first, then vision requirements, then safety |
| <b>Energy Management System for Manufacturing Plant</b> | Start with loads and usage patterns, then monitoring         |
| <b>Complete Machine Safety Assessment</b>               | Describe machine function, then hazards, then risk           |

17.11 Critical Evaluation of AI Outputs

AI can produce confident-sounding responses that are completely wrong. Engineers must develop critical evaluation skills.

17.11.1 Red Flags in AI Responses

Table 17.15: Red Flags in AI-Generated Responses

| Red Flag                      | Example                                                                 |
|-------------------------------|-------------------------------------------------------------------------|
| Calculations Without Units    | "The motor needs 150" - 150 what? HP? kW? Nm?                           |
| Missing Safety Considerations | Design has no E-stop, no guarding discussion, no lockout procedure      |
| Outdated Standards Referenced | References NEC 2014 when 2023 is current; cites withdrawn IEC standards |
| Overconfident Language        | "This will definitely work" or "This is the only solution"              |
| Generic Without Specifics     | "Use appropriate sensors" without specifying type, range, or model      |
| Ignoring Stated Constraints   | You specified 24VDC and AI designs for 120VAC                           |

17.11.2 Verification Checklist

Before using any AI-generated engineering content:

Table 17.16: AI Output Verification Checklist

| Category                        | Verification Item                                       | Verified                 |
|---------------------------------|---------------------------------------------------------|--------------------------|
| <b>Calculations</b>             |                                                         |                          |
| Calculations                    | All calculations include units                          | <input type="checkbox"/> |
| Calculations                    | Results are dimensionally consistent                    | <input type="checkbox"/> |
| Calculations                    | Order of magnitude is reasonable                        | <input type="checkbox"/> |
| <b>Standards Compliance</b>     |                                                         |                          |
| Standards                       | Referenced standards are current versions               | <input type="checkbox"/> |
| Standards                       | Applicable regional codes are addressed (NEC, CSA, IEC) | <input type="checkbox"/> |
| <b>Safety</b>                   |                                                         |                          |
| Safety                          | All identified hazards have mitigation                  | <input type="checkbox"/> |
| Safety                          | Failure modes are considered                            | <input type="checkbox"/> |
| Safety                          | Human factors are addressed                             | <input type="checkbox"/> |
| <b>Practical Implementation</b> |                                                         |                          |
| Practical                       | Specified components are available/not obsolete         | <input type="checkbox"/> |
| Practical                       | Cost and lead time are realistic                        | <input type="checkbox"/> |



## 17.12 When NOT to Use AI

AI is a tool, not a replacement for engineering judgment. Recognize its limitations.

### 17.12.1 AI Excels At:

- Generating first drafts and starting points
- Explaining concepts and standards
- Creating documentation templates
- Performing routine calculations (with verification)
- Suggesting troubleshooting approaches
- Comparing options and trade-offs

### 17.12.2 AI Struggles With:

Table 17.17: Tasks Requiring Human Engineering Judgment

| Limitation                   | Description                                                                | Example                  |
|------------------------------|----------------------------------------------------------------------------|--------------------------|
| <b>Site-Specific Context</b> | AI doesn't know your plant layout, existing equipment, or local practices  | Recommendations          |
| <b>Current Conditions</b>    | AI cannot inspect, measure, or observe actual equipment conditions         | Cannot diagnose          |
| <b>Ethical Judgment</b>      | Decisions balancing safety, cost, and risk require human judgment          | Choosing between options |
| <b>Novel Situations</b>      | Unprecedented problems may not have training data to draw from             | First-of-kind problems   |
| <b>Physical Intuition</b>    | Experience-based intuition about what 'feels wrong' in a machine           | Recognizing anomalies    |
| <b>Accountability</b>        | AI cannot stamp drawings, sign off on safety, or take legal responsibility | A P.Eng. must sign       |

Quick Check: AI or Human?

For each task below, decide whether AI assistance is appropriate, limited, or inappropriate:

1. **Drafting a maintenance procedure for a conveyor system** → *Appropriate with verification*
2. **Deciding whether to shut down a production line during a safety incident** → *Human only*
3. **Generating a bill of materials from a design specification** → *Appropriate with verification*
4. **Certifying a machine guarding design for compliance** → *Human only (requires P.Eng.)*
5. **Troubleshooting a motor that "sounds funny"** → *Limited - AI can suggest causes, human must inspect*
6. **Explaining the difference between ISO 13849 and IEC 62443** → *Appropriate*

**Key insight:** AI excels at information synthesis and drafting, but physical observation, legal certification, and real-time safety decisions require human expertise.

## 17.13 Ethical Considerations

Using AI in engineering work carries professional and ethical responsibilities.

### 17.13.1 Key Principles

1. **Transparency:** Disclose AI assistance where required by professional standards or employer policy
2. **Verification:** Never submit AI-generated work as final without verification by qualified personnel
3. **Accountability:** You remain responsible for work product regardless of AI assistance
4. **Confidentiality:** Do not input proprietary designs, client data, or trade secrets into public AI systems
5. **Competence:** AI assistance does not substitute for required qualifications or experience

### 17.13.2 Industry-Specific Considerations

Table 17.18: Industry-Specific AI Ethics Considerations

| Industry               | Key Ethical/Legal Concern                                        | Guidance                                |
|------------------------|------------------------------------------------------------------|-----------------------------------------|
| <b>Automotive</b>      | Functional safety (ISO 26262) requires traceable human decisions | Document AI use                         |
| <b>Food Processing</b> | Food safety regulations (FDA, CFIA) require validated processes  | AI can draft, but human review required |
| <b>Defense</b>         | ITAR/export controls may prohibit certain AI tool usage          | Consult legal counsel                   |
| <b>Mining</b>          | Worker safety regulations require qualified sign-off             | P.Eng. or equivalent required           |

Reflection: Your responsibilities as an AI-assisted engineer

Consider these questions for your own practice:

1. **Disclosure:** Does your employer have a policy on AI tool usage? What about your professional engineering body?
2. **Verification:** What specific steps will you take to verify AI-generated calculations before using them?

3. **Confidentiality:** What types of information from your work should NEVER be input into public AI tools?
4. **Professional development:** How will you continue building expertise rather than becoming dependent on AI for answers?
5. **Liability:** If an AI-assisted design fails, who is responsible? (Hint: It's always the licensed engineer who approved it.)

**Remember:** AI tools change rapidly, but professional engineering ethics remain constant. Your responsibility for the safety and quality of your work never transfers to a tool.

---

## 17.14 Hands-On Lab Exercise

### 17.14.1 Objective

Practice the complete AI collaboration workflow on a realistic engineering problem.

### 17.14.2 Scenario

You are tasked with designing an automated inspection station for a food packaging line. The station must:

- Inspect sealed packages for proper seal integrity
- Reject packages with defects
- Track inspection statistics
- Integrate with existing Allen-Bradley PLC system

### 17.14.3 Instructions

1. **Write your initial prompt** (5 minutes)
  - Apply the seven rules of effective prompts
  - Include role, context, specifications, constraints, and deliverables
2. **Conduct 3-4 iteration turns** (15 minutes)
  - Review each AI response critically
  - Identify gaps and request specific expansions
  - Push for component specifications and safety considerations
3. **Evaluate the final output** (5 minutes)
  - Use the verification checklist
  - Note any red flags
  - Identify what still requires human expertise
4. **Document your process** (5 minutes)
  - What worked well in your prompting approach?
  - What would you do differently next time?

- What human expertise was essential?

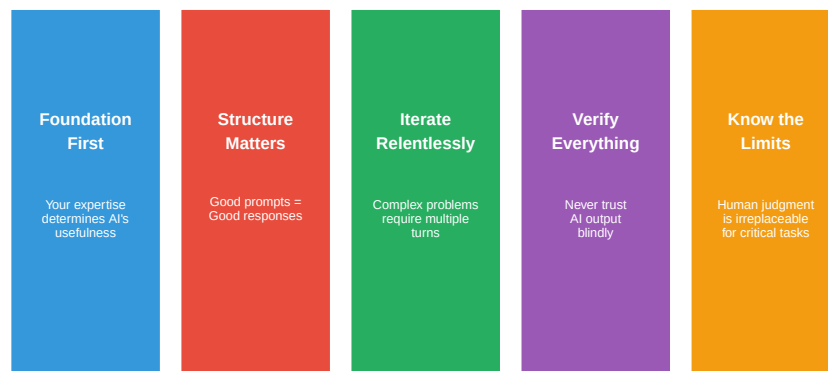
#### 17.14.4 Deliverables

- Initial prompt (graded on structure and completeness)
  - Iteration log showing refinement process
  - Critical evaluation notes
  - Reflection on lessons learned
- 

### 17.15 Summary

#### Five Principles for Effective AI Collaboration in Engineering

Master these concepts to leverage AI as a powerful professional tool



#### 17.15.1 Key Takeaways

1. **The Education Paradox:** Strong foundational knowledge is required to get quality AI assistance. “AI will do it for me” is a dangerous mindset.
2. **The Four Steps:** Effective AI use follows a consistent process: Define → Prompt → Iterate → Validate. Skipping steps leads to poor results.
3. **Prompt Engineering is a Skill:** Like technical writing, crafting effective prompts requires practice and follows learnable principles.
4. **Choose the Right Tool:** Different AI tools serve different purposes. Chat interfaces for exploration, NotebookLM for source-grounded research, Claude Code for development, and browser extensions for quick assistance.
5. **Iteration is Normal:** Professional engineers should expect 3-6 turns of AI conversation for complex problems.

6. **Critical Evaluation is Essential:** AI produces confident-sounding outputs that may be wrong. Verify everything.
  7. **Human Judgment Remains Central:** AI augments but does not replace engineering expertise, especially for safety-critical decisions.
- 

## 17.16 Review Questions

1. Explain the “education paradox” in AI usage and why it matters for engineering students.
  2. List and describe the four steps of effective AI use. Why is each step important?
  3. List the seven rules of effective engineering prompts and give an example of each.
  4. What are three red flags that should make you question an AI-generated engineering recommendation?
  5. Describe the iterative approach to AI collaboration and why single-prompt interactions are often inadequate.
  6. Compare and contrast the four main types of AI tools (chat interfaces, NotebookLM, Claude Code, browser extensions). When would you use each?
  7. For which types of engineering tasks is AI assistance most appropriate? Least appropriate?
  8. What ethical considerations apply when using AI for engineering work in safety-critical industries?
  9. Transform this bad prompt into a good prompt: “Help me size a pump.”
  10. You need to research a new ISO standard and create a summary for your team. Which AI tool(s) would you use and why?
  11. Why is it important to include standards references in engineering prompts?
  12. How does NotebookLM differ from general chat interfaces like ChatGPT or Claude.ai? When would this difference matter?
- 

## 17.17 References and Further Reading

- Anthropic Economic Index Report (2026) - AI usage patterns in profes-

sional work

- ISO 12100 - Safety of machinery, General principles for design
  - IEC 62443 - Industrial communication networks, Network and system security
  - Professional Engineers Ontario - Use of Artificial Intelligence guidelines
  - OpenAI Prompt Engineering Guide - General prompting techniques
  - IEEE Code of Ethics - Professional responsibilities in technology
- 

## 17.18 Glossary

**Prompt Engineering:** The practice of designing inputs to AI systems to achieve desired outputs.

**Iteration/Multi-turn Conversation:** Using multiple exchanges with AI to progressively refine and improve outputs.

**Augmentation:** Using AI to enhance human capabilities rather than replace them entirely.

**Automation:** Delegating complete tasks to AI with minimal human intervention.

**Hallucination:** AI generating plausible-sounding but factually incorrect information.

**Context Window:** The amount of text an AI can consider at once; important for complex problems requiring extensive background.

**Chat Interface:** Web-based conversational AI platforms (e.g., Claude.ai, ChatGPT, Gemini) for flexible, general-purpose AI interaction.

**NotebookLM:** Google's AI tool that provides source-grounded answers based on uploaded documents, with citation links and audio overview generation.

**Claude Code:** A command-line AI tool that can read, write, and execute code directly in project environments with full file system access.

**Browser Extension:** AI tools integrated into web browsers for instant assistance without switching applications; useful for webpage analysis and quick queries.

**Source-Grounded AI:** AI systems that base responses on specific provided documents rather than general training data, reducing hallucination risk.

# Chapter 18

## Labs

This chapter collects the lab instructions, prep work, rubrics, marking notes and supplementary information for every lab in the course, one sub-chapter per lab. The last section, Teacher preparation, is a term-planning aid for the instructor: it lists the room, equipment, software and a consolidated shopping list for every lab.

### 18.1 How to use this chapter

Every lab sub-chapter is broken into the **same fixed parts**, and each part has its own colour so you always know which part of the lab you are reading:

| Colour    | Part                                 | What it is                                                         |
|-----------|--------------------------------------|--------------------------------------------------------------------|
| Dark blue | <b>Overview</b>                      | What the lab is about and why you do it                            |
| Teal      | <b>Learning outcomes</b>             | What you should be able to do afterwards                           |
| Orange    | <b>Prep work</b>                     | Readings, videos and set-up to finish <i>before</i> the lab period |
| Blue      | <b>Materials &amp; equipment</b>     | What to bring / what is provided                                   |
| Purple    | <b>Procedure</b>                     | The steps you carry out during the lab                             |
| Green     | <b>Deliverables &amp; submission</b> | What you hand in, and when                                         |
| Slate     | <b>Rubric</b>                        | How the mark is calculated                                         |

| Colour      | Part                    | What it is                                                    |
|-------------|-------------------------|---------------------------------------------------------------|
| Red, dashed | <b>Instructor notes</b> | Marking guidance and prep — not required reading for students |

### ! How the labs are mapped to source handouts

The lab handouts in the course folder come from **two parallel lab streams** that have run under this course code in different terms:

- **Manufacturing / process-design stream** — matches this textbook and the current course plan (ENGR-3027, EMN program). **This is the stream scaffolded below.**
- **Instrumentation-trainer stream** — an older set authored for the B1033 trainer benches (Pressure Measurement for Level, Motor Control & VFD, PLC Introduction with Siemens, Flow Rate Measurement). These are summarised in Appendix: instrumentation-trainer lab stream but are **not** the primary labs.

Within the manufacturing stream the handout **numbering has drifted** between terms — the popsicle-stick bridge activity appears as both “Lab 1” and “Lab 2”; the “Engineering Design Process Documentation” handout is headed “Lab 2” internally but filed as Lab 3; “Safety in Automated Manufacturing – Part 1” exists as both “Lab 6” and “Lab 7”. **The mapping below follows the week order in the current course plan.** Where a lab has no current handout in the folder, the sub-chapter is a scaffold with TODO markers and whatever the course plan specifies.

| Lab | Title (this chapter)                                | Course-plan week | Primary source file(s)                                                                                                                                     |
|-----|-----------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Engineering Design Thinking – Popsicle-Stick Bridge | 2                | ENGR-3027 Lab 1 – Engineering Design Process2.pdf, Lab 1 Supplemental.pdf, Lab 1 – Rubric.docx, Lab- 1 – Engineering Design Process Flow – Lab Report.docx |



|   |                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---|------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | Logic Thinking<br>for Robot<br>Control –<br>LightBot &<br>Rocksi | 3 | Lab-2<br>-Introduction<br>to<br>Programming.docx,<br>Process<br>Engineering<br>Lab#2<br>Programming<br>Logic.pdf<br>ENGR-3027 Lab<br>3 -<br>Engineering<br>Design<br>Process<br>Documentation.docx,<br>Lab 3 -<br>Supplemental.pdf,<br>Lab 3-<br>Template- 3-<br>HSE Risk<br>Assessment<br>Report.docx,<br>BOM / Gantt<br>templates<br>ENGR-3027 -<br>Lab- 4-<br>Manufacturing<br>Facility<br>Design and<br>Simulation<br>Lab.docx,<br>ENGR-3027-<br>Lab-4-Supplemental<br>Material.pdf<br>ENGR-3027<br>-Lab 5 -<br>Lean-manufacturing-lab-worksheet.docx,<br>Lab 5 -<br>Lean-manufacturing-product-images.pdf,<br>Lab 5<br>-Lean-manufacturing-product<br>check<br>sheet.pdf |
| 3 | Engineering<br>Design Process<br>Documentation                   | 4 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4 | Manufacturing<br>Facility Design<br>& Simulation<br>(AnyLogic)   | 5 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5 | Lean<br>Manufacturing<br>Simulation                              | 6 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|    |                                                       |    |                                                                                                                |
|----|-------------------------------------------------------|----|----------------------------------------------------------------------------------------------------------------|
| 6  | Ergonomics in Automated Manufacturing (REBA & Inseer) | 8  | ENGR-3027<br>-Lab 6 -<br>Ergonomics.docx,<br>REBA-worksheet.pdf,<br>Cheatsheet -<br>Ergonomics<br>-REBA.pdf    |
| 7  | Safety in Automated Manufacturing – Part 1            | 9  | LAB 7 –<br>Supplementary<br>Material.pdf<br>(also LAB 6 –<br>Supplementary<br>Material-1.pdf,<br>same content) |
| 8  | Safety in Automated Manufacturing – Part 2            | 10 | LAB 7 –<br>Supplementary<br>Material<br>-2.pdf —<br><i>procedure not in<br/>folder,<br/>scaffolded</i>         |
| 9  | Process Safety & Functional Safety – Part 1           | 11 | <i>No handout in<br/>folder —<br/>scaffolded from<br/>course plan<br/>(prelab videos)</i>                      |
| 10 | Safety Instrumented System Architectures              | 12 | Lab- 10<br>Supplimentary<br>Material.pdf                                                                       |
| 11 | Review-week lab                                       | 13 | <i>No handout in<br/>folder —<br/>scaffolded from<br/>course plan</i>                                          |
| 12 | Take-home lab (video submission)                      | 14 | <i>No handout in<br/>folder —<br/>“at-home” lab,<br/>due any time<br/>before end of<br/>term</i>               |

Labs 1–10 are **mandatory to pass the course**. Labs are due at the end of the scheduled lab period; the late penalty is 25 % per lab session. Marks may be deducted for incomplete prelab work, missing safety training, not bringing required materials, not cleaning your station, or unprofessional conduct in the lab room.

## 18.2 Lab 1 — Engineering Design Thinking: Popsicle-Stick Bridge

You work through the full engineering design-thinking process — define the problem, research, generate concepts, analyse feasibility, prototype, test, iterate — by designing and building a popsicle-stick bridge that spans 30 cm and carries the largest possible load without failing. The lab is run in teams and mirrors how a real product moves from a napkin sketch to a tested prototype.

### 18.2.1 Learning outcomes

After this lab you should be able to:

- Apply the stages of the engineering design process to an open-ended build task.
- Produce initial and final design sketches and a materials inventory.
- Carry out a simple failure-mode analysis (FMEA) on your own design and justify the risks you accept.
- Load-test a structure, record its failure mode, and relate the result to tension, compression and load distribution.
- Reflect on team performance through a peer evaluation.

### 18.2.2 Prep work

Complete **before** the lab period:

1. Read **Lab 1 Supplemental.pdf** — overview of bridge types, technical vs. manufacturing/safety design aspects, the 8-stage engineering design process, and design thinking (the “FAIL = first attempt in learning” idea).
2. Watch the assigned prelab videos on FOL (bridge types; tension/compression).
3. Form teams of 2–3.
4. Come with a first rough sketch and a filled-in “Chart 1” (Identify the need & constraints → Research the problem → Develop possible design solutions → Select a promising solution).

### 18.2.3 Materials & equipment

Provided per team (you may use **only** provided materials):

- Small craft sticks — **maximum 100**
- Jumbo craft sticks — **maximum 20**
- White (PVA) glue
- Hot-glue gun — sticks joined with hot glue limited to **10**
- Masking tape — **maximum 50 cm total**
- String — **maximum 100 cm total**
- Ruler / measuring tape
- Camera or phone (to document the build)

Shared: load-test rig (two supports 30 cm apart), calibrated weights / sandbags, loading hook or basket.

**Size constraints:** span 30 cm between supports; height 10 cm; structurally sound and symmetrical.

### 18.2.4 Procedure

#### Phase 1 — Problem definition & initial design

1. Restate the problem: a bridge spanning 30 cm that holds the highest possible load. Populate your thought process using Chart 1.
2. Sketch your proposed design (beams, trusses, supports) and state the design goals (maximise load capacity, minimise material, efficient structure).
3. FMEA: list the failure modes your design could have and, for each, give a fix or a rationale for accepting it.
4. Materials inventory: list every material and quantity you will use.

#### Phase 2 — Prototyping & building

5. Build to the approved design. Photograph the build and note construction methods and any changes.
6. You may prototype **individual components** (not the whole bridge) to prove design intent. Stay within all material and size limits.

#### Phase 3 — Testing & evaluation

7. Place the bridge on the two supports. Add weight at the centre until it fails (breaks or deforms beyond use).
8. Record the maximum weight held and describe the failure mode.
9. Compare results across teams; discuss which structural elements helped or hurt, how your design evolved, and what you would change next time.

### 18.2.5 Deliverables & submission

Due at the **start of the next lab class** (see FOL). One group report plus individual components:

## 18.2. LAB 1 — ENGINEERING DESIGN THINKING: POPSICLE-STICK BRIDGE463

1. **Design report (group):** description of the design process; initial and final sketches; max load held and total bridge mass; discussion of failure modes and structural principles learned; any post-failure modifications and re-test results.
2. **Peer evaluation / reflection (individual, submitted separately, seen only by the professor):** what you would do differently, what worked and what did not in your group.
3. **Individual member report** requirements as outlined on FOL — done on your own.

Use the template Lab- 1 - Engineering Design Process Flow - Lab Report.docx.

**AI tools may not be used for any part of the group or individual report.**

### 18.2.6 Rubric

From Lab 1 - Rubric.docx — 11 criteria, 1 mark each (11 marks, rescaled to the lab's course weight):

| # | Criterion                                                                                                          | Marks |
|---|--------------------------------------------------------------------------------------------------------------------|-------|
| 1 | Objective & problem definition — clear statement; design targets (safety, scalability, modularity)                 | 1     |
| 2 | Research & concept generation — research on bridge designs/structural principles; multiple concepts with rationale | 1     |
| 3 | Feasibility analysis — materials, constraints, potential failure modes                                             | 1     |
| 4 | Initial design sketches — clear, detailed; planned materials & quantities documented                               | 1     |
| 5 | Construction procedure — step-by-step, logical sequence                                                            | 1     |

| #  | Criterion                                                                                                     | Marks |
|----|---------------------------------------------------------------------------------------------------------------|-------|
| 6  | Final design sketches & modifications — final sketches; changes after testing and their outcomes              | 1     |
| 7  | Failure modes & structural principles — 3 failure modes identified and explained                              | 1     |
| 8  | Prototype testing, results & observations — testing documented with photos; max weight and materials recorded | 1     |
| 9  | Peer evaluation & reflection — individual reflection; peer evaluation of teamwork                             | 1     |
| 10 | Conclusion — concise summary of findings                                                                      | 1     |
| 11 | Overall presentation — organisation, clarity, adherence to guidelines                                         | 1     |

### 18.2.7 Instructor notes

- **Prep:** cut/verify one class set of load-test rigs (supports exactly 30 cm apart); confirm the calibrated weight set and loading basket; count out material kits per team *before* class so the 100/20/50 cm/100 cm/10 limits are physically enforced.
- **Safety:** hot-glue gun burns are the main hazard — supervise, provide stands and a cool-down area. Eye protection during load-test-to-failure; keep spectators back from the rig; expect sticks to fly on failure.
- **Marking:** criteria 1–3 and 7 are where design-thinking (vs. “we just glued sticks”) is demonstrated — weight feedback there. Individual reflections are submitted separately and are confidential to the professor.
- **Time:** ~15 min briefing, 20 min design/FMEA sign-off, 60 min build, 30 min testing, 15 min cleanup.

## 18.3 Lab 2 — Logic Thinking for Robot Control: LightBot & Rocksi

An introduction to programming logic — sequencing, loops and conditionals — and to robot teach-pendant operation, using two free browser simulators: **LightBot** (a puzzle game that teaches sequence/loop/conditional thinking) and **Rocksi** (a teach-pendant simulator for the Sawyer robot). The focus is on *optimising* the number of programming steps.

### 18.3.1 Learning outcomes

After this lab you should be able to:

- Explain why programming logic matters for reliable, safe, repeatable automation.
- Use sequences, loops and conditionals to solve a control problem with the fewest steps.
- Describe the role of a teach pendant (jogging, teaching points, deadman/E-stop, reduced-speed teach mode, I/O testing).
- Program a simple pick-and-place-style sequence on a simulated teach pendant.

### 18.3.2 Prep work

- Read `Process Engineering Lab#2 Programming Logic.pdf` (importance of programming logic; loops; conditionals; teach pendants).
- Watch: *Rocksi robot simulator: Meet Sawyer*.

**Rocksi robot simulator: Meet Sawyer:** <https://www.youtube.com/watch?v=9oAFmTlMqzg>

- Check that **LightBot** and **Rocksi** load on your device.

### 18.3.3 Materials & equipment

- Computer or tablet with internet access — **one per student** (this is an individual lab)
- Access to **LightBot** (browser)
- Access to **Rocksi** teach-pendant simulator (browser)
- Lab worksheet (provided — the questions are reproduced in the Procedure below)

### 18.3.4 Procedure

#### Part 1 — Programming logic with LightBot

1. *Sequencing*. Complete the first levels. **Worksheet:** describe the sequence of commands you used to complete Level 1. (1 mark)

2. *Loops*. Complete levels that require loops. **Worksheet:** how did using loops let you complete the level more efficiently? (2 marks)
3. *Conditionals*. Complete levels that use conditional commands. **Worksheet:** give an example of a conditional you used and explain its purpose. (2 marks)
4. *Challenge — maximum level & optimisation*. Reach Level 15 with the minimum number of steps. **Worksheet:** describe your strategy and how you optimised. (2 marks)

### Part 2 — Robot teach-pendant simulation with Rocksi

5. *Basic movements*. Jog the robot to several positions. **Worksheet:** describe the steps to move the robot to Position A. (1 mark)
6. *Programming a sequence*. Program a sequence of moves with the teach pendant. **Worksheet:** list the command sequence you used. (1 mark)
7. *Simulating inputs & outputs*. Simulate an I/O to troubleshoot a robot operation. **Worksheet:** explain how you simulated the I/O and its effect. (1 mark)

Finish each part with a short reflection on how the skills apply to engineering design and safety.

### 18.3.5 Deliverables & submission

Completed worksheet (all questions above plus the two reflections), submitted on FOL by the end of the lab period. Assessment is based on task completion in both simulators and the quality of the written answers.

### 18.3.6 Rubric

From the handout's marking table — **10 marks total**:

| Task                                                  | Marks |
|-------------------------------------------------------|-------|
| Sequencing (LightBot)                                 | 1     |
| Loops (LightBot)                                      | 2     |
| Conditionals (LightBot)                               | 2     |
| Challenge task (LightBot) — reach Level 15, optimised | 2     |
| Basic movements (Rocksi)                              | 1     |
| Programming a sequence (Rocksi)                       | 1     |
| Simulating inputs & outputs (Rocksi)                  | 1     |

(The handout body lists the LightBot challenge at 2 marks; an earlier summary line totals the same tasks to 10. Use the table above.)



### 18.3.7 Instructor notes

- **Prep:** confirm both simulators are reachable through the lab network/firewall *the day before* — LightBot and Rocksi are third-party sites and are the single point of failure for this lab. Have an offline fallback (screenshots / a slide walk-through) ready.
  - Individual lab — one machine per student; no group submission.
  - **Marking:** award loop/conditional marks for *explanation*, not just level completion. For the challenge task, ask for the step count as evidence of optimisation.
- 

## 18.4 Lab 3 — Engineering Design Process Documentation

Using the bridge you built in Lab 1, you create the engineering documents that a real project needs to move a design into production: specification sheets, a bill of materials, a task list / process flow, a project schedule and Gantt chart, a production-facility layout drawing, and a health-safety-environment (HSE) risk assessment. The tools are Microsoft Visio/Word, Project and Excel.

### 18.4.1 Learning outcomes

After this lab you should be able to:

- Explain terms such as HSE, Basic Engineering Design Data (BEDD), Front-End Engineering Design (FEED), fast-tracking, project schedule and General Arrangement (GA) drawing.
- Produce a specification sheet and a bill of materials for a simple product.
- Build a task list / process-flow chart with durations and turn it into a project schedule and Gantt chart in Microsoft Project.
- Draft a production-facility layout drawing showing product in/out, stations, material and people flow.
- Complete an HSE risk-assessment for a build process.

### 18.4.2 Prep work

- Review Unit 2 content and **Lab 3 - Supplemental.pdf** (the “Lab-2 connection”, the detail-design/engineering process-flow diagram, and example templates).
- Watch: *3DEXPERIENCE Delmia — automated bridge manufacturing factory* (link in the supplement).
- Bring your Lab 1 bridge, sketches and photos.
- Confirm you can open Microsoft Visio/Word, Project and Excel on a lab PC.

### 18.4.3 Materials & equipment

- Lab PC with **Microsoft Word/Visio, Microsoft Project, Microsoft Excel**
- Templates from FOL:
  - `Bill_of_Materials_Template_for_Excel.xlsx`
  - `Gantt_chart.mpp` (Microsoft Project example file)
  - HSE risk-assessment Word template (`Risk assessment template`)
- Your Lab 1 bridge, sketches and build photos

### 18.4.4 Procedure

**Phase 1 — Specification sheets & bill of materials.** Examine the materials in your bridge and create specification sheets for the craft sticks and other materials, following the BOM examples in the supplement. Fill in the Excel BOM template.

**Phase 2 — Project schedule.** Build a chronological task list (flow chart) for making the bridge; add a column for the time each task needs; open the example `.mpp` file, enter the tasks and durations, and generate a schedule and Gantt chart. Use the task-numbering scheme (Task 1 → Sub-task 11 → Sub-sub-task 121...) from the supplement.

**Phase 3 — Production-facility layout drawing.** From the task list, allocate spaces in a hypothetical plant to perform those tasks. Sketch a first layout, discuss with your team and the instructor, then finalise it. Show product in / product out, stations, tool and material locations, and raw/waste/people flow.

**Phase 4 — Safety documentation (HSE report).** Identify at least **three** health-and-safety risks in the bridge-building process and document them in the risk-assessment template (hazard, who is harmed and how, current controls, further action, owner, deadline, done).

### 18.4.5 Deliverables & submission

The lab report is the **set of documents from all four phases**: specification sheets + BOM, task list/flow chart, Project schedule + Gantt chart, facility layout drawing, and the completed HSE report. Submit on FOL by the end of the lab period.

**AI tools may not be used for any part of the report.**

### 18.4.6 Rubric

From Lab 3 - Supplemental.pdf — **5 marks total**:

| Deliverable                            | Marks |
|----------------------------------------|-------|
| Specification sheets                   | 1     |
| Project schedule (Gantt)               | 1     |
| Production-facility layout drawing     | 1     |
| HSE report                             | 1     |
| Bridge-making instructions / task list | 1     |

### 18.4.7 Instructor notes

- **Prep:** verify Microsoft **Project** and **Visio** licences are live on every lab PC (these are the parts most likely to be missing from a standard Office install). Re-upload the three FOL templates each term and check the .mpp opens in the installed Project version.
- The internal heading of the source handout still reads “Lab 2” — the content is current; only the number changed.
- **Marking:** look for traceability — the BOM, task list, schedule and layout should all describe the *same* bridge and process, not generic examples.

## 18.5 Lab 4 — Manufacturing Facility Design & Simulation (AnyLogic)

You turn the paper facility layout from Lab 3 into an electronic drawing, then use **AnyLogic Personal Learning Edition** to run a discrete-event simulation of an order-fulfilment / warehouse model. By changing order and reorder parameters and reading the statistics screens, you see how inventory policy drives waiting time, picking time and throughput.

### 18.5.1 Learning outcomes

After this lab you should be able to:

- Produce an electronic facility layout showing stations, flows, manual vs. automated stations, and starvation/blockage risk points.
- Run an AnyLogic model, change input parameters, and capture the statistics output.
- Interpret order statistics (preorder waiting time, picking time, full processing time, order states) and goods statistics (unavailable goods, stored reserved/unreserved goods, storing time) and explain how they are linked.
- Explain how simulation supports facility design decisions.

### 18.5.2 Prep work

- Review Unit 3 content and ENGR-3027- Lab-4-Supplemental Material.pdf (order statistics, goods statistics, and the relationships between them).
- Watch the AnyLogic walk-through video (link in the handout).
- **Install AnyLogic Personal Learning Edition** on your machine *before* the lab and confirm it launches.
- Bring your Lab 3 layout drawing, spec sheets, schedule, BOM and flow documents.

### 18.5.3 Materials & equipment

- Lab PC with **AnyLogic Personal Learning Edition** and **Microsoft Visio/Word/Project/Excel**
- The AnyLogic demo model referenced in the video (order-picking / warehouse) — *confirm exact model and stage it on FOL / the lab PCs*
- Your Lab 3 documents

### 18.5.4 Procedure

**Phase 1 — Facility drawing.** Recreate your Lab 3 paper layout electronically (Visio or equivalent). Work through the checklist and record your findings: product in/out shown? work-sequence task list with timings? tools and material buckets shown? raw-material flow? waste-material flow? people-movement flow? how many stations, and are any automated? where could material *starvation* occur? where could material *blockage* occur? are warehouse / assembly / admin / changeover / tool / safe-assembly / social areas identified? Discuss with your team and the instructor and **get signed off** before Phase 2.

**Phase 2 — AnyLogic simulation.** Start AnyLogic and follow the video. Run the model for **3 minutes** at each of the four parameter sets below; screenshot the Statistics screen and write observations & analysis for each:

| Run | Min<br>goods/order | Max<br>goods/order | Order<br>interarrival<br>(s) | Reorder level |
|-----|--------------------|--------------------|------------------------------|---------------|
| 1   | 3                  | 5                  | 25                           | 45            |
| 2   | 3                  | 5                  | 25                           | 20            |
| 3   | 1                  | 5                  | 25                           | 45            |
| 4   | 3                  | 5                  | 45                           | 45            |

### 18.5.5 Deliverables & submission

Two deliverables in one report: (1) the electronic facility layout drawing, and (2) the AnyLogic activity report — four labelled statistics screenshots with

observations and analysis. Submit on FOL by the end of the lab period.

**AI tools may not be used for any part of the report.**

### 18.5.6 Rubric

From ENGR-3027- Lab-4-Supplemental Material.pdf — **10 marks total:**

| Deliverable                             | Marks |
|-----------------------------------------|-------|
| Manufacturing-facility drawing          | 4     |
| AnyLogic Run 1 — observation & analysis | 2     |
| AnyLogic Run 2 — observation & analysis | 2     |
| AnyLogic Run 3 — observation & analysis | 2     |
| AnyLogic Run 4 — observation & analysis | 2     |

(Handout lists drawing + three analysis blocks = 10 marks. If four runs are required, confirm the split — likely drawing 4, analysis  $1.5 \times 4$ .)

### 18.5.7 Instructor notes

- **Prep:** AnyLogic PLE install is the biggest failure point — push it to lab PCs ahead of the term and test a full model run. PLE has a model-size limit; confirm the demo model stays under it. Have the exact model file on FOL.
- Give the Phase 1 sign-off some rigour — students who skip it produce screenshots with no layout to tie them to.
- **Marking:** the drawing carries 40 % of the mark; analysis should reference the *specific* statistic that moved (e.g. “lower reorder level → more unavailable goods → longer preorder waiting time”).

## 18.6 Lab 5 — Lean Manufacturing Simulation

A hands-on paper-cutting production line. Two teams assemble four paper “triangle products” (house, tent, tree, umbrella) over **three timed trials**, changing the rules each trial to expose *muda* (waste). You measure time, count opportunities and defects, compute a defect ratio, and identify improvements — the core of a lean kaizen loop.

### 18.6.1 Learning outcomes

After this lab you should be able to:

- Identify value-adding vs. wasteful steps in a simple production line.

- Assign line roles (supervisor, supplies handler, assemblers, quality inspector) and run to a standard.
- Measure cycle time, count opportunities and defects, and compute defects per opportunity.
- Propose and test process improvements across successive trials, trading speed against accuracy.

### 18.6.2 Prep work

- Review Unit 4 content and the Lab 4 / lean supplementary documents.
- Have a **checklist and a timer** ready.
- Read the product check sheet and product images so you know exactly what each of the eight products must contain.

### 18.6.3 Materials & equipment

Per session (two teams of ~5):

- A4 paper — **pink**
- A4 paper — **yellow**
- **Blue** pens and **red** pens
- Scissors (several per team)
- Stopwatch / timer per team
- Printed **product check sheet**, **product images**, and **order tickets**

### 18.6.4 Procedure

Assign roles: line supervisor, supplies handler, assemblers (as many as needed), quality-control inspector.

**Prep the line:** (1) cut A4 sheets in two; (2) make three squares from one piece and cut each in half to get triangles; (3) cut triangles into smaller triangles. **Group 1** makes Yellow House, Pink Tent, Yellow Tree, Pink Umbrella; **Group 2** makes Pink House, Yellow Tent, Pink Tree, Yellow Umbrella. “Opportunities” = the number of drawn objects on each product (see the images).

**Trial 1 rules:** time the assembly of all four products; the supervisor may touch only the finished product and is the only one who moves product to inventory; every product must travel with the QC checklist.

**Trial 2 rules:** supervisor still only touches the finished product; only the supplies handler may fetch materials for assemblers; every product must travel with an order ticket; the supplies handler has a reference image; **switch product lines**.

**Trial 3 rules:** **switch product lines** again; same handler/supervisor rules; the supervisor must **cut at least two workers** (budget cut) and may re-assign roles to streamline further.

**Each trial, record:** time for four products; number of opportunities; number of defects; ratio of defects to total opportunities; and a list of wasteful steps / problems with proposed improvements.

### 18.6.5 Deliverables & submission

Group worksheet with the three-trial data table (time, opportunities, defects, defect ratio) and the improvement notes for each trial, plus a short conclusion on what changed between trials and why. Submit on FOL by the end of the lab period.

### 18.6.6 Rubric

TODO - no numeric rubric in the folder. Suggested split (confirm before the term):

| Element                                                          | Marks |
|------------------------------------------------------------------|-------|
| Trial 1 data complete + waste analysis                           | 3     |
| Trial 2 data complete + waste analysis                           | 3     |
| Trial 3 data complete + waste analysis                           | 3     |
| Conclusion — speed/accuracy trade-off, improvement across trials | 1     |

### 18.6.7 Instructor notes

- **Prep to buy each term:** 1 ream pink A4, 1 ream yellow A4, 1 box blue pens, 1 box red pens. **Print:** product check sheets, product image sheets, order tickets, QC checklists — one set per team plus spares.
- **Prep to stage:** class set of scissors; two stopwatches (or phone timers).
- Run with exactly two teams of ~5; you need room for two parallel lines with a supervisor “inventory” area each.
- **Marking:** reward *quantified* waste analysis (specific wasted motion / wait / rework) and evidence the team’s changes between trials were driven by the data.

## 18.7 Lab 6 — Ergonomics in Automated Manufacturing (REBA & Inseer)

You assess three workplace tasks for musculoskeletal risk two ways: manually with the **REBA (Rapid Entire Body Assessment)** worksheet, and automatically with **Inseer**, an AI video-based ergonomic-assessment tool. You then compare the two methods.

### 18.7.1 Learning outcomes

After this lab you should be able to:

- Explain why ergonomics matters in an automated manufacturing facility and name common ergonomic risks and their effects on workers.
- Score a task with the REBA worksheet and interpret the result.
- Generate an ergonomic risk-assessment report with Inseer.
- Compare manual vs. automated assessment — advantages and limitations of each.

### 18.7.2 Prep work

- Review the introduction-to-ergonomics and REBA material (video links in the handout, Procedure topics 1–2).
- Skim `Cheatsheet - Ergonomics -REBA.pdf` and the `REBA-worksheet.pdf`.
- Form groups of 3–4.

### 18.7.3 Materials & equipment

- **REBA Employee Assessment Worksheet** (printed, one per student/group)
- **Inseer** software tool — web-based; account/login required
- Computer with internet access
- Sample tasks from an automated manufacturing facility (videos / photo sets):
  - Task 1 — lifting boxes repeatedly
  - Task 2 — computer-workstation setup
  - Task 3 — floor cleaning with a buffer

### 18.7.4 Procedure

1. **Intro to ergonomics** — watch the overview videos (importance; common risks and impacts).
2. **Overview of REBA** — watch how to use the worksheet.
3. **Task evaluation** — each group is assigned one of the three tasks. Observe and analyse it, note ergonomic risks (repetitive motion, awkward posture, heavy lifting, vibration, glare, chair height ...), score it with the REBA worksheet, and interpret the score.
4. **Intro to Inseer** — overview of the tool; input the task data and generate a risk-assessment report.
5. **Data & observations** — record REBA scores and Inseer results for the task; note insights.
6. **Analysis** — compare the manual REBA assessment with the Inseer report; discuss advantages and limitations of each.
7. **Conclusion** — reflect on the importance of ergonomics in automated manufacturing and the benefits of AI tools like Inseer.



### 18.7.5 Deliverables & submission

One group report containing: the procedure, your group's REBA analysis, your group's Inseer report, and your group's observations and conclusions. Submit on FOL by the end of the lab period.

### 18.7.6 Rubric

TODO - no numeric rubric in the folder. Suggested split (confirm before the term):

| Element                                          | Marks |
|--------------------------------------------------|-------|
| Procedure documented                             | 1     |
| REBA analysis — correct scoring & interpretation | 4     |
| Inseer report generated & included               | 2     |
| Comparison / observations — manual vs. automated | 2     |
| Conclusion                                       | 1     |

### 18.7.7 Instructor notes

- **Prep:** set up **Inseer** access for the class *well ahead* — confirm the account type, seat count and whether students upload their own clips or use supplied ones. This is the lab's dependency risk.
- **Print:** REBA worksheets + the REBA cheat sheet, one per student.
- Stage the three task video/photo sets on FOL (**Lifting Boxes.pdf**, **Workstation.pdf**, **Cleaning.pdf** and the matching clips).
- **Marking:** REBA scoring is where students lose marks — check the trunk/neck/leg and arm/wrist sub-scores and the coupling/activity adjustments, not just the final number.

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## 18.8 Lab 7 — Safety in Automated Manufacturing, Part 1

An introduction to industrial-robot anatomy and safeguarding: the four robot components (end-effector, manipulator, controller, teach pendant), collaborative vs. non-collaborative robots, the range of safeguarding devices, and the robot application spaces (maximum, restricted, operating envelope).

### 18.8.1 Learning outcomes

After this lab you should be able to:

- Identify the four main components of an industrial robot and the function of each.
- Distinguish collaborative from non-collaborative robot applications.
- Describe the common safeguarding devices — fixed/perimeter guards, interlocked barrier guards, light curtains, pressure-sensitive mats, laser scanners, speed-and-separation monitoring, safety-rated monitored stop, hand-guided control, power-and-force limiting.
- Explain and sketch the maximum, restricted and operating space envelopes for a robot arm.

### 18.8.2 Prep work

- Read LAB 7 - [Supplementary Material.pdf](#) (robot components; robotic structures; joints vs. links; collaborative vs. non-collaborative; safeguarding devices; application spaces).
- Watch the prelab videos on robotic structures and on robot axes/joints (links in the supplement).
- Reference: OSHA OTM Section 4, Chapter 4 (Industrial Robots).

### 18.8.3 Materials & equipment

- Computer with internet access (videos, OSHA reference)
- LAB 7 - [Supplementary Material.pdf](#)
- *If available:* a robot cell / cobot demo with a light curtain or interlocked gate, for a walk-around of the space envelopes — **confirm equipment and room**

### 18.8.4 Procedure

**TODO** - the folder contains only the supplementary reading for this lab, not the step-by-step procedure or worksheet. Reconstruct from the prior-term worksheet or FOL. Expected shape, based on the supplement:

1. Label the four components on a supplied robot diagram.
2. Classify a set of example applications as collaborative or non-collaborative and justify.
3. For a given robot cell, select appropriate safeguarding device(s) and explain why.
4. From side and top views, construct the 3-D operating envelope and mark the maximum, restricted and operating spaces.

### 18.8.5 Deliverables & submission

**TODO** - **confirm**. Expected: a completed worksheet with the labelled diagram, the classification exercise, the safeguarding selection with rationale, and the envelope sketch. Submit on FOL by the end of the lab period.

## 18.9. LAB 8 — SAFETY IN AUTOMATED MANUFACTURING, PART 2477

### 18.8.6 Rubric

TODO - no rubric in the folder.

### 18.8.7 Instructor notes

- The same content also exists in the folder as LAB 6 - Supplementary Material-1.pdf (a prior term numbered it Lab 6). Use the Lab 7 copy.
  - **Action required:** locate the actual Lab 7 procedure/worksheet (FOL or prior term) and replace the TODO blocks above.
  - **Prep:** if a cobot/light-curtain demo is used, book the cell and confirm a qualified operator; otherwise this is a computer-room lab.
- 

## 18.9 Lab 8 — Safety in Automated Manufacturing, Part 2

TODO - scaffold. Course-plan week 10, following Unit 6 Part 2. Continues Lab 7: expected to move from *recognising* safeguards to *assessing* an automated cell — a structured safety assessment / risk assessment of a robotic or automated manufacturing station.

### 18.9.1 Learning outcomes

TODO. Likely: perform a task-based risk assessment of an automated cell; select and justify risk-reduction measures against the hierarchy of controls; document residual risk.

### 18.9.2 Prep work

TODO. Per course plan, Unit 6 Part 2 reference content; prelab readings/videos on safety assessment of automated manufacturing.

### 18.9.3 Materials & equipment

TODO - confirm room, equipment, software. Candidate source in the folder: LAB 7 - Supplementary Material -2.pdf, and the cross-course hand-out ENGR-3207-LAB-7-Safety Assessment in Automated Manufacturing - Part 2 (verify it belongs to this course before use).

### 18.9.4 Procedure

TODO - not in the folder.

### 18.9.5 Deliverables & submission

TODO. Expected: a completed safety-assessment report for the assigned cell.

### 18.9.6 Rubric

TODO.

### 18.9.7 Instructor notes

- **Action required:** confirm whether this lab uses the ENGR-3207 “Safety Assessment in Automated Manufacturing – Part 2” material or a manufacturing-stream handout, then populate this sub-chapter.
- 

## 18.10 Lab 9 — Process Safety & Functional Safety, Part 1

TODO - scaffold. Course-plan week 11, following Unit 7 Part 1 (Process Safety and Functional Safety). The course plan lists **prelab videos** for this lab. Expected focus: hazard identification for a process (HAZOP-style), and the idea of a safety instrumented function providing risk reduction.

### 18.10.1 Learning outcomes

TODO. Likely: apply a HAZOP guideword analysis to a simple process; identify where a safety instrumented function is needed; relate a required risk reduction to a SIL target.

### 18.10.2 Prep work

TODO. Per course plan: Unit 7 Part 1 reference content and the assigned prelab videos.

### 18.10.3 Materials & equipment

TODO - confirm room, equipment, software. Likely a computer room; a P&ID for the analysis; HAZOP worksheet template.

### 18.10.4 Procedure

TODO - not in the folder.

### 18.10.5 Deliverables & submission

TODO.

## 18.10.6 Rubric

TODO.

## 18.10.7 Instructor notes

- **Action required:** no handout for this lab is in the folder. Add the prelab video list and the worksheet/procedure from FOL.
- 

# 18.11 Lab 10 — Safety Instrumented System Architectures

An introduction to the architectures used in **Safety Instrumented Systems (SIS)** — why a safety integrity level (SIL) is *not* just a probability-of-failure calculation, and how architectural constraints (hardware fault tolerance, safe failure fraction) shape the structure. You work through the **MooN** family of architectures: 1oo1, 1oo1D, 1oo2, 1oo2D and 2oo2, comparing their dependability.

## 18.11.1 Learning outcomes

After this lab you should be able to:

- Define system, subsystem, element, component, channel and group as used in IEC 61508.
- Explain the three constraint types on a safety instrumented function — systematic capability, architectural, and probabilistic performance — and the parameters SFF, HFT and PFD.
- Describe the 1oo1, 1oo1D, 1oo2, 1oo2D and 2oo2 architectures and their effect on safe vs. dangerous failure behaviour.
- Select an architecture for a stated SIL / fault-tolerance requirement and justify it.

## 18.11.2 Prep work

- Read *Lab- 10 Supplementary Material.pdf* — *Structures of Safety Systems* (HIMA / Paul Hildebrandt GmbH): terminology, architectural constraints, the MooN architectures, and the comparison/evaluation section.
- Review Unit 7 Part 2 reference content.
- Watch the assigned prelab video.

## 18.11.3 Materials & equipment

- Computer with internet access

- Lab- 10 Supplementary Material.pdf
- *(Optional)* a reliability-block-diagram or PFD calculation sheet — **confirm**

#### 18.11.4 Procedure

TODO - the folder contains the supplementary reading only, not the exercise sheet. Expected, based on the supplement's "Comparison and evaluation" section:

1. For each of 1oo1, 1oo1D, 1oo2, 1oo2D and 2oo2, draw the block diagram and state HFT and the effect on spurious-trip vs. fail-to-danger behaviour.
2. Given a SIF with a target SIL and an HFT requirement, choose an architecture and justify it against the architectural constraints.
3. Discuss a trade-off case (e.g. 1oo2 vs. 2oo2: safety vs. availability).

#### 18.11.5 Deliverables & submission

TODO - **confirm**. Expected: a worksheet with the five block diagrams, the architecture-selection exercise with rationale, and the trade-off discussion. Submit on FOL by the end of the lab period.

#### 18.11.6 Rubric

TODO - no rubric in the folder.

#### 18.11.7 Instructor notes

- **Action required:** add the actual exercise sheet — the folder has only the HIMA reading.
  - Computer-room lab; no special equipment.
  - **Note:** an unrelated older handout, ENGR-3207-LAB-10- Introduction to Safety Instrumented Systems, covers similar ground for a different course — don't confuse the two.
- 

### 18.12 Lab 11 — Review-Week Lab

TODO - **scaffold**. Course-plan week 13 (Review). The course plan lists prelab readings and a video and a normal lab submission (~3.33 %). Topic not specified in the folder — likely a cumulative / integrative exercise or a make-up slot.

#### 18.12.1 Learning outcomes

TODO.

### 18.12.2 Prep work

TODO. Per course plan: course-review PDF; assigned prelab readings and video.

### 18.12.3 Materials & equipment

TODO - confirm room, equipment, software.

### 18.12.4 Procedure

TODO - not in the folder.

### 18.12.5 Deliverables & submission

TODO.

### 18.12.6 Rubric

TODO.

### 18.12.7 Instructor notes

- **Action required:** define this lab or confirm it is the make-up/catch-up slot.

---

## 18.13 Lab 12 — Take-Home Lab (Video Submission)

TODO - scaffold. A “take-home” / “at-home” homework lab: it can be done at any point before the due date (end of the term / week 14) and **requires a video submission**. Topic not specified in the folder.

### 18.13.1 Learning outcomes

TODO.

### 18.13.2 Prep work

TODO. Released in week 11 per the course plan — students have several weeks.

### 18.13.3 Materials & equipment

TODO. Student-supplied — done at home. A phone/camera for the required video is the only certain requirement.

### 18.13.4 Procedure

TODO - not in the folder.

### 18.13.5 Deliverables & submission

A written submission **plus a video**, on FOL, by the week-14 due date. Not subject to the in-lab late rule but still closes at end of term (grade of 0.1 after the Unit-7 cut-off for mandatory labs; Lab 12 is not in the mandatory 1–10 set — confirm).

### 18.13.6 Rubric

TODO.

### 18.13.7 Instructor notes

- **Action required:** add the take-home lab brief and video spec (length, content, what must be demonstrated on camera).
  - Release in week 11; due week 14.
- 

## 18.14 Teacher preparation

Everything the instructor needs to prep the term: the room, equipment and software per lab, then a consolidated shopping list, a software/accounts checklist, and a room-booking checklist. Items marked **[confirm]** are not stated in the source handouts.

### 18.14.1 Per-lab prep matrix



| Lab                                  | Room                                               | Equipment<br>(reusable)                                                                                                                                                                     | Software /<br>online                                                                                                                                                               | Consumables<br>& printing                                                                                                                                                                         |
|--------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Bridge                           | Activity<br>room with<br>large tables<br>[confirm] | Load-test rig<br>(supports 30<br>cm apart),<br>calibrated<br>weights /<br>sandbags +<br>loading<br>basket,<br>hot-glue<br>guns, scissors,<br>rulers/measuring<br>tapes, cam-<br>eras/phones | None (report<br>in Word)                                                                                                                                                           | Small craft<br>sticks<br>( 100/team),<br>jumbo craft<br>sticks<br>( 20/team),<br>white PVA<br>glue, hot-glue<br>sticks,<br>masking tape,<br>string/twine;<br>print rubric<br>+ report<br>template |
| 2 —<br>LightBot &<br>Rocksi          | Computer<br>lab, 1<br>PC/student<br>[confirm]      | PCs/tablets<br>with internet                                                                                                                                                                | <b>LightBot</b><br>(browser),<br><b>Rocksi</b><br>(browser);<br>prelab<br>YouTube<br>9oAFmTlMqzg                                                                                   | Print<br>worksheet<br>(optional)                                                                                                                                                                  |
| 3 — Design<br>documenta-<br>tion     | Computer<br>lab with full<br>Office<br>[confirm]   | PCs                                                                                                                                                                                         | <b>Microsoft</b><br><b>Word/Visio</b> ,<br><b>Microsoft</b><br><b>Project</b> ,<br><b>Microsoft</b><br><b>Excel</b>                                                                | None;<br>re-upload<br>FOL<br>templates:<br>BOM .xlsx,<br>Gantt .mpp,<br>HSE .docx                                                                                                                 |
| 4 — Facility<br>design &<br>AnyLogic | Computer<br>lab<br>[confirm]                       | PCs (check<br>AnyLogic<br>PLE<br>model-size<br>limit)                                                                                                                                       | <b>AnyLogic</b><br><b>Personal</b><br><b>Learning</b><br><b>Edition</b> ,<br>Microsoft Vi-<br>sio/Project/Excel;<br>AnyLogic<br>demo model<br>[confirm<br>which] +<br>prelab video | None                                                                                                                                                                                              |

| Lab                                          | Room                                                                                     | Equipment<br>(reusable)                                | Software /<br>online                                                                                                    | Consumables<br>& printing                                                                                                                                                                          |
|----------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 — Lean<br>simulation                       | Room for 2<br>parallel lines<br>+ tables<br>[confirm]                                    | 2 stop-<br>watches/timers,<br>class set of<br>scissors | None                                                                                                                    | <b>Pink A4,</b><br><b>yellow A4,</b><br><b>blue pens,</b><br><b>red pens;</b><br>print product<br>check sheets,<br>product<br>images, order<br>tickets, QC<br>checklists<br>(per team +<br>spares) |
| 6 —<br>Ergonomics                            | Computer<br>lab with<br>internet<br>[confirm]                                            | PCs                                                    | <b>Inseer</b> web<br>tool (class ac-<br>counts/login<br>— arrange<br>early);<br>browser;<br>ergonomics +<br>REBA videos | Print <b>REBA</b><br><b>worksheets</b><br>+ <b>REBA</b><br><b>cheat sheet</b><br>(per student);<br>stage task<br>clips/photos<br>(Lifting<br>Boxes,<br>Workstation,<br>Cleaning) on<br>FOL         |
| 7 — Robot<br>safety Part 1                   | Computer<br>room;<br>[confirm]<br>cobot/light-<br>curtain cell if<br>demoed<br>[confirm] | PCs; optional<br>cobot cell +<br>qualified<br>operator | Browser<br>(OSHA OTM<br>§4 ch.4);<br>prelab videos                                                                      | Print<br>worksheet<br>[confirm] —<br><b>not in</b><br><b>folder]</b>                                                                                                                               |
| 8 — Robot<br>safety Part 2                   | [confirm]                                                                                | [confirm]                                              | [confirm]                                                                                                               | [confirm] —<br>handout not<br>in folder                                                                                                                                                            |
| 9 — Pro-<br>cess/functional<br>safety Part 1 | Computer<br>room<br>[confirm]                                                            | PCs                                                    | Prelab videos<br>[confirm];<br>P&ID +<br>HAZOP<br>worksheet<br>[confirm]                                                | [confirm] —<br>handout not<br>in folder                                                                                                                                                            |

| Lab                        | Room                          | Equipment<br>(reusable) | Software /<br>online                                             | Consumables<br>& printing                                                                       |
|----------------------------|-------------------------------|-------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 10 — SIS<br>architectures  | Computer<br>room<br>[confirm] | PCs                     | Browser<br>(HIMA<br>reading);<br>optional<br>PFD/RBD<br>sheet    | Print<br>exercise sheet<br>[confirm —<br><b>only the<br/>reading is<br/>in the<br/>folder</b> ] |
| 11 —<br>Review-week<br>lab | [confirm]                     | [confirm]               | Course-<br>review PDF;<br>prelab video<br>[confirm]<br>[confirm] | [confirm] —<br>not defined<br>in folder                                                         |
| 12 —<br>Take-home<br>lab   | None (done<br>at home)        | Student<br>phone/camera |                                                                  | None —<br><b>video<br/>submission<br/>required</b>                                              |

### 18.14.2 Consolidated shopping list (buy / restock each term)

#### Lab 1 — bridge build

- ☐ Small (regular) craft/popsicle sticks — 100 per team  $\times$  (number of teams) + ~20 % spares
- ☐ Jumbo craft sticks — 20 per team  $\times$  (number of teams) + spares
- ☐ White PVA glue — 1 bottle per team
- ☐ Hot-glue sticks — 1 pack per 2 teams + spares
- ☐ Masking tape — 1 roll per team
- ☐ String / cotton twine — 1 ball per ~4 teams

#### Lab 5 — lean simulation

- ☐ A4 paper, **pink** — 1 ream
- ☐ A4 paper, **yellow** — 1 ream
- ☐ Ballpoint pens, **blue** — 1 box
- ☐ Ballpoint pens, **red** — 1 box

#### Shared / reusable — check stock, replace if missing

- ☐ Scissors — class set (Labs 1, 5)
- ☐ Rulers / measuring tapes — 1 per team (Lab 1)
- ☐ Stopwatches / timers — 1 per team (Lab 5; phones acceptable)
- ☐ Bridge load-test rig + calibrated weight set / sandbags + loading basket (Lab 1)
- ☐ Hot-glue guns — count vs. number of teams (Lab 1)

**Printing (per term)**

- ☐ Lab 1 rubric + group report template
- ☐ Lab 5 product check sheets, product image sheets, order tickets, QC check-lists
- ☐ Lab 6 REBA worksheets + REBA cheat sheet (one per student)
- ☐ Lab 2 worksheet (optional)
- ☐ Worksheets for Labs 7–11 once the procedures are confirmed

**18.14.3 Software, licences & accounts — set up before the term**

- ☐ **AnyLogic Personal Learning Edition** installed on every Lab 4 PC; run a full test of the demo model (watch the PLE size limit)
- ☐ **Microsoft Project** and **Microsoft Visio** licences live on Lab 3 / Lab 4 PCs (most likely gap in a standard Office image)
- ☐ **Microsoft Excel / Word** available on Lab 3 / Lab 4 PCs
- ☐ **Inseer** class accounts / login created and tested (Lab 6) — confirm seat count and upload workflow
- ☐ **LightBot** and **Rocksi** confirmed reachable through the lab network/firewall (Lab 2); offline fallback prepared
- ☐ FOL templates re-uploaded: BOM **.xlsx**, Gantt **.mpp**, HSE risk-assessment **.docx** (Lab 3)
- ☐ Task clips / photo sets staged on FOL for Lab 6 (Lifting Boxes, Workstation, Cleaning)
- ☐ AnyLogic demo model file staged on FOL / lab PCs (Lab 4)
- ☐ Prelab video links checked for Labs 1, 2, 3, 4, 5, 6, 7, 9, 10, 11

**18.14.4 Room bookings**

- ☐ **Activity room with large tables** — Labs 1 and 5 (Lab 5 needs space for two parallel lines with an “inventory” area each); Lab 3 if not done in the computer lab
- ☐ **Computer lab** with internet + full Microsoft Office + Project + Visio + AnyLogic PLE — Labs 2, 3, 4, 6, 10, and 7/9/11 (pending confirmation)
- ☐ **Robot cell / cobot demo area** with a qualified operator — Lab 7 only, if the light-curtain / envelope walk-around is run live
- ☐ **[confirm]** rooms for Labs 8, 9, 11

**18.14.5 Appendix: instrumentation-trainer lab stream**

An older set of labs for this course code was written for the **B1033** trainer benches (author “MZhang”). They are **not** part of the manufacturing stream above, but the handouts are in the folder and are summarised here in case the course runs them:

| Alt lab                   | Title                                  | Room  | Equipment                                                                                                                                                                          | Software                                                                            |
|---------------------------|----------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Alt — Pres-<br>sure/level | Pressure<br>Measurement<br>for Level   | B1033 | Trainer:<br>power-supply<br>module,<br>pump/tank<br>unit, RTD<br>temperature<br>transmitter,<br>pressure<br>transmitter,<br>float/reed<br>switch, 2 ×<br>DMM,<br>banana leads      | —                                                                                   |
| Alt — Motor<br>control    | Motor<br>Control &<br>Intro to<br>VFDs | B1033 | 3-phase<br>motor,<br>FVNR<br>(full-voltage<br>non-<br>reversing)<br>starters,<br>control-<br>relay/MCR<br>wiring,<br>E-stop, pilot<br>lights,<br>pushbuttons;<br>VFD               | —                                                                                   |
| Alt — PLC                 | PLC<br>Introduction<br>with Siemens    | B1033 | Siemens PLC<br>trainer, field<br>devices<br>(pushbuttons,<br>selector<br>switches,<br>pilot lights,<br>cylinders<br>with proxy<br>sensors), po-<br>tentiometer<br>0–10 VDC,<br>DMM | <b>Siemens<br/>TIA Portal;</b><br>reference<br>video<br><a href="#">Y0vQhvjaeqY</a> |

| Alt lab            | Title                    | Room  | Equipment                                                                                                                                                                    | Software |
|--------------------|--------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Alt — Flow<br>rate | Flow Rate<br>Measurement | B1033 | Trainer:<br>pump/tank<br>unit,<br>paddle-wheel<br>flow<br>transmitter,<br>rotameter<br>(LPM/GPM),<br>RTD<br>temperature<br>transmitter,<br>2 × DMM,<br>hoses/banana<br>leads | —        |

Prep for the alt stream: book **B1033**; confirm every trainer bench has a working power-supply module, pump/tank unit and the two DMMs; confirm **Siemens TIA Portal** licences for the PLC lab; print each handout's instructor sign-off sheet (4 sign-offs per lab).

# Bibliography